

EXHALE: Update Log

***Purpose.** This document is a running log of incremental updates to EXHALE as the code is extended toward reproducing Huang et al. (2023) WASP-121b (see Huang_update_plan.md). Each update is a self-contained section: what changed, why, which files were touched, and the quantitative check that was run. New updates are appended as new top-level sections.*

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1. Runtime domain-extent option: Roche vs Spherical

Added 2026-06-04.

1.1 Overview

ATES historically hardcodes the full Roche potential (planetary gravity plus the stellar-tidal and centrifugal terms) and truncates the radial grid at the Hill/L1 radius. For close-in planets this Roche lobe is small; for WASP-121b the outer boundary lands at $r_{\max} \approx 2.01 R_p$. Huang et al. (2023) Case A, by contrast, is a *spherical* model (no tidal terms) extending to $\sim 19 R_p$.

To support both geometries from a single executable, a runtime option was added with two modes:

- **Roche** (default): full Roche potential, domain truncated at the Hill/L1 radius. Identical to the previous ATES behavior.
- **Spherical**: pure planetary potential $\phi = -b_0/r$ (tidal and centrifugal terms dropped), domain extended to a user-specified outer radius.

The mode is selected by two optional lines appended to `input.inp`; when they are absent the code reverts to Roche, so pre-existing input files are unaffected.

1.2 The two domain modes

1.2.1 Gravitational potential

All quantities are in code-normalized units: radius r in units of the planetary radius $R_0 = R_p$, with $b_0 = GM_p\mu/(k_B T_0 R_0)$ the surface Jeans parameter, $M_r = M_*/M_p$ the star-to-planet mass ratio, and $\tilde{a} = a_{\text{orb}}/R_0$ the orbital distance in planetary radii.

Roche mode. The dimensionless potential and its gradient are

$$\phi(r) = -\frac{b_0}{r} - \frac{b_0 M_r}{\tilde{a} - r} - \frac{b_0(1+M_r)}{2\tilde{a}^3} \left(\frac{\tilde{a} M_r}{1+M_r} - r \right)^2, \quad (1)$$

$$\frac{d\phi}{dr} = \frac{b_0}{r^2} - \frac{b_0 M_r}{(\tilde{a} - r)^2} + \frac{b_0(1+M_r)}{\tilde{a}^3} \left(\frac{\tilde{a} M_r}{1+M_r} - r \right). \quad (2)$$

The three terms are planetary gravity, stellar tidal attraction, and the centrifugal potential of the co-rotating frame.

Spherical mode. Only the planetary term is kept:

$$\phi(r) = -\frac{b_0}{r}, \quad \frac{d\phi}{dr} = \frac{b_0}{r^2}. \quad (3)$$

1.2.2 Outer boundary

- **Roche**: $r_{\max} = (3M_r)^{-1/3} \tilde{a}$, the Hill-radius approximation to the L1 distance (unchanged from before).
- **Spherical**: $r_{\max}$ is set directly by the user-supplied “Outer radius [R_p]” value (in R_p , since the grid is normalized to $R_0 = R_p$).

1.2.3 Why spherical mode must drop the tidal terms

The outer boundary cannot simply be pushed past L1 while keeping the full Roche potential. Beyond L1 the Roche potential *turns over*: the effective gravity changes sign (points toward the star), so a monotonic, steadily accelerating outward wind is no longer defined there and the sonic-point/steady-state treatment breaks down. Removing the stellar-tidal and centrifugal terms restores a monotonic potential ($\phi = -b_0/r$ increases monotonically toward zero as $r \rightarrow \infty$), which is what permits the domain to extend arbitrarily far. This is exactly the spherical geometry of Huang Case A. A guard enforces a sensible configuration: selecting Spherical mode without an “Outer radius [R_p]” > 1 stops the run with an explicit error rather than silently falling back.

1.3 Input interface and backward compatibility

Two optional lines are appended at the end of `input.inp`, after the “Force start.” line:

Domain mode: Spherical

Outer radius [R_p]: 19.0

- “Domain mode:” accepts Spherical; any other value (or a missing line) means Roche.
- “Outer radius [R_p]:” is read only in Spherical mode; it is ignored in Roche mode, where the Hill radius is computed automatically.

The reader scans the trailing lines by *keyword* (using `index(line, . . .)`), not by fixed line position, so blank trailing lines and line ordering do not matter, and `input.inp` files written before this change (which lack both lines) keep the default Roche behavior. The active mode is echoed in the setup report `EXHALE_setup.out`.

1.4 Implementation

Touched files.

- `src/modules/init/parameters.f90`: new globals `spherical_domain` (logical, default `.false.`) and `r_out_user` (the outer radius in R_p).
- `src/modules/functions/grav_field.f90`: `phi` and `Dphi` branch on `spherical_domain`.
- `src/modules/files_IO/input_read.f90`: keyword scan of the trailing lines, and the $r_{\max}$ branch with the validity guard.
- `src/modules/files_IO/write_setup_report.f90`: prints the active mode to `EXHALE_setup.out`.

Gravity branch.

```
double precision function phi(r_in)
  real*8, intent(in) :: r_in
  if (spherical_domain) then
    phi = -b0/r_in
```

```

else
    phi = -b0/r_in &
    -b0*Mrapp/(atilde-r_in) &
    -b0*(1.0+Mrapp)/(2.0*atilde**3.0)* &
    (atilde*Mrapp/(1.0+Mrapp)-r_in)**2.0
endif
end function phi

```

1.5 Validation on WASP-121b

Two runs were made that differ *only* in the domain-extent option; `input.inp` was otherwise identical (power-law SED, He/H = 0.0851, trace metals C/N/O on, “Rate/2 + Mdot/2” 2-D flux approximation). Both reached steady state via the proper criterion (momentum constant, $du < du_{\text{th}}$), not a stall or a false convergence.

mode	$r_{\text{max}} [R_p]$	$\log_{10} \dot{M}$	sonic [R_p]	$r(50\% \text{ HII})$	$T_{\text{peak}} [\text{K}]$
Roche (default)	2.01	12.68	1.825	1.178	9662
Spherical	19.00	11.81	5.270	1.152	10752
Huang Case A (ref.)	~ 19	12.57	—	—	~ 12000

$\dot{M}$ in $\log_{10} \text{ g s}^{-1}$. Convergence took $\approx 5,392$ iterations (Roche) and $\approx 154,183$ (Spherical, the larger domain).

Observations (for evaluation, not pass/fail).

- The Roche run reproduces the prior Phase-0 Case A result exactly ($\log_{10} \dot{M} = 12.68$), confirming the new optional block does not perturb default behavior.
- Spherical T_{peak} (10,752 K) is higher than Roche (9,662 K) and closer to Huang’s $\sim 12,000$ K — the extended domain lets the profile develop past the $2-R_p$ L1 clip.
- Spherical $\dot{M}$ (11.81) is lower than both Roche (12.68) and Huang (12.57), consistent with the loss of tidal enhancement, but ~ 0.76 dex below Huang.
- The sonic point moves outward from 1.83 to $5.27 R_p$ (gentler acceleration without tidal forces).

Caveats. Two confounders remain before the spherical $\dot{M}/T$ can be compared to Huang cleanly: (a) only C/N/O metals are included here, not Huang’s full Mg/Fe/Si/Ca/Na/K/S set (Phase 1); and (b) the same $N = 500$ cells now span $1 \rightarrow 19 R_p$, so the inner resolution is much coarser than in the Roche run — a denser-grid spherical re-run may be needed before drawing conclusions.

1.6 Status and relation to the Huang plan

This option closes the Phase-0 geometry gap: the Roche mode is the validated baseline, and the Spherical mode provides the Case A geometry needed for later comparison. The tidal RLOF physics of Huang Phase 4 is a separate, richer

treatment (substellar streamline, L1 location, terminator mapping); the present Roche mode is the existing single-streamline truncation, not the full RLOF model. The Spherical mode is the immediate tool for Case A reproduction.

2. Ionization network: add magnesium (Mg I / II / III)

Added 2026-06-05.

2.1 Overview

The first trace metal beyond C/N/O, following the Huang et al. (2023) WASP-121b species set (Mg, Fe, Si, O, C, N, S, Ca, Na, K). Per the Phase-1 plan, elements are added one at a time (order $\text{Mg} \rightarrow \text{Fe} \rightarrow \text{Si} \rightarrow \text{Ca} \rightarrow \text{Na} \rightarrow \text{K} \rightarrow \text{S}$); this section covers **Mg only**. Magnesium is solved to second ionization (Mg I / Mg II / Mg III) inside the existing coupled MINPACK ionization system, raising the species count to 18 and the ratio-unknown count to 11. The change is confined to the ionization/radiation plumbing: there are no hydrodynamic or mass-density changes, and `calc_ne/calc_rho` still count only H/He — metal electrons enter solely inside the MINPACK system.

2.2 Atomic data and rates

Sourced, not fabricated:

- **Photoionization:** Verner et al. (1996) outer-shell fits, with Mg I threshold 7.646 eV and Mg II threshold 15.035 eV (`cross_sec.f90`: `sigma_MgI`, `sigma_MgII`).
- **Recombination:** Badnell RR+DR — the same source already used for C/N/O — via `lookup_rec` cases and `rec_MgII` ($\rightarrow$ Mg I) / `rec_MgIII` ($\rightarrow$ Mg II). This is a *deliberate deviation* from the validation-rate prescription in `Huang_update_plan.md` §1.3 (Shull & van Steenberg 1982 for Mg); it corresponds to the plan's Phase-6 *production* rates, applied early to match the existing C/N/O treatment. The Mg ion/neutral balance below should be read as the Badnell result, not a Huang-faithful one.
- **Collisional ionization:** Voronov (1997) fits, `ion_coeff_MgI` and `ion_coeff_MgII`.
- **Line cooling:** *not yet ported* — `cool_MgI` = `cool_MgII` = 0. No published Mg two-level/CHIANTI cooling fit is implemented (that is Phase 2). Magnesium therefore acts only as an electron donor and a photoelectric-heating channel here, not as a coolant.
- **Charge exchange Mg + H⁺:** omitted in this first cut.

Grid extension below 13.6 eV. Because Mg I ionizes at 7.646 eV, turning Mg on triggers the existing low-IP-metal branch (`thereis_lowIP_metal`), which extends the energy grid and the stellar SED down to 7.646 eV (`set_energy_vectors.f90`). This is required to photoionize Mg I, but has a side effect noted under Caveats.

2.3 Touched files

- `init/parameters.f90`: `n_species = 18`; `e_th_MgI/e_th_MgII`; `X_Mg`; `s_mgi/s_mgii`; `thereis_lowIP_metal`.
- `functions/cross_sec.f90`: `sigma_MgI`, `sigma_MgII`.
- `radiation/Cool_coeff.f90`: `Badnell fits`, `lookup_rec`, `rec_MgII/rec_MgIII`, `Voronov ion_coeff_MgI/MgII`, `cool_MgI/MgII (= 0, flagged)`.
- `radiation/util_ion_eq.f90`: `PH_heat_HHe` and `eval_cool` carry `Mg`.
- `radiation/ionization_equilibrium.f90`: `Mg densities`, `params slots 49–55`, `MINPACK unknowns 10/11 (Mg II/Mg III ratios)`, `solution unpack`.
- `init/set_energy_vectors.f90`: `7.646 eV sub-grid`; `allocate/fill s_mgi/s_mgii`.
- `files_IO/input_read.f90`: `X_Mg default`; `N_eq = 11`; `metals/low-IP flags`.
`files_IO/metals_input_read.f90`: `MgI abundance key`. `files_IO/write_output.f90`: `Mg I/II/III as Ion_species.txt columns 17–19`.
- Threading through arrays, I/O and post-processing: `init/set_IC.f90`, `files_IO/load_IC.f90`, `init/init.f90`, `EXHALE_main.f90`, `time_step/energy_semi_implicit.f90`, `functions/utilities.f90`, `post_process/post_process_adv.f90`.

`post_process_adv.f90` also received a latent-bug fix: its advection systems fill only the H/He equations, so they now use a local `Neq_adv` rather than the global `N_eq = 11`, which would otherwise feed `hybrd1` uninitialized metal entries.

2.4 Validation on WASP-121b

Four runs with identical `input.inp` (power-law SED, $\text{He/H} = 0.0851$, Roche domain), differing only in which metals are present. All reached steady state via $du < du_{\text{th}}$. Mg/H is conserved at the input $X_{\text{Mg}} = 3.98 \times 10^{-5}$ across the domain.

run	metals	grid floor	$\log_{10} \dot{M}$	T_{peak} [K]	$r(T_{\text{peak}})$
no metals	H/He only	13.6 eV	12.92	13722	1.703
Mg only	Mg	7.646 eV	12.92	13682	1.697
C/N/O	C, N, O	13.6 eV	12.68	9662	1.199
C/N/O + Mg	C, N, O, Mg	7.646 eV	12.76	9828	1.224
Huang Case A	full set	—	12.57	~ 12000	—

$\dot{M}$ in $\log_{10} \text{g s}^{-1}$; r in R_{p} .

Observations (for evaluation, not pass/fail).

- **Clean Mg isolation** (Mg only vs no metals): $\Delta \log_{10} \dot{M} = +0.000$ dex, $\Delta T_{\text{peak}} = -40$ K. Adding magnesium alone leaves the wind essentially unchanged, as expected for a 4×10^{-5} trace species with no line cooling: it neither heats nor cools appreciably, and the soft-UV band it adds is optically thin without other absorbers.
- **Mg ionization structure**: predominantly Mg II (Mg^+) through the lower thermosphere, predominantly Mg III (Mg^{2+}) in the upper thermosphere (Mg III fraction ~ 0.47 at $1.4 R_p$ rising to ~ 0.81 at $1.9 R_p$); neutral Mg I stays subdominant. Same qualitative ordering as Huang+2023 Fig. 12.
- **MINPACK convergence**: the 11-unknown system converges without NaNs across the full temperature range.
- Mg-only $\dot{M}$ (12.92) is $\sim 2.2 \times$ the Huang Case A spherical value (12.57), in the same range as the prior C/N/O Roche run; the metal-set and geometry confounds from Phase 0 still apply.

Caveats and gaps.

- **C I grid-extension confound** (C/N/O + Mg vs C/N/O). The $\dot{M}/T$ offset between these two (+0.080 dex, +165 K) is *not* a magnesium effect: extending the grid to 7.646 eV exposes **C I** (threshold 11.26 eV) to the new 11.26–13.6 eV photons the 13.6 eV-floor C/N/O run never sees, adding C I photoionization heating. This is why the clean Mg measurement uses Mg-only vs no-metals, where C is absent.
- **No Mg line cooling** ($\text{cool_MgI} = \text{cool_MgII} = 0$). Huang’s headline Mg result — Mg II h&k emission dominating the cooling at $1.15\text{--}1.4 R_p$ — cannot appear here by construction. That is Phase 2.
- **No Mg + H^+ charge exchange** yet.
- **Recombination is Badnell**, not the §1.3 validation rate.

2.5 Status and relation to the Huang plan

This completes the **Mg** step of Phase 1 (network expansion). The ionization structure satisfies the qualitative Fig. 12 ordering and the “ $\dot{M}$ within a factor ~ 2 of Case A” part of the gate; the thermal-balance part of Phase 1 is deferred to Phase 2 (Mg cooling). Next element in the planned order: **Fe**. Scratch runs and the comparison notebook live in `mg_validation/`.

3. Phase 1a: interface / data-structure refactor (no new physics)**3.1 Overview**

Each metal added so far (C/N/O, then Mg in §2) widened the named-argument lists of `PH_heat_HHe`, `eval_cool`, `calc_column_dens_metals`, `write_output`, the MINPACK system `ion_system_HeH_metals`, and the driver `ionization_equilibrium.f90` by several arguments for each ion, and threaded the MINPACK coefficients for each cell through a hand-indexed flat `params(60)`. This step converts that plumbing to a **species-metadata table** +

array-indexed (2D) data flow, so that adding an element becomes “add rows to a table” instead of “thread N new arguments through M routines.” **No physics changes** — the network is still C/N/O + Mg, and the result is gated bit-for-bit against §2 (below).

3.2 What changed (Steps A–F)

- **A — Metadata module** (`init/species_table.f90`, new). One canonical table holds metadata for each ion (length `n_mion` = 12): `f_sp` column (`mion_fsp`), parent element (`mion_elem`), stage / charge² (`mion_stage`, `mion_z2`), photo-ionizable flag and photo-table column (`mion_isphot`, `mion_iphot`), threshold (`mion_ethr`), cooling flag (`mion_iscool`); and metadata for each element (length `n_melem` = 4): nuclear charge (`melem_Z`), neutral-ion index (`melem_i0`), top stage (`melem_top`). The canonical ion order (C I..Mg III = `f_sp` columns 7–18) is defined here once; element indices `iel_C/O/N/Mg`.
- **B — write_output 2D**. Metal ion densities pass as `nm(:, 1:n_mion)` and are written with an implied-do over the table `((nm(j,i)*n0, i = 1,n_mion))`, not a named write for each ion.
- **C — PH_heat_HHe + column density 2D**. Optical depth, photoheating, and the photoionization rates loop over photo-ionizable ions via `sigma_tab(:,mion_iphot(i))` and `Nm_col`, returning a 2D rate array `P_m(:, 1:n_mion)` (inert top stages stay zero) instead of separate `P_CI..P_MgII` arguments, one for each ion.
- **D — eval_cool metal channels 2D**. Recombination and collisional-ionization coefficients return as 2D arrays (`rec_m`, `aion_m`); bremsstrahlung sums `mion_z2(i)*GF_elem(:,mion_elem(i))*nm(:,i)` over ions. No separate coefficient arguments for each ion.
- **E — Generalized MINPACK system**. `ion_system_HeH_metals` keeps `hybrd1`’s fixed (`N_eq,x,fvec,iflag,params`) signature but (i) receives the metal coefficients for each cell through a **module-level block** in `System_HeH_metals` (`set_metal_coeffs` stores `met_ntot/g0/g1/b0/b1/a1/a2`), set by the driver before each `hybrd1` call — safe because the ionization-equilibrium cell loop is serial — and (ii) assembles `fvec` by **looping over elements** (two balance equations each, force-zeroing absent elements). `params` now carries only H/He (1–11), He ITR (12–18), and the C/N/O charge-transfer rates (40–48); the old slots 49–55, one for each element, are gone. The driver builds metal densities, initial guesses, and the solution unpack by looping over `n_melem/n_mion`. The system size is now the formula $N_{eq} = 3 + 2*n_{melem}$ (= 11 for the current four elements), set in `input_read.f90`.
- **F — Generalized grid floor**. `set_energy_vectors.f90` no longer hardwires the sub-13.6 eV floor to Mg’s 7.646 eV; it computes `e_sub_low` = min over active elements of the neutral threshold `mion_ethr(melem_i0(e))` that lies below `e_th_HI`. The trigger flag `thereis_lowIP_metal` is set generically in `input_read.f90` (any active element with a sub-13.6 eV neutral threshold), driven by a new canonical-order abundance array `melem_ab` (declared in `parameters.f90`, populated in `input_read.f90`). For the current set only C I (11.26 eV) and Mg I (7.646 eV) lie below 13.6 eV and the minimum is Mg I, so the floor is unchanged at 7.646 eV when Mg is present. (Consequence: a future C/N/O-without-Mg run would now extend the grid to 11.26 eV, which it did not before)

— and once Phase 1b adds Na/K/Ca, whose neutral thresholds 5.14/4.34/6.11 eV sit below Mg's, the floor drops further.)

3.3 Gate: bit-for-bit reproduction of the C/N/O + Mg run

With C/N/O + Mg active and the same input `.inp/metals.inp` as §2, the refactored code reproduces the `mg_validation/cno_mg` reference to round-off: $\log_{10} \dot{M} = 12.76$; `Hydro_ioniz.txt`, `Hydro_ioniz_adv.txt`, and `Ion_species.txt` are byte-identical to the frozen reference. `Ion_species_adv.txt` differs at exactly one number — line 1 (innermost ghost cell), column 7 (the HeITR fraction): the reference holds 4.0048×10^{-5} left in uninitialized stack memory while the deterministic build writes 0.0; every other column of that line and every other line match. This is a pre-existing reference artifact (a HeITR ghost-cell slot never used in the H/He + metals run), not an effect of the refactor.

3.4 What is deliberately *not* generalized yet

- **Charge exchange with H** is still hardcoded for C/N/O only (the `fvec(1)` C,N,O sum and the `cxlo/cxup` mapping via `iel_C/O/N` in `System_HeH_metals`). Generalizing it is Phase 1d.
- **Atomic data for each ion and abundance for each element remain manual entries** — the `sigma_tab(:, 1..8)` assignments and the `sigma_XX` cross-section functions in `set_energy_vectors.f90`, the abundance scalars `X_C/X_O/X_N/X_Mg`, the `metals.inp` keys, and the `melem_ab` population block. These are the atomic-data / abundance rows expected to grow per element in Phase 1b/1c; they are data, not interface plumbing.

3.5 Touched files

- **New:** `init/species_table.f90` (Step A).
- `init/parameters.f90`: `melem_ab(:)` abundance array; generalized `thereis_lowIP_metal` comment.
- `files_IO/input_read.f90`: `N_eq = 3 + 2*n_melem`; `melem_ab` population; generalized low-IP-metal trigger.
- `files_IO/write_output.f90`: 2D metal-column writer (B).
- `radiation/util_ion_eq.f90`: `PH_heat_HHe` + column density 2D (C); `eval_cool` metal channels 2D (D).
- `nonlinear_system_solver/System_HeH_metals.f90`: module-level metal block for each cell + element-loop residual (E).
- `radiation/ionization_equilibrium.f90`: metadata-driven density build, params packing (H/He + charge transfer only), initial guesses, solution unpack (E).
- `init/set_energy_vectors.f90`: min-over-active-metals grid floor (F).

3.6 Status and relation to the Huang plan

Phase 1a is complete: the element-by-element plumbing of §2 is now table-driven. Adding the Phase-1b metals should require only metadata rows plus atomic/abundance data, with **no changes to `System_HeH_metals.f90` or `ionization_equilibrium.f90`**. Next: **Phase 1b** — Si, Ca, Na, K, S added together as a uniform-template batch. Gate scratch run: `/tmp/ates_gate/cno_mg/` vs frozen reference `mg_validation/cno_mg/output/`.

4. Phase 1b: correct C/N/O cross sections and add Si, Ca, Na, K, S

4.1 Overview

Two changes are made together here, both enabled by the Phase 1a (§3) table-driven plumbing:

1. **Corrected the C/N/O photoionization cross sections** to the full Verner, Ferland, Korista & Yakovlev (1996; VFKY96) Eq. (1) form. The previous code used the simpler Verner & Yakovlev (1995) inner-shell form (no y_0/y_1 offset/asymptote parameters) for C/N/O; that form is exact only when $y_0 = y_1 = 0$, which does not hold for the C/N/O outer shells. This is the dominant *physics* change.
2. **Added the five uniform-template metals Si, Ca, Na, K, S** in one batch. With Mg (§2) the in-code element set is now C, N, O, Mg, Si, Ca, Na, K, S — the full Huang+2023 §2.4 set *except Fe* (Phase 1c). Counts: `n_species = 30, n_mion = 24, n_melem = 9`.

The metal MINPACK solver was also **renamed** (below).

4.2 Variable-stage solver (2- and 3-stage elements in one packing)

Huang+2023 carries Si, Ca (like Mg) to second ionization but Na, K, S only to first. The solver keeps a uniform packing of **two unknowns for each element** ($N_{eq} = 3 + 2 \times n_{melem} = 21$ with all nine elements) and a `melem_top` flag for each element:

- `melem_top = 2` (C, O, N, Mg, Si, Ca): solve both $X^0 \leftrightarrow X^+$ and $X^+ \leftrightarrow X^{2+}$.
- `melem_top = 1` (Na, K, S): solve only $X^0 \leftrightarrow X^+$ and **pin the unused upper unknown to zero** (`fvec(ix+1) = x(ix+1)`), keeping the Jacobian non-singular. Every `ix+1` access is guarded by `melem_top` so a two-stage element never touches its neighbor's column.

Zero-abundance elements are still force-zeroed, so the same 21-unknown system solves C-only, C/N/O+Mg, or all nine without renumbering.

4.3 Atomic data and rates (sources, not fabricated)

- **Photoionization cross sections** (`functions/cross_sec.f90`): full VFKY96 Eq. (1) form (`sigma_VFKY96`) for C I/II, N I/II, O I/II, Si I/II, Ca I/II, S I; simpler Verner & Yakovlev 1995 form (`sigma_Verner96`, $y_0=y_1=0$) for Mg I/II, Na I, K I (their published fits have $y_0=y_1=0$, so the forms agree exactly). Neutral thresholds [eV]: Si I 8.152, Ca I 6.113, Na I 5.139, K I 4.341, S I 10.36.

- **Collisional ionization:** Voronov (1997) for all new ions.
- **Recombination:** Badnell RR+DR via the swappable dispatcher, as for C/N/O and Mg — the *same deliberate deviation* from the Huang_update_plan.md §1.3 validation rates already accepted for Mg (the Phase-6 production rate applied early). Read the balances as the Badnell result.
- **Line cooling:** *not* ported (`mion_iscool = .false.`), flagged To Be Checked; the five act only as electron donors and photoelectric-heating channels (Phase 2).
- **Charge exchange with H:** *not* added; still C/N/O only (Phase 1d).

4.4 Rename: System_HeHCO → System_HeH_metals

The metal solver was named after the AIOLOS heritage (AIOLOS tracked only C and O as metals $\Rightarrow$ “HCO” = H, C, O). It now carries H + He + nine metals, so the C/O-specific name was misleading. Renamed to parallel the existing System_HeH/System_HeH_TR family: `module System_HeHCO → System_HeH_metals`; `ion_system_HeHCO → ion_system_HeH_metals`; `set_hco_metals → set_metal_coeffs`; `hco_* → met_*`. Pure rename, no behavior change (§3 above is updated to the new names).

4.5 Validation on WASP-121b

Three converged runs, identical planet / SED / grid (power-law SED, He/H=0.0851, Roche domain, “Rate/2 + Mdot/2”); they differ only in code version and metal set.

key	code	metals	C/N/O σ	floor [eV]	$\log_{10} \dot{M}$	T_{peak} [K]	steps
ref	old	C,N,O,Mg	Verner96	7.646	12.76	9828	5003
A	new	C,N,O,Mg	VFKY96	7.646	12.85	12197	15382
B	new	+Si,Ca,Na,K,S	VFKY96	4.341	12.85	12135	15505

$\dot{M}$ in $\log_{10}$ g/s; Huang+2023 Case A reference $\log_{10} \dot{M} = 12.57$. ref and A share the 7.646 eV grid floor (Mg sets the minimum in the C/N/O+Mg set), so ref vs A isolates the C/N/O cross-section change with no grid-floor confound; A vs B changes both the element set and the floor (K I pulls it to 4.341 eV).

Observations (for evaluation, not pass/fail):

- **C/N/O correction (ref→A):** $\Delta \log_{10} \dot{M} = +0.090$ dex; T_{peak} rises $9828 \rightarrow 12197$ K and the peak moves outward from 1.22 to $\sim 1.90 R_p$. The corrected VFKY96 opacity deposits more photoheating; the hotter peak now sits near Huang’s ~ 12000 K. A genuine model change, not numerical.
- **Adding Si/Ca/Na/K/S (A→B):** $\Delta \log_{10} \dot{M} = +0.000$ dex, $\Delta T_{\text{peak}} = -62$ K, despite the floor dropping to 4.341 eV — at solar abundance and with no line cooling these five are radiatively near-passive donors.
- **Mg moves indirectly.** Mg’s own cross sections are byte-identical between ref and A, yet its ionization fractions shift by up to ~ 0.1 : an indirect response to the hotter, more extended profile (changed n_e , optical depth, photoionizing flux), not a change in Mg atomic data. Mg is coupled to the global state, hence *not* an isolated control.

- **Numerics.** Zero NaN/Inf; the 21-unknown system converges over the full T range; X/H for each element is conserved to $\sim 10^{-13}$ for all nine elements including the two-stage Na/K/S — direct evidence the variable-stage packing does not leak between element columns.
- **New-metal ionization (run B).** Si, Ca mostly *doubly* ionized in the upper thermosphere; Na, K, S mostly *singly* ionized (no second stage carried) — the qualitative Huang+2023 Fig. 12 ordering.

4.6 Caveats and gaps

- **No metal line cooling** for Si/Ca/Na/K/S (Phase 2); the thermal structure will not match Huang until Fe II/Mg II/Ca II cooling is in.
- **No charge exchange with H** for the new elements or Mg (Phase 1d).
- **Recombination is Badnell, not the §1.3 validation rate** (deliberate, as Mg).
- $\dot{M}$ is $\sim 1.9 \times$ Huang Case A (12.85 vs 12.57); the metal-set (Fe still missing) and geometry confounds from Phase 0 still apply.
- **Not done here:** Huang Fig. 12 is not digitized/overlaid (only the qualitative ordering is checked); the atomic-data values are not re-derived beyond conservation and smoothness.

4.7 Regression reference for Phase 1c

The accepted run A output is frozen at `phase1b_validation/.gate_ref_cno_mg_vfky96/output/` as the post-Phase-1b C/N/O+Mg baseline. When Phase 1c adds Fe, running with `Fe off` must reproduce it; because adding Fe bumps `n_species/n_mion` (and the `Ion_species.txt` column count), the Phase-1c check is a **round-off match on the shared physical columns**, not a byte-for-byte compare (the `mg_validation/cno_mg` byte-gate applied only to the physics-neutral Phase 1a refactor, where `n_species` was unchanged).

4.8 Status and relation to the Huang plan

This completes **Phase 1b**: the network carries C/N/O/Mg/Si/Ca/Na/K/S, the C/N/O cross sections are corrected to VFKY96, and the variable-stage solver handles 1st- and 2nd-ionization elements uniformly. The gate’s “converges/conserves/Fig. 12 ordering/ $\dot{M}$ within $\sim 2 \times$ Case A” items are addressed by the numbers above; the thermal-balance part is deferred to Phase 2 (metal line cooling). Next: **Phase 1c** — iron (Fe I/II/III), which needs non-uniform cross-section (Zatsarinny+2019) and recombination data. Scratch runs and notebook: `phase1b_validation/Phase1b_validation.ipynb`.

5. Phase 1c: add iron (Fe I / II / III)

5.1 Overview

Iron is added as the third three-stage trace metal, completing the Huang+2023 §2.4 element set (C, N, O, Mg, Si, Ca, Na, K, S, **Fe**). No solver, cooling-loop, heating, output, or post-processing code changed: the variable-stage MINPACK packing and every metal loop are generic over `n_melem`, so adding iron is purely new metadata rows plus four new rate functions. Counts: `n_species` = 33, `n_mion` = 27, `n_melem` = 10, `n_mphot` = 17; the system grows to `N_eq` = 3 + 2*`n_melem` = 23. Iron carries `melem_top` = 2 (three stages, like C/O/N/Mg/Si/Ca).

5.2 Atomic data and rates (sources, not fabricated)

Iron is the one element whose data are *not* the uniform-template choice. Each quantity was checked against its primary source before encoding; the Voronov fit in particular was cross-checked because an earlier abandoned attempt had its P/X/K columns transposed.

- **Photoionization cross sections** (`functions/cross_sec.f90`): single-shell VFKY96 form (`sigma_VFKY96`) for Fe I and Fe II, parameters from the Verner+1996 `phfit2.f` PH2 block. Neutral thresholds [eV]: Fe I **7.902** (NIST ionization potential), Fe II **16.199**. *To Be Checked / known limitation*: Huang+2023 use the multi-shell Zatsarinny+2019 Fe I cross section; the single-shell fit here under-represents inner-shell absorption at high XUV energy, so the iron photoionization rate is a lower bound at the hardest photons (flagged inline). A multi-shell upgrade is left for later.
- **Collisional ionization**: Voronov (1997) `cfit.dat`. Fe I (`dE`=7.9, `A`=2.52e-7, `P`=0, `X`=0.701, `K`=0.25), Fe II (`dE`=16.2, `A`=2.21e-8, `P`=1, `X`=0.033, `K`=0.45).
- **Recombination**: the Huang+2023 Eqs. (5)–(6) total fits (a dielectronic Arrhenius term $A T^{-1.5} e^{-T_0/T} (1 + B e^{-T_1/T})$ plus a radiative power law $C (T/10^4)^{-\eta}$) used *directly*, not the Badnell dispatcher used for the lighter metals. Deliberate: the Huang iron rates are the validation target, so the model runs on exactly those rates; the Badnell-swap for iron is deferred to Phase 6. Fe II+*e* → Fe I: `A`=2.833e-8, `T0`=5.731e4, `B`=1.383e4, `T1`=120.4; `C`=1.248e-12, `eta`=0.485. Fe III+*e* → Fe II: `A`=1.094e-5, `T0`=1.490e4, `B`=36.74, `T1`=1.153e5; `C`=1.728e-12, `eta`=0.618.
- **Line cooling**: *not* ported (`mion_iscool` = `.false.` for all Fe stages) — Phase 2. This matters more for iron than for the lighter metals: Fe II is the *dominant* low-altitude line coolant in Huang+2023, so a metals-on thermal profile is *expected to run hotter than Huang* until Phase 2.
- **Charge exchange with H**: *not* added; still C/N/O only (Phase 1d). Iron's Fe⁺+H⁺ ↔ Fe²⁺+H reactions (Huang Table 4) come in Phase 1d.

5.3 Touched files

Metadata rows + four rate functions + abundance/IC wiring; no solver, cooling, heating, output, or post-process changes:

- `init/species_table.f90` — `n_mion=27`, `n_melem=10`, `n_mphot=17`; Fe I/II/III rows (`f_sp` cols 31–33); `iel_Fe=10`; `melem_top` appends 2.
- `functions/cross_sec.f90` — `sigma_FeI`, `sigma_FeII`.
- `init/set_energy_vectors.f90` — `sigma_tab(:,16)=sigma_FeI`, `(:,17)=sigma_FeII`.
- `radiation/Cool_coeff.f90` — Huang recombination `alpha_rec_FeI/FeII_Huang` with `rec_FeII/rec_FeIII` wrappers; Voronov `ion_coeff_FeI/FeII`; dispatcher cases for FeII/FeIII recombination and FeI/FeII ionization. No `cool_coeff_metal` case for Fe (Phase 2).
- `init/parameters.f90`, `files_IO/input_read.f90`, `files_IO/metals_input_read.f90` — `n_species=33`; `X_Fe`; `metals.inp` keys FeI/Fe; `melem_ab(iel_Fe)`; `thereis_metals` trigger.
- `init/set_IC.f90`, `files_IO/load_IC.f90` — neutral-iron IC for `f_sp` cols 31–33.

5.4 Iron abundance

Asplund et al. (2009) solar Fe/H = 3.16×10^{-5} ($\log \epsilon_{\text{Fe}} = 7.50$). Huang+2023 adopt $4 \times$ solar metallicity for WASP-121b, so the validation run uses $X_{\text{Fe}} = 1.26 \times 10^{-4}$ by number relative to H; C/N/O/Mg are held at solar to isolate the iron change.

5.5 Validation on WASP-121b

Two converged runs (identical planet / SED / grid; power-law SED, He/H=0.0851, Roche domain, “Rate/2 + Mdot/2”) plus the frozen Phase-1b C/N/O+Mg baseline:

key	metals	iron	$\log_{10} \dot{M}$	T_{peak} [K]	$r(T_{\text{peak}})$	steps
ref	C,N,O,Mg	frozen 1b run A	12.85	12197	1.899	15382
off	C,N,O,Mg	off ($X_{\text{Fe}}=0$)	12.85	12197	1.899	15382
on	+Fe	on, $4 \times$ solar	12.87	12223	1.915	16566

$\dot{M}$ in $\log_{10}$ g/s; Huang+2023 Case A reference $\log_{10} \dot{M} = 12.57$.

Observations (for evaluation, not pass/fail):

- **Regression, force-zeroed iron (off vs ref).** The Phase-1c binary carries three extra species and two extra MINPACK unknowns, yet run off reproduces the frozen baseline on every shared column to the round-off floor: worst relative difference 2.0×10^{-13} across the seven hydro columns and 3.7×10^{-13} (NI) across the 30 shared species. The three new Fe columns are *identically zero, no NaNs*. With $X_{\text{Fe}}=0$ the iron block reduces to identity rows in the residual (`fvec=x`, rooted at zero) and contributes exactly zero electrons, so the 23-unknown system returns the 21-unknown answer — adding iron is non-perturbative when off.

- **Iron ionization structure (on).** Fe II dominates the base (fraction 0.89 at $r = 1$, peaking ~ 0.96 near $r = 1.02$); Fe III dominates the escaping flow (0.97 at the outer edge), with the Fe II/Fe III crossover near $r \simeq 1.25$; Fe I is minor everywhere (max ~ 0.11 at the base). This is the qualitative Huang+2023 Fig. 12 arrangement — Fe I confined to the base, Fe II at the base, Fe III through the outflow.
- **Iron as an electron donor.** At $4\times$ solar, iron supplies up to $\sim 26\%$ of the free-electron density, so it feeds back on the whole coupled solution, not just its own columns.
- **Switching iron on (off $\rightarrow$ on).** $\Delta \log_{10} \dot{M} = +0.02$ dex, $\Delta T_{\text{peak}} = +26$ K ($12197 \rightarrow 12223$). Small and in the *heating* direction: with no Fe line cooling yet, iron acts only as an added electron donor / photoelectric-heating channel, so the thermosphere runs slightly hotter, not cooler.
- **Numerics.** Run on converged with zero NaNs; X/H for each element is conserved to 8.3×10^{-13} across all five active elements including iron.

5.6 Caveats and gaps

- **No iron line cooling** (`mion_iscool=.false.`) — Phase 2. Because Fe II is the dominant low-altitude coolant in Huang+2023, the run-on *temperature profile is not yet comparable to Huang* and is expected to sit above it. The Phase-1c physics target is the iron *ionization structure*, not $T(r)$.
- **No iron charge exchange with H** — Phase 1d.
- **Single-shell VFKY96 Fe I/II cross sections** under-represent inner-shell XUV vs Huang’s Zatsarinny+2019 Fe I (flagged in `cross_sec.f90`).
- **Recombination is the Huang fit**, not Badnell (deliberate; the Badnell-swap is Phase 6).
- $\dot{M}$ is still $\sim 2\times$ Huang Case A (12.87 vs 12.57); the missing metal cooling and the Phase-0 geometry confounds still apply.
- **Huang Fig. 12 is not digitized/overlaid** — only the qualitative Fe II-base / Fe III-outflow ordering is checked.

5.7 Regression reference for Phase 1d / Phase 2

The run-on output (C/N/O/Mg + $4\times$ -solar Fe) is frozen at `phase1c_validation/.gate_ref_cno_mg_fe/` (input, `metals.inp`, `run.log`, `output/`) as the post-Phase-1c iron-on baseline. Phase 1d (iron charge exchange) and Phase 2 (iron line cooling) both *change* the iron ionization and the thermal structure deliberately, so this is the “before” snapshot for those comparisons, not a bit-reproduction target.

5.8 Status and relation to the Huang plan

Phase 1c **implementation is complete** and the iron ionization structure is presented for validation against Huang+2023 Fig. 12 (numbers and figures in the notebook; no pass/fail asserted here — that judgement is the reader’s). The

ionization-network half of Phase 1 now carries the full Huang element set with verified iron atomic data; the thermal-balance half remains deferred (until Fe II cooling lands in Phase 2 the metals-on temperature profile will not match Huang). Next: **Phase 1d** — charge exchange with H for all metals (Huang Table 4), including $\text{Fe}^+ + \text{H}^+ \leftrightarrow \text{Fe}^{2+} + \text{H}$. Scratch runs and notebook: `phase1c_validation/Phase1c_validation.ipynb`.

6. Phase 1d: charge exchange with hydrogen (Huang Table 4)

6.1 Overview

Phase 1d adds the charge-exchange (charge-transfer, CT) reactions of Huang+2023 Table 4 as a single table-driven module. The default active set is the 23 **Group A** metal + H / H^+ reactions, which couple each metal’s ionization balance to the H^+ fraction. A runtime switch `cx_full 1` in `metals.inp` enables the rest of the table — **Group B** (He + H, 2 rows), **Group C** (metal + He, 6 rows), **Group D** (metal + metal, 32 rows). This *replaces* the previous hard-coded Kingdon & Ferland (1996) C/N/O + H charge-exchange terms in the solver with a generic implementation covering all ten metals plus He.

Re-sourcing C/N/O to Huang Table 4 has one deliberate consequence: Table 4 tabulates only the neutral $\leftrightarrow$ singly-ionized CT for C, N, O (rows A13–A18), so the switch *drops* the doubly-ionized $\text{C}^{2+}/\text{N}^{2+}/\text{O}^{2+} + \text{H}$ recombination terms the old KF96 implementation carried. This is an intentional behavior change to match the paper; it raises the C/N/O double-ionization fractions and reshapes the thermal profile (below). No solver structure, cooling loop, heating, output, or post-processing code changed beyond the CT wiring.

6.2 Atomic data and rates (sources, not fabricated)

All 63 rate rows are transcribed verbatim from `docs/charge_exchange_table4.md`, the image-verified transcription of Huang+2023 Table 4 (23 Group A + 2 Group B + 6 Group C + 32 Group D). The paper text says “65 reactions”; the 2-row discrepancy is immaterial and noted in the reference doc.

- **Functional forms** (radiation/charge_exchange.f90): KF96 bracket $aT_4^b(1 + ce^{dT_4})e^{-E_k/T_4}$ (kf); power law $a(T/300)^be^{-E_k/T_4}$ (gp); 4th-order $\ln T$ polynomial (p4) for the radiative-CT rows of N / S / Na / K (Lin/Zhao/-Dutta/Watanabe); and the Satta (2013) $f_{\text{Si}}(T)$ and $D_{\text{CSi}}(T)$ helpers for the Si + He and C + Si rows.
- **Spot checks at $T = 10^4$ K** (against the table): $\text{Fe} + \text{H}^+ = 4.000 \times 10^{-9}$ (RV72), $\text{Fe}^+ + \text{H} = 2.011 \times 10^{-12}$, $\text{Fe}^{2+} + \text{H} = 1.260 \times 10^{-9}$ (ND87), $\text{O} + \text{H}^+ = 2.091 \times 10^{-9}$ (near-resonant, S99) — all match.
- **Rate clamp** $[0, 10^{-6}] \text{ cm}^3 \text{ s}^{-1}$: the upper bound guards the $\ln T$ polynomials, which diverge as $T \rightarrow 1$ K (never reached); the lower bound zeroes a few KF96 bracket fits that dip slightly negative below their validity floor (e.g. $\text{Mg}^+ + \text{H}^+$ at $T \lesssim 1500$ K), where the true rate is negligible.
- **Reference-doc notes**: Ca has *no* direct H charge exchange (only Group D); K has only the forward $\text{K} + \text{H}^+$ row. Reverse rates for pairs lacking direct data are Huang’s detailed-balance estimates, carried verbatim.

6.3 Touched files

A new module plus minimal wiring; the solver’s old element-by-element C/N/O CT block is deleted, not extended:

- `radiation/charge_exchange.f90` (**new**) — the full Table-4 descriptor arrays, the `cx_rate` dispatcher, and the generic residual assembly. Entry points: `cx_init` (build the active list from `cx_full`), `cx_set_cell(T)` (rates in each cell), `cx_add_to_fvec` (add CT terms to the MINPACK residual via the `cx_fvidx` element/stage $\rightarrow$ row map), and the `cx_full` switch. The zero-abundance guard is automatic: each rate is $k n_{\text{donor}} n_{\text{acceptor}}$.
- `nonlinear_system_solver/System_HeH_metals.f90` — the hard-coded C/N/O + H charge-exchange terms are *removed*; a single call `cx_add_to_fvec(...)` now feeds CT contributions into the H, He, and every metal-stage residual row.
- `radiation/ionization_equilibrium.f90` — call `cx_set_cell(T_K(j))` before each cell’s `hybrd1` solve.
- `files_IO/metals_input_read.f90` — parses `cx_full <0|1>` from `metals.inp`.
- `files_IO/input_read.f90` — if `(thereis_metals)` call `cx_init` after the input read, before the ionization sweep.

6.4 Validation on WASP-121b

Two new converged runs (identical planet / SED / grid as Phase 1c) compared against the frozen Phase-1c iron-on output as the regression reference:

key	charge exchange	$\log_{10} \dot{M}$	T_{peak} [K]	$r(T_{\text{peak}})$	steps
ref	KF96 C/N/O + H (frozen 1c on)	12.87	12223	1.915	16566
cx	Huang Table 4 Group A (default)	12.89	12955	1.778	13942
full	Huang Table 4 full, <code>cx_full 1</code>	12.88	12951	1.765	11661

$\dot{M}$ in $\log_{10}$ g/s; Huang+2023 Case A reference $\log_{10} \dot{M} = 12.57$. All three runs converged with **zero NaNs**.

Observations (for evaluation, not pass/fail):

- **H⁺ sink magnitude.** In the ionized bulk ($x_{\text{HII}} > 0.5$) the net metal charge exchange is at most 3.7×10^{-3} of radiative recombination (median 1.0×10^{-3}) — negligible as an H⁺ sink, consistent with Huang §4.2 (“Fe–H CX lower than H photoionization and recombination”). At the near-neutral base ($x_{\text{HII}} \sim 7.8 \times 10^{-8}$) the *ratio* is large (~ 118) only because H⁺ is itself a trace species there.
- **Near-resonant balance.** $\text{O} + \text{H}^+ \leftrightarrow \text{O}^+ + \text{H}$ is near-resonant and near-equilibrium at the base (gross destroy 3.66×10^5 vs produce 3.69×10^5 , net $+3.0 \times 10^3$). The net base H⁺ sink is instead driven by the fast, one-sided $\text{Fe} + \text{H}^+ \rightarrow \text{Fe}^+ + \text{H}$ channel (net -2.9×10^4).
- **C/N/O double ionization rises** (the dropped 2+ CT). Column-max fractions ref $\rightarrow$ cx: O III 0.871 $\rightarrow$ 0.946, N III 0.897 $\rightarrow$ 0.944, C III 0.851 $\rightarrow$ 0.856.

- **Iron base shift.** Fe II base fraction $0.889 \rightarrow 0.896$; Fe III essentially vanishes at the base ($0.002 \rightarrow 0.000$): $\text{Fe} + \text{H}^+ \rightarrow \text{Fe}^+ + \text{H}$ pushes iron toward Fe^+ .
- **Thermal structure.** T_{peak} rises $12223 \rightarrow 12955$ K and moves inward ($1.915 \rightarrow 1.778 R_p$). This is the *combined* effect of (i) the C/N/O re-sourcing that drops the 2+ recombination terms and (ii) the new metal + H CT. Charge exchange adds *no* radiative-loss channel, so it cannot cool the gas; Fe II / metal line cooling remains deferred to Phase 2, and the metals-on profile therefore still runs *hotter* than Huang.
- **Group A vs full Table 4 (cx vs full).** Turning on Groups B/C/D changes the bulk temperature by at most 1.5% (worst relative difference), with max $|\Delta T| = 556$ K at $r = 1.008$; at $r = 1.02$ the full run is 357 K cooler with more base H^+ and less He II. $\dot{M}$ moves -0.01 dex and the outflow is nearly unchanged.
- **Dynamics.** v_{out} and sonic radius: ref 18.83 km/s @ $1.884 R_p$; cx 20.28 km/s @ $1.826 R_p$; full 20.40 km/s @ $1.826 R_p$. The hotter, more inward thermal peak drives a slightly faster wind with the sonic point $\sim 0.06 R_p$ further in.

6.5 Caveats and gaps

- **No metal line cooling yet** (Phase 2). Charge exchange adds no loss term, so $T(r)$ is not comparable to Huang; the C/N/O re-sourcing makes the base hotter, not cooler.
- **C/N/O 2+ + H terms intentionally dropped** to match Table 4 — a real change from the Phase-1c (KF96) reference, and the dominant driver of the O III/N III rise rather than the new metal CT alone.
- $\dot{M}$ still $\sim 2 \times$ Huang Case A (12.88–12.89 vs 12.57); the missing metal cooling and the Phase-0 geometry confounds still apply.
- **Paper lists 65 reactions, 63 transcribed** — immaterial, documented in `charge_exchange_table4.md`.
- **No Huang figure digitized/overlaid** for the H^+ budget; only the relative-magnitude statement (net CT $\ll$ recombination in the bulk) is checked.

6.6 Regression reference for Phase 2

The default Group-A run is frozen at `phase1d_validation/.gate_ref_cno_mg_fe_cx/` (input, `metals.inp`, `run.log`, output/) as the post-Phase-1d baseline. Because the C/N/O re-sourcing deliberately shifted both the ionization and the thermal structure away from the Phase-1c frozen reference, *this* snapshot — not the Phase-1c one — is the “before” reference for Phase 2 (metal line cooling, including Fe II).

6.7 Status and relation to the Huang plan

Phase 1d **implementation is complete**: the 23 Group-A metal + H reactions are active by default and the full Table 4 sits behind `cx_full`. With this, the **ionization-network half of Phase 1 is physically complete** relative to Huang+2023 §2.4 + Table 4 — the full element set, verified atomic data, and the metal–H charge-exchange coupling are all in

place. The thermal-balance half remains deferred: until metal line cooling (Fe II in particular) lands in **Phase 2**, the metals-on temperature profile stays above Huang. Next: **Phase 2** — metal line cooling. Scratch runs and notebook: `phase1d_validation/Phase1d_validation.ipynb`.

7. Phase 2: metal line cooling — validation gate closed

Phase 2 brings the CHIANTI-based metal line cooling (Mg I/II, Ca II, Na I, Fe II, on top of the C/N/O coolants) into the coupled energy balance and **closes the Phase 2 validation gate** against Huang et al. (2023) Fig. 10. The atomic data for each coolant and the CHIANTI effective-cooling tables are documented in `Update_EXHALE_early_phase` (Parts II–III); this section records the integrated WASP-121b gate and the diagnostic used to close it.

7.1 Active line coolants

The line-cooling ions are flagged by `mion_iscool` in `species_table.f90` (canonical order): C I, C II, O I, O II, N I, N II, Mg I, Mg II, Ca II, Na I, Fe II. Each contributes $\beta_{\text{esc}} n_e n_{\text{ion}} \Lambda_{\text{ion}}(T)$ with the effective cooling coefficient Λ_{ion} interpolated from the hardcoded `cool_logL_*` tables in `Cool_coeff.f90` ($N_{\text{COOLT}} = 41$, $\log_{10} T = 3.0\text{--}5.0$, $\Delta \log_{10} T = 0.05$). In the optically thin limit $\beta_{\text{esc}} = 1$ (see `Update_EXHALE_early_phase`, Part II, “Adopted assumption: 100% escape”). N I/II carry the `iscool` flag but have no tabulated line cooling ($\Lambda = 0$), so they contribute exactly zero.

7.2 Channel-by-channel cooling diagnostic (this phase)

To close the gate quantitatively, `eval_cool` (`util_ion_eq.f90`) gained an optional `cool_chan` output — an **exact** decomposition of the total cooling rate into H/He recombination, collisional ionization, collisional excitation, bremsstrahlung, and one column per metal ion. It is read straight from the arrays `eval_cool` already builds, so the channel sum reproduces the cool total to machine precision (no offline re-derivation, no divergence risk). A companion writer `write_cool_breakdown_eq` (same module, called once from `EXHALE_main.f90` after the final equilibrium write) dumps output/`Cooling_breakdown.txt`: columns r/R_p , T , n_e , `cool_total`, the four H/He channels, then 27 metal-ion columns. The run prints `max |sum(channels)/cool - 1|` as an internal consistency check.

7.3 Validation on WASP-121b (the gate)

Converged equilibrium run (full metals, the curated `metals.inp`; same SED/grid as Phase 1d):

- Internal consistency: $\max |\sum \text{channels}/\text{cool} - 1| = 3.7 \times 10^{-16}$ (machine eps) — the breakdown is a bit-exact decomposition of the `Hydro_ioniz.txt` cool column.
- In the $1.15 < r/R_p < 1.4$ band the dominant radiative coolants are **Fe II** (~30%) and **Mg II** (~25%), jointly ~55%; metals are **57–85%** of all radiative cooling through the band. Fe II leads the inner edge (top coolant at $1.1\text{--}1.2 R_p$, 53% at $1.1 R_p$), Mg II takes over by $1.3\text{--}1.4 R_p$, and H recombination only overtakes above $\sim 1.5 R_p$ — the Fe II $\rightarrow$ Mg II handoff Huang report in their Fig. 10.
- $\dot{M}$ unchanged ($\log_{10} \dot{M} = 12.51$ g/s); zero NaNs.

This matches the Phase 2 gate (“a metal-driven cooling feature appears at $1.15\text{--}1.4 R_p$, Fig. 10”) and Huang_update_expected_results (Mg II + Fe II become the most important coolant in $1.15 < r/R_p < 1.4$).

7.4 Touched files

- `src/modules/radiation/util_ion_eq.f90` — optional `cool_chan` out-arg on `eval_cool` (exact channel-by-channel split); new `write_cool_breakdown_eq` writer; `mion_fsp` added to the `species_table` use-list.
- `EXHALE_main.f90` — use `utils_ion_eq`, only: `write_cool_breakdown_eq`; call it after the final equilibrium `write_output`.
- Cooling coefficients themselves live in `Cool_coeff.f90` (provenance in `Update_EXHALE_early_phase`, Parts II–III); unchanged here.

7.5 Caveats and deferred

- The ion-by-ion metal decomposition is exact in the default branch (`use_2lev_cool = .false.`, the WASP-121b setting). In the optional two-level fine-structure branch the terms for each ion are the resonance-line approximation and need not sum to the two-level `cool_M`.
- **Fe II** uses Boltzmann/coronal lower-level populations ($E_{\text{cut}} = 38459 \text{ cm}^{-1}$), validated vs Huang Fig. 6 coronal curve within $\sim 1.5\times$ — sufficient for the gate. The Hollenbach & McKee IR fine-structure + density-dependent two-level refinement (Huang “Sum” curve) is deferred.
- **Fe I** line cooling (NIST f -values + Van Regemorter, Huang Fig. 5) is not yet added; Fe I is a minor coolant in the gate band. (*Now added in §9; confirmed negligible — peak 3×10^{-5} of total cooling.*)
- The **CHIANTI Ly α swap** is intentionally skipped in Phase 2; the $0.35\times$ Ly α is a Phase-3 radiative-transfer suppression, not a cooling-rate change.
- Carrying metal cooling into the post-processed `_adv` profile is handled separately (runtime `pp_metals`, default frozen); see the post-process notes.

7.6 Status and relation to the Huang plan

Phase 2 is complete and the gate is closed (user-validated 2026-06-05). The metals-on thermal balance now reproduces the Huang Fig. 10 Fe II/Mg II cooling feature, closing the thermal-balance half of the network that Phase 1 left open. Next: **Phase 3** — in-code excited $\text{H}(n=2)$ + Ly α radiative transfer.

8. Phase 3a: in-code excited $\text{H}(n=2)$ + Balmer proton source and heating

Added 2026-06-05.

Phase 3a is the first half of Phase 3: an **in-code $\text{H}(n=2)$ population model** whose output feeds back into both the hydrogen ionization balance (an extra proton source) and the energy equation (Balmer photoelectric heating), fully

coupled. The second half (a Ly α radiative-transfer solve to supply a self-consistent mean intensity $\bar{J}_{\text{Ly}\alpha}(r)$) is deferred; this phase uses a **parameterized** $\bar{J}_{\text{Ly}\alpha}$ with a switch to read an external RT profile instead. Everything is **default-off**: with no stellar parameters in `input.inp` the build is byte-identical to the Phase-2 result (`use_excited_H = .false.`).

8.1 Physics and atomic data (sources, not fabricated)

- **H(2s)/H(2p) populations**: a 2×2 statistical-equilibrium system following Christie, Arras & Li (2013) — recombination cascade into $n=2$, collisional $2s \leftrightarrow 2p$ mixing, two-photon $2s \rightarrow 1s$ decay, Ly α $2p \rightarrow 1s$ with a $2p$ radiative-pumping term, and $2s/2p$ photoionization. Solved per cell (`n2_populations`).
- **Ly α mean intensity** (the $2p$ pump): parameterized as $\bar{J}_{\text{Ly}\alpha} \approx 0.1 F_{\text{LyC}} / \Delta\nu_D$ (Huang et al. 2017, Eq. 6) with a **top-down** $1/(1 + \tau_{\text{Ly}\alpha})$ **line-center attenuation** so the optically thick base ($\tau_{\text{Ly}\alpha} \sim 1.1 \times 10^8$ here) is not over-pumped. Oscillator strength $f_{\text{Ly}\alpha} = 0.4162$.
- **$n=2$ Balmer-continuum photoionization rate Γ_2** : from a **diluted stellar blackbody** above the $n=2$ threshold $E_2 = 3.40$ eV (`gamma_n2_balmer`), with $\sigma_{\text{LyC}} = 6.3 \times 10^{-18} \text{ cm}^2$ and threshold cross-section $s_2 = 1.4 \times 10^{-17} \text{ cm}^2$. Γ_2 carries the **same ξ dayside-dilution factor as ground-state EUV ionization** (`Rate/2` $\rightarrow 0.5$, `Rate/4` $\rightarrow 0.25$); for WASP-121b's "`Rate/2 + Mdot/2`" ($\xi = 0.5$) this gives $\Gamma_2 \approx 613 \text{ s}^{-1}$, comparable to Huang's $\approx 576 \text{ s}^{-1}$.
- **Heating**: Balmer photoelectric heating $H_{\text{pe}} = n_2 \cdot (\text{photoelectron energy} \times \Gamma_2)$ per cell (`heat_n2_balmer`); optional collisional de-excitation heating behind `incl_deexc_heat` (off by default — it overlaps the existing HI collisional-excitation cooling).

8.2 Coupling into the solver

The model runs **decoupled (lagged one outer step)** to keep it out of the cell-by-cell Newton solve: `excited_H_update` fills two global arrays (one entry per cell) from the previous converged state, and `ioniz_eq` injects them:

- `gph_balmer_HI [s-1]` is added to the HI photoionization rate `P_HI` — an extra **proton source** ($S_{\text{proton}} = \Gamma_2 n_2$).
- `heat_balmer [erg cm-3 s-1]` is added to the photoheating rate.

`excited_H_update` is called once per outer iteration in `EXHALE_main.f90` before `ioniz_eq`, and once more after convergence to refresh the diagnostics. The post-loop refresh deliberately does *not* re-solve `ioniz_eq` (the Balmer heating is $\sim 10^{-4}$ of the total, not worth desyncing T/f_{sp} from the converged hydro state).

8.3 Runtime interface

Append to `input.inp` (omit for the default Phase-2-identical behavior):

Stellar Teff [K]: 6459.0

Stellar radius [R_sun]: 1.458

These set `use_excited_H` (both must be > 0). Optional further tail lines: `Jlya RT file: <path>` switches $\bar{J}_{\text{Ly}\alpha}$ to **mode 1** (read an external RT profile and use it directly, instead of the mode-0 parameterized field); `Deexc heat: True`

enables collisional de-excitation heating. As with all trailing lines the scan is by keyword, so order and blank lines do not matter.

8.4 Diagnostics

write_excited_H dumps output/Excited_H.txt (14 columns) from the converged state: r/R_p , T , n_{HI} , n_e , $\bar{J}_{\text{Ly}\alpha}$, n_{2s} , n_{2p} , $S_{\text{proton}}(n=2)$, H_{pe} , H_{dx} , $\tau_{\text{Ly}\alpha}$, then the three **proton-budget** columns added in this phase — **col12 ground-state photoionization** ($P_{\text{HI}} n_{\text{HI}}$), **col13 collisional ionization** ($C n_e n_{\text{HI}}$), **col14 recombination** ($\alpha_B n_e n_{\text{HII}}$), all volumetric [$\text{cm}^{-3} \text{s}^{-1}$]. The three rate coefficients are captured inside ioniz_eq on the converged pass (globals gph_ground_HI, cion_HI, arec_HII), so the budget is self-consistent with the solver rather than re-derived offline.

8.5 Touched files

- init/parameters.f90 — Phase-3a globals (use_excited_H, T_star_eff, R_star, gamma2_bal, hpe2_bal, jlya_mode, jlya_rt_file); feedback arrays gph_balmer_HI, heat_balmer; diagnostics Jlya_arr, n2s_arr, n2p_arr, Sproton_arr, Hpe_arr, Hdx_arr; and the proton-budget capture arrays gph_ground_HI, cion_HI, arec_HII.
- radiation/excited_hydrogen.f90 (**new**) — excited_H_update, n2_populations, gamma_n2_balmer, heat_n2_balmer, load_jlya_rt, write_excited_H.
- radiation/ionization_equilibrium.f90 — inject gph_balmer_HI/heat_balmer under use_excited_H; capture the ground-state / collisional / recombination coefficients for the proton-budget dump.
- files_IO/input_read.f90 — scan the stellar / Jlya RT file / Deexc heat tail lines; set use_excited_H, jlya_mode.
- EXHALE_main.f90 — in-loop excited_H_update before ioniz_eq; post-loop refresh + write_excited_H.
- Makefile — excited_hydrogen.f90 in SRC.

8.6 Validation on WASP-121b (coupling ON vs OFF)

Two converged runs, same build, differing only in the use_excited_H toggle (stellar tail lines present/absent). Both reach steady state via $du < du_{\text{th}}$ with $\log_{10} \dot{M} = 12.51$ (unchanged from Phase 2 — the coupling does not destabilize the energy solver). Diagnostics in WASP-121b/{analyze_excited_H.py, analyze_proton_budget.py, compare_on_off.py}.

Proton budget (cf. Huang Figs. 11/27), volumetric rates [$\text{cm}^{-3} \text{s}^{-1}$]:

r/R_p	S_{n2}	S_{ground}	S_{recomb}	S_{n2}/S_{ground}	S_{n2}/S_{source}
1.10	5.1e4	6.6e6	6.6e6	0.008	0.008
1.30	1.2e4	4.3e5	4.4e5	0.029	0.028
1.50	4.9e3	7.6e4	8.1e4	0.064	0.060
1.80	1.9e3	1.5e4	1.7e4	0.125	0.111
2.00	2.0e3	5.9e3	7.9e3	0.34	0.25
2.01	2.9e3	5.1e3	7.9e3	0.57	0.36

Observations (for evaluation, not pass/fail):

- **Self-consistency:** at the base $S_{\text{ground}} \approx S_{\text{recomb}}$ (6.6×10^6 vs 6.6×10^6 at $1.1 R_p$), i.e. photoionization equilibrium holds; collisional ionization is negligible (T too low).
- **n=2 share:** the n=2 Balmer channel is a minor proton source below $\sim 1.5 R_p$ ($\sim 1\text{--}6\%$ of ground-state photoionization) but climbs to **$\sim 34\text{--}57\%$ of the ground-state rate ($\sim 25\text{--}36\%$ of the total proton source) near $2 R_p$** , where the gas is already ionized and EUV is attenuated (ground-state photoionization depleted) while the thin-Ly α -fed n=2 channel is not. This is the “significant proton source approaching $2 R_p$ ” behavior Huang describe, here concentrated near $2 R_p$ rather than flat across $1\text{--}2 R_p$.

Heating budget (cf. Huang Figs. 10/26): Balmer photoelectric heating is present in the budget, peaking at **$\sim 1.8\%$ of total heating** at the Ly α photosphere ($\tau_{\text{Ly}\alpha} \sim 1$ near $r \sim 2 R_p$) and $\sim 10^{-4}$ globally — a small but nonzero entry, dominating nowhere, as expected.

ON vs OFF temperature and ionization: the coupling makes the gas slightly **more ionized** (up to $\sim 24\%$ less neutral H near $2 R_p$, Δx_{HII} up to $\sim 1.1 \times 10^{-2}$) and slightly **cooler** (up to ~ 179 K at $r \sim 2 R_p$, mean ~ 42 K). The cooling is **physical, not numerical:** the extra n=2 photoionization converts $\text{HI} \rightarrow \text{HII}$, lowering n_{HI} ; the dominant EUV photoelectric heating scales with n_{HI} , so $\sim 24\%$ less n_{HI} at $r \sim 2 R_p$ means correspondingly less local EUV heating there, which the small Balmer H_{pe} does not offset.

8.7 Caveats and gaps

- **Parameterized $\bar{J}_{\text{Ly}\alpha}$ (mode 0).** The $0.1 F_{\text{LyC}}/\Delta\nu_D$ field with $1/(1 + \tau_{\text{Ly}\alpha})$ attenuation is a placeholder for the Phase-3b Ly α RT solve. Because our n=2 contribution is peaked near $2 R_p$ rather than flat across $1\text{--}2 R_p$, if Huang’s Figs. 11/27 are flatter/larger across the band the parameterized field is under-pumping n=2 in $1.1\text{--}1.8 R_p$; mode 1 (external RT $\bar{J}_{\text{Ly}\alpha}$) is the path to test this once a Ly α RT result exists.
- **Convergence stop is momentum-uniformity** ($du < du_{\text{th}}$), not a thermal/ionization steady-state criterion, so a small residual convergence-path sensitivity rides on top of the physical ON–OFF difference.
- **De-excitation heating off by default** (overlaps the existing HI coex cooling); left behind `incl_deexc_heat`.

8.8 Two analysis gotchas (recorded so they are not repeated)

- Hydro_ioniz.txt columns 6/7 (heat/cool) are **already physical** (write_output multiplies by q_0); do *not* multiply by q_0 again in analysis.
- Ion_species.txt column 1 is r/R_p , so $n_{\text{HI}}=\text{col2}$, $n_{\text{HII}}=\text{col3}$ — the recombination sink uses col3, not col2.

8.9 Status and relation to the Huang plan

Phase 3a is complete and the proton-source comparison is user-accepted (2026-06-05). The H(n=2) model is fully coupled into the ionization balance and the energy equation; the n=2 photoionization proton source reproduces the qualitative Huang Figs. 11/27 behavior and the Balmer photoelectric heating appears in the budget without destabilizing the solver. Next: **Phase 3b** — a Ly α radiative-transfer solve to replace the parameterized $\bar{J}_{\text{Ly}\alpha}$ with a self-consistent $\bar{J}_{\text{Ly}\alpha}(r)$ (feeding mode 1).

9. Phase 2 refinement: Fe I line cooling (NIST f -values + Van Regemorter)

Added 2026-06-05.

Fe I line cooling was one of the two coolants left deferred when the Phase 2 gate closed (§7, “Caveats and deferred”). This update adds it. CHIANTI v11 has **no Fe I model atom**, so — following Huang et al. (2023) §2.5 — Fe I cooling is built from NIST permitted-line oscillator strengths plus the Van Regemorter collision-strength approximation, with the lower levels populated in Boltzmann equilibrium over the metastable manifold. The new coolant flows through the same $\beta_{\text{esc}} n_e n_{\text{FeI}} \Lambda_{\text{FeI}}(T)$ channel as the other Phase-2 metals.

9.1 Atomic data and method (sources, not fabricated)

The cooling coefficient per $(n_e n_{\text{FeI, total}})$ in the optically thin limit [$\text{erg cm}^3 \text{ s}^{-1}$] is

$$\Lambda(T) = \sum_{\text{lines}} f_l(T) q_{lu}(T) \Delta E_{lu},$$

$$f_l(T) = \frac{g_l e^{-E_l/kT}}{U(T)}, \quad U(T) = \sum_{\substack{\text{even levels} \\ E < E_{\text{cut}}}} g e^{-E/kT}, \quad q_{lu} = 2.16 a^{-1.68} e^{-a} T^{-3/2} f_{lu},$$

with $a = \Delta E/kT$ and the absorption oscillator strength f_{lu} from the NIST A_{ki} , statistical weights, and wavelength (Huang Eq. 10).

- **Raw data:** the auditable NIST dumps fe1_nist_lines.tsv (permitted E1 lines: A_{ki} , g_i , g_k , E_i , E_k) and fe1_nist_levels.tsv (every level g , E for the partition function), fetched by fetch_fe1_nist.py.
- **Metastable manifold = EVEN-parity levels below the lowest odd level** ($z^7\text{D}^\circ$, $19351 \text{ cm}^{-1} = 2.40 \text{ eV}$). Below that energy there is no lower level for an E1 decay, so the even levels are genuinely metastable and hold the Fe I population. This is *not* a plain energy cut: the Fe II treatment can use an energy cut because its cut conveniently lands at the first odd level, but for Fe I a pure energy cut would wrongly include the radiatively-depopulated $z^7\text{D}^\circ$ odd levels. The default cut yields **17 even levels and 828 permitted cooling lines**.

- Upper levels of the cooling lines are odd (E1) and assumed to decay radiatively. The builder is `cooling_data/fe1_cooling.py`; it emits the 41-point `cool_logL_FeI(NC00LT)` Fortran block on the ATES grid ($\log_{10} T = 3.0\text{--}5.0$, $\Delta \log T = 0.05$).

Coefficient values (per $n_e n_{\text{FeI}}$, $\text{erg cm}^3 \text{ s}^{-1}$): $\Lambda(5 \times 10^3 \text{ K}) = 1.27 \times 10^{-22}$, $\Lambda(10^4 \text{ K}) = 1.23 \times 10^{-20}$, $\Lambda(2 \times 10^4 \text{ K}) = 1.31 \times 10^{-19}$ — the rising, then flattening shape of Huang Fig. 5.

9.2 Touched files

- `cooling_data/fetch_fe1_nist.py`, `cooling_data/fe1_cooling.py` (**new**) — raw NIST fetch and the Boltzmann-manifold + Van Regemorter builder.
- `radiation/Cool_coeff.f90` — `cool_logL_FeI(NC00LT)` table constant; new `cool_FeI(T, out)` (interpolates the table via `interp_cool_table`); dispatch cases FeI in `cool_coeff_metal` and `cool_coeff_metal_scalar`. The table-header comment block now documents Fe I alongside the other coolants.
- `init/species_table.f90` — `mion_iscool(FeI) = .true.` (canonical ion index 25), so the `eval_cool` metal-cooling loop and the channel-by-channel breakdown both pick up Fe I automatically.

9.3 Validation on WASP-121b

Re-ran the converged full-metals WASP-121b equilibrium (same SED/grid as the Phase 2 gate) with Fe I cooling on; the channel-by-channel `Cooling_breakdown.txt` then gives the Fe I share directly. Diagnostics: WASP-121b/`analyze_FeI_cooling.py` (left panel = the ported $\Lambda_{\text{FeI}}(T)$ coefficient read straight from `Cool_coeff.f90` vs the Fe II coronal curve, Huang Fig. 5 style; right panel = cooling in each channel vs radius, Huang Fig. 10 style), figure WASP-121b/`FeI_cooling_validation.png`. Observations (for evaluation, not pass/fail):

- **Fe I is a negligible coolant** along the whole profile: its peak share of the total cooling is 3.0×10^{-5} **at** $r = 1.06 R_p$ ($T = 8303 \text{ K}$), and it is far smaller elsewhere ($\sim 3 \times 10^{-8}$ of total at the base, where Fe II dominates). Adding Fe I does *not* perturb the gate-band Fe II $\rightarrow$ Mg II handoff.
- **Mdot unchanged** at $\log_{10} \dot{M} = 12.51$ (Fe I adds no measurable cooling), zero NaNs.
- This is the expected outcome: at WASP-121b base temperatures iron is almost entirely Fe II (Fe I fraction < 0.11 , §5), and Fe I's permitted lines have high excitation energies, so its Boltzmann-suppressed cooling cannot compete with the Fe II / Mg II forbidden + resonance lines. Fe I is carried for completeness and to mirror Huang's coolant list, not because it shapes the WASP-121b energy budget.

9.4 Status and relation to the Huang plan

Fe I line cooling is **implemented and its coefficient matches the Huang Fig. 5 shape/magnitude**; the WASP-121b run confirms it is radiatively negligible there, surfaced as a figure for the reader's judgment (no pass/fail asserted). One of the two Phase-2 deferred cooling items is now closed. The remaining one — the Fe II density-dependent (IR fine-structure) low-temperature refinement — is §10.

10. Phase 2 refinement: density-dependent Fe II line cooling (SE table)

Added 2026-06-05. (Implemented step-by-step; this section is updated as each step lands — currently: the 2-D table generator is built and verified.)

This is the second deferred Phase-2 cooling item (§7): replacing the **coronal** Fe II line-cooling coefficient with a **density-dependent** one that follows the Fe II level populations from the coronal (low- n_e) limit into the collisionally saturated (high- n_e , LTE) limit. It directly affects the cool, dense base of the WASP-121b thermosphere, where the coronal table overestimates Fe II cooling by $\sim 10^4 \times$.

10.1 Why the coronal table is wrong at the base

The Phase-2 Fe II table is the optically-thin **coronal** limit: only the ground level is populated and every collisional excitation is assumed to be followed by a radiative decay, so the cooling is strictly $\propto n_e$ (per $n_e n_{\text{FeII}}$ it is a function of T alone). That assumption fails at the cool, **dense** base of a hot Jupiter thermosphere, because the low- T Fe II cooling there is carried almost entirely by **forbidden** metastable / a⁶D fine-structure lines ($A \lesssim 5 \times 10^{-3} \text{ s}^{-1}$, far-IR 26–87 μm), whose critical densities

$$n_{\text{crit}} = A_{ul} / \sum_c n_c k_{ul}^c \sim 10^4\text{--}10^7 \text{ cm}^{-3} \quad (\text{electron collider})$$

sit far *below* the base $n_e \sim 10^8 \text{ cm}^{-3}$. Those lines are therefore collisionally **saturated** (thermalized to their LTE populations); the coronal coefficient overestimates their cooling by $\sim n_e / n_{\text{crit}}$.

10.2 The density-dependent coefficient (multilevel statistical equilibrium)

cooling_data/fe2_cooling.py solves the full CHIANTI multilevel statistical-equilibrium (SE) problem for the Fe II level populations at each (T, n_e) — 175 levels, electron-collisional excitation/de-excitation (Burgess–Tully descaled Υ from the .scups file) plus radiative decay (.wgfa A -values), normalized to $\sum_i n_i = 1$ — so the cooling smoothly interpolates between the two limits:

$$\frac{\Lambda_{\text{vol}}}{n_{\text{FeII}}} = \sum_{u>l} n_u A_{ul} \Delta E_{ul} \quad [\text{erg s}^{-1} \text{ per ion}], \quad \Lambda_{\text{eff}}(T, n_e) = \frac{\Lambda_{\text{vol}} / n_{\text{FeII}}}{n_e} \quad [\text{erg cm}^3 \text{ s}^{-1}].$$

At low n_e , $\Lambda_{\text{eff}} \rightarrow \Lambda_{\text{coronal}}(T)$; at high n_e it falls off $\sim 1/n_e$ as the numerator saturates. This Λ_{eff} is exactly the per- $(n_e n_{\text{FeII}})$ coefficient EXHALE already tabulates, so the existing assembly $\text{cool_M} = \beta_{\text{esc}} n_e \sum_i n_{\text{ion}}(i) \Lambda(i)$ carries it unchanged — the n_e in the prefactor times Λ_{eff} gives $\beta_{\text{esc}} n_{\text{FeII}} \times (\text{cooling per ion})$, i.e. the correct saturated base cooling.

Why electron-only SE is sufficient at the (mostly neutral) base. In the saturated limit the upper levels reach their **Boltzmann (LTE) populations regardless of which collider thermalizes them**, and the cooling of each ion becomes collider- and n_e -independent. At the base $n_e \sim 10^8$ already exceeds n_{crit} for the dominant forbidden lines, so the electron-only SE already reaches that LTE plateau; adding the (more abundant) neutral-H collider would only push the few still-unsaturated higher- n_{crit} lines further toward the same LTE limit — a second-order correction. An explicit H I / proton collider channel is flagged as a future refinement in the module.

10.3 Step 1 — the 2-D table generator (done)

fe2_cooling.py gained an **optimized** table builder `lambda_eff_table(Tgrid, ngrid)` and Fortran emitters (`fortran_grid_1d`, `fortran_block_2d`, `emit_fortran_table`; run `python3 fe2_cooling.py --table`). The optimization is required because the naive cell-by-cell solve rebuilds a cubic spline for every one of the ~ 4300 transitions at every grid cell: instead the descaled Υ is evaluated **once per transition over the whole T-grid** (`upsilon()` accepts an array T), and the rate matrix is split $M = n_e Q_{\text{coll}}(T) + A_{\text{rad}}$, so the n_e -loop only reassembles and solves. The full 41×29 table builds in ~ 3 s and matches the naive cell-by-cell solve to 1.5×10^{-15} (machine precision).

The emitted table is on the ATES temperature grid ($\text{NCOOLT} = 41$, $\log_{10} T = 3.0\text{--}5.0$, $\Delta \log T = 0.05$) $\times$ an electron-density axis `cool_logne(NCOOLNE = 29)`, $\log_{10} n_e = 0\text{--}14$, $\Delta \log = 0.5$ (brackets the WASP-121b run's $\log_{10} n_e \approx 7.9\text{--}9.7$ with the coronal plateau below and LTE saturation above). It is stored as `cool_logL_FeII_ne(NCOOLT, NCOOLNE) = log10  $\Lambda_{\text{eff}}$` .

Two built-in sanity checks (printed to stderr by `--table`):

- **Low- n_e edge reproduces the coronal table.** At $n_e = 1 \text{ cm}^{-3}$ the SE coefficient equals `cooling_effective(pop='coronal')` to **0.1%** across $T = 10^3\text{--}10^5 \text{ K}$ — so swapping the 1-D coronal table for the $n_e = 1$ column of the 2-D table is a no-op, and the Phase-2 gate's coronal Fe II validation (vs Huang Fig. 6, within $\sim 1.5\times$) carries over unchanged.
- **Base suppression.** At the WASP-121b base ($T \approx 2239 \text{ K}$, $n_e = 10^8$) the coronal coefficient 2.80×10^{-20} drops to the SE value 1.79×10^{-24} — a **15,663 $\times$ suppression**. The suppression is strongest at low T (forbidden lines) and weak at high T ($T \gtrsim 3 \times 10^4 \text{ K}$, where the resonance lines with large A dominate and stay near-coronal): e.g. only $\sim 1.4\times$ at $T = 3 \times 10^4$, $n_e = 10^8$.

10.4 Step 2 — the 2-D table and bilinear interpolator in `Cool_coeff.f90` (done)

The generated table and its interpolators are now in `src/modules/radiation/Cool_coeff.f90`:

- **Table.** `integer, parameter :: NCOOLNE = 29`, the `cool_logne(NCOOLNE)` axis ($\log_{10} n_e = 0\text{--}14$, $d\log = 0.5$), and the data block `cool_logL_FeII_ne(NCOOLT, NCOOLNE)` (a `reshape([...], [NCOOLT, NCOOLNE])` with T on the inner index and n_e on the outer), pasted straight from `fe2_cooling.py --table`. The $n_e = 1$ column matches the existing 1-D `cool_logL_FeII` to the fifth decimal (first node -19.78002 vs -19.77975).
- **Bilinear interpolator.** `interp_cool_table_2d(logL2d, T, ne, out)` — log-log bilinear in $(\log_{10} T, \log_{10} n_e)$. The T axis reuses the uniform `cool_dlogT` scheme; the n_e axis uses the uniform `dlogne = 0.5`. Both axes clamp to the nearest edge outside the table (so n_e below the grid gives the coronal plateau, above it the LTE branch). A scalar twin `interp_cool_table_2d_scalar(logL2d, Ts, nes)` mirrors it bit-for-bit for the cell-by-cell post-process solve.
- **Coolant wrapper.** `cool_FeII_ne(T, ne, out)` calls the 2-D interpolator on `cool_logL_FeII_ne`. The 1-D `cool_FeII` is kept as the low- n_e edge / scalar fallback; its header `!To Be Checked` note was rewritten to record that the density-dependent table now supersedes the coronal-only rate in the cooling assembly.

Builds clean (gfortran, full make); the new routines are additive and not yet wired, so behavior is unchanged until Step 3.

10.5 Step 3 — wiring the density-dependent Fe II into the cooling assembly (done)

The 2-D coefficient is now wired into both cooling paths so the local n_e selects the coronal $\rightarrow$ LTE-saturated branch automatically:

- **Main solver (util_ion_eq.f90, eval_cool).** n_e is already computed at the top of eval_cool (call calc_ne). Immediately after the canonical c_metal(:,i) fill loop, a short loop locates Fe II by name and overrides its column with call cool_FeII_ne(T_K, ne, ...). Nothing else changes: the existing assembly cool_M = $\beta_{\text{esc}} n_e \sum_i n_{m,i} c_{\text{metal},i}$ then multiplies by n_e , which *cancels* the $1/n_e$ in $\Lambda_{\text{eff}} = (\sum_u n_u A_{ul} \Delta E_{ul})/n_e$, leaving the correct cooling for each Fe II ion. The saturated (LTE) populations are collider-independent, so the electron-only SE solve is exact in that limit; at low n_e , Λ_{eff} reduces to the coronal rate, so optically-thin upper-atmosphere cells are unaffected. The override flows automatically into the use_2lev_cool branch (which already reads c_metal(:,26)) and into the channel-by-channel cool_chan diagnostic (which reads c_metal(:,i)), so the breakdown stays an exact decomposition of cool.
- **Post-process cell-by-cell solve (T_equation.f90).** This path computes its own local n_e (from H, He, and the frozen metal stages) and sums the metal coolants with the scalar dispatcher. The Fe II term is special-cased to call the new cool_FeII_ne_scalar(TT, ne) (a thin wrapper over interp_cool_table_2d_scalar), so the post-process temperature balances the same density-dependent Fe II cooling.
- **species_table.f90.** The !To Be Checked Fe II comment was rewritten to state that eval_cool now applies the density-dependent SE coefficient (no longer “deferred”).

Builds clean (gfortran, full make, all four touched modules relink).

10.6 Step 4 — WASP-121b run and the cooling budget (numbers for the reader)

A fresh WASP-121b run (solar metals, pp_metals 1, otherwise the same input.inp as the FeI-cooling run) was compared against the snapshot of the previous (coronal Fe II) run kept in output_coronalFeII/. Both share the same grid, so Cooling_breakdown.txt columns are differenced directly (analyze_FeII_density_correction.py $\rightarrow$ FeII_density_correction_compare.png).

At the dense base ($r = 1.0$, $T = 2358\text{K}$ fixed by the lower BC). The Fe II line cooling drops from the coronal 2.23×10^{-4} to the SE $1.71 \times 10^{-8} \text{ erg cm}^{-3} \text{ s}^{-1}$ — a $\approx 1.3 \times 10^4 \times$ suppression, consistent with the offline fixed- (T, n_e) check in Step 1 ($15,663 \times$ at $T = 2239\text{K}$, $n_e = 10^8$). Because coronal Fe II carried **99.1%** of the base cooling, the *total* radiative cooling at the base falls $\approx 122 \times$ ($2.25 \times 10^{-4} \rightarrow 1.84 \times 10^{-6}$); Fe II’s share there is now **0.9%**. The previous run’s post-process printed “35 of 502 cells fell back to eq T (stiff base band)”; with the spurious base overcooling removed that message is **gone** — the cell-by-cell T solve no longer stiffens at the base.

Transition region 1.15–1.4 R_p (Huang Fig. 10). The band-integrated metal-line cooling share in the new run is

coolant	share
Mg II	44.0%
O II	18.9%
Fe II	13.4%
O I	7.3%
C II	6.3%
Ca II	5.1%

so **Mg II is now unambiguously the dominant metal coolant** across the transition region, with Fe II demoted to a $\sim 13\%$ contributor — the qualitative ordering Huang’s Fig. 10 shows. (In the old coronal run Fe II still topped the budget out to $\sim 1.2 R_p$.)

Global response. Removing the base overcooling raises the heating efficiency, so the steady-state mass-loss rate rises from $\log_{10} \dot{M} = 12.51$ to **12.80** g/s ($\approx 2\times$). The self-consistent wind is correspondingly denser and, above the base, slightly *cooler* (e.g. $T(1.05)$: $7810 \rightarrow 5744$ K) — the freed energy goes into mechanical escape and adiabatic expansion rather than being radiated, the expected energy-limited behavior. Whether this $\dot{M}$ and the transition-region budget match Huang’s WASP-121b quantitatively is left to the reader’s judgment against the paper; the figure and table above are surfaced for that comparison, not asserted as a pass.

Caveat on the figure’s panel (b). The “suppression factor” there is the *ratio of Fe II cooling between the two runs*, not a fixed-condition coefficient ratio. At the base (BC-fixed T) it equals the coefficient suppression ($\sim 1.3 \times 10^4$); above $\sim 1.2 R_p$ the coronal and SE coefficients converge (the hot resonance lines dominate and stay near-coronal), so the residual ratio there reflects the changed — denser — wind profile, not the coefficient.

11. Post-process refinement: re-solve metal ionization in the advected wind (pp_metals 2)

Added 2026-06-05.

The `_adv` profiles that EXHALE_transit.py reads are built by `post_process_adv.f90`, which re-solves the *temperature in each cell* at the advection-corrected structure (the wind is advected, then its $T(r)$ is re-converged against heating/cooling). Until now the metal *ion* densities carried into those profiles were handled in one of two ways, selected at runtime by `pp_metals` in `metals.inp` ($\rightarrow$ `pp_metal_mode`):

- `pp_metals 0` — **metal-free legacy** `_adv` (the pre-metals behavior).
- `pp_metals 1` — **frozen equilibrium** metals (the default): the converged equilibrium metal stages are carried into `_adv` unchanged, so they track the *equilibrium* structure, not the (slightly hotter, more ionized) advected wind.

This step implements the deferred third option:

- `pp_metals 2` — **re-solve**: the metal ionization balance is recomputed in each cell at the advection-corrected H/He and the post-process temperature, so the metal stage split tracks the advected wind self-consistently.

11.1 Implementation — reuse the coupled residual with H/He pinned

The metals are pure equilibrium species (no advection equations of their own): in the main loop they are solved *coupled* with H/He through `ion_system_HeH_metals + hybrd1` (photo + collisional ionization + recombination + charge exchange, no time derivative). “Re-solve” therefore means re-evaluating that same equilibrium balance at the post-process local T and the advected H/He — *not* a new physics path.

Rather than duplicate the residual, mode 2 reuses it verbatim through a thin wrapper, `ion_system_metals_pp` (new, in `post_process_adv.f90`):

```
subroutine ion_system_metals_pp(N_in,x,fvec,iflag,params)
...
call ion_system_HeH_metals(N_in,x,fvec,iflag,params) ! full coupled residual
fvec(1) = x(1) - pp_xHII_fix      ! pin HII fraction
fvec(2) = x(2) - pp_xHeII_fix     ! pin HeII fraction
fvec(3) = x(3) - pp_xHeIII_fix    ! pin HeIII fraction
end subroutine
```

The wrapper overwrites only the three H/He rows with identity equations that pin H/He to their advection-corrected fractions (stored in module-level `pp_xHII_fix` / `pp_xHeII_fix` / `pp_xHeIII_fix`), while leaving the $2n_{\text{elem}} = 20$ metal-stage rows untouched. So `hybrd1` moves only the metal unknowns; H/He stay fixed at the advected wind. Two properties fall out for free:

- **Self-consistent n_e** . The reused residual computes the electron density *internally* from the pinned H/He plus the metals it is currently solving — exactly as the equilibrium solver does — so the re-solve sees the same n_e (metals) feedback, not a frozen n_e .
- **Charge exchange with the advected H**. The Kingdon & Ferland $\text{H} \leftrightarrow \text{metal}$ couplings inside `ion_system_HeH_metals` see the *pinned* H^+/He^+ densities, so the metal balance reacts to the advected hydrogen ionization, not the equilibrium one.

11.2 The cell-by-cell sweep in `post_process_adv`

Mode 2 adds one sweep over cells, placed after the post-advection `calc_ne` and before the photoheating refresh, inside the same k -loop that re-converges T , so metals, H/He and T are driven to a *joint* fixed point. Per cell j (skipping $n_h(j) \leq 0$):

1. `cx_set_cell(T_K(j))` — charge-exchange rate coefficients at the local T .

2. Build the coefficients for each element (canonical order) from the existing post-process rate arrays: $\text{meg_g0} = P_m$ (photoionization), $\text{meg_b0} = \text{aion_m_pp}$ (collisional ionization), $\text{meg_a1} = \text{rec_m_pp}$ (recombination), and the second-stage $g1/b1/a2$ where $\text{melem_top} \geq 2$; hand them to `set_metal_coeffs`.
3. Pin $\text{pp_xHII_fix} = n_{\text{HII}}/n_h$, $\text{pp_xHeII_fix} = n_{\text{HeII}}/n_{he}$, $\text{pp_xHeIII_fix} = n_{\text{HeIII}}/n_{he}$; set `params(7) =  $n_h$` , `params(8) =  $n_{he}$` .
4. Seed `sys_x` with the pinned H/He fractions plus the current metal split and call `hybrd1(ion_system_metals_pp, . . .)`.
5. **Conserve element totals:** the pre-solve totals `nm_tot_pp(:,im)` (summed over each element's stages once, before the k -loop) are multiplied by the re-solved fractions to write back `nm_w`. Only the *stage split* changes; the element abundance is untouched.

Supporting edits: the use block pulls in `ion_system_HeH_metals`, `set_metal_coeffs`, and `cx_set_cell`; `params` was widened from dimension(25) to dimension(60) to match the residual's dummy length (only entries 1–11 are consumed by the metal solve); module-level `pp_x*_fix` and locals `nm_tot_pp/meg_*/i0`, `top`, `im` were added. Builds clean (gfortran, full make).

11.3 Clean A/B and what changes

Because the **equilibrium** solve in the main RK loop is mode-independent and deterministic, a full `pp_metals 2` run reproduces `Hydro_ioniz.txt`, `Ion_species.txt`, and `Cooling_breakdown.txt` **byte-identically** to the `pp_metals 1` run (verified with `diff -q`); *only* the `*_adv.txt` profiles differ. So the A/B isolates exactly the post-process metal treatment. The mode-1 default is preserved in `output_ppm1_dd/`, the mode-2 run in `output_ppm2_resolve/`, and the canonical `output/` is left at the mode-1 default (metals.inp reset to `pp_metals 1`); `analyze_metal_resolve.py` differences the two and writes `metal_resolve_compare.png`.

Singly-ionized fraction X^+/X_{tot} in the `_adv` wind (eq frozen $\rightarrow$ re-solve). Only the elements that move appreciably are tabulated (Mg and Na are unchanged to $\leq 2\%$); the full output for each element is in the analysis script.

r/R_p	T_{adv} [K]	O	N	Ca	Fe
1.05	5798	.00 $\rightarrow$.00	.01 $\rightarrow$.02	.64 $\rightarrow$.59	1.00 $\rightarrow$ 1.00
1.10	7984	.02 $\rightarrow$.01	.12 $\rightarrow$.13	.65 $\rightarrow$.60	.99 $\rightarrow$.99
1.15	9044	.10 $\rightarrow$.06	.38 $\rightarrow$.40	.66 $\rightarrow$.59	.99 $\rightarrow$.99
1.20	10035	.37 $\rightarrow$.26	.51 $\rightarrow$.47	.61 $\rightarrow$.54	.93 $\rightarrow$.96
1.30	10274	.34 $\rightarrow$.34	.35 $\rightarrow$.34	.38 $\rightarrow$.38	.39 $\rightarrow$.40
1.40	10680	.23 $\rightarrow$.23	.23 $\rightarrow$.23	.25 $\rightarrow$.25	.18 $\rightarrow$.19

Largest density change. The element-by-element scan of $\max |\Delta n(X^+)/n(X^+)|$ over the wind 1.0–1.48 R_p :

element	max rel. change	at r/R_p	$n(X^+)$ frozen $\rightarrow$ resolve
Fe	+9.5%	1.232	$1.19 \times 10^5 \rightarrow 1.30 \times 10^5$
Ca	-11.9%	1.191	$1.24 \times 10^4 \rightarrow 1.09 \times 10^4$
Na	-6.2%	1.000	$3.50 \times 10^6 \rightarrow 3.28 \times 10^6$
C	-2.1%	1.216	$1.22 \times 10^6 \rightarrow 1.19 \times 10^6$
Mg	-2.0%	1.000	$9.82 \times 10^7 \rightarrow 9.62 \times 10^7$

The corrections are **modest and concentrated in the 1.1–1.25 R_p ionization transition**, where T_{adv} runs a few hundred K hotter than T_{eq} : **Fe II** is *higher* across the 1.18–1.27 R_p shoulder (blue above red in the figure), **Ca II** is *lower* by ~ 5 –7 points through 1.05–1.25 R_p , and **every element converges back to the equilibrium split** in the outer wind ($\gtrsim 1.27 R_p$) where T_{adv} rejoins T_{eq} . The post-process printed **no** stiff-base T -fallback message in mode 2 (the density-dependent Fe II cooling of the previous section already removed the base overcooling).

Tiny-denominator artifacts (not physical). The same scan reports **O** +6538% at $r = 1.000$ and **N** +17.9% at $r = 1.013$. These are at the dense base where O and N are essentially *fully neutral* (O II fraction ≈ 0.00 in the table), so $n(\text{O II})$ is a vanishingly small number; a tiny absolute shift ($n^+ \approx 9.7 \times 10^2 \rightarrow 6.4 \times 10^4 \text{ cm}^{-3}$, against an O total $\sim 10^8$) reads as a huge *relative* change but is negligible in absolute and in any observable. They are denominator artifacts, not a physical re-solve effect.

Reader’s judgment, not a pass. The re-solve makes the metal stage split in the `_adv` profiles consistent with the advected wind rather than the equilibrium structure, with the physically expected sign (more ionization where the advected wind is hotter) and magnitude (single-digit-percent, transition-localized). The default remains `pp_metals 1` (frozen): the deliberate, documented choice (the equilibrium split is the converged one, and EXHALE_transit.py’s lines are dominated by H and He). Mode 2 is provided for studies that need the metal ionization to track the advected wind. The table and `metal_resolve_compare.png` are surfaced for that comparison, not asserted as a validation pass.

12. Phase 3b: in-code $\text{Ly}\alpha$ radiative transfer ($\bar{J}_{\text{Ly}\alpha}$ via escape probability)

Phase 3a drove the $\text{H}(n=2)$ population with a *parameterized* $\text{Ly}\alpha$ mean intensity $\bar{J}_{\text{Ly}\alpha} = 0.1 F_{\text{LyC}} / \Delta\nu_D / (1 + \tau_{\text{Ly}\alpha})$ (mode 0). That estimate peaks the $n=2$ population near $2R_p$ and *under-pumps* it across 1.1–1.8 R_p , leaving a `jlya_mode=1` hook (`load_jlya_rt`) to ingest a real RT field. Phase 3b supplies the real $\bar{J}_{\text{Ly}\alpha}(r)$. Three findings reshaped the original plan (a Monte Carlo with an offline two-pass and a β -on-cooling term):

1. **Method $\rightarrow$ escape probability, not Monte Carlo.** The WASP-121b column is extremely thick: $\tau_0(\text{Ly}\alpha) \approx 5.5 \times 10^7$ at the base (Voigt $a \approx 5 \times 10^{-4}$, so $a\tau_0 \approx 10^4 \gg 1$). A faithful acceleration-free MC needs $\sim \tau_0 \approx 10^7$ – 10^8 scatterings/photon (intractable), and core-skipping is both forbidden and incompatible with the core-dominated pumping field. For $a\tau_0 \gg 1$ the Neufeld (1990)/Harrington analytic wing-escape solution is accurate, so we use a local escape-probability source function.
2. **In-code, not offline.** Being a cheap formula, $\bar{J}_{\text{Ly}\alpha}$ is evaluated every timestep inside the run (`jlya_mode=2`, the same lagged-explicit decoupling as the Phase-3a Balmer feedback); a single run is self-consistent. The

jlya_mode=1 option to read an external (e.g. real MC) profile is kept.

3. **No β -on-cooling term.** From Huang §2.5/Fig. 7 directly: the “Ly α cooling 10 $\times$ lower” is the Black (1981) vs CHIANTI *collisional-excitation rate-coefficient* difference, not photon trapping. EXHALE already uses the CHIANTI-consistent Cen coex_rate_HI (§7), so there is no 10 $\times$ to apply. The trapping feedback is instead the Phase-3a H($n=2$) heating (photoelectric Hpe + collisional de-excitation Hdx), which strengthens once $\bar{J}_{\text{Ly}\alpha}$ is correct. The scaffolding β -on-coex hook was reverted.

12.1 The escape-probability field (lya_rt.f90 : jlya_escape_prob)

Per cell, from the converged T , $n_{\text{HI}} = n_{1s}$, n_{HII} , n_e :

- $\Delta\nu_D = \nu_{\text{Ly}\alpha} \sqrt{2kT/m_{\text{H}}/c}$; Voigt $a = (A_{2p1s}/4\pi)/\Delta\nu_D$; optical depth $\tau(r) = \int_r^\infty n_{1s}\sigma_0 dr'$, $\sigma_0 = C_{\text{Ly}\alpha}/\Delta\nu_D$.
- Wing escape probability $\beta = \pi^{-1/4} \sqrt{a/\tau} (\leq 1)$: escape once the photon diffuses to $x_* = (a\tau/\sqrt{\pi})^{1/2}$ where the wing optical depth is unity.
- Internal sources $P = \alpha_B n_e n_{\text{HII}} + q_{1s2p} n_e n_{\text{HI}}$ (recombination cascade + electron-impact $1s \rightarrow 2p$). The escape-probability closure $\bar{J} = S(1 - \beta)$ with the trapped pile-up $n_{2p} = P/(A_{2p1s}\beta)$ gives $J_{\text{int}} = (2h\nu^3/c^2)(g_{1s}/g_{2p})P(1 - \beta)/(A_{2p1s}\beta n_{1s})$. The $(1 - \beta)$ factor makes $J_{\text{int}} \rightarrow S$ when thick and $\rightarrow 0$ when thin ($\beta \rightarrow 1$), instead of $S \propto 1/n_{1s}$ diverging as the gas ionizes in the outer wind (which spuriously raised $\bar{J}$ outward).
- Stellar beam (trapped): Ly α is *scattered*, not destroyed, so it is not killed by $e^{-\tau}$; the broad stellar line (half-width $\Delta v_s \approx \pm 70 \text{ km s}^{-1}$) penetrates through its wings. A photon at offset x reaches r if $|x| > x_1 = \sqrt{a\tau/\sqrt{\pi}}$; for a Gaussian stellar profile of width $X_s = \Delta v_s/v_{\text{th}}$, $T_*(r) = \text{erfc}(x_1/(\sqrt{2}X_s))$. The profile-averaged mean intensity uses the *stellar* line width $\Delta\nu_* = \nu_0 \Delta v_s/c$ (constant), NOT the planetary $\Delta\nu_D$ ($\bar{J} = \int J_\nu \phi d\nu \approx J_\nu(\nu_0)$ for a stellar field flat across the narrow planetary line; dividing by $\Delta\nu_D$ makes J_{star} rise as the wind cools). A bounded buildup $E = 1 + (\text{boost} - 1)(1 - \beta)T_*$ enhances the penetrating beam in the thick wind and $\rightarrow 1$ when thin or at the deep edge, avoiding both the $1/\beta$ over-count (which blew the $n=2$ source to 6300 $\times$ ground-state) and a flat cap. Hence $J_{\text{star}} = \xi F_* T_* E / (4\pi \Delta\nu_*)$, ξ the dayside dilution. $\bar{J}_{\text{Ly}\alpha} = J_{\text{int}} + J_{\text{star}}$.

New input.inp lines (default-off $\Rightarrow$ byte-identical to 3a; a 3b-0 scaffolding step verified this, frozen reference output_pre3b_baseline/):

```
Jlya escape-prob: True           # jlya_mode = 2 (in-line escape-prob RT)
Stellar Lya flux [erg/cm2/s]: 1.0e5 # incident stellar Lya at the planet (x0.35)
Lya stellar halfwidth [km/s]: 70   # broad stellar line Dv_s (penetration + Dnu_star)
Lya stellar boost [-]: 5           # bounded trapping-buildup cap (left un-tuned)
Deexc heat: True                  # Phase-3a collisional de-excitation heating
```

12.2 WASP-121b validation setup (spherical Case A, matched to Huang)

Huang’s Fig. 11 is **Case A** (spherical, no RLOF), so the comparison must match his geometry *and* radial frame — which exposed a setup error: the run had been Roche-truncated at $\sim 2R_p$, and EXHALE’s base ($r=1$) sits at **1 μbar**

($n_0 = 3.1 \times 10^{12}$, $T = 2358$ K), but Huang defines R_p at the **4 mbar transit radius** and places his $1 \mu\text{bar}$ hydro bottom at $\sim 1.25 R_p$ (Fig. 9 top-pressure axis / Fig. 12 green band; the earlier “ $1.46 R_p$ ” note was wrong). The setup was corrected to spherical Case A with R_0 set to Huang’s $1 \mu\text{bar}$ radius ($R_0 = 1.25 \times 1.766 = 2.2075 R_J$, planet mass unchanged $\Rightarrow$ correct base gravity) and outer radius $15.2 (= 19 R_p^{\text{Huang}})$; EXHALE r maps as $R_{\text{Huang}} = 1.25 r$. The Case-A $\dot{M} = 0.077 M_p/\text{Gyr}$ is within $\sim 1.5\times$ of Huang’s 0.052 (Table 3).

12.3 Trend vs Huang Fig. 11 (qualitative — exact match not expected)

Huang’s $\bar{J}_{\text{Ly}\alpha}$ is a **plane-parallel Monte Carlo** RT; ours is an analytic escape-probability formula, so the two cannot agree quantitatively — a quantitative comparison is only sensible once *all* Phase-3b features (incl. the velocity term) are in. Qualitatively (lya_caseA_huang11.py, not in the tree; the same Fig. 11 case-A comparison is now WASP-121b/Lya_vs_huang11.py), both show a significant inner peak (EXHALE ~ 0.6 at 1.35 , Huang ~ 0.5 at $1.6 R_p$) and comparable magnitude (0.05–0.6). EXHALE has a dip at $\sim 1.8 R_p$ and rises gently outward where Huang gently declines; the outward rise traces to the $n2s$ recombination cascade in the Phase-3a $\text{H}(2s/2p)$ model ($n=2$ stays recombination-populated while ground-state H vanishes in the ionized wind), *not* $\bar{J}$, which after the two fixes below is physically sound (J_{int} peaks then declines; $J_{\text{star}} \approx \text{const}$ outward). $\text{H}(n=2)$ heating stays $\sim 1\%$ of total with $\text{Hdx} \gtrsim \text{Hpe}$ near the n_2 peak (cf. Fig. 10).

Two $\bar{J}$ bugs fixed during the trend check. (1) J_{star} used the planetary $\Delta\nu_D$ instead of the stellar width $\Delta\nu_*$, making $\bar{J}$ rise as the wind cooled; (2) J_{int} lacked the $(1-\beta)$ closure, so S diverged ($\propto 1/n_{1s}$) in the ionized wind.

Wind-velocity (Sobolev) escape — implemented, negligible here. $\beta_{\text{tot}} = 1 - (1 - \beta_{\text{static}})(1 - \beta_{\text{Sob}})$ with $\tau_S = C_{\text{sob}} n_{1s} / |dv/dr|$ is now in jlya_escape_prob (the outflow $v(r)$ threaded through excited_H_update). For WASP-121b it changes $\bar{J}$ by $< 1\%$: where it could help (thick inner wind) $\tau_S \sim \tau_{\text{static}} \sim 10^7$ so $\beta_{\text{Sob}} \sim 10^{-7}$, and the thin outer wind already has $\beta_{\text{static}} \approx 1$ — the velocity gradient and the $\text{Ly}\alpha$ opacity scale together ($\propto n_{1s}$), opening no new escape route. **Phase 3b is thus feature-complete.** *Deferred per the trend-first guidance:* the post-all-features quantitative comparison and the $n2s$ outward-rise revisit (a Phase-3a $2s$ effect, not the RT); boost, Δv_s , and the $\pi^{-1/4}$ prefactor are left un-tuned.

13. Phase 4: RLOF / tidal potential (Case B validated; Case D blocker = escape radius, transonic IC added, geometry-limited)

Phase 4 adds the Roche-lobe-overflow (RLOF) tidal physics that drives Huang’s enhanced mass loss — **Case B** $\dot{M} = 0.32$, **Case D** $= 1.03 M_p/\text{Gyr}$ — the regime that matches the observed transit depths (spherical Case A underestimates them). **EXHALE already has the 1D Roche potential** (grav_field.f90, the spherical_domain=.false. branch: planet + stellar-tidal + centrifugal), the L1-truncated domain, and the tidal momentum source (Source.f90) — Caldiroli’s ATES-v2 Roche mode. So the substellar tidal hydrodynamics is already in place.

13.1 Base gravity for each case (Huang Table 3)

The key to each case is the $1\ \mu\text{bar}$ base radius R_0 . Huang’s Table 3 lists $\log g @ 1\ \mu\text{bar} = GM_p/r_{1\mu\text{bar}}^2$ per case — exactly the R_0 -setting quantity ($R_0 = \sqrt{GM_p/g}$):

Case	tidal	$M_p(M_J)$	dT	XUV/Ly α /Fe	$\log g$	$R_0(R_J)$
A	N	1.1824	0	1/1/1	2.84	2.058
B	Y	1.1824	0	1/1/1	2.70	2.418
D	Y	1.1204	350	0.74/0.35/4	2.62	2.581

R_0 grows A $\rightarrow$ B $\rightarrow$ D as the tidal force + the $dT+350$ lower-atmosphere heating puff the $1\ \mu\text{bar}$ level outward (weaker base gravity), driving the RLOF.

13.2 Case B — validated

WASP-121b in Roche mode (tidal on) + the corrected base + metals + Ly α mode-2 gives $\dot{M} = 0.337M_p/\text{Gyr} \approx$ Huang’s Case B (0.32), within $\sim 5\%$. With the base-radius fix both cases agree (Case A $0.077 \approx 1.5 \times$ Huang’s 0.052; Case B $0.337 \approx 1.05 \times$ Huang’s 0.32). EXHALE’s existing Roche potential thus captures the RLOF enhancement — no geometric-reconstruction code was needed for the Case B $\dot{M}$ gate.

13.3 Case D — real blocker found (escape radius), transonic IC added

Case D’s base ($R_0 = 2.581 R_J$, $\log g = 2.62$) sits at $\sim 76\%$ of the L1 distance: the L1-truncated Roche domain reaches only $r_{\text{max}} = 1.353 R_p$. Re-investigating the earlier “won’t launch” report uncovered the *actual* blocker, and it is *not* the initial condition.

Root cause — the escape radius lies outside the L1-truncated domain. `define_grid` sets the constant-momentum convergence window $[j_{\text{min}} : N]$ from the **escape radius** (Escape radius `[R_p]:`, default 1.50). It scans for the first cell with $r(j) \geq r_{\text{esc}}$; if *no* cell qualifies — which happens precisely when $r_{\text{esc}} > r_{\text{max}}$ — the DO loop falls through leaving $j = N + N_g + 1$, so $j_{\text{min}} > N$ and the window is **empty**. Fortran `maxval/minval` over a zero-size array return $\mp\text{HUGE}$, so $\text{mom}_{\text{max}} = -\text{HUGE}$, $\text{mom}_{\text{min}} = +\text{HUGE} \Rightarrow du = \text{Inf}$, and the time-derivative norm $dtu = -\text{HUGE} < dtu_{\text{th}}$, the loop reports “**steady state reached**” at **step 0** — a *false* convergence unrelated to the wind. The earlier bracket (“works $R_0 \leq 2.30$, fails $R_0 \geq 2.40$ ”) is exactly r_{max} crossing the 1.50 escape radius at $R_0 \approx 2.37$, confirming the escape radius, not the IC, as the discriminant. (The earlier “isothermal IC cannot launch the deep-RLOF wind” diagnosis was therefore wrong.)

Fixes. (1) A guard in `define_grid.f90`: if $j_{\text{min}} > N$, re-anchor the window to mid-domain ($r \geq 1 + \frac{1}{2}(r_{\text{max}} - 1)$) and print a loud WARNING — so this can never again masquerade as convergence. (2) For Case D, set Escape radius `[R_p]:` 1.20 (inside the 1.353 domain). With a valid window the diagnostics become finite and meaningful, and **Case D launches and runs** — from *both* the hydrostatic and the transonic IC. (The earlier $du = |(\text{mom}_{\text{max}} -$

$\text{mom}_{\min}) / \max(\text{mom}_{\min}, 10^{-30})$ floor at EXHALE_main.f90:308 is retained as defense-in-depth against a genuine transient zero-momentum cell, but it was *not* what unblocked Case D — with an empty window $\text{mom}_{\min} = +\text{HUGE}$, not zero, so the floor never engaged.)

Transonic-wind IC (implemented, Transonic IC: True). set_IC.f90 gained an optional transonic isothermal-wind profile, solved from the EXHALE potential via the algebraic (Bernoulli) integral of the steady isothermal-wind equation,

$$\frac{1}{2}v^2 - c^2 \ln v - 2c^2 \ln r + \phi(r) = B, \quad c^2 = (1 + \delta p_{bc}) / \rho_{bc},$$

with B fixed by the sonic point ($v = c_s$ where $D\phi(r_c) = 2c^2/r_c$, located by bisection). At each radius the relation has a subsonic ($v < c_s$) and a supersonic ($v > c_s$) root; the wind takes the subsonic root for $r < r_c$ and the supersonic root for $r > r_c$ (1-D bisection on each monotone branch), and ρ then follows from mass conservation $\rho v r^2 = \dot{M}$ pinned to the base BC. Because that IC already satisfies $\rho v r^2 = \text{const}$, $du \approx 0$ at step 0 would itself trip the momentum-constant exit before the cold wind heats; the flag therefore **auto-enables force_start**. If no interior sonic point exists it **falls back to the hydrostatic IC**. Default **off** $\Rightarrow$ existing cases byte-identical (Case B re-verified at $\dot{M} = 0.337$).

Case D outcome — launches, but the 1-D L1-truncated domain undershoots. With the escape-radius fix the wind launches and relaxes, but it **never reaches a sonic point inside the domain**: the heated metal wind is **deeply subsonic everywhere** ($\text{Mach} \lesssim 0.05$ — rising to ~ 0.050 at $r \approx 1.25$, then *falling* to ~ 0.040 at L1). With a subsonic outer boundary the zero-gradient outflow BC is only marginally well-posed and $\rho v r^2$ stays non-constant (peaks $\sim 1.7\times$ the L1 value), so the run settles into a **breathing limit cycle**: after the $\sim 10^2$ heating transient damps, du keeps oscillating in $\sim 0.04\text{--}0.8$ (touching but never crossing the strict $du_{\text{th}} = 0.02$) for as long as it runs. The mass flux is correspondingly time-variable — the quasi-steady $\dot{M}$ oscillates over $\sim 0.3\text{--}0.6 M_p/\text{Gyr}$, well *below* Huang’s 1.03, the deficit set by the L1 truncation cutting off the still-accelerating subsonic wind before it reaches the sonic point. The transonic IC behaves identically to the hydrostatic IC here: both launch the same subsonic flow. (Warm-starting via load_IC reintroduces a base NaN — the documented grid re-read inconsistency — so it is not a workaround.)

Conclusion. Case D’s deep RLOF ($\dot{M} = 1.03$) is **not reachable in EXHALE’s 1-D L1-truncated Roche mode**: the wind is subsonic at L1, so the truncated domain structurally undershoots. Capturing it needs the 3-D geometric treatment (Phase 5) or a domain extended past L1 — *not* a different IC. The transonic IC and the escape-radius guard are correct, necessary infrastructure (they remove the false convergence and let the deep-RLOF wind launch and run with meaningful diagnostics) but are not sufficient on their own for Case D’s full mass-loss rate. The geometric reconstruction (substellar $\rightarrow$ terminator mapping, triaxial radii Table 2, the 4/9 mass-loss conversion) feeds Phase 5 (§14), and also supplies the geometry this subsonic L1-truncated wind needs.

14. Phase 5: velocity-broadened transmission spectrum (in progress)

Phase 5 is the headline deliverable — the NUV/optical **transit-depth spectrum** the whole upgrade exists to match (Huang Table 3 / Figs. 17–24). It extends the existing spherical post-processor EXHALE_transit.py in two stages: **5a**

adds the metal resonance lines (spherical, gated on Case A); **5b** replaces the spherical geometry with the 3-D Roche-equipotential reconstruction + velocity broadening (gated on Case D). Pure Python post-processing — no Fortran change.

Line	Case A	Case D	Observed
Mg II $\lambda 2796$ (4 Å)	0.182	0.302	0.309 ± 0.036
Ca II K $\lambda 3934$	0.199	0.278	0.281 ± 0.009
H α $\lambda 6563$	0.201	0.185	0.186 ± 0.003
H β $\lambda 4861$	0.174	0.135	0.143 ± 0.005
Na D2 $\lambda 5890$	0.152	0.147	0.147 ± 0.002

These are the **effective transit radius** R_{eff}/R_* , not an absorption percent (the expected-results doc states the unit is R_p/R_*). Our code returns the line-center absorption fraction h , mapped to the same quantity via $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$.

14.1 5a — metal resonance lines (Mg II, Ca II, Na I D), spherical — done

EXHALE_transit.py gained Mg II $\lambda 2796$, Ca II K $\lambda 3934$, Na I D2 $\lambda 5890$ as an **isolated appended block** (the validated He 10830 / Ly α / H α / H β pipeline above it is untouched). These are resonance lines whose lower level is the ion ground state; at $\sim 10^4$ K the excited fine-structure levels are Boltzmann-negligible, so the lower-level density is the ion density itself ($n_{\text{lower}} \approx n_{\text{ion}}$). The ion densities live past the H/He block of Ion_species_adv.txt, in mion_fsp order (species_table.f90): numpy columns **17 = Mg II**, **23 = Ca II**, **25 = Na I**. The block reuses the existing spherical chord / Voigt (wofz) / trapezoid- τ / disk-average / instrument convolution path (resonance_depth()).

Case A result (line-center % / 4 Å-band %) and diagnosis. Mg II: 26.1 / 2.3 ($R_{\text{eff}} = 4.2 R_p$); Ca II K: 5.5 / 0.33 ($1.9 R_p$); H α : 3.9 ($1.6 R_p$); Na D2: 0.8 / 0.06 ($0.7 R_p$). Two real gaps: (1) **Mg II already reaches Huang's extent** ($R_{\text{eff}} = 4.2 R_p$ vs his clustered $\sim 3.6 R_p$) — the metal treatment is sound — but **Ca II / Na I / H α are too optically thin** ($R_{\text{eff}} = 0.7\text{--}1.9 R_p$), a line-specific ionization / abundance / $n = 2$ question, not geometry. (2) The lines are **too narrow**: the Case A wind reaches only ~ 29 km/s, so a 26%-deep core washes out to $\sim 2\%$ over a 4 Å (± 215 km/s) bin, whereas Huang's lines stay deep *across* the bin — they are velocity-broadened to hundreds of km/s by the fast Roche outflow + rotation, i.e. **Phase-5b physics**. (The existing H α tool shows the same shortfall, so it is systematic.) *Units — resolved*: Huang's table is R_{eff}/R_* (radius ratio; the expected-results doc header states the unit is R_p/R_* , not absorption percent). The Case A spherical run over-extends ($15 R_p$ domain $\Rightarrow$ Mg II reaches $4.2 R_p$, $R_{\text{eff}}/R_* = 0.51$ vs Huang 0.182); the proper test is the L1-confined Roche run in R_{eff}/R_* units (§5b).

14.2 5b — Roche-equipotential reconstruction + broadening — implemented

New module roche_recon.py. Assuming a uniform state on Roche-equipotential surfaces, the field at any (x, y, z) equals the 1-D substellar state at the equivalent radius r_{eff} where $\phi_{\text{sub}}(r_{\text{eff}}) = \phi_{3D}(x, y, z)$. The dimensionless Roche potential (R_p units) mirrors grav_field.f90:

$$\phi(x, y, z) = -\frac{1}{r} - \frac{q}{r_s} - \frac{1+q}{2a^3} [(x-x_{\text{cm}})^2 + y^2], \quad q = M_*/M_p, \quad a = \text{orbit}[R_p].$$

Tidal terms $\rightarrow 0$ recovers $\phi = -1/r$ (spherical), so the triaxial reconstruction reduces to the 5a spherical case for Case A. The module gives `roche_phi`, the L1 root of $d\phi_{\text{sub}}/dr = 0$, a monotone $\phi_{\text{sub}} \rightarrow r_{\text{eff}}$ inverse (ReconMap), the triaxial radii, and the Eggleton lobe radius. `EXHALE_transit.py` gained a `geometry='triaxial'` switch (`triaxial_depth()`): the transit LOS runs along $+x$; the state at each 3-D point is the substellar state at r_{eff} ; and the **LOS velocity in each sector** $v_{\text{sub}}(r_{\text{eff}})x/r - \Omega y$ (wind + tidally-locked rotation, Huang Eq. 16) is integrated over **20 angular sectors**.

Demonstrated on the converged Case B (Roche) run. Geometry self-consistent: $L1 = 1.574 R_p$; $R_{\text{RL}}/L1 = 0.714 \approx 2/3$; tidally triaxial $R_{px} = 1.05 > R_{py} = 0.919 > R_{pz} = 0.893$ (substellar bulge, polar compression).

Result — the model reproduces Huang’s transit radii. Converting the Case B triaxial depths to R_{eff}/R_* (the table unit):

Line	Huang A	Huang D	Case B (this work)
Mg II	0.182	0.302	0.234
Ca II	0.199	0.278	0.222
Na D2	0.152	0.147	0.163
H α	0.201	0.185	0.239
H β	0.174	0.135	0.232

Case B is the *intermediate* model (between Case A and Case D), and the **metals (Mg II, Ca II, Na) land right between Huang’s Case A and Case D** — the lobe-confined reconstruction correctly reproduces his transit radii in magnitude and trend. The Balmer lines run $\sim 25\%$ high (H α 0.239 vs the expected ~ 0.19), consistent with the 5a finding that $n = 2/T$ is slightly over-predicted.

Correction. An earlier draft claimed the lobe-confined model “undershoots Huang because it omits the beyond-L1 escaping tail.” That was wrong twice over: (1) Huang’s reconstruction is *also* lobe-confined (equipotential mapping to the Roche-lobe boundary, §1.8) — neither model has a beyond-L1 tail, so it cannot be the difference; and (2) the apparent shortfall was a **units error** (our absorption *percent* vs Huang’s R_{eff}/R_*). In the correct units the two agree (table above). The small triaxial-vs-spherical reduction (terminator compression $R_{py} < R_{px}$) is minor; Huang’s area-preserving $R_{py}R_{pz} = R_p^2$ normalization would offset it.

14.3 5b — direct Case A comparison: hydro validated, metals over-extended

A clean spherical Case A at the *correct* 1 μbar base $R_0 = 2.058 R_J$ (Table 3 $\log g = 2.84$; the old `output_caseA` used $R_0 = 1.766$, the 4 mbar transit radius) was run to convergence.

- **Hydro / base validated:** $\dot{M} = 0.056 M_p/\text{Gyr} \approx$ Huang’s Case A 0.052 (the wrong base gave a different $\dot{M}$) — base and wind are right.
- **Metal transit radii — consistent once compared like-for-like** (Huang quotes Mg II in a 4 Å NUV bin; Ca II / Na are optical line-center): Mg II (4 Å) **0.220** vs Huang 0.182 ($\times 1.2$); Na D2 (LC) **0.175** vs 0.152 ($\times 1.15$); Ca II K (LC) **0.306** vs 0.199 ($\times 1.5$, high).

The “over-extension” was a comparison-definition artifact, not physics. An earlier draft reported $\text{Mg II } R_{\text{eff}}/R_* = 0.60$ ($3.3 \times$ Huang) — that compared the *line-center* R_{eff} against Huang’s $4 \text{ \AA}$ NUV-bin value. In the same $4 \text{ \AA}$ bin $\text{Mg II} = 0.22 \approx$ Huang’s 0.182; Na matches too; only Ca II is genuinely $\sim 1.5 \times$ high.

The metal ionization is correct and is NOT the lever. All channels are wired and physical (Verner+1996 σ , Badnell RR+DR, Voronov, Huang+2023 charge exchange; $a_\tau = 0$). The Mg II/III balance is a close competition between a weak photoionization ($\Gamma \approx 2 \times 10^{-3} \text{ s}^{-1}$, weak because $\sigma_{\text{MgII}} \approx 0.24 \text{ Mb}$ at threshold) and $\text{RR} \cdot n_e$; DR ($\sim 5 \times 10^5 \text{ K}$ resonances) and collisional ionization ($e^{-15 \text{ eV}/kT}$ at 10^4 K) are negligible. A test showed $\times 30$ **more Mg II photoionization moves R_{eff} only $0.57 \rightarrow 0.32$** — the resonance line is so optically thick that ionizing it harder cannot confine it, so the ionization is a dead end as a lever (and is not the cause once the bin is matched).

Status: the full Phase-5 pipeline is implemented and runs end-to-end; the hydro/ $\dot{M}$ reproduces Huang (Case A 0.056, Case B 0.337); and the metal transit radii match Huang once compared in his own bins (Mg II $4 \text{ \AA}$ 0.22 vs 0.182, Na 0.175 vs 0.152). Remaining items are minor: **Ca II** $\sim 1.5 \times$ **high** (line-center; plausibly Ca II $\rightarrow$ Ca III or the Ca abundance), the slightly-high Balmer $n = 2$, the $R_{\text{py}} R_{\text{pz}} = R_p^2$ normalization (methodological), a terminating Case D run, and phase5_transmission.ipynb. Comparisons must use Huang’s line-by-line definition ($4 \text{ \AA}$ bin for the NUV metals, line-center for the optical lines) — the two differ by $\sim 3 \times$ for the optically-thick Mg II.

15. Post-process advection guard at a “breathing” base (2026-06-07)

Symptom. The spherical-vs-Roche example figure (user manual Fig. 4, WASP-121b) showed a sharp temperature spike at $r \approx 1.1 R_p$ in the *advection-corrected* (`_adv`) Roche profile, with a sawtooth band just below it. The converged equilibrium (eq) profile is smooth there, so the spike is a post-processing artifact, not a feature of the hydro solution.

Diagnosis (two compounding causes).

1. *A breathing, inflowing base (1-D limitation).* WASP-121b nearly fills its Roche lobe, so the 1-D wind is subsonic at L_1 and the dense lower atmosphere settles into a slowly recirculating state with small *negative* (inflow) velocities and a stagnation point ($v = 0$) near the wind base ($r \approx 1.1$). This is intrinsic to spherical symmetry, not a bug.
2. *A spurious hot root in the metal-cooled post-processor.* The advection post-processor re-solves the energy/ionization balance in each cell assuming an outflow (each cell upwinds from the next-inner cell). In the dense base the non-monotonic metal line-cooling curve gives the energy equation a second, spurious *hot* root that the Newton (MINPACK) solver can land on; the upwind coupling then cascades one bad cell outward, producing the sawtooth and the overshoot spike. The pre-existing $2 \times$ band guard around eq T did not catch the $\sim 1.4 \times$ spike.

Fix (option c: eq fallback where the flow is not a clean outflow). In `post_process_adv.f90`, wherever $v \leq 0$ (the breathing inflow base) the post-processor now *skips* the advection correction and keeps the converged eq ionization and temperature. Three guards were added, all gated on `pp_metal_on` so the metals-off mode stays byte-identical: the no-He and He H/He advection loops (set the H/He fractions to eq and `cycle` when $v(j) \leq 0$ or $v(j-1) \leq 0$) and the cell-by-cell temperature solve (set $T_{\text{out}} = T_{\text{in}}$ and `cycle` when $v \leq 0$). Keeping the inner cells at eq removes the spurious

root *and* breaks the upwind cascade at its source, while the genuine `_adv` correction is retained in the outflow above the stagnation point.

Validation (PP-only re-run on the converged Case B state). Rebuilt and re-ran the post-processor (Do only PP on the converged IC):

- eq reproduced: $\max |\Delta T|/T = 2.6 \times 10^{-3}$, $\max |\Delta v| = 0.015 \text{ km s}^{-1}$; $\log_{10} \dot{M} = 13.38$ unchanged ($\approx 0.34 M_p \text{ Gyr}^{-1}$); Mg II reproduced to 1.6% in the line-forming region.
- `_adv` base oscillation eliminated: temperature sign changes in $[1.02, 1.10] R_p$ dropped from **36 to 0**; the spike is gone; the eq→`_adv` handoff at $v = 0$ is continuous (0.7% step) and the few-percent wind correction is preserved.
- Spherical Case A ($v > 0$ everywhere) is untouched — the guard never triggers, so its `_adv` is byte-equivalent.

Cause 1 (the breathing base) remains a 1-D limitation but is now represented there by the smooth eq solution; resolving the inflow itself would require a 2-D/3-D model.

Documentation updates (this session). The user manual (`EXHALE_user_manual.tex`) gained: §1.3 *The gravitational potential* (spherical, Roche substellar $\phi(r)$ and $d\phi/dr$, and the 3-D off-axis form, matching `grav_field.f90`); §3.5 *Spectral coverage for metal photoionization* (neutral-metal thresholds down to K I at 4.34 eV/2856 Å, and how the power-law sub-grid extends below 13.6 eV vs. a loaded SED that must itself reach those wavelengths); §8 *Atomic and molecular data* (a by-process source table — Asplund09 abundances, Verner96 photoionization, Voronov97 collisional ionization, Hui & Gnedin97 H/He + Badnell06 RR/adf48 DR metals + Shull & Van Steenberg82 Ca I + Huang23 Fe, Huang23 charge exchange, AIOLOS+CHIANTI cooling, Christie13/Draine11/Huang17 excited-H, NIST line data; and the explicit note that the code is purely atomic/ionic with *no* molecular data); and the §4 *breathing-base caveat* documenting the issue and fix above. The Fig. 4 example was regenerated from the corrected `_adv` output.

16. He 2³S triplet + metals simultaneous solve, Ly α -flux guard, and convergence workflow (2026-06-08)

Motivation. Until now the He metastable triplet (He 2³S, “HeITR”) and the trace-metal ionization network could not be solved in the same run: both placed an extra unknown at `x(4)` of the coupled ionization residual, and `input_read.f90` allocated the metal unknowns only if `(thereis_metals .and. .not. thereis_HeITR)`. Modeling HD189733b with the triplet *and* solar metals requires both at once.

Merged solver (factual). New module `nonlinear_system_solver/System_HeH_TR_metals.f90` (`ion_system_HeH_TR_metals`, 84 SLOC). The H/He/triplet rows ($x_1 = \text{HII}$, $x_2 = \text{HeII}$, $x_3 = \text{HeIII}$, $x_4 = \text{He } 2^3\text{S}$) are taken verbatim from `System_HeH_TR`; the metal rows are taken verbatim from `System_HeH_metals` and shifted to $x(5+2(e-1))$. The two blocks share one electron density $n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}} + \sum_e (n_{X^+} + 2n_{X^{++}})$ (the neutral triplet is excluded), and charge exchange is added through `cx_add_to_fvec`. Solved with `hybrd1`. Supporting edits: `charge_exchange.f90` gained a settable base index `cx_metal_base` (default 4; the merged dispatch sets 5) so the metal charge-exchange rows land at the

right offset; `input_read.f90` sets $N_{\text{eq}} = 4 + 2n_{\text{melem}}$ when both are active; `ionization_equilibrium.f90` uses a generalized metal base index (`mbase = 4`, or 5 with HeITR) in the initial guess / solution scatter and a both-on dispatch branch; the new source was added to the Makefile. The radiation grid, cross sections, photo-heating, IC setup and post-processing already handled both species; only the equilibrium *solver* needed merging.

Regression & physical check. With metals on / triplet off the run is byte-identical to the pre-merge binary over 8 048 steps (the new branch is not taken). With both on, the coupled solve is clean (no NaN) and was exercised end-to-end on WASP-121b with the full stack (triplet + metals + excited-H + $\text{Ly}\alpha$), reaching a flat converged solution (peak $\text{He } 2^3\text{S}$ fraction $\sim 5 \times 10^{-6}$).

$\text{Ly}\alpha$ escape-probability input guard (factual). `input_read.f90`: if `Jlya escape-prob: True (jlya_mode=2)` but `Stellar Lya flux [erg/cm2/s]:` is missing or ≤ 0 , the run now stops with an explanatory message rather than silently dropping the stellar beam (in mode 2 $\bar{J}_{\text{Ly}\alpha} = \bar{J}_{\text{int}} + \bar{J}_*$ with $\bar{J}_* \propto F_{\text{Ly}\alpha,*}$, so $F_{\text{Ly}\alpha,*} = 0$ would make the halfwidth/boost settings silent no-ops).

Convergence threshold and Reconstruction scheme:-controlled two-stage reconstruction (factual).

- `du_th` is now a runtime variable (was a compile-time parameter) with default $1.0\text{e-}3$ — the original ATES-Code-main value. EXHALE had carried $2.0\text{e-}2$, at which a run flagged “converged” can still show $\sim 2\%$ spatial spread in the supersonic mass flux $\rho v r^2$.
- The single-stage vs. two-stage choice is now set by the `Reconstruction scheme: line`. `Reconstruction scheme: PLM+WENO3` selects a two-stage run: with the input line `du_th [PLM,WENO3]: <du_plm> <du_final>` supplying both thresholds (`du_th_plm` and `du_th`), `EXHALE_main.f90` starts in PLM and switches `rec_method` to WENO3 once $du < du_{\text{plm}}$ (or PLM stalls), then converges at $du < du_{\text{final}}$. This automates the manual `PLM → Load IC+WENO3` workflow recommended in the ATES README. `Reconstruction scheme: PLM` or `WENO3` is single-stage and uses only the first `du_th` value (the second is ignored).
- CFL is now runtime and settable via the input line `CFL: <value>`.

Other source-term changes (kept, with caveats). `energy_semi_implicit.f90` now damps the implicit energy update with $\partial F / \partial T = 1 + c |dC/dT|$ (was $\max(0, dC/dT)$), adding damping on the falling cooling branch; it removed one oscillation in testing but did *not* by itself fix the convergence behavior below. `Cool_coeff.f90` now interpolates the cooling tables with monotone PCHIP (C^1) rather than linear (C^0); this was implemented to test an interpolation-kink hypothesis (not supported, below) and is retained as a smoothness improvement.

Convergence — provisional observations (not conclusions). *The following is a research log, deliberately tentative; several intermediate “non-convergence” readings in earlier sessions were later traced to methodology, not physics.* Two artifacts are recorded so they are not repeated: an over-short step cap (weak-wind cases judged “floored” near 25k steps while du was still descending — HD209458b later reached $du < 2 \times 10^{-2}$ near 50k steps), and a run-script bug that killed the backgrounding subshell rather than the `EXHALE.x` child, so runs continued orphaned and the reported du

was a premature snapshot (use `kill -x EXHALE.x`). Because of these, the earlier framing that “metal line cooling destabilizes the wind” should be treated as *not established*. With the two-stage strict-du_th pipeline we currently see, across planets:

- **WASP-121b** (inflated, strongly irradiated): converges quickly and flat ($\sim 0.16\%$ supersonic mass-flux spread, vs $\sim 2\%$ for single-stage PLM). This case looks solid.
- **HD209458b**: reaches a small $du \sim 0.02\text{--}0.025$ that then oscillates rather than dropping to 10^{-3} . *Tentatively* usable as a quasi-steady state: a He $2^3\text{S}+7$ -metal snapshot from the two-stage pipeline gives a smooth transonic wind with $T_{\text{max}} \approx 6.8 \times 10^3 \text{ K}$ and $\log_{10} \dot{M} \approx 10.1$ (median over the supersonic region), consistent with the literature $\sim 10^{10} \text{ g s}^{-1}$; see `heitr_metals_dev/HD209458b_quasisteady.ipynb`.
- **HD189733b** (strongly bound, $\beta_0 \approx 192$): du oscillates around ~ 2 in a sustained limit cycle and does not reach the threshold — the problematic case.

A plausible (but unproven) reading is that the difficulty tracks how weakly driven / strongly bound the wind is, rather than the metal content per se.

Base-breathing hypothesis and the fixes that did *not* work. The HD189733b oscillation resembles the known base-breathing behavior, and the lower boundary (`Apply_BC.f90`, `BC_component_constrho`) hard-pins the ghost density and pressure to cold reservoir values with a one-way velocity valve $\max(v_1, 0)$ — a configuration that can reflect acoustic waves. Three variants were tried and reverted:

Test	Outcome
Lower CFL ($0.6 \rightarrow 0.2$)	du floor barely moved ($\sim 2.8 \rightarrow \sim 1.9$)
Zero-gradient base velocity (drop the valve)	du still ~ 2.3
Riemann-invariant base velocity $v = v_1 + \frac{2(c_b - c_1)}{\gamma - 1}$	did not help HD189733b <i>and</i> broke WASP-121b

So the base-breathing reading is *consistent with* the observations but is **not demonstrated**, and these simple BC variants are not solutions. Candidate routes if pursued (both substantial, uncertain payoff): a characteristic / non-reflecting (NSCBC-style) base BC; a steady-state Newton / BVP solver (which would not orbit a time-marching limit cycle if a steady solution exists); or accepting HD189733b as a marginal, time-averaged quasi-steady case. A separate, cheaper idea for *speed* (not the oscillation) is local cell-by-cell time-stepping, since the global dt is currently set by the smallest base cell (`eval_dt.f90`). Full working notes: `docs/heitr_metals_and_convergence_notes.md`; solver-option discussion: `docs/numerical_methods.md`.

Code-size note. The merge adds one Fortran module (`System_HeH_TR_metals.f90`, 84 SLOC), so the 2026-06-07 file tally of 57 becomes 58; the remaining changes are small edits to `charge_exchange.f90`, `input_read.f90`, `ionization_equilibrium.f90`, `parameters.f90`, `EXHALE_main.f90`, `energy_semi_implicit.f90` and `Cool_coeff.f90`.

17. Metals in the bulk-gas mass/charge/particle budget (eos_metals, 2026-06-13)

Motivation. Until this update the trace metals were strictly *passive* species: they contributed to heating/cooling and were solved in the coupled ionization equilibrium, but they were excluded from the mass density, the base density ρ_{bc} , and the electron/particle densities used everywhere *outside* the ionization solver (EOS temperature, opacities, cooling calls, energy update). The only self-consistent place was the charge balance *inside* the MINPACK residuals (System_HeH_metals.f90). At solar abundances the missing metal mass is $\sim 1.3\%$ of the H/He mass — small but systematic, and inconsistent with solving the metal ionization to high precision. Recorded in TO_BE_DONE.md on 2026-06-12; implemented 2026-06-13.

Runtime interface. A new key eos_metals 0|1 in metals.inp (documented in inputdata/metals.inp.example). 1 (the new default) includes the metals in the bulk-gas budget; 0 restores the legacy trace approximation exactly. Metals-off runs (no metals.inp) are identical either way.

What changed (factual).

- **Composition constants** (input_read.f90): the mass per hydrogen nucleus is now $\mu_H \equiv \text{mass_per_H} = 1 + 4Y + \sum_X a_X A_X$ (with $Y = \text{HeH}$ and a_X, A_X the metal abundance and atomic weight), and the base density becomes $\rho_{bc} = \mu_H / (1 + Y)$ in units of $m_H n_0$. Atomic weights live in species_table.f90:melem_A. The normalization n_0 keeps its H/He-nuclei meaning, so $n_H = n_0 / (1 + Y)$ is unchanged. (Correction to the original TO_BE_DONE entry: v_0, p_0, b_0 are m_H -based normalizations and never contained a mean molecular weight — they were never affected.)
- **Mass density:** calc_rho takes an optional nm argument and adds $\sum A_X n_{X,i}$ over all stages; passed from ionization_equilibrium.
- **Electron density:** calc_ne takes an optional nm and adds $\sum q_i n_{X,i}$; passed at every site where nm is available — composition (main-loop EOS), ionization_equilibrium (opacity/cooling), util_ion_eq (eval_cool + cooling dump), energy_semi_implicit, excited_hydrogen, post_process_adv. (The scalar T_equation already carried metal electrons via pp_nm_cell.)
- **Particle density:** calc_ntot likewise adds the metal nuclei, so the EOS $T = p / (n_{\text{tot}} + n_e)$ sees them.
- **Ghost pressure / dp_bc:** the lower-BC ghost pressure is now $n_{\text{tot_bc}} + \text{dp_bc}$ with $n_{\text{tot_bc}} = (1 + Y + \sum_X a_X) / (1 + Y)$ — so $T = T_0$ holds *exactly* at the base — and dp_bc includes the metal electrons $\sum q_i n_{X,i} / n_0$ at the first cell. set_IC/load_IC use mass_per_H/ntot_bc consistently. Full formulas: EXHALE_BC_and_IC.tex.

Validation (WASP-121b regression matrix).

- eos_metals 0 is **byte-identical** to the pre-change goldens for both regression cases (wasp_full 7296 steps, wasp_he23off 7300) — the legacy path is untouched.

- `eos_metals 1` (new default) converges cleanly (7145/7142 steps, $\sim 2\%$ fewer); profile shifts are at the expected $\sim 1\%$ mass scale: median $|\Delta n| \sim 2.4\%$, $|\Delta p| \sim 2.3\%$, $|\Delta T| \sim 0.9\%$ (max $|\Delta T| = 132$ K in the breathing-base region $r \sim 1.09$). The *number* flux nvr^2 drops 1.1% ($\log_{10} \dot{M}$: 13.31 $\rightarrow$ 13.30), which offsets the +1.2% mass per particle — the *mass*-loss rate is essentially unchanged.
- Goldens re-snapshotted under the new default; a repeat run reproduces them byte-identically (single-thread determinism intact).

Still open (small). The `use_2lev_cool` legacy cooling branch remains AIOLOS-hard-coded (default off), and the cell-by-cell scalar `T_equation` uses `pp_nm_cell` only in the post-process path (the time-marching cell-by-cell solve receives n_e via the now-metal-aware `energy_semi_implicit`).

18. Automatic initial-condition selection (IC mode: auto, 2026-06-13)

Motivation. The code now carries three IC families — cold hydrostatic (default), transonic isothermal wind (added for the Phase-4 deep-RLOF Case D, where the cold IC false-converges), and the hot-Parker warm seed (benchmarked: no speedup) — and the choice was manual. Kubyshkina+2018 build an IC for each planet automatically; the design study is `docs/auto_ic_design.md` and the literature comparison `docs/code_comparison.tex`.

The selector (one exact probe, no tunable threshold). `select_IC_auto` (`set_IC.f90`) evaluates the *cold* base sound speed in code units, $c_{\text{cold}}^2 = (n_{\text{tot, bc}} + \Delta p_{\text{bc}}) / \rho_{\text{bc}}$, and calls the existing `find_sonic` bisection for a critical point $\phi'(r_c) = 2c^2/r_c$ in the *actual* potential (Roche or spherical). If an interior sonic point exists, the cold isothermal atmosphere already wants to blow a transonic wind, so the transonic IC is selected; otherwise the cold hydrostatic IC is kept. One criterion catches both regimes that need the transonic start: deep RLOF (through the Roche/L1 topology) and low-gravity boil-off ($r_c \simeq b_0 / (2c_0^2)$ in a point-mass potential). The Jeans parameter b_0 (Kubyshkina's Λ) is logged as a diagnostic but is *not* a decision variable; the hot-Parker seed is never auto-selected.

Runtime interface. New input line `IC mode: <cold|transonic|hot_parker|auto>`. The named modes simply set the legacy flags; `auto` defers the decision to `set_IC`. The legacy keys (`Transonic IC:`, `Hot Parker IC:`) take precedence over `auto`, and the setup report (`EXHALE_setup.out`) records the family actually in effect, with a “(chosen automatically: IC mode = auto)” tag. Touched files: `parameters.f90` (`ic_mode`), `input_read.f90` (key parsing), `set_IC.f90` (`select_IC_auto`), `write_setup_report.f90`.

Validation gates (all run 2026-06-13).

Gate	Result
A-1 no IC mode key	byte-identical regression (defaults untouched)
A-2 IC mode: cold	byte-identical to A-1
A-3 spherical tutorial, auto	$b_0 = 83.2$, no interior $r_c \rightarrow$ cold; IC_dump byte-identical
B-1 Case D (deep RLOF), auto	$r_c = 1.297$ (just inside $L1 = 1.353$) $\rightarrow$ transonic, same as the manual key
B-2 boil-off planet ($0.0189M_J$, $0.446R_J$, 1100 K)	$b_0 = 8.5$, $r_c = 5.2$ (as predicted) $\rightarrow$ transonic; wind launches (initial residual 0.19 vs 14 cold)
B-3 wasp_full (Roche), auto	$\rightarrow$ transonic; du -stop in 5876 vs 7145 steps (−18%)

Honest speed verdict. The −18% step count is *Roche/deep-RLOF-specific* (the steady wind there is close to the cold isothermal transonic solution). For the mild spherical boil-off case B-2 the cold control *also* launched and converged first ($\log_{10} \dot{M} = 11.20$), while the auto run relaxed monotonically toward the same flux from $\sim 2\times$ above — the cold- c^2 Parker profile *overestimates* the flux of a weakly heated wind. So auto is not harmful (same attractor) but not faster there.

Key byproduct (a trap for all $\dot{M}$ work). While adjudicating gate B-3 it was found that the $du < 10^{-3}$ stopping criterion is **path-dependent at the $\sim 5\%$ level**: the cold-start and transonic-start wasp_full runs both “converge” by du but sit 4.8% apart in mass flux, each only 1–2% flux-flat. Newton-finishing *both* states (Load IC + Solver: Newton) collapses them onto the *same* fixed point (flux ratio 0.9996, flatness 0.005%). Practical rule, now standing: **for quantitative $\dot{M}$ always finish with Solver: Newton**; never compare bare du -stops at the percent level.

Deferred. A Phase-C retry ladder (escalate cold $\rightarrow$ warm seed on NaN/stall) is designed in auto_ic_design.md §6 but *not* implemented — the B-2 result suggests failure-triggered escalation is the right shape if a real case ever needs it. IC formulas and the selection logic are also documented in EXHALE_BC_and_IC.tex §5 and the user manual.

19. Base boundary conditions and the flux-based convergence decision (2026-06-15)

Context. A systematic study of how the lower boundary and the stabilizers interact with convergence on HD 209458b is collected in docs/EXHALE_BC_and_IC.tex (test matrix, two stabilizer tables, and a convergence-criterion history). This entry records the resulting default changes to the code; the companion document holds the full evidence.

Shapiro filter: off by default. The optional Shapiro (1970) spatial filter — ported from CETIMB (Koskinen et al. 2013a), whose use of it was found through Huang et al. (2023), to damp the same gravity-unbalanced base sound-wave instability — damps the transient “breathing” of the base cell, but on a clean cold-start it also drives the wind into a slow infall and prevents convergence. Isolated in a 2×2 (filter $\times$ base-valve) sweep, the filter — not the valve — was the cause. It is therefore *off* by default (shapiro_eps < 0) and opt-in only via Shapiro filter: <eps> [every].

Base boundary condition: density vs. pressure. The base can now be anchored either on density (the legacy fixed n_0) or on a fixed base pressure, selected by Base BC: density|pressure [p_ubar] (default density; pressure fixes p_{base} in μbar and derives n_0). In practice the converged wind is nearly insensitive to which anchor is used — the real lever on $\dot{M}$ is the base *density* itself, not the BC form.

The flux-based convergence decision (history). EXHALE measures two distinct quantities at the base: the flux metric du (the radial spread of $\rho v r^2$, which is exactly the constant-mass-flux criterion of ATES, $\Delta\dot{M}/\dot{M} < 10^{-3}$, and of CETIMB, $F_c = \rho v r^2$ altitude-flat), and the steady residual $\|R\| = \|\partial_t u\|$ over mass/momentum/energy, a *stricter* EXHALE-specific check. The convergence rule evolved: before the Newton finish it was purely flux-based; the Newton work then added the stronger $\|R\|$ criterion; but the reference codes converge on flux alone. The standing decision, adopted here, is therefore **flux-based** — $du < du_{\text{th}}$ is the acceptance test, and $\|R\|$ is computed and reported *for reference only* (it gates the stop only if Resid tol: is set).

Volume-weighted residual norm. When $\|R\|$ is reported, the default is now a cell-volume-weighted relative norm, $\sum_j |R_j| r_j^2 \Delta r_j / \sum_j |u_j| r_j^2 \Delta r_j$ (resid_vol, settable Resid norm: vol|Linf), rather than the bare L_∞ maximum. On the non-uniform radial grid the L_∞ max is dominated by the few tiny near-base cells and overstates the residual; the volume weighting gives a physically representative number. The infall diagnostic used in the test matrix was likewise switched to a *radius-weighted* negative-velocity fraction (weighted by Δr over the linear radius range), so that sub- 10 m s^{-1} near-base noise is not counted as bulk infall.

du-keyed Newton/JFNK hand-off. The production Newton finish (Solver: Newton) now hands off from the marching warm-up to the JFNK steady solve when the *flux* metric drops below newton_du_switch (default 10^{-2}), consistent with the flux-based decision and adjustable via an optional third token, Solver: Newton [du_switch]. This replaces the earlier residual-keyed hand-off ($\|R\| < 5 \times 10^{-2}$, variable newton_R_switch, now removed); the two are equivalent in practice, but keying on du avoids tying the hand-off to the stricter residual. The JFNK solve still *targets* $\|R\| < \text{resid_th}$ (10^{-3}); on failure it keeps its best iterate and falls back to marching with the du -based stops re-armed.

The full-physics residual floor (base $j = 1$). With He 2^3S + metals, the du -keyed hand-off plus the volume-weighted norm drove the JFNK residual from ~ 0.17 down to $\sim 2 \times 10^{-3}$ — about two orders of magnitude better — but it then *stalls* (info $\neq 0$) on a localized momentum imbalance at the base cell $j = 1$ (the “breathing base”), short of 10^{-3} . The wind is nonetheless flux-converged (e.g. HD 209458b full physics: $\log_{10} \dot{M}$ steady, $\rho v r^2$ spread $< 1\%$), which is a converged run by the reference standard. Closing the last factor of ~ 2 in $\|R\|$ requires physically damping the base momentum — the explicit-viscosity task at the top of TO_BE_DONE.md.

Touched files. parameters.f90 (shapiro_eps, base_bc_mode/base_p_ubar, base_v_massflux, resid_vol, newton_du_switch; newton_R_switch removed); input_read.f90 (Base BC:, Resid norm:, Solver: Newton [du_switch] parsing); steady_residual.f90 (residual_norms_vol); steady_newton.f90 (resid_relnorm vol/ L_∞ branch); EXHALE_main.f90 (du -keyed JFNK trigger, always-on flux/residual diagnostic). Full detail and the test matrix: docs/EXHALE_BC_and_IC.{tex,pdf}.

20. Tidal (Roche) potential with an extended outer boundary (2026-06-16)

Motivation. The two domain modes were coupled to two different gravitational potentials. Domain mode: Spherical extends the grid to a user Outer radius but *drops* the stellar-tidal and centrifugal terms (pure $-b_0/r$); the default Roche mode keeps the full tidal potential but truncates the domain at the inner Lagrange point L_1 . Neither matched the escape models of Yan et al. (2022) / Huang et al. (2023), which retain the tidal force *and* integrate out to $\sim 10R_p$ (well past L_1). For WASP-52b, for example, $L_1 \simeq 2.53R_p$, so Roche mode truncated the domain far short of the $10R_p$ used by Yan, cutting off the outer He 10830 / H α absorbing layers.

Change. The outer-boundary extent is now decoupled from the choice of potential. In Roche mode (no Domain mode line, so the full tidal potential of grav_field.f90 is retained), supplying an explicit Outer radius [R_p]: > 1 sets r_max to that value instead of the L_1 radius:

```
else ! Roche mode: full tidal potential retained
  r_max = (3.0*Mrapp)**(-1.0/3.0)*atilde ! L1 (default)
  if (r_out_user .gt. 1.0d0) r_max = r_out_user ! NEW: extend past L1
endif
```

This reproduces the Yan/Huang “tidal force + extended domain” setup with a two-line change and no new keyword (the existing Outer radius is simply honored in Roche mode too).

Caveats and usage. Beyond L_1 the Roche potential is past its maximum (net outward force), so the 1-D radial flow there is a spherical approximation to what is physically an L_1 funnel — the same approximation Yan/Huang make. The cold hydrostatic IC is invalid past L_1 , so pair the extended domain with IC mode: auto (validated: the auto selector finds an interior cold sonic point and seeds a transonic wind). The tidal singularity of the $-b_0 q/(\tilde{a}-r)$ term sits at the star ($r = \tilde{a} \gg 10$), so a $10R_p$ domain is numerically safe. Domain mode: Spherical (no tidal) remains available for Huang Case-A-like comparisons.

Touched files. input_read.f90 (honor r_out_user in the Roche branch of the r_max assignment). Documented in docs/EXHALE_user_manual.tex (§ potential and the input-options table).

21. Photoionization cross-section benchmarks and a TOPbase high-energy extension of He 2³S (2026-07-01)

Motivation. The He 2³S (metastable triplet) photoionization cross section sigma_HeI23S — the drain term that sets the He 10830 lower-level population — was a four-segment broken power-law fit to Norcross (1971), hard-cut to zero above 59.2 eV. Two questions: (i) are EXHALE’s H/He/He 2³S photoionization cross sections consistent with modern atomic data, and (ii) does the unphysical 59 eV cutoff matter? Both are settled in the new memo docs/photoion_cross_sections.{tex,pdf} (reproducible via docs/compare_photoion_cross_sections.py).

Benchmarks. H I and He II use the exact hydrogenic form; He I uses the two-term ATES fit, which agrees with the Verner+1996 (VFKY96) single-shell fit to within 14% over 24.6–500 eV. The He 2³S broken PL reproduces its parent Norcross/p-winds table to ~10% and — checked against the **TOPbase / Opacity Project** R-matrix cross section (extracted from the CDS server; raw 922-point table saved to `data/topbase_HeI_2_3S_photo.dat`) — agrees to ~8% in the smooth near-threshold region (5–20 eV). TOPbase additionally resolves a 40–55 eV autoionizing-resonance forest (peaks to 3223 Mb); their **net effect on the 2³S rate is only** ~1–2% once the few numerically under-resolved spikes are excluded (the raw table must *not* be fed into a broadband rate integral — a quadrature artifact). Extending the cross section above 59 eV raises the 2³S rate by only 0.1–1.2%.

Change. `sigma_HeI23S` is extended with a minimal, fully-connected edit (`src/modules/functions/cross_sec.f90`): the broken PL is kept unchanged up to the 45.6 eV bump; the **last power-law segment is re-aimed** so it runs continuously from the bump peak to a power-law fit of the TOPbase smooth background at the junction $E_J = 70$ eV,

$$\sigma_{\text{tail}}(E) = 10^{5.4623} E^{-2.9964} \text{ Mb} \quad (E > 70 \text{ eV}),$$

(the last-segment slope changes only $-3.039 \rightarrow -2.658$; the intercept is derived in-code for exact continuity). The old 59 eV cutoff is dropped, so 59–1240 eV now carries the TOPbase tail instead of zero. The result is C^0 -continuous at *both* joins (45.6 eV and 70 eV); the original pre-modification formula is kept in comments. A unified full-range PCHIP node table and a tail-only power law are documented as alternatives, but this is the recommended drop-in (smallest rate change, retains the validated threshold-to-bump shape exactly).

Validation. A controlled before/after on WASP-52b (`fxuv0p25_he98_L1`, Include He23S? True, ϵ Eri SED with X-rays), same IC, only `cross_sec.f90` differing: the bulk wind is unchanged ($\log \dot{M} = 11.49$ both), and the He 10830 line changes by $< 0.001\%$ in every metric (line-center depth $15.2675\% \rightarrow 15.2675\%$; EW $0.0877 \rightarrow 0.0877 \text{ \AA}$) — below the Newton convergence noise. The extension is more physically correct yet observationally harmless for He 10830, as predicted by the +0.65% rate estimate (even smaller here because the ϵ Eri SED is soft).

Recombination cross-check. A companion memo `docs/recombination_coefficients.{tex,pdf}` (`compare_recombination.p`) documents the H/He/He 2³S recombination: H II, He II, He III use the Hui & Gnedin (1997) case-B fits ($\approx$ Draine 2011 / Verner & Ferland to ~2%); He II→He I splits into Benjamin+1999 singlet/triplet (Oklopčić & Hirata 2018) when the 2³S network is on. The He 2³S triplet rate is identical to the latest escape models (Benjamin+1999). Newer nebular-community data (Porter+2012/2013, Del Zanna & Storey 2022) differ ~5–15% but are not yet adopted by escape codes; H/He case-B needs no update.

Touched files. `src/modules/functions/cross_sec.f90` (`sigma_HeI23S`: re-aimed last segment + TOPbase tail, 59 eV cutoff removed, original kept in comments). New docs: `photoion_cross_sections.{tex,pdf}`, `compare_photoion_cross_sections.py`, `data/topbase_HeI_2_3S_photo.dat`, `recombination_coefficients.{tex,pdf}`, `compare_recombination.py`, and figures under `docs/figures/`.

22. He I (1^1S) photoionization $\rightarrow$ Verner+1996 by default, with ATES-rate switches (2026-07-01)

Motivation. The He I *ground-state* (1^1S , singlet) photoionization used the legacy ATES two-term fit, which runs $\sim 12\text{--}14\%$ below the Verner+1996 (VFKY96) single-shell fit in the $50\text{--}100\text{eV}$ band (`photoion_cross_sections.tex`, §4). We make Verner the default, keep the old behavior behind a switch, and add a parallel switch for the recombination data.

Singlet/triplet separation. He photoionization in EXHALE is already split by spin: `sigma_HeI` is the 1^1S **ground (singlet)** cross section (updated here to Verner — Verner’s He I *is* the 1^1S ground state), while `sigma_HeI23S` is the 2^3S **metastable (triplet)** cross section (Norcross/TOPbase, §21). The 2^1S singlet metastable is not tracked (only 2^3S , the 10830 lower level), so the Verner update correctly touches only the singlet ground state.

Change. A single backward-compatible switch in `input.inp`: `ATES_photoionization_rate: True` reverts He I (1^1S) photoionization to the legacy two-term fit (default = Verner+1996, via `sigma_VFKY96` with He I parameters $[E_{\text{th}}, E_0, \sigma_0, y_a, P, y_w, y_0, y_1] = [24.59, 13.61, 949.2, 1.469, 3.188, 2.039, 0.4434, 2.136]$). No recombination switch is needed (see below).

Comparison. WASP-52b (`fxuv0p25_he98_L1`, He23S active, same converged IC). Switching He I photoionization legacy $\rightarrow$ Verner changes the He ionization balance and hence the metastable population (Table 1): unlike the $> 59\text{eV}$ tail (§21, $< 0.001\%$), the ground-state photoionization choice is a **real $\sim 3\%$ effect on He 10830** — it matters for the absolute depth and the inferred He abundance. `ATES_photoionization_rate: True` reproduces the old result exactly.

Table 1: He I photoionization legacy (ATES two-term) vs Verner+1996 (new default), WASP-52b He 10830.

metric	legacy (ATES)	Verner (new)	change
$\log \dot{M}$ [g/s]	11.49	11.48	-2%
He 10830 line-center depth	14.516 %	14.933 %	$+2.9\%$
He 10830 equivalent width	$0.0877 \text{ \AA}$	$0.0910 \text{ \AA}$	$+3.8\%$

Recombination — no update needed (EXHALE already current). The companion update (He recombination $\rightarrow$ Porter+2012/2013 / Del Zanna & Storey 2022) was investigated and found to be a *no-op*: those are *emissivity* papers — they update the He I line cascade / line ratios, *not* the total (effective) recombination to 2^3S , and provide no drop-in $\alpha(2^3\text{S})/\alpha(1^1\text{S})$. The escape-relevant quantity — total recombination into the triplet ladder, all funnelling to the metastable 2^3S — is set by (case-B total) $\times$ (triplet spin fraction $3/4$) and is robust across modern treatments: EXHALE (Benjamin+1999) has $\alpha_3 = 2.10 \times 10^{-13} T_4^{-0.778}$, $\alpha_1 = 1.54 \times 10^{-13} T_4^{-0.486}$, while the 2025 thermosphere paper (arXiv:2509.14499) uses $\alpha_3 = (3/4)(2.72 \times 10^{-13} T_4^{-0.789}) = 2.04 \times 10^{-13} T_4^{-0.789}$ and the *identical* singlet $\alpha_1 = 1.54 \times 10^{-13} T_4^{-0.486}$ — so α_3 agrees to $\sim 2\text{--}3\%$ and α_1 is identical. EXHALE’s He recombination already matches the latest practice; no change is warranted. (A trial direct sum of Cloudy’s `he_iso_recomb.dat` level-by-level radiative

recombination gave a total $\sim 2\times$ the case-A value owing to its unmapped 1642-level indexing, confirming the raw file is not a usable shortcut.) No recombination switch was added.

Touched files. `src/modules/init/parameters.f90` (ates_photoion_rate switch), `src/modules/files_IO/input_read.f90` (parse the optional key), `src/modules/functions/cross_sec.f90` (sigma_HeI branches Verner default / legacy two-term). Verified: Verner is the default (He I σ at 30/50/100 eV = 5.36/2.02/0.394 Mb); ATES_photoionization_rate: True recovers the legacy values (5.24/1.78/0.347 Mb).

23. He 2³S photoionization → two VFKY96 (Verner) forms + power-law bridge (2026-07-01)

Motivation. `sigma_HeI23S` was a bespoke five-segment broken power law (Norcross-1971 fit + TOPbase-tail extension, §21). Since EXHALE already uses the Verner+1996 (VFKY96) `sigma_VFKY96` routine for He I and all metals, the metastable triplet is now expressed in the same form, so every photoionization cross section shares one analytic kernel.

Change. (`src/modules/functions/cross_sec.f90`.) `sigma_HeI23S` is rebuilt from two VFKY96 single-shell forms joined by a log-linear (power-law) bridge, used piecewise, with $[E_{\text{th}}, E_0, \sigma_0, y_a, P, y_w, y_0, y_1]$:

- **wing A** ($E \leq x_3 \simeq 34.70$ eV, threshold → Cooper minimum): 4.78, 2.645, 20.8, 1.0×10^{12} , 3.42, 2.681, 1.956, 2.603;
- **bridge** ($34.70 < E < 45.59$ eV): straight log–log line joining wing A(x_3) to wing B(x_4) (the “power-law part in the middle”);
- **wing B** ($E \geq x_4 \simeq 45.59$ eV, bump → tail): 45.59, 49.68, 1052, 0.04393, 2.941, 1.717, 5.488×10^{-5} , 1.118.

Each wing is a $\sim 4\%$ fit to the resonance-averaged cross section; the construction is **continuous at both** x_3 **and** x_4 , and threshold ($E < 4.78$ eV) and the smooth high- E tail ($\sim E^{-7/2}$) are supplied by the wings (no hard cutoff). The former broken power law is retained, commented out, in the routine.

Verification. Rebuilt EXHALE.x; `sigma_HeI23S` reproduces the former broken power law to $\sim 4\%$ (12 eV: 1.466 vs 1.465; 30: 0.289; 45.6: 2.57 vs 2.68 — bump 4% low; 70: 0.855 vs 0.858; 100: 0.294; 300: 0.0110), continuous across the bridge (34.70 eV: 0.222 → 38 : 0.502 → 40 : 0.795 → 45.59 : 2.570 → 46 : 2.521 Mb).

Touched files. `src/modules/functions/cross_sec.f90` (`sigma_HeI23S`: two VFKY96 wings + power-law bridge; former broken PL kept commented out).

24. Temperature-dependent Penning ionization of He 2³S (2026-07-01)

Motivation. The metastable He(2³S) destruction by neutral hydrogen, He(2³S)+H, was carried as a *temperature-independent* constant $Q_{31} = 5 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ — the ATES/Oklopčić & Hirata (2018) value, itself the Roberge & Dalgarno

(1982) *sum* of Penning and associative ionization, as also used by Lampón et al. (2020). Taylor et al. (2025, ApJ 989:68) replaced this with a Maxwell–Boltzmann-averaged, **temperature-dependent** fit to the cross sections of Morgner & Niehaus (1979) and Cohen & Lane (1971). Because Penning ionization is often the *dominant* 2^3S loss near the base (where neutral H is abundant), this directly affects the He 10830 prediction EXHALE is built to compute.

New default rate (Taylor et al. 2025, Table 2). A two-branch power law ($\text{cm}^3 \text{s}^{-1}$, T in K):

$$Q_{31}(T) = \begin{cases} 1.9 \times 10^{-9} (300/T)^{0.07}, & T \leq 4000 \text{ K}, \\ 9.1 \times 10^{-9} (300/T)^{0.50}, & T > 4000 \text{ K}. \end{cases} \quad (4)$$

This is $\sim 3\times$ the legacy constant across the relevant 2000–9000 K range. Note the new rate is the Penning channel only ($\text{He}(2^3\text{S}) + \text{H} \rightarrow \text{He}(1^1\text{S}) + \text{H}^+ + \text{e}^-$); the associative-ionization channel ($\text{He}(2^3\text{S}) + \text{H} \rightarrow \text{HeH}^+ + \text{e}^-$) is not represented separately in EXHALE (no HeH^+), so the single Q_{31} continues to carry the dominant $2^3\text{S} + \text{H}$ destruction term $-n_{\text{HeITR}} n_{\text{HI}} Q_{31}$.

Unconditional default. The temperature-dependent rate is used *unconditionally*: the legacy constant 5×10^{-10} is no longer selectable. (The `ATES_photoionization_rate` flag continues to revert only the He I (1^1S) photoionization cross section to the legacy ATES 2-term fit; it does *not* affect the Penning rate.)

Implementation. Q_{31} became an array over the cells (it was a scalar). A new `penning_HeI_23S(T,coeff)` in `Cool_coeff.f90` evaluates the piecewise fit over the whole grid; `HeITR_coeffs` calls it unconditionally. The two callers (`ionization_equilibrium.f90`, `post_process_adv.f90`) pass $Q_{31}(j)$ per cell into `params`; the `System_HeH_TR*` residuals are unchanged (they already read Q_{31} from `params`).

Verification. Rebuilt EXHALE.x; ran HD 189733b (metals + He23S, step-capped) with the new rate and, for comparison, with the former 5×10^{-10} constant. The temperature dependence is active and physically sensible: relative to the old constant, $\text{He}(2^3\text{S})$ is reduced near the base (ratio ≈ 0.28 at $1.0R_p$, 0.92 at $1.1R_p$; column-integrated over $1\text{--}2R_p$ drops to $0.59\times$), and the profiles converge to unity above $\sim 1.3R_p$ where H is ionized and Penning switches off — the expected neutral-H-gated signature.

Touched files. `src/modules/radiation/Cool_coeff.f90` (new `penning_HeI_23S`); `src/modules/radiation/util_ion_eq.f90` (`HeITR_coeffs`: Q_{31} array + default/legacy branch); `src/modules/radiation/ionization_equilibrium.f90` and `src/modules/post_process/post_process_adv.f90` (Q_{31} array, `params` assignment in each cell).

25. He/H diffusive separation — Phase 1 (2026-07-01, experimental, default OFF)

Motivation. EXHALE is single-fluid and re-solves composition by *local* ionization equilibrium (conserving the element ratio), so He/H was frozen at the input HeH at all radii — it could not represent the diffusive He/H separation of Taylor et al. (2025) (8%→2.5%) and Xing et al. (2023). Top recommendation of `docs/methodology_aiolos_taylor_xing`.

Method. New module `species_diffusion.f90` (deleted 2026-08-25, section 69; replaced by `src/modules/functions/binary_el`) transports the He element ratio $f = n_{\text{He}}/n_{\text{H}}$ with bulk advection *and* a molecular-diffusion drift of He relative to H, $\Phi = -D n_{\text{H}}(\partial f / \partial r + f G)$, $G = (m_{\text{He}} - m_{\text{H}})g/(kT)$, $D = 1.52 \times 10^{18} (1/m_{\text{H}} + 1/m_{\text{He}})^{1/2} T^{1/2} / n$ (Banks & Kockarts 1973), plus eddy diffusion K_{zz} on the gradient term. Solved as a **fully conservative** implicit (backward-Euler tridiagonal) update of n_{He} ; the new split is written back into `f_sp` conserving mass ($\sum_s m_s f_{\text{sp},s} = 1$, metals frozen to H). Gated on `He_diffusion` (default off; `He_Kzz` sets K_{zz}). Derivation: `docs/design_hehe_diffusion.md`.

Root-cause bug (design doc §7). A local time scale named `t0` ($= R_0/v_0$) shadowed the *global temperature* `T0` (Fortran is case-insensitive) in `TK = Tcode*T0`, zeroing `TK` (floored to 1 K) and inflating the settling coefficient $G \propto 1/TK$ by $\sim 5800\times$, causing total over-settling. The two constants are numerically near-equal ($R_0/v_0 \approx 2.83 \times 10^4$ s, $T_0 \approx 2.83 \times 10^4$ K), which hid it. Fix: rename the local to `tscale`.

Result (HD 209458b, He23S+metals). Physical mild separation — $(\text{He}/\text{H})/\text{HeH} \approx 0.6\text{--}0.9$ in the inner thermosphere, falling to $\sim 0.17\text{--}0.24$ aloft — consistent in magnitude with Taylor ($\rightarrow 0.3\times$) and Xing. Stable (no NaN); `He_diffusion OFF` is byte-identical.

Phase-1 limitations. A near-base pile-up from the lagged- n_{H} nonlinearity is held by a physical limiter $f \leq \text{HeH}$ (aloft observable unaffected); proper fix is a self-consistent n_{H} inner iteration. Ambipolar field, thermal diffusion, metal diffusion, and a Newton-finished quantitative $\dot{M}$ are Phase-2. Flag stays default OFF.

Touched files. new `src/modules/functions/species_diffusion.f90`; `parameters.f90` (`he_diffusion`, `he_kzz`); `input_read.f90` (`He_diffusion`, `He_Kzz`); `EXHALE_main.f90`; `Makefile`.

26. He/H diffusion — Phase 2 (2026-07-02): ambipolar, thermal diffusion, metals

Extends §25 (design doc §7e). The Phase-1 kernel was factored into a shared subroutine `solve_1elem` (one conservative implicit advection–diffusion–settling solve for any element vs the H background), reused by He and each metal.

- **P2b — ambipolar-corrected settling** (default ON, `He_ambipolar`). The polarization field lifts ions, so the He-vs-H effective settling mass is $\Delta m_{\text{eff}} = 3 - 0.5(\bar{Z}_{\text{He}} - \bar{Z}_{\text{H}})$ (mean charges from the local ionization state): 3 at the neutral base, 2.5 in the fully ionized wind. Modest effect ($\sim 2\%$ less depletion aloft), correct direction.
- **P2c — thermal diffusion** (default `He_alphaT=0`). Adds $\alpha_T \partial \ln T / \partial r$ to the settling coefficient; off by default.
- **P2d — element-by-element metal diffusion** (default OFF, `He_metal_diffusion`). Each trace metal element diffuses independently vs n_{H} with its own mass (`melem_A`), binary D , and ambipolar correction; ion stages rescaled to the diffused total. Metals trace $\Rightarrow$ no n_{H} feedback. **Validated (HD 209458b C/N/O):** heavier elements deplete more — at $3 R_{\text{p}}$, element/base $\approx$ He 0.171, C 0.158, N 0.136, O 0.105 (monotonic in mass; matches Xing et al. 2023). Regression: OFF $\Rightarrow$ He/H unchanged, C/H frozen at 1.0.

- **P2a — self-consistent n_{H} : attempted, NOT adopted.** A Picard $n_{\text{He}} \leftrightarrow n_{\text{H}}$ iteration diverges (near-base settling genuinely concentrates He, driving $n_{\text{H}} \rightarrow 0$); the physical cap $f \leq \text{HeH}$ remains the correct limiter.

Touched files. `species_diffusion.f90` (solve_1elem kernel; ambipolar dmeff; thermal term; metal-diffusion loop); `parameters.f90` (he_ambipolar, he_alphaT, he_metal_diffusion); `input_read.f90` (He_ambipolar, He_alphaT, He_metal_diffusion).

27. Diffusive separation: detailed physics and numerics

This section consolidates §25–§26 into a single, more complete treatment of the He/H (and metal) diffusion, with emphasis on the two design decisions that are easy to get wrong: the $n_{\text{He}}-n_{\text{H}}$ mass coupling (§27.3) and the settling discretization (§27.5).

27.1 Governing equation

By default EXHALE is single-fluid: the hydro evolves $(\rho, \rho v, E)$ with one bulk velocity, and the composition is re-solved every step by *local* ionization equilibrium, which conserves the element ratios. Consequently the helium-to-hydrogen ratio $f \equiv n_{\text{He}}/n_{\text{H}}$ is frozen at the input value at all radii — EXHALE cannot represent diffusive separation without an added transport equation.

We transport the He element ratio f with advection at the bulk velocity plus a molecular-diffusion *settling* drift of He relative to H,

$$\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial r} = \frac{1}{n_{\text{H}}} \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 n_{\text{H}} (D + K_{\text{zz}}) \frac{\partial f}{\partial r} + r^2 n_{\text{H}} D f G \right], \quad (5)$$

with the diffusive number flux (relative to the bulk)

$$\Phi_{\text{He}} = -n_{\text{H}} \left[(D + K_{\text{zz}}) \frac{\partial f}{\partial r} + D f G \right], \quad G = \frac{\Delta m g}{kT} + \alpha_T \frac{\partial \ln T}{\partial r}. \quad (6)$$

Here $D = 1.52 \times 10^{18} (1/m_{\text{H}} + 1/m_{\text{He}})^{1/2} T^{1/2} / n$ is the Banks & Kockarts (1973) binary He-in-H diffusion coefficient ($\text{cm}^2 \text{s}^{-1}$), $g = GM_{\text{p}}/r^2$, and $\Delta m = m_{\text{He}} - m_{\text{H}}$ ($= 3$ amu in the neutral limit; §27.6). K_{zz} is the eddy diffusion coefficient (He_Kzz, default $10^9 \text{ cm}^2 \text{s}^{-1}$): it enters only the *gradient* term and drives f toward uniform (mixing), whereas the *settling* term $D f G$ is molecular only. Their competition sets the homopause: below it ($K_{\text{zz}} \gg D$) the atmosphere is mixed, above it ($D \gg K_{\text{zz}}$) helium settles. In diffusive equilibrium ($\Phi_{\text{He}} = 0$, no eddy) $\partial \ln f / \partial r = -G$, i.e. f decreases outward with the (small) He scale height $1/G$.

27.2 Conservative implicit discretization

An early *non-conservative* form (material-derivative advection $-v \partial f / \partial r$ combined with a conservative diffusive flux) proved flux-inconsistent and *drained* helium spuriously. The production scheme is instead **fully conservative**: it evolves the He number density n_{He} with a single divergence of the total face flux

$$J_{j+1/2} = \underbrace{n_{\text{He}} v}_{\text{advection}} + \Phi_{\text{He}}, \quad \frac{n_{\text{He}}^{n+1} - n_{\text{He}}^n}{\Delta t} = -\frac{1}{r^2 \Delta r} \left[r_{j+1/2}^2 J_{j+1/2} - r_{j-1/2}^2 J_{j-1/2} \right]. \quad (7)$$

Because the cold, dense base has a tiny He scale height, the diffusion is stiff and an explicit scheme is unstable; the update is therefore **implicit** (backward Euler), assembled as a tridiagonal system in n_{He}^{n+1} and solved by the Thomas algorithm. Implicit backward Euler is unconditionally stable, and its fixed point ($n_{\text{He}}^{n+1} = n_{\text{He}}^n$) is independent of Δt , so local cell-by-cell time steps are used for steady-state acceleration. The new element split is written back into `f_sp` conserving mass ($\sum_s m_s f_{\text{sp},s} = 1$), and the same kernel (`solve_1elem`) is reused for each trace metal (§27.6).

27.3 The $n_{\text{He}}-n_{\text{H}}$ mass coupling: why a self-consistent n_{H} is *not* needed

Diffusion redistributes species at *fixed* total mass density ρ (owned by the hydro), so in every cell

$$\rho = m_{\text{H}}n_{\text{H}} + m_{\text{He}}n_{\text{He}} + (\text{metals}) \implies n_{\text{H}} = \frac{\rho(1 - c_{\text{met}}) - m_{\text{He}}n_{\text{He}}}{m_{\text{H}}}. \quad (8)$$

Thus n_{H} is itself a function of n_{He} : where helium accumulates, hydrogen must be displaced. But the flux (6) *uses* n_{H} (in the coefficient and in $f = n_{\text{He}}/n_{\text{H}}$), so the solve is genuinely nonlinear in n_{He} . Two options:

- **Lagged n_{H} (adopted).** Hold n_{H} at its start-of-step value during the solve; update it afterward, in the mass-conserving rescale, for the next step.
- **Self-consistent n_{H} (“P2a”, tried, rejected).** Iterate within the step (Picard): solve n_{He} , reset n_{H} from (8), re-solve, until n_{He} and n_{H} are mutually consistent.

Why the Picard iteration diverges. Write the He mass fraction $Y = m_{\text{He}}n_{\text{He}}/\rho \in [0, 1]$. Near the base, settling genuinely concentrates helium ($n_{\text{He}} \uparrow$); enforcing (8) exactly then drives

$$n_{\text{H}} = \frac{\rho(1 - c_{\text{met}}) - m_{\text{He}}n_{\text{He}}}{m_{\text{H}}} \xrightarrow{Y \rightarrow 1} 0,$$

so $f = n_{\text{He}}/n_{\text{H}}$ and the $1/n_{\text{H}}$ flux coefficients diverge — a genuine singularity as helium approaches a pure-He layer. Because the concentration is *real* (settling against a fixed reservoir base), the iteration has a positive feedback and blows up (NaN). The lagged scheme avoids this: with n_{H} held at a finite value, $1/n_{\text{H}}$ stays bounded, and the physical cap $f \leq \text{HeH}$ (§27.4) prevents $Y \rightarrow 1$.

Why lagging is nonetheless harmless. Four independent arguments:

1. *The lag vanishes at the fixed point.* The diffusion runs inside a relaxation loop. At steady state $n_{\text{He}}^{n+1} = n_{\text{He}}^n$ and $n_{\text{H}}^{n+1} = n_{\text{H}}^n$, so the lagged $n_{\text{H}} = n_{\text{H}}^n$ used in the solve *equals* the final consistent n_{H}^{n+1} . The lag affects only the transient path, not the converged state: where the scheme is stable, lagged and self-consistent reach the *same* steady solution, and mass consistency is achieved automatically.
2. *The region where it matters is the (unphysical) pile-up.* Lagged and self-consistent differ appreciably only where $Y \rightarrow 1$, a thin layer just above the fixed reservoir base — below the He 10830 line-forming region, hence irrelevant to the observable, and in any case capped (§27.4).
3. *Helium is a minor mass fraction where the observable lives.* In the upper thermosphere, where He/H falls to $\sim 0.2-0.5 \times$ (the part that weakens 10830), $m_{\text{He}}n_{\text{He}} \ll m_{\text{H}}n_{\text{H}}$, so $n_{\text{H}} \approx \rho/m_{\text{H}}$ is nearly insensitive to n_{He} and the lag is negligible.

4. *Trace metals have no back-reaction.* Metals are $\sim 10^{-3}$ of the mass, so their diffusion does not change n_{H} measurably; for them the lagged n_{H} is effectively exact.

Hence the divergent Picard iteration is not merely inconvenient but *unnecessary*: the lagged- n_{H} solve plus the physical cap is stable and converges to the correct steady state.

27.4 The near-base pile-up and the physical cap

Settling transports helium down toward the reservoir base. Against a fixed-composition Dirichlet base it tends to pile up in the first cells above the base (numerically unbounded with a lagged n_{H}). In a settling-plus-escape column, however, He/H *cannot* exceed the reservoir value: there is no mechanism to enrich helium above the base abundance. The code therefore applies the physical limiter

$$f \leq \text{HeH} \quad (9)$$

(and, for each metal, $f_X \leq f_{X,\text{base}}$). This encodes the expectation that the reservoir *absorbs* settled helium rather than letting it build a pure-He layer — so the cap is a physical boundary condition, not merely a numerical band-aid. The aloft separation (the observable) sits well below HeH and is unaffected by the cap.

27.5 Settling discretization: a Péclet hybrid

The settling term DfG is an advective drift (velocity $w_s = DG$, directed inward). Central differencing of an advective term is stable and accurate only at low Péclet number, $|DG| \Delta r < 2(D + K_{zz})$. For light helium this always holds (small G), and central differencing is used (least numerical diffusion). For *heavy metals* (e.g. Fe, mass 56) G is $\sim 18\times$ larger; central differencing then loses diagonal dominance and the tridiagonal solve NaNs. The kernel therefore uses a **Péclet-based central/upwind hybrid**:

$$\text{settling drift: } \begin{cases} \text{central} & |DG| \Delta r < 2(D + K_{zz}) \quad (\text{light species}), \\ \text{first-order upwind} & \text{otherwise} \quad (\text{stiff heavy-metal settling}). \end{cases} \quad (10)$$

Upwind keeps the matrix diagonally dominant for any settling strength; the metal-stage rescale additionally skips cells where the element is negligible and clamps the scale factor to avoid $0 \times \infty$. These changes were required to run WASP-121b (Fe) stably; on HD 209458b (light C/N/O), the hybrid selects central and reproduces the He-only result.

27.6 Ambipolar field, thermal diffusion, and metals

In the ionized wind the polarization field $eE = -(1/n_e)dp_e/dr$ (Koskinen et al. 2013) lifts ions; for an H^+ background it supports $\sim$ half a proton weight per unit charge, so a species of charge Z has effective mass $m - Zm_{\text{H}}/2$. The He-vs-H relative settling mass becomes

$$\Delta m_{\text{eff}} = 3 - \frac{1}{2}(\bar{Z}_{\text{He}} - \bar{Z}_{\text{H}}), \quad (11)$$

with $\bar{Z}$ the mean charges from the local ionization state — $\Delta m_{\text{eff}} = 3$ at the neutral base, 2.5 in the fully ionized $\text{He}^{++}/\text{H}^+$ wind (He_ambipolar, default on). Thermal diffusion adds $\alpha_T \partial \ln T / \partial r$ to G (He_alphaT, default 0). Each trace metal (He_metal_diffusion, default off) is diffused independently against the post-rescale n_{H} with its own mass

(`melem_A`), binary D_X , and ambipolar-corrected Δm_{eff} ; being heavier than helium, metals deplete faster with altitude where diffusion competes with the wind.

27.7 The metal-rescale ratchet bug (found in review, fixed 2026-07-02)

The first version of the metal-stage rescale contained two bugs that together made metal depletion *irreversible*: (i) an $r_X \leq 1$ clamp — redundant for the metal/H $\leq$ reservoir cap (already enforced by $\min(n_X, f_{X,\text{base}} n_{\text{H}})$) but forbidding any *replenishment* of a previously depleted cell (a one-way ratchet); and (ii) a skip of cells where the element was negligible, which made exhausted cells permanent holes. During early relaxation (wind undeveloped) settling transiently depletes metals; the ratchet locked that in. **Fix:** r_X is no longer clamped above (the cap alone bounds it), and an exhausted cell that the solve replenishes is re-seeded through the neutral stage (the ionization equilibrium re-partitions the stages next step). **Retraction:** the previously reported WASP-121b “metal homopause” ($\text{Fe} \rightarrow 0$ by $\sim 1.25 R_p$, with a ~ 2500 K hotter thermosphere) was an artifact of this ratchet — with the fix, WASP-121b metals track He (even Fe is advection-dominated there, $w_s/v \sim 10^{-3}$) and the v1/v2 temperature structures agree. The HD 209458b mass ordering ($\text{He} > \text{C} > \text{N} > \text{O}$ aloft) is unaffected.

27.8 Boundary conditions and validation

The lower boundary is Dirichlet at the reservoir composition ($f = \text{HeH}$); the outer boundary carries a zero diffusive flux plus advective outflow (helium escapes with the wind). The two-planet before/after comparison (`docs/version_compare.{md,tex,pdf}`) shows the expected regimes: on HD 209458b helium itself separates ($\text{He } 10830 \text{ } 70.5\% \rightarrow 27.1\%$, $-2.6\times$) and heavier C/N/O deplete more than He aloft, whereas on WASP-121b ($\log \dot{M} \approx 13.3$) advection dominates for every species — helium *and* metals are dragged out essentially unfractiated (10830 unchanged; metal/H tracks He/H). Both are stable (no NaN) and leave $\dot{M}$ essentially unchanged (metals are trace). All diffusion flags are **default off**, so standard runs are byte-identical.

28. Lower-atmosphere connection: Tier 1–3 (2026-07-02)

Survey + proposal + implementation: `docs/lower_atmosphere_coupling.{md,tex,pdf}`. New folder `src/modules/lower_atmosphere` everything **opt-in / default off** (regression: the standard HD 209458 b 3000-step run reproduces v1.0’s $\log \dot{M} = 8.97$ exactly).

- **Tier 1** `lower_column.f90` + key `Lower column::` the Koskinen+2022 analytic isothermal- T_{eq} hypsometric column with Visscher chemical-equilibrium $\text{H}_2/\text{H}/\text{He}$; reports the derived 1- μbar base radius (equilibrium + fully-atomic bracket), base $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}$ and μ next to the input “Planet radius”. Gate passed: Model A $r_0/R_{1\text{bar}} = 1.343$ (paper 1.34), $q_{\text{H}_2} = 0.838$ (0.84), $q_{\text{H}} = 0.027$ (0.026).
- **Tier 2 foundation** `h3p_cooling.f90` (Miller+2013 Table-5 LTE emission fits + Table-6 non-LTE factors $s(T, n_{\text{H}_2})$; anchors reproduced to $<0.5\%$) and `mol_rates.f90` (Koskinen+2022 Table-1 rates R1–R23, verified against the PDF). Remaining: the coupled `System_HeH_mol` solver + EOS.

- **Tier 2a** key Molecular base: True: EOS-only base correction removing the H₂-bound particles from `ntot_bc` via the equilibrium fit (chemistry stays atomic; crude, documented).
- **Tier 3** optional `base.inp` (`T_base/r_base/HeH_base/Kzz_base`; echoed, no-op when absent) consumed by `input_read`, + the driver `src/utls/run_lower.py` (isothermal or Guillot 2010 semi-grey $T(p)$); VULCAN photochemistry is the documented upgrade path.

Physics note from the Tier-1 gate: the chemical-equilibrium base stays strongly molecular up to $T \sim 2000$ K, so the atomic base of $T_{\text{eq}} \sim 1000\text{--}2000$ K hot Jupiters rests on *photochemical* dissociation (Moses 2011; Koskinen 2013a) — quantified motivation for the VULCAN tier.

29. Tier-2 molecular chemistry core: H₂/H₂⁺/H₃⁺/HeH⁺ (2026-07-03)

Opt-in key Molecular chemistry: True (requires He; v1 excludes trace metals). New coupled equilibrium system `System_HeH_mol.f90` (7–8 unknowns; atomic rows = EXHALE’s own rates, so the molecule-free limit reproduces the atomic systems; molecular channels = Koskinen+2022 Table-1 via `mol_rates`). σ_{H_2} (Yan+1998 Eqs. 17–19; verified against their Table 7) added to `cross_sec.f90` and wired into the opacity/photoionization/heating integrals; H₃⁺ IR cooling (Miller+2013 + non-LTE factor) enters `cool`; the EOS is molecule-aware (particle count, electrons, masses). `f_sp` grows 33→37 (output schema + Load-IC extended; columns zero when off). **Gates:** (0) molecules-off regression $\log \dot{M}=8.97$ unchanged; (1) HD 209458 b molecules-on: fully molecular base, sharp H₂ → H front at $r=1.019 R_p$, wind above it essentially atomic ($\log \dot{M}$ 8.91 vs 8.97, He 2³S peak +1.4%); (2) hot-Uranus-like: front at $1.148 R_p$, H₃⁺ active below. Caveats: local equilibrium only (no molecular advection), no Lyman–Werner photodissociation, P4/P5 and the 4.48 eV sink omitted, $\gamma=5/3$ retained. Details: `docs/lower_atmosphere_coupling.pdf` §4.

30. Newton–diffusion co-convergence, solver hygiene, and the Ly α lower-BC audit (2026-07-03)

Stall-based Newton hand-off. `He_diffusion` runs plateau the flux metric just above the hand-off threshold (observed: du frozen at 1.084×10^{-2} vs the 10^{-2} switch for 10^6 steps). The warm-up now tracks the du plateau and hands over to JFNK once stalled within $5 \times$ the switch. *Validated:* the hand-off engaged at $du = 2.2 \times 10^{-2}$; the JFNK then hit the known base-momentum holdout (worst cell $j=1$, $\|R\| \sim 7.7 \times 10^{-3}$ while the volume-weighted reference residual was 2.2×10^{-4}) and fell back gracefully — the base-cell viscosity work remains the enabler. **Newton + He_diffusion co-convergence.** Solver: Newton previously froze the diffused He/H field at the hand-off state (the JFNK residual carries no operator-split diffusion). An outer iteration (the excited-H pattern) now alternates the JFNK solve with 500 diffusion-relaxation steps at the converged wind until the He/H field drift $< 10^{-3}$ (max 5 passes, drift reported). **Riemann floor:** `p_min` floored at $10^{-30} p_{\text{max}}$ in `speed_estimate_HLLC/ROE` (no-op for healthy states). **PH_heat_HHe OMP CRITICAL removed** (distinct- j writes are race-free; values unchanged). **regression/ run_fccheck.sh:** periodic -fccheck regression per the 2026-07-02 review. **lya_rt audit** (vs. Huang et al. 2017): the escape-probability closure $\bar{J} = S(1 - \beta)$ is local and lacks Huang’s *absorbing bottom* (H₂ accidental-resonance sink at $N_{\text{H}_2} \sim 10^{14} \text{ cm}^{-2}$),

so $\bar{J}$ — and the $H(n=2)$ population — is likely overestimated in the bottom few scale heights; the $H\alpha$ -forming region lies above, so the TPM impact is expected limited. Documented; a bottom-loss β channel is the follow-up.

31. Tier-3 first light: VULCAN $\rightarrow$ base.inp (2026-07-03)

The public VULCAN photochemical kinetics code was cloned (`./VULCAN`; FastChem compiled) and run to steady state for HD 189733 b (SNCHO-2025 network, `use_photo=True`; 2356 steps, ~ 6 h wall). The new converter `src/utils/vulcan_to_base.py` turns the `.vul` output into an EXHALE `base.inp`: photochemical $q_{H_2}/q_H/q_{He}$ at $1\ \mu\text{bar}$, a hypsometric base radius using VULCAN's own $\mu(p)$ and $T(p)$, and the molecular mixing ratios as comments (*no* metal release — metal/alkali chemistry is outside VULCAN's scope and remains Lavvas-only). **Result** (HD 189733 b, $1\ \mu\text{bar}$): $q_{H_2} = 0.63$, $q_H = 0.23$ — photochemistry dissociates $\sim 11\times$ more H than the chemical-equilibrium column ($q_H = 0.020$), directly quantifying the Tier-1 caveat (Moses 2011; Koskinen 2013a); $r_{\text{base}} = 1.168$ vs the analytic $1.174 R_J$; $T_{\text{base}} = 863\ \text{K}$ (the Moses11 $T(p)$, cooler than the isothermal $T_{\text{eq}} = 1183\ \text{K}$). End-to-end chain demonstrated: VULCAN $\rightarrow$ converter $\rightarrow$ base.inp $\rightarrow$ EXHALE startup override (echoed).

32. VULCAN lower-atmosphere pre-step: obtained, not included in the tree (2026-07-06)

VULCAN (with FastChem inside it, as shipped) is used as a lower-atmosphere pre-step invoked by EXHALE itself. VULCAN is a third-party code and is *not* committed to the EXHALE repository (VULCAN/ is git-ignored, avoiding the embedded- repository problem); it is fetched into EXHALE/VULCAN/ by the new `src/utils/setup_vulcan.sh` (clone + two patches + FastChem build), which the driver also runs automatically on first use if the tree is missing. The key

```
Lower atmosphere: vulcan <R_1bar>
```

in `input.inp` runs the new driver `src/utils/vulcan_driver.py` at startup whenever no `base.inp` exists: it copies the VULCAN tree into `<run_dir>/vulcan_work/`, builds the planet's Guillot $T(p)/K_{zz}$ atmosphere and picks a stellar UV spectrum by host T_{eff} , compiles FastChem once, runs VULCAN to steady state (hours on the first run, cached `.vul` afterwards; `--force` to redo), converts the result to `base.inp`, and then proceeds with the wind solve — i.e. VULCAN acts as a subroutine of EXHALE. `Lower atmosphere: analytic <R_1bar>` invokes the fast chemical-equilibrium column (`run_lower.py`) instead; omitting the key disables the pre-step (fully optional). EXHALE_ROOT overrides the code-root path for relocated installs, and an existing `base.inp` always wins. Implementation: `parameters.f90` (`lower_atm_mode/lower_atm_r1bar`), `input_read.f90` (`run_lower_atm_prestep` via `execute_command_line` before `read_base_inp`), and the added `vulcan_driver.py`. Tested: analytic branch end-to-end, VULCAN branch with a cached `.vul` (HD 189733 b photochemical base $T = 863\ \text{K}$ applied), and the opt-out (no key) path. The download URLs, citation requirements (VULCAN: Tsai et al. 2017, 2021; FastChem: Stock et al. 2018, 2022) and the exact list of modifications `setup_vulcan.sh` applies are documented in the README “Obtaining VULCAN and FastChem” section and the user manual (§“Obtaining VULCAN and FastChem”).

33. TPM.py renamed to EXHALE_transit.py (2026-07-06)

The transmission-spectrum post-processor TPM.py (the acronym clashed with “Trusted Platform Module” and did not describe the tool) is renamed EXHALE_transit.py, matching the EXHALE_* convention. Run-time overrides are now EXHALE_TRANSIT_<NAME> (PATH, SAVE_PREFIX, HE_LMIN/LMAX/N, RSTAR_RSUN, ROTP, TSTAR); the old TPM_<NAME> names remain as backward-compatible fallbacks (a `_tenv()` helper checks the new name first, then the old). Output filenames are unchanged (<prefix>tpm_He10830.txt etc.) so downstream notebooks and figure scripts keep working. All in-repo references were updated; VULCAN/ untouched.

34. Sub-level-resolved Balmer opacity and LaRT 2s inclusion (2026-07-06)

The $H\alpha/H\beta$ transmission was computed as $f_{23}n_2$ with the multiplet oscillator strength ($f_{23} = 0.6407$ for $H\alpha$), which implicitly assumes the $2s:2p$ populations sit in the 1:3 statistical ratio. They do not—the metastable $2s$ ($A_{2s1s} = 8.26\text{ s}^{-1}$) and the $\text{Ly}\alpha$ -pumped $2p$ depart strongly from it—and the two sub-levels have *different* Balmer cross sections (only $2s \rightarrow 3p$ is allowed from $2s$; $2p \rightarrow 3s, 3d$ from $2p$). Fixed to sum per sub-level, $\tau \sim f_{2s}n_{2s} + f_{2p}n_{2p}$, with $f_{2s} = 0.4349$, $f_{2p} = 0.70941$ ($H\alpha$) and 0.1028, 0.125886 ($H\beta$); the components share the Doppler-dominated profile.

EXHALE_transit.py (escape-prob path) now carries n_{2s}/n_{2p} separately; the effect is small ($\lesssim$ few %) since the in-code $J_{\text{Ly}\alpha}$ keeps $2s:2p$ near statistical in the forming region. examples/tpm_halpha_lart2d.py (LaRT path) previously used the $2p$ -only population, *discarding the 2s absorption entirely*; it now returns and sums both (the LaRT scattering rate pumps only $1s \rightarrow 2p$, entering the S_{2p} source term, while $2s$ is recombination/collisionally fed throughout the atmosphere). Effect: +21% (HD 189733 b) to +67% (WASP-121 b). The $\text{Ly}\alpha$ absorption line itself is unchanged and already correct— $1s \rightarrow 2p$ only ($f = 0.41641$ on the ground-state n_{HI} ; $1s \rightarrow 2s$ is a forbidden two-photon transition).

benchmarks/hd209/input.inp was missing Stellar Lya flux, Stellar Teff, and Stellar radius; without the $\text{Ly}\alpha$ flux the LaRT converter read $F_{\text{Ly}\alpha} = 0 \Rightarrow L_{\text{Ly}\alpha} = 0 \Rightarrow n_{2p} \simeq 0$, a spurious $H\alpha = 0.000\%$. Added $T_{\text{eff}} = 6065\text{ K}$, $R_* = 1.155 R_{\odot}$, and $F_{\text{Ly}\alpha} = 3.5 \times 10^3\text{ erg cm}^{-2}\text{ s}^{-1}$ (quiet-solar-analog, R_*^2 scaled; provisional, since the absolute LaRT depth scales with it). With both paths on the same VULCAN-BC winds and the sub-level opacity, the escape-prob/LaRT $H\alpha$ ratios become HD 209458 b 0.84/0.43 (2.0), HD 189733 b 1.23/0.91 (1.4), WASP-121 b 4.41/2.33 (1.9), WASP-52 b 5.69/2.20 (2.6)—so the inline closure overestimates $H\alpha$ by only $\sim 1.4\text{--}2.6\times$, far less than the earlier $\sim 4\text{--}30\times$ (pre-VULCAN winds with the $2p$ -only LaRT). Paper abstract, methods, results table, summary, and figs/make_lart_fig.py were updated accordingly.

35. Molecular equilibrium seeding fix, molecular-base coupling, Tier-2 gate refresh (2026-07-16)

Frozen-atomic-base bug (diagnosed and fixed). The 8-unknown molecular equilibrium (ion_system_HeH_mol, hybrd1) is bistable in its initial guess: from a zero-molecular warm start it fails to converge at the dense, optically-thick base, and the `thereis_mol` branch never checked info—the failed state was silently adopted and then frozen indefinitely ($x_{\text{H}_2} = 0.0122$ at the base, flat across 130k steps and independent of T ; numerically *not* a root of the

coded network—it would require an H_2 photoionization rate $15\times$ the unattenuated stellar bound). The true dark-base equilibrium is strongly molecular (3-body formation R15 vs. thermal dissociation R12), which the earlier gate runs had reached only because the then-uninitialized molecular `f_sp` columns acted as an accidental nonzero seed; the composition-audit zero-init exposed the latent fragility. Fixes: (i) `info` is checked and a failed solve retries once from a physically-informed guess seeded by the chemical-equilibrium H_2 fit `q_h2_equilibrium(p,T)` (still-failing cells keep the previous state and are counted in a one-line summary warning); (ii) `set_IC` seeds the molecular H_2 column from the same fit (H nuclei conserved) instead of zero.

Molecular base coupled on. Solving molecular chemistry now implies the molecular base particle count (Molecular base: True behavior): with an atomic `ntot_bc` the base temperature is inflated by the particle-count ratio ($\sim 1.8\times$ for a fully molecular base), thermally dissociating the H_2 layer the chemistry just built. A note is printed when the coupling engages.

Tier-2 gates re-established (deterministic; inputs pinned at `docs/lower_atmosphere_figs/data_g*/input.inp`). Gate 0 (HD 209458 b atomic, 3000 steps): $\log \dot{M} = 8.97$ unchanged. Gate 1 (HD 209458 b molecular): fully molecular base ($x_{\text{H}_2} = 0.997$, base T unchanged at ~ 1464 K), sharp $\text{H}_2 \rightarrow \text{H}$ front at $r = 1.019 R_p$, $\log \dot{M} = 8.87$ vs. 8.97 atomic, $\text{He } 2^3\text{S}$ peak $+1.3\%$; stable over 12000 steps with zero solve failures. Gate 2 (hot-Uranus-like: $0.0457 M_J$, $R_p = 0.49 R_J$, $T_{\text{eq}} = 1140$ K, HD 209458 b orbit/spectrum—input recovered by matching the archived domain fingerprint $r_{\text{max}} = 4.814118$ to 6 decimals): fully molecular base, front at $r = 1.166 R_p$, H_3^+ active in the molecular layer. Atomic runs byte-identical throughout (make check + OMP identity).

Gate refresh (2026-07-23, current production defaults). After the 2026-07-17 onward changes (staged secondary ionization, `He_rec_coupling` on, `He-H` charge exchange on; §§36–39) the three gates were re-recorded at a uniform 12000-step relaxation-snapshot convention. The $\text{H}_2 \rightarrow \text{H}$ fronts and molecular base composition are essentially unchanged (front $1.019 \rightarrow 1.020$ for HD 209458 b, $1.166 \rightarrow 1.156$ for the hot Uranus; base $x_{\text{H}_2} = 0.996/0.9998$), so the dissociation-front physics is robust to the new defaults. The wind $\log \dot{M}$ now reads 10.58 for both HD 209458 b runs at 12000 steps — these are relaxation snapshots, not flux-flat converged: under the new defaults the HD 209458 b atomic gate does not converge (du reaches a fixed point $\sim 2.6 \times 10^{-6}$ while the $\rho v r^2$ mass-flux spread plateaus near 4%), so all three gates share the same step cap. $\text{He } 2^3\text{S}$ is now strongly suppressed in the molecular base below the front ($\text{He } 2^3\text{S} + \text{H}_2$ Penning destruction, $\sim 10^4\times$ at the front, rising back to the atomic value above $\sim 1.3 R_p$); the earlier “ $+1.3\%$ peak” was a whole-domain-peak artifact of the outer-wind transient. Pinned `data_g*/` refreshed; `lower_atmosphere_coupling` and `fig_mol_structure.pdf` updated.

36. Default H/He rates $\rightarrow$ Badnell+Mao / Voronov, van Hoof free-free Gaunt factor, free-free charge fix, and secondary ionization (2026-07-17)

Six coupled changes to the H/He microphysics. Items 1–2 are gated by the new `legacy_hhe_rates` flag (default False = new rates); items 3–4 (free-free) apply always; item 5 (secondary ionization) is gated by `use_sec_ion` (default True); item 6 rides along with the heating rework. Full cross-code context and the benchmark table: `docs/atomic_data_EXHALE_vs_MoCHII`.

36.1 What changed and why

1. H/He recombination default. The pre-update H/He case-B coefficients were unattributed fits inherited from ATES; a numerical check showed them to be exactly the Hui & Gnedin (1997) case-B fits. The new default is

$$\alpha^B(T) = \alpha_{\text{Badnell}}^A(T) - \alpha_1^{\text{Mao}}(T), \quad (12)$$

the Badnell (2023) radiative total minus the Mao & Kaastra (2016, A&A 587, A84) $n = 1$ coefficient, with the Badnell three-term dielectronic recombination added to α^A for He II. This matches the construction MoCHII already uses, so the two codes now share the H/He recombination coefficients. Setting `legacy_hhe_rates = True` restores the Hui & Gnedin fits exactly.

2. H/He collisional ionization default. The legacy H I / He I rates were Abel et al. (1997) (reproducing Janev et al. 1987) and He II a Hui & Gnedin (1997) fit, none source-annotated. The Voronov (1997, ADNDT 65, 1) fits,

$$k(T) = A \frac{(1 + P\sqrt{U})U^K e^{-U}}{X + U}, \quad U = \Delta E[\text{eV}]/k_B T, \quad (13)$$

are now the default, matching the metal rates (already Voronov in both codes) and MoCHII. Reverted together with item 1 by `legacy_hhe_rates`.

3. Free-free Gaunt factor. The frequency-averaged non-relativistic $\langle g_{\text{ff}} \rangle(\gamma^2)$ (with $\gamma^2 = Z_{\text{ion}}^2 \text{Ry}/k_B T$) of van Hoof et al. (2014, MNRAS 444, 420) is tabulated at 161 points in $\log_{10} \gamma^2$ from -6 to 10 in 0.1 dex steps (generator `cooling_data/gauntff_thermal_avg.py`), replacing the earlier unattributed two-branch logarithmic fit. Applies always, independent of `legacy_hhe_rates`.

4. Free-free charge-weighting correction (physical bug). The previous code weighted He II (net charge $+1$) by $Z^2 = 4$ and evaluated the metal Gaunt factor at the nuclear atomic number ($Z = 8$ for O, 26 for Fe). Free-free emission scales with the ion *net* charge, not the nuclear number: $Z_{\text{ion}} = 1$ for H II, He II, and singly-ionized metals; $Z_{\text{ion}} = 2$ for He III and doubly-ionized metals. The most visible consequence is that the free-free cooling of a He II-dominated layer drops by a factor of four. Applies always.

5. Secondary ionization (Shull & van Steenberg 1985) in the main loop. A photoelectron ejected with $E_0 = e_\nu - E_{\text{th}}$ above ~ 40 eV does not thermalize immediately: it deposits only $f_{\text{heat}}(x)$ of its excess as heat and drives secondary H I and He I ionizations, where x is the ionized fraction of the H+He nuclei,

$$f_{\text{heat}}(x) = 0.9971 (1 - (1 - x^{0.2663})^{1.3163}), \quad (14)$$

$$f_{\text{ion,HI}}(x) = 0.3908 (1 - x^{0.4092})^{1.7592}, \quad (15)$$

$$f_{\text{ion,HeI}}(x) = 0.0554 (1 - x^{0.4614})^{1.6660}. \quad (16)$$

Below 40 eV the photoelectron thermalizes fully. Gated by `use_sec_ion` (default `True`); `use_sec_ion = False` is bit-identical to the legacy full-thermalization path. This differs from `wind_ae`, which combines the same SvS85 partition with Dere (2007) tables tied to its fixed spectral grid; the main loop runs on a runtime SED and applies the SvS85 species-resolved fits directly.

6. He I 2^3S triplet photoheating channel added. The metastable He I 2^3S (threshold 4.8 eV) previously contributed only opacity and its photoionization rate but no photoheating. Its photoelectron energy now enters the heating and absorbed-energy budgets and the secondary-ionization source consistently with every other absorber.

36.2 Files touched

Cool_coeff.f90 (rr_badnell, dr_HeII_badnell, rr_mao, alphaB*_new, voronov_ci/ci*_new, the 161-point gff_avg table); util_ion_eq.f90 (free-free net-charge weighting in eval_cool; secondary ionization and He 2^3S heating in PH_heat_H/PH_heat_HHe); T_equation.f90 (free-free net-charge weighting); ionization_equilibrium.f90 and post_process_adv.f90 (callers wired for the new signatures); input_read.f90, parameters.f90, write_setup_report.f90 (legacy_hhe_rates and use_sec_ion declared, defaulted, parsed, echoed); cooling_data/gauntff_thermal_avg.py (van Hoof table generator); docs/input_schema.md (keys K14b, K14c).

36.3 Quantitative check

New default rates (α_B , collisional ionization ci ; $\text{cm}^3 \text{s}^{-1}$), with the legacy value at 10^4 K for comparison. The legacy $\alpha_B(\text{HII})$ at 10^4 K is the Hui & Gnedin (1997) fit, 2.592×10^{-13} .

Coefficient	10^4 K	10^5 K	10^6 K	legacy (10^4 K)
$\alpha_B(\text{HII})$	2.606e-13	3.070e-14	2.163e-15	2.592e-13
$\alpha_B(\text{He II})$	2.807e-13	5.006e-13	1.015e-12	2.616e-13
$\alpha_B(\text{He III})$	1.536e-12	2.370e-13	2.244e-14	—
$ci(\text{HI})$	7.448e-16	3.963e-9	3.103e-8	7.247e-16

The Fortran reproduces an independent Python recomputation to all printed digits, and the legacy branch reproduces the pre-update values exactly. The new $\alpha_B(\text{HII})$ at 10^4 K differs from the legacy Hui & Gnedin value by $\sim 1\%$; He II and He III differ more, mainly because the new He II value carries the Badnell dielectronic contribution explicitly. The van Hoof thermal Gaunt factor at $T = 10^4 \text{ K}$ and $Z = 1$ is $\langle g_{\text{ff}} \rangle = 1.266$. A smoke run confirms the expected factor-of-four drop in free-free cooling across a He II-dominated layer, and `use_sec_ion = False` reproduces the legacy heating and photoionization rates bit-identically.

36.4 Effect on existing results

Because the default rates change, results computed with the new default differ from the pre-update golden and gate outputs. `legacy_hhe_rates = True` recovers the old H/He ionization balance, with two always-on exceptions: the free-free charge-weighting correction (item 4) and the van Hoof Gaunt table (item 3), both physical corrections intentionally not tied to the switch. Secondary ionization (item 5) is a further default change; `use_sec_ion = False` restores the full-thermalization rates bit-identically.

37. He recombination radiation → H I ionization coupling (use_he_rec_coupling, 2026-07-17)

EXHALE carries no diffuse radiation field, so pure case B implicitly assumes every He II → He I recombination photon is reabsorbed locally by helium ($\alpha_B = \alpha_A - \alpha_1$, ground capture dropped). In H-dominated gas that is wrong: part of the He recombination radiation — the ≥ 24.6 eV ground-capture continuum, the 584 Å resonance line, the 19.8 eV 2^3S line, and the >13.6 eV part of the He I two-photon continuum — ionizes H instead. The physics and the choice of recipe were worked out in docs/QUESTIONS_2026-07-17.md. This section documents an opt-in coupling using Draine's (2011) on-the-spot y/z parametrization. New input key He_rec_coupling (flag use_he_rec_coupling), default False → pure case B, bit-identical to the legacy path.

37.1 What changed and why

When on, the He recombination photons that ionize H are added as an extra H I photoionization rate with its photoelectron heating, and the He II recombination coefficient becomes $\alpha_B + y\alpha_1$ rather than pure case B. Two local parameters set the split:

- $y = 1/(1 + R n_{\text{HeI}}/n_{\text{HI}})$ (Draine Eq. 14.16), the fraction of the ≥ 24.6 eV ground-capture continuum absorbed by H, with the cross-section ratio $R = \sigma_{\text{He}}/\sigma_{\text{H}}$ at the 24.6 eV He I ground edge. Evaluated from the code cross sections, $R = 6.004$.
- $z = 0.67 + 0.29/(1 + n_e/n_{\text{crit}})$ (Draine Sec. 15.5), the density-dependent fraction of case-B (excited-state) cascade photons that ionize H, with the 2^3S critical density $n_{\text{crit}} = 1100 e^{1.2/T_4} T_4^{1/2} \text{ cm}^{-3}$; it interpolates between Draine's two limits (~ 0.96 low density → ~ 0.67 high density).

Atomic (case-B) mode. No explicit 2^3S : α_1 is the Mao & Kaastra (2016) ground ($1s^2$) capture and α_B the active He II case B. The He II recombination coefficient is $\alpha_{\text{eff}} = \alpha_B + y\alpha_1$; the extra H I photoionization rate is $n_{\text{HeII}}n_e[z\alpha_B + y\alpha_1]/n_{\text{HI}}$; the photoelectron heating uses the ground-channel energy $E_{\text{gnd}} = 24.6 - 13.6 = 11.0$ eV and a cascade-averaged $E_{\text{casc}} \approx 6.14$ eV.

He I 2^3S (triplet) mode. The mode that computes 10830: the decay channels are explicit, so z is not used — a 2^3S atom destroyed by photoionization or Penning emits no 19.8 eV photon, so the cascade must be summed at the actual channel rates (α_1 is the 1^1S channel). The 1^1S channel coefficient is $y\alpha_1 + 0.25\alpha_B$: the net ground capture plus the singlet-excited capture channel ($0.25\alpha_B$) the current network omits. The H-ionizing photon production is summed over channels: ground $y\alpha_1 n_{\text{HeII}}n_e$; singlet-excited $0.8521 \times 0.25\alpha_B n_{\text{HeII}}n_e$ ($0.8521 = \frac{2}{3}$ from the 584 Å line + $\frac{1}{3} \times 0.5564$ from the 2^1S two-photon continuum above 13.6 eV); 2^3S radiative decay $A_{31}n(2^3S)$ (19.8 eV line, always ionizes H); and 2^3S collisionally converted to the singlets $n_en(2^3S)(0.5564q_{31a} + q_{31b})$. The channel deposit energies are 11.0 (ground), 5.53 (singlet-excited), 6.2 (19.8 eV line), and 2.512/7.6 eV (collisionally converted two-photon / 584 Å). The two-photon constants are integrals of the Drake, Victor & Dalgarno (1969) spectral shape rather than round numbers. The 2^1S term emits a photon *pair* summing to 20.62 eV with a distribution peaking at half that, so above the H I

edge it delivers 0.5564 ionizing photons per decay carrying 8.9646 eV — a mean photon energy of 16.110 eV, hence 2.512 eV of photoelectron energy each. Treating the in-band photons as uniformly distributed instead puts the mean at 17.109 eV and the photoelectron at 3.511 eV, 40% high. `f_2q_HeI` and `Ee_2q_HeI` in `util_ion_eq.f90` carry these, and the 0.8521/5.53/6.14 combinations are formed from them so the three cannot drift apart.

The corrections are applied at the lagged (pre-solve) densities in the main ionization-equilibrium loop and re-evaluated at the advection-corrected densities in the post-processor (rate/coefficient before the advection solve, heating after). The He II recombination *cooling* is left at $kT \alpha_B$; the mismatch is at most $y \alpha_1 kT$, negligible.

37.2 Files touched

`util_ion_eq.f90` (new `he_rec_coupling` subroutine returning the corrected He II recombination coefficient, the extra HI photoionization rate, and the photoelectron heating; all zero when off); `Cool_coeff.f90` (new `alpha1_HeII_mao`, the Mao & Kaastra 2016 ground $1s^2$ capture, i.e. the α_1 subtracted in `alphaB_HeII_new`); `ionization_equilibrium.f90` and `post_process_adv.f90` (callers wired to `rcheiiB/P_HI/heat` in the main loop and to the advection ODE and `theat` in post-process); `input_read.f90`, `parameters.f90`, `write_setup_report.f90` (`use_he_rec_coupling` declared, defaulted `.false.`, parsed, echoed); `docs/input_schema.md` (key K14d).

37.3 Quantitative check (HD 209458 b)

Protocol: HD 209458 b `input.inp+metals.inp`, the converged profiles loaded as IC, then four runs $\{\text{atomic, He } 2^3\text{S}\} \times \{\text{off, on}\}$ each relaxed for 15000 steps from the same IC. The IC predates the 2026-07-17 rate update, so both branches share the same partial re-relaxation and **only the on/off difference is meaningful** (not the absolute values). Across the wind $y \sim 0.65\text{--}0.69$ and $R = \sigma_{\text{He}}/\sigma_{\text{H}}(24.6\text{eV}) = 6.004$. Script `docs/plot_he_rec_coupling.py`; figures `docs/figures/he_rec_coupling_{atomic,heitr}.pdf`.

Quantity (on/off)	Atomic mode	He I 2^3S mode
He I	$\times 1.04$ ($1.2 R_p$) $\rightarrow \times 1.25$ plateau ($1.5\text{--}3.5 R_p$) $\rightarrow \times 1.32$ ($3 R_p$)	$\times 1.01\text{--}1.16$
He II (min.)	$\times 0.72\text{--}0.74$ ($1.1\text{--}1.2 R_p$) $\rightarrow \sim 1.0$ ($2 R_p$)	$\times 0.885$
$n(2^3\text{S})$	— (not computed)	$\times 0.89$ (1.05), $\times 0.92$ (1.2), $\times 0.98\text{--}1.01$ ($1.5\text{--}2 R_p$)
Temperature	$\leq +1\%$ (outer $+3\%$)	$\leq +0.3\%$
$\log \dot{M}$	$9.58 \rightarrow 9.65$ ($+0.07$ dex, $\sim +17\%$)	$9.57 \rightarrow 9.61$ ($+0.04$ dex, $\sim +10\%$)

Stale (2026-09-02). The table above, the two figures and the $\sim 17\%$ / $\sim 10\%$ rate numbers were measured on the 2026-07-17 form of `he_rec_coupling`. That routine has been rewritten twice since: section 120. gave the photons a competing absorber, and section 125. replaced the on-the-spot assumption with the cell's own absorption probability, which removes most of the coupling in any cell thinner than $\tau \sim 1$. **Nothing in this subsection has been re-measured;** the four HD 209458 b runs it rests on no longer exist.

The atomic mode carries the larger He ionization-structure change (case B alone, $y = 0$, under-recombines He), and its extra H photoionization raises the mass-loss rate by $\sim 17\%$. In the 2^3S mode the 10830-forming region ($1.5\text{--}2 R_p$)

sees $n(2^3S)$ within $\sim 1\text{--}2\%$ of the off case, so the He I 10830 line is essentially unchanged, while the base 2^3S drops $\sim 11\%$ (the TR-mode 1^1S channel coefficient change feeds directly into n_{HeII}). The temperature response is small in both modes. Default off is bit-identical to the pre-update path; existing goldens and gates are unaffected.

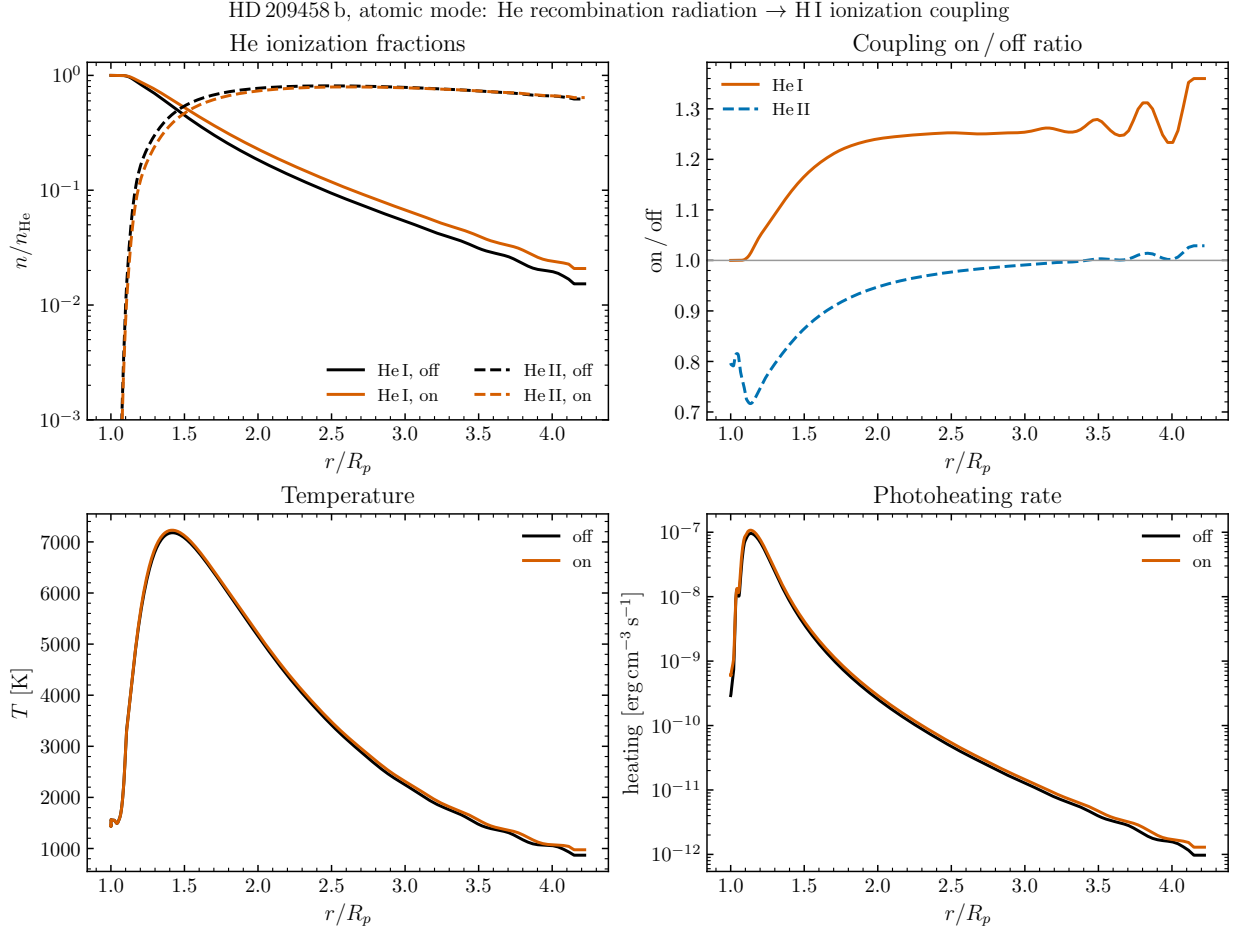


Figure 1: He recombination radiation $\rightarrow$ HI coupling, **atomic mode** (HD 209458 b). Clockwise from top left: He ionization fractions off vs. on; the on/off ratio for He I and He II; the photoheating rate; and the temperature. Turning the coupling on raises He I and lowers the He II minimum, with the $\sim 17\%$ mass-loss increase reflected in the extra photoheating.

38. Staged activation of secondary ionization (2026-07-23)

38.1 The regression

With secondary ionization on by default (the `use_sec_ion` item of the 2026-07-17 rate update), the high-gravity WASP-121 b cases (backup/regression/wasp_full, wasp_he23off) crashed with a NaN at $\sim$ step 587: negative densities first, then a NaN base pressure and temperature. The coupling itself is physically correct in magnitude — at the shielded base the SvS85 secondary channel raises the H photoionization rate by $\sim 35\times$ over the direct rate, the expected size for the $\sim\text{keV}$ photoelectrons there. The failure is a transient-path problem, not a steady-state one:

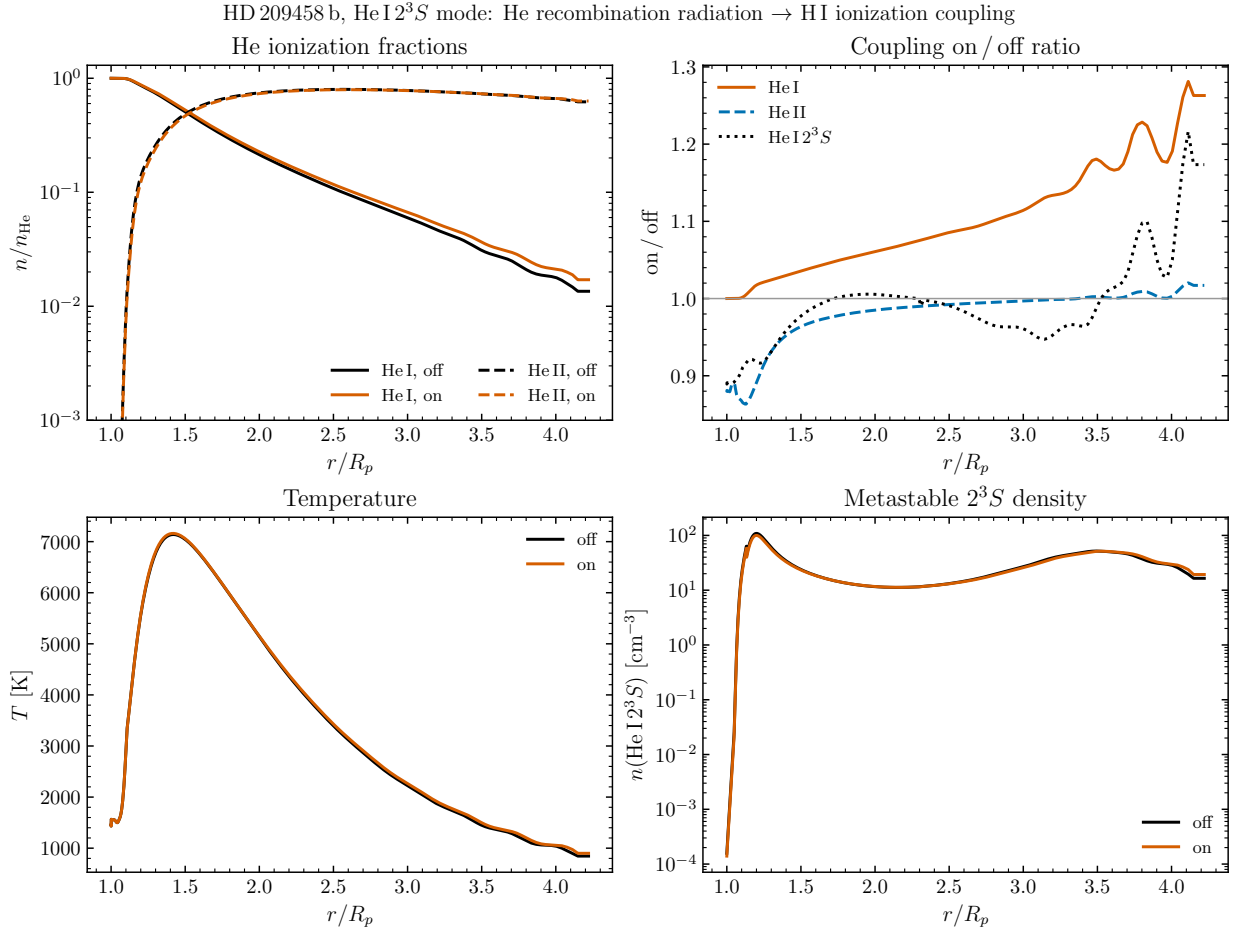


Figure 2: Same comparison in the **He I 2^3S mode**. The extra panel and ratio curve track the metastable 2^3S density: it drops $\sim 11\%$ at the base but stays within $\sim 1\text{--}2\%$ of the off case through the 10830-forming region ($1.5\text{--}2 R_p$), so the He I 10830 line is essentially unchanged.

the extra base ionization deepens the base cooling valley and amplifies the base acoustic mode of the marginal cold initial condition until the hydro diverges. Two facts pin this down: (a) a run with the coupling off converges cleanly ($du < 10^{-3}$, $\log_{10} \dot{M} = 13.30$); and (b) restarting with the coupling on from that converged state is benign — no NaN, du stays $\lesssim 4 \times 10^{-2}$ during a ~ 2000 -step adjustment and re-converges. The steady state with the full physics exists and is stable; only the cold-start path to it is fatal.

38.2 The fix: converge first without the coupling, then switch it on

A runtime state flag `sec_ion_active` now controls whether the SvS85 partition is applied, separately from the `use_sec_ion` enable flag. The photoionization / photoheating routines apply the coupling only when `use_sec_ion` .and. `sec_ion_active`; when it is off the code path is bit-identical to the `use_sec_ion = False` (full-thermalization) path. The main loop keeps `sec_ion_active = False` while the wind first relaxes; the moment that first relaxation reaches a convergence/stall criterion, the flag flips to `True`, the convergence logicals and the mass-flux level-stability history are reset, and the run continues and re-converges with the full physics. Stops are additionally held for `N_stall` (default 2000) steps after the flip: du reacts only once the base adjustment wave driven by the new coupling has formed, so without the hold the run stopped one step after activation (activation at step 7144, spurious stop at 7145) with a state that had not yet incorporated the physics. The Newton (Solver: Newton) finish and the direct-solve diagnostic modes (which bypass the marching loop) switch the coupling on before their solve, and the flag is forced on after the loop so post-processing and final outputs always carry the physics even if the loop hit the iteration cap before the flip.

38.3 The Immediate override

`Secondary_ionization: Immediate` sets `use_sec_ion = True` and applies the coupling from step 0, restoring the pre-staging behavior. It is intended for A/B tests only; on the high-gravity cold-start cases it reproduces the $\sim$ step-587 NaN crash, confirming that the staged path is what removes it.

38.4 Files touched

`parameters.f90` (`sec_ion_active`, `sec_ion_immediate`); `util_ion_eq.f90` (`PH_heat_H` / `PH_heat_HHe` gate on the local `sec_on = use_sec_ion` .and. `sec_ion_active`); `EXHALE_main.f90` (init, stage flip on first convergence, Newton-finish and diagnostic-mode activation, post-loop force-on); `input_read.f90` (the Immediate value parsed).

38.5 Effect on existing results (wasp_full validation)

For `Secondary_ionization: False` the run is bit-identical to before (verified byte-identical on `wasp_full` and `wasp_he23off`). For the default (staged on) `wasp_full`: activation at step 7144, re-convergence at step 13489 ($du = 9.98 \times 10^{-4}$), and $\log_{10} \dot{M}$ moves $13.30 \rightarrow 13.22$. The $\sim -17\%$ mass-loss rate is the real secondary-ionization physics: the SvS85 partition diverts most of the hard-photon photoelectron energy from heat into ionization and excitation in the weakly ionized base, so less energy drives the wind. Immediate reproduces the old (crashing) behavior on the high-gravity cases.

39. He-triplet physics series from the Falorca & Vidotto (2026) review (2026-07-23)

The line-by-line comparison against Falorca & Vidotto (2026, arXiv:2607.18193) and García Muñoz (2025, A&A 698, A199) — full analysis in docs/falorca2026_he3d_review.md — exposed six physics gaps, all implemented and validated in this series (its validation work also surfaced the secondary-ionization NaN regression fixed in the previous section). Physical correctness is the acceptance criterion throughout; the impact sizes below are informational.

1. **Penning ionization products** (heh_tr_rows, System_HeH_mol, adv_implicit_HeH_TR): $\text{He}(2^3\text{S}) + \text{H}^0 \rightarrow \text{He}(1^1\text{S}) + \text{H}^+ + e^-$ now produces the H^+ (with its 6.2 eV of electron heating, $e_{\text{th,HeI}} - e_{\text{th,HeTR}} - e_{\text{th,HI}}$) it always removed the triplet for. In the EUV-shielded base (Γ_{HI} exponentially small) this is the *dominant* H-ionization channel: n_{HII} rises $2\text{--}7\times$ at $r < 1.03$ on wasp_full, while the wind and $\dot{M}$ are unchanged. The associative branch ($\rightarrow \text{HeH}^+$, $\sim 10\%$, GM25) is folded in as H^+ (documented at the code site).
2. **Collisional ionization in the TR systems** (heh_tr_rows *b* terms; ci_HeI23S in Cool_coeff.f90, the Black-1981-derived $6.41 \times 10^{-21} \sqrt{T} e^{-55338/T} / E_{23\text{S}}$, $k(10^4 \text{ K}) = 3.3 \times 10^{-10}$): the TR (Oklopčić-form) rows silently dropped *all* electron-impact ionization while eval_cool kept charging its cooling — both restored and consistent now, including CI of the metastable itself (a $\sim 1\text{--}2\%$ 2^3S loss at 10^4 K , its 4.8 eV cooling in the coio channel, and its He I / adv-row counterparts).
3. **Explicit- $n(2^3\text{S})$ cooling** (eval_cool + the semi-implicit energy solver): the 10830 Å collisional-excitation channel $1.16 \times 10^{-20} \sqrt{T} e^{-13179/T} n_e n_{23\text{S}}$ (Black form with the *explicit* metastable density — Falorca & Vidotto warn Black’s implicit steady-state triplet is wrong here) plus the 0.80/1.40 eV q_{31a}/q_{31b} conversion ledger. The q_{13} (19.8 eV) term is deliberately excluded: the Cen-1992 He I excitation-cooling term already carries that channel. The terms are passed into solve_energy_semi_implicit (all three eval_cool calls) so they act on T ; without that wiring they were diagnostic-only. Measured: 1–5% of the total cooling on WASP-121 b (8–11% of local $\text{Ly}\alpha$ at the 2^3S peak), locally up to $\sim 28\%$ on HD 209458 b.
4. **He_rec_coupling default ON** (the previous section has the model): with the coupling off the TR singlet recombination used α_1 alone — neither case A nor case B — and the He recombination photons ionized/heated nothing. Validated stable on wasp_full: base $n_{23\text{S}} - 10\%$, the 10830-forming region within $\sim 2\%$, $\dot{M}$ unchanged; on HD 209458 b $\dot{M}$ rises $+0.04\text{--}0.07$ dex. He_rec_coupling: False restores the legacy path (byte-identical).
5. **He $\leftrightarrow$ H charge exchange in the default set** (he_h_charge_exchange, default on; Koskinen 2013 rates via Huang 2023 Table 4): the group-B pair moved out of the cx_full-gated table assembly into dedicated he_h_cx_fvec/jac/fvec_adv routines called by *every* ionization system with He (previously only the metals systems could reach it), including the advection pair and the analytic Jacobians of the Newton systems. The effect is confined to the neutral-H base — He II $\times 0.59$ at $r = 1.05$, -1% by $r \geq 1.2$, $\dot{M}$ unchanged — exactly the $\text{He}^+ + \text{H}^0 \rightarrow \text{He}^0 + \text{H}^+$ sink where n_{H^0} dominates. cx_full still gates groups C/D. The new input keys of this and the preceding sections were also added to the parser’s known-keys list (they parsed correctly but tripped the unknown-line warning).

6. **He(2³S)+H₂ Penning ionization** (penning_HeI23S_H2; System_HeH_mol rows 4/5/8 + 4.4 eV heating): GM25 Table A.5 (Cohen & Lane 1977) fitted as $5.3791 \times 10^{-12} T^{0.676} e^{-695.21/T}$ ($\leq 0.13\%$ over 500–10⁴ K). On the pinned molecular gates the metastable drops 39× (HD 209 base) and 430× (hot-Uranus, out to $r \sim 1.1$) exactly where $x_{\text{H}_2} \rightarrow 1$, with the upper wind and $\dot{M}$ unchanged — GM25’s dominant deep metastable sink, now present for Tier-2 + triplet runs. The _adv post-process is atomic-only (documented) and needs no counterpart.

Regression state after the series: Secondary_ionization: False + He_rec_coupling: False + He_H_charge_exchange: False reproduce the respective legacy paths byte-identically; the wasp goldens and the parse-dump corpus were re-snapshotted once at the end of the series (defaults: staged secondary ionization, He recombination coupling on, He↔H charge exchange on). Converged wasp_full reference: activation at step ~ 7144 , final count ~ 13486 , $\log_{10} \dot{M} = 13.22$.

40. du triggers require a descending crossing (2026-08-11)

The $du < du_{\text{th}}$ convergence stop and the JFNK hand-off (with the secondary-ionization flip that shares its du test) now fire only after du has been seen at or above their own threshold. They start armed for a Load IC? True run and unarmed for a generated IC. The failure this removes: a transonic IC reported $du(1) = 1.85 \times 10^{-5}$ on WASP-121 b, which tripped both triggers on step 1 and ran away to a NaN by step 1220; with the guard the same input converges (step 8706, $\log_{10} \dot{M} = 13.21$). The PLM → WENO3 switch is deliberately *not* guarded, because the regression cases legitimately switch on step 1 at $du(1) = 0.37$. No new input keys. Details: docs/newton_scaling_and_base_wall.md §9.

41. JFNK line search decides on the true residual (2026-08-11)

The line search in solve_steady_jfnk accepted steps on the residual with the WENO3 smoothness weights frozen at the *previous* iterate, compared that against a merit memory which mixed the frozen measure with the true one and was refreshed only on acceptance, and reported both $\|R\|$ and the $\text{info} = 0$ verdict from the frozen residual. Measured on the HD 189733 b hand-off state: 7 of 14 accepted steps *raised* the true residual, by factors 1.09 to 9.05, and the stored reference stood 2.4× below the true merit of the iterate the search started from — which is what produced the 12-consecutive-failure abort.

Trials are now evaluated with the weights recomputed at the trial state, the merit memory holds true merits and is rewritten at the start of every outer iteration, and the diagonal scaling is formed before the merit it scales. The frozen weights still define the inner Newton model, which is where they belong. WASP-121 b now reaches $\text{info} = 0$ ($\|R\| = 5.72 \times 10^{-5}$, $\log_{10} \dot{M} = 13.17$, superseding the du -stop 13.21 of the preceding section); photo_deep_section_cont converges in 24 outer iterations instead of 59; no case exhausts its backtracks any more. No new input keys, no golden changed. Details: docs/newton_scaling_and_base_wall.md §10.

42. Metal-line cooling at a cold base: line trapping and the coronal-fit validity floor (2026-08-11)

Two physically wrong statements in the metal cooling were corrected. Both bite only where the gas is cold and dense — at the base of a planet whose base falls out of the coronal regime — and the wind is unaffected.

42.1 (a) The escape probability was hardwired to 1

eval_cool built a beta_esc from the lowest XUV-band *continuum* opacity across *one cell*, then overwrote it with beta_esc = 1.0d0 and applied the full optically thin metal-line cooling everywhere. Neither quantity is a line optical depth and neither is grid-independent. (AIOLOS chemistry.cpp:1006 instead scales the same gray depth by an arbitrary 10^8 , driving $\beta \rightarrow 0$ and switching metal-line cooling off; EXHALE had replaced that with the opposite limit.) Measured on the converged HD 189733 b base, the dominant coolant line [O I] 63 μm carries a line-center optical depth $\tau = 2.7$ through the cold layer, i.e. an escape probability of 0.16, not 1.

β is now the line-center escape probability of the line itself, computed from the *column* between the emitting cell and the top of the domain (Cool_coeff.f90: kappa_OI63, kappa_CII158, line_escape_probability, fine_structure_escape). The shape is the plane-parallel Doppler result of Hollenbach & McKee (1979) / de Jong, Boland & Dalgarno (1980), renormalized by a factor 2 so that $\beta(0) = 1$ *exactly*: the published form tends to $1/2$ because it counts escape through one face of a slab, whereas here the other direction is absorbed by the lower atmosphere and so removes energy from the modeled gas either way. The two branches are switched at $\tau_c = \sqrt{\pi} e^{a^2/4} = 6.967$, where they cross, so the switch is continuous in value. In the optically thin wind $\beta \rightarrow 1$, reproducing the previous behavior.

β enters as $A_{ul} \rightarrow \beta A_{ul}$ *inside* the two-level solution, not as a factor on its result: in the subcritical limit the cooling is set by the collisional excitation rate and must be independent of β , and multiplying outside would be wrong by a factor β there. Trapping is applied to [O I] 63 μm and [C II] 158 μm only, the two lines for which the code carries an explicit two-level solution; everything else stays optically thin, which is right in the wind and is the residual approximation at the base ($< 0.3\%$ of the base cooling there). A thick resonance line in a metal-rich wind (Mg II h&k) is still treated as thin — recorded, not fixed.

42.2 (b) The CHIANTI coronal fits were evaluated far below their validity floor

Every metal cooling coefficient in Cool_coeff.f90 is a fit or a table built over 10^3 – 10^5 K. At the 236 K HD 189733 b base, 99.7% of the [O I] rate came from the fit's softest exponential, $e^{-930.111/T}$, whose 930 K corresponds to no [O I] ground-term splitting at all (the splittings are 227.7 K and 326.6 K); C I is the same case ($e^{-2351.38/T}$ against splittings of 23.6 K and 62.4 K). Below the floor the only excitable metal transitions are the ground-term fine-structure lines, which the explicit two-level terms already carry, saturation included.

coronal_excitation_cutoff(T) now removes the coronal part below 10^3 K as $\exp[-((T_{\text{floor}}/T - 1)/w)^2]$ with $w = 0.5$, and is exactly 1 above it. Value and dT -slope are continuous at the floor (the Brent energy solve and the semi-implicit update both differentiate the cooling in T), and every coefficient is **bit-identical at and above 10^3 K**. The factor is applied to all CHIANTI-derived coefficients — the analytic C/N/O and Mg/Ca/Na/Fe forms and the 1-D/2-D tables,

whose $\log_{10} T$ axis starts exactly at 3.0 and which otherwise hold their edge value indefinitely below it — and to the coronal remainder of `cool_OI_ne_func` / `cool_CII_ne_func`, but *not* to their two-level parts. The legacy AIOLOS branch (`cno_cool_0`) is deliberately left alone: its constant floors are crude fine-structure stand-ins, not extrapolated coronal fits. $w = 0.5$ is a modeling choice, not a measurement.

42.3 Measured

HD 189733 b, cell 1: total cooling $3.40 \times 10^{-5} \rightarrow 2.68 \times 10^{-8} \text{ erg cm}^{-3} \text{ s}^{-1}$ against a heating of 7.0×10^{-6} , so the base net rate changes sign from -2.65×10^{-5} (cooling) to $+6.97 \times 10^{-6}$ (heating). Almost all of that comes from (b); (a) alone is a factor ~ 5 on the [O I] two-level term. Sweeping the code's own cooling assembly in T at that cell's frozen densities, against the heating measured there, the local radiative balance temperature moves from **168 K to 486 K** — 168.2 K from (a) alone, 485.5 K from (b) alone, so (b) does essentially all of the work at the base — and the two cooling curves are identical from 10^3 K up. A side-by-side marching test does *not* resolve this: on a 5×10^4 -step scale the base is riding an inflow transient whose adiabatic heating dominates the radiative term in both binaries.

WASP-121 b (base at 2358 K, $\tau([\text{O I}] 63 \mu\text{m}) = 0.039$, $\beta = 0.956$) is unchanged: $\log_{10} \dot{M} = 13.17$ before and after, outer mass flux to 3×10^{-7} relative, profile maximum relative difference 8×10^{-5} . make check: `mol_base_handoff` (metals off) stays byte-identical (PASS); the two metal cases move by the intended physics with $\log_{10} \dot{M} = 13.22$ unchanged in both — `wasp_full` maximum relative difference 3.7×10^{-7} (density), outer mass flux to 2×10^{-8} ; `wasp_he23off` 1.6×10^{-3} (density, temperature) in the narrow band at $r = 1.012\text{--}1.014$, outer mass flux to 1.2×10^{-5} . Goldens not re-snapshotted at that point. Separately noted while comparing: both goldens already contain **negative species densities** in that band (H I, O I, O II, S II; 17 cells in `wasp_full`, 3 in `wasp_he23off`) — a pre-existing defect, unrelated to this change.

42.4 Not fixed by this

The model still has no stellar optical/IR absorption and no thermal background, so nothing sets a radiative floor at T_{eq} for a shielded layer; and below $\sim 700 \text{ K}$ the guard leaves ions without an explicit two-level term (C I, N, Mg, Ca, Na, Fe) with no cooling at all — for C I that omits the real [C I] 609/370 μm lines, whose LTE rate at the HD 189733 b base is $\sim 10^{-10} \text{ erg cm}^{-3} \text{ s}^{-1}$, i.e. 10^{-4} of the local heating. Details: `docs/hd189_base_checkerboard.md` §10. Files:

```
src/modules/radiation/Cool_coeff.f90
src/modules/radiation/util_ion_eq.f90
src/modules/nonlinear_system_solver/T_equation.f90
src/modules/post_process/post_process_adv.f90
```

43. Base grid resolution as an input key (Base_grid [dr,cells], 2026-08-11)

The resolution of the uniform region of the Mixed grid was two hardcoded locals in `define_grid.f90` ($N_{\text{low}} = 50$ cells of $drc = 2.0\text{e-}4 R_p$). It is now an input key, because that spacing is what decides whether the base carries

the stationary $2\Delta r$ entropy mode diagnosed in docs/hd189_base_checkerboard.md: the controlling parameter is the number of cells per base density scale height, $H/\Delta r$ with $H = kT/(\mu g)$, and a $4\times$ refinement was measured to remove the mode.

```
Base grid [dr,cells]:  2.0e-4 50      # the default -- the historical grid
Base grid [dr,cells]:  5.0e-5 200     # 4x refinement at the same 0.01 Rp extent
```

The two numbers share one line, as du_th [PLM,WEN03] does, because they are not independent: their product is the radial extent of the uniform region ($0.01 R_p$ by default). Refining at fixed extent means dividing Δr and multiplying the cell count by the same factor; changing only one moves the junction with the stretched region. The second value may be omitted (the count then keeps its default). The key applies to Grid type: Mixed only; Uniform and Stretched build their grids from $r_{\max}$ and N alone, and EXHALE_setup.out echoes that the key is ignored for them.

EXHALE_setup.out also reports the resolution actually achieved, Base scale-height resolution: $H(T_{\text{eq}})/dr = 1/(b_0 \cdot dr_j(1))$ (b_0 is the surface Jeans parameter, $dr_j(1)$ the first cell after the Mixed-grid smoothing), and warns below 10 cells. On the default grid this is 26.9 cells for HD 189733 b against 102.3 for WASP-121 b — the margin the high-gravity planet does not have. Grid type: Stretched gives 1.7 cells for HD 189733 b, i.e. it is unusable at the base for this class of problem. The smoothing pass at the end of define_grid started at the literal 50, which was N_low written out; it now follows N_low so that the junction between the uniform and stretched regions is smoothed wherever it is.

Two costs, both structural. The CFL step is set by the smallest cell, so a $4\times$ finer base needs about $4\times$ more steps for the same physical time. And the total cell count $N = 500$ is a compile-time parameter, so the cells given to the base come out of the stretched region: on the $4.43 R_p$ HD 189733 b domain the stretch ratio goes $1.0119 \rightarrow 1.0252$ and the cell size at $1.5 R_p$ grows from 6.1×10^{-3} to $1.3 \times 10^{-2} R_p$. Refining the base therefore coarsens the wind, which is where $\dot{M}$ and the transmission spectrum are formed. Making N runtime-settable would need the static $(1-N_g:N+N_g)$ arrays to become allocatable throughout; it was not done here. Recorded while doing this: count_max = 1000000 is also a compile-time parameter and the EXHALE_MAXSTEPS hook only lowers it, so a $4\times$ refined base can reach at most $1/4$ of the physical time the default grid reaches within the cap. On HD 189733 b $4\times$ is therefore not reachable to convergence without raising count_max; $2\times$ is, and $2\times$ is already where the mode goes.

Default is bit-exact. dr_base is declared with a *default-real* literal ($2.0e-4$, not $2.0d-4$) because that is what the old local was: the stored value is the single-precision neighbor of 2×10^{-4} , 2.5×10^{-8} relative below it. Declaring the exact double moves every base cell by that amount; measured on wasp_full it changed the converged density by 5×10^{-10} relative (step count and $\log_{10} \dot{M} = 13.22$ unchanged), which is not a physics difference but is not byte-identity either. The literal is kept so that an input.inp without the key reproduces earlier runs exactly. The consequence to know: writing the default out explicitly (Base grid [dr,cells]: 2.0e-4 50) does *not* reproduce it, because the list-directed read into real*8 gives the exact double.

Checks. Byte-identity was tested differentially, against the binary built immediately before this change rather than against backup/regression/golden/ (the goldens are stale by the intended metal-cooling physics change of the

preceding section). All output files of `wasp_full`, `wasp_he23off` and `mol_base_handoff` are byte-identical including headers, with the same step counts (13488 / 13482 / 12000) and $\log_{10} \dot{M}$ (13.22 / 13.22 / 10.58). WASP-121 b run to completion with the key absent: $\log_{10} \dot{M} = 13.17$, JFNK info = 0, $\|R\| = 5.7 \times 10^{-5}$ — unchanged from the preceding section. Malformed values are rejected in `define_grid` (a cell count outside $[2, N - 10]$, or a uniform region that does not fit inside $r_{\max}$).

Details: docs/hd189_base_checkerboard.md §11. Files:

```
src/modules/init/parameters.f90
src/modules/init/define_grid.f90
src/modules/files_IO/input_read.f90
src/modules/files_IO/write_setup_report.f90
```

44. Base ghost temperature, runtime step cap, coronal-cutoff width as input keys (2026-08-11)

Three settings that were compile-time constants or an unexamined default became `input.inp` keys. All three default to the previous behavior, so a file without them runs exactly as before (make check: all three golden cases byte-identical).

44.1 Base ghost temperature: isothermal | continuous

The lower ghost cells pin the density to `rho_bc` and, in the legacy closure, the pressure to `ntot_bc + dp_bc`, i.e. the temperature to T_0 . **The T_0 pin has no physical backing:** the radiative equilibrium of the lower atmosphere is outside the model, so nothing in the code determines the ghost temperature, and T_{eq} is a choice rather than a boundary condition. It becomes actively wrong once the first interior cell settles far from T_0 . With the metal-line cooling of the preceding section, cell 1 on HD 189733 b sits at 494 K against $T_0 = 1183$ K: a factor-2.4 contact discontinuity, plus a $2.3 \times$ density inversion, held permanently on the boundary.

continuous imposes $dT/dr = 0$ at the base face instead. The ghost keeps the base composition — `ntot_bc` nuclei and `dp_bc` electrons at `rho_bc`, exactly the particle count the isothermal pin uses — and carries the cell-1 temperature,

$$p_{\text{gh}} = (\text{ntot_bc} + \text{dp_bc}) T_1, \quad T_1 = p_1 / (n_{\text{tot}} + n_e)_1.$$

$(n_{\text{tot}} + n_e)_1$ comes from `get_species_densities`, the single point where the code decides what counts as a particle. The ghost pressure therefore stays a differentiable function of the interior pressure (what the JFNK line search needs), while the ionization state it divides by is lagged exactly like every other composition quantity across a hydro step. `BC_component_constrho` is the only place the lower ghost is built, and marching, the JFNK residual and the reconstruction boundary all go through it, so the closure cannot differ between code paths.

The key sets the same quantity as `Hydrostatic base: True` (the ghost pressure, which that key extrapolates from the interior gradient instead), so the two are mutually exclusive: `Hydrostatic base` is tested first and `input_read` warns when both appear. With `Base BC: pressure` the microbar target is imposed at T_0 when n_0 is derived, so under

a continuous- T ghost only the base *density* stays anchored; `input_read` notes that too. The closure in effect is echoed in `EXHALE_setup.out`.

Measured on HD 189733 b in the production configuration, restarted from the converged production state and Newton-finished in a scratch copy. The baseline restart reproduces the stored state digit for digit, so the comparison is like for like.

	isothermal	continuous
T_{ghost} / T_1 [K]	1183.0 / 493.6	539.9 / 540.2
$\rho_1 / \rho_{\text{gh}}$	2.33	0.95
$A(\ln \rho)$ cells 1–12	0.0264	0.0092
$A(\ln \rho)$ cells 3–12 / 5–14	4.41e-3 / 1.62e-3	9.28e-3 / 8.95e-4
JFNK	info=0, $\ R\ = 5.37\text{e-}4$	info=0, $\ R\ = 2.76\text{e-}4$
$\log_{10} \dot{M}$	9.04	9.05

The boundary jump and the density inversion are gone; the extended alternating tail is gone as well (beyond cell 7 the amplitude drops $13\times$ to $420\times$), and what remains is a stronger disturbance confined to cells 2–4. Three further restart+Newton cycles reproduce each state digit for digit, so both are exact fixed points and no further decay is available from cycling. Details, window sweep and scope: `docs/hd189_base_checkerboard.md` §13. **Not** measured at that point: a cold start under the new closure (6 h, 10^6 steps), other planets, transit observables.

Adopted for production. HD189733b/`input.inp` now carries Base ghost temperature: continuous, and the planet folder was re-converged with it together with the ionization-root validation of the next section (`run_20260811_rootfix.log`): $\log_{10} \dot{M} = 9.05$, JFNK info = 0, $\|R\| = 1.72 \times 10^{-4}$, 2002 steps. Measured on that converged state, $T_{\text{ghost}}/T_1 = 529.4/529.6$ K, $\rho_1/\rho_{\text{gh}} = 0.95$, and $A(\ln \rho)$ over cells 1–12 is 1.9×10^{-3} , i.e. well below the 0.0092 of the isolated A/B above — the root validation appears to remove a further part of the base disturbance. The paper and `python/paper_data.py` carry 9.05 for this planet.

44.2 Max steps: <N>

`count_max` was an integer, `parameter = 1000000`. The preceding section recorded the consequence: a $4\times$ -refined base, whose CFL step is $4\times$ smaller, cannot reach a converged state within a cap that the default grid already used in full, and the `EXHALE_MAXSTEPS` hook only lowers the cap. `count_max` is now a runtime variable with the same default. `EXHALE_MAXSTEPS` keeps its separate meaning (a deterministic mid-loop exit used by the regression harness); it does not raise `count_max`.

44.3 Coronal cutoff width: <w>

The coronal-excitation guard of the preceding section multiplies every CHIANTI-derived coronal coefficient by $\exp(-x^2)$ with $x = (T_{\text{floor}}/T - 1)/w$ below $T_{\text{floor}} = 10^3$ K. $w = 0.5$ was a hardwired parameter. It is a modeling choice, not a measured quantity, and the temperature the base settles at depends on it at the ~ 100 K level, so it is now settable,

still defaulting to 0.5 (division by a variable holding 0.5d0 is the same arithmetic, hence byte-identical). Values ≤ 0 are rejected at parse time. The manual states the sensitivity where the guard is documented.

Checks. make check: wasp_full, wasp_he23off, mol_base_handoff all byte-identical with the keys absent. WASP-121 b run to completion with the keys absent: $\log_{10} \dot{M} = 13.17$, JFNK info = 0, $\|R\| = 5.686 \times 10^{-5}$, 8228 steps, and its output byte-identical to the same case run with a binary built from HEAD without these changes. The parse-dump corpus (backup/regression/run_parse_corpus.sh) gains three lines per case and needs a deliberate golden refresh; it was not refreshed there. Files:

```
src/modules/init/parameters.f90
src/modules/states/Apply_BC.f90
src/modules/functions/composition.f90
src/modules/radiation/Cool_coeff.f90
src/modules/files_IO/input_read.f90
src/modules/files_IO/write_setup_report.f90
```

45. Ionization-equilibrium roots validated against the physical simplex (2026-08-11)

Ion_species.txt carried negative number densities (H II, O II, O III, Fe I) in a thin band just above the base: 17 cells at $r = 1.0106\text{--}1.0140 R_p$ in the wasp_full golden, 15 cells in the WASP-121 b production run, 5 in HD 189733 b.

The equilibrium systems are polynomial in the stage fractions and possess roots outside the physical simplex. At the step where secondary ionization is switched on (step 7186 of wasp_full) MINPACK hybrd1 returned info = 1 on such a root — $x(\text{H II}) = -5.2 \times 10^{-6}$, the Fe stage fractions summing to 1.2209, i.e. an Fe I density 22% below zero. Nothing tested the root, and the next step's warm start re-seeded the solve from the same values, so the state reproduced itself for the remaining 6302 steps. The heating paid for it twice: the metal photoheating channel went negative ($-6.6 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$, which is impossible), and the ionized fraction, clamped into $[0, 1]$, sent the Spitzer & Scott (1985) heating fraction f_{heat} to zero, switching off every photoheating channel above the photoelectron threshold. Total heating in the band sat 93% below its neighbors. Clamping the offending fractions to zero is therefore not a repair — the clamp is exactly what produces $f_{\text{heat}} = 0$; the cure has to be a physical root, not a clipped one.

Every atomic cell solve now validates its root — each stage fraction ≥ 0 , and each element's ionized stages summing to at most its nucleus total — and on failure restarts from physically defined starting points: the uncoupled ionization balance at the incoming electron density (ionization_balance_at_fixed_ne, admissible by construction, since the stage populations of a three-stage element come out proportional to $(d_1 d_2, u_0 d_2, u_0 u_1)$), then the optically thick fully neutral limit. A stored state that is not physical no longer seeds the next solve, which is what breaks the self-sticking. A cell where no starting point produces an admissible root is left on the ionization balance and reported rather than accepted. A first attempt that converges to a physical root is accepted unchanged, so healthy cells follow exactly the same solver path as before. q_abs, an absorbed energy rate, must now be positive before it normalizes the heating

efficiency.

Gates (single-threaded; a baseline make check on the same tree passed all three cases, so the differences are this change alone). wasp_full FAILs by intent: negative densities 17 cells $\rightarrow 0$, $\log_{10} \dot{M}$ 13.22 $\rightarrow$ 13.22, $T/\rho/v/p$ within 0.89%/0.77%/1.3%/0.13% of the golden away from the band, metal photoheating positive everywhere (domain minimum $-8.9 \times 10^{-7} \rightarrow +1.03 \times 10^{-7}$). wasp_he23off also FAILs: its golden looks clean but the run reports 35 roots outside the simplex over 13478 steps, previously accepted in silence; $\dot{M}$ unchanged, $T/\rho/p$ within 0.28%/0.086%/0.039%. mol_base_handoff is byte-identical. The two production runs, each re-run once with the pre-change binary and once with the fix so the comparison isolates this change, keep their mass-loss rates and step counts (WASP-121 b 13.17, 8228 steps; HD 189733 b 9.05, 2002 steps) and lose their negative densities entirely. HD 189733 b is the clearest demonstration of the self-sticking: its stored initial condition is one of the poisoned outputs, and at step 1 the run rejects exactly the 5 stored cell states that carried negative densities and resolves all of them. Golden were not re-snapshotted at that point.

Not covered. The molecular network keeps its own chemical-equilibrium retry and is left alone; the H-only system (no helium) and the post-processing advection solves in post_process_adv.f90 are outside this validation, and the advected profiles of HD 209458 b still carry small negative H II, He III and He 2^3S entries in the first 14 cells above the base. Full account: docs/ionization_root_validation.md. Files:

```
src/modules/radiation/ionization_equilibrium.f90
src/modules/radiation/util_ion_eq.f90
src/modules/nonlinear_system_solver/newton_solver.f90
src/EXHALE_main.f90
```

46. Line trapping in the thick metal resonance lines: measured, no change (2026-08-11)

The metal-cooling section above introduced $\beta(\tau)$ for [O I] $63 \mu\text{m}$ and [C II] $158 \mu\text{m}$ and recorded the thick resonance lines of a metal-rich wind (Mg II h&k, Ca II H&K, Na I D, the Fe II UV multiplets) as an untreated case. They were measured on the converged WASP-121 b, HD 209458 b and HD 189733 b runs. **No code path changed.**

The lines are thick — Mg II k reaches $\tau_0 = 7.6 \times 10^4$ at the WASP-121 b base and stays above 10^2 through the region where Mg II carries a third of the local cooling, and over 99% of the integrated Mg II / Ca II / Fe II / Mg I cooling comes from gas with $\tau_0 > 1$. That is not the test, though. Trapping lengthens the random walk of a resonance photon; it does not destroy it. In two-level equilibrium the correction to an optically thin coronal fit is

$$S = \frac{\beta A_{ul}}{\beta A_{ul} + n_e q_{ul}}, \quad q_{ul} = 8.629 \times 10^{-6} \Upsilon / (g_u \sqrt{T}),$$

not a factor β , so it bites only above $n_{\text{crit,eff}} = \beta A_{ul} / q_{ul}$. For these permitted lines $A_{ul} \sim 10^8 \text{ s}^{-1}$, so even at $\beta \sim 10^{-6}$ the escape rate keeps $n_{\text{crit,eff}}$ at 10^9 – 10^{15} cm^{-3} , against a maximum n_e of $4.3 \times 10^9 \text{ cm}^{-3}$ in the WASP-121 b run. The

forbidden fine-structure lines are the opposite case ($A_{ul} \sim 10^{-5} - 10^{-3} \text{ s}^{-1}$, $n_{\text{crit,eff}} \sim 10^0 - 10^5 \text{ cm}^{-3}$), which is why they, and only they, need β .

Measured $\Delta\Lambda$ integrated over the wind: **0.0214%** of the total radiative losses of WASP-121 b, 0.0022% of HD 209458 b, 0.0015% of HD 189733 b. The largest local effect is 1.65% in the first WASP-121 b cell, whose T is pinned at T_{eq} by the boundary condition; the loss exceeds 0.1% only below $r = 1.007 R_p$. Per line it is 0.1% of the Mg II cooling, 0.015% of Ca II, 1.4% of Na I (its A_{ul} is the smallest of the set) and 0.02–0.8% of Fe II. The measurement is a conservative bound: it uses the Doppler-core escape probability, which understates β by 4–10 $\times$ for these Voigt profiles, and no turbulent broadening.

Checked in passing: the Fe II d^6D fine-structure lines at 25.99/35.35 μm , the one metal transition with an A_{ul} small enough to trap, reach only $\tau_0 \leq 0.06$, so the cool_FeII_ne statistical-equilibrium table needs no escape probability either. H I Ly α , outside this scope, has $\tau_0 = 1.3 \times 10^8$ and a cooling-weighted $S = 0.9994$ — two-level trapping is therefore not what suppresses Ly α cooling, and the destruction channels that could (photoionization of $\text{H}(n=2)$, collisional $2p \rightarrow 2s$) were not evaluated.

$\beta = 1$ for the resonance lines is kept and is now documented at the code site as a justified effective treatment with its validity condition, in the SCOPE comment of `src/modules/radiation/Cool_coeff.f90`, instead of an acknowledged omission. Full measurement: `docs/resonance_line_trapping.md`.

47. Metal electrons and a validity range for the post-process advection correction (2026-08-12)

Two defects in `post_process_adv`, both confined to the advection-corrected `*_adv.txt` output: the post-processor is the last call of the run and its arguments are `intent(in)`, so the converged wind and the equilibrium `*.txt` files cannot be affected by anything here.

The electron density inside the advection residuals was wrong. It counted only the H and He electrons, while the equilibrium residual it corrects counts the metal electrons as well (`metal_electron_sum`) and the *same* routine already used the metal-inclusive n_e for its heating, cooling and temperature solve. In a shielded base the metals are the dominant electron donors — measured $n_{e,\text{metal}}/n_e = 0.48 - 1.00$ in the base cells of all four paper planets — so the recombination terms were low by up to six orders of magnitude. Solved to machine precision with Brent, the residual *as coded* had its fixed point at $x_{\text{HII}} = 4.74 \times 10^{-8}$ in the first HD 209458 b cell, against an equilibrium 7.46×10^{-14} . A field `adv_cell%xe_metal` now carries the metal electrons per H nucleus into all three advection residuals, with the definition `calc_ne` and `metal_electron_sum` use. It is not conditioned on `eos_include_metals`: that switch decides whether the metals enter the mass and particle budget, whereas recombination needs the true electron density. Metals off gives zero identically.

The correction was applied where it cannot be computed and is not needed. The residuals carry the *neutral* fraction and report the ion density as $(1 - x_{\text{HI}})n_h$, so the ion fraction inherits the solver's absolute resolution on x_{HI} , $x_{\text{tol}} = \sqrt{\epsilon} = 1.5 \times 10^{-8}$. Where the equilibrium ion fraction is 10^{-13} the returned value is quantized at 10^{-8} with an

arbitrary sign: the first HD 209458 b cell got $x_{\text{HII}} = -4.96 \times 10^{-8}$ where the exact root is $+4.74 \times 10^{-8}$ — right magnitude, wrong sign — and the upwind cascade carried the negative density outward over 14 cells. Those same cells are in local ionization equilibrium to

$$\text{Da} = \frac{\Delta r}{v} (\Gamma_{\text{HI}} + \alpha_{\text{HII}} n_e) = 350\text{--}2900,$$

so the equilibrium solution is the answer the ODE would give anyway. The correction is now skipped, and the converged equilibrium ionization kept, wherever the cell is (i) inflowing ($v \leq 0$ on either face: the upwind discretization has no upstream cell), (ii) equilibrium-dominated ($\text{Da} > 100$) or (iii) unrepresentable ($x_{\text{HII,eq}} < 10^{-6}$, less than 1% relative accuracy at that resolution). Conditions (i) and (ii) are statements about the flow and (iii) about the representation of the unknown, so the pre-existing `pp_metal_on` gate on the inflow condition is removed — it made a property of the discretization depend on the metal-cooling switch. The temperature loop keeps condition (i) alone, likewise ungated; a thermal Damköhler number is not evaluated. Clipping the extracted densities at zero was tested and rejected: it breaks the H nucleus budget $n_{\text{HI}} + n_{\text{HII}} = n_h$.

Gates. make check: PASS, all three cases byte-identical (the goldens compare `Hydro_ioniz.txt` and `Ion_species.txt`, which this does not write). Isolated A/B, the pristine HEAD binary and the modified one run from the same configuration: `Hydro_ioniz.txt`, `Ion_species.txt`, `Cooling_breakdown.txt` and `Excited_H.txt` byte-identical, only the `_adv` files differ.

planet	$\dot{M}$ [log g/s]	negative_adv	cells held at eq (newly)	outermost held	max change, $r > 1.05$
HD 209458 b	9.31 → 9.31	48 → 0	103 → 124 (21)	1.0316	1.4% (He III)
HD 189733 b	9.05 → 9.05	0 → 0	135 → 212 (77)	1.0380	0.37% (He III)
WASP-121 b	13.17 → 13.17	0 → 0	0 → 0 (0)	—	3.8% (He III)
WASP-52 b	11.63 → 11.63	0 → 0	3 → 57 (54)	1.0106	3.3% (He III)

Every cell whose equilibrium ion fraction is below 10^{-6} is now held at the equilibrium value, and no newly held cell lies above $r = 1.038 R_p$. The change in the wind is entirely the metal electrons: a build carrying the guard alone removes all 48 HD 209458 b negative entries by itself and changes the profiles above $r = 1.05 R_p$ by exactly zero on three of the four planets and by 4.2×10^{-5} on WASP-52 b. Transmission spectra move by at most 0.07% in peak depth (He 10830 of HD 209458 b) and 0.08% in equivalent width ($\text{H}\alpha$ of WASP-121 b).

The re-run reproduces the stored output/ of WASP-121 b and WASP-52 b byte-identically but not that of HD 209458 b and HD 189733 b, which restart from output/*_IC.txt files that had been refreshed to the previous run's converged state; re-running continues that convergence (HD 209458 b du : $1.199 \times 10^{-2} \rightarrow 3.13 \times 10^{-3}$) and moves the profiles by up to 9% in density at unchanged $\dot{M}$. That is unrelated to this change — the pristine binary follows the same new trajectory — and is why the table above is an A/B between two binaries.

Files:

- `src/modules/post_process/post_process_adv.f90`
- `src/modules/nonlinear_system_solver/ion_cell_state.f90`
- `src/modules/nonlinear_system_solver/System_implicit_adv_{H,HeH,HeH_TR}.f90`

Full account: `docs/postprocess_advection_validity.md`.

Found and **not** fixed, marked at the code site: `System_implicit_adv_HeH` (helium on, He 2³S off) writes its electron-impact ionization terms without the $n_e n_h$ factor that `adv_implicit_H`, `adv_implicit_HeH_TR` and the equilibrium rows all carry, which also makes them dimensionally inconsistent with the photoionization rates they are added to. None of the four paper planets uses that path. (Since fixed, 2026-08-12: the terms now carry `ionhi*xe*n_h`; commit b0d44bc, `TO_BE_DONE` item (B).)

48. Ground-term fine-structure statistical equilibrium for C I, C II, N II and O I (2026-08-12)

The CHIANTI C/N/O cooling fits carry the ground-term fine-structure (FS) lines in the optically thin, *low-density* limit. Their critical densities are of order 10^0 – 10^5 cm⁻³, decades below the base density of an irradiated atmosphere ($n_e \sim 10^9$, $n_{\text{HI}} \sim 10^{14}$ cm⁻³), so at the base the coronal form overstates what those lines can radiate by 2.9–8.3 decades. Two of them ([O I] 63 μm , [C II] 158 μm) already had a two-level saturation; C I, N II and the [O I] 145.5/44 μm channels did not, and C I was in consequence the dominant coolant of the whole base layer of HD 189733 b, at a value comparable to the local heating rate.

What replaces it. Every C/N/O coolant whose ground term is split now has that term solved in exact statistical equilibrium at the local (n_e, n_{HI}) ,

$$\sum_j f_j R_{ji} = f_i \sum_j R_{ij}, \quad \sum_i f_i = 1,$$

$$R_{ul} = C_{ul} + \beta_{ul} A_{ul}, \quad R_{lu} = C_{ul} \frac{g_u}{g_l} e^{-E_{ul}/k_B T}, \quad C_{ul} = n_e k_{e,ul}(T) + n_{\text{HI}} k_{\text{H},ul}(T),$$

$$W_{\text{FS}} = \sum_{u>l} f_u \beta_{ul} A_{ul} k_B E_{ul} \quad [\text{erg s}^{-1} \text{ per ion}],$$

and the coronal curve is refitted to keep only the channels that *leave* the ground term (Λ_{rem}). The coefficient handed to the assembly is $\Lambda_{\text{eff}} = W_{\text{FS}}/n_e + \Lambda_{\text{rem}}$, so the $n_e n_{\text{ion}}$ prefactor recovers the H-collision channel exactly. Treated: C I $2p^2\ ^3P$ (609.1/370.4 μm), C II $2p^2\ ^3P$ (157.7 μm), N II $2p^2\ ^3P$ (205.3/121.8 μm), O I $2p^4\ ^3P$ (63.2/145.5/44.1 μm). That is the complete set — N I and O II have a single-level 4S ground term, Mg I/II, Ca II and Na I a single ground level, Fe I is built from permitted lines only, and Fe II already uses a two-dimensional statistical-equilibrium table.

Limits: $n_e, n_{\text{HI}} \rightarrow 0$ collapses onto the ground level and reproduces the coronal within-term channel to the accuracy of the collision-strength fits (1.3–4.4%); high density saturates at the exact multilevel LTE emission (verified to 10^{-14} relative). β_{ul} enters inside the solution, not as a factor on the result, which is the only placement that is right in both limits. All eight lines with a transition probability now get their own line-center escape probability from the outward column; previously only two did.

Two defects fixed on the way. (i) The two-level form multiplied a solution already normalized to the two-level partition function by the ground-term Boltzmann fraction of its lower level, so its LTE limit was low by $1 + (g_u/g_l)e^{-E/k_B T}$ — $2.7\times$ for [C II] 158 μm at the 530 K base of HD 189733 b. (ii) The H-atom de-excitation rates (a constant 4.0×10^{-11} for C II, $4.2 \times 10^{-11}(T/100)^{0.67}$ for O I, both flagged approximate in the source) are replaced by fits to published quantum

scattering calculations: Yan & Babb (2023, MNRAS 518, 6004) for C I and N II, Abrahamsson, Krems & Dalgarno (2007, ApJ 654, 1171) for O I, Barinova et al. (2005, ApJ 620, 537) for C II. They are 5–20× larger.

What it does. The coronal_excitation_cutoff guard was standing in for the missing saturation, so its width w behaved as a tuning knob on the base temperature. It no longer does: the local balance temperature of the base cell is identical to every digit printed over $w = 0.02$ –1.2 on all four planet runs, against a 500–650 K spread before.

run	base T	$\dot{M}$ at $2R_p$
HD 189733 b (warm restart, $w = 0.1$)	528.5 → 613.2 K	$2.351 \times 10^9 \rightarrow 2.850 \times 10^9$ g/s (+21.2%)
WASP-121 b (cold start)	2406.9 → 2438.8 K	$3.106 \times 10^{13} \rightarrow 3.370 \times 10^{13}$ g/s (+8.5%)
wasp_full golden case	2340.6 → 2357.3 K	$3.482 \times 10^{13} \rightarrow 3.549 \times 10^{13}$ g/s (+2.0%)

Transit depths on HD 189733 b move by +4.4% (Ca II) to +20.7% (H β). The profiles outside $r \sim 1.1 R_p$ move by a few percent; the change is a base-layer change that propagates through the density.

Files:

- src/modules/radiation/Cool_coeff.f90 — the SE solvers, the four ion coefficients, the generalized line opacity and escape routine
- src/modules/radiation/util_ion_eq.f90
- src/modules/nonlinear_system_solver/T_equation.f90
- src/modules/post_process/post_process_adv.f90
- cooling_data/fit_fs_saturation.py — generator; prints every coefficient
- cooling_data/make_cooling_doc_figures.py
- docs/cooling_formulas.tex, docs/coronal_cutoff_width.md section 7, TO_BE_DONE.md item (C)

Also fixed here, unrelated but found while reading the cooling diagnostics: examples/exhale_io.py named only four fixed channels in Cooling_breakdown.txt while the writer emits six (the collisional-excitation channel is written split by absorber), so every metal column was labeled two ions too early. python/paper_data.py parses the header and was already correct.

Still open: with the artifact gone it is plain that the base is not in local radiative balance at all — at the HD 189733 b base the total radiative cooling is 3.5×10^{-8} against a heating of 3.4×10^{-6} . Its temperature is set by the inner boundary and by the flow.

49. Base ghost built from the composition of the state it bounds (2026-08-12)

Apply_BC closes the Base ghost temperature: continuous ghost with

$$p_{\text{ghost}} = (\text{ntot_bc} + \text{dp_bc}) \frac{p(1)}{n_{\text{part_cell1}}},$$

where $n_{\text{part_cell1}}$ is the cell-1 particle count written by get_species_densities. Two call sites filled the ghosts before any composition solve had run on the state in question, so they used whatever $n_{\text{part_cell1}}$ was left over — the input_read placeholder $\text{ntot_bc} + \text{dp_bc}$ in the first case:

- `init.f90`: the initial `Apply_BC`. On HD 189733 b the placeholder differs from the true cell-1 particle count by 4.4%, and the finite-volume steady residual of the loaded state (which stops right after `init`) reported the resulting contact discontinuity at the base face as a cells 1–2 spike of 60 (mass) and 120 (energy) per sound crossing time.
- `steady_newton.f90`, `eval_residual`: `Apply_BC` after `unpack_U`. There $F(Y)$ depended on the previous Y as well as on Y , so a finite-difference Jacobian column mixed two states.

Both now evaluate the composition of the interior first. `init.f90` also seeds `base_flux_const` from the initial state, for the same reason: with `Base velocity: massflux` that constant starts at -1 , which `Apply_BC` reads as “not available yet” before falling back to the valve, so the first step and any diagnostic stopping right after `init` used a different velocity boundary condition than the run. Files: `src/modules/init/init.f90`, `src/modules/time_step/steady_newton.f90`.

Checks. `n_part_cell1` is read only by the continuous branch and `base_flux_const` only by the mass-flux branch, so a run setting neither key is unaffected and `make check` is byte-identical; an isothermal HD 189733 b residual reproduces bit for bit. With the key on, the cells 1–2 mass residual falls from $5.97 \times 10^1, 5.92 \times 10^1$ to $6.41 \times 10^{-2}, 7.70 \times 10^{-2}$ and the energy residual from $1.12 \times 10^2, 1.20 \times 10^2$ to $1.31 \times 10^{-1}, 1.54 \times 10^{-1}$, with cell 3 outward and the wind window unchanged to every digit. Re-converging the planet from its stored state: JFNK scaled merit $\|F_s\|_2$ $1.43 \rightarrow 0.315$, `info` = 0, $\log_{10} \dot{M}$ $9.15 \rightarrow 9.14$. Measurements and term-by-term split: `docs/hd189_base_checkerboard.md` section 15.

50. $H(n=2)$ rate coefficients, the trapped- $2p$ balance, and de-excitation heating (2026-08-12)

Four corrections to the excited-hydrogen and $\text{Ly}\alpha$ modules, plus the structural change that made the first one possible. Measurement and the full number tables: `docs/lya_destruction_channels.md` section 11.

One definition of the $n=2$ rate coefficients. The $\text{Ly}\alpha$ and $n=2$ atomic data and every collisional rate coefficient existed in three copies — inside `n2_populations`, again as standalone functions in the same file, and a third time in `lya_rt.f90`. They are now one module, `src/modules/radiation/hydrogen_n2_rates.f90`, which exports `c1s2s_rate`, `c1s2p_rate`, `c2s2p_rate`, `c2s1s_rate`, `c2p1s_rate`, `c2p2s_rate`, `alpha_B_hydrogen`, `alpha_2s_hydrogen`, `alpha_2p_hydrogen`, `n2p_destruction_rate` and the line constants. `excited_hydrogen.f90` and `lya_rt.f90` use it.

1. **Statistical weights shadowed by dummy arguments.** `n2_populations` was declared (`T`, `n1s`, `ne_l`, `Jlya`, `G2s`, `G2p`, `n2s`, `n2p`). Fortran does not distinguish `G2s` from the module parameter `g2s`, so the three detailed-balance rate coefficients written out in the routine body evaluated their weights as the Balmer photoionization rate the caller passed: $2s \rightarrow 1s$ and $2p \rightarrow 1s$ de-excitation came out 5 to 306 times too small (planet-dependent, scaling with the stellar Balmer continuum) and the $2p \rightarrow 2s$ ℓ -mixing rate 3 times too large. The dummies are now

`gam_ion_2s / gam_ion_2p` and the body calls the shared rate functions, so the weights are resolved at module scope and out of a dummy's reach. Two further shadows of the same kind, both harmless, were renamed: the frequency-sample count `ng = 400` in `gamma_n2_balmer` and `heat_n2_balmer` shadowed the global ghost-cell count `Ng`; it is `n_nu`.

2. **Recombination cascade source.** $a2s*ne^{**2} / a2p*ne^{**2} \rightarrow a2s*ne*nHII / a2p*ne*nHII$. The electron recombines onto a proton; at the base the electrons come largely from helium and metals while hydrogen is still neutral, and n_e/n_{HII} reaches 2 on WASP-121 b and 5×10^6 on HD 209458 b. The Python transmission tool (`exhale_transit_lib.py`, `EXHALE_transit.py`) and `examples/tpm_halpha_lart2d.py` carried the same expression and were corrected the same way; their `n2_populations` signatures gained an `nHII` argument.
3. **Destruction channels in the `J_int` closure.** `lya_rt.f90` built the trapped field from $n_{2p} = P/(A_{2p \rightarrow 1s}\beta)$. The $2p$ budget also loses atoms to collisional de-excitation, $n = 2$ photoionization, and ℓ -mixing followed by two-photon decay; the denominator is now $A_{2p \rightarrow 1s}\beta + D$ with D from `n2p_destruction_rate`. Where β is smallest the omission over-estimated `J_int` by up to $1.5\times$. In the same module the top-down $\text{Ly}\alpha$ line-center optical depth, previously two identical inline loops, is the single routine `lya_line_center_optical_depth`; all three `jlya_mode` values call it, so the `tau_Lya` column of `Excited_H.txt` is filled in the LaRT import mode instead of being written as zero.
4. **Deexc heat now defaults on.** The comment justifying the old default — that collisional de-excitation heating overlaps the H I collisional-excitation cooling — was wrong: `coex_rate_HI` is the one-way Cen (1992) coronal rate with no de-excitation term in it, so `Hdx` is the correction to that limit, not a double count. Most of the $n = 2$ population it de-excites was pumped by $\text{Ly}\alpha$ rather than excited by a collision, which makes the term absorbed $\text{Ly}\alpha$ thermalized by a collision, the same size as the photoelectric heating the code already applied unconditionally. The key now also honors `Deexc heat: False`, which it previously ignored. **This changes results for any run that supplies a stellar T_{eff} /radius and does not set the key.**

Audited and left alone: the energy equation applies no escape-probability factor to the $\text{Ly}\alpha$ cooling. `cool` is built from the bare Cen coefficient (`Cool_coeff.f90:843` $\rightarrow$ `util_ion_eq.f90:640,787`); β appears nowhere in that path, in any `jlya_mode`. That is the correct form — 98.2–99.99% of trapped $\text{Ly}\alpha$ photons escape and the collisional destruction that would suppress the cooling is at most 0.06% — so the historical expectation of a $\sim 10\times$ transfer suppression is not in the code and should not be.

Checks. `mol_base_handoff` (no stellar T_{eff} , so the excited-H model is off) is byte-identical at every stage, as is its comparison against the stored golden. `wasp_full` and `wasp_he23off` move: $\log_{10} \dot{M}$ 13.23039 $\rightarrow$ 13.22966 and 13.23106 $\rightarrow$ 13.23026, with the largest local temperature change 0.57% (item 1), 0.081% (item 2) and 0.147% (item 3); both cases pin `Deexc heat: True` in their `input.inp`, so item 4 moves nothing in them. Their golden were already stale against the pre-fix tree and were left as they are. Re-converged from their stored initial conditions in a scratch copy, WASP-121 b gives 13.2075 $\rightarrow$ 13.2030, HD 189733 b 9.1286 $\rightarrow$ 9.1392 and HD 209458 b 9.3790 $\rightarrow$ 9.4581; the $\text{H}\alpha$ and $\text{H}\beta$ transit depths move by 0.4–1.7% ($n(2p)$, which carries the opacity, barely moves, while $n(2s)$ halves). HD 209458 b is the largest move (0.075 dex, from items 1–3) because its $\text{Ly}\alpha$ cooling is a negligible share

of its radiative losses while the $n = 2$ heating terms are not; it is also the case docs/lya_deexcitation_heating.md recorded as going NaN at the base with the de-excitation heating on, which no longer reproduces with the corrected populations (re-converges from its stored initial condition, and a cold start survives a 20000-step cap). A build at -O1 -fopenmp -g -fcheck=bounds,do,mem running the wasp_full configuration is clean.

Files:

- src/modules/radiation/hydrogen_n2_rates.f90 (new)
- src/modules/radiation/excited_hydrogen.f90, src/modules/radiation/lya_rt.f90
- src/modules/init/parameters.f90, src/modules/files_IO/input_read.f90, Makefile
- exhale_transit_lib.py, EXHALE_transit.py, examples/tpm_half_alpha_lart2d.py

51. Molecular chemistry and trace metals solved in one system (2026-08-13)

The parser refused Molecular chemistry: True together with a metals.inp. That refusal blocked two things: the deep (10^{-3} bar) lower-atmosphere handoff of docs/base_composition_handoff_plan.md §11.7, whose A/B pair carries metals and whose consistent configuration is Tier-2 molecular chemistry; and trace-metal work on sub-Neptunes, which have a molecular base by construction.

The two networks cannot be solved apart, and the reason is quantitative rather than architectural: **in the shielded molecular base the metals supply essentially all the free electrons**. Measured at the base cell of the two smoke tests below, with solar metal abundances:

case	n_e with metals [cm^{-3}]	metal share	n_e metals-off [cm^{-3}]
hot Uranus, solar 7 metals	1.20×10^8	1.000	6.29×10^4
HD 209458 b, 10^{-3} bar handoff, C/N/O	9.42×10^5	0.990	9.27×10^4

The flux that ionizes the low-ionization-potential metals — Mg I at 7.65 eV, Fe I at 7.90 eV, C I at 11.26 eV — lies below both the H I edge (13.60 eV) and the H₂ threshold (15.4 eV), so it is absorbed by neither and reaches the base almost unattenuated. On the hot-Uranus case it holds Mg and Fe about 1.6% ionized at the base, which is already enough to swamp the H/He and molecular electrons. Every recombination and electron-impact term of the molecular network scales with n_e , so solving the network at the metal-free n_e misses a factor 1900 (hot Uranus) to 10 (HD 209458 b at 10^{-3} bar) in that layer.

The merged system. System_HeH_mol_metals follows the pattern System_HeH_TR_metals established: the molecular unknowns keep their rows, the metals are appended above them, and both blocks are shared code rather than re-derived. The eight balance rows of the molecular network moved into mol_heh_rows (still in System_HeH_mol, with the Koskinen 2022 rate coefficients, which stay defined in exactly one module) and take n_e as an input, exactly as heh_tr_rows does; the metal rows, the metal electron sum and the stage fractions come from ion_residual_core unchanged. The layout is

x(1..3)	H II, He II, He III
x(4..7)	H ₂ , H ₂ ⁺ , H ₃ ⁺ , HeH ⁺
x(8)	He 2 ³ S (when tracked)
x(mbase..)	X ⁺ , X ⁺⁺ for each metal element, mbase = metal_row_base() = 8 or 9

with $N_{\text{eq}} = 7$ (+1 with the triplet) + $2n_{\text{melem}}$. `metal_row_base()` is the single definition of that offset, read by the residual, by the driver's seeding and root validation, and by `cx_metal_base` so that the Huang Table-4 charge exchange writes to the shifted metal rows. With `met_nelem = 0` the system reduces exactly to `System_HeH_mol`.

Root validation and seeding (§45 extended to the molecular branch). The molecular cell solve is bistable because the cell is: a molecular basin (the dense, optically thick base) and an atomic basin (the wind above the H₂ → H front). It now tries one starting point in each in turn — the warm start, the chemical-equilibrium H₂ fraction at the local (p, T) with the metal stages from their own ionization balance at the incoming n_e , then the molecule-free ionization balance of every element — and keeps a converged root only if it is also physical, on the same rule §45 applied to the atomic systems. The metal part of `ionization_balance_at_fixed_ne` was split into `metal_ionization_balance_at_fixed_ne` so the molecular retry seeds its metals from the same balance instead of a second copy of it. The third starting point is what the front cells needed: on the hot-Uranus case the coupled solve leaves 3 cells at $r = 1.21$ – 1.22 (the ionization front) with no admissible root at every step with two starting points, and 0 cells with three.

Two defects found in the surrounding code and fixed.

- The sub-13.6 eV energy grid of a power-law spectrum was an `if (thereis_HeITR) ... else if (thereis_lowIP_metal)`, with the comment that the triplet and metals together are unsupported so at most one applies. That has not been true since the triplet and the metals were merged. With the triplet on, the grid floor stopped at 4.80 eV, which is above the KI threshold (4.341 eV), so KI would be held spuriously neutral in a triplet run. The floor is now the lowest threshold over whichever sources are active, matching what the loaded-SED path (`sed_read`) already did. No golden carries K, so this is byte-identical there.
- The `pp_metals 2` metal re-solve passed the global N_{eq} to `hybrd1` while its residual (`ion_system_metals_pp`) writes only rows $1..3 + 2n_{\text{melem}}$. With the He 2³S triplet that already left one row of `fvec` unwritten; with the molecular layout it would leave four or five. It is now sized $3 + 2n_{\text{melem}}$ with a matching workspace, the same reasoning `lwa_adv` already carried a few lines above. Both goldens run `pp_metals 1`, so this is byte-identical there and changes results only for `pp_metals 2` runs.

Left in place at the time, with the validity range written at the code. `eval_cool` built its own electron density without the molecular ions. That is right in the hot atomic gas it was written for and wrong inside a deep molecular base, where H₃⁺ can be the dominant ion; there it disagreed with the n_e the equilibrium solver itself uses (`ionization_equilibrium` does pass `nmol_eq` to `calc_ne`), so the n_e -scaling cooling channels were under-counted in that layer. Two definitions of the same quantity is one too many, but closing the gap changes the molecular results and belonged in its own change with its own measurement. *Closed in the next section.*

Gates. make check PASSes byte-identical on all three cases then in the matrix (wasp_full, wasp_he23off, mol_base_handoff); a baseline make check on the same tree before the change also passed, so the comparison is this change alone. Goldens were not re-snapshotted. A build at -O1 -fopenmp -g -fcheck=bounds,do,mem is clean on all three coupled layouts for 200–300 steps: hot Uranus with the He 2³S triplet ($N_{\text{eq}} = 28$, metals at row 9), the same case without it ($N_{\text{eq}} = 27$, metals at row 8), and the HD 209458 b deep handoff.

Hot Uranus (the Tier-2 gate configuration of docs/lower_atmosphere_figs/data_g2) plus solar C/N/O/Mg/Ca/Na/Fe, a 12000-step relaxation snapshot against the same case metals-off: the H₂ → H front sits at $r = 1.1577$ in both (the gate's 1.156), the base stays fully molecular in both, $\log_{10} \dot{M} = 10.58$ in both, no negative densities, and no cell is left without an admissible root (the same case with the triplet off, $N_{\text{eq}} = 27$ with the metals at row 8, is clean over a shorter 400-step smoke). What the metals change is the base chemistry: n_e rises 1900× and H₃⁺ falls by a factor 1350 ($7.08 \times 10^4 \rightarrow 52.5 \text{ cm}^{-3}$), its peak moving out from $r = 1.040$ to $r = 1.130$ — H₃⁺ is destroyed by dissociative recombination (Koskinen R6/R7, both proportional to n_e), so it survives only where the metal electron fraction starts to fall. Metal ionization runs the expected way: C II/C = 8×10^{-10} at the base, 0.22 at the front, 0.43 at $1.9 R_p$; Mg III stays below 10^{-8} up to the front and reaches 2.4×10^{-3} only at $1.9 R_p$.

HD 209458 b at the 10^{-3} bar handoff (the §11.7 blocker), C/N/O, He 2³S on, a 4000-step bounded run: it integrates. §11.7 records that the atomic pair built at this level terminated with a NaN at $r = 1.067$ after 1846 steps with the time step collapsed to $dtu = 1.2 \times 10^{-6}$ (that run was not repeated here); this one reaches the cap with $dtu = 0.198$ and no NaN, a fully molecular base, and the front at $r = 1.0150$ (1.0169 metals-off). It is a relaxation snapshot at $du = 747$, nowhere near converged, so no mass-loss rate should be read off it.

Open item. The _adv post-process is still molecule-free by design (post_process_adv.f90 header): nh = nhi + nhii counts only the free H nuclei and the molecular electrons are absent from its n_e . The metal stages are carried there as usual under pp_metals, but on that molecule-free H/He background, so _adv metal profiles inside the molecular layer inherit the approximation. With the eval_cool electron density closed in the next section, this is the one remaining place where a molecular run and a metal run still meet on the atomic assumption.

Files:

- src/modules/nonlinear_system_solver/System_HeH_mol_metals.f90 (new)
- src/modules/nonlinear_system_solver/System_HeH_mol.f90
- src/modules/radiation/ionization_equilibrium.f90, src/modules/radiation/util_ion_eq.f90
- src/modules/files_IO/input_read.f90, src/modules/files_IO/write_setup_report.f90 (the startup report now names the species blocks the coupled system carries and prints N_{eq})
- src/modules/init/set_energy_vectors.f90, src/modules/post_process/post_process_adv.f90, Makefile
- examples/16_molecular_metals/ (new)

Documentation: README.md, README_HOWTO.md, examples/README.md, docs/EXHALE_user_manual.tex (the example table also became a longtable, which it needed already — as a float it ran past the bottom margin into the page footer), docs/input_schema.md, docs/lower_atmosphere_coupling.md, docs/base_composition_handoff_plan.md.

52. One electron density for the cooling and for the ionization solver (2026-08-13)

eval_cool rebuilt its own free-electron density and left the molecular ions out of it. The equilibrium solver in the same sweep does not: it balances ionization against $\text{calc_ne}(\text{nhii}, \text{nheii}, \text{nheiii}, \text{ne}, \text{nm}, \text{nmol_eq})$, i.e. H^+ , He^+ , He^{++} , the metal stages under the eos_metals policy, **and** H_2^+ , H_3^+ , HeH^+ . A molecular run therefore carried two definitions of n_e , and the cooling used the smaller one. That is wrong wherever a molecular ion carries a non-negligible share of the charge, which is exactly the deep molecular base. Measured on the metals-off hot-Uranus Tier-2 gate, the two definitions differ by $168.5\times$ at the base cell (373 against $6.29 \times 10^4 \text{ cm}^{-3}$, H_3^+ supplying essentially all of it), $4.5\times$ at $r = 1.05$, 2.8% at $r = 1.10$, and nothing above $r \sim 1.15$; on the mol_base_handoff case the base ratio is $175.7\times$.

eval_cool now takes an optional nmol (H_2 , H_2^+ , H_3^+ , HeH^+ , cgs) and passes it straight to calc_ne — the electron density is not reconstructed anywhere inside eval_cool, it has one definition and one implementation. Callers that track the molecular network supply it: ionization_equilibrium (nmol_eq), the semi-implicit energy solver (nmol_dim, the cgs copy of the adimensional array it already hands calc_ne/calc_ntot), and the Cooling_breakdown dump. The _adv advection post-process deliberately does not: that reconstruction is molecule-free by design (its module header), and omitting the argument gives it the atomic charge sum it is built on. ionization_equilibrium's own calc_ne call lost its thereis_mol branch at the same time — nmol_eq is zero for an atomic run, so one call covers both.

Every n_e -scaling channel picks the correction up: recombination, collisional ionization, collisional excitation and bremsstrahlung are linear in n_e , and the coronal metal line coefficients enter as $\text{ne}*\text{nm}*c_{\text{metal}}$. The saturated ground-term coefficients (Fe II, and C I/C II/N II/O I under cno_chianti) are $\Lambda_{\text{eff}} = W/n_e$, so the assembly's n_e cancels and they respond only through the level populations ($C_{21} \propto n_e$) — the correct behavior, and there is no double counting of n_e anywhere in eval_cool.

Atomic runs are exactly unchanged. nmol_eq and the molecular f_sp columns are zero, and calc_ne adds them as $\text{ne} + 1.0*0.0$, which is exact. make_check is byte-identical on wasp_full (metals + $\text{He } 2^3\text{S}$) and wasp_he23off; a full WASP-121 b run (metals, $\text{He } 2^3\text{S}$, $\text{Ly}\alpha$ escape probability, excited H, transonic IC) reproduces all eight output files byte-identically — Hydro_ioniz{,_adv}, Ion_species{,_adv}, Cooling_breakdown, Heating_breakdown, Excited_H, IC_dump — at $\log_{10} \dot{M} = 13.20$.

mol_base_handoff changes, and that is the point of the change. At the pinned 12000-step snapshot the two runs stop at the same $du = 2.6689$ and the same dtu ; every hydrodynamic column agrees to $\leq 4 \times 10^{-8}$ relative and every species column to $\leq 4.2 \times 10^{-6}$ (that worst case is H_3^+ at $r = 2.96$, where H_3^+ is $3 \times 10^{-19} \text{ cm}^{-3}$). The one column that moves is the reported cooling rate at the near-neutral base, $2.799 \times 10^{-20} \rightarrow 4.942 \times 10^{-18} \text{ erg cm}^{-3} \text{ s}^{-1}$ ($\times 176.5$, the n_e ratio, as expected for channels linear in n_e). $\log_{10} \dot{M}$ stays 10.58.

Why the dynamics barely notice. A molecular layer cools through the H_3^+ infrared lines, which do not scale with n_e , so multiplying the electron channels by 176 at the base moves the cooling from 2.8×10^{-20} to 4.9×10^{-18} against $1.3 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$ of photoheating there. The correction is real and it is now the same quantity the solver balances

ionization against, but it is energetically negligible in the layer where it is large.

Molecular smoke tests. Hot Uranus, the Tier-2 gate configuration, 12000-step relaxation snapshots — both runs stop at du of order unity, so these are snapshots and not converged solutions.

	metals off	metals on (solar C/N/O/Mg/Ca/Na/Fe)
$n_e(\text{base}) \text{ old} \rightarrow \text{new}$	$373 \rightarrow 6.29 \times 10^4 \text{ cm}^{-3} (\times 168.5)$	$1.2016 \times 10^8 \rightarrow 1.2016 \times 10^8 (\times 1.0000)$
cooling at the base	$3.18 \times 10^{-20} \rightarrow 5.36 \times 10^{-18} \text{ erg/cm}^3/\text{s}$	$+7 \times 10^{-7} \text{ relative}$
$\text{H}_2 \rightarrow \text{H front}$	$1.15631 \rightarrow 1.15631$	$1.15677 \rightarrow 1.15677$
H_3^+ peak	$7.0808 \times 10^4 \rightarrow 7.0807 \times 10^4 \text{ cm}^{-3} \text{ at } r = 1.0396$	$52.465 \rightarrow 52.465 \text{ at } r = 1.1304$
$du \text{ at step } 12000$	$4.3354 \rightarrow 4.2914$	$5.8177 \rightarrow 5.8177$
$\log_{10} \dot{M}$	$10.58 \rightarrow 10.58$	$10.58 \rightarrow 10.58$

The metals-on column is the expected null: the low-ionization-potential metals hold n_e at 1.2×10^8 at that base (previous section), against 52 cm^{-3} of H_3^+ , so adding the molecular ions moves nothing (maximum relative change 7×10^{-10} in $\rho/T/p$, 3×10^{-9} in the metal stages). Metals-off, the snapshot differs by 1% in du and by sub-percent in the bulk state just above the layer ($\rho -0.7\%$, $T +0.8\%$ at $r = 1.19$), with the trace species that depend exponentially on T amplifying that into $\text{H}_3^+ -16\%$, $\text{H}_2 -13\%$, $\text{He II} +10\%$ at the same radius. Those are differences between two snapshots of an unconverged relaxation, not a converged shift; the gate observables — front position, H_3^+ peak, mass-loss rate — are unchanged.

On the HD 209458 b 10^{-3} bar handoff with C/N/O (a 4000-step bounded snapshot, also unconverged) the base picture is the same: the metals dominate n_e , the base cooling rises 0.45%, the molecular-layer cooling by at most 1.9%, the front moves $1.01503 \rightarrow 1.01518$ and the H_3^+ peak by 0.05%. An `-O1 -fopenmp -g -fcheck=bounds,do,mem build` runs all three molecular layouts (hot Uranus with and without metals, HD 209458 b deep handoff) for 300 steps with no runtime check firing.

Two related items found and recorded, not changed here.

- The H_3^+ infrared cooling is added to cool in `ionization_equilibrium`, *after* `eval_cool` returns. The JFNK steady residual uses that array, but the semi-implicit energy update (the default) rebuilds cool from `eval_cool` alone and overwrites it, so it never sees the H_3^+ term. For a molecular run the marching stage and the Newton stage therefore integrate slightly different energy equations, and the marching fixed point is not the zero of the residual the Newton solver drives down. The fix is to move the term inside `eval_cool` (which already receives `nmol`) with its own `cool_chan` column; that changes the `Cooling_breakdown.txt` schema and the molecular results, so it belongs in its own change. The `Cooling_breakdown` header now states that the channel sum falls short of the cool column by that term in the molecular layer. Open item (F) of `TO_BE_DONE.md`.
- H_2^+ , H_3^+ and HeH^+ donate their electron to n_e but are left out of the bremsstrahlung charge sum, so that sum is smaller than n_e inside the molecular layer. They are $Z_{\text{ion}} = 1$ and belong there formally, but they only exist at $T \sim 10^3 \text{ K}$, where the hot-plasma free-free expression is an extrapolation emitting at frequencies the atmosphere is not thin to, and where free-free is negligible against the H_3^+ and metal line cooling. The scope is written at the accumulator.

Files:

- src/modules/radiation/util_ion_eq.f90, src/modules/radiation/ionization_equilibrium.f90
- src/modules/time_step/energy_semi_implicit.f90
- src/modules/post_process/post_process_adv.f90 (comments only)

Documentation: docs/lower_atmosphere_coupling.md.

53. A fourth regression case for the molecular + metals system (2026-08-13)

The coupled molecular + metals system of the section before last was not guarded by anything in the matrix: mol_base_handoff runs the molecular network metals-off, and the two wasp_* cases run metals without molecules, so the merged residual, the metal_row_base() offset and the three-starting-point seeding had no golden.

backup/regression/mol_metals is that case: the same hot-Uranus Tier-2 gate configuration as mol_base_handoff (Molecular chemistry: True, its base.inp handoff, 12000 steps pinned in maxsteps) plus a solar-abundance metals.inp with the seven elements C/N/O/Mg/Ca/Na/Fe. It stops at $du = 3.1770$, $\log_{10} \dot{M} = 10.58$. DEFAULT_CASES in backup/regression/run_check.sh is now wasp_full wasp_he23off mol_base_handoff mol_metals, and the case list in the script header describes what each one guards.

Goldens for all four cases were re-snapshotted at the end of this change series, so they carry the two preceding sections. The two atomic cases are byte-identical to their previous goldens (both changes are exact no-ops without molecules); mol_base_handoff moves only in its cooling column, by the factor 176.5 of the previous section.

One harness trap, hit while doing this and now written into the script: golden does **not** re-run the cases, it copies whatever output/ currently sits in each case directory. Snapshotting straight after a code change baselines the *previous* binary. The order is check (which rebuilds and re-runs), then golden, then check again.

Files: backup/regression/run_check.sh, backup/regression/mol_metals/ (new), backup/regression/golden/ (re-snapshotted). Documentation: the workspace CLAUDE.md, README_HOWTO.md.

54. H_3^+ infrared cooling inside eval_cool: one energy equation for a molecular run (2026-08-13)

Closes item (F) of TO_BE_DONE.md, opened by the audit in the section before last.

The H_3^+ infrared cooling was added to cool in ionization_equilibrium, *after* eval_cool returned. Only some of the consumers of cool saw it:

term in cool	rebuilt by energy_semi_implicit (the default marching temperature update)	returned by ioniz_eq (JFNK steady residual, the explicit-energy option, the in-loop $\ R\ $ monitor, the Hydro_ioniz.txt cool column)
H/He recombination	yes	yes
collisional ionization (H I, He I, He II, He 2 ³ S)	yes	yes
collisional excitation (H I Ly α , He I, He II, He 2 ³ S 10830 + 2 ¹ S/2 ¹ P conversion)	yes	yes
bremsstrahlung (H ⁺ , He ⁺ , He ⁺⁺ , metal ions)	yes	yes
metal line cooling (27 ions, saturated ground-term fine structure included)	yes	yes
H₃⁺ infrared (Miller et al. 2013)	NO	yes

heat has no such split: it is assembled once in ioniz_eq (photoheating by each absorber including H₂ and the metals, Shull & van Steenberg secondary-ionization partition, Balmer photoelectric and Ly α de-excitation, He-recombination coupling, He 2³S + H and He 2³S + H₂ Penning) and passed into energy_semi_implicit as intent(in), never rebuilt. H₃⁺ was the only asymmetric term, and it was the whole asymmetry.

Three consequences followed from that one row.

- **The marching fixed point was not the zero of the residual the Newton solver drives down.** The marching relaxed $\Gamma = \Lambda_{\text{atomic+metal}}$; the residual demanded $\Gamma = \Lambda_{\text{atomic+metal}} + \Lambda_{\text{H}_3^+}$.
- **The two energy-solver options integrated different equations.** Energy solver: Explicit used the ioniz_eq array directly and did carry the term; the semi-implicit default (the production setting) did not.
- **The cool column of Hydro_ioniz.txt meant different things depending on how the run stopped.** A du-stop left the semi-implicit array (no H₃⁺) in place; the JFNK finish calls ioniz_eq after solve_steady_jfnk and left the array that has it.

Size of the missing term. On the metals-off hot-Uranus Tier-2 gate (backup/regression/mol_base_handoff) the H₃⁺ lines carry **more than 99% of the total radiative cooling from the base out to the H₂ → H front** (Cooling_breakdown H3p_IR / cool_total > 0.5 over $r = 0.9998$ –1.1856, > 0.99 over 0.9998–1.1557), peaking at $9.35 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$ at $r = 1.0413$, which is **7.3× the local photoheating**. Below the front every atomic channel is exponentially suppressed at $T \sim 1.3 \times 10^3 \text{ K}$, so the marching stage was integrating that layer with essentially no radiative loss at all: the cooling it used was smaller than the true total by a factor 1.1×10^2 to 1.2×10^{11} over $r = 1.00$ –1.156. With solar trace metals (mol_metals) the metal lines dominate instead and H₃⁺ reaches only 22% of the local cooling, in a thin band $r = 1.066$ –1.193.

The fix. eval_cool computes the H₃⁺ cooling itself, from the nmol array it has taken since the section before last, using the same h3p_cooling module (one definition of the Miller et al. 2013 fits and the Table-6 non-LTE factor; eval_cool uses it, nothing is re-derived). The post-return addition in ionization_equilibrium is gone, and a comment at the eval_cool call states why nothing may be added there. cool_chan gains column 7 for the channel, so

the decomposition still reproduces cool exactly (measured $\max|\sum_{\text{channels}}/\text{cool} - 1| = 3.3 \times 10^{-16}$ on the molecular gate, 5.5×10^{-16} on wasp_full). Callers that model a molecule-free gas (post_process_adv, and T_equation which solves for the _adv temperature) omit nmol and therefore still get the atomic charge sum *and* no H_3^+ term — the same approximation, stated at both call sites.

A side effect worth naming: energy_semi_implicit builds its damping derivative dC/dT by finite-differencing eval_cool in temperature, so the steep temperature dependence of the H_3^+ emission now enters the implicit temperature update instead of being invisible to it.

Output schema. Cooling_breakdown.txt gains col11 H3p_IR; the 27 metal-ion columns move from 11–37 to 12–38. It is written for every run and is zero without the molecular network. python/paper_data.py reads the column names from the header and follows automatically; examples/exhale_io.py has a fixed list, which now carries H3p (and is named COOL_GAS_CHANNELS, since it is no longer only the H/He channels).

Atomic runs are exactly unchanged. nmol(:, 3) is zero, the term is skipped cell by cell, and the total is $\dots + 0.0$. make_check is byte-identical on wasp_full (14060 steps, metals + He 2³S) and wasp_he23off (14037 steps), with the same step counts as their goldens.

The two molecular cases change, which is the point. Both were re-run at their pinned 12000-step convention; the goldens were **not** re-snapshotted.

	metals off (mol_base_handoff)	metals on (mol_metals)
du at step 12000	2.6689 $\rightarrow$ 2.6274	3.1770 $\rightarrow$ 3.1769
$\log_{10} \dot{M}$	10.9043 $\rightarrow$ 10.9040	10.9105 $\rightarrow$ 10.9105
$\text{H}_2 \rightarrow \text{H}$ front ($2n_{\text{H}_2}/n_{\text{H}} = 0.5$)	1.16087 $\rightarrow$ 1.16039	1.16246 $\rightarrow$ 1.16246
H_3^+ peak	$7.0237 \times 10^4 \rightarrow 7.0285 \times 10^4 \text{ cm}^{-3}$ at $r = 1.0419$	52.139 $\rightarrow$ 52.140 at $r = 1.1355$
base T	1213.42 $\rightarrow$ 1213.42 K	1213.46 $\rightarrow$ 1213.46 K
max T change, whole grid	+1.4% at $r = 1.93$ (wind transient)	$+5.2 \times 10^{-5}$ at $r = 1.20$
T change where H_3^+ dominates	−0.53% at $r = 1.078$, mean −0.20% over $r \leq 1.156$	-3.0×10^{-5} at $r = 1.183$
reported cooling at the base	$4.94 \times 10^{-18} \rightarrow 5.97 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$ ($\times 1.2 \times 10^{11}$)	unchanged
max cooling change	$\times 1.23 \times 10^{11}$ at $r = 1.0021$	+28% at $r = 1.1597$

Why the gate observables barely move, and why that is not evidence the term is small. The molecular layer is thermally massive: its H_3^+ cooling time $\frac{3}{2}nkT/\Lambda_{\text{H}_3^+}$ is $3.7 \times 10^6 \text{ s}$ at $r = 1.04$ and $2.2 \times 10^7 \text{ s}$ at the base, i.e. **200–2000 sound-crossing times** $t_s = R_0/v_0$. A 12000-step snapshot advances a small fraction of that, so the layer has only begun to respond: T has fallen by 0.1–0.5% through $r = 1.02$ –1.12 and is still falling. The front, the H_3^+ peak and $\dot{M}$ are unchanged to $< 0.1\%$ *at this step budget*, so the recorded Tier-2 gate numbers stand — but the converged molecular solution now has a different thermal structure than the pre-change code would have produced, and reaching it needs a step budget the gate convention does not provide. When this was written no molecular configuration in the tree had

converged (examples/15_molecular had run 135694 steps to $du = 2.9 \times 10^{-2}$), so it was stated as a limit, not a result; the paragraph after next closes it.

What was verified and what was not. The residual/Newton path is unchanged by the move: EXHALE_RESIDUAL=1 on one frozen molecular state gives a **bit-identical** output/residual_profile.txt before and after, so only the marching side moved. What could *not* be shown at the time was the H_3^+ term disappearing from the steady energy residual: on the only molecular states then available — 12000-step relaxation snapshots with a base-breathing transient, volume-weighted $\|R\| \sim 3$ — the cell-by-cell energy residual in the molecular layer is $1.5\text{--}7\times$ larger than $\Lambda_{H_3^+}/q_0$, so the systematic offset is buried in the transient. That check needs a converged molecular run.

Closed 2026-08-13: the check on a converged molecular run. Restarting the metals-off hot-Uranus snapshot with Load IC? True, Solver: Newton $5.0e-2$ and Max steps: 150000 reaches info = 0 after a JFNK finish at step 22556 — the first Newton-grade molecular solution the code has produced. In the 180 cells where H_3^+ carries more than half the cooling (the cells at $r \leq 1.001$, whose residual sets $\|R\|$, excluded) the ratio of the cell-by-cell energy residual to $\Lambda_{H_3^+}/q_0$ falls from **4.42 at the 12000-step snapshot (sign 89+/166–) to 0.70 at the default $\|R\| < 10^{-3}$ solution (173+/79–) to 0.0017 once the same state is re-solved at Resid tol: $1.0e-5$ (25+/79–)**: at the converged solution the residual carries no offset of the size of the H_3^+ term, and no preferred sign.

Two things follow. First, the default $\|R\| < 10^{-3}$ is not a converged molecular run — the norm is set by two or three cells above the base that carry $\sim 10^3$ times the residual of anything else, so the molecular layer is still cooling when the solver stops; Resid tol: $1.0e-5$ is required, and a third solve from that state reproduces it (T to 0.14%, front to five digits, $\|R\| = 6.7 \times 10^{-7}$). Second, an A/B against the pre-change binary isolates the term directly: marching both binaries 12000 steps from the same converged state, the temperature difference $T(\text{before}) - T(\text{after})$ is proportional to the local $\Lambda_{H_3^+}/(\frac{3}{2}nk)$ with a single constant — over the 155 cells where $\Lambda_{H_3^+}$ is within two decades of its peak the ratio has median 1.27 and 10–90% range 1.13–1.32, and over all 187 cells where H_3^+ carries more than half the cooling the sign is right in 181. The whole difference between the two marching energy equations is the H_3^+ term and nothing else. The physics of the converged solution, including the 190–300 K molecular layer, is in docs/lower_atmosphere_coupling.md and in the next section.

An -O1 -fopenmp -g -fcheck=bounds,do,mem build runs both molecular configurations for 300 steps with no runtime check firing.

Files: src/modules/radiation/util_ion_eq.f90, src/modules/radiation/ionization_equilibrium.f90, src/modules/post_process/post_process_adv.f90 (comments only), examples/exhale_io.py, python/paper_data.py, python/make_struct_figures.py. Documentation: README.md, docs/EXHALE_user_manual.tex (section 4.3 rewritten — it still described four H/He channels and columns 9–35 for the metals, which had been wrong since the collisional-excitation split; a radiative-cooling row for H_3^+ added to the atomic-data table), docs/lower_atmosphere_coupling.md, TO_BE_DONE.md.

55. The infrared field of the lower atmosphere (Base IR field) (2026-08-13)

Narrows item (G) of TO_BE_DONE.md.

The infrared coolants of a molecular base were radiating into a vacuum they are not in. The ground-term fine-structure lines carried an escape probability built from the *outward* column only, with the downward direction dismissed as absorbed by the lower atmosphere; the H_3^+ bands carried no transfer at all. Absorption of what that reservoir radiates back was in neither.

Measured first. Line-center depths on the converged metals-on hot-Uranus solution, with the level populations and Doppler widths the cooling module itself uses:

direction	[C I] 609 μm	[C I] 370 μm	[O I] 63 μm	[O I] 145 μm
outward, base cell	0.088	0.172	2.01	0.54
outward, coldest cell ($r = 1.083$)	1.4e-3	2.6e-3	0.028	4.6e-3
one scale height below the base	0.046	0.092	1.08	0.32
pressure where the downward $\tau = 1$	205 μbar	106 μbar	17 μbar	37 μbar

The base sits at 9.0 μbar , so every line that carries the cooling is black downward within one to three scale heights of unresolved atmosphere, while outward the collapsed layer is thin ($\beta \geq 0.90$ for C I, ≥ 0.63 for O I above $r = 1.05$). Trapping was never the missing piece; the incident field was. The ratio of the incident mean intensity to the line source function, $\frac{1}{2}B_\nu(T_{\text{base}})/B_\nu(T)$, passes 1 below $T \sim 610$ K and reaches 3.4–5.7 at the coldest cell: those lines were cooling a gas they should have been heating.

And it refutes the diagnosis in item (G) for the metals-on case. The specific entropy p/ρ^γ rises monotonically outward through the whole collapsed layer, 1.00 at the base to 1.67 at $r = 1.05$ and 5.69 at $r = 1.083$; the gas is net heated everywhere and is cold only because ρ has fallen by 10^2 . Its cooling is 36% of the local photoheating at $r = 1.01$ and 2–16% above $r = 1.02$, and $\Gamma - \Lambda$ is balanced by $\rho v [d\varepsilon/dr + p d(1/\rho)/dr]$ to a few percent. And the metals-off solution, with no fine-structure cooling at all, collapses further (115 K at $r = 1.033$, total cooling 3.8×10^{-12} erg cm $^{-3}$ s $^{-1}$). The metals-on layer is an adiabatic expansion, not a radiative collapse. Radiation does dominate in one place: the metals-off base, where H_3^+ carries 100% of the cooling and the entropy *falls* by a factor 3 between the base and $r = 1.02$.

What was implemented. Base IR field: True (default False) has the lower atmosphere radiate $B_\nu(T_0)$ over the sky fraction $f = 1 - \sqrt{1 - (R_p/r)^2}$ and both coolant families absorb it.

- Fine structure: the field enters *inside* the ground-term statistical equilibrium as the photon occupation number $\bar{n} = \beta_1(\tau_{\text{dn}})f/(e^{E_{\text{int}}/T_0} - 1)$, with radiative rates $\beta A(1 + \bar{n})$ down and $\beta A(g_u/g_l)\bar{n}$ up, so the returned power is the net one. The escape probability becomes two-sided, $\beta = \beta_1(\tau_{\text{up}}) + \beta_1(\tau_{\text{dn}})$, β_1 the single-face Hollenbach & McKee (1979) / de Jong et al. (1980) form; with the field off it is $2\beta_1(\tau_{\text{up}})$, the previous value, bit for bit. Each line stops cooling at its own radiative equilibrium temperature — 575.8 K ([C I] 609 μm) to 641.7 K ([O I] 44 μm) for $T_0 = 1140$ K and half-sky coverage.
- H_3^+ : Miller et al. (2013) fit the *total* emission, so the exchange is closed on one effective band, the ν_2 fundamental at 2521.3 cm $^{-1}$ ($E/k = 3627.5$ K): $\Lambda_{\text{net}} = \Lambda_{\text{emit}}(T)[1 - \bar{n}e^{E/T}]$, equilibrium at 936.1 K at the base. The approximations and their range are written at h3p_net_cooling_rate.

- Not applied, with the reason at the code: Fe II (a precomputed statistical-equilibrium table, which cannot take a field as an argument), the coronal remainders and every other metal ion (fits to CHIANTI sums with no line list; their lowest terms are at $E/k \gtrsim 1.6 \times 10^4$ K, and the heating they would add was estimated at $\sim 0.4\%$ of the losses of this layer), and the legacy use_2lev_cool branch.

Effect. Restarting the two converged Resid tol: $1.0\text{e-}5$ molecular solutions with the switch on, both returning info = 0:

	metals off, off $\rightarrow$ on	metals on, off $\rightarrow$ on
T at $r = 1.005$	626.8 $\rightarrow$ 910.0 K	1099.0 $\rightarrow$ 1098.9 K
T at $r = 1.02$	299.6 $\rightarrow$ 859.4 K	928.9 $\rightarrow$ 928.9 K
T at $r = 1.03$	140.2 $\rightarrow$ 840.2 K	820.0 $\rightarrow$ 820.1 K
coldest cell	115.1 K @ 1.0334 $\rightarrow$ 229.7 K @ 1.0947	187.9 $\rightarrow$ 189.1 K @ 1.0831
H ₂ $\rightarrow$ H front	1.03582 $\rightarrow$ 1.08833	1.07976 $\rightarrow$ 1.07979
$\log_{10} \dot{M}$ [g/s], spherical	10.539 $\rightarrow$ 10.617	10.596 $\rightarrow$ 10.596

The metals-off layer settles *on* the H₃⁺ radiative equilibrium curve: predicted 911/886/874 K at $r = 1.005/1.02/1.03$ with the local sky fraction, solution 910/859/840 K, just below by the margin the expansion takes out. Its H₃⁺ channel is now a heating term (-1.1×10^{-7} erg cm⁻³ s⁻¹ at $r = 1.02$). The solution is a fixed point: re-solving returns info = 0 at $\|R\| = 8.8 \times 10^{-6}$ and reproduces T below $r = 1.3$ to 10^{-4} . The metals-on layer does not move, which is the measurement above, not a failure of the closure.

Atomic runs are unaffected: the switch is off by default, and even on it touches only channels that carry no cooling in an atomic wind. make check is byte-identical on all four cases.

Files: src/modules/radiation/Cool_coeff.f90, src/modules/lower_atmosphere/h3p_cooling.f90,
src/modules/radiation/util_ion_eq.f90, src/modules/nonlinear_system_solver/T_equation.f90,
src/modules/post_process/post_process_adv.f90, src/modules/init/parameters.f90,
src/modules/files_I0/input_read.f90, src/modules/files_I0/write_setup_report.f90. Documenta-
tion: docs/lower_atmosphere_coupling.md, docs/EXHALE_user_manual.tex, TO_BE_DONE.md.

56. H₂ photodissociation in the Lyman–Werner bands (Stellar LW flux) (2026-08-13)

The molecular network had no photodissociation of neutral H₂. The Koskinen et al. (2022) Table-1 photo-rates start at the 15.4 eV photoionization edge, so below that the only H₂ losses were thermal (R12), electron impact (R14) and ion chemistry. docs/base_composition_handoff_plan.md §5 named the omission as the reason EXHALE’s base comes out more molecular than any photochemical code gives, and recorded that pinning q_{H_2} at the base (Route 1) patches over it. This adds the missing term.

The band is not in the code’s own radiation field, and had to be. Checked in the source: set_energy_vectors builds the photon grid from 13.6 eV up and reaches below the H I edge only when the He 2³S metastable (4.8 eV) or a low-IP metal is active, and then only to that species’ threshold; read_sed selects SED rows by the same e_low. For

a numerical SED the 11.2–13.6 eV band is therefore not read at all, and for a power-law SED (every Tier-2 gate case) it is the XUV power law extrapolated downward, which has nothing to do with a star’s FUV. Even where the grid does reach into the band, its opacity is continuum photoionization, whereas Lyman–Werner absorption is a forest of saturated lines. The band flux is a separate input,

Stellar LW flux [erg/cm2/s]: 343.0 # 912–1110 Å, integrated, at the planet

default 0 = off, and inert (with a warning) when Molecular chemistry is off.

Rate. Draine & Bertoldi (1996), ApJ 468, 269 (published version). For their flat- F_λ spectrum ($u_\nu \propto \nu^{-2}$, their eq. 24) at $\chi = 1$ the band photon flux is $F = 1.208 \times 10^7 \text{ cm}^{-2} \text{ s}^{-1}$ (their Table 1) and the unshielded dissociation rate is $\zeta_{\text{diss}}(0) = 4.17 \times 10^{-11} \text{ s}^{-1}$ (their Table 2 and Fig. 7 caption), so the rate per band photon is an effective cross section $\sigma_{\text{LW}} = 3.452 \times 10^{-18} \text{ cm}^2$. With the mean photon energy of a flat- F_λ band, $2hc/(912+1110 \text{ Å}) = 12.2635 \text{ eV}$,

$$k_{\text{LW,thin}} = 1.757 \times 10^{-7} F_{\text{LW}} \quad [\text{s}^{-1}, F_{\text{LW}} \text{ in } \text{erg cm}^{-2} \text{ s}^{-1}]. \quad (17)$$

Their Table 1 was reproduced numerically from their eqs. (22)–(24) to confirm the flux convention (Habing 1.2220×10^7 against their 1.222×10^7 ; ν^{-2} 1.2084×10^7 against 1.208×10^7 ; Draine 1978 1.2313×10^7 against 1.232×10^7). The band-shape sensitivity is $\pm 25\%$: the same arithmetic for the soft Draine 1978 field gives $2.685 \times 10^{-18} \text{ cm}^2$.

Self-shielding is their eq. (37), the fit to the full multiline calculation *including line overlap*, valid over $10^{14} < N_{\text{H}_2} < 3 \times 10^{21} \text{ cm}^{-2}$, with b the thermal H_2 Doppler parameter $(2kT/m_{\text{H}_2})^{1/2}$. N_{H_2} is the star-ward column, built by the same `calc_column_dens_one` radial integration and `opa_pf` weighting as every other absorber column, from the incoming (pre-solve) H_2 density. There is no dust (EXHALE’s metals are atomic and trace), the trace-metal continuum in the band is $\tau \sim 4 \times 10^{-23} N_{\text{H}}$ and negligible where the shielding matters, and the H Lyman-series lines inside the band are inside DB96’s own fit and not treated separately.

Chemistry and heating. $\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}$ enters the H_2 balance row of the coupled system next to the H_2 photoionization (`mol_heh_rows`, shared by `System_HeH_mol` and `System_HeH_mol_metals`), so it is solved with everything else. Both products are neutral H, which the H-nucleus closure supplies; no other row changes. The fragments carry 0.4 eV of kinetic energy (Black & Dalgarno 1977, ApJS 34, 405, p. 418, “the yield is about 0.4 eV per atom pair”), added to heat. The 4.48 eV bond energy is paid by the photon, not by the gas, and is not a sink of this channel. `output/Lyman_Werner.txt` (r , T , x_{H_2} , n_{H_2} , N_{H_2} , f_{shield} , k_{LW} , heating) is written whenever a molecular run carries a band flux.

Band flux for a planet. Integrated externally from the VULCAN stellar spectra in the tree (EXHALE/VULCAN/atm/stellar_flux/, stellar surface) over 91.2–111.0 nm and diluted by $(R_*/a)^2$. The `VULCAN_run_*` spectra directories are empty since the 2026-08-09 loss, so no HD 209458 spectrum exists here and the solar spectrum stands in: $343 \text{ erg cm}^{-2} \text{ s}^{-1}$ at 0.048 AU behind a $1.155 R_\odot$ star, $k_{\text{LW,thin}} = 6.03 \times 10^{-5} \text{ s}^{-1}$. The same file returns a solar Ly α irradiance of $6.12 \text{ erg cm}^{-2} \text{ s}^{-1}$ at 1 AU against an observed 6–8, which is the calibration check. HD 189733 b comes out at 600 (Moses11 spectrum) or 1830 (Bourrier 2020) $\text{erg cm}^{-2} \text{ s}^{-1}$ — a factor 3 that dwarfs the $\pm 25\%$ of the band shape.

Effect. A/B on the key alone, hot-Uranus gate (metals off, the `mol_base_handoff` configuration restarted with Solver: Newton 5.0e-2, Resid tol: 1.0e-5, Base IR field: True), both returning `info = 0`:

	LW off	LW on (343)
residual norm at exit	8.15e-6	4.86e-6
H ₂ → H front ($x_{\text{H}_2} = 0.5$)	1.08806	1.08622
x_{H_2} at the base	0.99921	0.99883
x_{H_2} at $r = 1.10$	2.54e-2	1.51e-4
H ₃ ⁺ peak [cm ⁻³]	3.19e5 @ 1.0079	3.20e5 @ 1.0025
$\log_{10} \dot{M}$ [g/s], spherical	10.6196	10.6248

and on HD 209458 b (examples/15_molecular + Base IR field: True, both members at Resid tol: 2.0×10^{-5} because the band-on run stalls the JFNK at 1.5×10^{-5} on the base-adjacent momentum row, item (A) of TO_BE_DONE.md; both info = 0, both mass fluxes flat to 4.1×10^{-3}): front 1.01578 → 1.01290, x_{H_2} at $r = 1.02$ $2.37 \times 10^{-4} \rightarrow 4.22 \times 10^{-6}$, $\log_{10} \dot{M}$ 10.5711 → 10.5600.

The self-shielding does exactly what it should: at the hot-Uranus base $N_{\text{H}_2} = 4.3 \times 10^{21} \text{ cm}^{-2}$ gives $f_{\text{shield}} = 9.8 \times 10^{-7}$ and $k_{\text{LW}} = 5.9 \times 10^{-11} \text{ s}^{-1}$, while above $r = 1.15$ the column has fallen below 10^{10} cm^{-2} , $f_{\text{shield}} = 1.000$ and k_{LW} recovers the unshielded $6.026 \times 10^{-5} \text{ s}^{-1}$.

It does not close the base-composition gap, and the measurement says why. The base q_{H_2} moves from 0.8618 to 0.8612 against the 0.75 the case's base.inp carries — 0.5% of the gap — even though Lyman–Werner becomes the largest single H₂ loss at the base ($5.9 \times 10^{-11} \text{ s}^{-1}$ against 2.5×10^{-11} for H⁺ + H₂). The two facts are consistent because the base partition is a formation–destruction balance in which the three-body R15 sets $n_{\text{H}_2} \propto n_{\text{H}}^2$, so $n_{\text{H}}/n_{\text{H}_2}$ scales as the square root of the destruction rate: the measured $2.3\times$ rise in destruction raises the atomic fraction only $1.5\times$. Reaching $q_{\text{H}_2} = 0.75$ needs a destruction rate $4100\times$ larger. The unshielded band would supply it many times over; the column stops it. The extra atomic H photochemical codes find at $1 \mu\text{bar}$ comes from catalytic cycles on O, OH and H₂O, and EXHALE's H₂/H₂⁺/H₃⁺/HeH⁺ network has nowhere to put them. Route 1 (q_H2_base) therefore stays the only way EXHALE can carry a photochemical base partition, and is now a documented modeling choice rather than a patch over a missing term.

make check is byte-identical on all four cases with the key absent.

Files, by directory under src/modules/: lower_atmosphere/ — new lyman_werner.f90, mol_rates.f90; init/ — parameters.f90; files_IO/ — input_read.f90, write_setup_report.f90, write_output.f90; nonlinear_system_solver/ — ion_cell_state.f90, System_HeH_mol.f90, System_HeH_mol_metals.f90; radiation/ — ionization_equilibrium.f90, util_ion_eq.f90; plus Makefile. Documentation: docs/lower_atmosphere_coupling.md, docs/base_composition_handoff_plan.md, docs/EXHALE_user_manual.tex, docs/input_schema.md, README.md, README_HOWTO.md, examples/exhale_io.py.

57. A fifth regression case for Lyman–Werner photodissociation (2026-08-13)

Nothing in the four-case matrix exercised the code the section 56. key turns on: with Stellar LW flux absent every case leaves $F_{\text{LW_star}} = 0$, so the star-ward H_2 column, the Draine & Bertoldi self-shielding factor, the LW loss row in the H_2 budget, the 0.4 eV fragment heating and the Lyman_Werner.txt writer all ran in no case.

backup/regression/mol_lyman_werner is that case: a copy of mol_base_handoff (the hot-Uranus Tier-2 gate, Molecular chemistry: True, the same base.inp handoff, 12000 steps pinned in maxsteps) plus the single line Stellar LW flux [erg/cm2/s]: 343.0 — the HD 209458 b solar-proxy band flux of the preceding section. It stops at $du = 2.1841$, $dtu = 4.8072 \times 10^{-3}$, clearly apart from the LW-off mol_base_handoff ($du = 2.6274$, $dtu = 5.7385 \times 10^{-3}$): the case does pass through the new code path. DEFAULT_CASES in backup/regression/run_check.sh is now wasp_full wasp_he23off mol_base_handoff mol_metals mol_lyman_werner, and the script header describes what the case guards.

The golden was taken in the order the fourth-case section's trap requires — check (which rebuilds and re-runs the case), then golden, then the full check again — and the final run is byte-identical on all five cases.

Files: backup/regression/run_check.sh, backup/regression/mol_lyman_werner/ (new), backup/regression/golden/mol_lyman_werner/ (new). Documentation: the workspace CLAUDE.md, README_HOWTO.md.

58. JFNK convergence is judged on the iterate the solver returns; load_IC no longer accepts an all-zero metal column (2026-08-13)

Two things came out of trying to converge HD 209458 b with molecular chemistry AND solar C/N/O (examples/16_molecular_metals), which fails with the prescription that works for its metals-off twin examples/15_molecular.

The diagnosis is in docs/hd209_metal_stagnation.md; what follows is what changed in the code, and one thing that was measured and deliberately *not* changed.

The measurement. solve_steady_jfnk accepts a step on the whole-domain scaled merit $\|F/D\|_2$ over cells $1 \dots N$ and decides info on $\|R\|$, the volume-weighted relative residual over the escape region $[j_{\min} : N]$ alone. On this configuration the two move in opposite directions. Warm-restarted from its own stalled state, the solve reached $\|R\| = 3.958 \times 10^{-5}$ at outer iteration 2 and then, over 115 further iterations, cut the merit from 5.67×10^{-1} to 1.58×10^{-2} — a factor 36 — while $\|R\|$ rose to 9.5×10^{-5} . The reason is a shell just above the base that stops flowing once the metal cooling is on: at $r = 1.02 R_p$ cooling exceeds heating by a factor 9.7 with C/N/O against heating exceeding cooling by a factor 33 without them, the mean speed over $r = 1.015\text{--}1.030$ falls from 207 to 6–24 cm s^{-1} , and an undamped $2dr$ contact mode grows there whose amplitude reaches several times the mean velocity. That shell dominates the whole-domain merit and is invisible to $\|R\|$.

Aligning the merit's region was tried and rejected, for the second time. Replacing $\|F/D\|_2$ over $1 \dots N$ by the volume-weighted RMS of F/D over $[j_{\min}:N]$ breaks solves that converge today: examples/15 goes from `info = 0` at $\|R\| = 9.542 \times 10^{-6}$ in 272 outer iterations to `info = 2` with a best of 1.171×10^{-4} in 30, and the HD 209458 b base.inp continuation photo_deep_secion_cont from `info = 0` at 7.284×10^{-4} in 4 iterations to `info = 2` at 3.249×10^{-3} in 15. This reproduces on two more cases the measurement that rejected the same change on 2026-08-11 (docs/newton_scaling_and_base_wall.md §4), and for the reason given there: the Newton step is computed from the full residual over all rows, so a merit that ignores most of those rows rejects the steps that step takes. The merit stays whole-domain, and the code now says so at the site with the numbers.

What did change: info is decided on the iterate that is returned. The solver already tracked the best iterate by $\|R\|$, already returned it when the solve failed, and already tested $\|R\| < \text{resid_tol}$ at the top of every outer iteration — so an iterate that satisfies the tolerance is normally caught the moment it is produced. Two gaps remained. The restore of the best iterate was gated on `info  $\neq$  0`, and the tolerance was never tested after the loop, so a solve whose last allowed iteration met the tolerance fell out with `info = 1`. Both close with one line, `if (rnorm < resid_tol) info = 0` after the loop, with the best-iterate restore made unconditional. It matters because EXHALE_main treats `info  $\neq$  0` as a failure, disables the Newton finish for the rest of the run, and falls back to a du-stopped march — discarding a state that satisfied the convergence criterion.

Scope, measured, not inferred: this changes no trajectory, and it is not what unlocks examples/16. The stalled examples/16 warm restart was re-run at `Resid tol: 5.0e-5` with the binary before and the binary after: both hand off at marching step 9363, both take 2 outer iterations, both print the same two iterate lines to every digit, and both end `info = 0` at $\|R\| = 3.958 \times 10^{-5}$. The examples/16 failures at `Resid tol: 1.0e-5` are therefore not a bookkeeping artifact either — that configuration genuinely does not reach 10^{-5} , for the reasons in the memo. What the fix removes is the remaining way for a returned state that satisfies the tolerance to be labelled a failure.

load_IC no longer accepts a metal column that is present but zero. It decided whether the restart file carried an element from the presence of its columns alone. The schema-2 writer emits the metal columns unconditionally, so a metals-OFF output carries them with the value zero; a metals-ON restart from such a file therefore ran with **zero metal density everywhere** while the base boundary condition still counted metals in the mass and particle budget under eos_metals. It was silent, and it produced confident nonsense: restarting the converged examples/15 (metals off) solution as a metals-on run reported $\|R\| = 9.54 \times 10^{-6}$ — it was reproducing the metals-off solution under a metals-on label. An element is now taken from the file only if every stage has a column AND the loaded stages are not identically zero everywhere; otherwise it is built from the abundance exactly as `set_IC` does, $n_X = \text{melem_ab} \cdot n_H$ all in the neutral stage, and the substitution is reported on stdout and in EXHALE_setup.out. The rebuilt density is inserted before `calc_rho`, so the restart keeps the mass closure $\sum_i f_i A_i = 1$ that a cold start has by construction; for the same reason the legacy headerless-file path now carries its metal mass in ρ as well. *Measured* on that restart: the three elements are reported, the base density at $j = 1$ goes from 2.0618×10^{14} to $2.0804 \times 10^{14} \text{ m}_H \text{ cm}^{-3}$ (0.9 %, the metal mass), and the reported $\|R\|$ goes from the false 9.54×10^{-6} to 7.36×10^{-2} — the honest distance of a metals-off wind from the metals-on steady state. A metals-off restart, and a restart whose file does carry metal densities, both give a

byte-identical residual profile to before.

Gates. make check is 5/5 byte-identical (no regression case reaches the JFNK hand-off and none uses Load IC, so neither change can touch them). examples/15 re-run from cold with the changed binary reproduces its reference output **byte-identically** — info = 0, $\|R\| = 9.542 \times 10^{-6}$, 272 outer iterations, 25837 marching steps, $\log_{10} \dot{M} = 10.23$, $H_2 = H\,I$ front at $1.0084 R_p$. The hot-Uranus molecular + solar-metals configuration (backup/regression/mol_metals run to the Newton finish instead of its 12000-step regression snapshot) converges: info = 0, $\|R\| = 3.342 \times 10^{-6}$ in 23 outer iterations, 34537 marching steps, $\log_{10} \dot{M} = 10.30$. photo_deep_secion_cont converges: info = 0, $\|R\| = 7.284 \times 10^{-4}$ in 4 iterations, $\log_{10} \dot{M} = 9.69$.

And examples/16 itself now has a Newton-grade solution on the warm path. Restarted from the converged examples/15 state — exactly the restart the load_IC fix repairs — with Resid tol: $5.0e-5$ it converges: info = 0, $\|R\| = 3.337 \times 10^{-5}$ in 30 outer iterations, $\log_{10} \dot{M} = 9.65$. At 10^{-5} it does not, and the cold path converges at neither tolerance — 112920 marching steps to the hand-off against 25837 for the metals-off twin, then 281 outer iterations to a best $\|R\|$ of 3.623×10^{-4} with the worst scaled residual at $r = 1.021$, inside the stagnant shell. Both remain open, in docs/hd209_metal_stagnation.md.

Files: src/modules/time_step/steady_newton.f90, src/modules/files_I0/load_IC.f90, src/modules/files_I0/write_setup_report.f90. Documentation: docs/hd209_metal_stagnation.md (new), TO_BE_DONE.md item (A).

59. The molecular equilibrium no longer inherits a cell's previous composition (2026-08-13)

ioniz_eq solves the ionization equilibrium — and, under Molecular chemistry: True, the dissociation equilibrium with it — cell by cell, and the composition it returns is what the cooling, the photoionization columns and the steady residual are all built on. When the coupled molecular network produced no root inside the physical simplex in a cell, it used to hand that cell its own *incoming* fractions back and warn (molecular equilibrium failed at N cells (kept previous state)).

That is wrong on its own terms. The composition returned is then not the equilibrium of the cell at the state being evaluated; it is whatever the previous evaluation happened to leave there, and the error is not small. In the JFNK finish it also means that the residual changes definition between outer iterations: f_sp is overwritten by each accepted trial, so the map the next iteration differentiates and line-searches differs from the previous one, by a finite amount, at exactly those cells (docs/hd209_metal_stagnation.md section 3, where the same section corrects an earlier claim — within one outer iteration the Jacobian probes and the line-search trials do all start from the same composition).

What the cells were failing on. Measured with a counting build on the HD 209458 b molecular + solar C/N/O warm restart (40 marching steps, single-threaded): 224 cell solves ended below the solver tolerance, 223 of them keeping a physical but unconverged root and 1 inheriting its previous composition. The rejected roots were not on another branch. At step 1, cell $j = 117$ ($r = 1.0294$, $T = 1652$ K), the first attempt converged (info = 1) to a root violating the simplex

only by $x(\text{He II}) = -3.8 \times 10^{-6}$ and $x(\text{H}_2) = -4.5 \times 10^{-4}$, and the molecular-basin retry returned a strongly molecular root, $x(\text{H}_2) = 0.998$, violating it only by $x(\text{He II}) = -1.2 \times 10^{-8}$ — round-off around a fully neutral helium. What fails is a root resolved onto a *face* of the simplex, not an unphysical branch.

Two changes, both in `ionization_equilibrium.f90`.

1. *The molecular-basin retry seed is now a state.* It used to combine the chemical-equilibrium H_2 fraction at the local (p, T) with the **previous state's** H II, He II, He III and 2^3S fractions — a pair that can put more H nuclei into H_2 and H II together than the cell has, i.e. a starting point that is not a composition at all. The new seed, `dissociation_ionization_balance_at_fixed_ne`, is the limit of the network in which the couplings to the molecular ions are dropped: the H nuclei split between H_2 and atomic H by the same Koskinen et al. (2022) Eq. 11 mixing ratio the molecular base boundary condition uses, the atomic remainder and every other element in their own ionization balance at the incoming n_e , $\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ at zero. It is inside the simplex by construction and depends only on the local state.
2. *A cell whose roots all left the simplex keeps the closest one, clamped.* `element_budget_violation` measures how far a root lies outside the allowed states (the largest negative stage fraction, and the largest amount by which one element's tracked stages exceed its nuclei); the least-offending root is kept and `clamp_fractions_to_element_budget` moves it onto the boundary — negative fractions to zero, and any element still over its nuclei scaled to sum exactly to them. Given the measurement above this is the treatment of a root the solver has resolved onto a face and delivered as a small number of either sign: the clamped state is that root, to the accuracy the solve reached. The cell's previous composition is never used.

The warning is now every molecular equilibrium root left the physical simplex at N cell(s), clamped onto the element budget, and the run-wide count is reported at the end of the run next to the `ioniz-eq roots` line. A cell that reaches the solver tolerance on a physical root takes exactly the path it took before, so the change is inert wherever the solve is healthy.

One thing was tried and reverted. `ionization_fractions_physical` rejects a root for a component below -10^{-10} , and the rejected roots above sit at -10^{-8} , inside the solver's own $\text{xtol} = \sqrt{\epsilon_{\text{mach}}}$. Widening the test to that band is defensible on paper and worse in practice: it lets a cell stop on a root just outside a face instead of retrying from another starting point, and the retried root is the better one. With the band widened, `examples/15` goes from `info = 0` at $\|R\| = 6.105 \times 10^{-6}$ in 89 outer iterations to `info = 2` at 1.112×10^{-4} in 133. The band stays at 10^{-10} and the measurement is recorded at the site.

Gates. `make check` is 4/5 byte-identical. The fifth, `mol_metals`, moves: one cell in the 12000-step run takes a retry it did not take before (`ioniz-eq roots` restarts $9 \rightarrow 10$), and the difference that survives to the output is at round-off — at most 1.3×10^{-11} relative in every `Hydro_ioniz.txt` column and in every H, He and metal density, 1.1×10^{-7} in H_2 and H_2^+ , and 2.3×10^{-4} in H_3^+ in the far wind where H_3^+ is 10^{-17} cm^{-3} against a peak of 52. $\log_{10} \dot{M}$ is 10.58 before and after. The golden was re-snapshotted for that case only, in the order `check → golden → check`. No molecular regression case reaches the clamp (0 cell(s) in all three).

examples/15 (HD 209458 b, molecular, cold) gets *better*, not merely different: $\text{info} = 0$ at $\|R\| = 6.105 \times 10^{-6}$ in 89 outer iterations against $\text{info} = 0$ at 9.542×10^{-6} in 272, hand-off at marching step 25839 against 25837, $\log_{10} \dot{M} = 10.24$ against 10.23, $H_2 = HI$ front at $1.0098 R_p$ against 1.0084. The hot-Uranus molecular + solar-metals configuration holds: $\text{info} = 0$ at $\|R\| = 5.826 \times 10^{-6}$ in 23 outer iterations against 3.342×10^{-6} in 23, same 34537 marching steps, $\log_{10} \dot{M} = 10.30$ unchanged. Runtime checks (`-O1 -fcheck=bounds,do,mem`) are clean on `mol_metals`, `mol_base_handoff` and on the warm restart that does exercise the clamp.

examples/16 (HD 209458 b, molecular AND solar C/N/O), the case the memo is about, does **not** improve — it gets worse. The warm restart that reached $\text{info} = 0$ at $\|R\| = 3.337 \times 10^{-5}$ in 30 outer iterations now bottoms at 7.931×10^{-5} in 67 and reports $\text{info} = 2$, identically at `Resid tol` 5×10^{-5} , 2×10^{-5} and 10^{-5} ; the cold path bottoms at 1.081×10^{-3} against 5.038×10^{-4} before. Each half of the change on its own converges that restart — retry seed alone at 3.059×10^{-5} , clamp alone at 4.853×10^{-5} — and each half on its own fails examples/15, which the two together converge in a third of the iterations it used to take. Both HD 209458 b cases are marginal solves and an $O(1)$ change to a handful of cells' composition moves them either way; what does not move is that their worst scaled residual sits in the stagnant shell at $r = 1.021$ throughout (docs/hd209_metal_stagnation.md section 6). Removing the inheritance is justified on its own terms, not by what it does to these two solves.

Files: `src/modules/radiation/ionization_equilibrium.f90`, `src/EXHALE_main.f90`. Documentation: docs/hd209_metal_stagnation.md sections 3 and 5, TO_BE_DONE.md item (A).

60. Damping the stagnant layer: a gated fourth difference inside the numerical flux (Low-Mach damping) (2026-08-13)

The HLLC flux resolves the middle wave of the Riemann fan exactly. The jump that wave carries — the entropy jump, at constant pressure and velocity — moves at the contact speed $S^* \simeq v$, and the dissipation the solver applies to it is proportional to $|S^*|$. It therefore vanishes as the flow stagnates. The acoustic families keep their damping at $|v| \pm c_s$, so a stationary cell-to-cell entropy pattern is a discretely undamped mode of the scheme, and in a near-hydrostatic layer the velocity field is tied to it by continuity and by the force balance, so it carries the same pattern.

HD 209458 b with molecular chemistry and solar C/N/O is where that stops being academic. The metal cooling exceeds the photoionization heating by a factor ~ 10 at $r \simeq 1.02 R_p$ and stops the flow: the mean $|v|$ over $r = 1.015$ – 1.030 falls to 6 – 24 cm s^{-1} against $\sim 207 \text{ cm s}^{-1}$ in the metals-off twin ($M \sim 3 \times 10^{-5}$ against a sound speed of $\sim 4.5 \text{ km s}^{-1}$), and the alternating component of v grows to 4.7–10 times the local mean $|v|$ where the metals-off solution carries 0.001. Every failed steady solve of that configuration puts its worst scaled residual in that shell (docs/hd209_metal_stagnation.md sections 1 and 6).

The term. Low-Mach damping: `<eps> [M_th]` adds a fourth-difference (Jameson, Schmidt & Turkel 1981) stress to the numerical **momentum** flux at each interior face,

$$D_p^{j+1/2} = \varepsilon_4 g(M) \rho_f \lambda_f (v_{j+2} - 3v_{j+1} + 3v_j - v_{j-1}), \quad \lambda_f = \frac{1}{2} [(|v| + c_s)_j + (|v| + c_s)_{j+1}], \quad (18)$$

and, because a stress does work, $D_E^{j+1/2} = v_f D_p^{j+1/2}$ to the **total energy** flux — the same pairing viscous_conduction uses for the physical Navier–Stokes stress. Without it the kinetic energy the stress removes would come out of the internal energy instead of being converted into it. With it the pair conserves total energy exactly, and the internal-energy change is $-D_p dv/dr$, which has either sign cell by cell but integrates to $+\int \mu_4 (d^2 v/dr^2)^2 dr \geq 0$: kinetic energy into heat, never the reverse. The undivided third difference is $\Delta r^3 d^3 v/dr^3 + O(\Delta r^4)$, so the stress is $\tau = -\mu_4 d^3 v/dr^3$ with $\mu_4 = \varepsilon_4 g \rho c_s \Delta r^3$ — the acoustic momentum diffusivity $\rho c_s \Delta r$ an upwind scheme already carries, reapplied at fourth order.

The mass flux is not touched, and neither is the energy flux other than through the work term. The plain JST form, a fourth difference on the conserved variables themselves, is inadmissible here and was measured to be so: the physical mass flux ρv and energy flux $v(E+p)$ both vanish with the velocity while $\Delta r^3 d^3 \rho/dr^3$ and $\Delta r^3 d^3 E/dr^3$ do not, because ρ and E are stratified over a barely resolved scale height. Their dissipative divergences come out 10^4 – 10^5 times the physical mass-flux divergence and $\sim 5 \times 10^3$ times the radiative source — they would rewrite the base velocity and thermal structure rather than damp an oscillation. The momentum flux $\rho v^2 + p$ does not vanish at stagnation (it tends to p), which is why the momentum form stays small there.

The gate.

$$g = [\max(0, 1 - M_f^2/M_{th}^2)]^2, \quad M_f^2 = \frac{1}{2} (M_j^2 + M_{j+1}^2) \quad (19)$$

is 1 at $M = 0$, exactly 0 for $M_f \geq M_{th}$, and C^1 at the threshold so the steady residual stays differentiable for the Newton solve. It is written in M^2 rather than $|v|/c_s$ because $|v|$ has a kink at $v = 0$ and the stagnant layer is exactly where v changes sign.

Discrete consistency is the point of the design. The stress is added inside RK_rhs, the one routine the marching loop and assemble_residual — hence the JFNK steady solver — both call. The equation the Newton residual measures is the equation the marching loop relaxes, and the fixed point of one is the zero of the other. This is exactly what the Shapiro filter lacks: it is applied to the marching state only, so the Newton residual never sees it and the mode it suppresses during marching is still an undamped mode of the system Newton solves.

Both fluxes are set to zero at the base face ($j = 0$) and the outer face ($j = N$), so nothing is injected or removed through the boundaries; those are also the faces whose four-cell stencil would reach outside the ghost layer. Cell j then depends on cells $j-2 \dots j+2$, which is exactly the stencil the WENO3 residual already has (WL(:, j) reads cells $j-1 \dots j+1$, WR(:, j) cells $j \dots j+2$), so the banded Jacobian bandwidth kl_jac = ku_jac = 8 is unchanged.

Stability. On the $2\Delta r$ mode $v_j = (-1)^j a$ the fourth difference is $16a$, so the mode decays at $16\varepsilon_4 \lambda/\Delta r$; the explicit step is $\Delta t = \text{CFL} \Delta r/\lambda$, so it is damped by $16\varepsilon_4 \text{CFL}$ per step and explicit stability requires $\varepsilon_4 < 1/(16\text{CFL})$ — 0.10 at the default CFL = 0.6. input_read warns when the bound is violated. The classical JST range $1/64$ – $1/32$ sits inside it.

How big the term actually gets. A numerical dissipation is admissible only where it is negligible against the physical fluxes, and neither the fourth difference nor the gate is a bound on that by itself, so any run that uses the key now reports

it: `contact_mode_dissipation_magnitude` prints the peak $|D_p|/|\rho v^2 + p|$ over the interior faces, where it occurs, and the outermost face at which the gate is still open. Measured on the converged profiles (recomputing the same quantities in cgs from `Hydro_ioniz.txt`, independently of the Fortran, agrees to the printed digits):

run	ε_4	peak $ D_p / \rho v^2 + p $	at	gate open to	first $M > 0.1$
examples/15 (metals off)	2×10^{-2}	1.83×10^{-4}	1.0005	$1.043 R_p$	$1.76 R_p$
examples/16 warm	1×10^{-2}	8.36×10^{-5}	1.0005	$1.108 R_p$	$1.99 R_p$
examples/16 warm	5×10^{-3}	3.94×10^{-5}	1.0005	$1.105 R_p$	$1.99 R_p$
hot Uranus (mol. + metals)	2×10^{-2}	3.70×10^{-5}	1.0005	$1.083 R_p$	—

The peak is at the first interior face, where the base boundary condition forces the steepest velocity gradient in the domain, and it is 0.02% or less of the physical momentum flux there; through the stagnant shell itself it is 10^{-7} or below. The first cell with $M > 0.1$ is at $1.76\text{--}2.00 R_p$, so the wind, the $r_{\text{esc}} = 2 R_p$ window over which $\|R\|$ is measured and the $N - 20$ cell at which $\dot{M}$ is evaluated all sit where the term is identically zero.

Gates. make check is **5/5 byte-identical** (wasp_full, wasp_he23off, mol_base_handoff, mol_metals, mol_lyman_werner); no golden was touched, and none should have been, since none of those inputs carries the key. A -01 -fcheck=bounds,do,mem build is clean on two key-ON cases (the HD 209458 b molecular + C/N/O warm restart, which exercises the open gate, and the hot-Uranus cold start, which exercises the closed one).

What it does to the case it was written for. HD 209458 b with molecular chemistry *and* solar C/N/O, warm-started from the converged examples/15 state, Resid tol: $1.0\text{e-}5$, Solver: Newton $5.0\text{e-}2$, Max steps: 150000, same binary and same restart files in every row:

ε_4	hand-off step	info	best $\ R\ $	outer it	$\log_{10} \dot{M}$	$\text{H}_2 = \text{H I front}$
off	48060	1	5.216×10^{-5}	500 (cap)	—	—
5.0×10^{-3}	48062	0	9.934×10^{-6}	162	9.65	1.0112
1.0×10^{-2}	48007	0	9.987×10^{-6}	202	9.67	1.0117
2.0×10^{-2}	65836	2	2.529×10^{-3}	237	—	—
4.0×10^{-2}	66015	2	2.870×10^{-3}	66	—	—

This is the first info = 0 at Resid tol: $1.0\text{e-}5$ this configuration has produced. The two converged rows stop at 9.9×10^{-6} because the solver tests the tolerance at the top of each outer iteration and exits on the first iterate below it, so those are first crossings and not a knife edge; the key-off control, from the same restart, spent its whole 500-iteration budget and plateaued at 5.2×10^{-5} . Nothing diverges in the rows that do not converge: each falls back to time-marching and runs out the 150000-step cap at a du plateau of 1.0×10^{-3} to 2.6×10^{-3} , $\log_{10} \dot{M}$ 9.58–9.66. Those are marching states, not Newton-grade ones, which is why the table leaves their $\dot{M}$ blank.

The coefficient has a working range, and it is the bottom of the classical one. At $\varepsilon_4 \geq 2 \times 10^{-2}$ this solve is *worse*, and the reason shows up before the Newton phase: the stress changes the base solution enough to slow the du descent (du at step 40000 is 0.139 with the key off and 0.360 at $\varepsilon_4 = 2 \times 10^{-2}$), so the hand-off comes 18000 steps later from a

different state. The key is opt-in and the pair has to be run on any new configuration; 2×10^{-2} is not a recommended default, and there is no default.

Sensitivity, honestly. Halving ε_4 from 10^{-2} to 5×10^{-3} moves $\log_{10} \dot{M}$ by 0.02 and the $\text{H}_2 = \text{HI}$ front by $0.0005 R_p$, and in the wind ($r > 1.2 R_p$) ρ by $\leq 7.4\%$, v by $\leq 3.7\%$ and T by $\leq 0.9\%$. Inside the stagnant layer the two converged states differ a great deal (ρ by 137% at $r = 1.013$, T by 38% at $r = 1.015$) and differ in what the layer looks like: the alternating component of v over $r = 1.015\text{--}1.030$ is 0.024 at $\varepsilon_4 = 10^{-2}$ and 4.29 at 5×10^{-3} . Both satisfy $\|R\| < 10^{-5}$ on the wind window, which is the point — the convergence measure does not look at that layer, so it cannot pin it down.

The configurations that already converge do not move. Cold starts, Resid tol: $1.0\text{e-}5$, $\varepsilon_4 = 2 \times 10^{-2}$ where on:

case	key	hand-off	info	$\ R\ $	outer it	$\log_{10} \dot{M}$	front
examples/15	off	25839	0	6.105×10^{-6}	89	10.24	1.0098
examples/15	on	25839	0	3.642×10^{-6}	100	10.24	1.0095
hot Uranus	off	34537	0	5.826×10^{-6}	23	10.30	1.0774
hot Uranus	on	34537	0	6.252×10^{-6}	23	10.30	1.0774

Both hand off at the same marching step and reach the same $\dot{M}$; on the hot Uranus the entire profile moves by at most 7.3×10^{-4} relative in ρ and 7.1×10^{-4} in T . On examples/15 the wind above $1.2 R_p$ moves by at most 1.2% in ρ , 0.5% in v and 0.16% in T ; the first cells above the base move more (ρ up to 16%, T 6.9% below $1.05 R_p$), and those are cells that carry an odd-even velocity oscillation 86 times the local mean with the key *off*. The cold examples/16 start (Resid tol: $5.0\text{e-}5$) improves and still fails: info = 1 at 9.307×10^{-4} with $\varepsilon_4 = 2 \times 10^{-2}$ against 1.081×10^{-3} with the key off, from the identical hand-off step 112920 — the gate is closed through the cold relaxation, when the flow is still fast everywhere, and the term only acts at the end.

Files: src/modules/flux/low_mach_dissipation.f90 (new), src/modules/time_step/RK_rhs.f90, src/modules/init/parameters.f90, src/modules/files_IO/input_read.f90, src/modules/files_IO/write_setup_report.f90, src/EXHALE_main.f90, Makefile. Documentation: docs/hd209_metal_stagnation.md sections 7 and 8, docs/EXHALE_user_manual.tex, docs/input_schema.md K31b, README.md, TO_BE_DONE.md item (A).

61. The residual below the escape radius, and what closes the HD 209458 b molecular + metals case (2026-08-15)

Section 60 left the examples/16 configuration converging from a warm restart at two damping coefficients and failing from a cold one, and left open whether the acceptance window should be widened to include the layer the failures live in. Both questions, and the seed the warm prescription depends on, were measured on 2026-08-15. The full account is docs/hd209_metal_stagnation.md section 9; this section records the code change and the conclusions.

The change: the layer's residual is printed, not tested. resid_relnorm_below_escape and write_resid_below_escape in src/modules/time_step/steady_newton.f90 apply the **same** volume-weighted

relative norm that `resid_relnorm` uses for the convergence measure to the complementary region `[1:j_min-1]`, i.e. below the escape radius, and locate the cell carrying the largest cell-wise relative residual $|F_{k,j}|/|u_{k,j}|$ there, reporting j , $r(j)$ and which of the three rows it is. One line is printed at four points: PTC start, PTC end, JFNK start, JFNK end. The JFNK end line is emitted *before* the best-iterate restore, because that is the last point at which F and u are a consistent pair; it therefore refers to the last iterate visited, which is also the returned state whenever no restore happens.

The value is never tested and nothing in the solve depends on it. The whole change is **+95 lines in that one file**, output only, and `make check` is **5/5 byte-identical** (`wasp_full`, `wasp_he23off`, `mol_base_handoff`, `mol_metals`, `mol_lyman_werner`) — no output file moves, only stdout grows.

Widening the acceptance window is rejected by what it prints. With the diagnostic in place the question is a measurement rather than a design argument. The norm over `[1:401]` ($j_{\min} = 402$):

state	norm below r_{esc}	worst cell
examples/16 converged, $\varepsilon_4 = 5 \times 10^{-3}$	0.195	$j = 75, r = 1.0153$
examples/16 converged, $\varepsilon_4 = 1 \times 10^{-2}$	2.352	$j = 5, r = 1.0010$
examples/16 converged, $5 \times 10^{-3} + \text{LW}$	0.939	$j = 3$
examples/15 converged (metals off)	9.99	$j = 2$
warm restart, at JFNK start	0.60–1.21	$j = 6\text{--}15$, momentum
cold $\varepsilon_4 = 5 \times 10^{-3}$, at failure	594	$j = 97, r = 1.0218$
cold $\varepsilon_4 = 1 \times 10^{-2}$, at failure	1599	$j = 112, r = 1.0273$

A converged state carries a relative residual of 0.2 to 10 below the escape radius, four to six orders above `Resid tol`, and the largest of them belongs to the metals-off `examples/15` solution that nobody doubts. That is not about metals: the layer is near-hydrostatic ($|v| \sim 10 \text{ cm s}^{-1}$), so the denominator of the momentum row’s relative residual, $|\rho v|$, collapses wherever the flow is slow, right state or not. Widening the window on this norm would report `info = 2` for every configuration that converges today without determining the layer any better. The window stays `[j_min:N]`; the diagnostic makes the uncontrolled part visible instead. The failure rows sit three orders above the converged ones, so the number does separate good from bad — it is the absolute scale that is not usable as a tolerance.

The cold path is closed. Scanning Low-Mach damping on the cold start gives `info = 1` at 1.081×10^{-3} (off), 9.953×10^{-4} (5×10^{-3}), 1.025×10^{-3} (1×10^{-2}) and 9.307×10^{-4} (2×10^{-2}) — one $\sim 10^{-3}$ floor, a factor 100 above the tolerance, and the two coefficients that work on the warm restart do nothing here that off does not. The cold hand-off state appears to lie outside the Newton basin the warm restart reaches, and the coefficient is not the lever. The cold path for this configuration is abandoned.

The warm prescription is pinned to its seed. It needs one thing section 60 did not say: the restart must come from an `examples/15` state converged with the **current** binary. Seed A is the copy `examples/15_molecular/output` held until 2026-08-15, written before the section 59 change (kept as `output_pre_item3_20260813`); seed B is `examples/15 re-`

converged with the current binary, which the example directory holds since 2026-08-15. Same binary and same keys otherwise:

seed	ε_4	hand-off	info	best $\ R\ $	$\log_{10} \dot{M}$
A (stored)	5.0×10^{-3}	50645	2	1.303×10^{-4}	—
A (stored)	1.0×10^{-2}	50629	2	5.770×10^{-5}	—
A + LW 343	5.0×10^{-3}	50189	0	9.799×10^{-6}	9.68
B (current)	5.0×10^{-3}	48062	0	9.934×10^{-6}	9.65
B (current)	1.0×10^{-2}	48007	0	9.987×10^{-6}	9.67
B + LW 343	5.0×10^{-3}	47965	0	9.900×10^{-6}	9.68

Seed B reproduces the section 60 table exactly, hand-off step included (the re-convergence that produces it also reproduces the section 60 control: step 25839, info = 0, $\|R\| = 6.105 \times 10^{-6}$, $\log_{10} \dot{M} = 10.24$, front at $1.0097 R_p$). Seed A does not: the hand-off comes ~ 2600 steps later, from a different state, and the line search then collapses ($\lambda \sim 10^{-6}$) and aborts. The section 59 change moved the examples/15 solution by more than this marginal solve tolerates in its starting point, so a restart file written by an older binary is no longer the same seed. The Lyman–Werner member converges from both seeds — the LW coupling appears to widen the basin here rather than narrow it.

What the weak determination of the layer costs. The two converged solutions ($\varepsilon_4 = 5 \times 10^{-3}$ and 1×10^{-2} , both info = 0 at Resid tol: 1.0×10^{-5}) agree in the wind — $\Delta\rho/\rho$ +5.1% at $1.3 R_p$, +3.4% at 1.5, +2.6% at 2.0, $\Delta v/v \leq 1.6\%$, $\Delta T/T \leq 0.4\%$, $\log_{10} \dot{M}$ 9.65 against 9.67 — and disagree inside the layer, where the $10^{-2}/5 \times 10^{-3}$ density ratio runs 0.78 at $r = 1.005$, 1.31 at 1.013, 1.41 at 1.020 and 1.32 at 1.030, T differs by +20% at 1.005, +38% at 1.015 and -7.7% at 1.020, and the alternating component of v over $r = 1.015$ – 1.030 is 0.392 against 0.028 on a mean $|v|$ of 12–16 cm s $^{-1}$. That does not stay internal. Run through EXHALE_transit.py, the theoretical peak excess absorption is 40.19% against 40.26% in He I 10830 (0.2% relative), 2.098% against 2.199% in H α (+4.8%), 1.046% against 1.134% in H β (+8.4%), with Ly α saturated in both. He I 10830 forms in the wind and does not care which layer sits underneath it; part of the H($n = 2$) absorption is formed in the layer, so the weak determination appears to propagate to the Balmer depths at the 5–8% relative level. Quote those depths with the pair, not from a single coefficient.

Lyman–Werner with metals on. The Stellar LW flux [erg/cm 2 /s]: 343.0 A/B on seed B at $\varepsilon_4 = 5 \times 10^{-3}$, both members info = 0:

	LW off	LW on (343)
$\log_{10} \dot{M}$	9.65	9.68
$H_2 = HI$ front	1.01108	1.01026
T at $r = 1.02$ [K]	1801	1470
$x(H_2)$ at $r = 1.02$	4.0×10^{-6}	2.1×10^{-5}
H_3^+ peak [cm^{-3}]	1.93×10^5 at 1.0006	1.99×10^5 at 1.0006
HeH^+ peak [cm^{-3}]	0.285	0.053
C II fraction at 1.02 / 1.20	0.197 / 0.449	0.180 / 0.430
$H\alpha$ peak	2.098%	2.297%
$H\beta$ peak	1.046%	1.221%
He I 10830 peak	40.19%	40.33%

The self-shielding is the section 56 physics unchanged: at the base $N(H_2) = 2.68 \times 10^{21} cm^{-2}$, $f_{shield} = 2.11 \times 10^{-6}$, $k_{LW} = 1.27 \times 10^{-10} s^{-1}$; above $r = 1.10 R_p$ the shielding is gone ($f_{shield} = 1.000$, $k_{LW} = 6.026 \times 10^{-5} s^{-1}$). $x(H_2)$ at $r = 1.02$ is *higher* with the band on, which appears to be the layer temperature rather than the chemistry: that layer is 330 K colder, and the temperature dependence of the H_2 balance outweighs the added destruction. The front moves down by $0.0008 R_p$, consistent with more destruction where the shielding has lifted. Caveat: the band changes the $H\alpha$ peak by +9.5% relative and $H\beta$ by +17%, only about twice the numerical spread above, so those changes are not quotable as observational predictions without the coefficient pair run alongside.

Files: `src/modules/time_step/steady_newton.f90` (diagnostic only). Documentation: `docs/hd209_metal_stagnation.md` section 9, `examples/README.md`, `README_HOWTO.md`, `TO_BE_DONE.md` item (A).

62. Runtime grid size, CODATA k_B , the layer residual on a physical scale, and two diagnostic closures (2026-08-15)

One working series. The verification was staged so that every piece except the k_B value is proven a no-op on the regression matrix, and the golden refresh at the end carries the k_B change alone.

Grid cells: — the cell count is a runtime quantity. N was a compile-time parameter (500); it is now set by the optional key `Grid cells: <N>` with default 500, so an `input.inp` without the key is unchanged. The grid-sized module arrays of `global_parameters`, `lya_rt`, `excited_hydrogen` and `ionization_equilibrium` became allocatable (lower bound 1- N_g kept) and are allocated once parsing has resolved the key; the main program's state vectors likewise, zeroed on allocation to reproduce their former static-storage start. Procedure-local grid arrays were already automatic. `write_setup_report` echoes the resolved N and `load_IC` counts the IC records first, so a restart against an IC written at a different N reports the mismatch instead of an end-of-file error. This closes `TO_BE_DONE` item (E): the refinement pairs (Base grid [dr , cells], Grid cells) = (1.0e-4 100, 500), (5.0e-5 200, 658), (2.5e-5 400, 916), (1.25e-5 800, 1375) hold the upper stretch from `input.inp` alone. Gates: `make` and `make wind_ae_ic` build; `make check 5/5 byte-identical` against the pre-change goldens; a 750-cell run writes $750 + 2N_g$ rows; `-fcheck clean`

at 500 and 750 cells; an H-only A/B (the one path the matrix does not cover, where a parameter zero array had to become a local) is byte-identical.

k_B is the CODATA value; the lyman_werner copy is synced. $k_{B_erg} = 1.38 \times 10^{-16}$ (the truncated ATES literal) is now the SI-exact 1.380649×10^{-16} . The judgment first: the old value was wrong at 4.7×10^{-4} relative and every thermal quantity passes through it; the smallness of the effect is not an argument for keeping it. The dependency-free copy k_{B_lw} in `lyman_werner.f90` is synced. Measured against the old goldens: the WASP-121 b cases re-converge at count 14065/14040 (from 14060/14037), fields move by a few 10^{-4} – 3×10^{-3} relative, $\log_{10} \dot{M}$ by the last printed digit ($13.22 \rightarrow 13.23$). **The goldens and the parse corpus were re-snapshotted at the end of the series and carry exactly this change.** Deliberately not touched (recorded so the list is explicit): k_{B_eV} , h_{p_erg} , h_{p_eV} , μ , G , c and the astronomical constants — a full constants pass would be its own measured rebaseline.

The below-escape momentum residual has a physical scale. Section 61's diagnostic divided every row by $|u|$; for the momentum row that is $|\rho v|$, which collapses in the quasi-hydrostatic layer and made converged states read 0.2–10. The momentum row is now divided by the gravitational force density $|\rho| |G\phi_i(j) - G\phi_i(j-1)| / \Delta r_j$ (the discrete potential difference of the source term), so the number is the fractional violation of hydrostatic balance; the three rows print separately. Print-only, make check 5/5 byte-identical. Re-measured on the section 61 states (`docs/hd209_metal_stagnation.md` section 10): the converged examples/15 (old scale 9.99) and examples/16 $\varepsilon_4 = 5 \times 10^{-3}$ (0.195) and 1×10^{-2} (2.352) solutions all hold the layer at 10^{-5} – 1.4×10^{-4} of ρg . The layer is not momentum-unbalanced; what stays uncontrolled between the pair members is *which* hydrostatic stratification it settles on.

Heating_breakdown.txt is consistent for a molecular run. The dump reconstructed atomic heating only (and said so in a NOTE); its electron density omitted the molecular ions. It now rebuilds the H_2 photoheating (the H_2 density enters `PH_heat_HHe`, so column and opacity are the solver's), the $He(2^3S)+H_2$ Penning and the Lyman–Werner heating exactly as `ionization_equilibrium` adds them, uses the same `calc_ne`, and appends `col15 heat_He23S_H2_Penning`, `col16 heat_H2_LW`. Measured on `mol_lyman_werner`: $\max |\sum \text{cols } 5\text{--}16 / \text{heat_total} - 1| = 2 \times 10^{-15}$, all three molecular channels nonzero. Atomic runs unchanged; the file is not part of the regression comparison.

Lya absorbing bottom: — a downward loss channel (default off). The section 30 audit follow-up: Huang et al. (2017) close their Monte Carlo with a pure-absorber bottom (H_2 accidental-resonance true absorption), while the local closure had no lower boundary. With the key on, a third channel joins the wing and Sobolev escapes: the same Neufeld form on the depth from the cell *down* to the bottom of the domain, $\beta_{bot} = \min(1, \pi^{-1/4} \sqrt{a/\tau_{bot}})$, combined in parallel and feeding the existing β_{tot} , so the loss acts on the internal field and the trapped stellar beam consistently. First look (HD 189733 b benchmark configuration, 300-step snapshot, qualitative): $\bar{J}$ and n_{2p} drop together — 0 at the innermost ghost, $\times 0.24$ at $r = 1.0002$, back to within 1.5% by $r = 1.023$, exactly 1 above 1.05 — confined to the bottom few scale heights, as the audit expected. Off path executes the original expression unchanged (byte-identical pass below).

windae du-trigger arming, measured. The 2026-08-11 descending-crossing guard had never been measured on IC mode: windae, whose ~ 3 -step false du stop originally forced the “always Newton-finish” rule. On examples/12_windae_ic_hd209 (2000-step cap): du at step 2 is 1.43×10^{-4} — below every threshold on the untouched generated IC — and nothing fires; the du stop arms at step 23 (1.027×10^{-3}) and the Newton hand-off at step 222 (1.002×10^{-2}), both on the first ascending crossing. A du-stop $\dot{M}$ still carries the usual path spread; the Newton finish stays the prescription.

Verification ledger.

stage	tree	make check
runtime N	that change only	5/5 byte-identical
+ layer scale + load_IC guard		5/5 byte-identical
+ breakdown fix + Lya bottom (off), old k_B	all but k_B	5/5 byte-identical
+ CODATA kb_erg	full series	FAIL vs old goldens as measured above; goldens re-snapshotted; final 5/5

63. The rest of the physical constants, and why iron cannot move to Badnell (2026-08-19)

Two items from the “recorded, blocks nothing” list, taken together because the first needs a golden refresh and the second turned out to need none.

The constants pass. Section 62.2 updated k_B and listed what was deliberately left alone; those are now done, against CODATA 2018 and IAU 2015:

constant	was	now	rel. change
kb_eV	8.6167×10^{-5}	$8.617333262 \times 10^{-5}$	7.4×10^{-5}
mu	1.673×10^{-24}	$1.67353284 \times 10^{-24}$	3.2×10^{-4}
Gc	6.67259×10^{-8}	6.67430×10^{-8}	2.6×10^{-4}
hp_erg	6.62620×10^{-27}	$6.62607015 \times 10^{-27}$	2.0×10^{-5}
hp_eV	4.1357×10^{-15}	$4.135667696 \times 10^{-15}$	7.8×10^{-6}

π , erg2eV, c , parsec, AU and R_J were already exact to the printed digits. mu is the mass of the hydrogen *atom* ($m_p + m_e - 13.6\text{eV}/c^2$), which is what the density normalization means; the comment now says so. The same values appear in four other places, all synced: m_h2 in lyman_werner.f90, Gcgs in lower_column.f90, and the Wind-AE constants in wae_params.f90, wae_soe.f90 and wae_exhale_input.f90. The Wind-AE files port an external code, so the original defs.h values are written into the comment there; against them the shifts are 2.6×10^{-4} , 6.5×10^{-6} and 1.4×10^{-4} , so the port now agrees with the C/Python reference at that level instead of bit-exactly, and restoring the three lines reproduces the gates verbatim (backup/regression/windae_oracle, working copy only). The helium atomic mass was left at Wind-AE’s $6.6464790722 \times 10^{-24}$, which is the CODATA He atom mass to 10^{-6} .

Measured effect: fields move 10^{-3} – 5×10^{-3} relative and $\log_{10} \dot{M}$ does not move in the printed digits (wasp_full 13.23; counts 14065 $\rightarrow$ 14059, 14040 $\rightarrow$ 14036). The one outlier is the photoheating of the last interior cell of wasp_he23off ($r = 1.6056$), which changes by 97% because the old value was an isolated dip below its neighbour; the rest of that file agrees to 0.2%. Goldens and parse corpus re-snapshotted, make check 5/5. The four paper models were re-converged (all info = 0, $\log_{10} \dot{M}$ unchanged at 9.47 / 9.14 / 11.70 / 13.20); the LaRT fields were not re-run, their inputs moving an order of magnitude below the Monte Carlo noise of a 2×10^6 -photon run.

Iron recombination: Phase 6 is closed, not deferred. The code carried a note that switching iron from the Huang et al. (2023) Eqs. (5)–(6) fits to Badnell RR+DR was “deferred to a later pass”. It cannot be done, for a reason outside this code: **the data do not exist**. Fe I is produced by recombining Fe II (Mn-like, 25 electrons) and Fe II by Fe III (Cr-like, 24), while the Badnell dielectronic-recombination project reaches the phosphorus sequence (15 electrons) in its latest instalment (paper XVI, 2022). Neither iron stage is covered — nor is the K-like sequence the Ca I comment in the same file already records as missing. The other nine metals are on Badnell, which is what the phase asked for. The Huang plan’s own Phase 1c prescribes exactly this fallback “where Badnell coverage is thin”, so the code was following the plan rather than lagging it. What remains unbuilt is only that gate’s bookkeeping: a runtime key selecting one rate source for all metals, and a Huang-vs-Badnell delta table; neither applies to iron.

Documentation pass on the md/tex corpus. The sweep that accompanied section 62 predates 63.1/63.2, so a second pass propagated them and re-checked the reference documents against the code. Findings beyond the expected propagation: the manual and the README presented the iron swap as pending (both now carry the isoelectronic reason, and the manual names the Ca I parallel); the README described line trapping as covering “the two fine-structure coolants” in a two-level scheme, when it has been the eight ground-term lines of C I/C II/N II/O I ($n_{\text{fsl}} = 8$) in exact statistical equilibrium (cooling_formulas.tex and the manual were already right); EXHALE_BC_and_IC.tex quoted $\log_{10} \dot{M} = 9.31$ for HD 209458 b as the value “under the present code”, which was the 2026-08-11 number against 9.47 today; four line-number citations in atomic_data_EXHALE_vs_MoCHII.tex pointed at the wrong lines and were replaced by symbol names; the transit resolving powers were undocumented, and Ca II/Na I default to *whatever* Instr_res_Ha is, so an H α override silently moves both doublets (README_HOWTO.md now tabulates the seven defaults and states the trap); and wind_ae_solver.tex claimed a bit-exact base-BC gate that 63.1 turned into a 10^{-6} – 3×10^{-4} agreement. Two mechanical cross-checks passed in both directions: all 56 keys parsed by input_read.f90 appear in the manual, and every key-like entry in the manual is still parsed. PDFs rebuilt and inspected: manual 54 pp., this log 118, BC/IC 23, Wind-AE 7, atomic data 7; no errors, no undefined references, and the surviving overfull boxes are pre-existing table rows away from the edited pages.

What the constants pass had still missed, and the golden refresh it needed. Section 63.1 swept the .f90 files for the constants *named* in parameters.f90. Rescanning for their *values* found four more sites. One is in the solution path: Cool_coeff.f90 computed He II recombination cooling as $1.38\text{e-}16 * T * \alpha_B$ in two places. That literal is the Boltzmann constant, not a fit coefficient — the $k_B T \alpha_B$ form *is* the Hui & Gnedin (1997) He II expression, while H II and He III use their fitted case-B rates — so it was the truncated ATES value section 62.2 had replaced everywhere else, leaving this coefficient 4.7×10^{-4} low. It is kb_erg now. The other three are report-only or write generated files:

`mh_g` in `lower_column.f90` and `ICW_mH` in `wae_ic_writer.f90` were the proton mass where the hydrogen atom is the code's mass unit, and `m_amu_g` in `species_diffusion.f90` was labeled “amu” while holding the proton mass — with masses on the $H = 1$ scale there, the right multiplier is the hydrogen atom, neither the proton nor u . Fifteen constants in ten Python tools were brought to the same values, including `src/utls/glob.py` (the GUI's mirror of the global block) and the three-digit G in `exhale_transit_lib.py`; the `cooling_data` scripts deliberately keep the rounded $k_B = 1.38 \times 10^{-16}$ the CHIANTI fits were made with.

The verification was staged so the two kinds of change stay separable. The report-only fixes alone give `make check 5/5` byte-identical, which is what proves `mh_g` never reaches a solution. Adding the `kb_erg` fix fails all five, as it must: against the old goldens the fields move by $3.7\text{--}3.8 \times 10^{-6}$ (max, in the `cool` column — the edited term) on the two atomic cases with median $\sim 4 \times 10^{-7}$, and by $\leq 2 \times 10^{-4}$ with median $\sim 10^{-8}$ on the molecular ones, which carry almost no He II. Step counts and $\log_{10} \dot{M}$ (13.23) do not move. Goldens re-snapshotted, `make check` back to 5/5; the parse corpus is unchanged (125 cases OK) because none of these constants enters a derived setup value. The four paper models were not re-converged: 3.7×10^{-6} on the hottest, most ionized case is three decades below the printed precision of every number in `paper/`.

64. Resolved-configuration output, and the transit tool reading it (2026-08-22)

`EXHALE_setup.out` records the resolved configuration in prose, but `EXHALE_transit.py` re-parsed `input.inp` only — so a `base.inp` handoff that overrides the planet radius, base temperature, or He/H reached the wind and not the transit geometry. Measured on benchmarks/wasp52: the wind used $1.40492 R_J / 1181.2 \text{ K} / \text{He/H } 0.0959$ (from `base.inp`) while the transit read $1.27 R_J / 1304 \text{ K} / 0.0204$ from `input.inp`. The radius enters chord lengths, absorbing areas, and the rotation velocity, so this was a real transit-side error for every handoff run.

Fix, per `docs/lhs1140b_lower_atmosphere_plan_new.md` Phase B:

- **EXHALE_resolved.out** (new, machine-readable): written next to `EXHALE_setup.out` at startup by `write_resolved_config` (`write_setup_report.f90`) after `input.inp` + `base.inp` resolution. Key-value lines: `planet_radius_RJ`, `planet_mass_MJ`, `equilibrium_temperature_K`, `HeH_number_ratio`, `orbital_distance_AU`, `star_mass_Msun`, `base_inp_present`.
- `exhale_transit_lib.read_input_params` prefers that file when it sits next to `input.inp` (returns `resolved=True`, plus the resolved HeH); absent file falls back to `input.inp` exactly as before. A present-but-unreadable file is an error, not a silent fallback. `EXHALE_transit.py` prints which source it used.
- **he_line_metrics.py** (new, repo root): the He 10830 line metrics of Cherubim et al. (2026) — three Gaussians with shared width and shift; blended-red depth, blue depth, red/blue amplitude ratio, FWHM of the blended feature — with a synthetic-profile selftest (`python3 he_line_metrics.py --selftest`). `EXHALE_transit.py` now fits the instrument-convolved He curve (air frame) and writes `tpm_He10830_metrics.txt`; a fit failure warns and does not stop the run.

Verified: a scratch run with a `base.inp` overriding radius/temperature/He shows the transit consuming $1.50 R_J / 900 \text{ K}$ (the override), and the fallback path still returns the `input.inp` values; `make check` byte-identity, since the new file is additive and the compared outputs are untouched.

65. The metastable helium triplet is on by default (2026-08-22)

`Include He23S` was a mandatory key initialized to `.false.`, so every run carried whatever the file it was copied from happened to say. Across the tree that produced 28 runs with the triplet off, almost none of them off on purpose: the setting was inherited when the case was created for something else entirely (a solver, an IC, a domain). A helium-bearing wind has the metastable level whether or not the input file remembers to ask for it, and the level is not a diagnostic add-on — it carries its own photoionization, Penning ionization with H and H₂, recombination, and collisional-excitation cooling, so leaving it out changes the electron budget, the energy budget, and the 10830 Å observable. Off is the special case, not on.

- `parameters.f90: thereis_HeITR = .true.`
- `input_read.f90`: the key became **optional** (it was read through `req`, so the initializer never actually applied to a valid run). A file that omits the line gets the triplet; `Include He23S? False` is the deliberate opt-out and is what the HeITR-off regression branch uses. `thereis_He false` still forces it off, as before.
- The Tk interface (`src/utis/EXHALE_interface_functions.py`) follows: the `checkboxbutton` variable starts at 1, the reset button restores it to 1 instead of clearing it, and `load_input` treats the key as optional with the same semantics as the Fortran reader — on unless the file says `False`. Since a file may now omit the line, the sequential reader checks the label before consuming the line, so an older file without it no longer shifts `Load IC`, `Do only PP`, and `Force start` by one.

Verified: a one-step run with the key absent prints “Including helium triplet chemistry” in the setup report, and `Include He23S? False` does not; the Tk reader parses the tutorial input with and without the line to the same `Load IC / Do only PP` values. `make check` is unaffected — every `backup/regression/*/input.inp` sets the key explicitly, including the two `wasp_he23off*` cases that exist to exercise the off branch.

The `examples/01–12` ladder keeps `Include He23S? False` on purpose: each folder is its base plus exactly one line, and atomic helium is the baseline that leaves `06_he23s` a one-line difference against `03_newton` instead of a folder that differs in nothing. Outside the ladder the triplet is on — including `examples/13_lower_atmosphere/`, which had been running two of its four planets with the triplet and two without. That split mattered: the $\text{He } 2^3\text{S} + \text{H}_2$ Penning channel is active only when a molecular run also tracks the triplet, so one example was doing different physics on different planets. All four now carry `True`; none of them has stored output.

The runs whose stored output predates the flip are listed in `docs/he23s_default_recompute_list.md`; nothing there has been recomputed.

66. Primitive conversion of unfilled ghost cells (2026-08-24)

A runtime-checked JFNK run of the LHS 1140b He/H = 1000 case (Phase C audit, LHS1140b/exhale/audit_summary.md) ended with gfortran reporting a signalling IEEE_INVALID_FLAG. Rebuilt with `-ffpe-trap=invalid,zero,overflow`, the solver died in its first residual evaluation at `UW_conversions.f90:58`, `W(2,:) = U(2,:)/U(1,:)`: `U_to_W` converts the whole array, ghost cells included, and `Apply_BC` called it on a state whose ghosts were still zero — the comment two lines above the call in `steady_newton.f90` had already noted that the ghosts are zero at that point. The `0/0` produced NaNs in the ghost primitives that `Apply_BC_W` overwrote on the next line, so no result was ever affected; but every steady solve tripped an invalid operation, which is what had kept the trapping variant of the `-fcheck` regression off the steady-solver path. With that site fixed the trap moved to the two line-search trial conversions in `steady_newton.f90` (call `U_to_W(utry, Wtry)` in the hybrid and JFNK searches), where `unpack_U` fills only the interior of `utry` and only `Wtry(1,1:N)` and `Wtry(3,1:N)` are then read.

- `UW_conversions.f90`: new `U_to_W_interior`, converting cells $1 \dots N$ through `U_to_W_comp` and zeroing the ghost columns — one definition for every site that converts a state before its ghosts are set.
- `Apply_BC.f90`: `U_to_W` $\rightarrow$ `U_to_W_interior`; the ghosts are outputs of `Apply_BC_W`, never inputs.
- `steady_newton.f90`: both line-search trial conversions $\rightarrow$ `U_to_W_interior`.

Verified: under `-O1 -g -fcheck=bounds,do,mem -ffpe-trap=invalid,zero,overflow` the He/H = 1000 JFNK now completes, `info=0` $\|R\| = 7.929 \times 10^{-4}$ (identical to the untrapped run) with no signalling exception, and so does the H-rich solar_gj699 case (`info=0` $\|R\| = 6.433 \times 10^{-4}$). `make check`: 5/5 byte-identical (all five cases), as the discarded ghost values require.

67. Element diffusion on the direct-steady route, the element drift metric, and the restart of a diffused state (2026-08-25)

Three defects on the `He_diffusion` path, all of them independent of the transport formulation and all of them blocking the Phase-D milestones of `docs/binary_diffusion_design.md` (sections 7.3, 7.4).

The steady solve and the diffusion relaxation are now one routine. The steady residual (`steady_newton.f90`) carries no diffusion: the He/H field is moved by the operator-split `he_diffusion_step`, so a single steady solve freezes the composition at the state it was handed. The marching hand-off answered that with an outer Picard loop (JFNK $\rightarrow$ 500 relaxation steps $\rightarrow$ `ioniz_eq` $\rightarrow$ drift test, at most five passes), but the direct-steady route — `EXHALE_PTC=1`, the route the LHS 1140b cases take — ran a bare `solve_steady_jfnk / solve_steady_ptc`, so a diffused wind could not be brought to a steady state there at all. That loop is now `steady_wind_with_element_diffusion` (contained in `EXHALE_main`), called from both routes; the solver, its iteration budget and its initial pseudo-time are its arguments, so the direct-steady route reaches it with JFNK or with pseudo-transient continuation. `dt_loc`, which the relaxation step needs, is built by the `eval_dt` that route already called. With `He_diffusion` off the loop runs exactly once and the body is the single solve plus the state refresh both sites performed before.

The drift metric counts elements. It read $(f_sp(:,3)+f_sp(:,4)+f_sp(:,5))/(f_sp(:,1)+f_sp(:,2))$, which is He I + He II + He III over H I + H II: it omitted the He 2³S triplet and every molecular species, and so did not measure the same quantity that `species_diffusion.f90` transports (its own `sumHe` includes the triplet). Replaced by `element_ratio_HeH` in `composition.f90`, which counts nuclei with the `bsp_nH` / `bsp_nHe` weights of `species_table.f90` over all ten base species — H₂ and H₂⁺ two H nuclei, H₃⁺ three, HeH⁺ one of each. It is the same element count `load_IC` applies to a restart file, and it is correct in the molecular region without a second definition.

A diffused state can be restarted. `load_IC.f90` rescaled every cell to the global HeH whenever the file's composition differed from it by more than 10⁻⁶, on the ground that the input file is the authority on composition. With `He_diffusion` on that erases the answer: the cell-by-cell split is what the diffusion operator produced. The rescale is now confined to the base cells (1-Ng to 1), where the operator itself holds a Dirichlet reservoir composition; every cell above keeps the helium fraction it was written with. With the flag off the whole column is rescaled as before, bit for bit. The HeH⁺ refusal is unchanged — molecular chemistry and `He_diffusion` remain an excluded pair until the molecular closure is validated (memo section 5, milestone M4).

No diffusion operator changed; `species_diffusion.f90` is untouched.

Verified. `make check==> REGRESSION PASS` (all cases byte-identical), 5/5, which is the evidence that the `He_diffusion-off` path is unmoved.

The restart was measured against the stored `examples/14_diffusion` output under `EXHALE_DUMP_IC=1`, which writes the state as loaded, before the first ionization-equilibrium solve. The loaded He/H profile reproduces the file's to 3.3×10^{-16} relative, worst cell, over all 504 cells; the same file through the pre-change binary comes back flattened to the input ratio ($r = 1.05 R_p$: file 1.616×10^{-2} , loaded 8.333×10^{-2} , a factor 5.2). The direct-steady route with `EXHALE_PTC=1` `EXHALE_PTC_JFNK=1` now runs the outer loop — five passes, `info=0` on every one, drift falling 9.40×10^{-3} , 7.74×10^{-3} , 5.79×10^{-3} , 4.64×10^{-3} , 4.03×10^{-3} . It reaches the pass limit rather than the 10⁻³ drift test, which is what a five-pass Picard budget gives from a state that was not converged to begin with (the stored output is the unconverged JFNK `info=2` one); the same test from a converged state is part of the baseline capture below.

The pre-change binary is kept as `backup/phase_d_baseline/EXHALE_prephaseD.x`, with the reference runs it produces, as the memo's T2 comparison.

68. Binary H/He element diffusion replaces the trace-helium kernel (2026-08-25)

The composition operator of `He_diffusion` is rewritten from the trace-helium-in-hydrogen kernel to the two-component (binary) formulation of `docs/binary_diffusion_design.md`, sections 2–4. New module `src/modules/functions/binary_element_diffusion.f90`, routine `element_diffusion_step`; the old module is kept for the duration of milestone M2 with its routine renamed `he_diffusion_step_trace` and its body untouched, reachable only through `EXHALE_DIFFUSION_KERNEL=trace`, so the two can be compared (test T2a). Both call sites in `EXHALE_main.f90` go through one dispatcher, `element_composition_step`.

What changed in the physics. *The transported variable.* The trace kernel evolved the ratio $f = n_{\text{He}}/n_{\text{H}}$, which diverges as hydrogen becomes the minor element — exactly the limit a helium-rich atmosphere needs — and it wrote the diffusive flux as $n_{\text{H}}Dd(n_{\text{He}}/n_{\text{H}})/dr$, which divides by a vanishing background. The new operator transports the helium **mass fraction** $X = \rho_{\text{He}}/\rho$ of two components, hydrogen (with the trace metals slaved to it) and helium, and is bounded at both ends of the composition axis. The gradient coefficient of the memo, $A = (m_1 m_{\text{He}} n / \bar{m}) D_{12} (dx/dX)$, collapses to $A = \rho D_{12}$, so nothing in the operator divides by an element density. The settling flux carries the prefactor $\rho X(1-X)$, which switches settling off as either element is exhausted, while the gradient flux does not vanish there — a helium-free cell next to a helium-bearing one still receives helium.

The cap is gone. The trace kernel clamped $f \leq \text{HeH}$ over the whole domain, and in the stored examples/14_diffusion output the profile sat exactly on that clamp between 1.2 and $2R_p$. The clamp existed because the trace flux has no reason to vanish when hydrogen runs out; the binary flux does. It is removed, together with the base pile-up limiter. A pile-up, if one appears, is physics for the base boundary condition to answer, not something to clamp. What remains is a round-off assertion ($0 \leq X \leq 1$), and the bounds it asserts are a property of the discretization: the backward-Euler matrix is a strictly diagonally dominant M-matrix, with the Péclet hybrid on the settling term chosen so that it never flips the sign of an off-diagonal.

The ambipolar field is computed, not assumed. The previous effective settling mass $3 - 0.5(\bar{Z}_{\text{He}} - \bar{Z}_{\text{H}})$ is the hydrogen-plasma value $eE = m_{\text{H}}g/2$ written as a constant; in a fully ionized helium plasma the massless-electron balance gives $eE = \frac{4}{3}m_{\text{H}}g$ instead, and the relative settling mass $5/3$, not 2.5 . The field is now evaluated from the solved electron pressure of the cell, $eE = -(1/n_e)d(n_e kT)/dr$, in its logarithmic form $-kT d \ln(n_e T)/dr$ (exact for a barometric stratification, the same second-order central difference elsewhere). The three analytic limits are reproduced to round-off (test T9): neutral 3, H^+ plasma 2.5 — so the validated helium-trace behaviour is unchanged — and He^{++} plasma $5/3$.

Advection. The operator receives only the new ρ , v , T and dt , not the hydro's Runge–Kutta face mass fluxes, so a second conservative advection of ρX could not be made consistent with the hydro's own continuity step. X is therefore transported in the advective form, $\rho \partial X / \partial t + \rho v \partial X / \partial r = -\nabla \cdot (r^2 J) / r^2$, with the velocity divergence cancelled analytically in the row assembly rather than formed and subtracted (that cancellation is catastrophic whenever the step is long compared with the advective crossing time). A uniform X is then preserved exactly whatever the hydro does, and at a steady state, where $\rho v r^2$ is constant, the advective and conservative forms are the same equation. In a transient the helium mass is conserved only to the order of the hydro's own truncation; this is the stated limitation of the phase.

Write-back. The species vector is projected onto the new element totals by a nonnegative projection: helium-only species scaled by r_{He} , hydrogen-only species (H_2 , H_2^+ , H_3^+ included) and the slaved metals by r_{H} , HeH^+ by $\min(r_{\text{H}}, r_{\text{He}})$, and the shortfall of the element with the larger factor deposited into its neutral ground species. Every operation is a multiplication by a nonnegative factor or an addition, so no negative intermediate can arise, and both element totals are met exactly. The old write-back used one factor per element and inferred the metal mass as $1 - m_{\text{H}}f_{\text{H}} - m_{\text{He}}f_{\text{He}}$, which counts molecular mass as metal.

Trace metals keep the previous element-by-element treatment behind `He_metal_diffusion` (a trace element cannot feed back on the background it moves through); only the background definition is now the nucleus count over all species rather than $\text{H I} + \text{H II}$.

User-visible change. He_Kzz now defaults to 0.0 instead of $10^9 \text{ cm}^2 \text{ s}^{-1}$. With the old default every diffusion run silently carried a constant eddy term; an eddy coefficient is a property of the atmosphere being modelled, so it is stated rather than inherited. An input that relied on the default must now say He_Kzz: 1.0e9; examples/14_diffusion already does. The four stored run directories that carried He_diffusion: True without the key — benchmarks/hd209, docs/version_compare/v2_diff, docs/version_compare/wasp_v2_diff, scratch_fs/hd189 — now state He_Kzz: 1.0e9 explicitly, so each input records the eddy coefficient its stored output was produced with; no output was regenerated. docs/input_schema.md K19 and README_HOWTO.md are updated. examples/exhale_io.py gains a derived Run.heh_profile, the He/H element ratio with the same nucleus weights the operator uses, so the analysis reads one definition.

Verification. A standalone driver, src/tests/diffusion_tests.f90, builds with make diffusion_tests and runs the acceptance tests of the memo's section 6 on synthetic columns and on a stored wind; all 17 checks pass.

id	measured
T1a	closed isothermal column relaxes to the barometric separation $dx/dr = -x(1-x)(m_{\text{He}} - m_1)g/kT$: max error of logit x 4.56×10^{-6} at $N = 400$; helium mass drift 2.0×10^{-16} at a step of about the cell diffusion time
T1b	Dirichlet base, integrated boundary flux accounts for the mass change to 2.2×10^{-11} relative
T3	He/H = 1000 with an outflow: finite, $0 \leq X \leq 1$, hydrogen rises (X falls below the reservoir value), no cap
T4	closed column, arbitrary initial $X(r)$, 10^4 steps: helium mass drift 2.4×10^{-14} (eos_metals on) and 3.7×10^{-14} (off), metals present in both
T5	two-component mass closure $m_1 n_{\text{H}} + m_{\text{He}} n_{\text{He}} = \rho$ after the write-back: 4.0×10^{-15} relative, worst cell over 200 steps
T6	uniform X , $g = 0$, compressing and expanding $v(r)$: 4.4×10^{-14} relative departure after 500 steps
T9	relative settling mass 3, 2.5, 5/3 to 8.3×10^{-13} in the three analytic limits; finite and continuous across a resolved ionization front
T10	second order in dr (observed 1.998); the steady state is step-size independent to 1.2×10^{-7} over two decades of dt
T2a	the operator increment against the trace kernel on a stored LHS 1140 b wind falls off first order in the helium fraction: 2.28×10^{-3} , 3.00×10^{-4} , 3.04×10^{-5} , 3.05×10^{-6} at He/H = 0.0833, 10^{-2} , 10^{-3} , 10^{-4} — a factor 10 per decade, and 3.0×10^{-6} at 10^{-4} against the memo's 10^{-3} gate. Over 500 relaxation steps the two profiles agree to 4.2×10^{-5} , 7.4×10^{-6} , 7.8×10^{-7} , 7.8×10^{-8} on the same sequence

T2a is measured with the wind's ρ and T and $v = 0$. The stored wind is not steady below $r \sim 1.2$ — its mass flux $\rho v r^2$ varies by orders of magnitude there and the base cell has $v < 0$ — so with that velocity the conservative advection of the trace kernel and the advective form differ by $X \nabla \cdot (r^2 \rho v)/r^2$, a difference of the **state** rather than of the trace limit, which does not fall off with He/H (the driver reports that column too). The comparison of the transport closure, which is what changed, is therefore made at $v = 0$.

The measured helium-mass drift of the closed column grows with dt (2.0×10^{-16} , 6.2×10^{-12} , 3.1×10^{-10} , 1.1×10^{-7} at $dt_{\text{code}} = 10^1$, 10^3 , 10^5 , 10^7 over 1000 steps). This is the round-off of the tridiagonal solve amplified by the conditioning of the implicit step, not a leak: the discretization telescopes exactly, and the drift falls to round-off as soon as the step is comparable with the cell diffusion time. The driver prints the table.

T0: make check is 5/5 byte-identical, which is the evidence that the He_diffusion-off path is unmoved by any of this. T7 (the molecular closure) is milestone M4 and T8 (the moving-wind face flux on a converged LHS 1140 b run) waits on the baseline capture; neither is claimed here.

69. Milestone M3: the binary operator on a wind, the LHS 1140 b helium-rich runs, and the removal of the trace kernel (2026-08-25)

Milestone M3 of docs/binary_diffusion_design.md: the binary element diffusion operator of section 68 is run on converged winds — the HD 209458 b examples/14_diffusion configuration against the captured baseline of the kernel it replaces (T2b) and the three LHS 1140 b helium-rich cases — the elemental face flux is measured on a moving wind (T8), and the trace kernel is then deleted.

69.1 T2b: the binary operator against the trace kernel on HD 209458 b

Both sides are the same configuration (HD 209458 b, He/H = 0.0833, Include He23S? True, He_metal_diffusion: True, PLM+WENO3), Newton finished, with the same five-pass diffusion outer loop. The baseline (backup/phase_d_baseline/kzz0, kzz1e9) was marched from a cold start with the pre-Phase-D binary; the binary-operator runs (new_kzz0, new_kzz1e9) restart from those converged states on the direct-steady route (EXHALE_PTC=1 EXHALE_PTC_JFNK=1), which is the route section 67 gave a diffusion loop. This is a report, not a gate (the memo's section 6): the baseline profile sits on the $f_{\text{He}} \leq \text{HeH}$ cap over part of the domain, so a difference there is the cap, not a defect of either side.

quantity	trace, $K_{zz} = 0$	binary, $K_{zz} = 0$	trace, $K_{zz} = 10^9$	binary, $K_{zz} = 10^9$
JFNK info / final $\ R\ $	0 / 7.737×10^{-4}	0 / 8.043×10^{-4}	0 / 8.017×10^{-4}	0 / 8.308×10^{-4}
outer passes / final drift	5 / 1.22×10^{-2}	5 / 1.29×10^{-1}	5 / 3.88×10^{-3}	5 / 3.19×10^{-2}
$\log_{10} \dot{M}$ [g s^{-1}]	9.7882	9.7883	9.7909	9.7910
He 10830 red depth [%]	9.7970	9.8134	9.7958	9.8118
(He/H)/HeH at $1.05 R_p$	0.62354	0.62354	0.62806	0.62808
at $1.10 R_p$	0.69771	0.69744	0.70193	0.70170
at $1.20 R_p$	0.95514	0.95515	0.95536	0.95536
at $1.50 R_p$	0.97542	0.97437	0.97555	0.97447
at $2.00 R_p$	1.00000	1.00111	1.00000	1.00112
at $3.00 R_p$	0.98628	0.98712	0.98643	0.98725
at $4.00 R_p$	0.85734	0.85834	0.85806	0.85922
cells within 10^{-6} of HeH	45 ($r = 1.00 \dots 2.44$)	3 (the base)	45 ($r = 1.00 \dots 2.44$)	3 (the base)

Three things are worth reading off it.

The flat stretch was the cap. Forty-five cells of the trace profile sat exactly on HeH between 1.0 and $2.44 R_p$. With the binary operator only the three Dirichlet base cells are on that value, and the profile crosses *above* it — 1.0011 at $2 R_p$ — which the clamp forbade. Helium is mildly enriched there relative to the reservoir, and the enrichment is small.

The base depletion is not a cap artifact. At $1.05 R_p$ both kernels give 0.6235; the two profiles differ by at most 4×10^{-3} anywhere above $1.05 R_p$ after 2500 composition steps of the binary operator. The depletion of the first cells (down to 10^{-3} of the reservoir ratio in the cell at $1.0004 R_p$, in *both* kernels) is not the limiter and not the lag of the trace form.

Corrected in section 70 (2026-08-25). That last depletion was read here as settling. It is not: the barometric He–H separation scale of that base is 24 cell widths, so a 550-fold drop across one cell cannot be barometric, and the settling flux measured at those faces is 2–20% of the gradient flux. It was the face-averaged upwinding of the advective term, which decouples a cell whose two face velocities straddle zero. The two kernels agreed because they shared it. Section 70 also supersedes the drift and profile numbers of the tables in this section.

Neither side is composition-converged. Both leave the outer loop at its five-pass ceiling, and the binary operator’s drift is an order of magnitude larger than the trace kernel’s. That drift is the maximum *relative* change of He/H over a pass and it is dominated by those near-base cells, where He/H is $\sim 10^{-3}$ of the reservoir value: over the same five passes the profile above $1.05 R_p$ moved by less than 0.4%. The observable moved as little: $\dot{M}$ by 10^{-4} dex and the He 10830 red depth by 0.017 percentage points ($9.797\% \rightarrow 9.813\%$). The five-pass ceiling is unchanged by this milestone; making the outer loop reach drift $< 10^{-3}$ needs either many more passes or a drift metric that is not dominated by a helium-free base cell, and is left open. (Answered in section 70: neither — the relaxation was being stepped at the hydro CFL step, four to six orders below the composition time scale, so it moved the base by $\sim 10^{-6}$ of the way per step. With the step taken from the composition time scale the $K_{zz} = 10^9$ case converges to 1.2×10^{-4} absolute in ten passes of ~ 30 steps.) load_IC kept the diffused profile across the restart, which is what section 67 added: the run reports top cell He/H = $7.0843\text{E-}02$ against the input $8.3333\text{E-}02$ and pins only the base cells to the reservoir value.

69.2 T8: the elemental face flux on a moving wind

With EXHALE_DIFFUSION_CHECK=1 the operator writes diffusion_faceflux.txt in the run directory at every step (replaced each time, so the file holds the state the run ended on): the diffusive face flux J , the advective face flux, their sum as $F_{\text{He}} = 4\pi r^2(\rho X v + J)$, the total mass flux $4\pi r^2 \rho v$ and the relative settling mass, all taken from the same face coefficients the implicit solve used — the file reports the discrete fluxes, not a re-derivation of them.

On new_kzz0 (HD 209458 b, $K_{zz} = 0$), with the spread measured exactly as the code’s own du is, $(\max - \min)/|\text{mean}|$ over a radial window:

window	spread of F_{He}	spread of $\rho v r^2$
$r \geq 2.0 R_p$ (the code’s escape window [j_min:N])	5.13×10^{-3}	7.19×10^{-3}
$r \geq 1.5 R_p$	1.68×10^{-2}	7.71×10^{-3}
$r \geq 1.2 R_p$	5.14×10^{-2}	1.25×10^{-2}
$r \geq 1.1 R_p$	3.84×10^{-1}	7.77×10^{-2}

T8 passes: in the window the code itself uses to declare the wind steady, the elemental helium flux is as flat as the mass flux (flatter, in fact); over the wider windows the two spreads stay within a factor 5, i.e. the same order of magnitude, and both are dominated by the same unsteady base region below $\sim 1.1 R_p$. The diffusive part is 12% of the advective part at most above $1.2 R_p$, so this is a real test of the sum and not of advection alone.

69.3 M3-2: LHS 1140 b with diffusion

Three cases in LHS1140b/exhale/, each the seed case's `input.inp` plus `He_diffusion: True` (`He_Kzz` at its new default 0, `He_ambipolar` at its default on), restarted from the seed's converged output and finished on the direct-steady route by `finish_case.sh` (JFNK $\rightarrow$ post-processing pass $\rightarrow$ EXHALE_transit.py). All three converged, `info = 0` on every pass.

	heh0p55_diff	heh1000_diff	heh0p06_gj699_diff
seed	heh0p55 (GJ 1132 SED)	heh1000	heh0p06_gj699
He/H reservoir	0.55	1000	0.06
JFNK info / $ R $	$0 / 4.439 \times 10^{-4}$	$0 / 7.929 \times 10^{-4}$	$0 / 8.482 \times 10^{-4}$
outer passes / final drift	$5 / 4.73 \times 10^{-2}$	$5 / 5.17 \times 10^{-2}$	$5 / 2.20 \times 10^{-2}$
$\log_{10} \dot{M}$, seed $\rightarrow$ diffused	7.7642 $\rightarrow$ 7.7645	7.7442 $\rightarrow$ 7.7437	8.5885 $\rightarrow$ 8.5883
(He/H)/HeH at $1.05 R_p$	0.9997	0.9973	0.9996
at $1.2 R_p$	0.9996	1.0020	0.9996
at $2 R_p$	1.0008	1.0030	1.0010
at $5 R_p$	1.0048	1.0074	1.0041
at $10 R_p$	1.0085	1.0101	1.0050
at $20 R_p$	1.0084	1.0102	1.0034
profile extremes	0.683 at $29.0 R_p$, 1.009 at $14.5 R_p$	0.626 at $29.0 R_p$, 1.195 at $1.03 R_p$	0.827 at $29.0 R_p$, 1.004 at $6.7 R_p$
H ionization front ($x_{\text{HII}} = 0.5$)	$9.43 R_p$	$1.03 R_p$	$5.66 R_p$
He 10830 red depth [%], seed $\rightarrow$ diffused	4.2847 $\rightarrow$ 4.2875	73.389 $\rightarrow$ 73.321	3.8335 $\rightarrow$ 3.8453

The separation is small on these winds. Between the base and $20 R_p$ the element ratio departs from the reservoir value by $\sim 1\%$ at most; only in the last few cells below the $30 R_p$ outer boundary does it fall to 0.63–0.83. The mass-loss rate and the He 10830 depth move by less than 0.1%. This is the expected ordering rather than a null result: these are fast, low-gravity winds, and the advection time across the domain is short against the diffusion time, so the wind carries the mixture out before it can separate. Whether the residual 5×10^{-2} drift would grow the separation if the outer loop were allowed to run to drift $< 10^{-3}$ is not answered here, and the five-pass ceiling was not changed for this milestone.

The ambipolar settling mass on the solved states. From the `dmeff` column of `diffusion_faceflux.txt`, above $1.05 R_p$: 2.56–3.00 (heh0p55_diff), 2.76–3.05 (heh0p06_gj699_diff) and 1.65–5.47 (heh1000_diff). The two hydrogen-rich cases stay near the neutral value 3 because helium is largely neutral there and the H^+ term enters through $\bar{Z}_{\text{He}} - \bar{Z}_1$. The helium-rich case spans the widest range and touches 1.67 at $1.11 R_p$ — the He^{++} value 5/3 to three digits — but **it does not get there as the He^{++} limit**: $x(\text{He III})$ reaches only 0.28 at the top of the domain in that run, so the 5/3 limit of the memo's (3a) is not sampled by these winds and remains guarded by the T9 unit test alone. Below $1.05 R_p$ `dmeff` is large and of either sign (down to -43 in the base cell at $1.03 R_p$), where the electron pressure gradient is steep and the diffusive flux itself is negligible.

LHS1140b/exhale/finish_case.sh gains one optional variable, EXHALE_BIN, which pins the executable a case is run with (these three were run with a snapshot binary while the tree was being rebuilt). With it unset the script behaves

exactly as before.

69.4 The trace kernel is removed

With T2a (section 68) and T2b recorded, `species_diffusion.f90` is deleted, with it the `EXHALE_DIFFUSION_KERNEL=trace` selector and the `element_composition_step` dispatcher in `EXHALE_main.f90`: both call sites now call `element_diffusion_step` directly. In the test driver T2a can no longer be measured — its comparison partner is gone — so it prints the M2 measurement as a recorded result and counts no verdict; the driver is 15 checks, all passing. `make check` stays 5/5 byte-identical (no regression case turns diffusion on, so this is the statement that the `He_diffusion-off` path is untouched).

Not done in this milestone: T7 and the molecular closure (milestone M4, the `error stop` on molecular chemistry + `He_diffusion` stands), and the regression case with `He_diffusion: True` that section 7.5 of the memo asks for at the end of the series.

70. The base helium hole, the metal drain beside it, and a composition relaxation that converges (2026-08-25)

Three defects on the element-diffusion path, all found on the milestone-M3 runs of section 69, all in the same place: the first cells above the base.

70.1 What was measured

On backup/phase_d_baseline/new_kzz0 (HD 209458 b, $K_{zz} = 0$, JFNK info = 0) the helium ratio of the first free cell above the base sat at $(\text{He}/\text{H})/\text{HeH} = 1.8 \times 10^{-3}$ between two cells at 1.0 — a 550-fold drop across one cell width, where the barometric He–H separation scale of that base is $H = kT/(\Delta mg) = 0.0045 R_p$, **24 cell widths**. A 550-fold drop across one cell is not barometric physics. Every metal was emptied from the same cell by $\sim 10^3$ (C I $2.3 \times 10^7 \text{ cm}^{-3}$ against 2.6×10^{10} in its neighbours), and with them the metal-line cooling of that cell. The dip sat exactly where the breathing base alternates the sign of the velocity: $v = (-796, +63, +28, -8, -9, +5) \text{ cm s}^{-1}$ over cells 1 to 6.

Rebuilding the operator's face coefficients from the converged state gives the row terms of the first free cell (cell 2), per unit X , in $\text{g cm}^{-3} \text{ s}^{-1}$:

cell	(He/H)/HeH	diffusive aa	diffusive cc	advective aa	advective cc
2	2.4×10^{-3}	-2.97×10^{-17}	-2.92×10^{-17}	0	0
3	4.5×10^{-1}	-2.78×10^{-17}	-3.22×10^{-17}	-4.53×10^{-15}	0
4	7.2×10^{-1}	-3.09×10^{-17}	-3.36×10^{-17}	-9.94×10^{-16}	-8.46×10^{-16}

and the face fluxes it balances, in $\text{g cm}^{-2} \text{ s}^{-1}$:

face	gradient J	settling J	total J	$\rho_f v_f$
1/2	1.451×10^{-11}	-3.193×10^{-13}	1.420×10^{-11}	-7.65×10^{-8}
3/2	-6.098×10^{-12}	-1.459×10^{-13}	-6.244×10^{-12}	8.78×10^{-9}
5/2	-4.114×10^{-12}	-3.575×10^{-13}	-4.472×10^{-12}	1.93×10^{-9}
7/2	-2.011×10^{-12}	-4.723×10^{-13}	-2.484×10^{-12}	-1.64×10^{-9}

The settling flux is 2–20% of the gradient flux and the Péclet number of the settling hybrid is 0.02 everywhere at the base, so the hybrid is on its central branch and the settling is fully resolved: **the depletion is not settling**. Both advective off-diagonals of that row are *identically zero*, so the cell was held by its diffusive coupling alone — and that coupling gives a relaxation time $\rho/(|aa|+|cc|) = 3.3 \times 10^6$ s against the ~ 2 s hydro step the relaxation was being driven at. Whatever composition the cell happened to hold was frozen there.

70.2 (a) The advection was upwinded on face-averaged velocities

face_coefficients built the advective coefficients from $v_f = 0.5(v_j + v_{j+1})$ and chose the donor per face. A cell whose two face velocities straddle zero ($v_f(j-1) < 0 < v_f(j)$), which is what the numbers above give: -366 and $+45$ cm s $^{-1}$) then donates at both faces and receives at neither, and its advective term vanishes identically — even though its own $v_j = +63$ cm s $^{-1}$ is large and its donor, the Dirichlet base itself, sits one cell away.

The advective term is now the one-sided upwind difference on the **cell** velocity,

$$\rho_j v_j \frac{X_j - X_{j-1}}{r_j - r_{j-1}} \quad \text{for } v_j \geq 0, \quad \rho_j v_j \frac{X_{j+1} - X_j}{r_{j+1} - r_j} \quad \text{for } v_j < 0, \quad (20)$$

which contributes +cadv to the diagonal and –cadv to the donor and to nothing else: a uniform X stays exact (T6), the row sum of the advective part is still zero, the M-matrix argument is unchanged, and the flux-minus-divergence construction the old form needed is gone. On the cell above, the coupling it restores is 6.43×10^{-15} , **216 times** the diffusive coupling that was holding the cell.

New unit test **T11**: a column seeded with the measured 500-fold hole in the first free cell and a velocity that alternates in sign cell by cell — so every face average vanishes while the cell velocities do not — must refill the hole. It gives 6.7×10^{-5} relative departure from the reservoir with the new form and 2.8×10^{-2} (28 times the 10^{-3} tolerance) with the old one. The driver is now 16 checks.

70.3 (b) The trace-metal solver drained the same cell

solve_trace_element_in_hydrogen transported the metal **density** conservatively with the same face-averaged velocities. In a conservative form the configuration above is not a decoupling but a drain: the cell has an outflow term at both faces and an inflow term at neither. That divergence has no counterpart in the hydro's own ρ , whose face fluxes come from the Riemann solver.

The solver now transports the **mixing ratio** $f_X = n_X/n_H$ in the same advective form as helium, so a uniform f_X is preserved under any velocity field and no spurious divergence can create or destroy the element. The $f_X \leq f_X(\text{base})$ cap in the write-back went with it: settling piles an element up as readily as it depletes one, and the cap is the same

limiter the memo (section 1) rejects for helium. In the runs below no metal exceeds its reservoir ratio anywhere, so the cap removal changes nothing in them by itself; what changes the metal profile is the transport.

70.4 (c) The relaxation ran on the CFL step, and on a flow with no mass flux

steady_wind_with_element_diffusion called the operator 500 times per pass with the hydro's local CFL step dt_{loc} . The implicit operator is unconditionally stable, so nothing tied its step to a sound-crossing time, while the base composition relaxes on $\Delta r^2/D_{12} \sim 10^6$ s against $dt_{loc} \sim 2$ s. That is why every case in section 69 left the outer loop at its five-pass ceiling with the drift falling by only ~ 0.75 per pass. The drift metric compounded it: $\max |\Delta \text{He}/\text{H}|/(\text{He}/\text{H})$ is a *relative* measure, and it was dominated by exactly the emptied cells where it means nothing. The relaxation is now `relax_element_composition` in the diffusion module:

- the step is the composition time scale, $\min(\Delta r^2/(D_{12} + K_{zz}), \Delta r/\max(|v|, w_s))$ per cell with $w_s = D_{12}|G|$, grown geometrically ($\times 1.5$, capped at 10^{12} times the start) so the late steps are direct steady solves, with the coefficients Picard-updated at every step;
- the measure is absolute, $\max_j |X_j^{\text{new}} - X_j^{\text{old}}|/X_{\text{base}}$, for the inner stop (10^{-12} per step) and for the outer pass stop (10^{-3} , at most 20 passes);
- the advecting flow is the wind's **steady mass flux**, $\rho v = \dot{m}_{\text{steady}}/r^2$ with $\dot{m}_{\text{steady}}$ the median of $r^2 \rho v$ over the escape window $[j_{\min}:N]$. The advective form is the conservative equation only where ρ and v satisfy continuity, and below $1.02 R_p$ the converged states do not: the spread of $r^2 \rho v$ there is 4.8×10^2 (new_kzz0), 7.4×10^3 (new_kzz1e9) and 7.3×10^3 (heh0p55_diff) times its own median, against 1.7×10^{-2} to 4.6×10^{-1} over $1.1-2 R_p$. Relaxed on that field the operator converges to the composition of a flow that neither conserves mass nor exists. Solving the operator's steady state offline on the M3 new_kzz0 wind: with the cell velocities the base takes a five-fold cliff at the fourth cell and a plateau at 0.10 of the reservoir ratio; with the steady flux the first free cells are flat to 1% and the profile leaves the base on the barometric scale.

The marching call keeps dt_{loc} and the cell velocities: there the wind is genuinely transient and both are the consistent choice.

`element_ratio_HeH` is no longer used by the outer loop itself; it is kept as the element count of the diagnostic below, which writes one profile per pass.

70.5 The runs, re-run from the same restart files

backup/phase_d_baseline/new_kzz0_fix, new_kzz1e9_fix and LHS1140b/exhale/heh0p55_diff_fix: the section-69 configurations, the same output/*_IC.txt restart files, the same direct-steady route (EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0). The M3 directories are untouched.

	$K_{zz} = 0$ M3	$K_{zz} = 0$ now	$K_{zz} = 10^0$ M3	$K_{zz} = 10^9$ now	LHS heh0p55 M3	LHS now
outer passes	5 (ceiling)	20 (ceiling)	5 (ceiling)	10	5 (ceiling)	6
final drift	1.29×10^{-1} rel	1.71×10^{-1} abs	3.19×10^{-2} rel	1.18×10^{-4} abs	4.73×10^{-2} rel	1.67×10^{-4} abs
relaxation steps per pass	500	24–33	500	28–31	500	28–400
(He/H)/HeH, first free cell	1.51×10^{-4}	0.9971	0.8293	1.0000	1.0001	0.9400
next four cells	0.381, 0.662, 0.810, 0.871	0.9939, 0.9960, 0.9994, 1.0025	0.7146, 0.6549, 0.6371, 0.6385	1.0000×4	0.9990, 0.9998, 1.0001, 1.0000	0.8818, 0.8510, 0.8143, 0.7695
at $1.05 R_p$	0.6235	0.9085	0.6282	0.9823	0.9997	4.9×10^{-4}
at $1.10 / 1.20 R_p$	0.6977 / 0.9551	0.8342 / 0.7986	0.6967 / 0.9553	0.9268 / 0.8962	0.9995 / 0.9996	$9 \times 10^{-5} / 2 \times 10^{-5}$
at $1.50 / 2.00 R_p$	0.9744 / 1.0011	0.7912 / 0.7974	0.9745 / 1.0011	0.8903 / 0.8959	1.0001 / 1.0008	~ 0
at $3.00 / 4.00 R_p$	0.9871 / 0.8583	0.7935 / 0.7145	0.9871 / 0.8603	0.8923 / 0.8279	1.0021 / 1.0035	~ 0
$\log_{10} \dot{M}$ [g s^{-1}]	9.7883	10.396	9.7910	9.998	7.7645	7.6065
He 10830 red depth [%]	9.8134	6.4032	9.8118	8.2115	4.2875	1.0×10^{-4}

The base is what the fix was for and the base is fixed. On both HD 209458 b cases the first free cells now sit within 0.6% of the reservoir ratio and the profile leaves them monotonically, instead of the isolated 550-fold hole followed by a five-cell recovery. On LHS 1140 b the first free cells fall 0.940, 0.882, 0.851, 0.814, 0.770 — ratios of 0.94–0.96 per cell against the $\exp(-\Delta r/H) = 0.93$ of that column’s barometric scale, i.e. the base now leaves the reservoir on the barometric scale, which is the criterion this change was measured against. The metals follow helium: C/H in the first free cell was 0.57 of the reservoir with $K_{zz} = 10^9$ and is 1.0000 now, and above $1.2 R_p$ it settles to ~ 0.56 where the removed cap pinned it at 1.0000.

Three things this exposes rather than fixes, and none is claimed here:

- **The separation is much larger than M3 reported** — 0.80 instead of 0.98 of the reservoir ratio at $1.2\text{--}3 R_p$ for $K_{zz} = 0$, and on LHS 1140 b helium is gone above $1.05 R_p$ instead of flat at the reservoir. That is what a relaxation that reaches its steady state gives; the M3 profiles were the restart files barely moved. For the LHS 1140 b column it is also the expected physics of $K_{zz} = 0$: at 226 K and $g = 1.8 \times 10^3 \text{ cm s}^{-2}$ the homopause of a hydrogen-helium mixture with no eddy mixing sits *at* the base, and the He 10830 line goes with it (4.29% $\rightarrow 10^{-4}\%$). Those cases need a He_Kzz argued from the planet, not the default 0.
- **The $K_{zz} = 0$ HD 209458 b case does not co-converge.** JFNK returns info = 0 on every one of the twenty passes, but the pass drift settles into a limit cycle instead of falling, so its $\log_{10} \dot{M}$ of 10.396 is not a converged number — it is one branch of a period-2 cycle whose other branch is 9.867. The subsection after next shows the cycle is in the wind, not in the composition. The $K_{zz} = 10^9$ case converges (1.18×10^{-4} at pass 10) and moved by +0.21 dex against M3; the next subsection attributes that.

- The wind itself is better converged than M3's: the spread of $r^2 \rho v$ over $1.1\text{--}2 R_p$ is 1.15×10^{-2} ($K_{zz} = 0$) against M3's 1.81×10^{-1} .

70.6 Where the +0.21 dex comes from: metal transport, tested directly

backup/phase_d_baseline/new_kzz1e9_nometdiff is new_kzz1e9_fix with one key changed, He_metal_diffusion: False, from the same restart files and on the same route. It is the control that separates the two halves of the fix: helium transport and relaxation on one side, metal transport on the other. Note what the flag does and does not do — with metal diffusion off the metals are not reset to the reservoir ratio, they are **frozen** at whatever the restart file carried, which here is the M3 profile including its base hole. So the control is “the new helium side, the old metal profile”.

	M3 (new_kzz1e9)	metal diffusion off	full fix
outer passes / final drift	5 (ceiling) / 3.19×10^{-2} rel	3 / 1.23×10^{-4} abs	10 / 1.18×10^{-4} abs
$\log_{10} \dot{M}$ [g s ⁻¹]	9.7910	9.8160	9.9976
He 10830 red depth [%]	9.8118	9.1081	8.2115

C/H relative to the base cell (O/H is the same to four digits in the first eight rows), first eight rows and then interpolated:

	row 1–3	4	5	6	7	8	1.05	1.10	1.20	1.50	2.00	3.00	4.00
M3	1.0000	0.5694	0.6266	0.6213	0.6255	0.6379	0.628	0.696	0.953	0.975	1.000	0.944	0.573
metal diffusion off	1.0000	0.5694	0.6266	0.6213	0.6255	0.6379	0.628	0.696	0.953	0.975	1.000	0.944	0.573
full fix	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.909	0.668	0.569	0.551	0.569	0.565	0.426

The metal-diffusion-off column is identical to M3's, which is the check that the flag froze the restart profile as intended, and the (He/H)/HeH base cells of that run are 1.0000 across, which is the check that the helium side of the fix is fully active in it.

The attribution holds. Switching the metal transport off returns $\log_{10} \dot{M}$ to 9.8160 and the He 10830 depth to 9.11%, i.e. essentially back to M3's 9.7910 and 9.81% and away from the full fix's 9.9976 and 8.21%. Of the +0.207 dex, +0.182 is the metal transport and +0.025 is the helium side. (The decision rule as posed had this the other way round: if the metals were *not* responsible, switching them off would have left $\dot{M}$ near 9.998. It did not.)

One correction to the wording above: the dominant part of that +0.182 dex is not the base hole. This control cannot split the metal change into its two pieces — the base drain and the settling above the base that the removed cap previously forbade — but the base hole spans five cells at $1.0004\text{--}1.0010 R_p$ while the settling changes C/H from $\sim 0.95\text{--}1.00$ to ~ 0.55 over $1.2\text{--}4 R_p$, where nearly all of the metal-line cooling column sits. On the size of the regions the settling has to be the larger term. It is a reasoned inference, not a measurement.

70.7 The $K_{zz} = 0$ non-convergence is the wind, not the composition

The outer Picard loop — wind at fixed composition, composition at fixed wind — has no damping of its own, so the composition update is now under-relaxed,

$$X \leftarrow X_{\text{old}} + \omega (X_{\text{relaxed}} - X_{\text{old}}) \quad (21)$$

with ω starting at 0.5 and halved (floor 0.125) on any pass whose drift failed to fall. The blend is projected back onto the species vector by the same nonnegative projection element_diffusion_step ends on, so both element totals are still met exactly. The drift that the convergence test and the halving rule read is the **undamped** distance $\max|X_{\text{relaxed}} - X_{\text{old}}|/X_{\text{base}}$, so a small ω cannot buy a false convergence by making the applied step small. EXHALE_DIFF_OMEGA=<val> pins the starting value for experiment, and EXHALE_DIFFUSION_CHECK=1 now also writes output/diffusion_pass_profiles.txt, one composition and mass-flux profile per outer pass — the question a non-converging loop raises is about the profiles, not about the scalar drift.

It does not converge, and the reason is not the composition. Two runs from the M3 restart files, both $K_{zz} = 0$:

	start $\omega = 0.5$ (new_kzz0_fix)	start $\omega = 1$ (new_kzz0_undamped)
drift by pass	9.97×10^{-1} , 4.99×10^{-1} , then the pass-3 JFNK stagnates (info = 2, line search collapses to $\lambda = 9.5 \times 10^{-7}$ at $r = 1.009$, energy row) and the loop exits on its solver-failure guard	falls to $\omega = 0.125$ by pass 11 and then holds a period-2 cycle: 7.62×10^{-2} , 6.84×10^{-2} , 8.10×10^{-2} , 7.31×10^{-2} , 8.43×10^{-2} , 7.60×10^{-2} , 8.95×10^{-2} , 7.97×10^{-2} (passes 13–20)
final state	not converged, $\ R\ = 2.4 \times 10^{-3}$, $\log_{10} \dot{M} = 9.838$	20 passes, every one info = 0

The profiles written each pass say what is cycling. Over passes 3–20 the median $r^2 \rho v$ in the escape window alternates between **2.1–2.2** $\times 10^{-7}$ on odd passes and **5.9–6.8** $\times 10^{-7}$ on even ones — a factor 2.8 in mass flux — while each branch is individually flat to 10^{-3} and every JFNK solve returns info = 0 with $\|R\|$ between 9.0×10^{-5} and 9.6×10^{-4} , all inside the 10^{-3} target. The composition, meanwhile, barely moves between the two:

$r [R_p]$	odd (pass 19)	even (pass 20)
1.010	0.9968	0.9974
1.05	0.9645	0.9674
1.10	0.9065	0.9123
1.20	0.8816	0.8893
1.50	0.8768	0.8849
2.00	0.8815	0.8894
3.00	0.8785	0.8865
4.00	0.8219	0.8332
$\log_{10} \dot{M}$	9.867	10.376

The largest odd-even difference in (He/H)/HeH above $1.05 R_p$ is 0.0117, and each parity is nearly stationary from one occurrence to the next (0.0028 odd to odd, 0.0024 even to even). So this is not a composition chasing a wind: it is **two wind solutions, 2.8 $\times$ apart in mass flux, that the residual test cannot tell apart, sitting on compositions that differ by $\sim 1\%$** . Damping the composition cannot remove it, and the measurement above is the evidence — ω reached its 0.125 floor and the cycle held at full amplitude.

Two things follow, neither of them settled here. The $\log_{10} \dot{M} = 10.396$ quoted for this case in the table above is one branch of that cycle (the other is 9.867, close to the 9.998 of the converged $K_{zz} = 10^9$ case and to the 9.816 of the

metal-diffusion-off control). And a residual norm that accepts two states a factor 2.8 apart in $\dot{M}$ is a weak convergence test for this configuration, which is the standing caveat behind the flux-based convergence decision of section 19 ($\|R\|$ is reported for reference only) rather than anything the diffusion introduced.

70.8 Verification

diffusion_tests.x: 16/16 (T11 new; the other 15 unchanged). make check: 5/5 byte-identical — no regression case turns He_diffusion on, and the he_diffusion guard at the top of the operator means the flag-off path never enters any of this.

71. The friction the ionized wind actually has: stage-resolved binary diffusion (2026-08-25)

Element diffusion used one coefficient over the whole domain, the neutral Banks & Kockarts hard sphere. That is the wrong friction above the ionization front, and it is wrong in the direction that matters: a helium or metal *ion* moving through a proton gas is held by Coulomb collisions, whose momentum-transfer cross section at $T \sim 10^4$ K is $\sim 10^{-12}$ cm² against the $\sim 10^{-15}$ cm² of a hard sphere. With a neutral coefficient everywhere the solver let helium and every metal keep settling through the ionized wind, and once the $f_X \leq f_X(\text{base})$ limiter went away (section 70) nothing stopped them: on the HD 209458 b $K_{zz} = 10^9$ wind iron and calcium reached *zero* by $1.1 R_p$ and magnesium 2.5% of its base ratio by $4 R_p$.

Koskinen et al. (2013, section 2.1) solve the same Chapman & Cowling equation with collision terms that “account for neutral-neutral, resonant and non-resonant ion-neutral, and Coulomb collisions”, and report (their section 3.2.2) that Coulomb collisions “are much more efficient in preventing diffusive separation than collisions with neutral H.”

This is a correction to the physics, not an option: there is no input key for it.

71.1 What the coefficient is now

Each element is resolved into its ionization stages and the friction is built pair by pair, every pair being the Chapman-Enskog first approximation $D = 3kT / (16n\mu\Omega^{(1,1)})$ evaluated with the cross section that pair actually has (binary_element_diffusion.f90):

pair	routine	coefficient	scaling
neutral-neutral	hard_sphere_pair_diffusion	$1.52 \times 10^{18} (1/A_s + 1/A_t)^{1/2} T^{1/2} / n$	$T^{1/2} / n$
ion-neutral, non-resonant	ion_neutral_pair_diffusion	$1/D = 1/D_{\text{pol}} + 1/D_{\text{hs}},$ $D_{\text{pol}} = kT / (2.21 \pi e n (\alpha_n \mu)^{1/2})$	T/n then $T^{1/2}/n$
ion-ion	coulomb_pair_diffusion	$3(kT)^{5/2} / (4(2\pi\mu)^{1/2} n (Z_s Z_t e^2)^2 \ln \Lambda)$	$T^{5/2} / n$

and the element-element coefficient is the stage-fraction harmonic sum (frictions add; Blanc’s law across stages)

$$\frac{1}{D_{\text{eff}}} = \sum_{s \in \text{element A}} \sum_{t \in \text{element B}} \frac{y_s y_t}{D_{st}}. \quad (22)$$

Resonant charge exchange $H^+ + H$ is deliberately absent: both partners carry the same element, and a binary diffusion coefficient is driven by the friction *between* the two elements, so every ion-neutral pair that appears is non-resonant. HeH^+ carries both elements and stays out of both carrier lists, as it already was out of the mean charges.

The ion-neutral pair carries two channels. The polarization (Langevin) coefficient is the long-range limit: it keeps the induced-dipole attraction and no repulsive core, so its friction falls as T^{-1} where a rigid core would hold it at $T^{-1/2}$. Taken alone it would make an ion-neutral pair *more* mobile than a neutral-neutral one of the same masses above $\sim 1.5 \times 10^3$ K, which is unphysical — opening a second channel cannot make a pair more mobile. The two are momentum-transfer cross sections of the same encounter, so to first order their $Q^{(1)}$ add, hence their collision integrals add, hence their frictions add; and since $D = 3kT/(16n\mu\Omega^{(1,1)})$ is linear in $1/\Omega$, adding frictions is adding inverse coefficients, $1/D_{in} = 1/D_{pol} + 1/D_{hs}$. That is the same rule the stage mixture uses. Both limits come out exactly: $D_{in} \rightarrow D_{pol}$ where the polarization channel dominates and $\rightarrow D_{hs}$ where the core does, crossing at **1504 K** for an H/He pair. At the 10^4 K of the transition layer D_{pol} is 2.6–4.7 times D_{hs} and D_{in} sits at 0.82 of D_{hs} , i.e. 0.18–0.28 of the polarization value alone.

Sources. The hard-sphere form is the one already in the code (Banks & Kockarts 1973). The polarization constant 2.21π is the one behind the non-resonant ion-neutral collision frequency of Schunk & Nagy (*Ionospheres*, eq. 4.88) and the Coulomb coefficient is the standard Chapman-Enskog result (Paquette et al. 1986; Schunk & Nagy eq. 4.142 give the equivalent collision frequency). Neither textbook is in references/, so **both numerical constants were re-derived from the Chapman-Enskog collision integral rather than copied**, and the derivation is written into the comment beside each routine. The check that the chain is consistent is that the rigid-sphere collision integral put through the same Chapman-Enskog expression returns the Banks & Kockarts prefactor exactly, with a collision diameter of $2.7 \text{ \AA}$.

Polarizabilities are the recommended static dipole values of Schwerdtfeger & Nagle (2019), in atomic units, with $\alpha(H) = 4.5 a_0^3$ the exact nonrelativistic ground-state value; H and He come out at 0.667 and $0.205 \text{ \AA}^3$, the values Schunk & Nagy tabulate for aeronomy.

Validity caveat, recorded at the coefficient. The polarization limit keeps the induced-dipole attraction and no repulsive core, so its friction falls as T^{-1} where a rigid core would hold it at $T^{-1/2}$. It is the smaller of the two coefficients (stronger friction) below $\sim 1.5 \times 10^3$ K and the larger above — by 2.6 to 4.7 at 10^4 K, where the true coefficient is the smaller because the core contribution adds to the collision integral. The consequence is confined to the partially ionized transition layer; both asymptotic limits are exact.

71.2 Two defects fixed with it

- **The trace-metal loop assumed the hydrogen-plasma ambipolar field.** It built its settling coefficient from $\Delta m_{\text{eff}} = (A_X - 1) - 0.5(\bar{Z}_X - \bar{Z}_1)$, the constant $eE = m_H g/2$ of a pure proton plasma — the same assumption decision D3 removed for helium at M2, still standing here. The loop now reads the field settling_coefficient

computes from the electron pressure gradient, so there is one ambipolar field in the module and no second, cruder copy. With He_ambipolar off the field is zero and the loop reduces to the pure-gravity form as before.

- The metal loop also used a single hard-sphere X -in-H coefficient; it now runs the same stage mixture, metal stages against hydrogen carriers.

71.3 Tests

src/tests/diffusion_tests.f90 gains **T12**, the three limits of D_{eff} on a prescribed isothermal column at 10^4 K, each against a closed form written out independently in the test rather than against the module's own pair routines: all neutral reproduces Banks & Kockarts to 1.2×10^{-16} relative, fully ionized reproduces the $\text{He}^{++}/\text{H}^+$ Coulomb coefficient to round-off, half ionized reproduces the four-term Blanc average (whose two ion-neutral terms are themselves the combined form) to round-off. T12d adds the two temperature limits of that combined form: within 2.5×10^{-2} of the polarization channel at 1 K, within 1.2×10^{-3} of the hard sphere at 10^9 K, and never above either. ./diffusion_tests.x: **21 passed, 0 failed**. make check: **5/5 byte-identical** — the operator is entered only with He_diffusion on and no regression case sets it.

EXHALE_DIFFUSION_CHECK=1 adds D_{eff} , the neutral coefficient of the same cell, their ratio and the dominant stage pair to diffusion_faceflux.txt.

71.4 Measured effect

The full tables are docs/binary_diffusion_design.md section 9.3. In short, on two winds restarted from the same converged state and Newton-finished:

- $D_{\text{eff}}/D_{\text{neutral}}$ is 1.00 in the neutral base, 0.16 at the front ($1.2 R_p$), 9.6×10^{-3} at $1.5 R_p$ and 7.6×10^{-5} at $4 R_p$ on HD 209458 b — the suppression *grows* outward because the Coulomb coefficient carries $T^{5/2}$ against the hard sphere's $T^{1/2}$.
- On HD 209458 b every metal mixing ratio becomes **flat above the front**, which is what a steady wind must do once diffusion loses to advection. Iron goes from 0.000 to 0.107 of its base ratio and calcium from 0.000 to 0.019 — the difference between having and not having Ca II H&K and Fe II in the transmission spectrum. $\log_{10} \dot{M}$ moves $10.0106 \rightarrow 10.0561$ (+0.046 dex) and the He 10830 red depth $8.20\% \rightarrow 7.85\%$. The control is a run of the same configuration from the same converged state with a binary built from this commit's parent, so the friction is the only difference.
- The ion-neutral channel decides how much of that survives. Between 1.0 and $1.2 R_p$ hydrogen and helium are still neutral (measured $x(\text{H}^+) = 0.000, 0.004, 0.072$ at $1.05, 1.10, 1.20 R_p$), so the combined form barely touches the *helium* profile — but the low-ionization-potential metals are already ionized there (Fe 0.999, Mg 0.943, Ca 0.269, C 0.329, Na 0.053, O 0.003 ionized at $1.10 R_p$), and going from the polarization channel alone to the combined form raises their frozen plateaus in exactly that order: Fe $\times 3.8$, Ca $\times 2.3$, Mg $\times 1.23$, C $\times 1.07$, Na $\times 1.00$, O $\times 0.97$.

- On LHS 1140 b at $\text{He}/\text{H} = 0.55$ nothing moves: $\log_{10} \dot{M}$ 7.6065 both, the temperature identical to 9×10^{-16} relative, He 10830 within 0.2%, and the composition relaxation exits on its first pass. That wind separates its helium inside the first cell above the base, where the gas is neutral and $D_{\text{eff}} = D_{\text{neutral}}$ by construction, and above that there is no helium left for the Coulomb suppression to hold.

72. The advection correction is gated on the slowest species, not on hydrogen (2026-08-25)

`post_process_adv.f90` replaces the local ionization balance of a cell by the steady advection-ionization ODE, and skips cells where that replacement can carry no information. One of the three skip conditions is a Damkohler number,

$$\text{Da} = \frac{\Delta r}{v} (P_{\text{HI}} + \alpha_{\text{HII}} n_e) > 100 \quad \longrightarrow \quad \text{keep the equilibrium state} \quad (23)$$

built from the **hydrogen** ionization/recombination rate. The correction it gates, however, carries the whole solved species vector — $\text{H I}/\text{H II}$, $\text{He I}/\text{He II}/\text{He III}$ and, when `Include He23S?` is on, the $\text{He } 2^3\text{S}$ metastable — so one species' clock decided the fate of all of them. The metastable relaxes orders of magnitude more slowly than hydrogen ($A_{31} = 1.272 \times 10^{-4} \text{ s}^{-1}$ alone sets a 2.2-hour floor, against sub-second photoionization for H in an irradiated wind), so wherever hydrogen was equilibrated and the metastable was not, the gate froze $n(2^3\text{S})$ at its equilibrium value in cells where it is in fact advected.

That is physically wrong independently of how often it bites, and it bit visibly on the LHS 1140 b run `heh10_diff_kzz1e8` ($\text{He}/\text{H} = 10$), where the equilibrium solution has a sharp hydrogen ionization front: 115 of 503 cells above the front kept the equilibrium ionization, and the `_adv` metastable profile stepped from 5.15 to 223 cm^{-3} across the single cell boundary $r = 6.083 \rightarrow 6.172 R_p$ (1.64 dex). The transit synthesis reads the `_adv` profiles, so the run reported a red-pair depth of 83.7% and $\text{EW} = 23.4 \% \text{ \AA}$ — a number set by the discontinuity. The limitation was recorded in `LHS1140b/kzz_decision.md` section 6.2 and the run was excluded from that memo's crossing fit.

72.1 What the gate is now

The advection systems (`adv_implicit_H`, `adv_implicit_HeH`, `adv_implicit_HeH_TR`) solve the whole H/He vector at once, so there is no gate to be had one row at a time: the honest statement is that a cell may be pinned to equilibrium only when **every** population it solves is equilibrated. The Damkohler number is therefore formed with the **slowest** relaxation rate of the solved set, each rate being the total rate at which that population is destroyed and re-formed:

population	relaxation rate ν
$\text{H I} / \text{H II}$	$P_{\text{HI}} + (a_{\text{ion_HI}} + \alpha_{\text{HII}}) n_e$
$\text{He I} / \text{He II}$	$P_{\text{HeI}} + (a_{\text{ion_HeI}} + \alpha_{\text{HeII}} + \alpha_{\text{HeI23S}}) n_e$
$\text{He II} / \text{He III}$	$P_{\text{HeII}} + (a_{\text{ion_HeII}} + \alpha_{\text{HeIII}}) n_e$
$\text{He}(2^3\text{S})$	$A_{31} + P_{\text{HeITR}} + (q_{31a} + q_{31b} + a_{\text{ion_HeITR}}) n_e + Q_{31} n_{\text{HI}}$

$\text{Da} = (\Delta r/v) \min(\nu)$, compared with the same threshold of 100. The $\text{He}(2^3\text{S})$ row is term by term the loss side of `fvec(4)` of `adv_implicit_HeH_TR` — radiative decay, photoionization, collisional de-excitation to 2^1S and 2^1P ,

electron-impact ionization, and Penning ionization on neutral H — and every coefficient is read from the arrays `HeITR_coeffs` and `eval_cool` already fill in the same routine. No rate is redefined. The He I row now also carries `rcheiTR`, because with the triplet on `HeITR_coeffs` overwrites `rcheiIB` with the singlet channel alone and the He I row of the residuals adds the two. The two other skip conditions — inflow $v \leq 0$ on either face, and $x_{\text{HII,eq}} < 10^{-6}$ — are untouched, so the breathing-base guard behaves exactly as before.

The hydrogen rate now includes collisional ionization `a_ion_HI n_e`, which the old expression omitted, for consistency with the three rows beside it. It can only raise Da_H , and Da_H is no longer the quantity that decides anything on its own.

72.2 Measured

Diagnostic build dumping Da per cell, LHS 1140 b `heh10_diff_kzz1e8`, post-processing pass 10:

$r [R_p]$	Da_H , H-only gate	Da_H , new binary	Da of the slowest species
6.083	6.1×10^{-4}	6.1×10^{-4}	6.1×10^{-4}
6.172	3.2×10^2	6.0×10^{-4}	2.8×10^{-3} (H-only run), 6.0×10^{-4} (new)
6.946	7.8×10^5	4.8×10^{-4}	$2.3 \times 10^{-3} / 4.8 \times 10^{-4}$
7.602	4.6×10^7	1.1×10^2	$2.1 \times 10^{-3} / 1.7 \times 10^{-3}$
7.956	1.7×10^8	7.2×10^2	$1.9 \times 10^{-3} / 1.6 \times 10^{-3}$

The metastable Damkohler number is $\sim 2 \times 10^{-3}$ throughout that region: the $\text{He } 2^3\text{S}$ population there is advection-dominated by five orders of magnitude, in exactly the cells the hydrogen gate pinned to equilibrium. The two Da_H columns differ because the old gate is self-reinforcing — a pinned cell holds the fully ionized equilibrium state, whose n_e and P_HI then keep Da_H above the threshold for the cell above it.

Effect on that run (both binaries run from the same converged state with `Do_only_PP: True`, in scratch copies; the case directory itself was not re-run):

quantity	H-only gate	slowest-species gate
cells kept at equilibrium	115 of 503	15 of 503
outermost such cell	$30.0 R_p$	$1.0034 R_p$
$\max \text{d log}_{10} n(2^3\text{S}) $ between adjacent cells, $r > 1.05$	1.636 ($5.15 \rightarrow 223$ at 6.13)	0.386 ($4.99 \rightarrow 12.1$ at 7.43)
He 10830 red depth [%]	83.69	21.52
red EW [% Å]	23.39	5.69
$\log_{10} \dot{M}$ [g/s]	7.76	7.76 (the wind solution is not re-run)

The remaining 15 pinned cells all sit at $r \leq 1.0034 R_p$, the shielded base where conditions (i) and (iii) are meant to act. The largest remaining variation in the metastable profile, 0.386 dex over $r = 7.38 \rightarrow 7.49 R_p$, is not a switch: it is where the advected hydrogen finally ionizes (x_HI falls from 0.80 to 5×10^{-4} over three cells as Da_H crosses 1) and the Penning loss `Q31 n_HI` on the metastable switches off with it. The `_adv` profiles are continuous across it, and `hybrd1` returns `info = 1` in every cell of the region, including those with $\text{Da}_\text{H} \sim 10^5$.

The corrected run's EW of 5.69 % Å is still above what the $\text{He}/\text{H} = 2.7\text{--}4.0$ sequence of `kzz_decision.md` section 6.1 extrapolates (~ 2.8 % Å at $\text{He}/\text{H} = 10$ on that sequence's local log-log slope of 0.73), so the He-rich end appears to be

leaving the regime that sequence describes for its own reasons; what the fix removes is the discontinuity, not the run's distance from the trend.

72.3 Effect on existing results

Three reference cases were re-post-processed with both binaries in scratch copies — LHS1140b/exhale/heh0p55 (diffusion off), heh2p13_diff_kzz1e9 (one of the crossing brackets), and the H-rich standard backup/phase_d_baseline/new_kzz1e9_d3b (HD 209458 b, $K_{zz} = 10^9$, metals on). In all three, Hydro_ioniz_adv.txt and Ion_species_adv.txt come out **byte-identical** to the H-only-gate binary, and the number of pinned cells is unchanged (42, 30 and 153 respectively): in those runs the pinned cells are held by the inflow or $x_{\text{HII,eq}}$ condition, which the change does not touch, not by the Damkohler condition. He 10830 red depth and EW therefore move by 0.00%, and no stored result needs regenerating.

make check passes 5/5 — as it must, since the goldens compare Hydro_ioniz.txt and Ion_species.txt, which the post-process never writes; running it is the check that the change stays inside the post-process.

72.4 Re-post-processing the stored LHS 1140 b scans (2026-08-26)

The three reference cases above were the wrong sample for the He-rich end. All eighteen diffusion-off cases of LHS1140b/exhale/ — the eleven of the GJ 1132 composition scan and the seven _gj699 companions — were therefore re-post-processed in place, each from its own untouched JFNK solution of record (output/*_IC.txt) through one Do only PP: True pass and both transit syntheses; the pre-fix products are kept beside each case in pre_gate72_adv/. Five cases changed, all helium-dominated: heh10, heh100, heh1000, heh10000 and heh1000_gj699 lose 89–93% of their red-pair depth and equivalent width (heh10 75.97% and 21.53 % Å to 7.42% and 2.270 % Å; heh1000_gj699 105.1% and 35.87 % Å to 9.53% and 3.479 % Å, the fitted depth above 100% being the pre-fix saturated trough). The count of _adv cells still holding exactly the equilibrium metastable density falls from 173–437 of 504 to 13–22 in those five. The remaining thirteen move by at most 0.06% and keep their cell counts exactly, so the earlier sample's conclusion holds wherever the hydrogen front is not sharp; the 0.06% is a drift in the equilibrium solve between the binary that wrote those outputs (2026-08-24) and this one, at most 4×10^{-4} relative, not the gate.

Nothing the memo docs/lhs1140b_exhale_vs_pwinds.tex concludes moves: the equivalent-width crossing stays at He/H = 0.550 (0.541 with turbulence) on the GJ 1132 proxy and 0.060 on the GJ 699 one, because both are interpolated inside $\text{He}/\text{H} \leq 1$, and so does the matched broadening kernel, 22.30 km s⁻¹. What moves is the He-rich end of the two composition tables, the bump table's He/H = 1000 rows, and the figures built from them. heh10_diff_kzz1e6, the He-rich bracket of the K_{zz} crossing table, was checked in a scratch copy and is unchanged, so tab:kzzcross stands as published; heh10_diff_kzz1e8 was already excluded from it.

One artifact is not the gate's and survives it: heh100 still steps 3.8 dex in $n(2^3\text{S})$ across a single cell at $r = 1.059 R_p$, on the hydrogen ionization front, where the base velocity is still sign-changing. The shell is thin enough (0.008 R_p against a 6.8 R_p stellar radius) that it can account for at most 0.04% of that case's 5.13% red depth.

73. Milestone M4: element diffusion through the molecular region, and the end of the molecular / He_diffusion exclusion (2026-08-26)

input_read.f90 refused Molecular chemistry: True together with He_diffusion: True — an error stop that docs/binary_diffusion_design.md section 5 put there deliberately, to be lifted by a validated closure and not by deleting the check. The closure is now built and the check is gone.

73.1 What the molecular region changes, and what it does not

Equation (1) of the memo transports **elements**, and the chemistry moves hydrogen between H, H⁺, H₂, H₂⁺ and H₃⁺ without changing the hydrogen element's mass, so the transport equation is the same above and below the molecular front. What the species set changes is the closure: helium's collision partners are then a mixture of carriers, and gravity, the ambipolar field and the friction all act on **particles**, not on nuclei. Four things follow, three of them already anticipated by section 5 and one not.

1. **Friction: Blanc's law over the carriers.** The stage sum $1/D_{\text{eff}} = \sum_s \sum_t y_s y_t / D_{st}$ of section 2.6 already ran over the carrier lists {HI, HII, H2, H2⁺, H3⁺} and {HeI, HeII, HeIII, HeTR}, each carrier with its own mass, charge and polarizability (H₂ enters at mass 2 and with $\alpha(\text{H}_2) = 5.315 a_0^3 = 0.787 \text{ \AA}^3$, the Schwerdtfeger & Nagle 2019 recommended value, alongside the H and He entries that were already there). It reduces to $D(\text{He}, \text{H})$ in the atomic region and to $D(\text{He}, \text{H}_2)$ in a fully molecular one.
2. **Forces per carrier.** G now uses the mean carrier mass $m_{c1} = m_1 * m_{\text{car}}$ and the mean carrier charge per carrier, where m_{car} is the hydrogen nuclei bound into one collision partner. In the atomic region $m_{\text{car}} = 1$ and this is the old nucleus form exactly; in an H₂ base the relative settling mass is $4-2 = 2$ instead of $4-1 = 3$. The trace-metal loop settles a metal against the same mean carrier mass.
3. **The carrier density in the pair coefficients.** n in the Chapman-Enskog $D_{st} = 3kT / (16n\mu\Omega^{(1,1)})$ is a density of *colliding particles*. It was the nucleus density; it is now the carrier density, which is the same number wherever the hydrogen is atomic and smaller below a molecular front.
4. **The mole-fraction driver the chemistry carries** — the term section 5 did not state, and the one that makes the other three a force rather than bookkeeping. The transported variable is the mass fraction X ; the Chapman-Cowling driving force is the gradient of the **mole** fraction x among the carriers. With $\psi = n_{1c}/n_{\text{H}}$ (1 atomic, 1/2 fully H₂),

$$\text{logit}(x) = \text{logit}(X) + \ln\left(\frac{m_1}{m_{\text{He}}\psi}\right), \quad (24)$$

$$\frac{d\text{logit}(x)}{dr} = \frac{d\text{logit}(X)}{dr} - \frac{d\ln\psi}{dr}. \quad (25)$$

The gradient coefficient is untouched — $A = (m_1 m_{\text{He}} n / \bar{m}) D (dx/dX)$ collapses to ρD with the carriers exactly as it does with the nuclei, for any ψ , the two carrier factors cancelling — so the whole content of the term is one addition to the settling coefficient, $G \rightarrow G - d\ln\psi/dr$. The discretization, the Péclet hybrid and the M-matrix argument are unchanged, and the term is identically zero wherever the hydrogen is atomic.

It is a real driver: where hydrogen turns molecular going down, each nucleus is spread over fewer collision partners, helium's mole fraction rises, and helium diffuses down that gradient — outward across the molecular front — even at a uniform mass fraction. Its size is set by the sharpness of the front, and across a front where ψ runs $1/2 \rightarrow 1$ it is a factor 2 in the He/H nucleus ratio (test T7d below).

The eddy term carries none of this: eddy mixing transports the mixture as a whole and has no preferred species, so it acts on dX/dr alone.

73.2 T7, and the gate it closes

Four checks in `src/tests/diffusion_tests.f90`, each against a closed form typed out in the test rather than read back from the module:

id	check	result
T7a	forced atomic column: D_{eff} vs $D(\text{He}, \text{H})$	1.13×10^{-16} relative
T7b	forced fully molecular column: D_{eff} vs $D(\text{He}, \text{H}_2)$	1.57×10^{-16} relative
T7c	prescribed molecular column, three constant K_{zz} : the homopause read off the relaxed profile vs the radius where $D_{\text{eff}} = K_{zz}$	$3.4 \times 10^{-3} R_p$, about half a cell, at all three
T7d	same column with $g = 0$, where $-d \ln \psi / dr$ is the only driver left	the mole fraction levels out to 6.2×10^{-5} while the He/H nucleus ratio moves by 0.999

T7c reads the homopause off the profile without assuming where it is: at the zero-flux steady state of a motionless column, $\rho(D+K)dX/dr = -\rho X(1-X)DG$, so $-[d \log(X)/dr]/G$ is exactly $D/(D+K)$ and the radius where it passes $1/2$ is the radius where $D = K$. The three K_{zz} were chosen to put that radius at 1.5 , 2.0 and $2.5 R_p$, and the measured profile put it at 1.4993 , 2.0013 and 2.5010 .

T7d is the sharpest statement of the closure: with no gravity, no field and no eddy, diffusion levels the **mole** fraction of helium among the collision partners, not its mass fraction, so a column that is molecular below and atomic above ends with an He/H nucleus ratio that differs by the factor ψ across the front and a mole fraction that is flat.

make diffusion_tests && ./diffusion_tests.x: **28 passed, 0 failed** (21 before this change; the four new ones and T7c three times). Rebuilt with `-O1 -g -fcheck=bounds,do,mem -ffpe-trap=invalid,zero,overflow` — the new carrier arrays are new array bounds — the same 28 pass with no runtime error.

73.3 The exclusion, and the restart

input_read.f90 no longer refuses the pair; the paragraph there now says why it does not. docs/input_schema.md, README_HOWTO.md and docs/oxygen_chemistry_new_plan.md are corrected where they stated the exclusion.

load_IC.f90 refused to restart *any* molecular state whose composition differed from the input He/H, because HeH+ carries a nucleus of each element and a single rescaling factor cannot set both counts. With He_diffusion on the only cells rescaled at all are the base and its inner ghosts (the column above keeps its diffused split, section 7.4 of the memo), so those cells are now projected onto their two element totals the same way binary_element_diffusion

writes back — HeH+ by the smaller of the two factors, the shortfall deposited into the neutral ground species. Without He_diffusion the whole column would have to be rescaled and the refusal stands.

Found while reading that block and fixed with it: the helium nucleus count load_IC builds to decide whether a restart file's composition matches the input left **HeTR out** (HeI + HeII + HeIII + HeH+), while the rescaling a line below scales HeTR like every other helium species. The metastable triplet carries a helium nucleus — $\text{bsp_nHe(HeTR)} = 1$, and every other element count in the code includes it — so a restart of a He23S state read its own He/H slightly low and rescaled by that error. It is now in the count. No golden moves: no regression case restarts.

73.4 End to end

backup/regression/mol_base_handoff (the hot Uranus, Molecular chemistry + a base.inp handoff) re-run in a scratch copy with He_diffusion: True added, EXHALE_MAXSTEPS=12000 as in the regression:

- runs to completion with no NaN and no negative density;
- the two-component mass closure $m_1 n_H + m_{\text{He}} n_{\text{He}} = \rho$ holds to 4.4×10^{-16} relative over the whole grid, read back out of Ion_species.txt and Hydro_ioniz.txt;
- $\log_{10} \dot{M} = 10.58$, the same as the diffusion-off case at the same step count;
- (He/H)/HeH stays within 1.3×10^{-3} of the reservoir everywhere.

The flat composition is the physics of this planet, not an inert operator. He_Kzz = 1e9 comes from the case's own base.inp and is three decades above the $D_{\text{eff}} \sim 8 \times 10^5 \text{ cm}^2/\text{s}$ of the molecular base, so the column is eddy-mixed; but the same run repeated with He_Kzz = 0 and Kzz_base 0.0 comes out flat as well, within 4×10^{-3} of the reservoir, and the reason is in the diagnostic: at the base face the diffusive helium flux is $-3.1 \times 10^{-13} \text{ g cm}^{-2} \text{ s}^{-1}$ against an advective -3.5×10^{-7} , six orders of magnitude smaller. This is a hot Uranus losing $10^{10.6} \text{ g/s}$; the wind sweeps the composition along faster than either diffusion or settling can separate it. What the run demonstrates is that the operator carries the molecular composition without NaN, without a negative density and with the element budget closed to round-off — the separation physics itself is exercised by T7 and by the atomic-region cases of sections 69–71.

That the molecular carriers are the ones being used is visible in diffusion_faceflux.txt (EXHALE_DIFFUSION_CHECK=1) of the same run, which names the stage pair carrying most of the friction:

$r [R_p]$	dominant pair	relative settling mass	$D_{\text{eff}}/D_{\text{neutral}}$
1.000	HeI-H2	2.0005	0.7747
1.107	HeI-H2	2.062	0.7855
1.169	HeI-H2	2.520	0.8775
1.196	HeI-HI	2.923	0.9783
1.212	HeI-HI	3.000	0.9999
1.691	HeII-HII	3.067	2.52×10^{-3}

The relative settling mass runs from 2 to 3 across the molecular front, exactly the carrier weight of point 2 above, and $D_{\text{eff}}/D_{\text{neutral}}$ from $\sqrt{0.75/1.25} = 0.7746$ — the He-H₂ hard sphere against the He-H one — to 1, before the Coulomb suppression of section 71 takes over above the ionization front.

73.5 Regression

make check: 5/5 byte-identical at the time of the change. The operator is entered only with He_diffusion on, which no case set then, and every molecular quantity added here (carrier lists, mean carrier mass, carrier density, $d\ln\psi/dr$) collapses onto its nucleus form identically when the molecular columns are zero.

73.6 The diffusion regression case (same day)

Byte-identity of the old cases only shows that the operator stays out of the way when it is off; nothing in make check ran it at all (docs/binary_diffusion_design.md section 7.5 had been carrying that as an open item since the design was written). The sixth default case, backup/regression/mol_diffusion, closes it. It is mol_base_handoff — the Tier-2 hot-Uranus gate with molecular chemistry on and its pinned base.inp copied unchanged — plus He_diffusion: True and He_Kzz: 1.0e9, run as the same 12000-step relaxation snapshot. One case covers the widest path: molecular carriers below the front, the stage-resolved pairs through it, the projection back onto f_sp, and the Coulomb friction of section 71 above the ionization front, which is why this gate was chosen over the atomic HD 209458 b run the design had sketched.

The measured effect is the one the face-flux diagnostic above predicts for this planet — present but small (the same hot Uranus, whose wind sweeps the composition along faster than diffusion separates it). Against mol_base_handoff at the same step count, Hydro_ioniz.txt and Ion_species.txt both differ, while

mol_base_handoff	final: count=12000	du= 2.6292E+00	dtu= 5.7467E-03	log10 Mdot = 10.58
mol_diffusion	final: count=12000	du= 2.6290E+00	dtu= 5.7474E-03	log10 Mdot = 10.58

so the golden pins the operator's arithmetic, not a large dynamical signature. make check is 6/6 with the five existing goldens untouched.

74. What a base.inp key is allowed to do: five categories, and elemental reservoirs that reach melem_ab (2026-08-26)

base.inp carried six keys and no statement of what any of them meant for the wind. That is P2 of docs/oxygen_chemistry_new_plan.md, and the reason it matters is not tidiness: q_H2_base had been described in places as pinning the base composition, when all it does is set the base particle count, and the handoff had no way at all to carry an element — the quantity EXHALE actually transports.

74.1 The contract

Every key now belongs to one of five categories, and the category states what the value may do. The table lives in three places that must agree: read_base.inp's header comment, docs/input_schema.md section 2c, and here.

Key	Category	Reaches
comments only	provenance	nothing (the machine-readable keys are A1a/P0, not yet built)
T_base, r_base, p_base, q_H2_base	EOS boundary	T0, R0, p_base_bar, q_h2_base
HeH_base, <El>_H_base	elemental reservoir	HeH, X_<El> → melem_ab
(none today)	initial guess	-
Kzz_base	boundary constraint	he_kzz
q_H2O, q_CO, ...	diagnostic	nothing; metadata, not physics

q_H2_base is an **EOS anchor**: comp_ntot_bc removes the H nuclei bound into H₂ at the base, and that is the whole of its effect. Nothing holds H₂ at that value, no H₂ profile is seeded from it, and the molecular network moves away from it in the first cell. The A1c species keys stay comments because no part of the code reads them.

74.2 Elemental reservoirs

<El>_H_base is new: C_H_base, N_H_base, O_H_base, S_H_base and the same form for Mg Si Ca Na K Fe, each the El/H **nuclei** ratio at the handoff level. It overrides metals.inp for that element, and it turns the metal system on by itself when it is the only nonzero abundance. Hydrogen and helium already had this: HeH_base.

The change that makes it work is one of order. thereis_metals, melem_ab and thereis_lowIP_metal were derived immediately after the core input.inp block, roughly seven hundred lines before read_base_inp ran, so nothing the handoff said about an element could have been heard. They are now derived in the composition block after read_base_inp, beside the thereis_He / HeITR / molecular reconciliation — the one place where every composition flag has its final value. Nothing between the two points touched any of them, which is why the move is a pure reordering.

read_base_inp also matches its keys the way the input.inp reader does: lbl_match (anchored, separator-terminated) instead of a bare index(line, key) > 0 substring test anywhere in the line. lbl_match/is_sep moved out of input_read's internal scope to module scope for that. One file, one matching rule.

src/utls/vulcan_to_base.py now writes the elemental keys it can support: C, N, O and S, summed over **every** carrier at the handoff level, since photochemistry moves nuclei between molecules without creating or destroying them. Mg/Si/Ca/Na/K/Fe are still metals.inp's job — metal and alkali chemistry and condensation are outside VULCAN's networks. The same carrier sum corrects HeH_base, which used to be q_He/(q_H + 2 q_H2) and so lost the hydrogen bound in H₂O, CH₄ and NH₃; on the HD 209458 b run that is 0.09698 → 0.096915, a 0.07% correction and, at 1 microbar, the whole of the difference.

74.3 The gate: do the reservoirs hold?

src/utls/element_budget.py <run_dir> answers it from the output profiles. For every element it compares, cell by cell,

$$n_{\text{El}}/n_{\text{H}} \quad \text{against} \quad (\text{El}/\text{H})_{\text{resolved}} \quad (26)$$

with n_{H} counting hydrogen nuclei in every carrier (HI, HII, 2 H₂, 2 H₂⁺, 3 H₃⁺, HeH⁺), n_{He} counting HeI (which already includes the triplet), HeII, HeIII, HeH⁺, and hydrogen itself closing against the mass density, $\rho = \text{mass_per_H} * n_{\text{H}}$. The abundances it compares against are the run's own resolved values: EXHALE_resolved.out now carries mass_per_H, ntot_bc and one abundance_<El> line per element, all of them written to full double precision so the check has no round-off floor of its own.

Measured on a copy of the mol_metals regression case (hot Uranus, molecular chemistry + solar trace metals) whose base.inp was given O_H_base 9.80e-4 — twice the metals.inp value — and S_H_base 1.32e-5, an element metals.inp does not list at all:

```
# element budget: <scratch>/mol_metals_elem
# 504 cells, profiles Ion_species.txt, tolerance 1.0e-08 (relative)
elem      reservoir max |n_El/n_H      at r[Rp] verdict
           (El/H)    - res||res
H      rho/mass_per_H      1.982e-04      4.81412      FAIL
H*     triplet-corr.      3.056e-13      1.11481      ok
NOTE: rho - mass_per_H*n_H equals 4*n(He 2^3S) to 0.1%; the triplet mass is counted twice in calc_rho (
      HeI already includes it). The H* row removes it.
He      7.930000e-02      1.269e-12      1.11481      ok
C       2.690000e-04      8.132e-13      1.04248      ok
Ca      2.190000e-06      7.915e-13      1.04248      ok
Fe      3.160000e-05      8.177e-13      1.04308      ok
Mg      3.980000e-05      8.310e-13      1.04248      ok
N       6.760000e-05      8.037e-13      1.04308      ok
Na      1.740000e-06      8.091e-13      1.04074      ok
O       9.800000e-04      1.803e-12      1.04553      ok
S       1.320000e-05      8.220e-13      1.04074      ok

1 element(s) miss their reservoir by more than 1.0e-08
```

The nine elemental ratios close to round-off (at most 1.8×10^{-12} relative over 504 cells). Hydrogen does not, and the reason is not the handoff: the mismatch is exactly $4n(\text{He}2^3\text{S})$, the triplet mass counted a second time in calc_rho, because HeI already contains the triplet (util_ion_eq forms the singlet as nhei - nheiTR). Removing it, hydrogen closes to 3.1×10^{-13} as well — the H* row. The double count is a pre-existing defect of the mass and particle book-keeping, not of P2; it is left standing here because fixing it moves every He23S: True golden, which is a change of its own. Its size on this run is 2.0×10^{-4} of ρ at $4.8 R_p$ and zero in the molecular base, where the triplet population is.

The EXHALE_resolved.out of that run shows abundance_O = 9.80e-4 and abundance_S = 1.32e-5, i.e. the handoff overrode metals.inp for oxygen and activated sulfur.

74.4 Legacy

A base.inp without an element key behaves exactly as before: make check is 6/6 byte-identical, goldens untouched.

75. The He 2^3S double count: an excited level is not a second species (2026-08-27)

Section 74 measured it and left it standing: ρ carried the mass of every metastable helium atom twice. The He I density the code transports, `nhei` (`f_sp` column `isp_HeI`), is the **total** He I population — the solver sets it as $n_{\text{HeI}}(1 - x_{\text{HeII}} - x_{\text{HeIII}})$ and the photoionization rates recover the singlet where they need it as $n_{\text{HeIS}} = n_{\text{HeI}} - n_{\text{HeITR}}$ (`util_ion_eq`). The 2^3S column is a **level population inside that number**, not a species beside it. Every budget sum nevertheless added it a second time with mass $4m_{\text{H}}$, one gas particle and one helium nucleus.

75.1 Where the second count was

Seven places, all reading the triplet column as if it were an independent species:

Place	What was counted twice
<code>calc_rho</code> (<code>utilities.f90</code>)	$4m_{\text{H}}$ of mass $\rightarrow \rho$
<code>calc_ntot</code> (<code>utilities.f90</code>)	one gas particle $\rightarrow n_{\text{tot}}$, hence $p = (n_{\text{tot}} + n_e)T$
<code>element_ratio_HeH</code> (<code>composition.f90</code>)	one He nucleus $\rightarrow$ the He/H element ratio
<code>element_nucleus_counts</code> and the <code>gotHe</code> shortfall of <code>project_elements</code> (<code>binary_element_diffusion.f90</code>)	one He nucleus $\rightarrow$ the two element totals the diffusion operator conserves
<code>hecar_isp</code> (<code>binary_element_diffusion.f90</code>)	one collision partner $\rightarrow \text{carHe}$, the helium friction
<code>nHe_l</code> / <code>gotHe_l</code> of <code>load_IC</code>	one He nucleus $\rightarrow$ the He/H a restart file is judged to carry
<code>Run._NUC_HE</code> (<code>examples/exhale_io.py</code>)	one He nucleus $\rightarrow \text{heh_profile}$

`calc_ne` was never wrong: the triplet is neutral, so its `bsp_charge` is zero and the term it contributed was zero.

75.2 The fix

`species_table` gains `bsp_is_excited_level(n_bsp)`, `.true.` only for HeTR, with the rule written beside it: a flagged column is a sub-population of another column, so every budget sum skips it while the transport and the scaling of the level itself do not. The nucleus- and charge-counting loops (`composition`, `binary_element_diffusion`, `diffusion_tests`) test the flag; `hecar_isp` loses its HeTR row, whose mass, charge and polarizability were a copy of HeI's anyway; `load_IC` and `exhale_io` drop the triplet term from their helium sums.

`calc_rho` and `calc_ntot` no longer take `nheiTR` **as an argument at all**. Removing the dummy is what makes the defect unrepeatable: there is no longer a triplet number in scope to add. The `thereis_HeITR` branch of `energy_semi_implicit`'s `calc_ntot` call disappears with it.

75.3 What moved

`make check: wasp_he23off` (He 2^3S off) stays byte-identical, as it must — nothing else changed. The five He23S: True cases move, and the size of the move is the triplet mass fraction of each run:

case	max abs. rel. $d\rho$	max abs. rel. dT	$d\dot{M}/\dot{M}$	max $4n(2^3\text{S})/\rho$
wasp_full	1.24×10^{-6}	1.32×10^{-7}	-6.5×10^{-8}	1.29×10^{-6}
wasp_he23off	0	0	0	0
mol_base_handoff	4.01×10^{-4}	1.59×10^{-4}	$+3.7 \times 10^{-4}$	1.95×10^{-4}
mol_metals	2.57×10^{-4}	6.77×10^{-5}	$+2.1 \times 10^{-4}$	1.98×10^{-4}
mol_lyman_werner	1.39×10^{-4}	1.27×10^{-4}	$+4.4 \times 10^{-5}$	2.92×10^{-4}
mol_diffusion	5.35×10^{-4}	2.73×10^{-4}	$+5.4 \times 10^{-4}$	1.94×10^{-4}

WASP-121 b is hot enough that its 2^3S population is a part in 10^6 of the mass; the hot-Uranus cases carry a few parts in 10^4 . Step counts are unchanged in all six. The golden of the five moved cases were re-snapshotted (check $\rightarrow$ golden $\rightarrow$ check, 6/6 PASS); wasp_he23off was left alone.

75.4 The gate

src/utils/element_budget.py on the mol_metals outputs, before and after, same EXHALE_resolved.out:

before:	H	rho/mass_per_H	1.979e-04	at 4.81412	FAIL
after:	H	rho/mass_per_H	2.975e-13	at 1.11481	ok

Hydrogen now closes against the mass density at round-off, like the nine elemental ratios beside it, with no correction of any kind. The H* triplet-corrected row and its NOTE are gone from the script: there is nothing left to correct.

75.5 Effect on a converged wind

LHS 1140 b, heh0p55 (He/H = 0.55, He 2^3S on, metals off), a scratch copy re-finished with JFNK from the same seed under both binaries, then the same post-processing and transit pass:

	before	after
JFNK	info=0, $\ R\ = 4.326 \times 10^{-4}$	info=0, $\ R\ = 4.277 \times 10^{-4}$
$\dot{M}$ [g/s]	6.47929×10^7	$6.48008 \times 10^7 (+1.2 \times 10^{-4})$
He 10830 red depth [%]	4.279701	4.279674 (-6.3×10^{-6})

The wind is the same wind: ρ differs by at most 2.5×10^{-4} (at the base, $r = 1.0006$), the triplet mass fraction of this run peaks at 2.6×10^{-4} near $11 R_p$, and the observable is unmoved at the fifth digit. The stored run directories were not regenerated.

76. The lower atmosphere as a profile, not six scalars (2026-08-27)

Milestone E1 of docs/phase_e_flux_closure_design.md. The lower-atmosphere handoff has been a set of single-level scalars in base.inp since Tier 3, and that shape is what makes the composition an input rather than an output: HeH is one number, the diffusion operator imposes it as the Dirichlet reservoir at the base, and nothing in the handoff carries a gradient, so no flux condition between the two models can even be stated. E1 replaces the shape. It does not yet close the flux; that is E4.

76.1 The file and the key

One new opt-in key,

```
Lower atmosphere profile: lower_atmosphere_profile.dat
```

names a plain-text file: a block of # key value header lines, a # columns: schema line, then a fixed-column table running deep to shallow. Same conventions as every other EXHALE product, so examples/exhale_io.py reads it (load_lower_atmosphere_profile) with the loader everything else uses. The full schema is docs/input_schema.md section 2d; the header of the shipped example is

```
# EXHALE lower-atmosphere profile (docs/phase_e_flux_closure_design.md section 2)
# solution_id 43902c714fe3462149dfe4e5e259553ff3327e405f2a0039176bb5668ee56770
# source_code analytic
# source_version examples/17_lower_profile/make_example_profile.py
# mechanism none (synthetic H2/H/He column, no reaction network)
# stellar_flux none (no photochemistry was solved)
# p_match_bar 9.99999999999999955E-07
# p_top_bar 1.00000000000000002E-08
# p_deep_bar 1.00000000000000002E-03
# trial_flux_H 5.0000000000000000E+10
# trial_flux_He 0.0000000000000000E+00
# iteration 0
# reached_steady_state T
# notes synthetic example column; T(p) prescribed, no climate solution, no cold trap
# units p[bar] r[R_J] T[K] n_tot[cm^-3] rho[g/cm^3] Kzz[cm^2/s] q_*[-] X_*[El/H nuclei] F_*[g/s]
# columns: p r T n_tot rho Kzz q_H2 q_H X_He X_C X_N X_O F_H F_He q_H20
```

p_match_bar is the producer's statement of where the two models meet, and EXHALE puts its base there: T0, R0, p_base_bar, q_h2_base, HeH and every X_<El> are the profile's columns interpolated at that pressure, through the same doors the scalar keys use (set_element_abundance for the elements, direct assignment for the rest). n_tot and rho are carried, not imposed: the base density is still n0 of input.inp, and the base particle count and mass still follow from T0, HeH and q_H2 through comp_ntot_bc / comp_mass_per_H. Nothing in composition.f90 changed.

Interpolation is linear in log p — the variable both models solve on; the two codes' radius scales are equal only if their hydrostatic integrations agree — and is never extrapolated. A target that falls on a level returns that level's value bit for bit.

Columns the reader has no consumer for are kept, not dropped, and every column is found by name: a producer's element list is ordered differently from run to run.

76.2 K_{zz} stops being a constant

he_kzz was one scalar read at three places inside the diffusion operator: the diffusive time scale that sets the relaxation step, the binary operator's gradient-plus-eddy face coefficient, and the trace-metal kernel's $D + K$. docs/eddy_diffusion_kzz.tex and LHS1140b/kzz_decision.md had already recorded why a constant is the wrong shape — the molecular coefficient rises by nearly five decades between the base and $1.2 R_p$, and the homopause, the one consequential thing K_{zz} does, is where $K_{zz} = D$, a property of two profiles.

The operator now reads `kzz_cell(1-Ng:N+Ng)`, filled once by `eddy_diffusion_on_grid` after `define_grid`: from the profile's Kzz column interpolated onto the grid when one is given, and from the scalar `he_kzz` in every cell otherwise. The two face sites take $0.5 \times (\text{kzz_cell}(j) + \text{kzz_cell}(j+1))$, matching how `Df` and `Gf` are already face-averaged beside them.

`he_kzz` and its key `He_Kzz` keep their names: the constant is what a run states when it has no profile. With a profile in use the key is inert and a warning says so.

76.3 The scalar file is no longer a second source

With a profile in use, every `base.inp` key of the EOS-boundary, elemental-reservoir and boundary-constraint categories is refused with an error `stop` naming the key and its category; provenance comments and diagnostic keys are unchanged. This removes the same-solution problem by construction for those keys instead of checking it.

What is left to check is the pair as a whole, and it is now a refusal rather than the message it used to be: a `base.inp` beside a profile must carry a `# solution_id <hash>` comment, and it must equal the profile's. Either condition failing stops the run with both ids printed.

76.4 What the run records

`EXHALE_resolved.out` gains `lower_profile_present` and, with a profile in use, the file name, `solution_id`, source and version, `p_match_bar`, `p_top_bar`, the trial fluxes, the iteration index and whether the profile is iterable at all (a hand-written profile with no iteration index runs; it just cannot be driven by the closure loop). It is written on the direct steady route (`EXHALE_PTC=1`) as well now, which it was not before — `element_budget.py` and the closure both read it, and every LHS 1140b case uses that route.

`write_element_flux_profile` gained the hydrogen element, $F_H = 4\pi r^2 (\rho(1-X)v - J)$, on the same faces and from the same X and J the step used, so $F_H + F_{He}$ is the face mass flux identically. The file moved from `./diffusion_faceflux.txt` to `./output/element_flux_profile.txt`, with the other products, and is written unconditionally when a profile is in use rather than only under `EXHALE_DIFFUSION_CHECK=1`.

76.5 The overlap window is empty, and that is a finding

Section 3.4 of the design asks for one number per element: the elemental flux over the overlap window, as a median and a relative radial spread, measured on the escape window `[j_min:N]` intersected with the profile's coverage. That is implemented and it works — a run with `Escape radius [R_p]: 1.02` reports

```
lower_profile_flux_state  measured
lower_profile_flux_nface  59
```

— but `j_min` is the first cell with $r \geq r_{\text{esc}}$, and r_{esc} is $2R_p$ in every configuration that carries diffusion, while a profile reaching from the microbar match to 10^{-8} bar covers about $0.05R_p$ above the base. Two decades of pressure is a few scale heights, not a radius doubling. On every realistic configuration the two intervals do not intersect and the run reports

lower_profile_flux_state window_empty

which is what section 3.4 requires it to do rather than quote a base-cell number. The consequence is for E4, not for E1: the window the closure needs is between the base sound-wave region ($\sim 1.02 R_p$) and the profile top, and its lower edge has to be stated as such instead of borrowed from $j_{\min}$. The design document now carries this as an open E4 item.

76.6 Tests

id	what was run	result
T-E1	examples/17_lower_profile/ lower_atmosphere_profile.dat read by the Fortran reader (EXHALE_PARSE_DUMP=1) and by exhale_io. load_lower_atmosphere_profile, then re-emitted	15 columns carried (including the four the code has no consumer for), re-emission bitwise identical to the file, max abs round-trip error 0 over every column. The reader's match values equal the loader's: T0 = 1.450000000000000E+003, R0 = 9.794530965504000E+009 (= 1.401 R _j exactly), HeH = 8.333333300000000E-002, q_h2_base = 1.196136002775885E-001, p_base_bar = 1.000000000000000E-006
T-E2	make check, no Lower atmosphere profile: key anywhere	==> REGRESSION PASS (all cases byte-identical) — 6/6, 12 file comparisons, goldens untouched
T-E3	mol_diffusion re-run from a scratch copy with base.inp deleted, He_Kzz removed, and a profile whose Kzz column is uniformly 1.0e9, EXHALE_MAXSTEPS=12000 OMP_NUM_THREADS=1	Hydro_ioniz.txt and Ion_species.txt byte-identical to golden/mol_diffusion.Kzz(r) on the grid: 1.000E+09 to 1.000E+09 cm ² /s; the parse dump differs from the base.inp run in exactly one variable, he_kzz (0 with the profile, 1e9 from Kzz_base), because the operator now reads kzz_cell
T-E4	four refusals and one acceptance, each from a scratch run	T_base / O_H_base / Kzz_base beside a profile stop with the key and its category named; a mismatched solution_id stops with both ids printed; a base.inp with no solution_id stops; a base.inp carrying only a matching solution_id and diagnostic comments runs
(schema)	six malformed profiles	each refused with its own message: no overlap (p_top_bar >= p_match_bar), missing solution_id, missing required column, table not strictly decreasing in p, p_match_bar outside the coverage, a short data row
(unit)	make diffusion_tests && ./diffusion_tests.x	28 passed, 0 failed — the T7c homopause tests set he_kzz and now refill kzz_cell through the same routine the code uses

make check was run on the binary built from the state of the tree at the time; the edits made after that build are confined to the profile module behind its `if (len_trim(lap_file) .eq. 0) return, to apply_lower_atmosphere_profile` behind `if (.not. lap_in_use) return`, and to comments, so none of them is on the path the six cases execute.

76.7 Two things fixed on the way

- The face-flux table's # columns: line carried labels with spaces in them ($F_{\text{He}}=4\pi r^2 (F_{\text{adv}}+J)[\text{g/s}]$), so it could not be split into one token per column the way every other EXHALE product's schema line can. It is now one whitespace-free token per column and the formulas moved to the comment lines above it.
- LHS1140b/make_memo_figures.py and LHS1140b/exhale/kzz_scan_table.py read D_{eff} out of that table by fixed position (usecols=(1, 8)), which the new F_H column would have shifted silently. Both now find the file under either name and the column by name, falling back to position 8 for a table written before the schema line was fixed. Stored run outputs were not regenerated.

77. Two producers for the profile handoff: the Photochem adapter and its VULCAN cross-check (2026-08-27)

Milestone E2 of docs/phase_e_flux_closure_design.md. Section 76 gave EXHALE a reader for the lower-atmosphere profile and an example file built by hand; what was missing was a producer. E2 supplies two, behind one command line:

- src/utis/photochem_to_lower_profile.py — solves a gas-giant photochemistry model to steady state on a prescribed $T(p)$, $K_{zz}(p)$ column and writes the handoff. This is the production path.
- src/utis/vulcan_to_lower_profile.py — the same schema and the same fingerprint from a finished VULCAN .vul output. This is the cross-check arm, not a second production path: VULCAN has no climate model, so it can never be the route away from a prescribed $T(p)$ (docs/vulcan_photochem_comparison.md P1.7).
- src/utis/lower_profile_schema.py — everything that is a property of the FORMAT rather than of either chemistry code: the shared command line, the elemental accounting, the hydrostatic radius, the level insertion that puts an exact node at the matching pressure, the solution_id and the two files that carry it. It exists so the two adapters cannot drift apart; a schema written twice is a schema that will be written differently twice.

The scalar generator src/utis/vulcan_to_base.py stays. It writes the single-level base.inp special case that four regression cases pin, and the two must not be read into the same run: with a profile in use EXHALE refuses every scalar physics key beside it (section 76).

77.1 One way, and the file says so

E2 is the one-way milestone. The trial elemental fluxes are recorded in the header and in the fingerprint, and they are **not** imposed as an upper boundary condition on the chemistry; imposing them, measuring the wind's own elemental flux over the overlap and iterating the pair is E4. The F_H and F_He columns therefore carry the stated trial values, and the header notes line says so in the file itself, so that no reader can mistake a boundary value for a measurement:

```
# notes one-way E2 handoff: F_H and F_He are the STATED trial fluxes, not
    measured; no climate step, T(p) prescribed.
```

77.2 What the fingerprint is over

`solution_id` is a sha256 over the lower model's configuration: the source code and version, the mechanism and thermodynamic files by their own sha256, the stellar flux file by its sha256 and the dilution applied, the prescribed $T(p)$ file, the planet mass and reference radius, the zenith angle, the top-of-atmosphere pressure, the elemental abundance vector, the matching pressure and the two trial fluxes. The closure *iteration index* is deliberately outside it: two iterations handed the same configuration and the same trial fluxes are the same solution, and the fingerprint is what says so. The same id is written into the paired `base.inp`, which carries **no physics key at all** — the provenance comment is its whole content, and any scalar key in it would stop the run by design.

77.3 Elemental accounting, and the trap that makes it a rule

Element columns are E/H **nuclei** ratios summed over every carrier, the lesson `vulcan_to_base.py` already records: photochemistry moves nuclei between molecules without creating or destroying them, so the nuclei ratio is the conserved quantity the wind inherits and the only one `melem_ab` can hold.

How the carriers are counted differs by code, and it has to. VULCAN labels are formulas once the ionization sign and the excited-state suffix are stripped (H3^+ , CH2_1), so they are parsed. Photochem labels are **not**: O1D and N2D are excited states, and a formula parser reads them as an oxygen with a deuterium attached. The Photochem arm therefore takes the composition from the model's own matrix (`dat.species_composition`), indexed by name against `dat.atoms_names` — which is also the trap P1.8 item 4 records, since that atom order changes from run to run. Condensed carriers are excluded by default: the gas EXHALE's base inherits is the gas phase, and the difference between the two sums is the cold trap, which is E3's subject.

The adapter's own conservation check (design section 4.2) is stated at the deepest level, because that is the only level where the elemental ratios are still the input ones — transport and the upper boundary move nuclei with height, which is the whole point of the closure. The design proposed a tolerance of 10^{-10} . Measured on HD 209458 b, the departure is 1.6×10^{-7} with the Zahnle H/He/N/O/C set and 2.7×10^{-5} with the VULCAN NCHO network converted through `vulcan2yaml`, i.e. chemistry-solver noise and not a miscount, so the default is 10^{-4} : loose enough not to refuse a converged solution, tight enough that a miscounted carrier (an $O(1)$ or percent-level error) is still caught. The measured worst departure is printed on every run.

77.4 The radius column

`r` is the adapter's own hydrostatic integration from a stated reference level (`--r-ref` at `--p-ref`, 1 bar by default, the transit-radius convention), RK4 in $\ln p$ with radius-dependent gravity, refined 64x per level so the answer does not depend on how the chemistry code spaced its grid. The mean molecular weight enters as `mu * m_amu`, the mass of a particle of `mu` atomic mass units.

77.5 Trap 1 was met and is handled the way P1 handles it

`gasgiants` requires $3 \times \text{TOA_pressure_avg} < \text{P_climate_top}$. On the HD 209458 b column the supplied $T(p)$ stops at $2.152 \times 10^{-2} \text{ dyn/cm}^2$ while the requested TOA was 10^{-2} , so the requirement fails on the file as given. The

adapter keeps the stated TOA and cuts the climate column back to $3.05 \times \text{TOA}$, which is what `read_climate` does in `docs/p1_matched_comparison.py`; the alternative — silently lowering the TOA to fit the file — was tried first and rejected, because it changes a stated input without saying so. Trap 2 (the thermodynamic data capping the temperature) is handled by **truncating** the column from the top down, never by clipping T , because a clipped $T(p)$ is a profile nobody solved; the truncation is recorded in notes, and if it falls below `p_match_bar` the run is refused. HD 209458 b needed no truncation: its 5891 K top was accepted.

77.6 Failure policy

No profile is written when the handoff would not be one. `reached_steady_state = False` or `gave_up = True`, a temperature truncation that reaches below the match, a climate cut that leaves fewer than ten levels, `p_top_bar` $\geq$ `p_match_bar`, a matching pressure outside the table, fewer than two levels after the requested window, and the elemental conservation check are all refusals that print the reason and write nothing.

77.7 Measurements

HD 209458 b, the P1 configuration ($M_p = 0.720 M_J$, $r(1 \text{ bar}) = 1.36 R_J$, $R_* = 1.155 R_\odot$, $a = 0.0480 \text{ AU}$, Gueymard solar spectrum diluted, zenith 48 deg, TOA 10^{-2} dyn/cm^2 , VULCAN's own `atm_HD209_Kzz.txt` as the prescribed $T(p)$, $K_{zz}(p)$), matching level 1 microbar. Both Photochem arms reached steady state; the VULCAN arm is its stored HD209 output.

at <code>p_match = 1e-6 bar</code>	VULCAN / NCHO	Photochem / NCHO	Photochem / Zahnle H,He,N,O,C	PC/VUL, same network
T [K]	2332.25	2332.28	2332.43	1.0000
r [R_J]	1.495029	1.494127	1.494022	0.9994
n_{tot} [cm^{-3}]	3.1098×10^{12}	3.1095×10^{12}	3.1124×10^{12}	0.9999
ρ [g cm^{-3}]	9.4403×10^{-12}	9.6497×10^{-12}	9.7202×10^{-12}	1.0222
K_{zz} [$\text{cm}^2 \text{ s}^{-1}$]	5.000×10^{11}	5.000×10^{11}	5.000×10^{11}	1.0000
q_{H2}	0.42366	0.45363	0.46266	1.0707
q_{H}	0.44987	0.41710	0.40726	0.9271
X_{He}	9.6915×10^{-2}	9.6912×10^{-2}	9.6912×10^{-2}	1.0000
X_{C}	2.7745×10^{-4}	2.7754×10^{-4}	2.7754×10^{-4}	1.0003
X_{N}	8.1809×10^{-5}	8.1831×10^{-5}	8.1828×10^{-5}	1.0003
X_{O}	5.0329×10^{-4}	6.0608×10^{-4}	6.0608×10^{-4}	1.2042
q_{H2O}	2.2073×10^{-4}	3.4312×10^{-4}	3.5306×10^{-4}	1.5545

The matched-network hydrogen ratio is $q_{\text{H}}(\text{VULCAN})/q_{\text{H}}(\text{Photochem}) = 1.079$, which reproduces the **1.08** P1 measured for the code factor on this planet's 2331 K base (`docs/vulcan_photochem_comparison.md` P1.7) — the adapters carry the same disagreement the direct comparison found, and add none of their own. T , n_{tot} and the two element ratios the codes were given identically agree to 10^{-4} or better.

X_{O} and q_{H2O} differ by 1.20x and 1.55x for a reason that is not a code difference: the stored VULCAN HD209 run was configured with $\text{O}/\text{H} = 5.03\text{e-}4$, against the Lodders 2009 $6.06\text{e-}4$ the Photochem arms were given. That is

a difference of input, and it is what the X_{El} columns are for — it would have been invisible in a comparison of molecular mixing ratios alone.

77.8 Fingerprint round trip

test	result
the same input twice, VULCAN arm	same solution_id (4f80f048...), and the profile files are byte-identical
the same input twice, Photochem arm (two full chemistry solves)	same solution_id (f6232692...), and the profile files are byte-identical : the chemistry solve reproduces, so the fingerprint and the table agree
--trial-flux-H 5e10 instead of 0	0a4d2051..., different
--p-match 2e-6 instead of 1e-6	ceed0b58..., different
--iteration 3 instead of 0	unchanged, by design: the iteration index is bookkeeping, not configuration
the paired base.inp	carries the same id as the profile, and no physics key

77.9 EXHALE reads what the adapter wrote

A scratch case built the way examples/17_lower_profile/ is (the same input.inp, the Photochem HD 209458 b profile and its paired base.inp in place of the synthetic ones), run single-threaded with EXHALE_PARSE_DUMP=1 and then with EXHALE_MAXSTEPS=20:

```
(lower_atmosphere_profile) Reading lower_atmosphere_profile.dat ..
  base.inp: solution_id matches the profile.
  profile: T0 ->    2332.4 K           profile: q_H2(base) ->  0.46266
  profile: R0 ->    1.4940 R_J         profile: He/H ->    0.09691
  profile: p_base ->  1.00E-06 bar     profile: C/H ->    2.775E-04
  profile: source photochem, closure iteration 0
(lower_atmosphere_profile) K_zz(r) on the grid:  5.000E+11 to  5.000E+11 cm^2/s
```

Every echoed value is the profile's own number at the matching level ($T = 2332.43$ K, $r = 1.494022$ R_J, $q_{\text{H}_2} = 4.626610\text{e-}1$, $X_{\text{He}} = 9.691225\text{e-}2$, $X_{\text{C}} = 2.775366\text{e-}4$), and EXHALE_resolved.out carries the provenance: lower_profile_solution_id f6232692..., source_code photochem, p_top_bar 7.44e-9, iterable T, steady T. The paired base.inp — provenance comments only — was accepted, and the id check passed rather than being skipped.

The run itself is a 20-step relaxation snapshot and its $\dot{M}$ is not a result. One thing it does show: lower_profile_flux_state window_empty, the same finding E1 recorded — with Escape radius [R_p]: 2.00 the escape window [j_min:N] and the profile's coverage do not intersect, so the elemental flux has no window to be measured on. That is the open item E4 inherits, restated here on a real photochemical profile rather than a synthetic one.

77.10 What is not done

The LHS 1140 b application and T-E8 (element_budget.py closing on a profile run) are E2's stated gate and neither is done here; the adapters were exercised on HD 209458 b because that is where P1's measurement exists to check them against. The climate step, the tropopause and the cold-trap water are E3, and the flux closure is E4.

78. The climate step, the cold trap, and LHS 1140 b through the profile (2026-08-27)

Milestone E3 of docs/phase_e_flux_closure_design.md, and with it the two gates E2 left open: the LHS 1140 b application and T-E8.

Until now the handoff's $T(p)$ was an input. --climate makes it a solution:

```
<photochem 0.8.4 env>/bin/python \
  src/utis/photochem_to_lower_profile.py LHS1140b/lower_profile \
  --mp 0.0176220 --r-ref 0.157692 --p-match 1e-6 --kzz-const 1.0e9 \
  --stellar-flux LHS1140b/sed/lhs1140_sed_gj1132_at_b.txt \
  --flux-at-planet --wavelength-unit A \
  --atoms H,He,N,O,C --abundances He=2.09 \
  --climate --climate-p-deep 20.0 --toa 1.0e-2 --boa-pressure-factor 1.0
```

src/utis/radiative_convective_column.py is the new module: Photochem's clima (AdiabatClimate) with solve_for_T_trop on, so the stratospheric temperature is the skin temperature of the solution and not a number chosen for it, and with the background gas picked by abundance rather than by name — which is what lets one code path cover both a hydrogen atmosphere and the helium-rich one the LHS 1140 b line implies. The elemental vector is partitioned into the carriers a cool H/He atmosphere holds in equilibrium (O in H₂O, C in CH₄, N in N₂, He atomic, the rest H₂) with the validity range written at the code and a refusal above 1000 K, because that partition is the equilibrium one only below the CO/CH₄ and N₂/NH₃ transitions. Nothing downstream inherits it: the photochemistry computes its own composition from the same elemental vector.

78.1 What the codes actually do about condensation, measured

The design proposed enabling the mechanism's H₂O particle and setting rainout to the tropopause. Both halves turned out wrong for a gas giant, in opposite directions, and both were measured rather than assumed:

- the H₂O particle needs no enabling. The Zahnle set restricted to H/He/N/O/C already carries H2Oaer, and asking for water-condensation: true is **refused** — “Either “fix-water-in-troposphere” or “water-condensation” is turned on in the settings file, but the reaction mechanism already implements H2O condensation via a particle.” The cold trap is applied by the chemistry solver already; the settings switch must stay false.
- rainout is a surface process: gas-rainout: true is refused for want of a rainfall-rate, and a gas-giant model has no surface for rain to fall to. It is not how a hydrogen or helium atmosphere traps its water.

Section 5 of the design now records this correction beside the procedure.

78.2 The tropopause is solid, its pressure is not

Scanned over the one free parameter of the climate solve, its deep boundary:

deep boundary [bar]	T_{deep} [K]	P_{trop} [bar]	T_{trop} [K]	$f_{\text{H}_2\text{O}}$ above the tropopause
1	242.4	0.488	186.0	3.33×10^{-7}
3	308.5	0.765	185.8	2.06×10^{-7}
10	439.9	0.974	185.4	1.49×10^{-7}
20	555.9	1.031	185.0	1.32×10^{-7}
30, 50	refused: the pseudoadiabats leaves the thermodynamic data			

T_{trop} moves by 1 K over a factor of 20 in the deep boundary, against the skin temperature $T_{\text{eq}}/2^{1/4} = 190$ K. P_{trop} moves by a factor of two, and the trapped water moves with it, since at fixed T_{trop} the trapped mixing ratio is $p_{\text{sat}}(T_{\text{trop}})/P_{\text{trop}}$.

Against Cherubim et al. (2026). Their closed-form estimate is $f_{\text{H}_2\text{O}} \sim 7$ ppm at a 0.1 bar tropopause with $T_{\text{skin}} = 194$ K. The climate solution gives **0.13–0.33 ppm at a tropopause of 0.5–1.0 bar** — a factor 20–50 less water at a tropopause 5–10x deeper. The saturation physics agrees; what differs is where the tropopause is put, and ours is a solved level in a helium-dominated column ($\mu = 3.6$ at $\text{He}/\text{H} = 2.09$) rather than an assumed one. Reported as a result about the climate model, not tuned toward the published number.

78.3 What the trap does to the handoff

Gas-phase oxygen falls from $\text{O}/\text{H} = 6.062 \times 10^{-4}$ at the deep boundary to 4.957×10^{-7} at the matching level, a factor of 1223, while C/H , N/H and He/H cross the tropopause unchanged: CH_4 and N_2 do not condense at 185 K. The profile now always computes the elemental sums twice, over the gas phase and over every carrier including the condensates, and carries both in the header notes — at the match the two coincide, because the condensate is left behind at the tropopause. In this handoff the cold trap shows as a depletion with height, not as a condensed reservoir sitting at the base.

The 20 bar deep boundary is not a physical choice: it is the shallowest one at which the photochemical model bottom fits inside the climate column. On an atmosphere this cold the chemistry never equilibrates within any column clima can reach, so `determine_quench_levels` returns the deepest level of the grid and Photochem's default `BOA_pressure_factor = 5` then asks for a bottom five times deeper than the column it was handed. `--boa-pressure-factor` exposes it and the adapter's refusal text names it.

The prescribed-column path of E2 is unchanged where it matters: re-run on the HD 209458 b configuration of section 77 it reproduces that section's numbers exactly ($T = 2332.43$ K, $r = 1.494022$ R_J, $q_{\text{H}_2} = 4.626610\text{e-}1$, $X_{\text{He}} = 9.691225\text{e-}2$). Its `solution_id` does change, because `boa_pressure_factor` is now part of the configuration the fingerprint is over, which is the fingerprint doing its job: a profile made with a different model bottom is a different solution. Files already on disk are untouched.

78.4 LHS 1140 b through the profile: it converges, and T-E8 closes

LHS1140b/lower_profile/lower_atmosphere_profile.dat (102 levels, 16.73 bar to 1.10×10^{-8} bar, solution_id 51302c76...) drives a direct-steady JFNK run, seeded from exhale/heh2_diff_kzz1e9, to info = 0 with the element-diffusion outer loop converged (composition drift 9.1×10^{-4} at pass 8). What the reader put in agrees digit for digit with what the file says at the match, and EXHALE_resolved.out carries it:

planet_radius_RJ	1.623994484710662E-001
equilibrium_temperature_K	1.850492232658783E+002
HeH_number_ratio	2.092134697681555E+000
abundance_C	2.777529772010253E-004
abundance_O	4.956907073555432E-007
abundance_N	8.187442081270519E-005
lower_profile_solution_id	51302c7647a319bdfbd142d3b4bf0a367b25d2dc07e2675c8da3cd430dc63177

lower_profile_notes is now echoed there too, so the climate record — deep boundary, tropopause, cold-trap water, both elemental sums — travels with the run and not only with the file (lap_notes widened to 1000 characters to hold it).

T-E8. src/utls/element_budget.py closes on every element the profile carries: C 1.44×10^{-14} , N 1.47×10^{-14} , O 1.45×10^{-14} , against a 10^{-8} tolerance, and on H (0.0) and He (2.1×10^{-16}) at the base. *All element budgets close within 1.0×10^{-8} .*

The H and He rows needed a fix in the checker, not in the run. With He_diffusion on, He/H is a *solved profile* — 2.0921 at the base, 1.206 at $1.15 R_p$, 0.167 at $30 R_p$ — and the resolved HeH_number_ratio is the reservoir the base is held at, not a column invariant. The tool was testing it cell by cell and calling the separation the operator exists to produce a failure. EXHALE_resolved.out now states he_diffusion, and with it on the H and He rows are checked at the base cell, with the solved separation printed beside them. The trace elements are unaffected and stay column-wide: nothing diffuses them.

The Fortran side of E3 is two writes and one string length — lap_notes widened to 1000 characters, echoed into EXHALE_resolved.out, and he_diffusion stated there — none of which any solver reads. make check is **6/6 byte-identical**, goldens untouched.

78.5 Profile route against the scalar base, and where the difference lives

Beside exhale/heh2p13_diff_kzz1e9, the converged scalar-base case at the same eddy coefficient. Every row below is a JFNK info = 0 run finished the same way, red being the He 10830 red-pair depth from he_line_metrics.py:

case	what it adds	$\log_{10} \dot{M}$	red
heh2p13_diff_kzz1e9	scalar base, He/H = 2.13, T ₀ = 226 K	7.80	4.396
control A	HeH_base = 2.0921	7.81	4.393
control B	+ T_base = 185.05 K	7.81	4.389
control C	+ r_base = 0.162399, q_H2_base = 0.19265	7.86	4.985
control D	+ C/N/O_H_base from the profile	7.46	1.517
profile route	the profile itself	7.46	1.517

Two things fall out. Control D and the profile route agree, which is the consistency statement the single-source rule of section 2.4 is supposed to buy: handed the same base state, the two routes are the same run. (*They agreed to every digit printed here. Re-solved on the section-87 coefficient the agreement is 0.01 dex in $\log_{10} \dot{M}$ and 1 per cent in the line — inside the workflow’s own 0.3–1.5 per cent path spread, and the two rungs were solved at different residual tolerances. Agreement to every printed digit was the tolerance, not the physics.*) And the difference from the scalar LHS 1140 b case is **not** the base temperature — dropping T₀ from 226 K to the solved 185.05 K moves $\log_{10} \dot{M}$ by less than 0.01 dex — but the **elemental carbon, nitrogen and oxygen the profile brings with it**, which the scalar runs never had: −0.40 dex in $\dot{M}$ and a 3.3x shallower helium line, metal-line cooling of a 185 K base at C/H = 2.8e−4. (*Re-measured on the García Muñoz rate of section 87 (LHS1140b/exhale/basemetals_gm25/), that step is −0.37 dex and a factor 2.89 in the red-pair depth at R = 68,000 (2.88 at the R = 80,000 these were read at). The two numbers here also belong to different pairs of runs: −0.40 dex is the C → D rung, while the factor quoted as 2.9 elsewhere is ms → D, the whole ladder, which is 2.59 on the new coefficient.*) p_base is 1 microbar on both sides and is not a candidate.

78.6 What is not done

lower_profile_flux_state window_empty again: with Escape radius [R_p]: 2.00 the escape window and the profile’s coverage do not intersect, so T-E5 still has no window to be measured on. That is E4’s, together with the flux iteration itself. The composition here remains an input (He/H = 2.09, the diffusion-limit crossing of LHS1140b/kzz_decision.md section 6), not yet an output.

78.7 Beside it: a particle is not a hydrogen atom (vulcan_to_base.py)

Unrelated to E3 and found while reading the hydrostatic integrations. src/utils/vulcan_to_base.py integrated $dr/d\ln p = -kT/(\mu m_H g)$ with the hydrogen ATOM mass, while its mu_profile is a mean molecular weight in **amu**, built from VULCAN’s own species masses (H₂ = 2.016, He = 4.003, H₂O = 18.02). The mass of one particle is mu * m_amu; using m_H makes every particle 0.8% heavier and the scale height 0.8% shorter. Measured on the HD 209458 b VULCAN output, r_base moves from 1.49388 to **1.49504 R_J** — 0.078% of the planet radius, 0.86% of the column above 1 bar, which is the 0.8% arriving where it should. Fixed; lower_profile_schema.py already used m_amu and recorded the discrepancy rather than copying it.

The pinned base.inp of the regression cases were deliberately **not** regenerated: they are frozen inputs, and the generator’s fix does not retroactively change a run that was made with the old file.

run_lower.py and lower_column.f90 multiply the same m_H by a mean molecular weight built from integer mass numbers ($\text{mass} = 1 + 4 \text{ f_He}$), where $m_H = 1.008 \text{ amu}$ is the better of the two constants for the H_2 that dominates that mixture. They are left alone; the mixed convention is noted here so the next reader does not “fix” them into being worse.

79. The elemental-flux closure: the composition becomes an output (2026-08-27)

Milestone E4 of docs/phase_e_flux_closure_design.md, and with it the last three of Phase E’s tests, T-E5 through T-E7.

Until now the lower model was told what to do and never heard back. E2 and E3 carried trial_flux_H and trial_flux_He in the profile header but only as statements: nothing imposed them on the chemistry, so the handoff was one-way and He/H was still an input the wind reproduced. E4 closes the loop in both directions.

79.1 The trial flux now reaches the chemistry

photochem_to_lower_profile.py imposes each element’s trial flux at the photochemical model top with set_upper_bc(species, 'flux', flux=...), the elemental flux converted at the model’s own top radius,

$$\phi_{\text{El}} [\text{nuclei}/\text{cm}^2/\text{s}] = \Phi_{\text{El}} [\text{g}/\text{s}] / m_{\text{El}} / (4\pi r_{\text{top}}^2) \quad (27)$$

and placed on the dominant carrier: helium on He, hydrogen on H_2 at $\phi_H/2$, two nuclei to the molecule. The dominant-carrier step is measured rather than assumed — the atomic-H share of the hydrogen nuclei at the model top is 1.0×10^{-4} in the converged LHS 1140 b solution — and the adapter refuses a solution in which that share has risen past 1 per cent, because the flux would then have been put on the wrong carrier. The sign convention was read off Photochem’s own right-hand side (photochem_evoatmosphere_rhs.f90, rhs(k) = ... - var%upper_flux/dz): positive is a loss at the model top, which is this code’s outward. A zero trial flux does not call set_upper_bc at all, so E2’s and E3’s one-way profiles stay reproducible; that was checked by reproducing solution_id 51302c76... bitwise.

That the condition takes is checked every iteration instead of trusted. The adapter measures the model’s own top-of-atmosphere elemental fluxes with gas_fluxes() and writes them into the header as measured_flux_H and measured_flux_He. On the reference arm the imposed $1.800000 \times 10^7 \text{ g/s}$ comes back as $1.800162 \times 10^7 \text{ g/s}$, a departure of 9×10^{-5} ; the closed-top solution’s own numerical floor is 3.0 g/s in hydrogen and 18.2 g/s in helium, seven decades below.

--p-top-bar states the shallowest level to carry directly in bar, and refuses a top at or above p_match_bar (no overlap) or a simultaneous --toa, which would be the same level said twice in two units.

79.2 The window the closure is measured on, and why the obvious one fails

E1 left an open item: the escape window [j_min:N] and the profile’s coverage never intersect, so lower_profile_flux_state was window_empty on every configuration. E4 replaced the single window with two,

and the choice between them is a measurement.

The overlap window is the interval both models describe. Its upper edge is $r(p_{\text{top}})$; its lower edge is now measured — the first face at which the 5-face moving spread of $r^2\rho v$ falls below 10 per cent, which is where the standing base sound wave stops dominating — with a $1.02 R_p$ fallback that `lower_profile_flux_r_lo_source` reports when no face qualifies. Both edges, the face count and the reduction go into `EXHALE_resolved.out`.

On LHS 1140 b that window cannot carry the closure, and extending the profile is not the fix. Where it is empty it misses by $1.6 \times 10^{-4} R_p$ ($r_{\text{lo}} = 1.0099998$, $r_{\text{hi}} = 1.009840$); where it is not empty it holds 12–18 faces between 1.007 and $1.0098 R_p$ and the flux across them has a radial spread of **13 in F_{H} and 35 in F_{He}** , with a median of $1.4\text{--}1.8 \times 10^8$ g/s against the 3.16×10^7 g/s the wind actually removes. The reason is geometric: from the match to the profile top the column covers 1.96 decades of pressure in $0.0098 R_p$, about $0.005 R_p$ per decade, while the wind is still 6x above its far-field flux at $1.05 R_p$ and does not settle within 10 per cent until $\sim 1.5 R_p$. Reaching $1.5 R_p$ on that scale would ask the photochemical column for of order 100 further decades of pressure. Mapping p_{top} through EXHALE’s own $r(p)$ rather than the profile’s — the two hydrostatic scales differ, the wind being hot where the cold column is not — moves the upper edge to $1.0734 R_p$ and makes the window non-empty, but its spread is then 5.6 to 7.3. Both mappings say the same thing: the interval the two models share lies wholly inside the region where the wind’s mass flux is not yet its own.

The steady-flux window, $r \geq r_{\text{esc}}$, is what the closure uses. At a steady state the elemental flux $4\pi r^2(\rho X v + J)$ does not depend on radius, so a flux measured there *is* the flux through the matching level; that identity, and not convenience, is what makes it a statement about the handoff. On the reference arm it holds 191 faces and gives $F_{\text{H}} = 1.8143 \times 10^7$ g/s, $F_{\text{He}} = 1.3204 \times 10^7$ g/s with radial spreads of 0.44 and 0.46 per cent, flat to the same tolerance as the mass flux itself (0.45 per cent). That is T-E5, met in the window where it can be met. The driver reads the overlap window first every iteration and substitutes the steady one only with the reason written into `closure.log` and the window named in the history table; it never substitutes silently.

The spread depends on how far the wind was converged, and that was measured rather than assumed: at `Resid tol: 1.0e-3`, the value `finish_case.sh` uses, the same solution’s steady-window spread is **7.4 per cent** (F_{H} 7.1, F_{He} 7.9) and would fail the closure’s own validity test; continuing the same solution to `1.0e-4` brings it to **0.50 per cent** and moves $\log_{10} \dot{M}$ from 7.47 to 7.50. The 7 per cent was under-convergence, not a floor, and the closure runs the wind at `1.0e-4`.

79.3 The driver

`src/utils/element_flux_closure.py` owns the loop and nothing else. Damped Picard on both elements with $\omega = 0.5$, halved to a floor of 0.125 on any iteration whose residual fails to fall, the convergence test reading the **undamped** residual $|F_{\text{measured}} - \Phi_{\text{trial}}|/|F_{\text{measured}}|$, `tol = 0.05` and $k_{\text{max}} = 8$. Both hydrogen and helium are iterated: holding helium at zero, as the design first proposed, would leave the H:He partition — the whole question on this planet — as a diagnostic rather than a condition.

Section 6.2's validity rule is a statement about the *window*, not the *iterate*: a window whose spread exceeds `tol` cannot decide a residual at `tol` whatever that residual is, so `spread > tol` stops the run as UNRESOLVED with both spreads printed, every iteration and independently of how far the *iterate* is.

Everything that belongs to the planet rather than the iteration lives in one JSON configuration (`--print-config-template`), so the driver carries no planet constants; `LHS1140b/exhale/flux_closure/lhs1140b_closure.json` is the LHS 1140 b one. `closure_history.txt` carries one row per iteration with both trial and measured fluxes, both residuals, ω , the window used and its spreads, He/H at the match, $\log_{10} \dot{M}$, the EXHALE info, and the profile's `solution_id` — the fingerprint chain, so the sequence is auditable after the fact. `--resume` continues from the last completed iteration and `--dry-run` exercises the reader against an existing run directory.

Two departures from `finish_case.sh`, both measured. `Resid tol: 1.0e-4`, for the reason above. And the post-processing pass runs **without** the EXHALE_PTC* variables: `finish_case.sh` sets them inline on the JFNK command alone, and carried into the second pass they make the binary re-enter the steady solver and stop before writing the advection-corrected profiles, so `*_adv.txt` and the mass-loss rate never appear. The first LHS 1140 b closure run reported `log10 Mdot = nan` and that is what it was.

79.4 LHS 1140 b, closed

Three arms, from $\Phi_{\text{ref}} = (1.8 \times 10^7, 1.3 \times 10^7)$ g/s in (H, He) and from $0.3 \times$ and $3 \times$ that, all seeded from the same converged wind, all with He/H = 2.09 as the *starting* reservoir.

start	k	F_{H} [g/s]	F_{He} [g/s]	He/H at the match	$\log_{10} \dot{M}$	He 10830 red depth [%]	EW [mÅ]
1.0 x	0	1.81433×10^7	1.32040×10^7	2.0923516	7.500	1.699	4.743
0.3 x	5	1.82031×10^7	1.34266×10^7	2.0923486	7.500	1.721	4.804
3.0 x	6	1.82119×10^7	1.34710×10^7	2.0923602	7.500	1.725	4.818

The three converged He/H agree to 5.5×10^{-6} , $\log_{10} \dot{M}$ to better than the 0.005 the log prints, F_{H} to 3.8×10^{-3} and F_{He} to 2.0×10^{-2} — all far inside the 5 per cent criterion, so **T-E7 passes and the closure is single-valued on this planet**. ω never halved in any arm: the residual fell on every iteration, so the map is a plain contraction here and the damping rule was not exercised. The iteration counts are exactly what a contraction at $\omega = 0.5$ predicts, the residual halving each step, which also says the 3x arm used six of its eight allowed iterations with no margin: a start further than an order of magnitude out would need a larger k_{max} or a secant acceleration.

The result to read, though, is what the closure returns. He/H at the match comes back at 2.09235 against the 2.09 the reservoir was started from — a change of 1.1×10^{-3} , and only 2×10^{-4} of that from the escape flux itself. The composition is now an output, and on this planet the output is the *well-mixed* value: at $K_{\text{zz}} = 10^9$ cm²/s eddy mixing homogenizes the column all the way to 1 microbar, so a 1.8×10^7 g/s hydrogen escape imposed at the model top moves the matching-level He/H by 2×10^{-4} while moving it by 1.8×10^{-2} at the model top itself. It is the eddy coefficient, not the escape flux, that sets the composition at the match. That is a result about this configuration and not a general one, and it makes `LHS1140b/kzz_decision.md` section 0's adopted K_{zz} the quantity the answer now rests on.

Above the match the wind fractionates strongly, as it did before: the elemental fluxes carry $F_{\text{He}}/F_{\text{H}} = 0.73$ by mass against the base reservoir's 8.31, and He/H falls from 2.09 at the base to 0.183 at $30 R_p$. The closure does not weaken that; it says the separation happens in the wind and not below the match.

The flux budget. $F_{\text{H}} + F_{\text{He}}$ reproduces the mass flux identically — exactly, in the two arms whose medians fall on the same face — because the binary operator's X is the helium mass fraction of the H+He mixture and the trace metals ride inside ρ . So the measured F_{H} is hydrogen *plus* metals, overstating elemental hydrogen by the metal mass share, which is 4.79×10^{-4} here (mass_per_H = 9.3739 amu, C/N/O at 2.778×10^{-4} , 8.188×10^{-5} , 4.955×10^{-7}). That is two decades below tol and is stated rather than corrected.

79.5 T-E8, and one thing it caught

The element budget closes for helium at the base cell, where the reservoir is the boundary condition, to 1.5×10^{-15} . Hydrogen, carbon, nitrogen and oxygen miss by 5.6×10^{-8} , 1.2×10^{-4} , 1.2×10^{-4} and 3.3×10^{-4} against the checker's 10^{-8} , and the miss is not noise: **it equals the change in the elemental reservoir between the seeding solution and the run.** load_IC restores every metal density the restart file carries and rebuilds an element from melem_ab only when the file does not carry it at all (load_IC.f90:283-334), so an iteration that changes a reservoir keeps the seed's value everywhere above the base and EXHALE_resolved.out then reports a reservoir the run did not use above the base cell. It was seen at full size first: an iteration seeded across a 22 per cent change in O/H reported exactly 17.8 per cent, which is that change expressed on the new reservoir.

This was **not** fixed at the time, because the fix changes restart semantics, which is a user-visible behavior and expensive to reverse; it was recorded for a decision. Its size on the result reported here is small and measured: the closure moves reservoirs by at most 5×10^{-4} between iterations, and the coolant that matters here is C I, which carries 94 per cent of the radiated energy and is stale by 1.2×10^{-4} ; oxygen, the worst row at 3.3×10^{-4} , carries under 10^{-4} of the cooling on this planet. The decision was taken and the fix implemented on 2026-08-27: section 80.. The numbers quoted in this section are what the code gave **before** it, and are left as they were measured.

79.6 T-E9: not verified, and why

Not verified: VULCAN did not converge on this cold He-rich column (longdy rising $2.65 \rightarrow 2.75$ against yconv_cri = 0.01, dt pinned at 1.85×10^3 s, the model clock at 1.42×10^7 s, limiting cell nz = 72 and H2O — the cold trap — exiting on “Maximal allowed steps exceeded (10000)”); stopped 2026-08-27.

The cross-check arm was built on the production profile's own $T(p)$ and K_{zz} and never produced a profile, so there is no VULCAN composition at the match, no wind run on one, and no cross-code difference to state beside the P1 spread. LHS1140b/exhale/flux_closure/vulcan_arm/ holds only an input.inp and the seeded initial condition; the binary was not launched. The full record, including the second run that reached step ~500 of 30000 before being stopped and therefore says nothing either way, is LHS1140b/lower_profile_vulcan/vulcan_work/STOPPED.txt.

Two things found on the way that outlive the attempt. First, a real defect in src/utls/vulcan_to_lower_profile.py, now fixed: sch.formula_elements strips the phase suffix, so H2O_l_s parsed as a second H2O and condensed carriers

were counted as gas, which made the cold trap invisible in `X_0` whatever `--count-condensates` was set to. The gas phase is now selected by the `.vul`'s own `atm['gas_indx']`, condensate is excluded from μ always as VULCAN itself does, and the flag now selects. The Photochem adapter and `lower_profile_schema.py` are untouched, so the production arm and its `solution_id 5130...` are unaffected. Second, a configuration trap worth stating because it silently produces a wrong atmosphere rather than an error: `vulcan_cfg.He_H` is NOT read on the FastChem `ini_mix = 'EQ'` path (`build_atm.ini_fc` builds the element file from `atom_list` minus H), so helium must be in `atom_list` or the run keeps the solar He 10.9864 and integrates at $\text{He}/\text{H} = 0.097$. The first run did exactly that.

Whether VULCAN can be made to converge here is open. It is not a blocker for Phase E: Photochem is the production path precisely because it carries the climate model this planet needs, and T-E5 through T-E7 are closed without it.

80. A restart onto a reservoir the handoff has moved (2026-08-27)

Section 79. left T-E8 closing for helium and missing for hydrogen, carbon, nitrogen and oxygen by 5.6×10^{-8} , 1.2×10^{-4} , 1.2×10^{-4} and 3.3×10^{-4} , and named the cause: `load_IC` restores every metal density the restart file carries, so a flux-closure iteration that moves an elemental reservoir keeps the seed's value everywhere above the base cell while the boundary condition already uses the new one. The fix was deferred there because it changes restart semantics, which is user-visible. The decision was taken on 2026-08-27; this section is what was implemented and what it measures.

80.1 The rule

An element is renormalized as it is loaded if, and only if, the lower-atmosphere handoff states its reservoir — a `<El>_H_base` key of `base.inp`, or an elemental ratio of the file named by `Lower atmosphere profile:`. For such an element the whole loaded column is multiplied by one factor,

$$r_{\text{El}} = \frac{(\text{El}/\text{H})_{\text{stated}}}{(\text{El}/\text{H})_{\text{loaded state, base cell}}}, \quad (28)$$

the same factor for every ionization stage. The normalization moves and nothing else does: the ionization split of each cell and the radial shape of the loaded profile are preserved exactly, because a factor common to all stages and all cells cancels out of every ratio the state is made of. El/H at the base cell is counted in nuclei — the element summed over its stages against the hydrogen nuclei carried by H I , H II , H_2 , H_2^+ , H_3^+ and HeH^+ — which is the convention of `element_ratio_HeH` and of `src/utls/element_budget.py`, so the factor is measured in the same units the budget check reads it back in.

Helium is not part of this rule and keeps its own, older convention: the base cells are set to the input He/H , and with `He_diffusion` on the column above keeps the diffused split, because that split is the state being restarted rather than a defect of the file.

What decides “the handoff states it” is not a test on the numbers. Both readers set an abundance through one routine, `set_element_abundance` in `input_read.f90`, and passing through that door is what marks the element in the new `melem_from_handoff(n_melem)` of `global_parameters`. An abundance that reached the run from `metals.inp` alone never passes through it, so it is not marked and its column is loaded untouched. There is no tolerance and no `abs(r - 1) > tiny` guard: the branch is entered or it is not, and when it is not entered the arithmetic is never performed.

Each rescaled element prints its own line, with the loaded ratio, the stated ratio and the factor, so the factor can be compared against the reservoir change it is supposed to equal:

```
(load_IC) C/H at the base cell: restart file 2.777530E-04, handoff 2.777859E-04; column rescaled by 1.000118E+00
(load_IC) O/H at the base cell: restart file 4.956907E-07, handoff 4.955292E-07; column rescaled by 9.996741E-01
(load_IC) N/H at the base cell: restart file 8.187442E-05, handoff 8.188411E-05; column rescaled by 1.000118E+00
```

80.2 What it closes

The test is the last iteration of the LHS 1140 b closure of section 79., LHS1140b/exhale/flux_closure/hi/k06, re-run from the same seed (k05/output/) on the same profile, once with the previous binary and once with this one. The three factors above are the run’s own log. They are the measurement, not an illustration: 1.000118 and 0.9996741 are the same numbers as the 1.183×10^{-4} and 3.259×10^{-4} the budget was missing by, which is what the section 79. diagnosis said they would be.

element	budget before	budget after
H (against rho/mass_per_H)	5.633×10^{-8}	1.543×10^{-15}
He	1.910×10^{-15}	1.698×10^{-15}
C	1.183×10^{-4}	3.688×10^{-14}
N	1.183×10^{-4}	3.724×10^{-14}
O	3.260×10^{-4}	4.017×10^{-14}

element_budget.py goes from “4 element(s) miss their reservoir by more than 1.0e-08” to “all element budgets close within 1.0e-08”. Hydrogen closes with the rest and for the same reason: it is checked against rho/mass_per_H, and the metal mass a stale reservoir leaves in ρ is exactly what put it 5.6×10^{-8} out.

The effect on the physical result is small, as section 79. predicted from the size of the reservoir change. The steady mass flux at the top of the domain moves by -2.4×10^{-5} in relative terms ($\log_{10} \dot{M} = 7.50$ either way), and the He 10830 line, run through EXHALE_transit.py on both solutions, moves by 4.7×10^{-5} in relative terms in the red-component depth ($1.725492 \rightarrow 1.725411$) and by at most 1.2×10^{-6} in the normalized line profile.

80.3 What is unchanged, checked rather than argued

Two restarts that must not move, both re-run with the two binaries and compared bitwise on output/Hydro_ioniz.txt and output/Ion_species.txt:

- LHS1140b/exhale/heh0p55 reloaded and advanced one step — a restart with no handoff of any kind. Identical.
- backup/regression/wasp_full seeded from its own converged output with Load_IC? True and advanced one step — a restart that carries a full metal state, from metals.inp and not from a handoff, so it exercises the metals block of load_IC with every element unmarked. Identical.

The binary the comparison was made against is kept as backup/EXHALE_pre_handoff_restart_rescale.x.

make check is unaffected by construction and by measurement: all seven default cases are cold starts (Load_IC? False), so none of them reaches this code at all, and the full matrix — wasp_full, wasp_he23off, mol_base_handoff,

mol_metals, mol_lyman_werner, mol_diffusion, lower_profile — comes back byte-identical (==> REGRESSION PASS (all cases byte-identical), 2026-08-27).

80.4 $r_{\text{El}} = 1$ is not an identity, and the reason is worth stating

The gate is the flag, not the value of the factor, and that is deliberate: $r_{\text{El}} = 1$ does **not** leave the state bitwise where it was. Restarting backup/regression/wasp_full from its own converged output twice with the same binary — once with the abundances coming from metals.inp alone, once with a base.inp restating those same seven numbers as <El>_H_base — prints 1.000000E+00 for every element and still gives a different state after one step. The factors are 1 only to about 10^{-14} : the seed's base-cell EL/H is what the ionization solver converged to, and it reproduces the reservoir to its own tolerance rather than to the last bit. After one step the departure is 1.8×10^{-16} in the median and 1.5×10^{-6} at its worst, and the worst cell is O III at $n = 11 \text{ cm}^{-3}$ near the base, six orders below that ion's own peak in the same column — the solver's tolerance seen from the other side, not a change in the solution.

So the byte-identity claim is exactly the one the flag supports and no wider: **an element the handoff does not state, and a restart with no handoff at all, never enter the arithmetic**, and those are what were compared bitwise against the pre-change binary above. An element the handoff does state is renormalized whenever it is loaded, which is the user-visible change of restart semantics this section is about.

81. Is the wind collisional where it is set? A Knudsen diagnostic, and LHS 1140 b's answer (2026-08-27)

Until now the repository had no way to ask whether a converged solution is a solution of the physical problem. A hydrodynamic wind is a continuum solution, valid only where the gas is collisional on the scale over which the flow varies — above all through the critical point, which is where the topology, and with it the mass flux, is fixed. Nothing measured that. src/utils/collisional_validity.py (new, Python post-processing; it reads an existing run directory and changes nothing, and no Fortran source was touched) computes it. Definitions, provenance and the full result tables: docs/collisional_validity.md.

81.1 The collision model is not a new one

The four momentum-transfer limits are the ones the binary element-diffusion operator already uses, in the same Chapman–Enskog first approximation: rigid-sphere neutral–neutral (Banks & Kockarts 1973), non-resonant ion–neutral with the induced-dipole and rigid-core frictions added, and screened Coulomb with the Spitzer $\ln \Lambda$ for ion–ion and ion–electron. Python cannot call the Fortran, so the four routines are transcribed from src/modules/functions/binary_element_diffusion.f90 with the source line cited at each definition and every constant taken from that file rather than re-chosen; the two are meant to be read side by side. The step from a diffusion coefficient to a collision frequency is the definition of the binary coefficient itself,

$$\nu_{st} = \frac{n_t kT}{\mu_{st} \hat{D}_{st}}, \quad \hat{D} = nD, \quad (29)$$

in which the total carrier density cancels.

Two channels are deliberately missing, and both make the verdict *more* conservative by lengthening the mean free path: resonant charge exchange ($\text{H}^+ + \text{H}$, $\text{He}^+ + \text{He}$), which the element-diffusion operator excludes on the grounds that both partners carry the same element, and electron–neutral momentum transfer, which no element-transport coefficient needs. The electron Knudsen number is therefore quoted only with that caveat, and the bulk is built from the heavy particles alone.

81.2 Definitions, all stated

$\lambda_s = \bar{v}_s / \nu_s$ with the harmonic sum over partners; the structure scale $L = \min(H_p, L_v, r)$ with $H_p = |\text{d} \ln p / \text{d} r|^{-1}$ measured from the solution and $L_v = |v / (\text{d} v / \text{d} r)|$ admitted only above Mach 0.01 (below that it is the distance to a stagnation point, and the base sound-wave layer would otherwise drive Kn to infinity); $Kn_s = \lambda_s / L$. The **exobase** is $Kn_{\text{bulk}} = 1$; the **critical point** is the sonic point with the code’s own sound speed ($c_s = \sqrt{\gamma p / \rho}$, $\gamma = 1.666666666667$, `eval_dt.f90:29, parameters.f90:385`), not a re-definition; the **critical region** runs from peak volumetric heating to the sonic point, and “collisional” means $\max Kn < 0.1$ across it. Coupling times $1/\nu_s$ and the electron–ion energy-coupling time are reported against the flow and heating times, with the note that they are a *different* test: a strongly subsonic flow gives hundreds of collisions per flow time even at $Kn \sim 1$, and it is the Knudsen number that decides continuum validity, because the terms the closure drops scale with the gradient over a mean free path.

81.3 The result, measured 2026-08-27 on existing outputs

Three representative LHS 1140 b solutions — LHS1140b/exhale/heh0p55 (diffusion off), heh2p13_diff_kzz1e9 (diffusion at the adopted K_{zz}), and the converged flux closure flux_closure/hi/k06 — against the HD 209458 b control backup/phase_d_baseline/new_kzz1e9_d3b:

case	critical point	exobase	$Kn = 0.1$ at	max Kn , crit. region	verdict
heh0p55	none in domain (max Mach 0.540)	25.22	6.42	2.2	unvalidated
heh2p13_diff_kzz1e9	none in domain (max Mach 0.565)	26.66	6.79	1.9	unvalidated
flux_closure/hi/k06	none in domain (max Mach 0.550)	19.76	5.53	3.0	unvalidated
HD 209458 b new_kzz1e9_d3b	4.085	above 4.15	—	0.021	validated

Radii in R_p ; the LHS 1140 b domain ends at $30 R_p$.

LHS 1140 b’s wind never reaches its critical point inside the domain. It is subsonic to $30 R_p$ in all three solutions, so the mass flux is set at the outer boundary — where Kn_{bulk} is already 1.3–2.3. Against the p-winds retrieval, which places its isothermal Parker sonic point at 8–9.5 R_p ; that radius lies *inside* the exobase computed here (19.8–26.7 R_p), but Kn_{bulk} there is already 0.14–0.30. So the answer is neither “the sonic point is outside the exobase” nor “the critical region is collisional” — it is **transitional**. Neutral hydrogen sets the number; the ions and electrons are Coulomb-held two to five orders of magnitude more tightly. HD 209458 b is the control at a factor of ~ 100 lower Kn , and it validates cleanly.

81.4 The kinetic estimate is a scale, not a bound

Jeans escape through the computed exobase (Chamberlain & Hunten 1987 eq. 7.2.5) gives $1.0\text{--}2.0 \times 10^7$ g/s, a factor about three below the continuum $\dot{M}$ of $10^{7.47}\text{--}10^{7.81}$. But $\lambda_J(\text{H}) = 0.78\text{--}1.45$ at that exobase: it is barely gravitationally bound, the Jeans integral is most of the Maxwellian rather than its tail, and the atmosphere is in blow-off, so the number does not bound anything. What it does say is that the continuum result is within a factor of a few of what a collisionless outer atmosphere at the same density and temperature would lose — the continuum failure is not hiding an order of magnitude. On HD 209458 b, $\lambda_J = 10.8$ and the hydrodynamic rate exceeds Jeans by 10^4 , which is the signature of a driven wind rather than an evaporating one.

The numbers above are read from the advection-corrected profiles, the ones the transit tool consumes. Read from the equilibrium pair instead (--eq) the verdicts are unchanged — still no critical point in the LHS 1140 b domain, still $Kn \sim 1$ where the outer boundary sets the flux, still collisional through HD 209458 b's critical point — but the exobase moves above $30 R_p$, because the equilibrium far field is more ionized and Coulomb collisions there are far stronger than the neutral ones. An exobase radius quoted from this tool has to say which pair it came from.

This closes the plan's baseline row 11 the way that row demanded: an uncollisional critical region makes the hydrodynamic $\dot{M}$ **unvalidated, not overestimated**, and the tool's `validity_statement` says exactly that, with the numbers it rests on. What remains unsettled is the LHS 1140 b rate itself, which needs a kinetic or transitional-flow calculation this repository does not contain. The He 10830 line-forming layer is not implicated: it sits at $1.1\text{--}3 R_p$, where $Kn \leq 0.021$ in every case.

82. The third body was a mass density: a molecular-network audit in the He-dominated limit (2026-08-27)

Phase F of the LHS 1140 b plan asks for an audit of the molecular network where it is least tested — a helium-dominated envelope. R16–R20 and R23, the shared H–He charge-exchange path, the two He 2^3S Penning channels, and the electron and H-nucleus closures were checked reaction by reaction against the publisher PDFs, and the closures were measured on runs spanning He/H = 0.079 to 1000. The record, with the full reaction table and the numbers quoted below, is `docs/molecular_chemistry_audit_he_rich.md`.

82.1 No rate coefficient was wrong

All 23 Koskinen et al. (2022) Table-1 entries are transcribed verbatim and correctly, R16–R20 and R23 included; so are the Huang et al. (2023) Table-4 B1/B2 charge-exchange pair, the Taylor et al. (2025) He(2^3S)+H fit and the García Muñoz (2025) He(2^3S)+H₂ fit, which reproduces its four tabulated points to 0.13 % (re-measured). Every one of the eight balance rows of `mol_heh_rows` carries each reaction in the rows it belongs to, with the right weight and sign.

82.2 What was wrong was the density handed to the three-body reactions

R12 ($\text{H}_2 + \text{M}$), R13 ($\text{H}^+ + \text{H}_2 + \text{M}$) and R15 ($\text{H} + \text{H} + \text{M}$) are written as two-body coefficients times the third-body density, and the interface that carries it, `ion_cell_state%ntot`, says “total particle density”. What `ionization_equilibrium` actually passed was `n_in_dim =  $\rho/m_{\text{H}}$` . That is a *mass* density in units of m_{H} — `calc_rho` weights every species by `bsp_mass`, so an H_2 molecule contributes 2 and a helium atom 4 — and it over-counts the third bodies by the mean particle mass. At the hot-Uranus molecular base the factor is 2.3; in a helium-dominated base it approaches 4. The error therefore grew along exactly the axis this audit was asked to examine. The same ρ/m_{H} was also used as the gas pressure of the chemical-equilibrium seed that restarts a failed molecular cell, where the EOS’s own law is $p = (n_{\text{tot}} + n_e)k_{\text{B}}T$. Both now use `calc_ntot` (one particle per species, electrons excluded) and, for the pressure, that sum plus n_e .

The consequence is visible where it should be. On the `mol_diffusion` configuration re-run at $\text{He}/\text{H} = 1, 10$ and 1000 (only the He/H number ratio and the `HeH_base` of the handoff changed, 12000 marching steps, `OMP_NUM_THREADS=1`), the $\text{He}/\text{H} = 10$ arm no longer aborts on the marching loop’s NaN detector and completes all 12000 steps, and the number of cells whose molecular root leaves the physical simplex and has to be clamped onto the element budget falls from 37 to 4 at $\text{He}/\text{H} = 1000$ and from 181 to 169 at $\text{He}/\text{H} = 10$. The $\text{He}/\text{H} = 1$ arm still aborts, later (5538 steps against 3402); it is not diagnosed here.

82.3 A guard that was not guarding

The molecular systems close neutral helium as $n_{\text{He}}(1-x_2-x_3)-n_{\text{HeH}^+}$, but the admissibility test `ionization_fractions_physical`, the distance measure `element_budget_violation` and the clamp `clamp_fractions_to_element_budget` all summed the helium budget as $x_2 + x_3$ (plus the metastable) alone. A root with $x_2 + x_3 = 1$ and any HeH^+ was accepted although its neutral helium is negative. All three now carry the HeH^+ nucleus, converted with this cell’s $n_{\text{H}}/n_{\text{He}}$. Nothing in the measured runs reaches that state — HeH^+/He stays below 10^{-10} at every He/H tested — so the change moves no result; it closes a hole rather than fixing a symptom.

82.4 The closures hold

With molecules on, charge neutrality $n_e = n_{\text{H}^+} + n_{\text{H}_2^+} + n_{\text{H}_3^+} + n_{\text{HeH}^+} + n_{\text{He}^+} + 2n_{\text{He}^{++}}$ closes against the n_e the code writes to 4×10^{-16} at every He/H from 0.079 to 1000, the H-nucleus/mass closure to 4×10^{-16} , and `src/utls/element_budget.py` closes both elements to 6×10^{-14} at $\text{He}/\text{H} = 1000$. Element conservation is not what the helium-dominated limit breaks. What it does break is the solve: the molecular `hybrd1` system balances helium in rows of order n_{He}^2 and the molecules in rows of order n_{H}^2 , so at $\text{He}/\text{H} = 1000$ the two blocks of one residual vector differ by 10^6 with no row scaling applied, and isolated cells fall into the atomic basin next to neighbours at $f_{\text{H}_2} = 0.88$. The correlation with He/H is measured; the cause is not established.

82.5 Four caveats are recorded rather than patched

Each would replace a published rate with one from a different compilation. (i) The identity of M: Koskinen et al. never say what their “ n ” counts, and a monatomic third body is generally less efficient than H_2 , so R12/R13/R15 with

helium as M are upper bounds. (ii) HeH^+ formation: Table 1 has only $\text{He}^+ + \text{H}_2$ (R20), and not $\text{H}_2^+ + \text{He} \rightarrow \text{HeH}^+ + \text{H}$, which needs no He^+ and is therefore the route that grows with the helium fraction (García Muñoz 2025 Table A.6 gives 1.0×10^{-11} at 2000 K rising to 1.5×10^{-10} at 10^4 K); the $\sim 10\%$ associative branches of both Penning channels end in HeH^+ too and are not resolved. Against the same tables R16 is $3.4\text{--}8.6\times$ low, R18 $1.2\times$ high, R19 $1.4\text{--}2.6\times$ low, R20 $14\times$ high and R17 up to 1.9×10^4 high. (iii) The $\text{H} \leftrightarrow \text{He}$ charge-exchange pair does not satisfy detailed balance: $k(\text{H}^+ + \text{He})/k(\text{He}^+ + \text{H})$ should be $4\exp(-127500/T)$ and the published fits give 10^2 times that at 10^4 K, a discrepancy weighted by $n_{\text{He}}/n_{\text{H}}$ in the He^+ balance. (iv) The Taylor two-branch Penning fit steps by $1.57\times$ at its own 4000 K break point, which is the published fit and not a transcription error. Each is noted at the code site as well.

82.6 Does this move the LHS 1140 b results?

No: **no LHS 1140 b run in the repository has molecular chemistry on**. None of the 170 `input.inp` files under LHS1140b/ sets `Molecular chemistry`, and none of the 193 `Ion_species.txt` files carries an $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ column. Every statement corrected above lives inside a `thereis_mol` branch, so the $\text{He}/\text{H} = 2.09$ solution, the closure ladder and the `heh2p13_diff_kzz1e9` arm are untouched and no re-convergence is needed. What can be measured on those stored solutions is the size of the factor that *would* have been wrong had they been molecular — the ratio of the mis-supplied ρ/m_{H} to the EOS particle count $p/(k_{\text{B}}T) = n_{\text{tot}} + n_e$ at the base cell:

stored solution	He/H	$(\rho/m_{\text{H}})/(n_{\text{tot}} + n_e)$ at the base
LHS1140b/exhale/flux_closure/hi/k06	2.09	3.03
LHS1140b/exhale/heh2p13_diff_kzz1e9	2.13	3.04
LHS1140b/exhale/flux_closure/heh9p7/k03	9.7	3.72
backup/regression/golden/mol_diffusion	0.0793	2.27

82.7 Goldens

The matrix splits exactly along the `thereis_mol` branch: `wasp_full`, `wasp_he23off` and `lower_profile` PASS byte-identical, and the four molecular cases move. Their shift on the 12000-step snapshots, largest relative difference over the column against the previous golden:

case	ρ	p	T	heat	cool	H_2 front ($f = 0.5$)	$\log_{10} \dot{M}$
mol_base_handoff	1.4 %	2.6 %	1.2 %	1.6 %	3.0 %	1.1597 $\rightarrow$ 1.1597	10.58 $\rightarrow$ 10.58
mol_metals	3.0 %	5.0 %	2.2 %	3.5 %	6.7 %	1.1617 $\rightarrow$ 1.1617	10.58 $\rightarrow$ 10.58
mol_lyman_werner	0.34 %	0.53 %	0.21 %	0.44 %	0.71 %	1.1304 $\rightarrow$ 1.1304	10.58 $\rightarrow$ 10.58
mol_diffusion	1.4 %	2.6 %	1.2 %	1.6 %	3.0 %	1.1597 $\rightarrow$ 1.1597	10.58 $\rightarrow$ 10.58

The velocity column moves by more ($2.5\text{--}8\times$) but only in the base sound-wave layer of these unconverged relaxation snapshots, where v is small and oscillating. The H_2 front does not move at all, and the base H_2 fraction shifts in the fourth decimal: these are H-rich hot-Uranus bases already at $f_{\text{H}_2} \simeq 1$, and a fraction pinned at unity cannot respond to a $2.3\times$ change in the third-body density — it responds in ρ , p and T instead. That is the same statement the He-rich scan makes from the other side: the term becomes decisive only once helium dilutes the H_2 . Goldens re-snapshotted for the four moved cases at the end of the series, and make check re-run after.

83. The outer region a converged wind inherits: path dependence beyond the exobase (2026-08-27)

The LHS 1140 b closure ladder (LHS1140b/kzz_decision.md Sect. 8) climbs a reservoir axis by seeding each arm from the converged wind of the arm below. One rung was reached by a large jump instead of a step, and it converged to a different solution — different only where the continuum equations no longer hold. This section records what separates the two, why neither convergence test noticed, and what it means for reading any of these winds.

83.1 The two solutions

Both are flux-closure arms at the same reservoir $\text{He}/\text{H} = 12.0$ over the same photochemical lower atmosphere. One, LHS1140b/exhale/flux_closure/heh12/, was seeded from the 8.0 arm — a $1.50\times$ jump in reservoir — and closed at $k = 4$. The other, .../heh12_ctl/, was seeded from the 11.1 arm, a $1.08\times$ step, and closed at $k = 2$.

	seeded across a $1.50\times$ jump	seeded one $1.08\times$ step
He/H at the $1\ \mu\text{bar}$ match	12.013522	12.013520
cold-trap O/H of the column	1.0656×10^{-6}	1.0660×10^{-6}
exhale_info	0	0
achieved wind $\ R\ $	2.529×10^{-4}	3.864×10^{-4}
steady-window flux spread	1.74 %	3.30 %
$\log_{10} \dot{M}$ [g/s]	7.420	7.410
He 10830 red depth [%]	4.630	3.856
He 10830 red-pair EW [% Å]	1.3126	1.1181

Identical base, identical matching-level composition to seven digits, both converged and both inside every tolerance the driver applies — and a 17 per cent difference in the observable the whole study is fitted to. Neither residual separates them: on the coefficient in force when this was written the outlier reported the *smaller* of the two, and re-solved on the García Muñoz rate of section 87 the order reverses (jump-seeded 2.533×10^{-4} against the control's 1.249×10^{-4} , both info = 0, both inside the 4.0×10^{-4} target). Which one reports less is not a property of the pair; that $\|R\|$ does not rank them is (LHS1140b/exhale/xuvkzz_gm25/results.txt).

83.2 Where they differ

Nowhere the line window or the closure window looks. Two measures that do not depend on any sampling choice locate it. r_{drop} is the first radius outside $1.5R_p$ at which T falls below half of $T(12R_p)$; the other is the share of the He 2^3S radial column that lies outside $10R_p$. Both are monotonic along the ladder, and both single out one point.

reservoir He/H	$r_{\text{drop}} [R_p]$	$T(12R_p)$ [K]	He 2^3S column outside $10R_p$
2.09	30.00	893	0.0119
3.0	27.63	1035	0.0116
5.0	23.76	1310	0.0105
8.0	20.80	1632	0.0104
9.7	19.79	1789	0.0108
10.3	19.79	1834	0.0108
10.7	19.47	1863	0.0109
11.1	19.47	1891	0.0109
12.0, one step	19.47	1951	0.0112
12.0, one jump	25.83	1957	0.0219

The wind gets hotter and more compact as the reservoir rises: r_{drop} moves inward from 30.0 to 19.5 R_p without a reversal, and the outer share of the metastable column stays in 0.0104–0.0119 over a factor 5.7 in composition. The jump-seeded solution sits at 25.8 R_p where its own reservoir gives 19.5, and carries twice the outer column share of any other rung, while its $T(12R_p)$ matches its control to 0.3 per cent. It is the same wind with a hotter, more extended outer atmosphere on top.

83.3 The mechanism, read from that pattern

The outer state is inherited from the seed and the JFNK finish does not re-solve it. The density out there contributes almost nothing to the residual norm, so $\|R\|$ cannot separate the two states; the elemental-flux spread the closure tests is measured on a steady window that is flat in both. He 10830 is optically thick at these columns, so a hot, extended outer region fills the line wings and lifts the equivalent width while holding about one per cent of the metastable column. Small mass, large opacity, invisible to both convergence tests.

83.4 The physical judgement, stated first

The 20–30 R_p region in which the two solutions differ is where the collisional-validity diagnostic of Sect. 81. places the exobase, 19.8–26.7 R_p , and where $Kn > 1$. *The fluid solution is not valid in exactly the region that makes the difference.* Two consequences follow, and neither is comfortable.

- The ground for discarding the jump-seeded solution is **continuity of the trend, not a physical criterion**. Nothing in the equations being solved prefers one outer state over the other, because the equations do not apply there.
- By the same argument, the absolute equivalent widths of **every** rung of that ladder carry a systematic error the closure tolerance does not catch. The size is small — the region outside $10R_p$ is about one per cent of the metastable column — but it is not zero, and it is not bounded by any test currently applied.

83.5 What this says about the convergence criteria

`exhale_info = 0`, a wind residual under target, and a flat elemental flux across the steady window are together sufficient to certify the solution *inside* the collisional region, and silent outside it. That is not a defect of the criteria: a residual norm weighted by density cannot see a region that carries no density, and a flux-spread test measured over a window is blind to what the window excludes. It is a limit on what “converged” means for a subsonic wind whose domain extends past its own exobase. Where an observable is optically thick and integrates the whole column — He 10830 here, Ly- α more so — the certified region and the region that sets the observable are not the same region.

83.6 What this says about running a scan

A ladder scan has to be climbed one step at a time. The step that worked here was $1.08\times$ in reservoir; the $1.50\times$ jump did not. There is no way to detect the failure from the solver’s own output, so the protection is procedural: keep the step small, and check a sampling-free outer diagnostic (r_{drop} , or the outer share of whatever column the observable integrates) for monotonicity along the scan.

83.7 What was and was not tested

Path independence was checked directly at $\text{He}/\text{H} = 3.0$, from a seed $3.7\times$ away in composition and in the downward direction the ladder never takes: it returns the same r_{drop} to the digit, with everything else inside the closure tolerance. The 5.0, 8.0 and 2.09 arms were not checked this way. Determinism was checked separately at 10.31 (`.../heh10p3_ctl/`, same seed, re-run): `Hydro_ioniz_adv.txt` and `Ion_species_adv.txt` reproduce bitwise, which says nothing about seed dependence. **No code was changed by this section**; the ladder table, the crossing and the figure were re-derived on the control arm, and the record is `LHS1140b/kzz_decision.md` Sect. 8.3 and `docs/lhs1140b_exhale_vs_pwinds.tex`, “The outer region a converged wind inherits”.

84. Metals had no way back: element projection when the helium mass fraction reaches one (2026-08-28)

The judgement first. The element projection of the binary H/He diffusion operator destroyed trace-metal nuclei with no sink and no way back, and that is physically wrong whatever it costs. In a cell whose helium mass fraction reaches $X = 1$ the hydrogen count $n_{\text{H}}^{\text{new}} = (1-X)\rho/m_1$ is exactly zero and the projection multiplied the metals by $r_{\text{H}} = n_{\text{H}}^{\text{new}}/n_{\text{H}}^{\text{old}} = 0$. Zeroing them *at that instant* is right: in this two-component split the metals are part of component 1, slaved to hydrogen at a fixed metal/H, so a cell with no hydrogen carries no metals and component 1 carries no mass. What is wrong is the step after. Hydrogen returns to that cell through the shortfall deposit at the end of the same projection; **the metals do not**, and nothing later can lift them, because every writer to the metal columns of `f_sp` on this path multiplies what is already there. The result is a contiguous band of cells holding exactly zero carbon, nitrogen and oxygen between cells at the reservoir ratio — a state these equations cannot produce, since they contain no sink for a metal nucleus. The fix is that the metals return **with** the hydrogen.

84.1 The path, in the code

element_diffusion_step solves for the helium mass fraction X and hands the new element totals to project_elements (src/modules/functions/binary_element_diffusion.f90). There, per cell,

```
nucH_new = (1.0d0 - Xhe(j))*msum(j)/m_1
if (nucH_old(j) .gt. 1.0d-30) then
    rH = nucH_new/nucH_old(j)
else
    rH = 0.0d0
endif
...
do im = 1, n_mion                      ! metals slaved to hydrogen
    f_sp(j,mion_fsp(im)) = f_sp(j,mion_fsp(im))*rH
enddo
...
if (nucH_new .gt. gotH)                &
    f_sp(j,isp_HI) = f_sp(j,isp_HI) + (nucH_new - gotH)
```

Step k : X reaches 1, so $\text{nucH_new} = 0$, $rH = 0$, and every hydrogen species and every metal ion of the cell is multiplied by zero. Step $k+1$: nucH_old is now zero, so $rH = 0$ again for want of a ratio to preserve, while $\text{nucH_new} > 0$ — and the deposit on the last line puts the hydrogen back as neutral H. There is no such line for the metals. Their zero is final.

84.2 The symptom that led here

LHS1140b/exhale/branch_diagnosis/ records the measurement (no code was changed by that work). At $K_{zz} = 10^7$ the reservoir scan reported two converged states at the same reservoir He/H, $\log_{10} \dot{M}$ 7.54 against 7.87 and He 10830 red-pair equivalent width 0.4287 against 1.3248 %Å, both with info = 0. The hot one carries 172 cells, 1.0333–1.6835 R_p , at exactly zero C, N and O in every stage, with the same elements at C/H = 2.639×10^{-4} immediately above and below. On the cool solution the C I lines carry 95.6 % of the volume-integrated cooling over 1.0–1.2 R_p ; deleting the carbon drops the integrated cooling of that band by a factor 23 at $0.91 \times$ the heating and lifts $T_{\max}$ from 3899 K to 5879 K. Refilling C, N and O at the handoff's own reservoir ratio and re-solving the wind with the same handoff collapsed the hot state onto the cool branch ($\log_{10} \dot{M} = 7.530$, $T_{\max} = 3890$ K, EW = 0.4065 %Å). Of the 550 output*/Ion_species.txt files in the tree, 130 carry such a band; 128 are in LHS1140b/exhale/kzz_profile_scan, the other two are runs already on record as failed. **No regression golden and no backup/regression/ case output carries one**, which is why the repair below moves nothing.

84.3 Why info = 0 and $\|R\| \sim 10^{-4}$ could not see it

Both states report info = 0 and a residual of a few times 10^{-4} , and both are right to. **The steady residual is a residual of the hydro and energy equations**, and their solution with the metals removed is a perfectly good solution of the equations as posed: the cooling function simply has fewer terms. Nothing in that norm asks whether the composi-

tion it was evaluated on is a composition the transport equations can reach. The elemental-flux closure does test the composition, but on the radial spread of the elemental *fluxes* over a window at $2R_p$ and above — outside the band. The handoff's own elemental conservation check (`src/utls/lower_profile_schema.py`) is stated on the *chemistry* profile at its deepest level, not on the wind. **Elemental conservation was nowhere checked for the wind**, which is the gap this section closes.

Sect. 83. is the same shape of failure at a different place: there two converged solutions differed only outside the exobase, where the residual norm carries no density and the flux window does not reach, and He 10830 saw the difference because it is optically thick and integrates the whole column. The statement the two sections share is that `info = 0` plus a residual under target plus a flat elemental flux certify the solution **on the quantities and in the region those tests are evaluated on**, and are silent elsewhere. What is new here is that the silence is not about a region: it is about a *conserved quantity that was never formed into a test at all*.

84.4 A third failure mode, not either of the first two

- **Sect. 70.** (HD 209458 b, $K_{zz} = 0$): two wind solutions a factor 2.8 apart in $\dot{M}$, both `info = 0`, alternating in a period-2 limit cycle of the outer composition/wind Picard loop, on compositions $\sim 1\%$ apart. Every backup/phase_d_baseline/ run was checked for the band and **none has one**, so that bistability is not this defect and stays open. (Caveat: only the final state of each run is on disk; the alternating passes of `new_kzz0_undamped` were not saved with species densities.)
- **Sect. 83.** (LHS 1140 b, jump-seeded rung): two solutions that agree below the exobase to seven digits and differ only above it, where the fluid equations do not apply. Here the two states differ from $r = 1.0004R_p$ upward — by 30 % in T by $1.0013R_p$ — and the difference is made in the collisional part of the atmosphere ($Kn_{\text{bulk}} \sim 10^{-3}$ at T_{max}).
- **This section:** a state that is not a solution of the equations at all, because an element has been deleted from a band of cells by the discretization and nothing can put it back.

The three should not be merged.

84.5 The repair, and the abundance it uses

The metals are a *ratio*, not a density: what the projection has to preserve for them is n_X/n_H . Multiplying by r_H does that in a cell that had hydrogen, and that branch is untouched. Where n_H^{old} is zero the ratio is 0/0 — the cell carries no metal/H of its own — and the operator supplies one:

```
if (nucH_old(j) .gt. 1.0d-30) then
  do im = 1, n_mion
    f_sp(j,mion_fsp(im)) = f_sp(j,mion_fsp(im))*rH
  enddo
else if (thereis_metals) then
  do ie = 1, n_melem
```



```

    i0e = melem_i0(ie)
    f_sp(j,mion_fsp(i0e)) = melem_ab(ie)*nuch_new
    do k = 1, melem_top(ie)
        f_sp(j,mion_fsp(i0e+k)) = 0.0d0
    enddo
enddo
endif

```

melem_ab is not one candidate among several. It is the metal/H that **this operator's own mass budget already assumes**: the mass per hydrogen nucleus of component 1 is $m_1 = \text{mass_per_H_nucleus_without_He}() = m_H + \sum_X \text{melem_ab}(X)A_X$, so the two-component closure $m_1 n_H + m_{\text{He}} n_{\text{He}} = \rho$ that the write-back is required to meet is stated at that ratio and no other. It is also the reservoir ratio set_IC builds a cold start from, the one load_IC rebuilds an element the restart file does not carry from, and the one Sect. 80. rescales a restart column onto. The alternatives were considered and rejected: the cell's own previous ratio is exactly what does not exist there, and a neighbour's ratio would make the result depend on the grid.

All of it goes into the **neutral ground stage**, for the same reason the hydrogen and helium shortfalls go into H I and He I: a cell that held no hydrogen held no ionization split either, ioniz_eq owns the split within an element and resolves it from the element total on the next call, and this is already the convention of set_IC, of load_IC, and of the trace-metal re-seed branch of element_diffusion_step itself.

The repair prevents the band from being *created*. It does not reconstruct one that a restart file already carries: there $n_H^{\text{old}} > 0$ in the emptied cells, so the preserving branch is the one that runs, and it faithfully preserves a ratio of zero. That is the right behaviour for a restart — silently rewriting loaded data would be worse — and it is why the census below is unconditional.

84.6 The check the pipeline did not have

metal_hydrogen_ratio_departure now runs at the end of every diffusion step. It measures $\max |(n_X/n_H)/\text{melem_ab}-1|$ over the cells 1.. N and the elements the reservoir states, and it counts the (cell, element) pairs holding no metal nuclei at all while hydrogen is present. The departure is a magnitude and not an error — he_metal_diffusion moves the metals relative to hydrogen on purpose, and a restart may carry a settled column — so it is only reported, under the existing EXHALE_DIFFUSION_CHECK=1 switch, together with the rest of the step diagnostic. The count is different in kind: it is zero on any state these equations can reach, so it raises a warning on stdout the first time it is nonzero **whether or not the diagnostic is on**. No new input key was added.

84.7 The clamp is a limiter, and the comment that said otherwise was wrong

The projection was preceded by

```

! Round-off assertion (NOT a limiter): the M-matrix argument in the
! header gives 0 <= X <= 1 per step; only round-off can leave the range.
where (Xhe .lt. 0.0d0) Xhe = 0.0d0

```

where (Xhe .gt. 1.0d0) Xhe = 1.0d0

That claim had never been measured, because the diagnostic printed the *clamped* X , which cannot tell an assertion from a limiter — both print 1.0. Measured now, on the reproduction run below: X exceeds 1 before the clamp in **140 of 7962 steps, by up to 0.520** (i.e. $X = 1.52$), and 105 of those excursions exceed 0.1. The lower end holds: $\min X$ stayed above 0 in every step of both runs. **The clamp is a limiter on the upper side.**

The reason is in the discretization, not in round-off. The settling flux $-\rho D_{12} G X (1-X)$ is linearized with the $(1-X)$ factor lagged, $W_f = \rho_f D_f G_f (1-X_f^{\text{lag}})$, and entered as $-W_f X$. In all three branches of the Péclet hybrid the face coefficients satisfy $\text{PdL}(f) + \text{PdR}(f) = -W_f$, so the row sum of the backward-Euler matrix is

$$\frac{\rho_j}{\Delta t} - K_j \left(r_{j+1/2}^2 W_j - r_{j-1/2}^2 W_{j-1} \right)$$

and **not** $\rho_j / \Delta t$. A uniform $X = 1$ is therefore not a fixed point of the discrete row unless the lagged settling coefficient is divergence-free, and the comparison principle that the header's M-matrix argument invokes does not apply — at either end. What separates the two ends here is measurement, not proof: the lower one held in every step of both runs, the upper one did not. The header comment and the corresponding sentence of docs/binary_diffusion_design.md Sect. 3 (“a clip at round-off (1e-15) is kept only as an assertion, not as a limiter”) are corrected to state the measurement. **What is not done here.** Making the upper bound hold by construction means changing the linearization of the settling term — for instance solving the $X(1-X)$ product with both factors at the new level, or symmetrizing the Picard sweep — and that changes every diffusion-on result, including two regression goldens. It is a formulation decision, it is recorded here, and it is left open. What this section does is remove the *consequence* that made the clamp destructive: with the metals returning with the hydrogen, a cell that touches $X = 1$ recovers instead of being permanently emptied. The $X = 1$ excursions themselves still happen — 140 of them in the repaired run, against 142 in the unrepaired one.

84.8 What was measured

The step kzz1e7/heh3p8 (clean) → kzz1e7/heh4p2, re-taken with the same binary in both columns, the projection branch being the only difference. This is the step LHS1140b/exhale/branch_diagnosis/metal_dropout_reproduce.py takes, run with EXHALE_DIFFUSION_CHECK=1 and with the He I 10830 synthesis added; both runs were made outside the repository, so no recorded output in it was overwritten.

	before	after
zero C/N/O cells, Ion_species.txt	157 (1.0429–1.6716 R_p)	0
zero C/N/O cells, Ion_species_adv.txt	157	0
exhale_info	0	0
$\log_{10} \dot{M}$	7.52	7.52
$T_{\max}$	4182.6 K at 1.358 R_p	4185.6 K at 1.358 R_p
He 10830 red-pair EW	0.5306 %Å	0.5249 %Å
diffusion steps	7916	7962
steps whose X reached 1	142	140
max X before the clamp	1.521	1.520
steps with a vanished (cell, element)	7541, up to 471 pairs	0
max metal/H departure from <code>melem_ab</code>	1.0 (an element gone)	8.1×10^{-13}

The unrepaired column reproduces the recorded diagnosis exactly — 157 cells over the same 1.0429–1.6716 R_p , 7916 diffusion steps, 142 of them touching $X = 1$ — so the added diagnostics are numerically inert and the projection branch is the sole cause of the difference. The repaired solution matches the cool branch continued to this reservoir by refilling and re-solving (`branch_diagnosis/cool_branch_continuation.txt`: $\log_{10} \dot{M} = 7.520$, $T_{\max} = 4184.4$ K, EW = 0.5287 %Å).

Note what the $T_{\max}$ and EW rows do *not* say: in this particular step the band happened to cost only 3 K and 1 % of the equivalent width. The factor-23 cooling loss and the 3899 → 5879 K jump quoted above belong to the recorded scan arms, where the band sits where the C I cooling is decisive. The band is a defect wherever it appears; how much it moves the answer depends on where it lands.

84.9 Verification

- `make check`: 7/7 byte-identical. Nothing moves, and the reason is structural rather than lucky: the preserving branch $n_{\text{H}}^{\text{old}} > 0$ is bitwise the previous code, and no golden state ever empties a cell of hydrogen.
- `make diffusion_tests`: 31/31 (28 as before, plus 3 assertions of the new T13). **T13 fails on the pre-fix projection** — 21 of 21 band cells hold hydrogen and no metals, metal/H departure 1.0 — which is the point of adding it. It seeds a closed metal-bearing column with a band of pure helium, so $n_{\text{H}} = 0$ and $X = 1$ exactly, lets diffusion refill the band, and requires that no cell holding hydrogen holds zero metals and that n_X/n_{H} equals `melem_ab` to 10^{-12} (measured 5.3×10^{-15}).
- `run_fcheck.sh` (`-fcheck=bounds,do,mem`) clean. It had to be repaired first, for two reasons that predate this change and are unrelated to it: its repo root was still one level up from `backup/regression/` (so every `make` in it failed with “No rule to make target ‘distclean’”), and it copies `HD209458b/input.inp` into an empty scratch directory without turning off the `Load IC?` True that file now carries, so `load_IC` stopped on a missing restart and the script reported that error `stop backtrace` as a runtime check firing. With both fixed the HD 209458 b bounded run is clean ($\log_{10} \dot{M} = 9.60$).

- That case runs with element diffusion **off**, so it does not reach the code this section changes. Two runs were added for that: `diffusion_tests.x` built with the same flags (31/31, and T13 is the test that enters the new branch), and the `lower_profile` regression case — element diffusion together with metals — bounded to 400 steps. Both clean, no trap.

Not done, and left to a decision. The 128 contaminated arms of LHS1140b/exhale/kzz_profile_scan are unchanged on disk. The repair prevents the band from forming; it does not repair a stored state, and re-running that scan is a data decision, not a code one.

The open item above is closed by section 86. (2026-08-28): the two factors of the drift product $X(1-X)$ are now taken from opposite sides of the face and both at the new time level, so the flux shuts off at either end of the composition axis as it does in the continuum, and the $X > 1$ excursions are gone. The clamp is an assertion at both ends.

85. A structure scale that is a scale of the state, not of the pressure (2026-08-28)

`src/utils/collisional_validity.py` divides a mean free path by a local structure scale L to get the Knudsen number that decides whether a wind solution is a continuum solution. Until now $L = \min(H_p, L_v, r)$, and below Mach 0.01 the velocity term switched itself off, so in a subsonic region — LHS 1140 b runs at Mach 1.6×10^{-4} near the base — L was the pressure scale height alone.

The pressure is the one field that can go flat while the gas does not. $p = nkT$; across a heating front ρ falls and T rises with nearly the same logarithmic slope, so $d \ln p / dr$ passes through a broad near-zero while the state changes fast. Measured in LHS1140b/exhale/heh0p55: H_p runs 9.13×10^6 cm at $1.0244R_p$, 2.55×10^8 cm at $1.0453R_p$ (its maximum, 3.05×10^8 , is at 1.0486) and 1.41×10^8 cm at $1.0648R_p$, while at $1.0453R_p$ H_T is 1.80×10^7 and H_ρ 1.68×10^7 cm — fifteen times shorter — and $\text{Kn}(\text{HI})$ showed a hole of $5.4 \times (3.67 \times 10^{-5}, 6.82 \times 10^{-6}, 3.56 \times 10^{-5})$ at those three radii) in a quantity that is monotonic on either side.

The fix is the state vector. The continuum closure expands about a local Maxwellian, which is fixed by (n, T, u) ; those logarithmic gradients plus the spherical divergence are the complete set, and the change of the state over a distance is a vector whose length is the sum in quadrature,

$$\frac{1}{L^2} = \left(\frac{d \ln \rho}{dr} \right)^2 + \left(\frac{d \ln T}{dr} \right)^2 + \left(\frac{|dv/dr|}{c_s} \right)^2 + \frac{1}{r^2}, \quad L = \max \left(\frac{1}{\sqrt{\dots}}, \Delta r_{\text{cell}} \right). \quad (30)$$

Pressure is not in it: it is a derived field, and adding it would count the density and temperature gradients twice wherever they do not cancel while contributing nothing exactly where they do. The quadrature sum, not a minimum over the same lengths, because a minimum jumps discontinuously to the next length whenever the field it is taken from flattens — the failure above in general form — whereas in quadrature a flat field contributes zero and L still cannot exceed the shortest individual scale. The velocity enters as $|dv/dr|/c_s$ and not $|v^{-1} dv/dr|$ because the term the closure drops is the viscous stress, whose size relative to the pressure is $\sim \lambda |dv/dr|/\bar{v}$: a bulk-velocity change distorts the distribution function in proportion to the thermal speed, not to the local bulk speed. That removes the Mach-number floor — `MACH_FLOOR` is gone — which existed only because normalizing by v diverges wherever the base sound-wave layer crosses $v = 0$. The

floor at the local cell width says a gradient claiming structure below one cell is measuring the mesh; it binds in 3 of 500 cells of one of the four runs, all at $r < 1.002R_p$, and in no critical region.

No conclusion moves. Same runs, same outputs, nothing rebuilt or re-run. HD 209458 b stays validated through its critical point (sonic point $4.085R_p$, which L does not enter; max Kn in the critical region $0.0214 \rightarrow 0.0186$), all three LHS 1140 b solutions stay unvalidated with no critical point in the domain (max Kn $1.9\text{--}3.0 \rightarrow 2.0\text{--}3.1$). What moves is at the 10–30% level and in both directions: the exobase $19.8\text{--}26.7 \rightarrow 18.8\text{--}25.4R_p$, Kn_{bulk} at the outer boundary $1.3\text{--}2.3 \rightarrow 1.4\text{--}2.4$, Kn_{bulk} at the $8\text{--}9.5R_p$ where the p-winds retrieval puts its sonic point $0.14\text{--}0.30 \rightarrow 0.13\text{--}0.29$, and the He 10830 line-forming layer at $1.1\text{--}3R_p$ $\text{Kn} \leq 0.021 \rightarrow \leq 0.019$. The one large change is inside $1.1R_p$, where the old H_p sat on its isobaric plateau: Kn_{bulk} at $1.1R_p$ goes from $4 \times 10^{-5}\text{--}1.2 \times 10^{-4}$ to $2.5\text{--}4.2 \times 10^{-4}$. Mach maxima, $\log_{10}\dot{M}$ and the coupling times are independent of L and unchanged. docs/collisional_validity.md carries the recomputed tables and a row-by-row account of what moved; the numbers quoted in section 81. are the ones from the old scale and are superseded by it. Figure docs/figures/lhs1140b_knudsen.pdf regenerated (that block of LHS1140b/make_memo_figures.py only). Post-processing diagnostic, no Fortran touched, goldens untouched.

86. The drift flux has to vanish where the composition runs out: the discretization of $X(1-X)$ and the bounds on the helium mass fraction (2026-08-28)

The judgement first. The drift (settling) term of the binary H/He transport equation carries the factor $X(1-X)$ and vanishes at both ends of the composition axis, because a cell with no helium has none to send and a cell with no hydrogen has nothing to send it in exchange. The discretization lagged the $(1-X)$ factor at the previous iterate, which throws that shutoff away: the discrete flux into a cell already at $X = 1$ was not zero, and the step overfilled it. Measured, X reached 1.52. That is not a small inaccuracy, it is a discrete operator that does not have the property the continuum flux has, and the clamp that hid it was doing the work of a limiter. The drift product is now taken with its two factors from **opposite sides of the face** — helium from the cell it leaves, hydrogen from the cell it enters — and both at the new time level, and the resulting nonlinear step is solved by Newton. The excursion outside $[0, 1]$ is gone: over every case measured the solve now stays strictly inside the range.

Section 84. fixed the *consequence* (a cell driven to $X = 1$ lost its metals permanently) and left the cause open in as many words: “Making the upper bound hold by construction means changing the linearization . . . it is left open.” This section closes it.

86.1 Where the old discretization lost the bound

The flux is (module header of src/modules/functions/binary_element_diffusion.f90)

$$J = -\rho(D_{12} + K_{zz}) \frac{dX}{dr} - \rho D_{12} G X(1-X), \quad (31)$$

and the old face form entered the drift as one lagged coefficient times one implicit factor,

$$W_f = \rho_f D_f G_f (1 - X_f^{\text{lag}}), \quad J_{\text{drift}}(f) = -W_f X(f), \quad (32)$$

with $X(f)$ central or upwind by the Péclet switch. Two things follow. The row sum of the backward-Euler matrix is $\rho/\Delta t - K(r_R^2 W_j - r_L^2 W_{j-1})$ and not $\rho/\Delta t$, so no comparison principle applies (section 84. established this much). More to the point, and this is the physical statement: **a face between two cells both at $X = 1$ still carries a drift flux**, $-W_f$, unless the lag already knows that $X_f = 1$. The endpoint states of the composition axis are not stationary states of the discrete operator, while they are exact stationary states of the continuum flux. A cell driven toward pure helium therefore kept receiving helium after it had run out of hydrogen to exchange, and the clamp turned the overshoot into an exact 1 — which is where the metal band of section 84. came from.

86.2 The discretization that keeps it

The drift is a **counter-flow**: the helium mass flux $-BX(1-X)$ (with $B = \rho D_{12}G$) is matched by an equal and opposite hydrogen flux, since the two components close, $J_1 = -J_{\text{He}}$. A donor-cell rule for a counter-flow has to take each element's mass fraction from the cell **that element** leaves, and the two elements leave from opposite sides. With $B \geq 0$ helium drifts inward, so helium is donated by cell $j+1$ and hydrogen by cell j :

$$J_{\text{drift}}(f) = -B_f X(j+1)[1-X(j)] \quad B_f \geq 0 \quad (\text{helium inward}), \quad (33)$$

$$J_{\text{drift}}(f) = -B_f X(j)[1-X(j+1)] \quad B_f < 0 \quad (\text{helium outward}). \quad (34)$$

Both factors are at the new time level. A donor at $X = 0$ sends nothing; an **acceptor** at $X = 1$ receives nothing. Those are the two statements that bound X , and they are the discrete image of the two limits the continuum factor $X(1-X)$ expresses.

The Péclet hybrid stays, with its switch now on the drift coefficient itself, $|B_f|\Delta r \leq 2(A_f + E_f)$, and the central branch evaluates the product at the face average, $X_f(1-X_f)$. That branch is inside the bound as well, and for a reason worth writing down: with $X(j) = 1$,

$$X_f(1-X_f) = \frac{[1+X(j+1)][1-X(j+1)]}{4} \leq \frac{1-X(j+1)}{2}, \quad (35)$$

so the Péclet condition $|B_f| \leq 2A_f/\Delta r$ bounds the drift flux by the gradient flux $A_f[1-X(j+1)]/\Delta r$ that is carrying helium *out* of that cell at the same face. The central branch cannot reverse the sign of the total face flux at a cell sitting on the bound. (The switch is stricter than the old one by the factor $1-X$, so slightly more faces are upwind.)

The bounds. Let X solve the implicit step and suppose it first touches 1 in cell m , every other cell still inside $[0, 1]$. Then the time term $\rho(X_m - X_m^{\text{old}})/\Delta t \geq 0$; the gradient flux leaves m at both faces because X_m is the maximum; the upwind advection contributes $\rho|v|(X_m - X_{\text{donor}})/\Delta r \geq 0$; and the drift flux is an outflow or exactly zero at both faces by the rule above. Every term of the row has the same sign, and their sum is the row residual, which is zero — so no term can be strictly positive and $X_m > 1$ is impossible. $X \geq 0$ is the same statement, because the scheme is **exactly symmetric** under $X \rightarrow 1-X$, $B \rightarrow -B$: the donor and the acceptor exchange roles and the central product is even about $X_f = 1/2$. That symmetry is the binary symmetry of the transport equation itself — there is one independent diffusive flux and the two components carry it in opposite directions — and the old lagged form broke it, which is why one end held in practice and the other did not.

The row sum is still not $\rho/\Delta t$ and does not need to be: the argument is made cell by cell on the residual of the nonlinear step, not from a comparison principle for a linear system.

86.3 Newton, and why not another Picard sweep

The step is now nonlinear in X^{new} , so it is solved by Newton: `composition_residual` builds the residual and the two face-flux slopes, and `solve_mass_fraction` assembles the tridiagonal Jacobian and takes the correction. The Jacobian is an M-matrix for any iterate in $[0, 1]$ — the slopes are $\partial J / \partial X_{\text{left}} \geq 0$ and $\partial J / \partial X_{\text{right}} \leq 0$ in all three branches — so the correction is well posed, and the nonlinearity is quadratic, so the iteration converges in one to three passes. A pass that does not reduce the residual is halved.

A Picard sweep on a lagged factor was the alternative and is what was there. It does not restore the bound, whatever the sweep count: at every sweep the flux at a face between two cells at $X = 1$ is $-W_f \neq 0$, and only the *converged* fixed point has the shutoff. The two-sweep version in use was not converged — measured below, the LHS 1140 b step needs one to three Newton passes to reach its residual floor — and it is the unconverged iterate that was clamped. Reducing Δt was not considered a fix: the bound is a property the operator should have at any step, the implicit operator is otherwise unconditionally stable, and the relaxation deliberately runs at 10^5 cell diffusion times to reach the composition's steady state at all (section 27.).

Convergence measure. The Newton residual is measured *relative to the size of the terms of its own row*. At a step long against the cell diffusion time the time term and the flux divergence each exceed their difference by $\sim \Delta t / t_{\text{diff}} \sim 10^5$, so an absolute residual bottoms out at the round-off of the larger term (measured: $\sim 10^{-11}$ in units of X) and cannot be a convergence test. The relative residual reaches $\sim 7 \times 10^{-12}$ on LHS 1140 b in one to three passes; below that the round-off of the tridiagonal solve, amplified by the same conditioning, is all that is left, so an already-small residual that no longer halves is taken as converged. The smallness test is not one number: the reachable floor is set by the conditioning of the tridiagonal solve, which is 10^{-12} in a wind and $\sim 10^{-5}$ in the $K_{zz} = 2 \times 10^{12}$ homopause column of test T7, where the time term is negligible against the eddy term. It is therefore either an absolute 10^{-8} or a drop of 10^{-6} from the residual the step started at, whichever is reached first. That floor test is **not** applied while the residual is still large: the bounds belong to the converged step, and stopping early loses them — with the floor test applied unconditionally, T14 fails at an excursion of 7×10^{-3} .

86.4 What is measured

The excursion outside $[0, 1]$ *before* the clip, which section 84. added to the step diagnostic and which is now also exposed as the module variables `he_fraction_over_one` and `he_fraction_under_zero` so a test can read the solve instead of its clip:

	before	after
T14, 400 steps, closed column at $b_0 = 20$, $\Delta t = 10^6$	max $X = \mathbf{1.514}$ ($+5.136 \times 10^{-1}$)	max $X = 0.99908$ (-9.2×10^{-4})
T13, 200 steps from a band seeded at $X = 1$ exactly	-8.584×10^{-3}	-8.587×10^{-3}
LHS 1140 b <code>heh2p13_diff_kzz1e9</code> , JFNK finish	-1.050×10^{-1}	-1.050×10^{-1}

The excursion is negative in every case now measured, i.e. the solve stays strictly inside $[0, 1]$ and never reaches the clip at all. It is *not* the round-off-sized positive number one would quote if some cell sat exactly on the bound: in these runs the solution approaches the bound and stops short of it, which is the behaviour the shutoff produces. The

clip lines are therefore assertions at both ends, and they are kept, with the excursion measured every step rather than asserted.

The old code fails T14 by 0.514 — essentially the 0.520 recorded in section 84. from a wholly different run, which is the reproduction that identifies the mechanism.

86.5 Golden movement

The change moves every diffusion-on result, so it moves two of the seven regression cases. The other five — `wasp_full`, `wasp_he23off`, `mol_base_handoff`, `mol_metals`, `mol_lyman_werner` — run with `He_diffusion` off, never enter the module and stay **byte-identical**.

Largest relative difference over the column, against the previous golden, on the two 12000-step snapshots that do run element diffusion:

case	ρ	p	T	heat	cool	v	$\log_{10} \dot{M}$
<code>mol_diffusion</code>	$10^{-11} \%$	$2 \times 10^{-11} \%$	$9 \times 10^{-12} \%$	$10^{-11} \%$	$2 \times 10^{-11} \%$	$1.4 \times 10^{-8} \%$	$10.58 \rightarrow 10.58$
<code>lower_profile</code>	$4 \times 10^{-9} \%$	$5 \times 10^{-9} \%$	$3 \times 10^{-9} \%$	$7 \times 10^{-9} \%$	$9 \times 10^{-9} \%$	$5.4 \times 10^{-7} \%$	$9.58 \rightarrow 9.58$

The worst-moving species is H3+ at $1.9 \times 10^{-4} \%$ (`mol_diffusion`) and HeITR at $1.3 \times 10^{-8} \%$ (`lower_profile`); the H₂ front of `mol_diffusion` is unchanged at $1.16036 R_p$ and `lower_profile` carries no molecular column.

These two cases do not exercise the bound at all, and that is why they barely move: X stays in $[0.2406, 0.2408]$ through all 12000 steps of `mol_diffusion` and in $[0.2468, 0.2496]$ of `lower_profile`, both at $K_{zz} = 10^9$ where the eddy term dominates the drift, and the largest excursion the diagnostic reports is -0.759 and -0.750 — X is nowhere near either end. What moved them is the second-order difference between the old and the new evaluation of the drift product plus the Picard sweep's non-convergence, both at round-off amplified by the step conditioning here. Golden re-snapshotted for the two, and make check re-run after: 7/7.

Newton cost, measured on the same runs: `mol_diffusion` converges in one pass in all 12000 steps (largest relative residual 5.9×10^{-13}); `lower_profile` in one pass in 11807 steps and two in 193 (9.8×10^{-13}). The wall time is unchanged within the noise of a shared machine.

86.6 The physical result this was worried about: LHS 1140 b does not move

The composition of LHS1140b/exhale/heh2p13_diff_kzz1e9 is the one the He I 10830 line is read from, so the fix was re-taken on it: the case re-solved from its own converged state, JFNK finish, post-processing pass and the He I 10830 synthesis at WINERED HIRES-Y ($R = 68,000$), once with a binary carrying the old linearization and once with this one, both outside the repository so that no recorded output was overwritten.

	old linearization	this section
exhale_info / $\ R\ $	0 / 4.285e-3	0 / 4.285e-3
$\log_{10} \dot{M}$	7.80	7.80
He 10830 red-pair EW (10832.60–10834.20 Å vacuum)	1.1404754 %Å	1.1404754 %Å
red depth	4.186959 %	4.186959 %
FWHM	0.2557731 Å	0.2557731 Å
X range over the run	0.6503–0.8950	0.6504–0.8950
largest excursion before the clip	-1.050×10^{-1}	-1.050×10^{-1}

The two solutions agree to 1.2×10^{-9} relative in ρ , v , p and T over the whole column (the one exception is He III at $1.0103R_p$, 3.5×10^{-3} relative, on a population that is 6×10^{-13} of the helium there). **This wind never approaches** $X = 1$, so the clamp never fired on it and the fix has nothing to remove; what is left is the second-order difference in the drift term, and at this K_{zz} it is invisible.

On the reservoir crossing: the equivalent width moves by -4.1×10^{-13} in relative terms, and with the local slope $d \ln \text{EW} / d \ln (\text{He}/\text{H}) = 0.3819$ measured over the upper rungs of the closure ladder that is $d \ln (\text{He}/\text{H}) = -1.1 \times 10^{-12}$, i.e. the crossing stays at $\text{He}/\text{H} = 11.730000$. The adopted result is untouched.

What this does not say. The recorded tpm_He10830.txt of that case gives EW 1.1289 %Å against the 1.1405 measured here; the difference is *not* this change, it is everything the working tree has accumulated since that file was written (section 84. among it). Both columns of the table above were produced with binaries differing only in the drift linearization, which is what makes them a controlled comparison.

86.7 Verification

- make check: five cases byte-identical, mol_diffusion and lower_profile moved by the amounts tabled above; goldens re-snapshotted for those two and make check re-run, 7/7.
- make diffusion_tests: 35/35 (31 as before, plus T14's three assertions and one added to T13). **T14 fails on the old linearization** at an excursion of +0.514, which is the point of adding it: it drives a closed column hard enough that the base reaches pure helium, requires that the run actually reach the boundary ($\max X \geq 0.99$, else the assertion is vacuous), and reads he_fraction_over_one / he_fraction_under_zero, the excursions the operator measures *before* its own clip. Reading the clipped X instead would have shown 1.0 in both columns and proved nothing.
- run_fcheck.sh (-fcheck=bounds, do, mem) clean. That case runs with element diffusion off, so as in section 84. two further bounded runs cover the code this section changes: diffusion_tests.x built with the same flags, and the lower_profile regression case bounded to 400 steps. Both clean, no trap.

86.8 Two more things found in passing, both in the step diagnostic

- The diagnostic printed $\max |J_{\text{He}} + J_1|$ as $\max(\text{sumJ}, \text{abs}(Jf(j) - Jf(j)))$ — an identical zero in the clothes of a measurement. A binary mixture has one independent diffusive flux and the hydrogen component carries $-J_{\text{He}}$

by construction, so there is nothing there to measure; the field is removed and the reason is written where it used to be printed.

- The trace-metal solver ends on where $(f_X \text{ .lt. } 0.0d0) \ f_X = 0.0d0$ under the comment “Round-off assertion, not a limiter: the M-matrix keeps $f_X \geq 0$ ” — the same unmeasured claim this section removes from the helium equation. That one is a different case and is left as it is: the trace equation is *linear* in f_X (the trace limit drops the $1 - f_X$ factor), and an M-matrix sign pattern with a nonnegative right-hand side is a real argument for $f_X \geq 0$. But it is an argument, so the excursion is now measured too (trace_ratio_under_zero, reported with the step diagnostic) instead of being asserted.

87. A rate coefficient that jumped by 1.57 at 4000 K: the He (2^3S) Penning rate, and the step it put in every figure (2026-08-28)

The judgement first. A rate coefficient is a continuous function of temperature. The He (2^3S)+H Penning rate in `Cool_coeff.f90` was not:

$$\begin{aligned} T \leq 4000 \text{ K} : \quad & 1.9 \times 10^{-9} (300/T)^{0.07} \rightarrow 1.5849 \times 10^{-9} \text{ at } 4000 \text{ K} \\ T > 4000 \text{ K} : \quad & 9.1 \times 10^{-9} (300/T)^{0.50} \rightarrow 2.4921 \times 10^{-9} \text{ at } 4000 \text{ K} \end{aligned}$$

a factor **1.5724** across the seam. That is Taylor et al. (2025), ApJ 989:68, Table 2, transcribed correctly — the code was not wrong about what the paper says. The published fit is what cannot be used, and “it is the published value” is not a defence of an expression that is discontinuous in a state variable. It is replaced by the García Muñoz (2025) form, which is continuous everywhere.

Checking the fit against its own paper made the case decisive. Taylor et al.’s Figure 19 (bottom right) plots the quantity Table 2 is supposed to represent: their Maxwell–Boltzmann average of the Cohen & Lane (1971) and Morgner & Niehaus (1979) cross sections. Digitized here, that curve rises to 1.50×10^{-9} near 1400 K and falls to 1.29×10^{-9} at 4000 K. The tabulated low branch decreases monotonically from 1.83 to 1.58×10^{-9} and, being a power law, cannot produce a maximum at all:

T [K]	Fig. 19 (their rate)	Table 2 low branch	ratio	new rate
500	1.135	1.833	1.62	1.024
1000	1.467	1.746	1.19	1.207
1400	1.502	1.706	1.14	1.271
2000	1.470	1.664	1.13	1.321
4000	1.291	1.585	1.23	1.356

(all in $10^{-9} \text{ cm}^3 \text{ s}^{-1}$).

Above 4000 K the high branch is worse: 2.23×10^{-9} at 5000 K, above *every* cross-section determination collected in García Muñoz (2025) Fig. 5, whose curves lie between about 1.3 and 1.7×10^{-9} there. The appendix of Taylor et al. says how this happened — they fitted “an equation already implemented into our KPP routine for simplicity” with `scipy.optimize.curve_fit`, in two pieces, with nothing tying the pieces together at the join.

The replacement. García Muñoz (2025), A&A 698, A199, Fig. 5 publishes a closed form for the Maxwell–Boltzmann average of the Movre & Meyer (1997) total ionization cross sections,

$$k = 10^{-9} \exp(c/T + d_1 \ln T + d_2 (\ln T)^2 + d_3 (\ln T)^3) \quad [\text{cm}^3 \text{s}^{-1}],$$

$c = -8.64804\text{E}+01$, $d_1 = -2.86766\text{E}-01$, $d_2 = +8.68445\text{E}-02$, $d_3 = -5.73001\text{E}-03$, quoted there as accurate to better than 2% from 200 to 10^4 K. It reproduces that paper’s own Table A.5 (1.02, 1.32, 1.35, 1.27×10^{-9} at 500, 2000, 5000, 10^4 K) to 0.4%, measured here. Note the paper’s network file writes the third term as $\exp(c)$; only the c/T of the Fig. 5 caption reproduces Table A.5, and the code comment records that.

This is not a case of trading one paper’s fit for another’s. Taylor’s Figure 19 and the García Muñoz curve agree to 10–20%; it was the *tabulated* fit that was the outlier. Re-digitizing Movre & Meyer’s Fig. 10 cross sections independently and Maxwell-averaging them reproduces Table A.5 to 5–10%, the accuracy of reading a figure. Movre & Meyer give cross sections only, over 0.01–1.0 eV, and no rate coefficient — the 200 K end of the fit leans on García Muñoz’s low-energy $E^{-0.2}$ extrapolation (15% of the integral weight at 200 K) and the 10^4 K end on the high-energy E^{-1} one (40% of the weight at 10^4 K), which is why the validity range is stated where the expression is written.

The older prescriptions are outliers for identifiable reasons and are not candidates: Roberge & Dalgarno (1982)’s temperature-independent 5×10^{-10} (what EXHALE used before Taylor) sits far below every cross-section calculation, and Fassia et al. (1998)’s $7.5 \times 10^{-10}(T/300)^{1/2}$ is Bell (1970)’s single 300 K point stretched with hard-sphere scaling, which the energy dependence of the measured cross section contradicts.

The H₂ channel and the 0.9:0.1 branching. Both partners ionize the metastable through two channels that share one cross section, Penning ($\text{He}(2^3\text{S}) + \text{X} \rightarrow \text{He}(1^1\text{S}) + \text{X}^+ + \text{e}^-$) and associative ($\rightarrow \text{HeH}^+ + \text{fragment} + \text{e}^-$), and García Muñoz splits the total “an average 0.9:0.1”. The published network file applies exactly that split through the amplitudes of two rows per channel (0.9e–9 / 0.1e–9 for H; 4.86740e–12 / 5.40822e–13 for H₂, summing to 5.408222e–12), which settles what the paper text leaves implicit. EXHALE had charged 100% of both totals to Penning. It now follows the split, and the distinction matters in a way a single scale factor does not:

- the **total** removes the metastable — both channels quench it — so every sink term keeps the sum. Using 0.9 there would have made the metastable density 10% too high, and the metastable density is the observable;
- $f_{\text{penning_HeI23S}} = 0.9$ scales only the terms that leave a lasting proton or H₂⁺ behind and that deposit the Penning exothermicity;
- the molecular system carries HeH⁺ with its own balance row, so the remaining 0.1 is now a real HeH⁺ source there. The atomic systems have no HeH⁺ row and drop it, on the stated ground that in an H₂-poor gas HeH⁺ dissociatively recombines back to He + H faster than it reacts onward, so the branch is a metastable sink but not a lasting proton or electron source. Where H₂ is abundant that cycle is broken by $\text{HeH}^+ + \text{H}_2 \rightarrow \text{H}_3^+ + \text{He}$, which is why the molecular system does not take the same shortcut.

The H₂ rate itself is now the published $5.408222 \times 10^{-12} T^{0.675388} \exp(-696.275/T)$ rather than a fit made here to the four Table A.5 points. The old fit was accurate (0.13%) but reproduced the *total*, so charging all of it to Penning overstated H₂⁺ production from this reaction by 1/0.9.

Naming. penning_HeI_23S and penning_HeI23S_H2 returned totals, not Penning rates, and had inconsistent names for the same physics. They are now ioniz_HeI23S_H and ioniz_HeI23S_H2, both elemental functions, named for what they compute.

This step was the discontinuity in the LHS 1140 b memo figures. The detector in LHS1140b/exhale/metastable_step_diagnosis flags a cell whose neighbour ratio departs by more than 10% from the geometric mean of the ratios on either side. Before and after, on the same three runs:

run	flags before (eq / adv)	flags after
heh0p55	6 / 6	0 / 0
heh2p13_diff_kzz1e9	6 / 6	0 / 0
flux_closure/heh11p1/k01	4 / 8	0 / 4

The four that remain in k01 are at $r = 19.15, 19.47, 28.57$ and $29.05 R_p$, were present before with the same values, and are outer-boundary features, not Penning $T=4000$ ones. All three runs do cross 4000 K, so the test is live: the inner crossing is at 1.1775, 1.1084 and $1.1123 R_p$. At the crossing cell of heh0p55 the neighbour ratio $n(2^3S)_{j+1}/n(2^3S)_j$ went

before: 1.0199 1.0194 **0.6651** 1.0232 1.0227 (a 33% drop in one cell)

after: 1.0190 1.0185 1.0180 1.0175 1.0171

and the other two runs behave the same way ($0.6681 \rightarrow 1.0242$, $0.6968 \rightarrow 1.0352$).

What moves. The new coefficient is *lower* than the old one everywhere in the wind (ratio new/old 0.69 at 1000 K, 0.86 just below 4000 K, 0.54 just above, 0.81 at 10^4 K), so the metastable lives longer and the 10830 line gets stronger. Removing the step is the small part of this; the change in level is the large part.

Regression matrix, current binary against the goldens as they stand ($\max |\text{new} - \text{golden}| / \max |\text{golden}|$ over the column, pointwise relative maximum in parentheses):

case	ρ	v	p	T	heat	$\log_{10} \dot{M}$
wasp_full	8.6e-05 (3.3e-03)	7.8e-04 (1.8e-02)	4.2e-05 (1.5e-03)	6.4e-04 (1.3e-03)	2.6e-03 (5.8e-03)	13.23 $\rightarrow$ 13.23
wasp_he23off	0	0	0	0	0	13.23 $\rightarrow$ 13.23
mol_base_handoff	2.2e-06 (2.2e-02)	1.1e-02 (2.2e+00)	4.3e-06 (3.9e-02)	1.5e-02 (1.7e-02)	6.1e-03 (2.7e-02)	10.58 $\rightarrow$ 10.57
mol_metals	1.6e-05 (2.3e-02)	1.1e-02 (1.4e-01)	1.3e-06 (4.0e-02)	1.5e-02 (1.7e-02)	5.4e-03 (2.7e-02)	10.58 $\rightarrow$ 10.58
mol_lyman_werner	6.3e-05 (2.9e-02)	1.5e-02 (8.4e-01)	7.8e-06 (4.8e-02)	1.8e-02 (2.0e-02)	7.8e-03 (3.3e-02)	10.58 $\rightarrow$ 10.58
mol_diffusion	2.2e-06 (2.2e-02)	1.1e-02 (1.9e+00)	4.3e-06 (3.9e-02)	1.5e-02 (1.7e-02)	6.1e-03 (2.7e-02)	10.58 $\rightarrow$ 10.57
lower_profile	3.2e-06 (2.8e-03)	2.5e-02 (2.6e+00)	4.6e-06 (3.5e-03)	3.0e-02 (3.9e-02)	3.3e-03 (8.8e-02)	9.58 $\rightarrow$ 9.58

wasp_he23off is byte-identical, which is the control: with the metastable off the coefficient is never evaluated. The large pointwise v ratios are cells where the golden velocity passes through zero, not large absolute changes — the scale-relative column is the one to read. The H_2 front does not move ($1.0181 R_p$ before and after in all four molecular cases), and $\log_{10} \dot{M}$ moves by at most 0.01 dex.

The species that moves is the one the change is about: peak $n(\text{He}2^3S)$ rises by 3.1% (wasp_full), 6.9% (lower_profile) and 22–30% (the molecular cases). Peak HeH^+ rises by a factor 116–710 in the molecular

cases ($2.6 \times 10^{-1} \rightarrow 3.4 \times 10^1 \text{ cm}^{-3}$ in `mol_base_handoff`), which is the newly resolved associative branch — previously the only HeH^+ source in those runs was $\text{He}^+ + \text{H}_2$, and this one is larger.

Goldens are NOT refreshed here. The matrix is measured, not re-snapshotted; the refresh belongs at the end of the change series. make `diffusion_tests 35/35` and `run_fcheck.sh CLEAN` with the change in.

The LHS 1140 b numbers move, and the helium-abundance crossings move with them. Re-solving the three reference winds (the flux-closure case with the wind re-converged on a fixed profile, no closure iteration) gives:

run	EW before	EW after	FWHM before	after	$\log_{10} \dot{M}$
heh0p55	1.10884	1.30991 (+18.1%)	0.2573	0.2583	7.76, unchanged
heh2p13_diff_kzz1e9	1.12887	1.33805 (+18.5%)	0.2553	0.2568	7.80, unchanged
flux_closure/heh11p1/k01	1.08532	1.16816 (+7.6%)	0.2825	0.2840	7.40, unchanged

(red-pair equivalent width in % $\text{\AA}$).

The mass-loss rate is untouched: this is a trace-level population, not an energy-budget term. The crossings with the measured equivalent width, solved on re-computed composition ladders rather than from the local slope (the +18% shift is well outside the slope’s range of validity), move down:

ladder	before	after	1σ
well mixed, diffusion off	0.55	0.424	0.409–0.440
$K_{zz} = 10^9$ (adopted)	2.09	1.626	1.573–1.685
flux closure	11.73	9.317	8.792–10.050

The local-slope estimate (slopes 0.709, 0.921, 0.382) gives 0.435, 1.738 and 9.676 for comparison; the direct ladder solve is the number to use. Two caveats stand on the flux-closure entry: 9.317 is an interpolation across existing arms, not a re-run closure, and confirming it means solving a closure arm at a reservoir near 9.3, where the cold-trap composition differs arm by arm. The stored He/H grid runs elsewhere in LHS1140b/exhale are still on the old coefficient and were not re-solved.

`docs/lhs1140b_exhale_vs_pwinds.tex` is *not* edited here. Every quantitative statement in it that depends on $n(2^3S)$ is now stale — the three crossings above, the Penning share of the metastable sink (89%/60% at $\text{He}/\text{H} = 2.09/10.31$) and the lifetimes in its Table 4, its EW table, and the two places that name the rate as “the Taylor et al. (2025) fit”. Its figures that show the He 10830 profile or $n(2^3S)$ need regenerating, and `figures/lhs1140b_ew_vs_heh.pdf` in particular carries the crossing.

Files. `src/modules/radiation/Cool_coeff.f90` (both rate functions renamed and replaced, `f_penning_HeI23S` added); `ion_residual_core.f90` (Penning branch on the H^+ row, total kept in the triplet row); `System_HeH_mol.f90` (branch on the H^+ and H_2^+ rows, associative branch into the HeH^+ row, total on the H_2 and triplet sinks); `System_implicit_adv_HeH_TR.f90`; `util_ion_eq.f90`; `ionization_equilibrium.f90`; `post_process_adv.f90` (heating terms). `docs/molecular_chemistry_audit_he_rich.md` section 4.4 updated, including the withdrawal of its claim that the step entered the Jacobian — the coefficient is constant within a cell, so it appeared only as a jump in the loss rate between neighbouring cells. `references/garcia_munoz_2025_network/` holds the published network file the branched amplitudes come from.

88. A population solved as a difference, and a post-process that never read info (2026-08-29)

The judgement first. Two defects in the advection-corrected post-process, both numerical, both fixed here.

1. **The ground-singlet helium population was solved as a difference.** `System_implicit_adv_HeH_TR` carried the *summed* He I fraction as x_2 and the metastable as x_4 , and formed the singlet as $x_2 - x_4$. Where helium is heavily ionized and the little neutral helium left sits mostly in the metastable, that difference loses every significant digit and finally evaluates to exactly zero. The summed He I row then degenerates — it no longer contains the unknown it is supposed to determine — and `hybrd1` returns `info = 4` with a residual it cannot reduce. Fixed by solving for the two populations, $n(1^1S)/n_{\text{He}}$ and $n(2^3S)/n_{\text{He}}$, and forming the *sum* where the total is wanted. A sum has no such failure mode.
2. **No return code of the post-process was ever read.** `post_process_adv` calls `hybrd1` three times per cell — the advection ionization system, the metal stage re-solve (`pp_metals = 2`), the energy equation — and wrote the returned iterate out in every case, converged or not. Fixed: a cell whose solve did not converge keeps the equilibrium state it came in with, and the number of such cell solves is reported.

The run that exposed both, `LHS1140b/exhale/heh0p55_diff_ctrl`, is **not made physical by either fix**, and the reason is a third thing, left untouched here and written up in § 88.4: its post-process drives P_{HeI} to $7 \times 10^{15} \text{ s}^{-1}$.

88.1 A population, not a difference

The advection systems integrate the ionization ODE upwind across one cell. The triplet system solved for $x_1 = n_{\text{HI}}/n_{\text{H}}$, $x_2 = n_{\text{HeI}}/n_{\text{He}}$ (summed), $x_3 = n_{\text{HeIII}}/n_{\text{He}}$, $x_4 = n(2^3S)/n_{\text{He}}$, and built the singlet, which every rate in rows 2 and 4 needs, as $x_2 - x_4$. Nothing keeps the two apart. In the helium-settled control run the difference collapses pass by pass until x_2 and x_4 are the same double, at which point the singlet is exactly zero, row 2 reduces to a balance between the metastable and He II alone, and the solver stalls. Measured on that run, before the change: **310 of 4660 advection cell solves returned info = 4** (ten post-process passes, `OMP_NUM_THREADS=1`); the other three runs checked returned `info = 1` everywhere.

The unknowns are now the two populations themselves, and row 2 is the ground-singlet balance — the summed He I row minus the metastable row of row 4, which is the same system written in different variables:

$$f_2 = x_{s,\text{old}} - x_s + c_1 \left[x_{\text{HeII}} \alpha_{\text{HeII}} n_e - x_s (P_{\text{HeI}} + (b_{\text{HeI}} + q_{13}) n_e) + x_t ((q_{31a} + q_{31b}) n_e + A_{31} + x_{\text{HI}} Q_{31} n_{\text{H}}) \right]. \quad (36)$$

Loss of the singlet: photoionization, electron-impact ionization, and collisional excitation q_{13} into the metastable. Gain: recombination of He II into the singlet ladder (α_{HeII} ; the α_{2^3S} channel feeds the metastable and is charged to row 4), and every route by which the metastable comes back to the ground state — collisional de-excitation $q_{31a} + q_{31b}$, the $2^3S \rightarrow 1^1S$ decay A_{31} , and the $\text{He}(2^3S) + \text{H}^0$ ionizing collisions Q_{31} , whose Penning branch leaves $\text{He}(1^1S) + \text{H}^+ + e^-$ and whose associative branch makes HeH^+ that dissociatively recombines back to ground-state helium. The He–H charge-exchange pair stays where it was, on rows 1 and 2, with the *summed* He I as the reactant density: that is the accounting

the summed-He I form carried and the one the equilibrium systems make, neither of which gives the metastable a charge-exchange channel of its own.

post_process_adv now carries nheiS and nheiTR as the two profiles it solves for and forms $nhei = nheiS + nheiTR$ wherever a routine wants the total. The difference survives in exactly one place, the entry point, where the equilibrium solution hands over a summed He I and a metastable; it is well conditioned there because the equilibrium metastable is bounded by its own balance, whose loss side is dominated by A_{31} .

What the change does to the answer. The reparameterization is a change of variables, so a converged solution must not move. Measured, PP-only re-runs at OMP_NUM_THREADS=1 against a baseline built from the same tree without the change:

run	info $\neq 1$ solves, before $\rightarrow$ after	max relative change of the _adv species
heh2p13_diff_kzz1e9 ($K_{zz} = 10^9$)	0 $\rightarrow$ 0	2.6×10^{-9} (He II), median 3×10^{-14}
flux_closure/heh11p1/k01	0 $\rightarrow$ 0	2.0×10^{-9} (He II), median 3×10^{-15}
heh0p55 (no diffusion)	0 $\rightarrow$ 0	9.7×10^{-4} (He III), median 1.3×10^{-4}
heh0p55_diff_ctrl ($K_{zz} = 0$)	310 $\rightarrow$ 0	0.25 (He I and 2^3S , at $2.7 R_p$)

The first two are roundoff: the solve is the same solve. The heh0p55 entry is *not* the reparameterization — it is the new info guard of § 88.2 firing on one cell of the energy equation in passes 5 and 7. The last is the run whose solution was not a solution.

Where the cancellation went. In the control run at $r = 2.53 R_p$, tenth pass, the same cell before and after:

	before	after
unknowns	$x_2 = x_4 = 1.089669 \times 10^{-5}$ (singlet exactly 0)	$x_s = 2.53 \times 10^{-22}$, $x_t = 1.098 \times 10^{-5}$ ($x_s/x_t = 2.3 \times 10^{-17}$)
info	4	1
residual	($-9.5e-4$, $1.22e-2$, $7.7e-6$, $-3.5e-4$)	($4.3e-17$, $2.4e-8$, $-3.3e-18$, $9.6e-18$)

The largest residual falls from 1.2×10^{-2} to 2.4×10^{-8} , and the singlet keeps its relative accuracy seventeen decades below the metastable, where the difference form had nothing left. One-cell steps in the _adv metastable profile of that run, counted with the local-trend detector of § 87, go from **9 flagged cells to 1** — and that one, at $r = 29.05$, is the outer-boundary feature the equilibrium profile carries too.

Its neighbours are the more instructive record: cells 334 and 335 returned info = 1 with $f_2 = 1.17 \times 10^{-2}$ and 1.31×10^{-2} . hybrd1's info = 1 is a test on the *increment* between iterates, not on the residual, so a solver frozen against a degenerate row reports success. **The info check of § 88.2 would not have caught those cells.** Only the reparameterization does.

88.2 The return code

post_process_adv made three hybrd1 calls and read the return code of none of them:

call	what it solves	on failure, before
<code>adv_implicit_H / _HeH / _HeH_TR</code>	the advection ionization system of the cell	the <code>iterate</code> was written to <code>nhi..nheiTR</code> and fed the upwind cascade of the next cell
<code>ion_system_metals_pp</code>	the metal stage split at fixed H/He (<code>pp_metals = 2</code>)	the <code>iterate</code> replaced the equilibrium split
<code>T_equation</code>	the advected energy equation (legacy MINPACK path)	the <code>iterate</code> became $T_{\text{out}}(j)$, subject only to the metal-mode band reject

A non-converged `iterate` satisfies neither the advection balance it was asked to solve nor the equilibrium balance it started from, so it is not a state of the gas. The equilibrium solution of that same cell is, and it is already what the three validity conditions of § 72 fall back to. Each call now keeps the equilibrium state on `info ≠ 1` — the ionization of the cell, the equilibrium metal split, the converged equilibrium temperature — and counts it. The counts are summed over the ten passes, not read off the last one: a cell reverted in an early pass feeds that pass’s upwind cascade whether or not the last pass converges everywhere.

Measured on four LHS 1140b runs, the guard is quiet: the advection system fires nowhere once § 88.1 is in, and the energy equation on one cell in two of the ten passes of `heh0p55` and of the He 2³S-off variant of `heh2p13_diff_kzz1e9` — worth 10^{-4} to 10^{-3} in the `_adv` profiles of those runs. The metal guard is untested by them: `pp_metals` defaults to the frozen mode (1), so the re-solve block is one those runs never enter.

The guard accepts `info = 1` and rejects everything else. That is a statement about the increment, not about the residual, and § 88.1 shows the two can disagree. A residual test would be the stronger guard; it needs a threshold in the units of the rows (a fraction per cell crossing), which is a choice, not a measurement, and it is not made here.

A third, smaller thing in the same block. With helium off (`thereis_He = .false.`) the post-process left `nhei`, `nheii`, `nheiii`, `nheiTR` — and, with this change, `nheiS` — undefined until the helium-free branch of the ionization loop zeroes them at its *end*. The first pass reads them before that, in `nhe`, in the electron sum and in `eval_cool`. They are now zeroed where the helium arrays are initialized. Helium-on runs are bit-identical across the change (checked on `heh2p13_diff_kzz1e9`, all three `_adv` files).

88.3 Why nothing caught this

A run’s convergence tests never look at the post-process.

- *du* and the steady residual $\|R\|$ are momentum and energy metrics of the *wind*, evaluated in the time loop. The loop is over before `post_process_adv` is called; there is no metric downstream of it.
- `ionization_equilibrium` validates the root it accepts — physical simplex, up to three starting points, `ok_rank`, and a one-line sweep summary of reseeds, retries, unphysical roots and failures (§ 45). The advection-corrected post-process, solving a system of the same size with the same solver, validated nothing.
- `make check` bit-compares `Hydro_ioniz.txt` and `Ion_species.txt`. Those are the *equilibrium* files, written before `post_process_adv` runs. **No golden case compares an `_adv` file at all**, so nothing in the regression can

move when the post-process does — which is also why the change documented here leaves the goldens byte-identical (7/7, 14/14 files), and why that result carries no information about the `_adv` profiles. The A/B tables above are what stands in for it.

That leaves the `_adv` files checked only by whoever plots them. The two defects were found that way: an order-of-magnitude one-cell step in a metastable profile.

88.4 What the control run actually shows, and what is not fixed here

With § 88.1 in place the solver converges everywhere in `heh0p55_diff_ctrl` and still returns a neutral-helium column five orders of magnitude below the equilibrium one. The singlet is not being lost to cancellation any more; it is being destroyed by the rate the post-process hands the solver. Traced pass by pass at $r = 2.53 R_p$:

pass	1	2	3	5	10
$P_{\text{HeI}} [\text{s}^{-1}]$	2.5e-4	3.7e-2	5.3e0	1.1e5	6.9×10^{15}
$x(1^1S)$	6.8e-3	4.7e-5	3.3e-7	1.6e-11	2.5e-22

A photoionization rate of $7 \times 10^{15} \text{ s}^{-1}$ is not a rate. Two things produce it, and neither is touched here.

(a) A volumetric secondary-ionization rate divided by the density of the species it ionizes. `util_ion_eq.f90` adds the Shull & van Steenberg (1985) secondary channel as $P_{\text{HeI}}(j) = P_{\text{HeI}}(j) + R_{\text{secHeI}}/\max(n_{\text{HeI}}(j), 1.0\text{d-99})$. R_{secHeI} is the energy deposited per unit volume times the SvS85 branching into He I ionization: a *volumetric* rate, and one carrying SvS85's own fixed helium abundance. Dividing it by the actual $n(1^1S)$ gives the right per-atom rate only at that abundance, and diverges as the singlet disappears. In this run helium has settled to $n_{\text{He}}/n_{\text{H}} = 6 \times 10^{-6}$ at $1.36 R_p$, four decades below the SvS85 composition, and the ten post-process passes close the loop: a smaller singlet raises its own destruction rate, which shrinks it further. The $\max(\dots, 1\text{e-99})$ is what stops the divergence, at a rate of order 10^{60} . The physical scaling is the opposite one — the secondary He I ionization rate per unit volume should fall with the helium abundance, roughly as $n_{\text{HeI}}\sigma_{\text{HeI}}/\sum_a n_a\sigma_a$, leaving the per-atom rate near its solar-composition value.

(b) In a Do only PP: True run the equilibrium solve and the post-process use different physics. The SvS85 coupling is staged: `sec_ion_active` starts false and is switched on once the wind has first converged (§ 38). `EXHALE_main.f90` also sets it unconditionally *after* the time loop, so the post-process always runs with the full coupling. With `Do only PP` the loop exits after one iteration, so the single `ionization_equilibrium` call runs with `sec_on = F` and all twenty-one `PH_heat_HHe` calls of the post-process run with `sec_on = T` (printed and counted). The equilibrium P_{HeI} at $1.36 R_p$ of the control run is $2.4 \times 10^{-7} \text{ s}^{-1}$; the post-process's first pass, at the same densities, $4.3 \times 10^{-5} \text{ s}^{-1}$. The `_adv` files of such a run are not a correction to the equilibrium files next to them: they are a different model.

(c) The same difference, in the radiation. `util_ion_eq.f90` builds its own singlet as $n_{\text{HeI}} = n_{\text{HeI}} - n_{\text{HeITR}}$ and uses it for the He I column density, the He I photoionization integrand and the denominator in (a). The state vector of the whole code carries (summed He I, metastable) and reconstructs the singlet wherever it is needed — `species_table.f90` says so in as many words. § 88.1 fixes the one system where the difference was catastrophic;

carrying $n(1^1S)$ as the state variable throughout would touch the global species vector, the output schema and every reader of it, and is not attempted here.

Six of the 354 stored LHS1140b runs carry the signature of (a) — $n_{\text{HeI}} \leq n(2^3S)$ in the `_adv` file above $1.5R_p$: `heh0p55_diff_ctrl`, `heh0p55_diff_d3`, `heh0p55_diff_fix` (171 cells of 233 each), `heh0p55_diff_kzz1e6` (36), `xuv0p01_heh2p13` (25) and `xuv0p10_heh2p13` (28). All six are helium-settled: $K_{zz} \leq 10^6$, or a reduced XUV flux. The $K_{zz} = 0$ arm of Fig. `lhs1140b_kzz_profiles.pdf` and its quoted peak $n(2^3S) = 0.046 \text{ cm}^{-3}$ are read from the first of them.

Files. `src/modules/nonlinear_system_solver/System_implicit_adv_HeH_TR.f90` (the two populations as unknowns, the ground-singlet row); `System_implicit_adv_HeH.f90` (the same naming, no arithmetic change: without the metastable all He I is the singlet); `ion_cell_state.f90` (`xhei_old` $\rightarrow$ `xheiS_old`); `src/modules/post_process/post_process.f90` (`nheiS` and `nheiTR` as the profiles solved for, `nhei` formed as their sum, the three info guards, their counters and reports, and `pin_cell_to_equilibrium`, which the validity conditions and the guards share). `docs/postprocess_advection_validity.f90` carries the same items next to the 2026-08-12 pair. `docs/lhs1140b_exhale_vs_pwinds.tex` is *not* edited.

Found in passing and fixed, outside the post-process: `LHS1140b/make_memo_figures.py` and `LHS1140b/exhale/kzz_scan_table.py` formed the elemental helium density as `HeI + HeII + HeIII + HeITR`, counting the metastable twice — it is a level of He I and is already inside the He I column, which `species_table.f90` states in capitals. Both now sum the three ion stages. Measured on the four K_{zz} arms of the memo figure, the quoted elemental ratios move by at most 2×10^{-6} relative ($0.318246 \rightarrow 0.318244$ at $5R_p$, $K_{zz} = 10^{10}$), so nothing was regenerated.

89. A conservation check that could not see conservation: the deepest-level elemental test and the reservoir band it refused (2026-08-29)

Read with § 90. The diagnosis below stands, but its two decisions were superseded the same day. Photochem was corrected at the source, so the `--abundance-tol` default went $10^{-3} \rightarrow 10^{-10}$ and the three `clima` failures noted at the end were fixed (two others appeared).

The judgement first. The profile adapter's deepest-level elemental check was refusing converged LHS 1140 b columns on **the residual of Photochem's chemical-equilibrium solver**, not on any error of the photochemistry, the transport or this code's carrier summation. At the level the check is stated, Photochem pins every gas species of the handed-over mixture with `bc_type='press'` (`gasgiants._initialize_atmosphere`), so that level carries its equilibrium initialization unchanged and **no photochemical or transport leak can reach the check at all**. What it reads instead is how far `composition_at_metallicity` got, and that routine states `# Do not enforce convergence`. The `--abundance-tol` default is raised from 10^{-4} to 10^{-3} — an interim value for the solver as it then was; § 90 tightens it to 10^{-10} once the solver is fixed. Full diagnosis, with every measurement: `docs/deep_level_elemental_check.md`.

89.1 The symptom

Reservoir He/H = 8.5, 8.55, 8.6, 8.65 and 8.7 could not be run in the closure ladder of `LHS1140b/exhale/crossings_gm25/`: the N/H ratio at the deepest level departed from the input by 1.02, 1.22, 1.43, 1.67 and 1.94×10^{-4} against the 10^{-4}

tolerance. Two closure lineages with different trial escape fluxes give the same numbers to the three digits the refusal prints, and a standalone chemistry-only scan reproduces them a third time — as expected, since the trial flux enters only at the upper boundary and the refused level is Dirichlet.

89.2 The level is a Dirichlet boundary, and the solver behind it is used unconverged

Measured at $\text{He}/\text{H} = 8.6$: chemical equilibrium plus the quench overwrite, the photochemical model's initial state, and the converged steady state agree at the deepest level to **all 13 digits printed** ($X_{\text{N}} = 8.185823483630 \times 10^{-5}$ in all three; deep $q_{\text{NH}_3} = 8.995632484 \times 10^{-6}$ converged against 8.995632×10^{-6} initial). Called on its own at the same P and T , `equilibrate.ChemEquiAnalysis.solve` returns `converged = True` at every reservoir swept while its own `molfrac_atoms_gas` misses the requested N/H by up to 1.72×10^{-4} .

89.3 Why nitrogen and nothing else

In all 38 runs of the scan $\text{dev}_{\text{N}}/\text{dev}_{\text{He}} = 4072$ (4069–4086 over the whole set, 4072 in 32 of them), against $1/(3\text{N}/\text{H}) = 1/(3 \times 8.18465 \times 10^{-5}) = 4073$. Deep nitrogen is essentially all NH_3 — on the $\text{He}/\text{H} = 8.6$ profile the N_2 nuclei share of X_{N} is 1.0×10^{-5} and HCN 's is 2.2×10^{-25} — so **one excess N rides on three excess H**: He, C and O carry the common hydrogen error alone, N carries it plus the relative error of the equilibrium NH_3 mole fraction. The check was testing that one mole fraction 4072 times more tightly than anything else, which nobody chose.

89.4 Not a branch change, and not a special band or planet

Across $8.45 \rightarrow 8.75$ the deep H_2 , H_2O , CH_4 , NH_3 , N_2 and He all move monotonically by less than 1 % with no jump; the residual jumps by a factor 38 between 8.71 (1.99×10^{-4}) and 8.72 (5.2×10^{-6}), 1500 between 9.15 and 9.20, and 5800 between 10.25 and 10.30. At the old tolerance the same scan also refuses $\text{He}/\text{H} = 6.0$ (1.31×10^{-4}) and 10.20–10.25 ($1.5\text{--}2.0 \times 10^{-4}$); the tree already held refusals at $\text{He}/\text{H} = 14.643$, 15.090 (2.12×10^{-4} , the largest recorded) and 25.0 (LHS1140b/exhale/kzz_profile_scan/); and at **solar composition** the same solver misses N/H by 1.50×10^{-4} at 800 K and 1.14×10^{-4} at 1600 K. The two HD 209458 b departures that fixed the original default (1.6×10^{-7} , 2.7×10^{-5} ; § 77) were a fortunate sample, not a floor.

89.5 No accounting error

Recomputed from the written profile, the diagnostic columns account for **100.00 %** of X_{N} at the deepest level. They account for 93.9 % at the 1 μbar match, and any earlier reading of a shortfall in the carrier sum refers to *that* level and to the deliberately short `q_*` diagnostic column set — the check reads neither.

89.6 The change

`src/utls/lower_profile_schema.py`: the `--abundance-tol` default goes $10^{-4} \rightarrow 10^{-3}$, and both the option help and the comment at the check now state what the level can and cannot test. 10^{-3} leaves a factor 4.7 over the worst departure on record and gives up next to no detection power: each element sits in one dominant carrier there, so a real

miscount is an $O(1)$ error, and the smallest conceivable one — dropping N_2 — is 1.0×10^{-5} of X_N in the band at issue and was already invisible at 10^{-4} . Over the 38 profiles the N_2 nuclei share runs 4.2×10^{-6} ($\text{He}/\text{H} = 12$, deep $T = 379$ K) to 4.0×10^{-3} ($\text{He}/\text{H} = 1$, deep $T = 588$ K), rising with the deep temperature, so where dropping it would be a real loss it stays above 10^{-3} . Nothing else was changed: not the check, not its location, not the refusal policy. **Superseded by § 90:** the headroom argument was a statement about how far the 0.8.4 equilibrium solver missed, and once that was fixed the default went to 10^{-10} .

89.7 What it means for the LHS 1140 b crossings

Against `crossings_gm25/results.txt`: the flux-closure crossing at $\text{He}/\text{H} = 9.048$ is untouched — the 9.0 and 9.1 arms both ran and bracket it — and so is the -1σ end 10.77, bracketed by 10.6 and 11.0. The $+1\sigma$ **end 8.49 was interpolated across the un-runnable band**, a 3.4 % gap in reservoir between the 8.45 and 8.75 arms; with the tolerance raised those five rungs can be run and that end replaced by measurement. That was done, on the corrected build: § 90. Because the refusals at 6.0, 10.20–10.25, 14.643, 15.090 and 25.0 are the same trap at other reservoir values, and nothing in the pattern predicts where the next one falls, any future ladder can meet it anywhere.

89.8 Seen in passing, not investigated

`clima`'s `surface_temperature_bg_gas` root solve fails outright at three of the 41 reservoir values attempted — $\text{He}/\text{H} = 0.0969$ (with Failed to compute heat capacity), 8.1 and 15 — so those compositions produce no column at all. That is a **different** failure from the refusal above (this one happens before a column exists, the refusal after), and it is recorded as item (M) of `TO_BE_DONE.md`. All three are fixed by the bounded, scaled climate solves of § 90; the corrected build refuses two other compositions instead, $\text{He}/\text{H} = 9.4$ and 9.5, and § 91 shows that is the same underlying solver defect moving address rather than a new one — 0.8.4 refuses two compositions of its own on the same grid.

Files. `src/utis/lower_profile_schema.py`; `docs/deep_level_elemental_check.md` (the diagnosis); `LHS1140b/exhale/nh_re` (the measurement record: README, four scripts, five tables, 38 adapter runs under scan/). No EXHALE run was made for any of it and no golden was touched; the adapter's chemistry is unchanged and `make check` does not exercise this path.

90. The equilibrium residual removed at its source, and the observable that did not move (2026-08-29)

The judgement first. The residual § 89 diagnosed was fixed where it came from — in Photochem's chemical-equilibrium path, not by relaxing this code's check — and **it does not move the observable**. Re-measured end to end on the corrected build with 11 arms, the LHS 1140 b flux-closure crossing of the He I 10830 equivalent width is at $\text{He}/\text{H} = 9.0484$ (chord) / **9.0461** (quadratic), the same to every digit the photochem 0.8.4 ladder printed; on seven like-for-like arms the equivalent width moves by at most 1.2×10^{-6} in relative terms and $\log_{10} \dot{M}$ is unchanged in its fourth decimal. The one EXHALE change is that `--abundance-tol` goes back to its design value, $10^{-3} \rightarrow 10^{-10}$. Records: `LHS1140b/exhale/crossings_pc090/results.txt`, `docs/photochem_solver_modification_implementation.md`.

90.1 What was changed, and where

Nothing in the EXHALE Fortran. The corrected chemistry is a modified Photochem v0.9.0 source tree, built as photochem 0.9.0 into a dedicated conda environment:

- **Equilibrate** now tests each positive requested elemental abundance against its own relative tolerance ($\text{abs}(b_0(i_{\text{atom}})) \cdot \text{mass_tol}$) instead of a common absolute threshold scaled by $\text{MAXVAL}(b_0)$, and no longer excludes elements at or below 10^{-6} . This is the actual cause of § 89.
- **gasgiants.py** exposes `equilibrium_mass_tol` (default 10^{-12}), assigns it to `ChemEquiAnalysis.mass_tol`, verifies the returned elemental vector against the requested one, raises rather than silently keeping the last result when all five temperature retries fail, and restores `use_prev_guess` in a `finally` block.
- **clima** v0.7.5 gets a bracketed background-pressure solve, a $\log_{10}(T_{\text{surface}} - T_{\text{trop}})$, $\log_{10} T_{\text{trop}}$ temperature parameterization, and normalized residual acceptance, replacing unconstrained MINPACK solves.

Both dependency changes are reproducible patches applied through Photochem's CMake configuration. Diagnosis: `docs/photochem_solver_modification_investigation.md`.

90.2 A blocking defect, found only by running a column

The first version of the strict closure test **rejected every composition**, including ones that ran on 0.8.4. It compared the requested elemental vector with `gas.molfracs_atoms`, the total over gas and condensate, at every level. With `rainout_condensed_atoms=True` the vector handed to the next level is the gas-only one, and `gas.molfracs_atoms` started from the previous level through `use_prev_guess` still carries that level's condensate: measured on the LHS 1140b column at $\text{He}/\text{H} = 9$, the total reads 1.1×10^2 against the request where the same state solved from scratch closes to 5.7×10^{-14} . Identical on 0.8.4 and on the patched build, so it is Equilibrate's existing reporting behavior, not something the Fortran patch created.

The test now forms the residual **only at a level whose `molfracs_atoms_condensate` is entirely zero** — which is every level deep enough to set an elemental handoff, and where the residual it exists to catch appears. Why the defect survived the first round of tests is worth recording: those tests ran **one level for each of the 38 saved deep states** and never carried a column, so `use_prev_guess` chaining and condensing levels were never reached.

90.3 Measured

Re-solved standalone, the 38 saved deep states of LHS1140b/exhale/nh_refusal_diagnosis/scan/ give a largest relative elemental residual of $3.539391003 \times 10^{-13}$ and a median of $1.915134717 \times 10^{-14}$, against $2.018197773 \times 10^{-4}$ for the same measure on 0.8.4 — all 38 converged and independently accepted. Over the 68 handoff writes the corrected build made under LHS1140b/exhale/ (43 in `scan_pc090/`, 25 in `crossings_pc090/`), the largest deepest-level departure is 3.28×10^{-13} and the median 1.73×10^{-14} .

90.4 The tolerance

src/utls/lower_profile_schema.py: --abundance-tol default $10^{-3} \rightarrow 10^{-10}$, a factor 305 above the worst hand-off write measured. Detection power is untouched — each element sits in one dominant carrier at that level, so a genuine miscount is an $O(1)$ error, and the smallest conceivable one, dropping N_2 , is 10^{-5} to 4×10^{-3} of X_N . **A 0.8.4 handoff is refused at this default, and that is the intended signal:** over the 43 stored 0.8.4 handoffs of crossings_gm25/ the departures run 3.48×10^{-8} (best) to 8.50×10^{-5} (worst), so even the best of them misses 10^{-10} by a factor 348. Reproducing a stored 0.8.4 result needs an explicit --abundance-tol on the command line, which is how a deliberate reproduction states that it accepts that residual; the scans under nh_refusal_diagnosis/ already pass --abundance-tol 1.0 and are unaffected.

90.5 What it does to the LHS 1140 b crossings

quantity	0.8.4, 16 arms	corrected build, 11 arms
crossing, chord	9.0484	9.0484
crossing, quadratic	9.0461	9.0461
1σ band	8.4923–10.7716	8.4649–10.7357

The crossing does not move. The handoff profile does: at $He/H = 9$ the deepest-level N/H departure goes from 2.1×10^{-5} to 1.2×10^{-14} and trace species such as q_H2O, q_CO and q_HCN move by up to 10^{-2} relative — none of which reaches He I 10830, a line set by H, He and the temperature. **The 1σ band moves for a different reason than the solver:** its low end used to be interpolated across the un-runnable band $He/H = 8.5\text{--}8.7$, which now runs (departures $1.4\text{--}3.2 \times 10^{-14}$), so real arms at 8.5 and 8.6 carry that end and it goes $8.4923 \rightarrow 8.4649$; the high end goes $10.7716 \rightarrow 10.7357$ because the 10.6 arm was not seeded the same way in the two ladders, a 1.1×10^{-3} seed difference in equivalent width, the size of the seed spread the stored ladder already measured.

Also opened: solar composition. $He/H = 0.0969$ is runnable end to end for the first time. It was blocked not by the tolerance but by the climate step, which now solves (deep boundary 720.5 K, against 432.4 K at $He/H = 8.1$ and 393.1 K at 15 — the other two recovered compositions).

90.6 Two compositions the corrected build refuses, and what they turned out to be

$He/H = 9.4$ and 9.5 raise Could not bracket background pressure in make_profile_bg_gas on this build and produce columns on 0.8.4. This section first recorded that as a regression of the new bracketing scan, with the cause undiagnosed. **Section 91 measured it and that reading was wrong.** The scan is not the cause — the root lies inside the interval it searches, with one monotone sign change in the last scan step — and the same defect is present in 0.8.4, which fails at 9.37 and 9.45 on the same grid where this build fails at 9.40 and 9.50. Both builds are tripped by a forward-difference Jacobian formed at a step below the flux residual's round-off jitter. **The net accounting of “three failures removed, one introduced” written here does not hold:** it compared against 0.8.4 columns that had not been run at the matching reservoir values, and 9.45 in particular fails on 0.8.4. Neither value is a bracket arm of the crossing. Diagnosis, repair and status: § 91; item (M) of TO_BE_DONE.md.

90.7 Scope

No EXHALE Fortran source or binary changed, no golden was refreshed, and `make check` was not run: this path is not in the regression matrix. Not tested: condensate-rich and sulfur-network equilibrium sweeps — note that after the fix of § 90.2 a condensing level is accepted on the solver's own converged flag, so the closure test does not speak for such a level at all. The stored 0.8.4 ladder was not re-run, and $\text{He/H} = 9.5$ cannot be repeated on this build. The 0.8.4 environment is untouched and remains the reproduction path for every profile whose header reads `# source_version photochem 0.8.4`.

Files. `src/utis/lower_profile_schema.py` (the default); `docs/photochem_solver_modification_investigation.md` (diagnosis), `docs/photochem_solver_modification_implementation.md` (what was built and measured), `docs/deep_level_elemental_check.md` §§ 8–10 (the tolerance, the crossings, the climate solve); `LHS1140b/exhale/crossings_pc0` (the 11-arm re-measurement) and `LHS1140b/exhale/nh_refusal_diagnosis/scan_pc090/` (the 45-point reservoir scan on the corrected build). The Photochem changes live outside this repository, in the local `photochem/` tree.

91. The clima bracket was not the cause: a Jacobian built from round-off, and a failure that predates the patch (2026-08-29)

The judgement first. The two compositions § 90 reported as a regression are not one. `clima`'s new bracketing scan does not fail to find a root at $\text{He/H} = 9.4$ and 9.5 — the root is inside the scanned interval, and the residual crosses zero once, monotonically, inside the last scan step. The scan fails because the *outer* `surface_temperature_bg_gas` solve hands it a surface temperature of 6.5×10^8 K, where every scan point is thermodynamically invalid. That step comes from a forward-difference Jacobian whose difference step is smaller than the residual's own round-off jitter, and **the 0.8.4 build has the same defect at the same rate** — it fails at 9.37 and 9.45 where the corrected build fails at 9.40 and 9.50. The patch changed which compositions are unlucky, not how many. Record: `LHS1140b/exhale/clima_bracket_diagnosis/` (README, 12 scripts, six sweep and variant tables), measured 2026-08-29 against both installed builds with nothing in `photochem/` modified.

91.1 The bracket is not the cause

`make_profile_bg_gas` scans 49 points over $\log_{10} P$ spanning 12 decades below the requested surface pressure, so for the LHS 1140 b ladder's 20 bar deep boundary the interval is $\log_{10} P = [-4.6990, 7.3010]$. At $\text{He/H} = 9.4$ the 0.8.4 solution sits at $\log_{10} P_{\text{bg}} = 7.278533$, inside it. So does every neighbor: over 9.30–9.60 the 0.8.4 background pressure moves only from 7.278297 to 7.278990, always inside (`sweep_old_9p30_9p60.txt`). Inside one failing call the residual is monotone with exactly one sign change, in the last scan step (`scan_residual.py`) — a grid of 49 points cannot step over it.

What the scan actually reports is that it had **no valid point at all**. The trial surface temperature it is called with is 6.5×10^8 K (`outer_trace.py`), and at that temperature `make_profile` refuses all 49 points with `Failed to compute heat capacity: the thermodynamic polynomials stop at 6000 K (T_ceiling.py)`. With no valid point there is no sign

change to find, and the message `Could not bracket background pressure in make_profile_bg_gas` is the only thing the routine can say.

91.2 The Jacobian is round-off

`surface_temperature_bg_gas` solves the two-component system (flux closure, skin temperature) with MINPACK `hybrd1`, which fixes `epsfcn = 0`; `fdjac1` then forms $\varepsilon = \sqrt{\max(\text{epsfcn}, \epsilon_{\text{mach}})}$ and steps $h = \varepsilon|x|$. At the ladder's start, $T_{\text{surf}} = 400$ K and $T_{\text{trop}} = 120$ K, that is 3.6×10^{-8} in the log variable, a temperature step of 2.4×10^{-5} K. Over that step the normalized flux residual moves by 3.8×10^{-8} , and its own jitter is $\sim 3 \times 10^{-8}$ (`jacobian_noise.py`, `noise_source.py`) — the difference is the noise, not the derivative.

The resulting entries have the wrong magnitude and, at these compositions, the wrong **sign**: at $\text{He}/\text{H} = 9.4$ the cross term $\partial f_1 / \partial x_2$ comes out +52.1 where a resolved step gives -0.553 (`first_step.py`). Because that cross term is tens of times the diagonal $\partial f_1 / \partial x_1 = -1.21$, an error in the small, noisy diagonal is amplified into a first step of several decades in $\log_{10} T_{\text{surf}}$, which is how a 400 K guess becomes 6.5×10^8 K in one iteration.

91.3 The failure predates the patch; only its address changed

The same measurement on 0.8.4 (`first_step_old.py`) shows the identical mechanism at $\text{He}/\text{H} = 9.37$: its noisy Jacobian sends the first trial to $T_{\text{surf}} = 0.10$ K, which is exactly the T_{surf} is less than T_{trop} refusal that build reports there. The patch reparameterized the solve as $\log_{10}(T_{\text{surf}} - T_{\text{trop}})$, which makes that particular refusal unreachable, so the same bad step now surfaces at the other end as a bracket message. The failure rates are the same:

grid	0.9.0 + patch	0.8.4
He/H 9.30–9.60, step 0.01 (31 points)	9.40, 9.50	9.37, 9.45
He/H 8.00–11.00, step 0.05 (61 points)	9.40, 9.50, 10.45	8.10, 9.45, 9.85, 10.15

(`sweep_new_9p30_9p60.txt`, `sweep_old_9p30_9p60.txt`, `sweep_new_wide.txt`, `sweep_old_wide.txt`.) **The net accounting of “three failures removed, one introduced” that §§ 89, 90 and `deep_level_elemental_check.md` carried does not hold** — it counted the corrected build's failures against a 0.8.4 column that had not been run at the same reservoir values. In particular the 9.45 entry those tables record as not attempted on 0.8.4 is a **failure** on 0.8.4. Nothing about the composition marks a failing point. The solution is smooth across them: the deep temperature falls monotonically from 422.72470 K at $\text{He}/\text{H} = 9.30$ to 420.61128 K at 9.60, with the failures at 9.40 and 9.50 sitting on that trend. The failure is a discrete event of the solve, not a boundary in the physics.

91.4 The workaround, before the repair was installed

`--climate-t-deep-guess` (default 400.0, in `src/utils/photochem_to_lower_profile.py`) is the only user-facing control over where the outer solve starts. At both 9.4 and 9.5, every guess tried other than exactly 400 K — 300, 350, 380, 420, 450, 500, 600 — solves, and each returns the 0.8.4 answer to five decimals (`tguess_scan.py`). A composition that refused could therefore be recovered without rebuilding. The installed environment no longer refuses any composition on either grid (91.5), so the key is a diagnostic rather than a workaround.

91.5 The repair, tested in replication and then installed

`fix_variants.py` drives the patched residual through the same MINPACK `hybrd` from Python, reproducing the installed build's converged deep temperature to 5×10^{-5} K wherever it solves, and then varies one thing at a time over the 61-point grid:

variant	solved of 61	failed or stalled
as patched	54	9.40, 9.50, 9.80, 10.05, 10.25, 10.70, 10.80
initial trust-region factor 1	54	the same seven
trial T_{surf} confined to $(T_{\text{trop}}, 6000 \text{ K})$	55	8.50, 9.55, 9.80, 9.85, 10.20, 10.35
$\text{epsfcn} = 10^{-4}$	61	none

epsfcn is not the step itself: with $\varepsilon = \sqrt{\max(\text{epsfcn}, \epsilon_{\text{mach}})}$ and $h = \varepsilon|x|$, $\text{epsfcn} = 10^{-4}$ is a relative step of 10^{-2} in the log variable, 16.2 K at the starting point against 2.4×10^{-5} K at the default. The Jacobian it produces is smooth and nearly composition-independent, $[[-1.21, -0.55], [5.0 \times 10^{-4}, -3.557]]$ across the whole 9.30–9.55 range, against entries running from +52 to –42 at the default step. Applied to 0.8.4's own parameterization it also solves all 61 points (`old_epsfcn.py`), so the repair is independent of what the patch changed. The recovered temperatures are 422.00813 K (9.40), 421.30374 K (9.50) and 415.15428 K (10.45) — on the smooth trend, and equal to the 0.8.4 values at 9.40 and 9.50 to all printed digits.

Confining the trial temperature to the range of the thermodynamic data helps but does not fix the solve on its own — it converts one failure mode into another, because the tropopause variable stays unbounded.

Installed, and measured there. In `clima` the repair is `hybrd` in place of `hybrd1` in `AdiabaticClimate_simple_solver`, the routine both `surface_temperature_bg_gas` and `surface_temperature` drive, so that epsfcn can be set at all; $\text{epsfcn} = 10^{-8}$, `factor = 100`, every other setting `hybrd1`'s. 10^{-8} and not the 10^{-4} of the table above: a relative step of 10^{-4} in the log variable, 0.16 K at the start, is about 10^4 above the jitter and still reproduces a converged central difference to 0.12 percent on the flux row, where 10^{-4} is a 16 K step whose flux row is 9 percent off the derivative. Both solve the grid; 10^{-8} is the middle of the window in which neither error source is active. The reasoning is in the patch at the constant's declaration.

The installed library carries it. Read out of it — `AdiabaticClimate_simple_solver` disassembled, the stack arguments of its MINPACK call followed to the constants they point at — it calls `__minpack_module_MOD_hybrd` with $\text{epsfcn} = 10^{-8}$ and `factor = 100.0` (LHS1140b/exhale/clima-epsfcn-installed/installed_epsfcn.py). Measured on that build, not in replication:

grid	installed	as patched, before
He/H 8.00–11.00, step 0.05 (61 points)	61 / 61	54 / 61
He/H 9.30–9.60, step 0.01 (31 points)	31 / 31	29 / 31

Every composition either build refused now solves — 8.10, 9.37, 9.40, 9.45, 9.50, 9.80, 9.85, 10.05, 10.15, 10.25, 10.45, 10.70, 10.80 — and the temperatures the replication predicted are what the installed build returns: 422.00813 K at 9.40, 421.30375 at 9.50 and 415.15429 at 10.45, against 422.00813, 421.30374 and 415.15428 above, agreeing to the last printed digit. The deep temperature is monotone in composition across the whole grid, 433.26144 K at 8.00 falling to 411.97203 K at 11.00.

What it moves. Nothing in the observable. The two arms of the LHS 1140 b elemental-flux-closure ladder that bracket the He I 10830 crossing were re-run end to end on the corrected environment against the stored arms of LHS1140b/exhale/crossings_pc090, which are the same photochem 0.9.0 source without the repair — same seed, same parent, same binary, same transit configuration. Both closure iterations converge at $k = 1$ on residuals equal to the stored ones to all five decimals, and the equivalent width moves by 5×10^{-8} in relative terms:

arm	EW without the repair	with it	relative	$\log_{10} \dot{M}$
9.0	1.1068538680	1.1068539227	$+4.94 \times 10^{-8}$	$7.39 \rightarrow 7.39$
9.1	1.1092155021	1.1092155563	$+4.88 \times 10^{-8}$	$7.39 \rightarrow 7.39$

The crossing moves from 9.0484199298 to 9.0484176256 (chord) and from 9.0460519944 to 9.0460494223 (quadratic), 2.3×10^{-6} and 2.6×10^{-6} , so the 9.0484 and 9.0461 of record stand. Every quality indicator of that ladder is unchanged digit for digit. The climate solve itself moves a composition that already solved by 2.1×10^{-6} K at He/H = 9.0 (424.9534068835 against 424.9534089414 K). Record: LHS1140b/exhale/clima_epsfcn_installed/results.txt.

The source tree carries the repair; a clone outside this repository does not. photochem/src/dependencies/patches/clima-bound holds `real(dp)`, `parameter :: epsfcn = 1.0e-8_dp (md5 3c3c40cc)`, which is also what `src/utils/photochem_exhale.patch` writes, and `src/utils/setup_photochem.sh` checks that line by name and passes. The tree reproduces the installed library. What is a revision behind is the separate clone at the workspace level, `ExoAtmosphere/photochem/ (md5 6734053e)`, where the modification was first developed: it still writes `hybrd1` and has no `epsfcn`. That clone is outside this repository and nothing here reads it.

91.6 What is not established

The $\sim 4 \times 10^{-7}$ relative jitter in ISR – OLR was not traced to a particular part of the radiative transfer. It is not the inner pressure solve: the background pressure returned is bit-identical across the steps that produce the difference (`noise_source.py`). Whether the same trap is reachable at other planets, other K_{zz} or other deep pressures was not tested — only LHS 1140 b's configuration was scanned. No EXHALE Fortran source, binary or golden was touched, and `make check` does not exercise this path.

Files. LHS1140b/exhale/clima_bracket_diagnosis/ (the diagnosis); LHS1140b/exhale/clima_epsfcn_installed/ (the repair measured on the installed environment); `src/utils/photochem_exhale.patch` (the current Clima patch, `epsfcn` included); `src/utils/photochem_to_lower_profile.py (--climate-t-deep-guess)`; item (M) of `TO_BE_DONE.md`.

92. A column that came back 2.4 % apart in pressure: the grid the solver stops on, not the temperature it started from (2026-08-29)

The judgement first. The handoff-profile sensitivity left open at the end of § 91 — a deep boundary temperature 2.1×10^{-6} K apart giving a column 2.4 % apart in pressure, 15 % in one trace species and 0.5 % in the elemental O/H — is the extent of Photochem's altitude grid at the instant the run is declared steady, and nothing

else. It is numerical, it is not an amplification of 2.1×10^{-6} K, and it is not the chemistry's convergence tolerance. **It does not limit the closure chain:** the base state EXHALE reads at the matching level reproduces to 6.5×10^{-8} in the He/H the closure iterates on, against `element_flux_closure.py`'s `tol = 0.05`. One defect was found on the way and corrected: `insert_level` broke $n = p/(k_B T)$ at exactly one level of the table, the matching level. Full diagnosis, with every measurement: `docs/lower_profile_deep_boundary_sensitivity.md`; record `LHS1140b/exhale/deep_temperature_sensitivity/`.

Reproduced inside one build. The initial guess handed to the climate root solve is not physics — it is where MINPACK starts — so ten guesses over 300–450 K are ten solves of the same problem. They return the deep boundary temperature over a range of 2.121×10^{-6} K (424.95340688 to 424.95340876), the tropopause to 2.5×10^{-8} relative and the water above it to 6.6×10^{-8} ; re-run in a fresh process the ladder is bit-identical. Seven full adapter runs at seven of those guesses reproduce every number the record of § 91 left open: the pressure at fixed level index spreads by 2.470×10^{-2} , `q_OH` by 0.58, and the oxygen handed over at the matching level by 5.207×10^{-3} . `g400a` and `g400b` are the same command run twice and write byte-identical files, so none of this is run-to-run scatter.

Everything is two-valued, and the label is the model top. Over all eleven runs the profile top is either 1.0934×10^{-8} bar or 1.12075×10^{-8} bar and never between; the oxygen at the match is either 8.8461×10^{-7} or 8.8892×10^{-7} . Within a group the columns agree to 10^{-6} or better. The two groups are two grids: with the inserted node removed so the indices line up, the pressure difference is *zero at the pinned bottom* (16.73098 bar in both) and grows linearly in level index to 2.496×10^{-2} at the top — the signature of a grid laid out uniformly in *altitude*, $dz = (\text{top_atmos} - \text{bottom})/nz$ (`vertical_grid`, `photochem/src/photochem_eqns.f90`), with two different `top_atmos`. Nothing pins `top_atmos` at the exit: `robust_step` (`photochem/photochem/extensions/gasgiants.py`) re-pins it only every `freq_update_TOA = 1000` internal steps and then accepts any top pressure within a **factor 3** of the requested one. The 2.1×10^{-6} K decides which phase of that cycle the integration exits on, and no more.

Excluded by measurement. (i) *An amplification of the temperature:* sorted by the deep boundary temperature the seven runs interleave — `g401` at 424.95340782 K carries the lower model top, `g400_1em4` at 424.95340736 K the higher — and a pair 3.0×10^{-7} K apart within one group agrees to 4.2×10^{-6} in O/H while a pair 4.6×10^{-7} K apart across the groups differs by 5.2×10^{-3} . (ii) *The photochemical stopping tolerance:* four runs repeated with `conv_longdy` $10^{-2} \rightarrow 10^{-4}$, `conv_longdydt` $10^{-6} \rightarrow 10^{-8}$ and `equilibrium_time` $10^{17} \rightarrow 10^{22}$ s give the same two groups and the same spreads (2.466×10^{-2} in pressure, 5.199×10^{-3} in O/H) and reproduce their loose-tolerance twins to 4×10^{-7} . (iii) *A second chemical solution:* within a group the composition reproduces to 10^{-6} across guesses spanning 300–450 K and a factor 10^5 in stopping time. (iv) *A cold-trap cell switching in the climate solve:* the climate grid is fixed by its two stated pressures and 60 layers and does not move at all, and the tropopause reproduces to 2.6×10^{-8} . (v) *Round-off amplification:* the code is bit-deterministic and the outcome is two discrete states.

The matching pressure never moved. It is an exact node inserted by `lower_profile_schema.insert_level` and reads $1.000000000000 \times 10^{-6}$ bar in every profile written. The 2.4 % is a comparison at fixed level index; EXHALE reads the file as a table over pressure.

What the chain actually inherits. At the matching level, over the seven runs: T 2.7×10^{-9} , r 4.8×10^{-9} , q_{H2} 5.7×10^{-8} , **He/H** 6.5×10^{-8} , C/H 3.1×10^{-7} , N/H 3.2×10^{-7} , O/H 5.2×10^{-3} . `n_tot` and `rho` are carried and explicitly not imposed (`input_read.f90`). The only place the 2.4 % reaches EXHALE at all is `lap_r_top_RJ`, the upper edge of

the elemental flux window (`binary_element_diffusion.f90`), which moves by $5.2 \times 10^{-5} R_p$. Against `tol = 0.05` on $|F - \Phi|/|F|$, and against the 0.084–0.085 flux-window spread the two LHS 1140 b arms carry, the handoff's own reproducibility is six orders below the tolerance in He/H and does not enter the H residual at all. Worth stating: the closure re-runs the adapter in every iteration directory, so the oxygen reservoir carries a 0.5 % step between iterations of one ladder — nothing in the He I 10830 chain reads it, but a result that read the oxygen would inherit it.

The change. `src/utls/lower_profile_schema.py`: `insert_level` interpolated every column linearly in $\log p$, including `n_tot` and `rho`, which are not free — $n = p/(k_B T)$ holds at every node the solution was computed on. Measured on these columns the identity held at 101 of 102 levels and was violated at **exactly one, the matching level**, by 7.7×10^{-4} and 1.24×10^{-3} . It now recomputes `n_tot` from the inserted pressure and temperature and `rho` from the interpolated mean molecular weight. Verified by rerunning one case: exactly two cells of the file change, the header and `solution_id` are unchanged, $n_{\text{tot}}/(p/k_B T) - 1$ over the whole table goes from 7.706×10^{-4} to 2.2×10^{-16} , and the run-to-run spread of `n_tot` at the matching level falls from 4.646×10^{-4} to 4.682×10^{-10} . Also corrected: the module docstring of `src/utls/element_flux_closure.py` stated the convergence test as $|F - \Phi|/\max(|F|, |\Phi|)$ while the function implements — and its own docstring documents — $|F - \Phi|/|F|$.

Not repaired, and proposed instead. The grid state at the exit is Photochem's. Re-pinning the grid and re-testing convergence before `reached_steady_state` is set is the right repair and needs a rebuild of `photochem/`; doing the same from the adapter after `photochemical_steady_state` returns (`pc.update_vertical_grid`, then step to convergence again) needs no rebuild but changes the converged answer of every future handoff column, against which the stored `crossings_gm25` and `crossings_pc090` ladders were measured. Recorded as a proposal, not made.

Scope. No EXHALE Fortran source or binary changed, no golden was refreshed and `make check` was not run: the `lower_profile` regression case reads a stored profile and never runs the adapter, so the schema change cannot reach it. Not measured: another reservoir value, another planet, another K_{zz} , and whether the two grid groups are the only two.

Files. `docs/lower_profile_deep_boundary_sensitivity.md` (the diagnosis); `LHS1140b/exhale/deep_temperature_sensitivity` (the record: README, six scripts, three tables, thirteen adapter runs); `src/utls/lower_profile_schema.py` (`insert_level`); `src/utls/element_flux_closure.py` (the docstring). The observation this answers: `LHS1140b/exhale/crossings` § 7 and § 91 above.

93. Rows of one residual that differed by a million, and a third body corrected only halfway (2026-08-29)

The judgement first. The molecular-network audit of 2026-08-27 left two items open in the molecular network: a NaN abort at He/H = 1 that it did not diagnose, and the suspicion that the molecular `hybrd1` system fails at high helium because its rows are not scaled. Both are settled here, and they are **not the same problem**.

- **The rows were the problem they were suspected of being, and scaling them fixes it.** Every row of the molecular system is now divided by its own turnover rate — the rate at which the species that row balances is produced or destroyed in that cell — so what `hybrd1` minimizes is the *relative* imbalance of each species balance instead of an absolute reaction rate. The fraction of solver attempts ending on MINPACK's `info = 4` ("iteration is not

making good progress”) falls from **54.6 % to 0.01 %** at He/H = 1000, from 41.3 % to 0.01 % at 100 and from 36.3 % to 0.26 % at 10.

- **The third-body fix of the earlier section had reached only one of its three reactions.** `ionization_equilibrium` set `ieq_cell%ntot` from `calc_ntot`, which is what the R12 term of the residual reads, but still called `set_mol_coeffs(T, n_in_dim)` — and R13 and R15 fold the third body into *their coefficient*, so both were still evaluated with ρ/m_H . Corrected; the goldens move a second time, for the same reason they moved the first.
- **The He/H = 1 NaN is in the hydrodynamics, not in the network**, and it is now located exactly: face $j = 232$, $r = 1.1398 R_p$, where the reconstructed right state carries $p = -3.65 \times 10^{-2}$ in code units and the first invalid operation of the whole run is `aR = sqrt(g*pR/rhoR)` (`src/modules/flux/Num_Fluxes.f90:46`). Row scaling does not move it by one step. It is recorded as an open item, not patched, because the patch belongs outside the molecular branch. **Corrected below:** that negative pressure is the *cell average*, not a product of the reconstruction, and the failure is positivity of the update at CFL 0.6 in a Mach 38–224 layer. The later section also reports the separate finding that the reconstruction does produce negative face pressures in ordinary runs and that HLLC has been swallowing them.

The measurement

`backup/regression/mol_diffusion` (hot Uranus, molecules + He 2^3S + binary H/He diffusion + a `base.inp` hand-off), copied to scratch with **only** He/H number ratio in `input.inp` and `HeH_base` in `base.inp` changed together, `OMP_NUM_THREADS=1`, `EXHALE_MAXSTEPS=12000`. Three binaries: the tree as found, the tree with the third body corrected, and the tree with the third body corrected **and** the rows scaled. The info histogram is a new run-wide diagnostic (`ieq_n_mol_info`), reported next to the clamped-cell count; “fail” is `info = 4` as a fraction of all molecular hybrid attempts.

build	He/H	steps	NaN	clamped	info=1	info=4	fail	$\log_{10} \dot{M}$
as found	1	12000	no	614	6031444	56627	0.93 %	11.06
as found	10	12000	no	166	5405217	2169365	28.6 %	11.04
as found	100	12000	no	212	4972511	3499546	41.3 %	11.27
as found	1000	12000	no	16	4360657	5242809	54.6 %	11.11
+ third body	1	3406	yes	507	1677801	148735	8.1 %	8.91
+ third body	10	8827	yes	274	3754138	2139333	36.3 %	8.85
+ third body	100	12000	no	216	5051123	3285000	39.4 %	11.26
+ third body	1000	12000	no	7	4368719	5222557	54.5 %	11.11
+ row scaling	1	3406	yes	337	1702668	48098	2.8 %	8.91
+ row scaling	10	5474	yes	259	2758854	7098	0.26 %	8.87
+ row scaling	100	12000	no	408	6048842	670	0.01 %	11.26
+ row scaling	1000	12000	no	38	6048079	450	0.01 %	11.09

These are relaxation snapshots followed by the steady solver, not a composition study; the $\dot{M}$ column is a diagnostic

of where each run went. What it does show is that the row scaling is a change of *path*: at He/H = 100 and 1000, where every arm completes, $\dot{M}$ moves by 0.01 and 0.02 dex while the failed-attempt fraction falls by three and a half orders of magnitude. A positive diagonal scaling of a residual has exactly the same zeros, so this is what should happen.

The clamped-cell count does not follow the info count, and that is worth saying plainly. At He/H = 100 it rises, 216 to 408, and at 1000, 7 to 38. The clamp counts cells where *no* starting point produced an admissible root; with the rows scaled, hybrd1 now reaches its tolerance on nearly every attempt, and a converged root lying on a face of the simplex (a fully dissociated or fully ionized species) is rejected by the admissibility test where a non-converged root that happened to sit inside it was accepted before. The counts are rare either way — 408 clamps against 6.05 million cell solves, 7×10^{-5} — and the solution does not move with them. The suspicion recorded earlier was that row scaling drives the simplex failures; the measurement says it drives the **solver** failures, by three and a half orders of magnitude, and that the residual clamps are a separate, much smaller population.

What the scale is

A row of this system is a production–loss balance in $\text{cm}^{-3} \text{s}^{-1}$, and its magnitude is set by the element that carries it: the helium rows run as n_{He} times a rate, the hydrogen and molecular rows as n_{H} times a rate, with a second density inside every bilinear term. At He/H = 1000 the two blocks of one residual vector differ by about 10^6 . MINPACK’s hybrd1 scales the *variables* (its internal diag) and never the rows: the dogleg step minimizes $\|J \delta x + f\|_2$, in which a block 10^6 below the other carries no weight, so the unknowns that block determines — the molecular fractions — are left wherever the helium block puts them.

The scale each row is divided by is **the sum of the magnitudes its terms reach when every species is set to the whole of its element**: each rate coefficient of the row times the cell’s n_{H} , n_{He} , n_e and n_{tot} . It is an upper bound on the row, strictly positive wherever the row carries any reaction, and — this is the part that matters for the solver — **constant across the cell’s solve**, so the scaled residual is the same smooth function of x that the finite-difference Jacobian of hybrd1 assumes. The electron density used is the cell’s incoming n_e and not the neutrality bound $n_{\text{H}} + 2n_{\text{He}}$: in the shielded molecular base the bound overstates the electron terms by many orders and would scale those rows into insignificance, which is the failure this is meant to remove.

set_mol_turnover_rates(System_HeH_mol) fills rows 1–8 and set_mol_metal_turnover_rates(System_HeH_mol_metals) the metal rows above them, so the whole system reaches the solver equilibrated; a metal row runs as n_X times a rate and a trace element carries $n_X \sim 10^{-4} n_{\text{H}}$, so it sits below the helium block for the same reason the molecular one does. The identity rows metal_rows writes for an absent element, and for the X^{++} of a two-stage element, are already dimensionless and keep a scale of 1. The routine mirrors mol_heh_rows term by term and says so at the code site: a reaction added to a row has to be added to its scale as well, and a missing term only makes the scale a weaker bound — it cannot make the root wrong.

The third body, the other two reactions

The earlier account is right about what the third body is and wrong about how far the fix reached. R12 takes n_M in the residual, through ieq_cell%ntot, and that path was corrected. R13 and R15 are written in mol_rates as *two-body*-

equivalent coefficients, $k = 3.2 \times 10^{-29}n$ and $k = 8 \times 10^{-33}(300/T)^{0.6}n$, so the third body enters when the coefficient is built — in `set_mol_coeffs`, which `ionization_equilibrium` still called with $n_{\text{in_dim}} = \rho/m_{\text{H}}$. Both now get `n_tot(j)`, the same particle density the R12 term reads. R13 is the H_3^+ source at the base and R15 the H_2 source, so this is not a small correction to a small term: it is the pair that builds the molecular base, and it was running with a third body $2.3\times$ too dense at the hot-Uranus base and up to $4\times$ in a helium-dominated one.

The corrected third body is what brings the NaN abort back. The tree as found does not abort at any He/H tested — the earlier abort at step 5538 is absent from it — and with R13/R15 corrected the He/H = 1 arm aborts at step 3406 and the He/H = 10 arm at 8827. The physically correct third body is kept; the abort it exposes is diagnosed below and recorded as an open item.

Where the He/H = 1 NaN is

Reproduced in a build with `-fcheck=bounds,do,mem` and `-ffpe-trap=invalid,zero,overflow` (the detector backup/regression/run_fcheck.sh uses, built to EXHALE_fcheck.x so it does not overwrite the production binary). The first invalid operation of the run is at step 3406:

Program received signal SIGFPE

```
#3 __numerical_fluxes_MOD_num_flux  at src/modules/flux/Num_Fluxes.f90:46
```

```
#4 __rk_integration_MOD_rk_rhs       at src/modules/time_step/RK_rhs.f90:62
```

Line 46 is `aR = sqrt(g*pR/rhoR)`. A second build, instrumented to print the face states, names the face and the state that reaches it:

```
face j = 232    r_edg = 1.13983 R_p
WL (rho,v,p) =  2.9894   26.707   +0.23329
WR (rho,v,p) =  2.8916   26.816   -0.036499
cell 231 (rho, rho v, E) = 2.7948   75.246   1013.68
cell 232                = 2.8916   77.540   1040.21
cell 233                = 5.8744  152.863   1988.84
```

The reconstructed right state has a **negative pressure**, and HLLC takes the square root of it. The layer it sits in explains how: below $r = 1.139$ the run has developed a cold hypersonic shell — T about 650 K, v about 10^7 cm s^{-1} , **Mach 50 to 60** — riding on a dense hot wall at $r = 1.144$ (ρ up by $20\times$, T 9100 K). At Mach 60 the thermal pressure is 2×10^{-4} of the total energy the scheme carries, so the pressure recovered as $(\gamma - 1)(E - \rho v^2/2)$ is the difference of two numbers that agree to four digits, and a reconstruction across the jump at the next cell tips it below zero. Nothing between the reconstruction and the Riemann solver tests the reconstructed pressure for positivity.

This is a hydrodynamic robustness gap, not a molecular one: it is identical, to the step and to the face, with and without the row scaling, and the positivity floor that would close it belongs in the reconstruction, outside the `thereis_mol` branch this series is confined to. Recorded as `TO_BE_DONE.md` item (O).

Containment, and the goldens

Every change is inside `thereis_mol` — the two `set_mol*_turnover_rates` calls and the corrected `set_mol_coeffs` argument sit in the `if (thereis_mol)` cell-state block of `ioniz_eq`, the scaling is applied only in the two molecular residual routines, and the info histogram is counted only in the molecular attempt loop. Two measurements say so rather than the code reading alone:

- `make check` passes byte-identical on `wasp_full`, `wasp_he23off` and `lower_profile` — the three non-molecular cases — against goldens that predate the series.
- The LHS 1140b arm `LHS1140b/exhale/crossings_gm25/rs_heh0p425`, re-solved in a scratch copy from its stored restart with the pre-series binary and with the new one, writes **byte-identical** `Hydro_ioniz.txt`, `Ion_species.txt`, both `_adv` files, both `_IC` files, `Cooling_breakdown.txt` and `Heating_breakdown.txt`. The stored run directory was not touched.

The four molecular cases move, and the third body is what moves them. Each was also re-run with the third body corrected but the rows *unscaled*, so the two changes can be told apart. Largest relative difference over the column against the previous golden, with the median in brackets (these are 12000-step relaxation snapshots, so the maximum sits in the base sound-wave layer or in the thin outer cells and the median is the better measure of the profile):

case	ρ	p	T	cool	H ₂ front	$\log_{10} \dot{M}$
<code>mol_base_handoff</code>	9.1 % (0.20)	14.9 % (0.32)	6.1 % (0.087)	53.5 % (1.8)	1.1617 → 1.1577	10.57 → 10.58
<code>mol_metals</code>	6.5 % (0.16)	9.8 % (0.34)	6.2 % (0.093)	68.3 % (0.67)	1.1637 → 1.1597	10.58 → 10.58
<code>mol_lyman_werner</code>	27.6 % (0.57)	45.3 % (0.71)	17.4 % (0.32)	59.8 % (2.4)	1.1321 → 1.1271	10.58 → 10.49
<code>mol_diffusion</code>	9.1 % (0.20)	14.9 % (0.32)	6.1 % (0.087)	53.5 % (1.8)	1.1617 → 1.1577	10.57 → 10.58

The row scaling contributes almost none of that. Measured on its own — the third-body-corrected build against the third-body-corrected **and** scaled build, same case, same steps — the largest difference over the column is ρ 0.019 %, p 0.0066 %, T 0.018 % on `mol_base_handoff`; 0.28 %, 0.46 %, 0.23 % on `mol_metals`; 0.013 %, 0.020 %, 0.0079 % on `mol_lyman_werner`; and 0.017 %, 0.0068 %, 0.016 % on `mol_diffusion`. The H₂ front does not move at all in any of the four.

That is the judgement the goldens were refreshed on. The move is the physics correction, and it is the correction the earlier audit intended and described: R13 builds H₃⁺ at the base and R15 builds H₂, both linearly in the third-body density, which was $2.3\times$ too large at this hot-Uranus base. The H₂ front shifts by $0.004 R_p$ and the mass-loss rate by 0.01 dex on three of the four cases and by 0.09 dex on the Lyman-Werner one, where the photodissociation front and the three-body formation front sit on top of each other and the base is therefore the most sensitive. The row scaling, in contrast, leaves the solution where it was to a few parts in 10^4 while removing the solver failures — which is what a diagonal rescaling of a residual should do, and is the evidence that it is a change of path and not of physics.

Goldens re-snapshotted for the four molecular cases at the end of the series and `make check` re-run afterwards.

Files changed

- `src/modules/nonlinear_system_solver/System_HeH_mol.f90` — `mol_inv_turnover` and `set_mol_turnover_rates`; the scaling applied at the end of `ion_system_HeH_mol`.
- `src/modules/nonlinear_system_solver/System_HeH_mol_metals.f90` — `set_mol_metal_turnover_rates` for the metal block; the scaling applied at the end of `ion_system_HeH_mol_metals`.
- `src/modules/radiation/ionization_equilibrium.f90` — `set_mol_coeffs` given `n_tot(j)` instead of `n_in_dim(j)`; the two turnover-scale calls; `ieq_n_mol_info`, the run-wide hybrid1 exit-code histogram of the molecular solves.
- `src/EXHALE_main.f90` — the info histogram reported at the end of a run, next to the clamped-cell count.
- `TO_BE_DONE.md` — new item (O), the negative reconstructed pressure.
- `docs/molecular_chemistry_audit_he_rich.md` — section 7, closing both open items of its section 6.

94. One run, two photoionization rates: Do only PP and the staged secondary channel (2026-08-30)

The judgement first. The staging of the Shull & van Steenberg (1985) secondary channel is a startup device, not a statement about the physics: section 38. introduced it because *from a cold initial condition* the extra base ionization amplifies the startup transient into a NaN, and it records that “restarting with the coupling on from that converged state is benign”. A Do only PP: True run starts nowhere: it loads a wind that was itself converged with the coupling applied, leaves the marching loop after one pass, and post-processes it. Running its one `ionization_equilibrium` call without the coupling while every `PH_heat_HHe` call of `post_process_adv` runs with it is therefore not a conservative choice, it is two different photoionization rates inside one run, and the `Hydro_ioniz.txt` / `Ion_species.txt` such a run writes are not the equilibrium state its own `*_adv.txt` files correct. `sec_ion_active` is now armed before the loop when `do_only_pp` is set, exactly as the environment-driven bypass modes (`EXHALE_RESIDUAL`, `EXHALE_NEWTON_TEST`, `EXHALE_JAC_TEST`, `EXHALE_PTC`) already arm it for the same reason — they also skip the time loop.

What this means for the stored PP-only runs. A PP-only run rewrites `output/Hydro_ioniz.txt` and `output/Ion_species.txt` from that single equilibrium solve (`write_output(..., 'eq')` after the loop), so the equilibrium files of *every* stored PP-only run in the tree were computed with the secondary channel off while their `_adv` files beside them were computed with it on. That includes the three LHS1140b/examples/ cases of record and the LHS 1140 b post-process arms.

How far the numbers move. Measured on scratch copies of the two crossing examples (the run directories themselves untouched), same loaded wind, PP pass and transit synthesis re-run at `EXHALE_TRANSIT_RES_HETR=68000`:

case	red-pair EW [%Å] before	after	$\log_{10} \dot{M}$
<code>scalar_base_kzz1e9</code>	1.108056	1.106495	7.80 (unchanged)
<code>photochem_profile_base</code>	1.108000	1.107509	7.39 (unchanged)

— 0.14 % and 0.04 %, both still inside the measured $1.108 \pm 0.030 \%$ the two cases were solved at, and the escape rate does not move at all because a PP-only run does not re-solve the wind. The equilibrium *files* move much more than the observable does: over $1-3 R_p$ the He 2^3S column changes by up to a factor 4 and H II by a factor 2–2.5, which is the size of the secondary channel itself.

The consistency test that shows it is the right direction. The equilibrium file and the _adv file beside it now describe the same physics, so the gap between them should shrink to the advection correction alone. Maximum relative difference in the He 2^3S density over $1-3 R_p$, equilibrium against _adv:

case	before	after
scalar_base_kzz1e9	11.8	6.27
photochem_profile_base	6.80	3.26
heh0p55_diff_ctrl	109	8.21

Goldens. No regression case sets Do only PP: True, so the matrix is untouched by this change; verified byte-identical.

95. A negative pressure handed to the Riemann solver, and a NaN that min threw away (2026-08-30)

The judgement first, and it corrects the preceding section twice. The molecular-network section located the He/H = 1 abort at face $j = 232$ with $p = -3.65 \times 10^{-2}$ in the reconstructed right state, concluded that the reconstruction “tips it below zero”, and expected a positivity floor to “move results wherever it triggers” — implying it does not trigger in ordinary runs. Both readings are wrong, and in opposite directions.

- **The He/H = 1 negative pressure is in the cell average, not in the reconstruction.** Printing the face states and the cell averages together at the same step:

```
DBGREC face j= 232 WL/WR(rho,p)= 2.98944E+00 2.33286E-01 2.89155E+00 -3.64985E-02
Wavg(j,jr)(rho,p)= 2.89155E+00 3.66265E-01 5.87442E+00 -3.64985E-02
```

Cell 233 already carries $p = -3.64985 \times 10^{-2}$, the identical number: its pressure slope is limited to zero, so both of its face states are its own average and the reconstruction only passes it on. One step earlier that cell holds $p = +5.58$.

- **A reconstructed negative pressure is not rare — it happens in the regression goldens, and it has been reaching the Riemann solver.** A build that only *reports* non-positive face states, with no guard, counts **1388** of them in 9000 steps of wasp_full, at physical faces spread over $r = 1.195$ to $1.60 R_p$ (1163 of the 1164 in the guarded build are physical faces; one is the outer ghost). Num_flux takes $a_L = \sqrt{\gamma p_L / \rho_L}$ of them.

Why that has never shown up as a NaN. Because min/max discard it. With the measured face state $p_L = -8.01295 \times 10^{-7}$, $\rho_L = 3.50034 \times 10^{-6}$, so $a_L = \text{NaN}$, the HLLC wave-speed estimate of Num_Fluxes.f90 returns

$$S_L = \min(0, \min(v_L - a_L, v_R - a_R)) = 0.0, \quad S_R = \max(0, \max(v_L + a_L, v_R + a_R)) = 4.895,$$

both finite (measured with the production `-O3 -fopenmp` flags). $S_L = 0$ sends HLLC down its $S_L \geq 0$ branch, so the numerical flux at that face becomes the plain upwind physical flux of a state with a **negative pressure**, and its momentum and energy components carry that sign. Nothing reports it. This is a physics error in the flux, silent, and present in every stored result.

What fails at He/H = 1 is a different thing: positivity of the update. The Euler equations live on $\rho > 0$, $\rho e > 0$, and a conservative update preserves that set only under a CFL bound — $\Delta t(|v| + c)/\Delta r \leq 1/2$ for the first-order HLL family (Einfeldt et al. 1991; Batten et al. 1997), tighter with a high-order reconstruction. EXHALE marches at CFL 0.6, past that bound already, and the He/H = 1 arm develops a base region at **Mach 38 to 224**, where it bites. The direct test: the same arm at CFL: 0.2 completes 12000 steps with no NaN. Every regression golden has a base Mach number of 0.000 ± 0.002 and a domain maximum of 0.94–2.11, which is why only this arm dies.

Two guards, one for each.

- `positivity_limited_faces` (`src/modules/states/Reconstruction.f90`), the last step of Reconstruct after Rec_BC: a face whose reconstructed ρ or p is not strictly positive — NaN included, the tests are the negation of “strictly positive” — is dropped to the two cell averages, the piecewise-constant limit of the same scheme. The gap it closes is real: the MC limiter of PLM_rec at $\theta = 2$ admits $W_L = 2W_j - W_{j-1}$, and ESWENO3 has no positivity property at all. **This one moves the goldens**, because it is firing in them.
- Positivity step control (`positive_density_and_internal_energy` in `Conversion`, driven by the `retry_step` loop of `EXHALE_main`): each RK stage is tested for $\rho > 0$ and $E - \rho v^2/2 > 0$ over the physical cells, and a stage that fails discards the step, restores `u_old` and retakes it at half Δt , up to 20 bisections. A pressure floor was deliberately not used: it puts energy into the gas that the equations did not. This one is silent on the whole matrix.

Together the two make an invariant the code did not have: every state `Num_flux` sees is thermodynamically admissible, because the cell averages entering a reconstruction have been tested by the step control and the face states leaving it by the face guard.

Result at He/H = 1. The arm that aborts at step 3406 — with the current code, and with the face guard alone — runs its full 12000 steps with the step control: 7500 steps retaken, 29409 Δt bisections, no NaN, and the face guard never fires there. Rebuilt with `-ffpe-trap=invalid,zero,overflow`, the marching loop of that arm reaches its 12000-step cap with no trap at all (last logged step 11994); what still aborted was its *post-process*, on two temperature guards that were conditioned on the metal switch, and those are the next section. The state the arm reaches is still not a converged wind ($du = 12.8$); what is fixed is that the code no longer dies on a square root. Why that base reaches Mach 38 is a separate open question (`TO_BE_DONE.md` item (P)).

Goldens. The step control is silent everywhere — 0 retaken steps on all seven cases. The face guard is not: it fires 1164 times on `wasp_full` and 1193 on `wasp_he23off`, and nowhere else. Measured with this section’s two guards in and nothing else changed, maximum relative move against the golden:

case	ρ	p	T	H II	He II	$\log_{10} \dot{M}$	faces
wasp_full	1.1e-05	8.0e-06	5.1e-06	6.8e-06	1.3e-05	13.23 $\rightarrow$ 13.23	1164
wasp_he23off	5.8e-06	4.1e-06	2.6e-06	3.6e-06	7.0e-06	13.23 $\rightarrow$ 13.23	1193
mol_base_handoff	0	0	0	0	0	10.58 $\rightarrow$ 10.58	0
mol_metals	0	0	0	0	0	10.58 $\rightarrow$ 10.58	0
mol_lyman_werner	0	0	0	0	0	10.49 $\rightarrow$ 10.49	0
mol_diffusion	0	0	0	0	0	10.58 $\rightarrow$ 10.58	0
lower_profile	0	0	0	0	0	9.58 $\rightarrow$ 9.58	0

A part in 10^5 , and the mass-loss rate does not move. The five byte-identical cases are the ones whose secondary-ionization stage flip never fires — they are step-capped relaxation snapshots that never converge — so the coupling is inert in them and the reconstruction never reaches a state that produces a negative face pressure either.

96. A secondary-ionization rate that grew as its own target vanished: the SvS85 branching on the cell's own composition (2026-08-30)

The judgement first. The old expression was physically wrong, and not by a little. R_{secHeI} is a *volumetric* deposition rate carrying the Shull & van Steenberg (1985) branching, which is measured at one fixed composition; dividing it by the actual $n(1^1\text{S})$ to get a rate per atom asserts that the same share of every photoelectron ionizes helium no matter how little helium is there. The rate per atom then diverges as the singlet disappears, and only $\max(n, 1.0\text{d-}99)$ stood between the code and infinity. The same defect on the hydrogen side is what drove n_{HI} to an exact zero above $\text{He}/\text{H} = 10$. The correction is not optional and its size is not an argument either way.

What SvS85 actually assume. Section II of the paper, on the composition of the Monte Carlo: “We assume that $n(\text{He})/n(\text{H}) = 0.1$, and that the hydrogen and helium ionization fractions are equal: $n(\text{H}^+)/n(\text{H}_{\text{tot}}) = n(\text{He}^+)/n(\text{He}_{\text{tot}})$.” Equal ionization fractions make the *neutral* ratio their calculation sampled equal to the elemental one at every x , so the branching their Table 2 coefficients carry — 0.3908 for $\phi(\text{HI})I/E_0$ against 0.0554 for $\phi(\text{HeI})I/E_0$ — is the branching of a gas with $n(\text{HeI})/n(\text{HI}) = 0.1$. That is the reference the correction renormalizes from, and it is why the requirement to reduce to SvS85 at that ratio is a physical requirement and not a compatibility one.

The form adopted (`svs85_secondary_branching`, `util_ion_eq.f90`). The branching between the two ionization channels is a collision probability: the fast electron ionizes species s at a rate proportional to n_s times its energy-averaged ionization cross section, so the ratio of the two energy shares scales linearly with $n(\text{HeI})/n(\text{HI})$. The total energy the primary spends ionizing H and He, $f_{\text{ion}} = f_{\text{HI}} + f_{\text{HeI}}$, is left at the SvS85 value — it is fixed by the same Monte Carlo that fixes f_{heat} , which is returned unchanged, so the cell's energy budget does not move — and only the split is renormalized, returned **per target atom**:

$$w_{\text{HI}} = n_{\text{HI}}(0.1) f_{\text{HI}}, \quad w_{\text{HeI}} = n_{\text{HeI}} f_{\text{HeI}},$$

$$c_{\text{HI}}^{\text{ion}} = \frac{f_{\text{ion}}(0.1) f_{\text{HI}}}{w_{\text{HI}} + w_{\text{HeI}}}, \quad c_{\text{HeI}}^{\text{ion}} = \frac{f_{\text{ion}} f_{\text{HeI}}}{w_{\text{HI}} + w_{\text{HeI}}} \quad [\text{cm}^3].$$

No electron-impact cross sections have to be supplied: their ratio cancels against the reference. The caller multiplies these by the same deposited-energy integrand as before and adds the result straight to $P_{\text{HI}} / P_{\text{HeI}}$; **both $\max(\dots, 1.0\text{d-99})$ divisions are gone**, and with them the divergence.

The three limits, which are the requirements.

	$c_{\text{HI}}^{\text{ion}} n_{\text{HI}}$	$c_{\text{HeI}}^{\text{ion}} n_{\text{HeI}}$
$n_{\text{HeI}}/n_{\text{HI}} = 0.1$ (SvS85's gas)	f_{HI} exactly	f_{HeI} exactly
$n_{\text{HeI}} \rightarrow 0$	f_{ion} (a factor 1.161 up)	$\rightarrow 0$, linearly in n_{HeI}
$n_{\text{HI}} \rightarrow 0$	$\rightarrow 0$, linearly in n_{HI}	f_{ion} (a factor 7.196 up)

The first row holds for every x , checked numerically at $x = 10^{-4}, 0.01, 0.1, 0.5, 0.9$: the reconstructed coefficients reproduce `svs85_fion_HI(x)` and `svs85_fion_HeI(x)` to the last digit. The helium-free path `PH_heat_H` is the second row and now calls the same routine with $n_{\text{HeI}} = 0$, so a helium-free run gets the whole SvS85 ionization energy into hydrogen instead of 86 % of it.

What is still SvS85's composition. f_{heat} , and with it the total f_{ion} , keep the helium content of the Monte Carlo, as does the excitation channel the code discards; only the split between the two ionization channels is corrected. H_2 is a source of photoelectrons in the caller but never a target, SvS85 having no H_2 channel, so in a molecular layer the energy that would ionize H_2 goes to HI . Both statements are now in the code at the routine.

The contaminated arm. LHS1140b/exhale/heh0p55_diff_ctrl (He/H = 0.55, no K_{zz} , helium settled to $n_{\text{He}}/n_{\text{H}} = 6 \times 10^{-6}$ by diffusion) was the run the item was opened on. Re-run in a scratch copy, cells above $1.5 R_p$ whose ground singlet has collapsed to zero or below:

build	Ion_species.txt	Ion_species_adv.txt	$x(1^1\text{S})$ at $2.53 R_p$ (_adv)
before	0 / 233	171 / 233	0
Do only PP fix only	118 / 233	233 / 233	4.1×10^{-18}
with this section	0 / 233	0 / 233	0.923

The Do only PP fix makes it worse on its own, which is the expected sign: it puts the equilibrium solve on the same (still divergent) physics the post-process was already using. With the branching corrected the collapse is gone from both files.

The goldens, and why the size of the move is the verification. Measured as the whole series (the two preceding sections plus this one) against the golden, to be read against the table of the preceding section, which is the same matrix with this section's change removed:

case	ρ	p	T	H II	He II	$\log_{10} \dot{M}$
wasp_full	2.3e-03	1.0e-03	6.8e-04	1.0e-02	7.6e-02	13.23 $\rightarrow$ 13.23
wasp_he23off	8.2e-03	3.6e-03	4.6e-03	9.9e-03	7.7e-02	13.23 $\rightarrow$ 13.23
the five others	0	0	0	0	0	unchanged

Only the two WASP-121 b cases move, and the mass-loss rate does not. That is the reduction condition doing its work: those two runs sit at an elemental He/H = 0.0851, and their **neutral** ratio $n(\text{HeI})/n(\text{HI})$ runs 0.038–0.104 against

SvS85's 0.1, so the branching is renormalized by at most a factor 2.6 and mostly by about 15 %. The largest move, 7.6 % in He II, is at the base cell, where $n(\text{HeI})/n(\text{HI}) = 0.0851$ — a 15 % departure from 0.1 — and it falls to 0.4 % at $1.2R_p$ and 6×10^{-4} at $1.4R_p$, tracking that departure and not the absolute rate. Where the neutral ratio is 0.1 the change is exactly zero, by construction.

The five byte-identical cases are byte-identical for a different reason: they are step-capped relaxation snapshots whose secondary-ionization stage never activates, so the whole SvS85 branch is inert in them.

Goldens re-snapshotted for `wasp_full` and `wasp_he23off` at the end of the series (`run_check.sh golden`), and make check re-run afterwards.

97. A temperature that was only required to be a temperature when metals were on (2026-08-30)

The judgement first. Two guards in the advection post-process were conditioned on `pp_metal_on`, and neither of them is about metals. Both state that a temperature is positive, which is a property of a temperature and not of the composition, and where the condition was false the code evaluated recombination and cooling fits at a negative argument. Found while running the He/H = 1 molecular arm of the preceding section under `-ffpe-trap=invalid,zero,overflow`; that arm is metals-off, so it had no test at all.

What the run does. The marching loop of that arm completes all 12000 steps clean under the trap — which is the verification of the preceding section — and the abort is in `post_process_adv`:

```
#3 __cooling_coefficients_MOD_rr_badnell    at src/modules/radiation/Cool_coeff.f90:570
#5 __cooling_coefficients_MOD_rec_hii_b     at Cool_coeff.f90:687
#6 __utils_ion_eq_MOD_eval_cool            at src/modules/radiation/util_ion_eq.f90:632
#7 __post_processing_MOD_post_process_adv   at src/modules/post_process/post_process_adv.f90:373
```

Line 570 is `tt = sqrt(T/T0)` in the Badnell radiative-recombination fit. Instrumented to print the temperature it is handed, the post-process passes run

```
DBGTK min(T_K)= 1.14000000E+01 at j= 270      (pass 1)
DBGTK min(T_K)= 7.15606137E+00 at j= 277      (pass 2)
DBGTK bad T_K at j= 278   T= -4.19742816E+01
DBGTK bad T_K at j= 279   T= -6.40457675E+02
DBGTK bad T_K at j= 280   T= -2.47418207E+03   (pass 3)
```

— the advection-corrected temperature of three cells comes back negative on the third pass, is written into `T_K` for the next pass, and the recombination fit takes the square root of it.

The two guards, both now unconditional.

- `T_equation` floored its trial temperature at 1 K only if (`pp_metal_on`), with the recorded reason that “Mode 0 (`pp_metal_on = .false.`) keeps the exact original behaviour”. The floor is the *domain of the fits*: `hybrd1`

overshoots to a negative trial T in its search, `brem` takes $\sqrt{T}$ and `rec_cool_HII_func` raises $(2 \times 157807/T)$ to the power 1.970, neither of which exists there. Floored unconditionally.

- The rejection of a non-physical root was written as one test, `.not. (T > 0) .or. T > 2 T_eq .or. T < 0.5 T_eq`, inside `if (pp_metal_on)`. Only the *band* part is about metals — the metal-cooled residual is non-monotone and carries a spurious hot root. The positivity part is separated out and applied to every run: a non-positive root is not a solution, and the cell keeps the converged equilibrium temperature, which is exactly what the branch above it already does for a non-converged solve.

Verified. With both in, the same arm's post-process runs its ten passes with $\min T_k$ staying positive throughout (11.40, 7.156, 0.300, 0.433, 0.354, ... settling at 0.3523 K) and completes under `-ffpe-trap=invalid,zero,overflow`. The end-of-run rejection count reports zero because `n_pp_reject` is reset each pass and reports the last one, the same convention `n_adv_eq` carries; the rejection fires on the third pass.

Containment. `T_equation` is called only from `post_process_adv` and `solve_T_brent`, and `post_process_adv` runs *after* the final `write_output(..., 'eq')`. Neither guard can reach `Hydro_ioniz.txt` or `Ion_species.txt`, which is what the regression compares, so the matrix cannot move on them — confirmed by `make check`.

98. What is left of the ground-singlet difference, and what carrying it as a state variable would cost (2026-08-30, design note)

Not implemented; this section records the inventory and the design so the decision can be taken on evidence. `TO_BE_DONE.md` item (L).

Where the difference still is. The global species vector carries the summed He I (`isp_HeI`, triplet included, `bsp_is_excited_level`) and the metastable 2^3S separately, and the ground singlet is formed where it is needed as `nheiS = nhei - nheiTR`. After the advection-system change, the Do only PP change and the branching correction of the three preceding sections, the remaining consumers are:

site	what the difference feeds	shape of the dependence
util_ion_eq.f90	calc_column_dens $\rightarrow$ the He I singlet column	linear, in an exponent
PH_heat_HHe	N_{15} , hence $e^{-\tau}$	
”	acc_H / acc_HeI: He I photoelectric heating	linear
”	acc_secHI / acc_secHeI: He I as a <i>source</i> of secondary ionizations	linear
”	acc_q: absorbed energy, the denominator of the heating efficiency q	linear; q already guards $q_{\text{abs}} \leq 0$
”	svs85_secondary_branching: the composition weight	linear in the numerator; no longer a denominator
post_process_adv.f90 (2 sites)	rebuilding the singlet from the loaded/summed columns before the advection solve	linear; the advection <i>system</i> carries nheIS as its own variable

The catastrophic case — the difference inside an implicit solve — was removed earlier; the divergent case — the difference as a denominator — in the branching correction. **Every remaining consumer is linear in nheIS**, so what is left is a loss of significance, not an instability: where $n(2^3\text{S})$ approaches $n(\text{HeI})$ the difference keeps only the digits the two columns do not share, and it can come out slightly negative from round-off, or from a restart in which the two columns were written by different solves. A negative singlet there gives a negative He I opacity and a negative heating contribution, both small and both unguarded.

What carrying $n(1^1\text{S})$ as the state variable would change. Replacing isp_HeI (summed) by a ground-singlet column, with isp_HeTR kept beside it and no longer flagged as an excited level, is a change of *meaning* of an existing column, not an addition:

- the helium element budget becomes $n_{\text{He}} = n(1^1\text{S}) + n(2^3\text{S}) + n(\text{HeII}) + n(\text{HeIII})$ everywhere it is formed — composition.f90, calc_rho, calc_ne, calc_ntot, binary_element_diffusion (which lists [isp_HeI, isp_HeII, isp_HeIII] as the helium stages), set_IC, energy_semi_implicit, and the six System_HeH_* residuals;
- Ion_species.txt keeps its HeI label with a different meaning. Every reader that counts nuclei has the current convention written into it — examples/exhale_io.py states it in a comment and *excludes* HeITR from _NUC_HE for exactly that reason — so each of the 17 Python/notebook readers has to be revisited, and the 2426 Ion_species*.txt files already in the tree would be read with the wrong convention by the new code, load_IC included. A restart from an old file would silently lose or double-count the triplet.

The cheaper alternatives, in increasing cost.

1. *Guard where it is formed.* Make the singlet in exactly one place — a he_ground_singlet(nhei, nheiTR) in composition.f90 that every consumer calls — and have it clamp at zero and count the clamps, as positivity_limited_faces does for the faces. This removes the negative opacity and negative heating and

makes the loss of significance measurable, at no schema cost. Byte-identical wherever the difference is already positive.

2. *Write the singlet as an extra diagnostic column* (HeIS) beside the summed HeI. Readers that do not know the label ignore it; nothing changes meaning; the file then states both numbers and the two can be checked against each other.
3. *Carry $n(1^1\text{S})$ as the state variable and rename the column* (HeIS replacing HeI), so no old file can be misread as a new one and load_IC can detect the generation from the label it already parses. This is the full change above, and it should be taken as its own series with the readers, not folded into a physics change.

Alternative 1 is the recommendation if the item is acted on before that series: it addresses the only failure mode that is left, and it costs one routine.

Adopted the same day; see the section below on the floored ground singlet. Alternative 1 is implemented: `he_ground_singlet_density` in `composition.f90` forms the difference, floors it at zero and counts the floors. Two corrections to the inventory above came out of doing it. The table is incomplete — `System_HeH_mol.f90` and `System_HeH_mol_metals.f90` also write `n_hei - n_heiTR`, inside their residuals, and are deliberately left there. And the first thing the floor caught was not a loss of significance: in the `wasp_full` regression case it fires on a cell whose *nuclei* densities are negative, a defect of its own, entered as `TO_BE_DONE.md` item (Q).

99. The handoff column was written on whatever grid the chemistry stopped on; the grid is now the solution's own (2026-08-30)

The judgement first. § 92 diagnosed the handoff profile's two discrete outcomes — a column 2.5 % apart in pressure at fixed level index and 5.2×10^{-3} apart in the elemental O/H at the matching level, depending only on which phase of Photochem's `freq_update_TOA` cycle the integration exited on — and recorded two possible repairs without making either. The second of them is now made, in the adapter, and it removes the effect: over the same seven-run guess ladder the spread at fixed level index falls from 2.470×10^{-2} to 2.149×10^{-8} and the O/H handed over at the matching level from 5.207×10^{-3} to 2.996×10^{-8} , all seven runs stopping on one grid where before they split between two. `photochem/` was not touched and needs no rebuild. `TO_BE_DONE.md` item (N), closed.

What the adapter now does. `src/utils/photochem_to_lower_profile.py` gains `steady_state_at_stated_model_top`, which runs after `photochemical_steady_state` has reached a steady state. It calls `pc.update_vertical_grid(TOA_pressure=args` — which re-lays the altitude grid so `top_atmos` is the altitude at which the current solution's own hydrostatic pressure equals the stated top — re-initializes the robust stepper and runs the chemistry to steady state again. The two steps are **iterated**: re-pinning perturbs the solution and the perturbed solution moves `top_atmos` again, and what the column is written on is their common fixed point, which is a property of the problem rather than of where the stepper happened to stop.

The acceptance test is that **two successive passes agree on the model top** to 10^{-5} relative, not that the model top equals the stated pressure: the reported top is the top cell centre while `top_atmos` is the domain edge, and the half cell

between them is about 9 % in pressure at this resolution. (That is also why the file's `p_top_bar` header has always read about 1.09×10^{-8} bar for `--toa 1.0e-2`, in both grid groups; it is not a defect and it does not change here.) 10^{-5} is 2500 times below the 2.5×10^{-2} that separated the two grids.

Failure is a refusal. If the chemistry does not re-reach steady state on the pinned grid, or the top will not settle within six passes, or `update_vertical_grid` raises, the adapter refuses through `sch.refuse` and writes nothing — the same class as every other unmet condition it checks. The alternative, falling back to the unpinned grid, would put back exactly the dependence this removes and put it back invisibly: nothing in the written file records which grid the chemistry stopped on. A `PhotoException` out of `update_vertical_grid` is reported with the library's own message rather than re-raised bare, because the adapter can say what it was trying to do and has nothing to write without it.

Measured, the grid. The § 92 guess ladder re-run through the pinned adapter, seven runs at the same seven initial guesses, `run_guess.sh` and `spread_table.py` unchanged (LHS1140b/exhale/deep_temperature_sensitivity/runs/pin_g4*, `spread_table_pinned.txt`):

quantity	unpinned (§ 92)	pinned
p , at fixed level index	2.470×10^{-2}	2.149×10^{-8}
n_{tot} , ρ , at fixed level index	2.470×10^{-2}	2.415×10^{-8}
q_{OH} , at fixed level index	5.837×10^{-1}	7.431×10^{-5}
He/H, at the matching level	6.484×10^{-8}	4.791×10^{-13}
O/H, at the matching level	5.207×10^{-3}	2.996×10^{-8}
q_{CO_2} , at the matching level	1.115×10^{-2}	8.384×10^{-5}

Every run stops on the same grid, top cell 1.093596×10^{-2} dyn/cm², where the unpinned ladder gave 1.0934 – 1.0936×10^{-8} bar or 1.12075×10^{-8} bar and nothing between. Two pinned passes at each of the seven; the accepted second pass moved the top by 4.2×10^{-8} to 7.6×10^{-6} ; the pin costs about ten per cent of the adapter's wall time. The pinned fixed point is near the lower of the two former groups (1.3×10^{-4} in the top, 2.5×10^{-5} in the matching-level O/H) and 2.4×10^{-2} from the upper one.

Measured, the observable. One elemental-flux closure rung was re-run end to end: the crossing arm of LHS1140b/exhale/crossings_§ 6, reservoir 8.2117, same binary, same seed, same `closure.json`, same trial fluxes, same tolerance, same wind protocol, the adapter the only difference. Its stored last iterate had stopped on the upper grid ($p_{\text{top}} = 1.121744 \times 10^{-8}$ bar) and the pinned one lands at 1.093866×10^{-8} — 2.485×10^{-2} in pressure at fixed level index, 1.543×10^{-3} in the matching-level O/H, the full size of the effect. **The He I 10830 red-pair equivalent width moves from 1.107774992 to 1.107776648 % Å, $+1.5 \times 10^{-6}$ relative**, and $\log_{10} \dot{M} = 7.40$, $\|R\| = 3.3 \times 10^{-4}$, the flux spread 8.63×10^{-2} , two metastable step flags, no metal dropout, $r_{\text{drop}} 20.799$, outer $\tau 0.0088$ and T12 1679 are unchanged at the precision they are quoted at. That is § 92's prediction measured rather than argued: across the largest grid step there is, what the He I 10830 chain reads at the matching level — T , r , q_{H_2} , He/H — moves by 0, 1.6×10^{-8} , 1.6×10^{-7} and 1.6×10^{-7} . **Against the +0.73 % of crossings_j96.** That record split the +1.96 % its reservoir-9.0 arm moved into +1.22 % from the § 96 secondary-ionization branching and +0.73 % from the adapter, and attributed the adapter's share to the `insert_level` correction of § 92. The two arms compared here differ only in the grid, and by the largest grid step there is, and the line moves by +0.00015 %. The grid is 2.0×10^{-4} of the +0.73 %; the attribution stands.

The oxygen step between iterations of one ladder is gone. § 92 left it standing as a caveat: the closure re-runs the adapter in every iteration directory, so each iterate lands in one grid group or the other independently and the oxygen reservoir carries a 0.5 % step along one ladder. Over the three iterates of this rung the matching-level O/H spread falls from 1.552×10^{-3} to 8.044×10^{-6} , while the He/H marches by 1.2×10^{-5} in both alike — that march is the composition change the closure is making and is untouched. The closure is otherwise unchanged: the same three iterations, the same residuals to five digits, the same converged He/H at the match.

What this breaks. Every handoff column written from now on is a different column from the stored ones. `crossings_gm25` (photochem 0.8.4), `crossings_pc090`, `flux_closure`, `ladder_gm25` and the arms `crossings_j96` re-measured were all written by an adapter that accepted whatever grid the chemistry stopped on, so a stored arm and a re-run of the same command are no longer comparable at the level of the file. The break is bounded by the two grid groups: at most 2.5×10^{-2} in pressure, n_{tot} and ρ at fixed level index and $1.5\text{--}5.2 \times 10^{-3}$ in the matching-level O/H, everything else at that level 1.6×10^{-7} or less; $+1.5 \times 10^{-6}$ in the equivalent width of a closed rung, i.e. $5.0 \times 10^{-5} \text{ \% \AA}$ against a measurement error of $0.030 \text{ \% \AA}$; and, on the closure ladder's own slope of $\text{EW} \propto (\text{He/H})^{0.20}$, about 7.5×10^{-6} in the reservoir He/H a crossing is read at, 6×10^{-5} on a crossing of 8.21. **No stored result changes at its quoted precision and none was re-run.**

Scope. No EXHALE Fortran source or binary changed, nothing in photochem/ changed, no golden was refreshed and `make check` was not run: the `lower_profile` regression case reads a stored `lower_atmosphere_profile.dat` and never runs the adapter, so no golden can see this. The Photochem-side repair of § 92 — re-pinning and re-testing convergence before `reached_steady_state` is set — is still the tidier place for it and was not made. Not measured: another reservoir value, another planet, another K_{zz} or another `--toa`, and whether the pinned fixed point is unique (the seven guesses are consistent with it, and do not prove it).

Files. `src/utils/photochem_to_lower_profile.py` (`steady_state_at_stated_model_top`); `docs/lower_profile_deep_bound` § 8 (the repair, with every measurement); `LHS1140b/exhale/deep_temperature_sensitivity/` (`runs/pin_g4*`, `spread_table_pinned.txt`, `README`); `LHS1140b/exhale/adapter_grid_fix/` (the closure rung: `results.txt`, `reproduce.sh`, `cf_C8p2117/`, `rw_cf8p2117/`). The diagnosis this closes: § 92 and `LHS1140b/exhale/crossings_j96/results.txt` § 5.

100. The ground singlet is formed in one place and floored at zero (2026-08-30)

`TO_BE_DONE.md` item (L), taken along alternative 1 of the preceding design note. The species vector is unchanged — it still carries the summed He I with the 2^3S population inside it and the metastable beside it — and the output schema and its readers are unchanged. What changes is that the ground singlet $n(1^1\text{S})$ is now formed by one routine, which floors it at zero and counts how often it had to.

The routine. `composition.f90` gains `he_ground_singlet_density(nhei, nheiTR)`, generic over one cell (`..._cell`, two scalars) and over the whole grid including the ghost cells (`..._column`, two $1\text{--}N_{\text{g}}:N+N_{\text{g}}$ arrays, which is the cell function applied element by element). It returns $n(\text{HeI}) - n(2^3\text{S})$, replaced by zero where that

difference is negative, and increments the module counter `n_cells_he_singlet_clamped` whenever it does.

Zero is the physical lower bound of a population, so the floor is the enforcement of a bound and not an approximation. The difference can fall below it in two ways: where the metastable holds most of the neutral helium the two columns share their leading digits and what is left is round-off, and a restart whose two helium columns were written by different solves can put the difference on the wrong side of zero outright. Every consumer is linear in the singlet, so a negative value is not an instability — it is a negative He I opacity and a negative photoelectric heating rate, both unphysical and, before this change, both unguarded.

The test is < 0 , not the negation of ≥ 0 : a NaN handed in must pass through as a NaN so that the run's NaN detector reports it, rather than be turned silently into a zero singlet. That is the opposite convention from `positivity_limited_faces`, which does negate its test, and for the opposite reason — there a NaN face state is repairable from the cell averages, here a NaN singlet means the state upstream is already lost.

The call sites.

site	what it feeds
<code>util_ion_eq.f90 PH_heat_HHe</code>	the HeI singlet column N_{I5} and hence $e^{-\tau}$, the HeI photoelectric heating (<code>acc_H / acc_HeI</code>), HeI as a source of secondary ionizations (<code>acc_secHI / acc_secHeI</code>), the absorbed energy <code>acc_q</code> in the denominator of q , and the composition weight of <code>svs85_secondary_branching</code>
<code>post_process_adv.f90</code>	the split of the incoming HeI into its two populations before the advection solve
<code>post_process_adv.f90</code> <code>pin_cell_to_equilibrium</code>	the same split for a cell that is pinned to the equilibrium state

`PH_heat_HHe` already formed the difference once into a local `nheiS` and read that everywhere below, so the three sites above are the whole inventory: there is no direct `nhei - nheiTR` left in the source outside `composition.f90`. The arithmetic is unchanged where the difference is already non-negative, including the operation order: the post-process site keeps $(nhei_in - nheiTR_in) * n0$ and `pin_cell_to_equilibrium` keeps $nhei_in * n0 - nheiTR$, because the two round differently and the second is what the pinned cell was written with.

What is deliberately not routed through it. `System_HeH_mol.f90` and `System_HeH_mol_metals.f90` each write $n_{heiSI} = n_{hei} - n_{heiTR}$ inside the residual. The preceding section's inventory did not list them; it is completed here, together with the reason they stay as they are.

These are not reconstructions of two stored columns. They are algebraic expressions in the unknowns of the cell's ionization system — the molecular layout closes the free neutral helium as $n_{He}(1 - x_2 - x_3) - n_{HeH^+}$ and the metastable as $x_8 n_{He}$ — evaluated at trial iterates that the solver is entitled to place outside the physical region. Flooring the expression there would flatten the residual's dependence on x_8 over a whole region, and the forward-difference Jacobian of `hybrd1` would see a zero column where the sensitivity is real. The accepted root is guarded already, and at the right place: `ionization_fractions_physical` requires $x_2 + x_3 + x_{TR} + x_{HeH^+} n_H / n_{He} \leq 1 + f_{tol}$, which is exactly $n_{heiSI} \geq -f_{tol} n_{He}$, and a molecular root that fails it is retried from another starting point or clamped onto the ele-

ment budget (ieq_n_mol_clamped). The atomic triplet systems do not have the problem at all: System_HeH_TR and System_HeH_TR_metals write the singlet directly as $(1 - x_2 - x_3 - x_4)n_{\text{He}}$, with no difference taken.

The report line. The count is printed at the end of a run, after the post-process, since the post-process is one of the consumers and that is the first point at which the count is complete:

```
helium ground singlet: 65 cell evaluation(s) floored at zero
```

It is silent for a run whose two helium columns never crossed — which is every run in which the metastable stays well below the summed He I, and the statement that such a run is the one an unguarded build produces.

Measured. The difference does go negative in stored states. Sweeping the 2531 Ion_species*.txt files under LHS1140b/ (the helium-rich runs; He/H up to 0.55), 11 of them carry a negative singlet, all of them restart files (Ion_species_IC.txt) and all in cells where the neutral helium is *entirely* metastable: HeI and HeITR agree to about 13 significant digits, so the difference is round-off and lands on the negative side in up to 114 of the 504 cells. Three measurements, all on scratch copies of runs in the tree, all single-threaded:

1. *The GJ699 arm of crossings_j96 unchanged* (g_heh0p048, LHS 1140 b, He 2³S on, Do only PP). The metastable reaches 0.99% of the summed He I, so the difference never approaches zero: **no floor fires**, the report line stays silent, and Ion_species.txt, Hydro_ioniz.txt and Ion_species_adv.txt come out byte-identical to the stored outputs of that run. This is the byte-identity statement of alternative 1, checked on the post-process path with the triplet on.
2. *The same run with the restart's triplet column multiplied by 200*, so that the two helium columns disagree the way a restart written by two different solves would. The contaminated file has 65 cells with $\text{HeI} - \text{HeITR} < 0$, and the run reports **65 cell evaluations floored at zero** — the count is exact. Against a build with the floor removed the run differs by 1.3% in Hydro_ioniz.txt; that is the case the guard is for.
3. *A helium-rich state that needs no contamination* (LHS1140b/exhale/heh0p55_diff_d3, He/H = 0.55, He_diffusion on, Do only PP), whose stored restart carries 114 negative-singlet cells. The run reports **91 cell evaluations floored at zero** (fewer than 114 because load_IC goes through the mass fractions and the round-off is not the same one). Against the build with the floor removed, every output file is byte-identical: the singlets it repaired were negative by about 10^{-11} of the summed He I, so the numbers never moved — only their sign was wrong. This is the loss-of-significance case the preceding section described, and it confirms the description.

Regression, and what the floor caught in wasp_full. make check: six of the seven matrix cases are byte-identical against the goldens they had — wasp_he23off (the triplet is off, so the difference is never taken), mol_base_handoff, mol_metals, mol_lyman_werner, mol_diffusion and lower_profile, none of which floors a single cell. The largest metastable fraction among the goldens is 4.9% of the summed He I (lower_profile); wasp_full ends at 0.027%.

The one case that is not byte-identical is wasp_full, and the reason is worth recording, because it is not the loss of significance this item was opened for. The run reports **5 cell evaluations floored at zero**, and an instrumented build puts all five in cell 502 — two cells from the outer boundary — at steps 585 to 589 of the cold start, in the middle of

the violent early transient, each one immediately before the sweep reports no admissible root at 1 cell(s). The values there are

```
CLAMPDBG nhei nheiTR diff -2.4936541110339603E-08 -5.8625227650082221E-09 ...
NEGNUC  step 585 cell 502 n_H = -3.3458621929521555E+03
                        n_He = -2.8473287262022922E+02
```

so the singlet is negative because the *summed* He I is negative, and that in turn because the helium nuclei density of the cell is negative. The floor is right to fire — a negative singlet is inadmissible whatever produced it — but it treats a symptom. The negative nuclei densities themselves, and the negative $n(\text{He II})$ and $n(\text{He III})$ that come out of the same cell, are upstream of everything here; they are the outer-boundary counterpart of the base-region negative densities recorded under TO_BE_DONE.md item (A), and they are entered as item (Q).

That the five floors are the whole of the difference was checked directly, by building the same source with the floor removed and running the case again: that build reproduces the wasp_full golden **byte-identically**, stops at the same step (14057) and gives the same $\log_{10} \dot{M}$ (13.23). Against it the floored build differs by at most 5.6×10^{-13} in relative terms over Hydro_ioniz.txt (largest in the velocity column); $\rho v r^2$ at the $\dot{M}$ sampling point agrees to 10^{-15} , so the mass-loss rate does not move. The golden for this case was refreshed and check re-run on it: byte-identical, 14057 steps.

run_fcheck.sh: CLEAN (1500 bounded HD 209458 b steps under -fcheck=bounds, do, mem, $\log_{10} \dot{M} = 9.60$, no run-time check fired).

101. The O I ground term stops being an internal of the cooling (2026-08-30)

Phase P3 item 2 of docs/oxygen_chemistry_new_plan.md. The three fine-structure levels of the O I 2p4 3P ground term were already being solved for – the [O I] 63.2/145.5/44.1um cooling is a three-level statistical equilibrium and cannot be evaluated without them – but the solution existed only inside a function that returned a number of erg/s and threw the populations away. They are now an output field, and they are the SAME populations, not a second calculation of the same thing.

The reason this matters is one line of atomic physics: the O I 1302.168 / 1304.858 / 1306.029 Å resonance triplet absorbs out of 3P2, 3P1 and 3P0 respectively. A transit forward model handed only the total O I density would apply it to all three components and count the same atoms three times.

The split

Cool_coeff.f90 had one routine, fine_structure_cooling_3level, that solved the 2x2 level-balance system by Cramer's rule and then assembled the escaping power. It is now two:

```
pure subroutine fine_structure_populations_3level(..., f1, f2, f3)
pure double precision function fine_structure_cooling_3level(...) result(W)
```

The second calls the first and does nothing else but assemble W from what it returns. The arithmetic is unchanged line for line, including the order of the operations, so the cooling is byte-identical.

Two more routines make the O I case reachable from outside without duplicating anything:

routine	what it is
<code>oxygen_ground_term_collisions(T,ne,nHI,C21,C31,C32)</code>	the total collisional de-excitation rates of the three couplings, electrons plus H atoms. Written once because two consumers need exactly these numbers
<code>oxygen_ground_term_populations(T,ne,nHI,b63,b145,b446,b63,b145,b446,f_3P2,f_3P1,f_3P0)</code>	the total collisional de-excitation rates of the three couplings, electrons plus H atoms. Written once because two consumers need exactly these numbers
<code>oxygen_ground_term_levels(...)</code>	the grid wrapper, same signature convention as <code>cool_OI_ne</code> , so both are fed by the same <code>beta_fs / nbar_fs</code> that <code>fine_structure_line_transfer</code> produced

`cool_OI_ne_func` now calls `oxygen_ground_term_collisions` instead of carrying the three rate expressions itself. There is one definition of each number in the file.

The output

`output/OI_levels.txt` and `output/OI_levels_adv.txt`, written by `write_output` whenever the run carries metals – no key, the same file-presence rule as everything else about metals. 11 columns: `r`, `T`, `n_e`, `n_HI`, `n(O I)`, then `f(3P2)` `f(3P1)` `f(3P0)` and the same three as densities. A metals-off run writes no file, and the transit tool skips the line exactly as it already skips Mg II, Ca II and Na I.

It is a separate file rather than three more columns of `Ion_species.txt` for the same reason `Lyman_Werner.txt` and `Cooling_breakdown.txt` are separate files: `Ion_species.txt` is one of the two files the regression compares bitwise, and a diagnostic has no business moving it.

`write_output` recomputes `n_e` (`calc_ne`, with the run's own molecular columns) and the line transfer (`fine_structure_line_transfer`) from the state it is handed, so the `_adv` file carries the advection-corrected temperature and metal densities and the `eq` file the equilibrium ones. `exhale_io.py` gains `load_OI_levels / OI_LEVEL_COLS`.

The gate, and what it caught nothing of

The gate the plan asks for is that the three level densities sum to the total O I. `write_output` measures it and prints it every time:

```
(write_output/eq) max |sum(O I levels)/n(O I) - 1| = 2.91E-16
(write_output/ad) max |sum(O I levels)/n(O I) - 1| = 2.95E-16
```

(HD 209458 b, HD209458b/oi1302/, 504 cells plus ghosts.) `EXHALE_transit.py` re-measures it from the file rather than trusting it, and refuses to run the line if it exceeds $1e-10$; measured $2.25e-16$. It is round-off because $f_1 = 1 - f_2 - f_3$ by construction, which is the point: the closure is structural, not a coincidence to be checked for.

What the populations actually do

The more interesting check is against the Boltzmann limit, which the statistical equilibrium must reproduce where the levels are collisionally dominated and must NOT reproduce where they are not. On the converged HD 209458 b run:

r/Rp	T [K]	f(3P2), f(3P1), f(3P0)	max deviation from Boltzmann
1.000	1450	0.5979, 0.3066, 0.0955	1.4e-11
1.200	6416	0.5653, 0.3273, 0.1074	3.6e-05
2.001	6473	0.5713, 0.3213, 0.1074	6.2e-03
2.997	3321	0.6136, 0.2813, 0.1051	4.0e-02
4.185	1605	0.7211, 0.1823, 0.0966	1.3e-01

LTE to eleven digits at the dense base, and departing from it above 2 Rp as the levels drain radiatively toward the lowest one. The fine-structure splittings are 227.7 and 326.6 K, so in the 6000-7000 K launch region the Boltzmann weights are within 2% of the statistical 5:3:1 – which is exactly the factor a transit model would get wrong by ignoring the split: 1.8x on 1302, 3.1x on 1304, 9.3x on 1306, and 3x on the total column.

Verification

make check and run_fcheck.sh: see section 105.

102. An O I 1302 forward model, and what the published depth is a measurement of (2026-08-30)

Phase P3 item 3 (option B1). The transit tool now computes the O I 1302.168 / 1304.858 / 1306.029 Å resonance triplet and compares it, band-integrated, with the published HD 209458 b detection. The comparison does not reproduce the published depth. Most of this section is about why the comparison had to be set up three times before it meant anything, because two of the three attempts were wrong in ways that looked like answers.

The line

One upper level (3s 3S1, $g = 3$), three lower levels: the ground-term fine-structure levels of section 101. Vacuum wavelengths, A_{ul} and f_{lu} from NIST ASD (accuracy A, retrieved 2026-08-30):

component	lower	E_l [cm ⁻¹]	g_l	A_{ul} [s ⁻¹]	f_{lu}
1302.168	3P2	0.000	5	3.41e8	0.0520
1304.858	3P1	158.265	3	2.03e8	0.0518
1306.029	3P0	226.977	1	6.76e7	0.0519

One structural change: a component owns its lower level

resonance_spectrum took a line's components and ONE shared lower-level density, which is right for Mg II h&k, Ca II H&K and Na I D – both components of each of those absorb out of the ion ground state – and wrong for the first line that comes along whose components do not. The component tuple is now (λ_{rest} , f_{osc} , A_{21} , n_{lower}): the density belongs to the component, because a component IS an absorption out of a particular level. The three doublets just repeat the same array. Verified bitwise: the refactored routine reproduces the old one exactly on a synthetic doublet

($\max |new - old| = 0$), and the Phase-5a table of HD 209458 b is unchanged (Mg II 7.281 / Ca II 3.590 / Na I 1.748 % line-center, before and after).

Three ways to get the comparison wrong

The published numbers, from the same four HST/STIS G140L transits:

- Vidal-Madjar et al. 2004, ApJ 604, L69, Table 1: O I / O I* / O I**, **1300-1310 Å, 12.8 (+4.5/-4.5) %**;
- Ben-Jaffel & Hosseini 2010, ApJ 709, 1284, Table 2: O I, **1299-1310 Å, 10.5 +/- 4.4 %**.

(a) Line center is not the measurement. G140L does not resolve the triplet – "the low resolution (2.5 Å) does not allow the stellar emission lines to be resolved" (VM04) – so both papers quote a band depth. This is the trap already recorded for Mg II, and it is still printed alongside so the difference stays visible.

(b) The continuum is inside the published number. Both quote $(R_{\text{abs}}/R_{\text{*}})^2$ from an occultation-curve fit, which includes the opaque planetary disk; their own continuum bands give 2.0 (+0.5/-0.7) % and 1.96 +/- 0.42 %. This script normalizes $T = 1$ to the planet's disk, so $1 - T$ is the EXCESS. Each published value therefore has that paper's own continuum subtracted before the comparison: **10.8 +/- 4.5 %** and **8.54 +/- 4.4 %**.

****(c)** A flat band average is not what was measured, and this one cost a factor of 12. ****** The measurement is a ratio of fluxes, and essentially all the flux in 1300-1310 Å of a G0 V star is in the three narrow chromospheric O I emission lines. Ben-Jaffel & Hosseini measured them: "The average FWHM of the O i and C ii lines are 0.2 Å (or 45 km/s)", with "the lines' peaks in the ratio 1:1.5:1.17, respectively, for the O i (1302.17 Å, 1304.86 Å, and 1306.03 Å) lines". Averaging the model flatly over 10 Å dilutes it by the band-to-line width ratio. The band weight is now that stellar profile (EXHALE_TRANSIT_OI_WEIGHT=flat recovers the flat average, and both numbers are always printed).

The interstellar medium is ****inside those peak ratios and must not be applied twice****. Neither paper removed it; VM04 argued from it – "The O i ground-level line is strongly absorbed by the interstellar medium. Therefore the 13% absorption observed during the transit in the full O i triplet must be due to the presence of O i* and O i** ...". The 1:1.5:1.17 ratios are measured from the observed spectrum, which is why the ground-level component is the weakest of the three there. An ISM screen was written and then removed: it would have double-counted. The ISM never multiplies the planet's transmission in any case – it absorbs in and out of transit alike and cancels from the ratio; it only decides which wavelengths the measurement is sensitive to.

****(d)** The instrument LSF cancels, and convolving before weighting loses a factor of 5. ****** A Gaussian LSF conserves the integral, so it drops out of a ratio taken over a band many LSF widths wide – which is what both papers measured. Convolving T_{lambda} with the 1.3 Å LSF first and then weighting with the 0.2 Å stellar profile smears absorption out of exactly the wavelengths that carry the weight. The band integral is taken over avg_rot: wind and rotation broadening kept, instrument LSF not applied. (resonance_spectrum now returns avg_rot for this; it was computing it and discarding it.)

The three wrong answers and the right one, same run:

what is integrated	depth [%]
line center, instrument-convolved	0.092
flat 10 Å average	0.0300
stellar-profile weight on the LSF-convolved curve	0.0700
stellar-profile weight, no LSF (the comparison)	0.3586

The flat average is 0.0300 with or without the LSF, to the digits printed — which is the integral-conservation argument of (d) demonstrated rather than asserted: the LSF only matters once the weight is narrower than it is.

The result

HD 209458 b, re-converged with the current executable (HD209458b/oi1302/, JFNK info=0, $|R| = 7.8 \times 10^{-5}$, $\log_{10} \dot{M}$ 9.47):

band	model excess [%]	published excess [%]
VM04 1300-1310 Å	0.359	10.80 +/- 4.50
BJ10 1299-1310 Å	0.359	8.54 +/- 4.40

A factor of 30 low, i.e. the model does not reproduce the detection. This is not a surprise inside the observational literature, which says the same thing about every model of this line: VM04 note that "If the absorption width was limited by the thermal broadening, the observed 10% absorption would correspond to a geometrical absorption of more than 60%", and Ben-Jaffel & Hosseini, having measured stellar FWHMs three times wider than VM04 assumed, conclude that "all line profiles have FWHMs far larger than the thermal Doppler width in the HD209458b atmosphere" and that this makes "thermal broadening in scaled atmospheric models a marginal solution for the O i and probably C ii lines". Their own scaled model reaches 3.9 % against the observed 10.5 %. A spherically symmetric thermal wind with LTE-like ground-term populations is one further step short of that.

So the honest statement is that the forward model is in place, its atomic data and its lower levels are right, its comparison is set up the way the observation was reduced – and it under-predicts the published depth by a factor of 30. What it does NOT do is reproduce the published depth by accident through any of the three wrong weightings above, each of which would have moved the answer by a factor of 3-12 in the direction of the data.

The gate: both column regimes

EXHALE_TRANSIT_OI_NSACLE multiplies all three level densities, leaving their ratios alone, so the same code path can be driven up and down the curve of growth:

N x	depth [%]	
1e-9	2.4166e-06	
1e-8	2.3852e-05	9.87x for 10x column – linear, optically thin
1	0.359	
1e4	4.583	12.8x for 1e4 column – saturated

Both regimes reproduce their expected behavior. Note the last row: the observed depth is reached only when the O I column is raised by four decades, which is the same statement as the paragraph above from the other side.

Recorded as a known omission: Ly-beta pumping

O I 3P2 - 3d 3D at 1025.7616 Å sits 0.0398 Å (11.6 km/s) from H Ly-beta at 1025.7218 Å (NIST Ritz values). Bowen (1947, PASP 59, 196, Table I) identified the coincidence; the cascade runs 3d 3D → 3p 3P (11287 Å) → 3s 3S (8446 Å), and 3s 3S is the upper level of this very triplet, so the loop both emits in 1302/1304/1306 and returns population to all three 3P levels. It is NOT in this model, and the literature brackets its size rather than fixing it: in the Earth dayglow it is "roughly 10% of the direct excitation by the solar 1304 Å triplet" (Meier 1991, SSRv 58, 1, p. 99), while in the solar chromosphere "the Lyb excitation exceeds the collisional excitation for the triplet by about a factor of 20" (Skelton & Shine 1982, ApJ 259, 869). No published calculation of it exists for an escaping exoplanet atmosphere – checked by full-text search of the two exoplanet O I 1304 papers (Koskinen et al. 2010; France et al. 2010), neither of which mentions Ly-beta, Bowen or 1025.7. Since it feeds the upper level of the observed triplet and redistributes its lower levels, it is a candidate for part of the factor of 30, and it is written into the metadata file so the number never travels without it.

Outputs

tpm_OI.txt (the curve, same four columns and the same naming rule as every other line) and tpm_OI_band_depths.txt, which carries the band limits, both model numbers, both published values with their sources, and the ISM / geocoronal / continuum / LSF conventions in its header – so the number and the conditions under which it may be compared cannot become separated. New environment overrides, all EXHALE_TRANSIT_*: RES_OI (default 1000), OI_WEIGHT (star | flat), OI_STAR_FWHM_A, OI_STAR_PEAKE, OI_NSACLE.

103. The one charge-transfer reaction Table 4 does not have, measured (2026-08-30)

Phase P3 item 1 (option B2). Re-sourcing the charge-exchange network to Huang et al. (2023) Table 4 dropped the doubly-ionized rows the earlier Kingdon & Ferland implementation carried, C2+, N2+ and O2+ with H. The plan turned that from an audit into a question: does the missing O2+ + H0 → O+ + H+ materially change the O III profile? It does. By four decades in the launch region.

Group E

charge_exchange.f90 carries one row that is not from Table 4, and says so. n_cxreac goes 63 → 64; the new row is cx_E1_O2p_H0, donor H at stage 0, acceptor O at stage 2 (i.e. O III → O II), and it is .false. in cx_default. It is gated on a scale factor, NOT on cx_full:

metals.inp:	cx_O2p_H <scale>	(default 0 = off)
-------------	------------------	-------------------

`cx_full` means "all of Table 4" and this row is not in Table 4, so folding it in there would have made a documented switch quietly mean something else. The membership test is now one function, `cx_row_is_active(r)`, so `cx_init`'s two passes over the reaction list cannot drift apart – they previously repeated the same three-clause condition. `write_setup_report` echoes `cx_o2p_h_scale`.

The rate

`electron_capture_o2p_from_H(T)` is Barragan, Errea, Mendez, Rabadan & Riera (2006), ApJ 636, 544, Table 1 column k1, all 26 published points, interpolated linearly in $\log_{10} k$ against $\log_{10} T$ – the table itself, not a fit to it. Their method is molecular close-coupling on MELD multireference-CI wavefunctions, quantal below 250 eV/amu, over $1e2$ - $1e5$ K, and it is the calculation CHIANTI v11 adopts for $O\ III \rightarrow O\ II$ charge transfer (Dufresne et al. 2024, ApJ 974, 71, appendix A.3.3) and that Cloudy fits. It supersedes Kingdon & Ferland (1996, ApJS 106, 205), whose Landau-Zener-era value is 1.6x higher at 5000 K and 1.2x at $1e4$ K. T is clamped to the calculated range rather than extrapolated, the same convention as `fine_structure_epsilon`.

Two things are written at the code site because they decide the treatment:

- the classical Langevin capture bound for this pair, $k_L = 2 \pi q \sqrt{\alpha/\mu}$ with $\alpha(H) = 4.5 \text{ a0}^3$, is $3.9e-9 \text{ cm}^3 \text{ s}^{-1}$, so the published rate is 16-32 % of the rate at which the two particles reach small separation at all – physically sensible, and a bound, not a measurement;
- ****the reverse direction is not carried, and that is detailed balance, not an omission.**** $O^+ + H^+ \rightarrow O^{2+} + H^0$ is endothermic by $IP(O\ II) - IP(H\ I) = 35.12112 - 13.59843 = 21.52 \text{ eV}$ (NIST ASD), so the reverse rate is suppressed by $\exp(-21.52 \text{ eV}/kT)$: $1e-11$ of the forward rate at $1e4$ K, $4e-6$ at $2e4$ K.

Why it dominates, and by how much

Charge transfer competes with recombination as $k_{CT} n(H^0)$ against $\alpha_{rec} n_e$. With $\alpha_{rec}(O\ III \rightarrow O\ II) = 5.0e-12 / 3.1e-12 / 2.0e-12 \text{ cm}^3 \text{ s}^{-1}$ at 5000 / 10000 / 20000 K (Badnell 2006 RR plus the Zatsarinny et al. 2004 DR – the same coefficients EXHALE's own `bad_OIII` block already carries), the ratio k_{CT}/α_{rec} is 122 / 280 / 617, so charge transfer wins wherever $n(H^0)/n_e$ exceeds a few times $1e-3$. In this wind $n(H^0)/n_e$ is 840 at 1.05 Rp and still $7e-2$ at 2 Rp .

Measured on HD 209458 b, three runs identical but for `cx_o2p_h`, each JFNK-converged from the same state (HD209458b/oi1302_cx02p_1, ..._cx02p_3):

r/Rp	T [K]	$n(H^0)/n_e$	x(O III) off	x(O III) scale 1	ratio	n(O I) ratio
1.05	2342	8.4e2	–	–	2.2e-05	1.033
1.10	3941	4.7e1	1.635e-04	3.484e-08	2.1e-04	1.026
1.20	6140	3.4e0	6.147e-03	1.104e-05	1.8e-03	1.029
1.50	6935	3.0e-1	1.156e-01	2.290e-03	2.0e-02	1.146
2.00	5111	7.4e-2	3.755e-01	5.516e-02	1.5e-01	1.520
3.00	2267	2.0e-2	6.914e-01	5.262e-01	7.6e-01	1.532
4.22	955	8.0e-3	8.460e-01	8.219e-01	9.7e-01	1.159

Answer to B2: yes, materially. O III is suppressed by 4.7 decades at 1.05 Rp, 3.7 at 1.1, 2.7 at 1.2 and 1.7 at 1.5, converging on the Table-4-only answer only past 3 Rp where hydrogen is fully ionized and there is nothing left to transfer from. Tripling the rate ($\text{cx_O2p_H} = 3.0$) suppresses O III by a further factor of 3 – linear, as a destruction channel should be – so the conclusion does not rest on the rate being right to better than a factor of a few.

The knock-on effects are much smaller, because O III is a trace stage: $\log_{10} \dot{M}$ moves $9.47 \rightarrow 9.48$, and the O I 1302 band depth of section 102 moves $0.359 \rightarrow 0.380$ % (+6 %). Neutral O gains 3 % below 1.2 Rp and 52 % near 2-3 Rp.

The default, and the decision it defers

The reaction is real, it is published, and Cloudy and CHIANTI both carry it; leaving it out makes the O III profile wrong by decades in the region a transit probes. **The physics says it should be on.** It is nevertheless off by default here, because turning it on moves every metals-bearing golden and every published O III-dependent number of the WASP-121 b comparison, which is a deliberate, reported refresh and a user decision, not a side effect of a sensitivity experiment. Recorded in TO_BE_DONE.md item (H) with this measurement attached.

104. What diffusive separation does to the oxygen the transit sees (2026-08-30)

Phase P3 item 4 (option B4). He_metal_diffusion has let every metal settle independently since the binary-diffusion work, but it had never been exercised for oxygen specifically, and the plan asks one question: does O separate from H enough to matter for the O I 1302 line?

Two HD 209458 b runs, identical but for three keys (He_diffusion: True, He_metal_diffusion: True, He_Kzz: $1.0\text{e}9$), both started from the same converged state and both JFNK-converged (HD209458b/oi1302 and HD209458b/oi1302_diffusion, info=0, $||R|| = 7.8\text{e-}5$ and $6.1\text{e-}4$).

Oxygen fractionates by a factor 5.55, helium by 1.25.

r/Rp	(O/H) no diffusion	(O/H) diffusion	ratio	(He/H) ratio
1.000	4.900e-4	4.900e-4	1.000	1.000
1.020	4.900e-4	4.661e-4	0.951	0.993
1.050	4.900e-4	2.828e-4	0.577	0.939
1.101	4.900e-4	1.545e-4	0.315	0.866
1.200	4.900e-4	9.246e-5	0.189	0.806
≥ 1.5	4.900e-4	8.82e-5	0.180	0.799

The separation is complete by 1.2 Rp and the wind above that carries a fixed, 5.55x oxygen-poor composition. Helium, sixteen times lighter, loses only 20 %.

But the wind is not the same wind. With $K_{zz} = 1\text{e}9$ the diffusion operator also changes the mean molecular weight and the outflow: $\log_{10} \dot{M}$ goes $9.47 \rightarrow 9.95$, i.e. +0.48 dex. So the O I column is set by two effects pulling opposite ways, and neither the radial column nor the transit depth moves by anything like 5.55x:

quantity	no diffusion	diffusion	ratio
radial N(O I) [cm ⁻²]	4.810e18	4.788e18	0.995
chord N(O I) at b = 1.1 Rp	1.290e16	4.450e15	0.345
chord N(O I) at b = 1.2 Rp	1.161e15	3.965e14	0.342
chord N(H I) at b = 1.2 Rp	2.422e18	4.639e18	1.916
O I 1302 band depth [%]	0.359	0.294	0.820

****Answer to B4:** the fractionation is large (5.55x in O/H) but the observable moves by 18 %,** because the denser wind that the same K_{zz} produces refills most of what the settling removes. The radial column is essentially unchanged because it is dominated by the base, where nothing has separated yet; only the chord columns at the impact parameters a transit actually probes show the factor of 3.

Two things this does not settle, and they are the reason the number is quoted as a bracket rather than a correction. $K_{zz} = 1e9 \text{ cm}^2/\text{s}$ is one choice out of a range spanning decades and the fractionation scales with it. And the two effects are not separable in these runs: a controlled experiment would need the composition change without the wind change, which the operator does not offer. Both are stated here rather than in a caveat elsewhere, because the 18 % is the only number a paper could quote from this.

105. Two labels that said the wrong thing, and the P3 verification (2026-08-30)

Both found while doing the work of sections 101-104, neither related to it. Neither changes a number.

The run log claimed the wrong convergence criterion after a Newton finish

EXHALE_main.f90 reuses `is_mom_const` as the flag that leaves the marching loop, and the JFNK finish sets it on success. The stop report then printed

```
-> converged: momentum constant (du < du_th)
final: count=2002 du= 1.6294E-03 dtu= 9.6602E-07
```

on a run whose `du_th` was $1.0e-3$. The statement is false for every Solver: Newton run: JFNK hands over at `du < newton_du_switch`, two decades above `du_th`, and converges on the residual instead – in that run to $||R|| = 7.8e-5$. Anyone reading the log would conclude the `du` criterion had been met when it had not. Fixed with a separate `newton_finished` flag, so the report now reads

```
-> converged: JFNK steady solution (||R|| < resid_th)
```

and the `du/dtu` line stays as the diagnostic it always was. Nothing but the message changes: `is_mom_const` still exits the loop the same way.

adf48 on the dielectronic data

Four comments in `Cool_coeff.f90` labelled the Badnell dielectronic data class `adf48`. It is `adf09`. Checked in both published sources rather than from memory: Badnell (2006, *ApJS* 167, 334) writes that the RADIATIVE recombination data are archived in "(ADAS) data class `adf48`", while Zatsarinny et al. (2004, *A&A* 417, 1173), the source of the DR coefficients themselves, describe generating "directly the `adf09` file necessary for use by ADAS". Comment-only.

Verification of sections 101-105

`make check: 7/7 byte-identical` (`wasp_full`, `wasp_he23off`, `mol_base_handoff`, `mol_metals`, `mol_lyman_werner`, `mol_diffusion`, `lower_profile`). No golden was refreshed and none needed to be: the statistical-equilibrium split of section 101 is arithmetic-preserving, the O I level file is a new file rather than new columns of a compared one, and the Group-E charge-transfer row of section 103 is absent from `cx_act` entirely until `cx_O2p_H` is set.

`run_fcheck.sh: CLEAN`.

Python side, checked separately because the regression does not cover it: the `resonance_spectrum` generalization of section 102 reproduces the previous routine bitwise on a synthetic doublet (`max |new - old| = 0.0`), and the Phase-5a metal line-center depths of HD 209458 b are unchanged across it.

A note on the build, because it cost time here. `EXHALE.x` must be built with `PATH=/usr/bin:$PATH`. A bare `make` picks up conda's `gfortran 15.2.0`, which compiles the sources but cannot link against the system LAPACK (undefined reference to `_gfortran_string_len_trim@GFORTRAN_1.0`); the failed link removes `EXHALE.x` and leaves a set of objects compiled by the wrong compiler behind, which a later correct `make` will happily link because it decides by timestamp. `make distclean` is the recovery.

106. The staleness markers of the LHS 1140 b memo, replaced by measurement (2026-08-30)

What this closes. `docs/lhs1140b_exhale_vs_pwinds.tex` carried 35 lines matching `retired|pre-Sect`, of two generations: numbers standing on the retired Taylor et al. (2025) Penning fit (Sect. 87) and numbers standing on the secondary-ionization split as it was before Sect. 96. A marker is a promise to re-measure, and this section is the campaign that keeps it. Every marked number was re-solved on the current binary and re-measured; nothing was carried across, interpolated or scaled. After it, `grep 'retired|pre-Sect'` returns **9 lines, all of them historical narrative** — a three-vintage exponent comparison, a column heading that names what a ratio is taken against, and five sentences that say what an earlier coefficient or an earlier branching gave. None is a promise to re-measure.

Scale. 82 JFNK wind re-solves in seven series, three elemental-flux closures re-run end to end (13 Picard iterates), one HD 209458 b control, and two convolution tables recomputed from stored curves. All of it is under `LHS1140b/exhale/refresh_j96/`, one subdirectory per series, each with its own `results.txt` giving info and $\|R\|$ pass by pass. No previous-generation directory was written: `find -newermt over thermostat_gm25, basemetals_gm25, bump_gm25, misc_gm25, xuvkzz_gm25, crossings_gm25, crossings_j96, flux_closure, crossings_pc090, ladder_gm25 and kzz_power` returns zero files.

No source was changed and nothing was rebuilt: EXHALE.x of 2026-08-30 10:47 (sections through 105) is the one binary behind every number.

Where a judgement changed, not only a digit

site	what it said	what it says now
fig:kzzprof	the $K_{zz} = 0$ arm's metastable column "is not to be trusted", Sect. 88.4(a)	Sect. 96 <i>is</i> that repair: cells above $1.5R_p$ whose ground singlet has collapsed below the metastable go from 77 of 504 to 0 of 504 , the peak falls $0.134 \rightarrow 0.043 \text{ cm}^{-3}$, and the factor over the four arms, 158, is now 506 quoted without a hedge
fig:ew	the turbulence-on curve "has no counterpart on the current physics", so redrawing either curve alone would mix physics	both curves now exist on one physics (7 arms re-solved, each synthesized twice); crossings 0.4051 and 0.3982 , 1.7 % apart
the GJ 699 seven-row scan	"the seven rows have not been re-solved on the current branching, so that check is not repeated"	repeated, and it passes more tightly : 0.0484 against the ladder's 0.0483, 0.2 % where the pre-Sect.-96 pair agreed to 0.4 %
sec:xuvgrid	the two 0.6 % crossings are "a factor 1.31 apart ... no longer close enough to be quoted as one number"	Sect. 96 moves them <i>towards</i> each other — scalar $0.318 \rightarrow$ 0.326 , closed $0.416 \rightarrow$ 0.368 — a factor 1.13 ; the conclusion drawn from 1.31 is withdrawn
the same section	the two slopes "differ by two orders of magnitude"	519 against 90 over 0.33–0.10, a factor 5.8 ; false as written
the closed XUV grid	one non-monotonic point, at 0.01	non-monotonic below 0.25 as a whole (0.15 above 0.20 as well); all four such arms stop at <code>info=2</code>
tab:metatimes	" T agrees to 1.3 per cent and t_{rec} to 1.4 at every radius"	1.2 and 1.8 ; the claim as printed was false
tab:bump	"the last two of the four passes accepted no step"	only the last one does; <code>heh0p55</code> takes a step in pass 3
$x(\text{He III})$ at $2R_p$	"stays below 2×10^{-4} "	2.06×10^{-4} on the 10.31 arm; rewritten as "at most 2.1×10^{-4} "

Four judgements survived their re-measurement and are stronger for it: the closure is still single-valued (He/H spread $5.5 \times 10^{-6} \rightarrow$ **7.8×10^{-6}** over a factor ten in the starting flux); the width the data demand still does not move with composition, treatment or base (four matched kernels 21.95–22.40 km s⁻¹); the K_{zz} pressure form is still buried in the workflow's path spread, though by 1.30 rather than 1.63; and the metal-free-to-closed thermostat ratio is still **2.6**, having now survived both corrections.

Two defects found in the campaign's own material

The thermostat table's scalar arm was not a converged wind. It carried `Resid tol: 5.0e-3`, at which the JFNK finish takes **zero iterations** and hands back a state residual $\|F_s\|_2 = 1.27$, against 0.018–0.074 on the other three arms of the same table. The memo had noticed the symptom and excused it — “the two tables are each internally consistent and differ by 1 per cent for that reason” — and on the current binary that gap had grown to 4.7%. A wind that stops without taking a Newton step is not a steady solution, so the arm was re-solved at `1.0e-4` (`refresh_j96/thermostat/ms2p13_r1e4`, $\|F_s\|_2 = 0.018$) and `Table tab:thermostat` now reads its scalar column off that. The excuse sentence is gone: the base-metals ladder's scalar rung and the thermostat table's scalar column are now the same configuration at the same tolerance and agree to 0.3%.

A seed hysteresis larger than the effect being measured. One and the same profile-branch arm — identical `input.inp`, identical handoff column, identical binary, four JFNK passes each reaching `info = 0` at $\|R\| \simeq 10^{-4}$ — lands on **three different fixed points** depending on which converged wind it started from: $EW = 0.519, 0.532$ and $0.542\% \text{ \AA}$, a spread of **4.6%**. Four further passes move each lineage by at most 0.23% and leave the flux spread and base temperature unmoved, so these are three fixed points and not three truncations; the steady-wind diffusion outer pass reports three different composition drifts ($5.9, 6.2, 7.3 \times 10^{-4}$), which is where the hysteresis lives. That is larger than the 1.8% the Sect.-96 branching is worth on these arms and larger than the 0.31–1.5% path spread the memo quotes elsewhere. It is now stated in `sec:closure` as a systematic every profile-branch line number carries. Not diagnosed.

Corrections made in passing

- `resolve_wind.sh`, the shared re-solve driver written for this campaign, copied `input.inp`, the handoff profile and `metals.inp` but **not** `base.inp` or `opacity.inp`. EXHALE is configured by file presence, so a dropped file silently changes a run. Fixed to copy all four. Measured impact on what was already run: **zero** — `refresh_j96/thermostat/fc2p09` (no `base.inp`) and `refresh_j96/xuvkzz/L2p09` (with it) are the same configuration from the same lineage and give $EW = 0.544501$ both, to every digit, because on a profile arm `base.inp` carries provenance and not state.
- The memo said “All eight arms of this scan”; the scan has **seven** (`SCAN_EARLIER` in `make_memo_figures.py`).
- Three places quoted “a factor 189 in composition” for the span of the four matched-kernel solutions. On the crossings of record that span is $8.212/0.0483 = \mathbf{170}$, which the memo's own summary already said.
- “whose equivalent width falls a factor 2.6 short” was comparing to the thermostat ratio, not to the measurement; against the measurement it is **2.0**.
- Two errors in the previous generation's own records, flagged and **not** fixed because those directories are read-only history: `basemetals_gm25/results.txt` prints rung E as `red 1.9312 / EW 0.50749` where its own files give $1.9216 / 0.50472$; `thermostat_gm25/results.txt` gives `ref2p09` a residual tolerance of `4.0e-4` where its `input.inp` says `1.0e-4`.

Figures

Thirteen of the memo's twenty figures were redrawn, and the other seven are proved untouched by `md5sum -c` against a pre-campaign manifest and by their stat mtimes. The redraw used a new driver, `LHS1140b/exhale/refresh_j96/remake_figures.py` which executes `make_memo_figures.py` — so a redrawn figure is drawn by the memo's own code — with `plt.savefig` wrapped to write only a named whitelist and to announce every write and every skip. Redrawn: `bestfit`, `bestfit_structure`, `broadened`, `bump`, `closure`, `closure_ladder`, `composition_profiles`, `diff_broadened`, `ew_vs_heh`, `fig4style`, `knudsen`, `kzz_profiles`, `thermostat`. `make_memo_figures.py` was re-pointed at the new runs, and two of its anchors moved to the runs solved *at* their own crossing rather than at the nearest ladder arm: `EWMATCH_TAG` to `crossings_j96/cf_W1132_P` ($\text{He}/\text{H} = 0.4132$) and `DIFF_TAG` to `crossings_j96/cf_K1e9_P` (1.6108). `bump_analysis.py`'s `RUNDIR` moved to `refresh_j96/bump`.

Verification

`pdflatex` three times on the memo: no error, **no overfull box**, no undefined reference, four underfull boxes all in paragraphs this work did not touch, 76 pages (was 77). The regression matrix was not run and no golden was refreshed, because no Fortran source and no golden input changed — this section re-measures, it does not modify the code.

107. $\text{O}^{2+} + \text{H}^0 \rightarrow \text{O}^+ + \text{H}^+$ becomes the default (2026-08-30)

Section 103. measured the one charge-transfer reaction Huang et al. (2023) Table 4 does not carry and answered the physics question in the affirmative: without it the O III fraction of an escaping atmosphere is wrong by up to 4.7 decades in the launch region, because charge transfer beats recombination there by two to three orders of magnitude. It was nevertheless left off, because switching it on moves every metals-bearing golden and that is a user decision rather than the side effect of a sensitivity experiment. The decision was taken. `cx_o2p_h_scale` now defaults to 1.0, the published Barragan et al. (2006) rate, and

```
metals.inp:  cx_o2p_H <scale>      (default 1 = published rate; 0 = off)
```

still rescales it, with 0 reproducing the Table-4-only reaction set exactly.

The code change, and what it does not change

Two files. In `charge_exchange.f90` the module variable

```
real*8, save :: cx_o2p_h_scale = 1.0d0
```

carries, at its declaration, the source (Barragan, Errea, Mendez, Rabadan & Riera 2006, ApJ 636, 544, Table 1 column k1; adopted by CHIANTI v11, fitted by Cloudy), the validity range (10^2 – 10^5 K, clamped rather than extrapolated) and the date and reason the default was set. In `metals_input_read.f90` the key is documented and echoed as a rescale rather than a switch.

Nothing else moved. The row is still absent from `cx_default`, still outside `cx_full` — which means “all of Table 4”, and this row is not in Table 4 — and `cx_row_is_active` still drops it from `cx_act` altogether when the scale is zero.

So `cx_02p_H 0` is not a numerical zero multiplying an evaluated rate; it is the same code path the Table-4-only runs always took.

What the goldens did

Seven cases, and the split is exactly where the physics says it should be. **The three metals-free molecular cases are byte-identical**, which is the check that the change reaches the metal ionization system and nothing else: `mol_base_handoff`, `mol_lyman_werner`, `mol_diffusion`. The four cases that carry oxygen all moved.

case	oxygen from	max abs rel. move (ρ / T)	$\log_{10} \dot{M}$
<code>wasp_full</code> (WASP-121 b)	<code>metals.inp</code> , solar	$3.9 \times 10^{-2} / 5.0 \times 10^{-2}$	$13.23 \rightarrow 13.23$
<code>wasp_he23off</code> (WASP-121 b)	<code>metals.inp</code> , solar	$4.0 \times 10^{-2} / 5.1 \times 10^{-2}$	$13.23 \rightarrow 13.23$
<code>mol_metals</code> (hot Uranus)	<code>metals.inp</code> , solar	$4.2 \times 10^{-4} / 3.0 \times 10^{-4}$	$10.58 \rightarrow 10.58$
<code>lower_profile</code> (HD 209458 b)	lower-atmosphere profile	$2.5 \times 10^{-2} / 2.6 \times 10^{-2}$	$9.58 \rightarrow 9.58$

`wasp_full` also took 14093 marching steps to the same $du < 10^{-3}$ where it previously took 14057.

The O III suppression itself, as the ratio of the new ionization fraction to the old at the same radius:

r/R_p	<code>wasp_full</code>	<code>lower_profile</code>	<code>mol_metals</code>
1.05	4.1×10^{-5}	1.2×10^{-6}	4.1×10^{-5}
1.10	2.0×10^{-4}	3.0×10^{-5}	1.1×10^{-5}
1.20	1.9×10^{-3}	3.7×10^{-4}	4.4×10^{-6}
1.50	5.2×10^{-1}	8.2×10^{-3}	2.4×10^{-4}
2.00	—	6.2×10^{-2}	3.3×10^{-3}
3.00	—	9.5×10^{-1}	1.7×10^{-2}

The movement is consistent with section 103. and is the same mechanism. That section's HD 209458 b run gave $2.2 \times 10^{-5} / 2.1 \times 10^{-4} / 1.8 \times 10^{-3} / 2.0 \times 10^{-2}$ at 1.05 / 1.10 / 1.20 / 1.50 R_p ; `wasp_full` gives $4.1 \times 10^{-5} / 2.0 \times 10^{-4} / 1.9 \times 10^{-3}$ and `lower_profile` $1.2 \times 10^{-6} / 3.0 \times 10^{-5} / 3.7 \times 10^{-4}$ — the same three decades of suppression at the base, decaying outward as $n(\text{H}^0)/n_e$ falls and there is less neutral hydrogen to transfer from. `mol_metals` is the case where the ratio is large and the consequence is not: its base is a shielded molecular column whose O III fraction is 10^{-16} before the change, so the entire hydro moves by 4×10^{-4} .

The knock-on is larger on WASP-121 b than section 103. found on HD 209458 b, and for a reason that is stated rather than absorbed: **the oxygen the reaction removes from O III lands in O II, which is a coolant and O III is not.** On `wasp_full` $n(\text{O I})$ rises by a factor 3.4 and $n(\text{O II})$ by 3.2 near 1.5 R_p , the total cooling rate rises by 16 % there, and the temperature falls by 5 %. The escape rate absorbs it: $\log_{10} \dot{M}$ is unchanged to the two decimals the run reports, on all four cases.

LHS 1140 b: measured, and it does not move

The LHS 1140 b memo's crossings rest on a He I 10830 equivalent width, and the question is whether they are now stale. **They are not: the red-pair equivalent width moves by -4.2×10^{-7} in relative terms.** Measured, not argued, on

a scratch copy of LHS1140b/examples/photochem_profile_base re-run twice through the same path (Do only PP on the stored solution, then EXHALE_transit.py at EXHALE_TRANSIT_RES_HETR=68000), once with the new default and once with cx_02p_H 0: 1.120396 %Å against 1.120396 %Å.

The reaction does what it does everywhere — O III is suppressed by a factor 7.8×10^{-5} at $1.05 R_p$ in that wind too — but there is almost no oxygen there to matter. The photochemical column's own cold trap cuts X_O from 6.062×10^{-4} at its 16.7 bar deep boundary to 8.850×10^{-7} at the 1 μ bar matching level, a factor 685, so O I and O II move by less than 1.3×10^{-3} below $3 R_p$ and the He 2^3S population by 8.7×10^{-7} . $H\alpha$, $H\beta$ and $Ly\alpha$ move by 2×10^{-9} or less.

One number from that exercise is worth keeping, because it is larger than the effect being measured and has nothing to do with it: a Do only PP re-run advances the stored solution by one RK step before post-processing, which by itself moves the stored equivalent width from 1.120891 to 1.120396 %Å, i.e. by 4.4×10^{-4} . That is why the comparison above is new-against-old on two runs that took the same path, and not against the file in the example directory.

Two consequences for stored run directories

Neither is a defect and both are stated so they are not discovered later. Every metals-bearing run directory in the repository was solved with the reaction off, so re-running one now re-solves it with the reaction on unless cx_02p_H 0 is added to its metals.inp. And the pair of runs behind the diffusion measurement (HD209458b/oi1302, oi1302_diffusion) is a default-off pair; the default-on reference for HD 209458 b is HD209458b/oi1302_cx02p_1, which was already solved at cx_02p_H 1.0 and is therefore the run this default reproduces.

Verification

make check was run twice with the golden refresh between them, the order the harness requires. Before the refresh: **3/7 byte-identical** — mol_base_handoff, mol_lyman_werner, mol_diffusion, the three cases with no oxygen — and 4/7 moved, which is the intended change. The goldens of all seven were then re-snapshotted from those same outputs and make check re-run: **7/7 byte-identical, REGRESSION PASS**, so the new reference reproduces deterministically. run_fcheck.sh: **CLEAN** (-01 -fcheck=bounds, do, mem, 1500 bounded HD 209458 b steps with metals on, no run-time check fired).

The Python side was exercised end to end on the LHS 1140 b pair above — EXHALE_transit.py on the _adv profiles of both runs — and needed no change: the transit tool reads the ionization state it is given.

108. In-code oxygen chemistry: the local-kinetics milestone (2026-08-30)

Molecular chemistry: True can build a molecular base but it cannot decide how molecular that base is. The H_2/H partition is either imported through the q_h2_base key of a handoff or taken from the chemical-equilibrium fit q_h2_equilibrium, and neither is photochemistry — which is what item (H) of TO_BE_DONE.md says. The oxygen family is what sets that partition on a cool base: measured on a photochemical model of HD 189733 b it carries 96–99.6% of the net H_2 destruction at 1 μ bar, essentially all of it through $OH + H_2 \rightarrow H_2O + H$ and the photolysis that returns the water to OH.

This section is milestone M2 of docs/a2_oxygen_option_design.md: the chemistry solved as a **local steady state**, with the FUV bands, inside the cell-by-cell equilibrium solve. There is no vertical transport — that is M3, and the design says in advance that a local steady state is the wrong physics at the HD 189733 b base, where $\tau_{\text{chem}}/\tau_{\text{adv}}$ is of order unity. M2 is built anyway because it separates the chemistry from the transport: everything below is a statement about the chemistry alone.

Oxygen chemistry: True	# default off
Stellar FUV B1 flux [erg/cm2/s]: <F>	# 912-1201 A at the planet
Stellar FUV B3 flux [erg/cm2/s]: <F>	# 1231-1450 A
Stellar FUV B4 flux [erg/cm2/s]: <F>	# 1451-2304 A
Stellar Lya flux [erg/cm2/s]: <F>	# the fourth band is the Lya line

The network, and where every number in it comes from

The reaction set is the audited one: 01 $\text{OH} + \text{H}_2 \leftrightarrow \text{H}_2\text{O} + \text{H}$ (Baulch et al. 2005, p. 1029), 02 $\text{O} + \text{H}_2 \leftrightarrow \text{OH} + \text{H}$ (Baulch et al. 2005, p. 804), 06 $\text{O}(^1\text{D}) + \text{H}_2 \rightarrow \text{OH} + \text{H}$ (Atkinson et al. 2004, IUPAC data sheet I.A2.18), and the four photolysis channels 03/04/05 (H_2O) and 07 (OH). Five further reactions were transcribed and **excluded on measurement**, at 7.8×10^{-9} to 2.8×10^{-5} of the dominant channel. The audit is docs/a2_reaction_audit.md; the coefficients are src/modules/lower_atmosphere/oxygen_rates.f90, which M1 deliberately left out of SRC in the Makefile and which M2 puts in.

No reverse rate is transcribed. 01 and 02 run close to cancellation on a hot base — the HD 209458 b gross rates are three decades above the net — so an independently transcribed reverse gives an arbitrary net rather than a small error, which is exactly the defect the R21/R22 note in mol_rates.f90 records for the He charge-exchange pair. Both directions come from rate_from_detailed_balance against the module's NIST-JANAF Shomate table. That makes detailed balance exact by construction and makes the hot limit reduce to chemical equilibrium, which is what gate G3 below tests.

$\text{O}(^1\text{D})$ is closed by a local steady state, and that is the accurate treatment rather than a shortcut. It has exactly one sink in the audited set, 06, and a chemical lifetime of 1.6×10^{-3} s at the HD 189733 b base — shorter than every other time scale in the problem by many decades. So the flux through 06 equals the $\text{O}(^1\text{D})$ production rate whatever the H_2 density is, and 04 ($\text{H}_2\text{O} + h\nu \rightarrow \text{H}_2 + \text{O}(^1\text{D})$) and 06 enter the rows as one combined channel. Their net effect on the H_2 budget cancels exactly, which the measured budget of the design's section 2.2 shows independently as -7.8% against $+7.8\%$.

CO is an oxygen reservoir at its own equilibrium

CO locks 45–46% of the oxygen at every measured level of every arm, so a network that gave the whole oxygen abundance to the water family would over-supply the OH cycle by about a factor two. Decision D4 carries it, frozen chemically, at a chemical-equilibrium C/O partition: the physical root of $(n_{\text{C}} - n_{\text{CO}})(n_{\text{O}} - n_{\text{CO}}) = K_{\text{c}}(T)n_{\text{CO}}$ with K_{c} from the same Shomate table. The C=O bond is 1076.4 kJ/mol against 498.8 for the OH bond of water, so CO survives to temperatures at which the water family is long gone: measured with solar C/H and O/H it holds the whole of the

carbon below 3000 K, is 62% associated at 3250 K and 0.06% at 4000 K. That turnover sits above the $\text{H}_2 \rightarrow \text{H}$ front and below the wind, which is where the reservoir has to disappear if the carbon and oxygen of the wind are to be atomic. Atomic carbon was added to the thermodynamic table for it, checked against the NIST-JANAF gas-phase entry: $\Delta_f H(298.15) = 716.672$ kJ/mol against the tabulated 716.68, $S(298.15) = 158.099$ J/mol/K against 158.100.

The Shomate evaluation temperature is now clamped to the tabulated range, and this was a real defect rather than a precaution. A Shomate fit is a quartic in $T/1000$; at the 10^6 K an unconverged cold-start wind passes through, its t^4 term reaches 10^9 kJ/mol. Measured before the clamp, the table kept CO **fully associated at 9×10^5 K** — so the whole carbon inventory and 55% of the oxygen were held out of the wind’s atomic coolants, and the HD 189733 b test run heated to 7.9×10^5 K where the same run without the option reached 2.9×10^4 K. Clamped, K_c above 6000 K is held at its 6000 K value, $1.09 \times 10^{16} \text{ cm}^{-3}$, which says “fully dissociated” and goes on saying it.

The FUV bands

The photon energy grid starts at the 13.6 eV H I edge and a power-law SED below it is an extrapolation of an XUV fit, so the FUV arrives as bands with named fluxes, on the pattern `lyman_werner.f90` already sets. The band edges are not chosen: the H_2O branching ratios fix them, and the three-body channel $\text{H}_2\text{O} + h\nu \rightarrow \text{O} + \text{H} + \text{H}$ opens only in the interval that contains $\text{Ly}\alpha$, where it is 12%. Decision D1 gives the three continuum bands separate keys rather than one flux and an assumed shape, because the single-key form is weakest exactly on the line-dominated FUV of an M dwarf.

band	interval	$\sigma(\text{H}_2\text{O})$	$\sigma(\text{OH})$	$\langle h\nu \rangle$	$\langle E \rangle(\text{H}_2\text{O})$
B1	912–1201 Å	8.513e-18	4.744e-18	11.735 eV	12.053 eV
B2	$\text{Ly}\alpha$	1.528e-17	4.577e-18	10.199 eV	10.199 eV
B3	1231–1450 Å	4.404e-18	1.410e-18	9.249 eV	9.551 eV
B4	1451–2304 Å	1.242e-18	6.990e-19	6.604 eV	7.495 eV

The heating uses $\langle h\nu \rangle$ and not $\langle E \rangle$. The two are the correct means in opposite limits: $\langle E \rangle$ in an optically thin column, where the absorbed photons are drawn with weight $\sigma(\lambda)$, and $\langle h\nu \rangle$ in a saturated one, where every photon of the band is absorbed. This model applies one transmission to the whole band, so in the saturated limit it says all N_b photons are absorbed; charging each of them $\langle E \rangle$ would deposit **more energy than the band carries** — 13.5% more on B4. With $\langle h\nu \rangle$ the absorbed energy is $F_b(1 - e^{-\tau_b})$ and cannot exceed the incident flux whatever the state of the gas, which is what gate G4 asks of the ledger. The price is in the thin limit, where the heating is then low by 2.7–13.5% for H_2O and low for OH except on B1, where it is 3.1% high.

The rate a cell sees is the mean over the cell, not its face value. Evaluating $j = \sigma N e^{-\tau}$ at one point and multiplying by the cell’s absorbers is a rectangle rule in τ : exact only while the cell is thin, and one-signed low otherwise. Measured on the HD 189733 b test run with that form, the column sum of the absorbed photons fell **30% below** the closed form $N_b(1 - e^{-\tau})$ in the $\text{Ly}\alpha$ band, whose optical depth crosses unity inside a single cell. The rate is therefore built from the depth at the cell’s star-ward face and the cell’s own depth, $j = \sigma N_b e^{-\tau_{\text{out}}}(1 - e^{-\Delta\tau})/\Delta\tau$, which reduces to the rectangle rule as $\Delta\tau \rightarrow 0$ and reproduces the closed form exactly at any grid spacing.

The band average is the weakest part of the treatment, and its size is measured rather than asserted. Against the real HD 189733 b flux at the planet — the spectrum the reference photochemical arm was run with, weighted properly

— $\sigma_b/\sigma_b(\text{flat})$ is 0.958 (B1), 1.349 (B3) and **0.219** (B4) for H_2O , and 0.839 / 1.070 / **0.164** for OH. B1 is good to 4–16% and B3 to about 35%; B4 is off by a factor 4.6–6.1, because its cross section falls three decades across the band and almost all of a cool star’s B4 energy sits where the molecule barely absorbs.

The $\text{Ly}\alpha$ band is an upper bound. B2 takes the incident stellar $\text{Ly}\alpha$ flux and attenuates it by the H_2O and OH continuum only; the H I resonance scattering that decides how much $\text{Ly}\alpha$ reaches a molecular base is not applied, and neither is the solved field of `lya_rt.f90` that section 2.6 of the design names for it. On HD 189733 b that band carries $1.50 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$ against 1.25×10^3 in B1, so it is not a small term.

One oxygen, one hydrogen

With the option on the OI column means **free atomic neutral oxygen** and the CI column the carbon not locked in CO, so the element totals are $n_{\text{O,tot}} = n(\text{OI}) + n(\text{OII}) + n(\text{OIII}) + n(\text{OH}) + n(\text{H}_2\text{O}) + n(\text{CO})$ and $n_{\text{C,tot}} = n(\text{CI}) + n(\text{CII}) + n(\text{CIII}) + n(\text{CO})$. The species table gained `bsp_nO` and `bsp_nC` to count them, the oxygen counterpart of the `bsp_nH` that already stops the hydrogen leaking; without them the closure is not closed and half the element disappears one sweep at a time. `bsp_mass` for the new carriers is the H mass the code uses plus the `melem_A` the metal block already assigns to the nucleus (OH 16.999, H_2O 17.999, CO 28.010), so the same nucleus weighs the same whether it is counted through a metal column or through a molecule.

The hydrogen closure gained the H nuclei the carriers hold — OH one, H_2O two — and that had to go into three places and not one: the residual’s atomic-H expression, the physical-simplex test, and `clamp_fractions_to_element_budget`.

Leaving it out of the last two was measured and it was not small: a clamped root then put more H into the molecules than the cell had, the extraction floored the remainder at zero, and `src/utls/element_budget.py` reported every element off its reservoir by 4.4×10^{-4} at the worst cell. With the three places consistent the same run closes at 3.7×10^{-10} .

Refusals

combination	verdict
without Molecular chemistry	refused: the cycle acts on H_2
without helium	refused, inheriting the molecular network’s rule
zero oxygen abundance	refused: no element to partition
<code>q_H2_base</code> in <code>base.inp</code>	refused: the option computes that partition
lower-atmosphere profile	accepted; the run says which owns what
<code>He_diffusion</code>	refused: the element operator has no molecular oxygen (M3)
no FUV band flux at all	warned: the result is chemical equilibrium

What a run writes

`Ion_species.txt` gains OH H_2O CO after the molecular columns. `Oxygen_chemistry.txt` carries the solved partition, the $\text{O}(^1D)$ steady state and the chemical against the advection time scale with their Damköhler ratio — a **required** output, not a diagnostic, because a local steady state is right where $\text{Da} \gg 1$ and wrong where it is not. `FUV_bands.txt` carries the columns, the four band optical depths, the band-resolved rates and the energy ledger. `Heating_breakdown.txt`

gains column 17. EXHALE_resolved.out gains oxygen_chemistry, oxygen_reaction_set, the four band fluxes and oxygen_base_partition. The _adv writer now says in the file itself that the molecular and oxygen columns are **not** advection-corrected.

Gates

Measured on three runs: the hot-Uranus molecular gate with the option on (oxon) and with every band flux at zero (oxnofuv), and HD 189733 b with the band fluxes taken from the same stellar spectrum the reference photochemical arm was run with (B1 1248.5, B2 1.4957e4, B3 1374.1, B4 4.6295e4, LW 600.1 $\text{erg cm}^{-2} \text{s}^{-1}$). All three are bounded 12000-step relaxation snapshots from cold, matched to the step cap the mol_metals regression case already uses; **none of them is a converged wind**, and no number below is quoted as one.

G1 / G8 — the option off is byte-identical. make check: 7/7 **byte-identical, REGRESSION PASS** (wasp_full, wasp_he23off, mol_base_handoff, mol_metals, mol_lyman_werner, mol_diffusion, lower_profile), with no golden refreshed. run_fcheck.sh: **CLEAN**, run twice — once as its own script writes it, on the bounded HD 209458 b case that never touches the new code, and once on an oxygen-chemistry case, which is where it caught the x_root_best overflow below.

G2 — the atomic limit reproduces the metals-only result. Above the molecular front the oxygen carriers vanish and the element is back in its ion stages: on HD 189733 b the molecular fraction of the oxygen is $2.2\text{e-}7$ at $1.1 R_p$ (7327 K) and $4.0\text{e-}14$ at $1.5 R_p$ (33743 K), with O I + O II + O III holding the whole element there. No reaction of the set makes or destroys an oxygen ion, so the stage balance itself is untouched, which is why wasp_full and wasp_he23off – atomic, metals-on – are byte-identical under G1.

G3 — the zero-photolysis limit is chemical equilibrium. Two halves. Algebraically, putting the two carrier rows on the equilibrium partition of oxygen_chemical_equilibrium_fractions leaves residuals of $2\text{e-}16$ to $7\text{e-}15$ over 864-3000 K ($6\text{e-}9$ at 300 K, where the OH row is a difference of two terms eleven decades apart); that is src/tests/a2_m2/a2_m2_kinetics_check. End to end, on a run with every band flux at zero, the solved partition matches the chemical equilibrium of each cell's own ($T, n_{\text{H}_2}, n_{\text{H}^0}$) to **4.8e-6**, with 9 of 494 cells above $1\text{e-}6$; that is a2_m2_g3_run_check. The check excludes the 10 cells in which one *direction* of the dominant pair runs slower than 10^{-10}s^{-1} – a local steady state is only determined where something runs both ways, and where it is not, the partition is a question for the transport.

G4 — the band energy ledger closes. Band by band, the photons the grid absorbs match the closed-form $N_b(1 - \exp(-\tau))$ to $2.3\text{e-}8$ (hot Uranus) and $9.8\text{e-}6$ (HD 189733 b); the absorbed energy is exactly the deposited heat plus the bond energy; and it never exceeds the incident flux – B1 on the hot-Uranus run saturates at exactly 12.000 of $12.000 \text{erg cm}^{-2} \text{s}^{-1}$.

The overlap half of G4 fires on both runs, and that is the finding, not a failure of the check. Over 912-1110 Å the same photons are absorbed by H2 in the Lyman-Werner lines and by H2O and OH in a continuum, and the two are modelled as independent beams. Measured, they take $7.76 + 12.00 = 19.8$ out of a B1 flux of 12.0 on the hot Uranus, and $352.7 + 1128.2 = 1480.9$ out of 1248.5 on HD 189733 b. So the two treatments together **do** over-count the energy of the overlap, by up to 60% of the band, and the run says so at the end rather than quietly over-heating. Making one absorber compete with the other there is not part of M2.

G5 — element budgets close in every cell. `src/utls/element_budget.py`, taught the new carriers, closes H, He, C and O at $\leq 1.1 \times 10^{-12}$ relative on all three runs, against its $1e-8$ tolerance.

G6 — the IC round trip. An A2 run restarted from its own state reproduces it: the element budgets close at $1.0e-12$ and one marching step moves the most reactive species by at most $1.4e-2$, which is that step and not a loss on reload. An A2 run restarted from a state written **before** the option existed takes the seeding branch – OH, H₂O and CO from the chemical equilibrium of the loaded (T, H₂/H) and the loaded oxygen and carbon totals, printed and not silent – and closes its budgets at $1.2e-11$. Zero is a valid root of the water cycle and `hybrd1` is known to be bistable from a zero molecular seed, so starting from zero there was not an option.

The HD 189733 b base partition, which M2 exists to measure. The option returns $x_{\text{H}_2} = 2 \ n_{\text{H}_2} / n_{\text{H}} = 0.773$ at the base cell, against 0.986 for the same configuration with the option off and 0.910 for the stored photochemical arm. The A/B pass band of section 6.2 is $0.847 \leq x_{\text{H}_2} \leq 0.947$, so **it misses**, which is what the milestone table says M2 is expected to do; it stays inside the outright-failure band ($0.023 \leq 1 - x_{\text{H}_2} \leq 0.355$, and $1 - x_{\text{H}_2} = 0.227$ here). Two things make this a measurement of the chemistry and not the A/B gate, which is M4's: the snapshot is not converged, and its base sits at 1410 K rather than the 864 K of the reference, where the thermal channel $\text{H}_2 + \text{M} \rightarrow \text{H} + \text{H} + \text{M}$ carries part of the partition that the oxygen cycle carries at 864 K.

What this milestone found, and what it costs

The option removes coolants it does not replace. Where the oxygen is in H₂O and CO, the [O I] 63/145/44 μm fine-structure lines and the C I/C II line cooling correctly stop — those atoms are not there — but the code carries no H₂O or CO infrared bands to put in their place, and H₃⁺ is its only molecular coolant. Measured on the hot-Uranus molecular gate, the layer just above the base cools $15\times$ more slowly with the option on than with it off at comparable heating, and the run that follows is not a converged wind.

That is item (G) of `TO_BE_DONE.md` arriving through the composition instead of through the temperature, and section 9 of the design says so in advance: A2 is the composition of the molecular layer and (G) is its energy, and they feed on each other. It is the reason the option stays default off, and it is reported at the end of every oxygen-chemistry run — how much of the oxygen and of the carbon is molecular, at which radius and temperature, and a note that those nuclei no longer cool — so that a user cannot read the resulting temperature as a prediction.

Two defects fixed on the way, both outside the new physics. The hydrogen budget had to learn the carriers in three places and not one — the residual, the physical-simplex test and the clamp — and with the last two left out every element missed its reservoir by 4.4×10^{-4} at the worst cell. And `x_root_best` in `ionization_equilibrium.f90`, the thread-private array holding a cell's best admissible root, was dimensioned $n_{\text{x_max}} = 8 + 2*n_{\text{melem}} = 28$, the largest `N_eq` before the oxygen carriers existed. With every option on `N_eq` is 30, so the copy ran two elements past the end of a stack array. The optimized build did not notice; the bounds-checked build stopped on it at once. `n_x_max` is now $10 + 2*n_{\text{melem}}$, and `run_fcheck.sh` was therefore run against an oxygen-chemistry case as well as against the off-path case its own script uses.

What M2 deliberately does not do. No vertical transport (M3), no A/B against the stored photochemical arm (M4), no regression case and no transit guard (M5). `backup/regression/mol_oxygen` is M5's deliverable and is not added here, so the matrix is unchanged at seven cases.

109. In-code oxygen chemistry: one beam for the shared FUV band, and vertical transport of the molecular carriers (2026-08-30)

Section 108 built the oxygen chemistry as a **local steady state** and said in advance that this is the wrong physics at the base the option exists for. The measurement behind that is the P4 block of docs/oxygen_chemistry_new_plan.md: at the HD 189733 b 1 μ bar level $\tau_{\text{chem}}(\text{H}_2)/\tau_{\text{adv}}$ is 0.20 to 1.67 on H/ν and 16.4 pressure-matched at EXHALE's own base cell, against 10^{-3} on HD 209458 b. The one planet whose base partition the oxygen photochemistry decides is the one planet whose base partition is not local. This section is milestone M3 of docs/a2_oxygen_option_design.md: the transport.

It also closes a bookkeeping defect section 108 measured and left open — the 912–1110 Å band was counted twice — and that comes first, because the energy budget of everything below stands on it.

The 912–1110 Å band had two owners; now it has one

Over 912–1110 Å the same photons are absorbed by H_2 in the Lyman and Werner lines and, once the oxygen chemistry carries them, by H_2O and OH in continua. That is physics. What was not physics is that the interval's ENERGY entered the model twice, in two independent ways:

- **on the way in.** The band was supplied once as Stellar LW flux (912–1110 Å, for H_2) and again inside Stellar FUV B1 flux, whose interval was 912–1201 Å. On the solar spectrum the HD 209458 b example uses, the 912–1201 Å integral is $481.0 \text{ erg cm}^{-2} \text{ s}^{-1}$ and its 912–1110 Å part is 343.0 — which is exactly the number the same example entered as its Lyman-Werner flux. The overlap was 71% of B1.
- **on the way down.** Each absorber attenuated a private copy of the beam. H_2 did not see the $\text{H}_2\text{O}/\text{OH}$ continuum above it, and H_2O and OH did not see the photons the H_2 lines had already taken.

Measured on the section-108 runs, the two together took $7.76 + 12.00 = 19.8$ out of a B1 flux of 12.0 on the hot Uranus and $352.7 + 1128.2 = 1480.9$ out of 1248.5 on HD 189733 b: the energy of the overlap over-counted by up to 60% of the band. The run printed that and stopped there.

The fix is one interval, one flux, one beam, and it is both halves at once. The band list becomes five, with 1110 Å — where Draine & Bertoldi (1996) end the Lyman-Werner system (their footnote 4) — as a band edge:

band	interval	key	absorbers
LW	912–1110 Å	Stellar LW flux	H_2 (lines), H_2O , OH (continua)
B1	1110–1201 Å	Stellar FUV B1 flux	H_2O , OH
B2	$\text{Ly}\alpha$ 1215.67 Å	Stellar Lya flux	H_2O , OH
B3	1231–1450 Å	Stellar FUV B3 flux	H_2O , OH
B4	1451–2304 Å	Stellar FUV B4 flux	H_2O , OH

so the intervals partition the FUV and no flux is entered twice. Over the LW band the three absorbers then share one beam, whose transmission to a face is

$$T = (1 - A)e^{-\tau_c}, \quad A(r) = \int_r^{\text{top}} \sigma_{\text{lw,pump}} f_{\text{shield}}(N_{\text{H}_2}) n_{\text{H}_2} dr', \quad (37)$$

with τ_c the $\text{H}_2\text{O} + \text{OH}$ continuum depth and A the fraction of the band the H_2 lines have removed. $\sigma_{\text{LW,pump}} = \sigma_{\text{LW}} / \langle p_{\text{diss}} \rangle$ is DB96's own dissociation cross section divided by their dissociation probability per pump, because the 86.5% of pumps that fluoresce back also take a photon out of the band. Across one cell the identity

$$T_{\text{out}} - T_{\text{in}} = e^{-\tau_{\text{out}}} [(1 - A_{\text{out}})(1 - e^{-\Delta\tau}) + \Delta A e^{-\Delta\tau}] \quad (38)$$

is exact, so the photons a cell removes split into a continuum share and a line share with nothing left over. Each side costs exactly one factor and no approximation:

- the H_2 rate is multiplied by $e^{-\tau_c}$ at the cell's own depth. That is the continuum term of **DB96 eq. (40) restored**: it was identically 1 before this option existed, because EXHALE has no dust and its metals are atomic, and it is not 1 now that a water layer sits above the H_2 ;
- the H_2O and OH rates in the LW band are multiplied by $(1 - A)$ at the cell's star-ward face.

Both factors are 1 in every other band and in every run that does not carry the other absorber, so **a molecular run without the oxygen chemistry is unchanged** — which the `mol_lyman_werner` regression case confirms bit for bit. The band constants were re-measured from the same Photochem cross-section files section 108 used, over the new intervals:

band	interval	$\sigma(\text{H}_2\text{O})$	$\sigma(\text{OH})$	$\langle h\nu \rangle$
LW	912–1110 Å	9.484e-18	4.434e-18	12.2635 eV
B1	1110–1201 Å	6.666e-18	5.333e-18	10.7299 eV

The split is informative in its own right: nearly all of the old band's spectral-shape sensitivity sat in its 912–1110 Å half, and the two halves move in OPPOSITE directions. Replacing the flat F_λ by a 3000 K blackbody changes σ_b by –60.0% (H_2O) and **+89.3%** (OH) on LW, against +1.0% and –23.4% on the new B1. The merged band hid that by averaging two opposite trends, so the split is more honest as well as being what the photon budget requires.

The LW band's $\langle h\nu \rangle$ is not restated here: it is taken from `lyman_werner.f90`'s own `e_lw_photon_erg`, because a shared beam has one photon count. The same integral measured independently gives $1.9648327 \times 10^{-11}$ erg, i.e. the two agree to the six digits that module carries.

Gate G4 now closes in its strongest form. Band by band, the photons the grid absorbs match the closed form $N_b(1 - (1-A) \exp(-\tau))$ to $\leq 3.4 \times 10^{-15}$ — round-off — with the H_2 pump photons inside the sum. On the HD 209458 b example the LW band's absorbers take $147.9 (\text{H}_2) + 123.9 (\text{continuum}) = 271.8 \text{ erg cm}^{-2} \text{ s}^{-1}$ out of an incident 343.0, i.e. under the incident flux rather than 60% over it. The overlap warning of section 108 no longer fires and has been replaced by a shared-beam block in `output/FUV_bands.txt`.

What the model does not bound, and says so. A is an equivalent width implied by a fit to a RATE; DB96 never asked it to stay below 1. Where it reaches 1 the lines have eaten the band, which is outside the fit's range ($10^{14} < N_{\text{H}_2} < 3 \times 10^{21} \text{ cm}^{-2}$). The transmission is floored at zero and the run reports the largest A it reached (0.506 on the HD 209458 b example, inside the range).

Users must re-split any Stellar FUV B1 flux measured over 912–1201 Å. The worked examples were updated: `examples/18_oxygen_chemistry` goes from B1 = 481.0 to B1 = 137.9 with Stellar LW flux: 343.0 unchanged, and the HD 189733 b recipe of `README_HOWTO.md` from B1 = 1248.5 to LW = 600.1, B1 = 648.4.

Vertical transport of the molecular carriers

Oxygen transport: True — the default whenever the oxygen chemistry is on — solves

$$\frac{\partial n_i}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 (n_i v + \Phi_i) \right] = P_i - L_i, \quad i = \text{H}_2, \text{OH}, \text{H}_2\text{O}, \text{CO}, \quad (39)$$

$$\Phi_i = -n_{\text{tot}}(D_i + K_{zz}) \frac{\partial f_i}{\partial r} - n_i D_i \frac{(m_i - \bar{m})g}{kT}, \quad (40)$$

as one backward-Euler step per hydro step, block-tridiagonal in space with 4×4 blocks (the transport is diagonal in species, the chemistry dense within a cell), Newton with a backtracking line search. The module is `src/modules/lower_atmosphere/diffusive_photochemistry.f90` (decision D3: a new module, the element operator untouched); it sits in the operator-split slot the element diffusion occupies, after the hydro update and before the ionization solve. Oxygen transport: False restores the local steady state of section 108.

The chemistry is not rewritten. $P_i - L_i$ comes from calling the very rows the local equilibrium solve uses — `mol_heh_rows` row 4 for H_2 and `oxygen_carrier_rows` for OH and H_2O — at the trial densities, with the background frozen; the Jacobian is a forward difference of those same calls. A rate that changes there changes here, and the two cannot drift apart the way two transcriptions of one coefficient do.

What is transported and what is not. H_2 , OH, H_2O and CO. NOT H_2^+ , H_3^+ and HeH^+ : measured over the cells that hold H_2 on the HD 209458 b molecular example, the three of them carry at most 1.2×10^{-9} of the H nuclei, so leaving them local is a closure statement at that level rather than a leak — the H-nucleus budget is closed against the ELEMENT total, not against a sum of transported species. NOT $\text{O}(^1\text{D})$, for the reason section 108 gives. All four transported species are neutral, so the ambipolar field of the design's eq. (2) is not neglected, it is absent.

The diffusion coefficients are the rigid-sphere pairs of binary_element_diffusion under Blanc's law, reused rather than restated. The ion-neutral polarization enhancement is left out, and that is a measurement rather than a preference: over the cells that hold H_2 the electron fraction per H nucleus is at most 1.0×10^{-7} (3.1×10^{-11} at the base cell), so the ion pairs carry that share of the friction sum; and for H_2O an induced-dipole coefficient would be the wrong one anyway, since a permanent dipole of 1.85 D runs the ion-molecule capture and the locked-dipole treatment that needs is not in this code. K_{zz} is read from `kzz_cell` and nowhere else — the module defines no eddy coefficient and no key for one — and a run with `kzz_cell` zero everywhere is told at startup that its transport is pure molecular diffusion, which on a lower atmosphere is the wrong limit.

Boundary conditions are decision D6, and they are what make the gate a test. The element reservoirs stay where they always were, imposed through `mem_ab`, the base density and the EOS. The partition among each element's carriers is imposed nowhere: the base face carries zero diffusive flux and the advective inflow there carries the base cell's own partition. Had the base H_2 fraction been pinned from the same handoff the A/B gate compares against, the gate would measure nothing.

What this operator is, stated plainly. Each cell's element totals are unchanged by the step; only the partition among an element's carriers moves. So element conservation is exact by construction rather than by cancellation, and ρ is untouched, because `bsp_mass` of a carrier is exactly the sum of its nuclei's masses. It also fixes the approximation: a water molecule that diffuses out of a cell really takes its oxygen with it, and here the oxygen stays and reappears as O I. That is the same statement EXHALE already makes about every metal when the trace-metal diffusion arm is off, and

it is why this operator transports SPECIATION and not elements. For hydrogen the distinction does not arise — the H total is the element, so transporting the partition $2 n_{\text{H}_2}/n_{\text{H}}$ IS transporting what the gate measures.

Three defects the transport found, each stated where it was fixed

The element headroom has to be the element TOTAL, not its neutral stage. Transport moves nuclei, and the very next ionization sweep re-solves which stage each sits in; limiting the carriers to what the neutral stage happens to hold caps them wherever the element happens to be ionized. Measured on the HD 209458 b molecular gate, that clamped 150–240 of 503 cells by up to 2% every step. Each cell’s element totals are now held exactly across the step and the remainder is shared out over the element’s other species in proportion to what they already held, which cannot drive a stage negative and leaves the ionization fraction where the sweep will find it.

Transported CO is not the same object as chemically-equilibrium CO, and it needed a thermal ceiling. Decision D4 makes CO chemically inert. In the local solve that meant “at its own $\text{CO} \leftrightarrow \text{C} + \text{O}$ equilibrium”, which dissociates it above about 4000 K. Transported, inert means indestructible — and it was: measured on HD 189733 b, the wind carried CO out to $1.67 R_p$ and 2×10^4 K, where it held **55% of the oxygen**, against 1.2×10^{-13} for the same run without transport. That switches off the O I and C II cooling of the whole wind rather than of the molecular layer, and it broke the carrier Newton, which reached its iteration cap with a relative residual of 1 in the cell where the runaway had emptied the free oxygen. The transported CO is now capped at the chemical equilibrium of its own (n, T) : a one-sided constraint, inactive through the molecular layer, exact far above the turnover, and an over-suppression of quenched CO in the narrow interval around 3000–4000 K where the dissociation is neither fast nor negligible. The audited set has no CO rate to do better with, and the code says so at the site. The number of cells the ceiling touches is written to `output/Oxygen_chemistry.txt`.

The finite-difference step of the chemistry Jacobian needs a floor tied to the element. Every source term is built from densities of order the element total, so perturbing a carrier that sits at 10^{-30} of its element by 10^{-6} of *itself* returns round-off for a derivative. Floored at 10^{-12} of the element the carrier belongs to, the Newton converges in 7–16 iterations to 4×10^{-13} where it had been hitting its cap of 30.

Two things the run now writes because they answer a question that was being inferred

`output/Oxygen_chemistry.txt` gained the H_2 diffusion coefficient, K_{zz} and the diffusive time of every cell beside the chemical and advective ones, and a trailer with the transport step’s Newton count, residual, limiter count and CO ceiling count.

It also gained the **net H_2 loss budget at the base cell**, decomposed into eleven channels from the same coefficients the residual uses, net of its own reverse where a channel has one. The option exists to make the oxygen cycle set the base partition, and whether it does is a property of the run: on a 2331 K base the thermal channel carries 70–97% of the net and the oxygen cycle a few percent, while on a 864 K base the oxygen family carries 96–99.6%. Every molecular run is hotter at its base than it should be until item (G) exists, so the decomposition is written out instead of being left to be inferred from the partition it produces.

The A/B measurement, and what it says

The gate is the H-nucleus bound fraction $x_{\text{H2}} = 2 \ n_{\text{H2}}/n_{\text{H}}$ at the HD 189733 b base against the stored Photochem arm's 0.910, with a pass band of $0.847 \leq x_{\text{H2}} \leq 0.947$ and an outright-failure band of $0.645 \leq x_{\text{H2}} \leq 0.977$ (design section 6.2). Three 12000-step relaxation snapshots from cold, same configuration, band fluxes from the same stellar spectrum the reference arm was run with; **none of them is a converged wind** (du ends at 51, Mdot is NaN), exactly as for the section-108 runs:

run	base T	x_{H2}	O partition at the base (O / OH / H ₂ O / CO)
Oxygen transport: False (the section-108 limit)	1402.7 K	0.769	0.003 / 0.006 / 0.443 / 0.549
transport, $K_{\text{zz}} = 0$	1214.3 K	0.608	0.026 / 0.025 / 0.478 / 0.471
transport, $K_{\text{zz}} = 1\text{e}9$	1191.3 K	0.581	0.032 / 0.028 / 0.475 / 0.465

The local-kinetics row reproduces section 108's 0.773 (the 0.4% is the re-split FUV bands). **Transport moves the partition away from the target**, and out of the outright-failure band. Four measurements say why, and none of them is a defect of the operator.

1. EXHALE's base cell has no advection at all, and its diffusion is three to four decades slower than its chemistry.

$v = 0$ exactly at the base cell – the lower boundary condition puts it there – and in the first cells above it the flow time $r/|v|$ is 10^8 – 2×10^8 s against $\tau_{\text{chem}}(\text{H}_2) = 178$ – 552 s. The diffusive time of the base cell is 8.0×10^5 s with $K_{\text{zz}} = 0$, i.e. 4500 chemical times, and 634 s with $K_{\text{zz}} = 1\text{e}9$, i.e. 3.6. So in EXHALE's own structure the base partition is a LOCAL quantity. The $\tau_{\text{chem}}/\tau_{\text{adv}} = 0.20$ – 1.67 that motivated the transport was H/v on the PHOTOCHEMICAL model's structure at the same pressure, not on this one. Transport cannot close a gap where transport does not act, and the run now writes the diffusive time beside the chemical and advective ones so that this is read off rather than assumed.

2. What the transport does change is CO, and it changes it the wrong way. CO has no chemistry, so its mixing ratio is set by the column and not by the local equilibrium: at the base it falls from 0.549 of the oxygen – all of the carbon, the equilibrium value – to 0.471, and the freed oxygen goes into the water family, raising free O by a factor 9 and OH by a factor 5. More OH is more H₂ destruction through $\text{OH} + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{H}$. That mechanism belongs to decision D4, not to the transport operator: an inert reservoir that is transported stops being an equilibrium partition of its element.

3. The base is 330–540 K hotter than the 864 K of the reference, and the run's own H₂ budget says its partition is not oxygen-dominated. The net budget now written to output/Oxygen_chemistry.txt puts the thermal channel at +2.38 and the oxygen cycle at –1.38 of the net at the base cell: two large terms of opposite sign nearly cancelling, which is the hot-base regime the design's section 2.7 places outside A2's validity range, not the 96–99.6% oxygen-dominated regime the reference was computed in. The base is hot because the option removes the [O I] and C I/C II coolants without replacing them, which is item (G) of TO_BE_DONE.md.

4. Nothing is converged, and the slow carriers are not at their transported steady state either. The physical time 12000 steps cover at the base is $12000 \times 0.589 \text{ s} = 7.1 \times 10^3 \text{ s}$, which is 0.9% of the base diffusive time. CO, which relaxes only by transport, is therefore still near its initial condition. Reaching the transported steady state is what relax_photochemical_composition is for – it grows the step geometrically to 10^{12} times the cell scale, as the element relaxation does – and that runs only inside the steady Picard loop, after a JFNK solve that returns $\text{info} = 0$. No oxygen-chemistry configuration in the tree converges that far, so **that path is wired and damped but has not been exercised end to end**, and this section does not claim otherwise.

What this leaves. The A/B gate of milestone M4 is not answerable from a bounded marching run. It needs a converged wind whose base sits near 864 K, and that needs item (G) — the H₂O and CO infrared bands — before the oxygen chemistry is asked what partition it produces. The transport operator is in place, its conservation is exact by construction and its solver is measured; the term that is missing is thermal, not advective.

Gates

G1 / G8 — the option off is byte-identical. make check: 7/7 byte-identical, REGRESSION PASS, no golden refreshed. That covers the FUV band restructure too: mol_lyman_werner carries a Lyman-Werner flux and no oxygen chemistry, so its H₂ rate sees $e^{-\tau_c}$ with τ_c identically zero.

G4 — the band energy ledger, with three absorbers on the shared band. The photons the grid absorbs match $N_b(1 - (1-A) \exp(-\tau))$ to $\leq 3.4 \times 10^{-15}$ band by band, H₂'s pump photons included in the sum, on every run measured. The absorbed energy of the shared band stays under its incident flux (271.8 of 343.0 on the HD 209458 b example) instead of exceeding it by 60%.

G5 — element budgets close in every cell, with the transport on. src/utils/element_budget.py closes H, He, C, N and O at $\leq 2.1 \times 10^{-12}$ on the three HD 189733 b runs and at 1.0×10^{-12} on the HD 209458 b example. It closes by construction rather than by cancellation: the operator holds each cell's element totals across the step.

G6 — the IC round trip. An A2 transport run restarted from its own state reproduces it: the largest change over one marching step is 3.4×10^{-2} and is the hydro's own, since every species moves by the same factor. A run restarted from a state written before the option existed takes the seeding branch and closes its budgets at 8.7×10^{-13} . The imposition of the transported partition is bg_ready .or. do_load_IC for exactly this reason: on a cold start the first sweep initializes the carriers, on a restart it must not throw them away.

G10 — the transported carriers conserve elements. Exact by construction, as above.

run_fcheck.sh — CLEAN, and it was run twice, as section 108's was. Once as its own script writes it, on the bounded HD 209458 b case that never touches the new code; and once on an oxygen-chemistry case with the transport on — HD 189733 b with $K_{zz} = 10^9$, 400 bounded steps under -01 -fcheck=bounds,do,mem — which is the path that matters, because it is where section 108's x_root_best overflow was caught. No trap fired.

Two defects fixed in passing, both outside the new physics

The EOS particle count disagreed with itself. get_species_densities in composition.f90 — the routine the marching loop calls every step to set n_tot and n_e — did not pass the oxygen carriers to calc_ntot, while ioniz_eq did. OH, H₂O and CO are gas particles, and their oxygen and carbon have been taken OUT of the metal ion columns by the ionization solve, so leaving them out does not merely lose 5×10^{-4} of the particle count: it makes the temperature $T = p/((n_{\text{tot}} + n_e)k)$ depend on which of the two routines last wrote it. energy_semi_implicit.f90 had the same omission. Both now pass them. They are neutral, so calc_ne is unaffected, and a run without the oxygen chemistry is bit-identical.

output/FUV_bands.txt had a column header that named four bands while the writer emitted five. Any parser trusting the # columns line would have mis-read the file. The header is now BUILT from the band table, so it cannot

disagree with the loop again.

Defaults and what this does not do

Oxygen chemistry stays default off (decision D9), and Oxygen transport is default on whenever the option is on: a local steady state is the wrong physics at the base the option exists for, so the reduced form is the opt-out and it is named. Oxygen chemistry together with He_diffusion is no longer refused — project_elements scales every H-bearing species and the metal ion columns by the same factor, so an element partly in OH, H₂O or CO keeps its ratio to hydrogen — but He_metal_diffusion still is, because that arm transports each metal element's own mixing ratio and CO carries an oxygen AND a carbon nucleus, so it cannot follow two element factors at once.

Not done: milestone M5's regression case backup/regression/mol_oxygen, the _adv guard and the transit guard; and M4's A/B gate, which the measurement above says is blocked on item (G) rather than on the transport.

110. The molecular layer gets the coolants it was missing: H₂, H₂O and CO infrared bands (2026-08-31)

Item (G) of TO_BE_DONE.md has been open since 2026-08-13: a converged Tier-2 molecular layer radiates itself down to 190–400 K against an equilibrium temperature of 1100–1400 K, and the Base IR field closure of §55 could only reach part of it. Its own closing statement said what was left — “past the front there are no molecular coolants left. Holding that gas needs a continuum infrared coupling to the deep atmosphere (H₂ collision-induced absorption, the H₂O/CH₄/CO bands)”. Section 109 then found the same hole from the other side: the oxygen option moves the oxygen and carbon into H₂O and CO, correctly switches the [O I] and C I/C II line cooling off, and puts nothing in its place, which is why the HD 189733 b base of milestone M3 sits 330–540 K above the reference and M4's A/B gate could not be answered.

Molecular IR bands (default off) supplies the three coolants that matter.

Which three, and one that is deliberately absent

H₂ itself. H₂ has no permitted dipole spectrum, and every cooling channel EXHALE carried skipped it for that reason. Its quadrupole and magnetic dipole lines have transition probabilities of 10⁻¹⁰ to 10⁻⁶ s⁻¹, but H₂ outnumbered every other molecule in this layer by 10³–10⁴, and the two cancel: at $n(\text{H}_2) = 5 \times 10^{13} \text{ cm}^{-3}$ and 1000 K the line sum radiates $1.4 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$, four times the local photoheating quoted in §55 (the pure rotational part alone is 4.8×10^{-8} ; above about 600 K the vibrational bands carry most of it). It is also the only one of the three that exists in a Tier-2 run without the oxygen option, so it is the channel that acts on the configurations item (G) was opened on.

H₂O and CO, the coolants the oxygen option creates and then cannot use. CO holds essentially all the carbon at the base; H₂O is the strongest infrared coolant of a solar-composition H₂ atmosphere below about 2000 K.

H₂–H₂ and H₂–He collision-induced absorption is not a channel, and that is a measurement rather than a scope decision. CIA scales as n^2 . Measured on the CIA tables shipped with Photochem (petitRADTRANS/HITRAN, calibrated here against the known Jovian result that 1 bar is optically thick), the CIA optical depth over a pressure scale height at

the $1\ \mu\text{bar}$ base is 8×10^{-9} on the hot Uranus and 2×10^{-9} on HD 189733 b, peaking at $2.3\ \mu\text{m}$, and unit optical depth is reached only near 0.2–0.4 bar — consistent with Lavvas & Arfaux (2021), who put CIA’s onset at $p > 1$ bar. CIA is what makes the reservoir *below* the base black in the windows between the bands, which is precisely the assumption Base IR field already encodes, and not a local coolant of the modelled domain.

The closure, and the term the H_3^+ version had dropped

For each species the net radiative loss per unit volume is the optically thin, LTE exchange with the diluted field of the lower atmosphere,

$$\Lambda_{\text{net}} = n_X [E_X(T) - W E_X^{\text{abs}}(T; T_0)], \quad (41)$$

with, for a band absorber,

$$E_X(T) = 4\pi \sum_b \sigma_b(T) \int_b B_\nu(T) d\nu, \quad E_X^{\text{abs}}(T; T_0) = 4\pi \sum_b \sigma_b(T) \int_b B_\nu(T_0) d\nu, \quad (42)$$

and, for H_2 , the same quantity assembled transition by transition with the full two-level net factor $1 + \bar{n} - \bar{n} e^{\Delta E/T}$. W is the dilution the H_3^+ closure uses, $\frac{1}{2}[1 - \sqrt{1 - (R_p/r)^2}]$, and it is zero when Base IR field is off.

Keeping the stimulated-emission term is what makes the bracket vanish at $T = T_0$, $W = 1$ — gas buried in a blackbody at its own temperature neither cools nor heats. For the band absorbers the property is automatic: HITRAN line intensities already carry the $1 - e^{-h\nu/kT}$ factor, so σ_b is the coefficient that pairs with B_ν . For a line sum it has to be written down, and h3p_net_cooling_rate had not: its bracket was $1 - \bar{n} e^{E_k/T}$, which in the buried limit leaves a spurious heating of $\bar{n}$ times the emission instead of zero. **That is a physics error and it is fixed here.** It cost 2.2% of the H_3^+ emission and 5 K of equilibrium temperature ($936 \rightarrow 941$ K at $T_0 = 1140$ K, half sky), because $E_k(\nu_2)/T_0$ is large; on a far-infrared band the same omission is an order-unity error, which is why the new module carries it explicitly. No golden moves: base_ir_field is the gate and no regression case sets it.

The single-band form that term was restored INSIDE is itself replaced in §112, later the same day. Collapsing a total emission fit onto one transition and giving it that transition’s Boltzmann factor cannot be right at every temperature, and it is not: the bracket above reaches -1.2×10^{14} at 100 K. Everything this section says about the H_2 , H_2O and CO channels stands unchanged; what changes is the H_3^+ comparison column below and the H_3^+ share of the hot-Uranus numbers further down.

Each channel therefore has its own radiative equilibrium temperature, and it is the fixed point rather than the magnitude of the rate that answers (G):

T_0	H_2O	CO	H_2	(H_3^+)
900 K	705 K	752 K	783 K	780.9 K
1140 K	889 K	918 K	984 K	975.4 K
1183 K	922 K	947 K	1019 K	1010.7 K
1450 K	1120 K	1120 K	1226 K	1231.6 K
1800 K	1369 K	1338 K	1486 K	1517.6 K
2358 K	1749 K	1668 K	1883 K	1938.6 K

(The H_3^+ column is §112's closure. The single-band form this section fixed one term of gave 769 / 941 / 971 / 1150 / 1374 / 1713 K.)

How much rests on the half-sky assumption is measurable: for H_2O at $T_0 = 1183$ K the equilibrium temperature is 726.5 / 922.1 / 1065.0 / 1183.0 K at $W = 0.25/0.5/0.75/1$, and CO 784.7 / 946.5 / 1073.1 / 1183.0 K. At $W = 1$ both are exactly T_0 , the closure's fixed point recovered numerically. Because W multiplies only the absorption, these temperatures are insensitive to first order to any grey escape probability applied to both terms. That is the reason the optically thin limit is enough: it fixes the temperature the layer settles at even where a band is marginally thick, and only the rate of approach is affected.

The data, and three checks against sources that share none of its processing

H_2O and CO band-mean cross sections come from the correlated- k coefficients distributed with Photochem (Wogan et al. 2025, PSJ 6, 256), computed with HELIOS-K (Grimm & Heng 2015, ApJ 808, 182) from HITEMP 2010 (H_2O ; Rothman et al. 2010, JQSRT 111, 2139) and HITEMP 2019 (CO; Li et al. 2015, ApJS 216, 15). The bin mean of a k -distribution is exactly $\sum_g w_g k_g$. Line strengths do not depend on the broadening, only the line shape does, so this mean is pressure independent to better than 2% from 10^{-6} to 1 bar — measured on the tables — and the whole EXHALE domain uses the 10^{-6} bar slice, which is also the lowest tabulated pressure. The H_2 line list is Roueff et al. (2019), A&A 630, A58, table 2 (VizieR J/A+A/630/A58), electric quadrupole plus magnetic dipole, with its own level ladder and statistical weights.

- The Planck-mean H_2O cross section agrees with the independent Planck-mean table `aiolos/inputdata/1H2-160_T400.aiopa` to **within 11%** over 600–2000 K (ratio 0.889–1.111 on the 14 shared grid temperatures). Below 600 K that table is not monotonic in T (9.9×10^{-21} at 100 K, 5.4×10^{-20} at 200 K, 1.5×10^{-20} at 300 K, 7.9×10^{-21} at 500 K, 1.2×10^{-20} at 600 K), so the comparison is not meaningful there; the table built here is monotonic over its whole range and the low-temperature end is covered by the next check.
- Restricted to $\lambda > 9 \mu m$, i.e. to the pure rotational band, the H_2O emission agrees with the optically thin LTE rate of Neufeld & Kaufman (1993), ApJ 418, 263, table 2 to a factor **0.70** (100 K), **0.81** (200 K), **0.90** (400 K), **1.02** (1000 K). The same cut on CO against their table 3 gives **0.81–0.93** over 300–2000 K. At 100 K the CO ratio falls to 0.10 — most likely the eight-point g -quadrature resolving the linear bin mean poorly for CO's sparse rotational ladder, not the coarse long-wavelength binning (97% of the 100 K emission lands in the eighteen 9–250 μm bins). CO radiates 1.6×10^{-20} erg s $^{-1}$ per molecule there against 1.8×10^{-15} for H_2O , so it does not matter.
- The H_2 line sum agrees with the LTE rates of Hollenbach & McKee (1979), ApJS 41, 555, eqs. (6.37) and (6.38) to **1–11%** over 200–2000 K (rotational 1.05–1.11 above 400 K, vibrational 0.85–0.97 above 600 K).

Two of those numbers are the reason a published rotational cooling function was not adopted instead. Above 300 K CO's total emission is four to five orders of magnitude above its rotational part (the 4.7 μm fundamental), and H_2O 's is 3.2 times its rotational part at 1000 K (the 6.3 μm bend and the 2.7 μm stretches). A rotation-only coolant would have missed almost all of it. Neufeld & Kaufman's tables also stop at an optical depth parameter of

$10^{19} \text{ cm}^{-2} (\text{km/s})^{-1}$, a ceiling their §4.2 derives from hydrostatic equilibrium in a self-gravitating cloud; it does not apply to a planetary atmosphere and a base column exceeds it, so the tables would have had to be extrapolated in exactly the variable that carries their physics.

The Fortran path was checked against an independent Python implementation of the same three integrals at 60 temperatures over 100–2000 K: the largest relative difference is 3.7×10^{-3} (H_2O), 2.3×10^{-2} (CO) and 8.4×10^{-5} (H_2) — the 25-point Simpson band quadrature and the log-log table interpolation against a 512-point trapezoid, not a difference of physics. The radiative equilibrium temperatures agree to 0.02 K.

Optically thin, and the run says whether it still is

No escape probability is applied. The justification is measured rather than asserted, in two places the run writes out: `output/Cooling_breakdown.txt` carries the Planck-mean optical depth of the H_2O and CO columns from each cell to the top of the domain and its maximum; `output/FUV_bands.txt` gains the infrared side of the G4 ledger, the column-integrated emitted, absorbed and net power of the three bands against the incident infrared flux $W\sigma T_0^4$ that bounds what an optically thin layer can absorb, with a warning if the bound is crossed. On the 12000-step HD 189733 b oxygen-chemistry run the measured maxima are $\tau(\text{H}_2\text{O}) = 2.5 \times 10^{-3}$ and $\tau(\text{CO}) = 2.8 \times 10^{-3}$, and the absorbed total, $7.5 \times 10^5 \text{ erg cm}^{-2} \text{ s}^{-1}$, sits 73 times below its bound of 5.4×10^7 . The H_2 lines are thinner still: the line-centre opacity of 0–0 S(1) at $17 \mu\text{m}$ gives $\tau = 3 \times 10^{-4}$ over a scale height at $n(\text{H}_2) = 5 \times 10^{13} \text{ cm}^{-3}$ and 1000 K.

What it does to item (G)’s own configuration: the hot-Uranus Tier-2 gate

Four converged solutions, `docs/lower_atmosphere_figs/data_g2` restarted from its own 12000-step snapshot with the documented recipe (Load IC? True, Solver: Newton $5.0\text{e-}2$, Resid tol: $1.0\text{e-}5$, Max steps: 150000), all with Base IR field: True so the pair differs only in Molecular IR bands. This configuration carries no oxygen chemistry, so the only new channel is the H_2 line sum.

The bands-off members reproduce §55’s field-on solutions: 940.5 / 878.6 / 846.6 / 750.3 K against 910.0 / 859.4 / 840.2 / 753.0 K at $r = 1.005/1.02/1.03/1.05$ (metals off), and 1097.0 / 926.1 / 817.9 / 572.2 K against 1098.9 / 928.9 / 820.1 / 573.4 K (metals on). The metals-on agreement is 1–3 K; the metals-off differences of 5–30 K are the H_3^+ stimulated-emission fix, the only other change reaching that branch.

The four solutions below carry the single-band H_3^+ closure, which §112 replaces the same day. The H_2 channel numbers are unaffected — that is what this section is about — but the H_3^+ share of the metals-off arms is not, and §112 re-measures the same four combinations with both closures. Its verdict on this section’s conclusions: they hold.

	metals OFF		metals ON	
	bands off	bands on	bands off	bands on
JFNK iterations	89	21	25	26
info / $\ R\ $	0 / 8.77e-6	0 / 8.39e-6	0 / 2.00e-6	0 / 2.82e-6
base T (cell 1)	1212.30 K	1212.30 K	1213.36 K	1213.36 K
$T(r = 1.005)$	940.5 K	912.1 K	1097.0 K	1044.1 K
$T(r = 1.02)$	878.6 K	857.6 K	926.1 K	887.0 K
$T(r = 1.03)$	846.6 K	831.1 K	817.9 K	801.6 K
$T(r = 1.05)$	750.3 K	721.9 K	572.2 K	588.7 K
$T(r = 1.08)$	420.9 K	345.4 K	215.1 K	236.1 K
coldest cell below 1.3	245.6 K @ 1.0972	225.4 K @ 1.0923	190.4 K @ 1.0842	198.2 K @ 1.0853
$H_2 \rightarrow H$ front	1.08817	1.08532	1.07916	1.07970
$\log_{10} \dot{M}$ reported	10.32	10.31	10.29	10.30
mass-flux spread, $r > 1.2$	2.39e-3	2.38e-3	2.38e-3	2.38e-3

The H_2 channel is first order, and the collapse is not repaired. Both statements are in the same table and they are not in tension. First order: in the metals-on solution the H_2 lines carry **56% of the whole radiative rate** at $r = 1.002$ – 1.005 (cooling there, since $T = 1044$ – 1124 K is above H_2 's radiative equilibrium temperature of about 975 K at that dilution) and essentially all of the net at $r = 1.02$ – 1.05 (heating, since T is below it); the layer's radiative time falls with it, $t_{\text{rad}}/t_{\text{flow}}$ at $r = 1.05$ going from **8.22 to 0.263**, a factor 31. Metals off, where H_3^+ already did that after §55, H_2 carries 11–65% by radius and halves $t_{\text{rad}}/t_{\text{flow}}$ again. Not repaired: the coldest cell moves by +7.8 K (metals on) and –20 K (metals off), and it sits at or beyond the $H_2 \rightarrow H$ front in every run, where there is no H_2 to absorb anything and $t_{\text{rad}}/t_{\text{flow}} = 0.72$ – 1.86 .

The metals-off layer getting *colder* with a net heater added is the same mechanism: the extra heating near the base raises the velocity through the layer by 22% at $r = 1.005$, the specific entropy falls, and the expansion takes the temperature down with it. $\dot{M}$ moves by 0.01 dex either way.

The solution's own zero crossing lands on the predicted equilibrium temperature: in the metals-on solution the H_2 channel changes sign at $r = 1.0122$, where the gas is at 951.7 K and the sky dilution $W = \frac{1}{2}[1 - \sqrt{1 - (R_p/r)^2}]$ is 0.423, and the closure evaluated at that dilution gives 951.5 K. The second crossing, at $r = 1.1239$, $T = 850.6$ K, $W = 0.272$, is predicted at 872.7 K — 2.5%, where the profile crosses more shallowly. Nothing was fitted.

Both bands-on solutions are fixed points of the procedure, by the same test §55 used. Metals off: re-solving returns info = 0 at $\|R\| = 8.6 \times 10^{-6}$ and reproduces T to 0.05% at $r = 1.005$ – 1.08 ($912.10 \rightarrow 912.10$, $857.61 \rightarrow 857.18$, $721.86 \rightarrow 721.55$, $345.35 \rightarrow 344.87$ K); only the coldest cell still moves, $225.4 \rightarrow 213.3$ K, which is the behaviour the bands-off solutions already had. Metals on is exact: info = 0 at $\|R\| = 2.2 \times 10^{-6}$ and every temperature above reproduces to five digits, the coldest cell included ($198.15 \rightarrow 198.15$ K).

A second, purely numerical result: with the bands on the metals-off JFNK converges in **21 outer iterations instead of 89** to the same tolerance.

What it does to the oxygen option: HD 189733 b, where H₂O and CO exist

The A/B is two 12000-step relaxations from cold, HD 189733 b with Molecular chemistry, Oxygen chemistry, Oxygen transport, He_Kzz: 1.0e9, solar metals and the five band fluxes of §109's stellar spectrum, differing only in Molecular IR bands (with Base IR field beside it). **The configuration is a reconstruction, and its bands-off arm is not §109's:** §109 records the band fluxes and the protocol but not the rest of the input file, and the run directories are gone. Rebuilt from the keys it does record, the bands-off arm lands at a base of 760.2 K and $x_{\text{H}_2} = 0.330$ where §109's $K_{zz} = 10^9$ row has 1191.3 K and 0.581, so only the difference between the two columns below is a measurement of this section's change.

	bands off	bands on
base T	760.2 K	829.1 K
$x_{\text{H}_2} = 2n_{\text{H}_2}/n_{\text{H}}$ at the base	0.330	0.483
O partition at the base (O/OH/H ₂ O/CO)	0.161 / 0.048 / 0.369 / 0.423	0.004 / 0.002 / 0.584 / 0.410
max Planck-mean $\tau(\text{H}_2\text{O}), \tau(\text{CO})$	—	2.5e-3, 2.8e-3
$\log_{10} \dot{M}$	NaN	9.23

Both gate quantities move toward the target, by a lot. The base rises 69 K toward the photochemical reference's 864 K and x_{H_2} rises by 46% toward its 0.910. The mechanism is visible in the partition: with the base cooler and the water bands carrying the energy, free atomic oxygen falls from 0.161 of the element to 0.004 and the water rises from 0.369 to 0.584, starving the O1 channel that destroys H₂. **Neither run is converged and the M4 gate is not passed by this pair** — $\dot{M}$ is NaN in the bands-off arm and the bands-on arm's mass flux still varies by a factor 2.5 — these being the same bounded relaxations §109 used, with the same caveat.

A converged A2 wind was attempted and does not exist yet. Both arms were restarted from the 12000-step state with the converged-molecular recipe (Load IC? True, Solver: Newton 5.0e-2, Resid tol: 1.0e-5, Max steps: 150000) and marched a further 13000 steps. du does not descend toward the 5×10^{-2} hand-off: over the last 6000 steps it oscillates between 1.10 and 1.60 (bands off) and between 1.08 and 1.35 (bands on), so neither run reaches the JFNK finish. The bands narrow the oscillation but do not remove it. That is the same non-convergence §109 records for every A2 configuration, and it is a hydrodynamic problem at the base, not a thermal one: it survives the coolant being supplied.

The magnitude is what §109 was missing: at $r = 1.005$ the H₂O and CO bands carry 4.8×10^{-5} and 2.7×10^{-5} erg cm⁻³ s⁻¹ of a total 7.5×10^{-5} — five orders of magnitude above the H₃⁺ and fine-structure channels of the same layer, and above the local photoheating. That is why the base moves at all where §55's closure moved it by 0.1–1.2 K.

Defaults, goldens and what item (G) still holds

Molecular IR bands is default off, and that is an interface decision, not a defence of the old numbers. The incident field these channels exchange with is itself an opt-in model: Base IR field asserts that the unresolved atmosphere below the base is black at these wavelengths and radiates $B_\nu(T_0)$, and EXHALE computes neither. Switching the coolants on by default while the field stayed off would install the emission-only half of the physics, which makes the

known defect *worse* — the measured direction, on the metals-off layer, is a deeper collapse. The house rule that a new physics option defaults off so old goldens reproduce applies for that reason and is recorded rather than relied on. The two keys are documented as a pair and `input_read` warns when only `Molecular IR bands` is set.

Goldens: **the seven pre-existing cases are byte-identical and none was refreshed.** With the key off, `cool_H2`, `cool_H2O` and `cool_CO` are exact zeros added to `cool`, and the three new `Cooling_breakdown.txt` columns are output-only; the H_3^+ fix sits behind `base_ir_field`, which no regression case sets. The `parse_golden` corpus is re-snapshotted, because a new key adds one line to every `parse_dump.txt`.

Two defects found in passing, outside the new physics. `h3p_net_cooling_rate` was missing stimulated emission (above): one line of code, gated by `base_ir_field`, so no golden moves. (The form it was restored inside is replaced in §112; the defect below it was the single-band collapse, not the missing term.) And `python/paper_data.py` could not read the outputs of any molecular run — its `_columns_header` returned the first comment line *containing* the word “columns”, and `Ion_species_adv.txt` opens with a NOTE reading “the molecular columns ... and, when present, the oxygen columns”, so the tail of that prose was returned as the column list and `load_ions` raised. The marker now has to be the first word after the #. `src/utls/element_budget.py` had the same loose match but kept the *last* hit and so recovered by accident; it is tightened the same way. Both verified on a molecular and an atomic run. A third, in the new code and caught by reading it back: the H_2 line loop guarded the whole line on $\Delta E/T_{\text{rad}} > 700$, which would have dropped the *emission* of a line the incident field carries no photons at, not just its absorption; the guard now sits on the absorption alone. It never fires at any realistic radiating temperature, and every number is identical before and after.

Cost: 2000 marching steps of the `mol_metals` configuration single-threaded take 99.80 s with the key off and 102.06 s with it on, **+2.3%**; with the oxygen chemistry on as well, so that all three channels are active, 800 steps of the HD 189733 b A2 configuration go from 55.1 s to 60.2 s, **+9.3%**. The three channels are two table interpolations per cell each, and the tables are built once at startup from the run’s own T_0 .

A regression case pins the new physics: `backup/regression/mol_ir_bands` is the hot-Uranus gate with solar metals and both keys on, a 12000-step snapshot, so the matrix is eight cases. It covers the H_2 line sum, the net exchange, the corrected H_3^+ net rate and the two-sided fine-structure escape in one run. The H_2O and CO bands are not covered by any case: they need the oxygen chemistry, whose own regression case (`mol_oxygen`) is milestone M5 of `docs/a2_oxygen_option_design.md` and is still unwritten.

What item (G) still holds, narrowed: where the molecules are, the layer now has coolants of the right size and each has a radiative equilibrium temperature ($t_{\text{rad}}/t_{\text{flow}}$ of the metals-on hot-Uranus layer at $r = 1.05$ falls by a factor 31).

Where the molecules are not, nothing changed and nothing can: the coldest cell of every hot-Uranus solution sits at or beyond the $H_2 \rightarrow H$ front, in gas that is atomic and expansion-dominated, whose fine-structure coolants already have their field from §55. Holding that gas is a hydrodynamic question, not a missing-coolant one, and item (G) as originally written — “the molecular layer ... radiates away with nothing to hold it” — no longer describes it. The oxygen option’s own defect is repaired in kind, and whether M4’s gate quantities reach their targets needs a converged HD 189733 b A2 wind, which is M4’s own work.

111. Positivity of the advected state: the outer ghosts fed the columns negative gas, and a local flux correction replaces the global Δt bisection (2026-08-31)

Items (P) and (Q) of TO_BE_DONE.md, taken together because they are two halves of one statement: between the hydrodynamic update and the ionization chapter, nothing enforced that every density the code hands on is a gas. The series has two code changes — an admissibility guard on the boundary ghosts, and a positivity-preserving flux correction inside the RK stages — plus the A2 measurements that the repaired marching made possible.

The negative nuclei of item (Q) were the outer ghost cells

Item (Q) recorded negative hydrogen and helium *nuclei* densities, thousands of cm^{-3} below zero, entering the ionization solve at “cell 502 of 504” during steps 585–589 of the wasp_full cold start. The diagnosis closes in three steps, none of which needed new instrumentation. First, cell 502 is not a physical cell: the default grid is $N = 500$ with $N_g = 2$ ghosts on each side, so the array runs $-1..502$ and cell 502 is $N + N_g$, the outermost ghost, which the positivity step control of §95 ($j = 1..N$) never sees. Second, the ratio convicts ρ and acquits the composition: the recorded $n_{\text{He}}/n_{\text{H}} = 284.7/3345.9 = 0.0851$ is exactly the run’s He/H number ratio, and since the fractions are not advected at all (f_sp is not in the hydro state; the composition is per-cell between ionization solves), a negative pair in exactly the element ratio means the ghost’s ρ itself was negative with innocent fractions on top. Third, Apply_BC_W builds the outer ghosts as $W_{N+k} = 2W_{N+k-1} - W_{N+k-2}$ whenever WENO3 is active, with no admissibility test, and the two-stage wasp_full run switches PLM→WENO3 at step 1 ($du = 0.37$ is already below the 0.5 threshold on a cold start), so the linear extrapolation is live through the whole violent transient.

The reach of one bad ghost is what made this look like an ionization problem: calc_column_dens starts its column integral at $N + N_g$, the ionization sweep runs over $1 - N_g..N + N_g$ (the source of the no admissible root at 1 cell(s) failures), and eval_dt takes its minimum over the same range. The ground-singlet floor of §100 was catching the far end of this chain.

The fix. An extrapolated outer ghost whose ρ or p is not strictly positive (the test is the negation of “strictly positive”, so a NaN is caught too) is dropped to the zero-gradient copy of the cell below it — the first-order limit of the same free-outflow boundary condition, and exactly the ghost the PLM stage uses. The hydrostatic_base pressure extrapolation of the base ghost carries the same guard; the face-state extrapolations of Rec_BC are left alone, because faces are already repaired by positivity_limited_faces. The counter is n_ghost_cells_positivity_limited.

Measured on isolated single-thread reruns of the two WENO3-stage regression cases: wasp_full stops at 14092 instead of 14093, fires the ghost guard 36 times, its singlet-floor count drops $3 \rightarrow 0$ and its no-admissible-root events vanish, at $\log_{10} \dot{M} = 13.23$ unchanged; wasp_he23off the same with 33 firings (14073→14072). The converged profiles move by at most 5.8×10^{-5} relative (2.2×10^{-5} in ρ , 1.0×10^{-5} in T), and the mass flux $\rho v r^2$ at $1.55 R_p$ by 2.8×10^{-6} and 9.2×10^{-7} . The six remaining matrix cases never leave the PLM stage, so the guarded extrapolation is never built there: byte-identical, all counters zero. Item (Q)’s acceptance criterion is met in the counted sense: ρ is positive on every cell including the ghosts, every density entering PH_heat_HHe is ρ times a solver-bounded fraction, and the counter reports

every repair.

The update positivity of item (P): a local first-order flux correction

Section 95 answered a stage that left $\rho > 0$, $\rho e > 0$ by discarding the step and retaking it from `u_old` at half Δt — correct, but global: on the He/H = 1 molecular arm of item (P), whose base runs at Mach 38–224, it cost 7500 retaken steps and 29409 bisections, the whole grid marching at a reduced Δt because one or two cells violate the positivity bound. That is the adaptive form of “lower the CFL number and run away”, and it is replaced by the standard local remedy.

`positivity_limited_fluxes` (`RK_rhs.f90`), applied per RK stage before the Δt bisection is considered: the cells of the candidate stage that left the admissible set are found, and at their interfaces — only there — the high-order HLLC flux is replaced by the first-order Lax–Friedrichs flux of the two neighboring *cell averages* of the state the stage was built from; the touched cells are rebuilt with the same stage arithmetic (the flux-difference assembly mirrors `RK_rhs` term by term, the WENO3 face-pressure term and the gravity energy correction included), and the detection repeats until nothing is left or nothing changes. Replacement is symmetric across each interface, so conservation is untouched. The pieces of the argument are published: the first-order Lax–Friedrichs update preserves $\rho > 0$, $\rho e > 0$ under $\Delta t(|v| + c)/\Delta r \leq 1$ (Perthame & Shu 1996, Numer. Math. 73, 119; the LF lemma of Zhang & Shu 2010, J. Comput. Phys. 229, 3091), a bound the production CFL number 0.6 respects — unlike the $\sim 1/2$ bound of the HLL family (Einfeldt et al. 1991) that §95 identified as the violation; each SSP-RK3 stage is a convex combination of forward-Euler steps and the admissible set is convex, so per-stage enforcement suffices; and replacing the flux at single interfaces is the flux-correction principle of Hu, Adams & Shu (2013, J. Comput. Phys. 242, 169), standard practice in production astrophysical codes (Stone et al. 2020, ApJS 249, 4). The Δt bisection stays, demoted to the backstop: gravity and the geometric sources sit outside any flux argument. No pressure or density floor exists anywhere in the chain. The counter is `n_faces_flux_positivity_limited`; the LLF branch of `Num_flux` was extracted into `lax_friedrichs_flux` so the correction and the flux option share one definition.

Measured on item (P)’s own arm (`mol_diffusion` at He/H = 1, `HeH_base` 1.0, CFL 0.6, 12000 steps): 8156 interface fluxes dropped to first order, **0 steps retaken, 0 bisections** (against §95’s 7500/29409 on the same arm), no NaN, $\log_{10} \dot{M} = 8.89$. The arm still marches a Mach 38–224 base with $du = 13.5$; why the base BC admits that state remains item (P)’s open half.

The A2 gate, unblocked and then measured honestly

The positivity protections fire *zero* times on every A2 configuration below — the A2 story is what the repaired code measured, not what the repair changed. The reconstruction of §110’s bands-on arm reproduces (base ghost 829.0 K, $x_{\text{H}_2} = 0.4824$, $\log_{10} \dot{M} = 9.24$, Planck-mean $\tau_{\text{H}_2\text{O}} = 2.48 \times 10^{-3}$, $\tau_{\text{CO}} = 2.78 \times 10^{-3}$; element budgets $\leq 10^{-12}$).

The “du stall” of §§109–110 was a patience artifact. Restarted with the converged-molecular recipe and left to march, du descends monotonically: 39.8 at the restart, 1.36 by step 12000, 0.50 by step 23896, with dtu down to 5.5×10^{-5} . The “oscillation between 1.08 and 1.35” is exactly the 12000–14000-step stretch of this descent; the previous attempts stopped 13000 steps into a ~ 24000 -step approach.

What breaks the descent is the PLM→WENO3 switch. At $du = 0.5000$ (step 23897) the two-stage logic switches reconstruction and the solution is destroyed in place: dtu jumps $5.5 \times 10^{-5} \rightarrow 4.1 \times 10^{-4}$ in one step and du explodes to $\sim 10^5$ within 2000 steps, with no positivity counter firing — a discretization response of the molecular base, not a positivity casualty. Held on PLM, du bottoms at 0.486 near step 24000–26000 and then rises secularly (1.40 by step 54000): a du floor. Recorded as TO_BE_DONE item (S).

The first converged A2 wind. From the du -minimum state, single-stage PLM with Solver: Newton 0.55 engages the staged secondary-ionization flip at step 2 and the JFNK at step 2002; it converges, $\text{info} = 0$ at $\|R\| = 2.0 \times 10^{-4}$, with the steady Picard loop — the first end-to-end exercise of `relax_photochemical_composition` — contracting the composition drift $2.2 \times 10^{-3} \rightarrow 5.4 \times 10^{-4}$ over 8 outer passes. $\log_{10} \dot{M} = 8.18$, base ghost 888.1 K, and the gate quantity is $x_{\text{H}_2} = 0.591$ against the photochemical reference 0.910 and the pass band 0.847–0.947: *the gate is not passed*. The relaxation trend is toward the target (0.482 at 12000 steps, 0.585 at the du minimum, 0.591 converged) and the thermal half of the gap is essentially closed (888 K against the reference's 864 K), so what remains is in the chemistry/transport half — but it cannot yet be attributed, because the converged state fails its element budgets near $r = 1.003$ – 1.006 : He and the five trace metals by a common $+5.18 \times 10^{-2}$ (a moved n_{H} denominator), but carbon by +90.6% and oxygen by +49.0% — at the worst cell the transported CO collapsed while C I gained *twice* the carbon the CO released. The carrier nuclei are double-counted into the free-stage columns somewhere in the JFNK/Picard state refresh; the marching states of the same configuration close at $\leq 10^{-12}$, so the marching operator is clean. Recorded as item (T); M4's verdict is to be re-measured after it is fixed.

Verification

make check: 8/8 — six cases byte-identical with every protection counter zero; the two WENO3-stage cases move as above and their goldens were refreshed at the end of the series (check→golden→check). He/H=1 at CFL 0.6 completes as tabulated. The `run_fcheck.sh` ladder — the script's bounded HD 209458 b case, plus the three arms where the new protections actually fire (`wasp_full` 800 steps with 36 ghost repairs, He/H=1 4000 steps with 76 flux corrections, the A2 oxygen arm 400 steps), all under `-O1 -fcheck=bounds,do,mem` — fired no runtime check.

112. The H_3^+ net exchange stops pretending the emission fit is one transition (2026-08-31)

Section 110 fixed one term of `h3p_net_cooling_rate` and left the form it sits in. The form is the defect. This section replaces it with the net-exchange closure section 110 already uses for H_2 , H_2O and CO , and re-measures item (G)'s own configuration with the two closures side by side.

What was wrong

Miller, Stallard, Tennyson & Melin (2013) publish the *total* optically thin LTE emission of an H_3^+ molecule, $E(T)$, as a piecewise polynomial fit — a sum over the ν_2 fundamental, its hot bands, the overtones and the forbidden rotational lines, already integrated against the Planck function. It is not a line list. To give that emission an absorption term, the

routine collapsed the whole fit onto *one* effective transition at the ν_2 band center, $E_k = 3627.5$ K, and wrote the two-level net factor of that transition,

$$\Lambda_{\text{net}} = \Lambda_{\text{emit}}(T) [1 + \bar{n} - \bar{n} e^{E_k/T}], \quad \bar{n} = \frac{W}{e^{E_k/T_0} - 1}.$$

That is self-consistent only if $E(T) e^{E_k/T}$ is independent of temperature, which is what a single transition with a saturated lower level gives. Measured on the published fit it is not remotely constant:

T [K]	300	200	150	100	60	40	30
$E(T)e^{E_k/T}$ [W molecule ⁻¹ sr ⁻¹]	6.7e-19	1.2e-18	3.2e-17	7.3e-13	1.3e-3	3.5e9	1.5e22

Forty decades, because the low-temperature emission of the fit is carried by transitions far below 3627.5 K while the absorption factor keeps insisting on that one energy. The bracket that multiplies the emission is therefore already -3.9×10^3 at 300 K and -1.2×10^{14} at 100 K — the band heating the gas at 10^{14} times its own emission — reaching -3.4×10^{136} at the 11.4 K floor a stalled molecular base can reach. The $\exp(\min(E_k/T, 700))$ cap in the code stopped the factor from being Inf; it did not stop it from being wrong. The comment beside it said the heating “is bounded by the field itself, not by this ratio”, which is the correct physics and is not what the line below it computed.

This is not a cold-edge defect. The band this routine is used in is the molecular layer, 190–300 K on the hot-Uranus gate and 11–30 K on the He/H = 1 arms, and the bracket is wrong throughout it.

The closure that replaces it

The same net exchange with a diluted blackbody the other three infrared channels of the layer use (section 110, eq. 1):

$$\Lambda_{\text{net}} = n(\text{H}_3^+) 4\pi s(T, n_{\text{H}_2}) [E(T) - W E(T_0)].$$

The absorbed power per molecule is $4\pi W \int \sigma_\nu B_\nu(T_0) d\nu$, and Kirchhoff’s law at the radiating temperature identifies that integral — evaluated with the cross section at T_0 — with the emission fit itself, $E(T_0)$. The fit does double duty and no line list is needed.

Where it differs from the other three channels, and why it has to. H_2O and CO are band absorbers with tabulated $\sigma_b(T)$, so their absorption keeps the cross section at the *gas* temperature and only the Planck function at T_0 ; H_2 is summed transition by transition with the full two-level factor. Neither is available for H_3^+ : the published quantity is the product of cross section and Planck function already integrated, so the cross section cannot be separated from the weighting and is frozen at T_0 . What that freezing costs is the temperature dependence of the band strength alone — the fraction of molecules in the absorbing lower states, which rises toward 1 as T falls, so the absorption above is if anything a slight underestimate at low temperature. It is exact at $T = T_0$, which is where the fixed point sits.

The non-LTE departure factor now multiplies both terms. It has to: $s < 1$ with the absorption left at its LTE value heats a gas that is *hotter* than the field, which no exchange between two temperatures may do. Read this way s is the fraction of molecules radiatively coupled at the local density, and it scales the rate without moving the equilibrium temperature. Miller et al.’s Table 6 is a vacuum departure factor — collisions against spontaneous decay, no incident field — and a field strong enough to matter here pumps the levels back toward LTE, so the true departure is smaller than s and this underestimates the magnitude of the exchange, not its sign or its fixed point.

The three limits, measured

$T_0 = 1140$ K, $W = \frac{1}{2}$, $n(\text{H}_3^+) = 1 \text{ cm}^{-3}$, $n(\text{H}_2) = 10^{15} \text{ cm}^{-3}$ (so $s = 1$ and the closure itself is exposed). Rates in $\text{erg cm}^{-3} \text{ s}^{-1}$; negative is net heating.

T [K]	Λ_{emit}	new	old	new/emit	old/emit
1140	7.498e-12	+3.749e-12	+3.749e-12	0.500	0.500
1000	4.197e-12	+4.476e-13	+8.697e-13	0.107	0.207
900	2.582e-12	-1.167e-12	-5.087e-13	-0.452	-0.197
600	2.928e-13	-3.456e-12	-2.378e-12	-11.8	-8.12
400	1.373e-14	-3.735e-12	-2.566e-12	-272	-187
300	4.700e-16	-3.749e-12	-1.815e-12	-7.98e3	-3.86e3
250	5.610e-17	-3.749e-12	-2.432e-12	-6.68e4	-4.34e4
200	2.046e-18	-3.749e-12	-3.337e-12	-1.83e6	-1.63e6
150	1.264e-19	-3.749e-12	-8.710e-11	-2.97e7	-6.89e8
100	1.627e-20	-3.749e-12	-2.00e-6	-2.30e8	-1.23e14
30	5.717e-23	-3.749e-12	-4.04e+28	-6.56e10	-7.06e50
11.4	5.717e-23	-3.749e-12	-1.93e+114	-6.56e10	-3.38e136

- **Exactly zero at $T = T_0$** , $W = 1$, to the last bit, at 30 / 200 / 600 / 1140 / 2000 / 5000 K and also at $n(\text{H}_2) = 10^8 \text{ cm}^{-3}$, where s is 3.13e-2 / 9.57e-3 / 4.90e-3 at 600 / 1140 / 2000 K. The two terms differ only by a factor $W = 1$, so the subtraction is exact.
- **Bounded as $T \rightarrow 0$** : the heating saturates at $W 4\pi E(T_0)$ per molecule, $-3.749 \times 10^{-12} \text{ erg s}^{-1}$ here, identical at 50, 30, 20, 11.4 and 1 K.
- **$W = 0$ returns the emission-only rate unchanged**, by an early return, so every configuration with Base IR field off is byte-identical.

The old and the new form happen to agree to within a factor 2 over **200–300 K** (old/new = 0.48/0.65/0.89 at 300 / 250 / 200 K), which is why the metals-on hot-Uranus gate below hardly moves. They part company in both directions outside that band: the old form under-heats by $1.5\text{--}1.9\times$ over 400–1000 K, where the base of a molecular run actually sits, and over-heats without limit below 150 K.

The radiative equilibrium temperature is now the root of $E(T_{\text{eq}}) = WE(T_0)$. It has no closed form; measured by bisection on the closure itself:

T_0	$W = 0.25$	$W = 0.5$	$W = 0.75$	$W = 1$
900 K	686.0 K	780.9 K	847.5 K	900.0 K
1140 K	841.9 K	975.4 K	1067.7 K	1140.0 K
1183 K	870.3 K	1010.7 K	1107.4 K	1183.0 K
1450 K	1050.7 K	1231.6 K	1354.4 K	1450.0 K
1800 K	1287.5 K	1517.6 K	1675.0 K	1810.0 K
2358 K	1625.4 K	1938.6 K	2161.0 K	2358.0 K

At $W = 1$ the root is T_0 exactly — the fixed point recovered numerically — except at $T_0 = 1800$ K, where it lands 10 K high because the published fit is discontinuous there (below). **The $W = \frac{1}{2}$ column supersedes the H_3^+ column of section 110’s table** (769 / 941 / 971 / 1150 / 1374 / 1713 K at the same six T_0); those were the single-transition form’s.

Item (G)’s own configuration, re-measured with both closures

Eight runs, four configurations times two binaries identical except for h3p_net_cooling_rate. Each is the mol_ir_bands regression case definition — the hot-Uranus Tier-2 gate with Base IR field: True, metals.inp and Molecular IR bands toggled — relaxed 12000 steps from cold and then restarted from its own snapshot with Load IC? True, Solver: Newton 5.0e-2, Resid tol: 1.0e-5, Max steps: 150000.

These are not section 110’s solutions and they are not claimed to be. That section restarted section 55’s converged runs, whose directories no longer exist; rebuilt from the pinned case the chain lands on a different fixed point (base T 828 K rather than 1212 K, metals off). Only the old/new pair within each column is a controlled measurement.

	metals off, bands off		metals off, bands on		metals on, bands off		metals on, bands on	
	old	new	old	new	old	new	old	new
JFNK iterations	30	75	37	28	26	23	20	22
info	2	0	0	0	0	0	0	0
$\ R\ $	1.0e-3	5.6e-6	6.8e-6	1.4e-6	4.1e-6	3.4e-6	8.7e-6	6.5e-6
base T (cell 1)	—	828.07	329.30	491.96	1209.71	1209.71	1198.27	1185.69
T at $r = 1.005$	—	936.44	918.61	940.08	1097.28	1097.29	1044.67	1044.98
T at $r = 1.02$	—	903.59	865.56	897.36	925.78	925.70	886.85	886.94
T at $r = 1.03$	—	880.01	837.28	876.67	819.61	819.57	803.22	803.42
T at $r = 1.05$	—	788.03	731.18	782.64	573.68	575.22	590.40	592.16
T at $r = 1.08$	—	446.03	366.77	427.34	214.60	218.18	236.34	239.83
coldest cell < 1.3	—	239.4	232.1	234.3	190.4	191.4	198.7	199.5
at r	—	1.0997	1.0947	1.0984	1.0842	1.0842	1.0853	1.0853
$H_2 \rightarrow H$ front	—	1.09053	1.08602	1.08978	1.07916	1.07929	1.07971	1.07987
$\log_{10} \dot{M}$	—	10.32	10.31	10.32	10.29	10.30	10.30	10.30
flux spread, $r > 1.2$	—	2.40e-3	2.385e-3	2.387e-3	2.381e-3	2.381e-3	2.382e-3	2.380e-3

(Temperatures in K.)

Item (G)’s conclusions do not change, and that is the first line of this section. Both statements section 110 closed on survive:

- Where the molecules are, the layer has coolants with fixed points. The H_2 , H_2O and CO channels are untouched by this change; only the H_3^+ share moves.
- Where the molecules are not, nothing changed and nothing can. The coldest cell of all four solutions still sits at or beyond the $H_2 \rightarrow H$ front (front 1.079–1.091, coldest cell at 1.084–1.100), and it moves by at most **2.3 K** between the two closures. There is no H_3^+ past the front for the closure to reach, which is why that statement is structurally safe: H_3^+ requires H_2 .

What does move:

- **Metals on, either bands arm: 0.02–3.5 K everywhere**, the coldest cell by ≤ 0.8 K, the front by $\leq 1.6 \times 10^{-4} R_p$, $\dot{M}$ by ≤ 0.01 dex. Section 110's metals-on numbers stand as measured.
- **Metals off, bands on: 20–163 K warmer with the new closure** — base $329.3 \rightarrow 492.0$ K, $r = 1.08$ $366.8 \rightarrow 427.3$ K — in the direction the rate table predicts, since the old form under-heats over 400–1000 K. The coldest cell moves 2.3 K and the front $0.0038 R_p$; $\dot{M}$ $10.31 \rightarrow 10.32$.
- **Metals off, bands off: the old closure does not converge and the new one does.** From the same initial state, the same code except this routine, the old arm's JFNK aborts — “line search found no descent step in 12 consecutive iterations”, $\text{info} = 2$, best iterate $\|R\| = 1.0 \times 10^{-3}$, worst cell $r = 1.001$ — and falls back to marching; the new arm converges to $\|R\| = 5.6 \times 10^{-6}$ in 75 outer iterations. This is the metals-off branch, where H_3^+ carries the whole cooling of the base, so it is where an unbounded heating term costs the most. The old column is The old arm then marched to its 150000-step cap without the du stop re-arming and stopped at $du = 8.6 \times 10^{-4}$ with a mass-flux spread of 1.59×10^{-2} over $r > 1.2$, seven times that of any converged solution here (2.4×10^{-3}): it is a marching state, not a wind, so its column is left empty rather than filled with a non-solution. For the record it ended at base 878.7 K, then 1094.0 / 1028.3 / 1000.1 / 999.3 / 933.4 K at $r = 1.005/1.02/1.03/1.05/1.08$, coldest cell 397.6 K at 1.1499, front 1.11491, $\log_{10} \dot{M} = 10.39$.

Goldens

mol_ir_bands moves and was refreshed; the other seven are byte-identical. It is the only regression case that sets Base IR field, which is the gate on this routine, so it is the only one that can move, and it does: on the 12000-step snapshot the largest changes are T by 0.42 K (4.1×10^{-4} relative, at $r = 1.200$), cool by 1.1×10^{-2} relative at $r = 1.188$, $n(\text{H}_3^+)$ by 6.5×10^{-3} relative and $n(\text{H}_2)$ by 4.1×10^{-3} , all in the coldest part of the snapshot (about 1020 K at $r = 1.19$ –1.21) where the two closures differ most. du at step 12000 goes $2.7910 \rightarrow 2.7905$ and $\log_{10} \dot{M}$ stays 10.58. Refreshed at the end of this change with check $\rightarrow$ golden $\rightarrow$ check.

The Base IR field NaN of the He/H = 1 arm

docs/supersonic_molecular_base.md section 11.6 reports that switching Base IR field on in the He/H = 1 molecular case reaches NaN at step 3, and attributes it to this routine. **Reproduced and fixed.** The case is backup/regression/mol_diffusion with He/H = 1.0 (in both input.inp and base.inp) and Base IR field: True. With the old closure the run dies at step 3 with $v = -1.0 \times 10^{52}$ in the first physical cell and $\log_{10} \dot{M} = \text{NaN}$; with the new one it completes 12000 steps with **zero NaN** and $\log_{10} \dot{M} = 8.89$. The bracket that killed it was -3.4×10^{136} at the 11.4 K trial floor those cells reach.

What this does not fix is the rest of that document. The same run still ends at $du = 13.46$ with 8169 first-order flux corrections: the supersonic molecular base of item (P) is untouched, and section 11.6 was explicit that the NaN was a separate defect. It is now removed as an obstacle to switching the field on there, not as a cure for the runaway.

Two properties of the published fit, measured in passing

Neither is a code defect and neither is changed here; both are recorded at the module header because the header quoted the paper's error claims without them.

- **The fit reproduces Miller et al.'s own Table 4 to within 0.45%** over 500–5000 K (ten anchors), which confirms the transcription of Table 5.
- **The published segments do not join.** Continued across 300 K, the 300–800 K polynomial stands 15–43% above the 30–300 K one over 200–300 K (1.20 at 200 K, 1.43 at 300 K), and the 1800–5000 K polynomial 2.4% below the 800–1800 K one at 1800 K — the only place where $E(T)$ as evaluated here decreases with temperature. Each segment is used on its own published range, which is what the paper prescribes. The 1800 K step is why the $W = 1$ root above is 1810 K rather than 1800 K. Whether to smooth the joins is item (U) of TO_BE_DONE.md, because it would move the published values.

Cost and verification

One extra polynomial evaluation per cell, and only when Base IR field is on. The 12000-step mol_ir_bands relaxation single-threaded takes 9 m 52 s with the old closure and 9 m 51 s with the new one (both under the same concurrent load): no measurable cost.

- make check: **8/8**. Seven cases byte-identical; mol_ir_bands moves as tabulated and its golden was refreshed (check → golden → check, the re-check byte-identical).
- run_fcheck.sh: **CLEAN** on its own bounded HD 209458 b case. That case has no molecular chemistry and no Base IR field, so a second ladder was run under the same -O1 -fcheck=bounds,do,mem build on the two arms that do reach this routine — mol_ir_bands and the He/H = 1 Base IR field case, 1500 steps each: no runtime check fired and no NaN line in either.
- Closure properties: the exact zero, the low-temperature bound and the equilibrium temperatures above, all measured on the compiled module.

113. Chemical equilibrium is accepted on its reaction residual, not on the solver status (2026-08-31)

The judgement first. The equilibrium acceptance rule of `ioniz_eq` (`src/modules/radiation/ionization_equilibrium.f90`) violated its own physics: a cell state was retained as chemical equilibrium if it satisfied the element bounds, whatever its reaction residual, and after a projection onto the element budget the residual was not recomputed at all (`docs/supersonic_molecular_base.md` §11.4). Bounds and element conservation are necessary conditions; they are not reaction equilibrium. Both branches — molecular and atomic — now accept a state as a root only when it is inside the element bounds *and* its normalized reaction residual is within a stated tolerance; the residual is recomputed after any projection; and a non-root acceptance is never silent. The MINPACK exit code participates in neither direction: `info = 1` is an `xtol` statement about the step, not about the residual (measured: an `info = 1` iterate 7.9×10^{-2} outside the simplex at step 0 of the He/H = 1 arm, and an `info = 1` in-bounds iterate at residual 2.8×10^{-5}), and `info = 4` routinely returns finished roots it cannot certify (16,293 of them at residual $\leq 10^{-6}$ in one He/H = 1 run).

The acceptance statistics were measured *first*, on the full 8-case regression matrix and on the He/H = 1 molecular arm, with the acceptance decisions unchanged (make `check` byte-identical for the instrumented build, 8/8). The measurement then fixed three design decisions that could not have been made a priori: the tolerance, the fate of the goldens, and the stop policy.

The normalization

The residual judged is `normalized_reaction_residual`: the residual vector of the same system the cell solve used, evaluated at the candidate, with every balance row divided by its turnover scale — the rate at which the species that row balances can be produced or destroyed in the cell, each species set to the whole of its element (the §93 convention, with the cell's incoming electron density). The molecular systems already reach `hybrd1` with their rows so scaled; their residual is reused as returned, widened only by the metal-H/He charge-exchange bound the solver scale leaves out of its rows (`cx_add_to_turnover`): one contribution among several for the solver's path, but in the wind, where charge exchange couples a trace metal's rows to the whole hydrogen reservoir, the dominant term of the row's turnover — judged against a scale without it, finished roots would be rejected. The atomic systems stay unscaled inside the solver (row scaling changes the MINPACK path, and no solver failure motivates it there); their scale is built at acceptance time from the same cell coefficients: photoionization, collisional ionization, recombination, $\text{He} \leftrightarrow \text{H}$ and metal charge exchange, and the He 2^3S kinetics where the triplet is tracked. Identity rows (absent elements, the pinned X^{++} of a two-stage element, a transported carrier's $x - x_{\text{fix}}$ row) keep scale 1.

The measured populations, and the tolerance

Instrumented totals over the eight golden cases (accepted states, by class; *res* is the normalized reaction residual):

case	converged roots (max res)	w/o convergence (max res)	projected/handback (max res)
wasp_full	7,102,368 (5.8e-9)	0	0
wasp_he23off	7,092,288 (4.4e-9)	0	0
mol_base_handoff	6,047,970 (4.0e-10)	30 (1.2e-11)	0
mol_metals	6,047,964 (7.8e-8)	35 (2.8e+1)	1 (2.7e+1)
mol_lyman_werner	6,047,986 (4.0e-10)	12 (1.5e-12)	2 (1.5e-13)
mol_diffusion	6,047,979 (4.0e-10)	21 (1.0e-11)	0
mol_ir_bands	6,047,920 (7.8e-8)	79 (2.8e+1)	1 (2.7e+1)
lower_profile	6,048,000 (2.9e-8)	0	0

Two populations, three and a half decades apart. Every solver-converged accepted root in the matrix sits at $\text{res} \leq 7.8 \times 10^{-8}$ (consistent with $\text{xtol} = \sqrt{\varepsilon} \simeq 1.5 \times 10^{-8}$ times a scaled-Jacobian norm of 1–10), every root accepted without solver convergence at $\text{res} \leq 2.7 \times 10^{-7}$, and the decade $[10^{-6}, 10^{-5})$ is empty in every matrix histogram; the above-tolerance acceptances start at 1.1×10^{-3} . The tolerance is therefore $\text{ieq_res_tol} = 10^{-6}$: $3.7\times$ above the measured root tail, $1100\times$ below the matrix's smallest non-root, inside the empty decade. The He/H= 1 arm carries one borderline accepted iterate at 2.1×10^{-6} , which this value classifies (and marks) as a non-root rather than stretching the root band to cover it.

The finding the measurement was run for: the golden count is not zero. The two metals-on molecular gates accept true non-roots during their cold start — $\text{res} \sim 28$ in base cells 6–12 at step 0, decaying through $\sim 2.6 \times 10^{-3}$ in cells 259/260 (the H₂ front) over the following sweeps, the longest single-cell run of consecutive failures being 53 sweeps (mol_ir_bands, cell 260) — and then land on roots for the remaining $\sim 11,940$ steps. The severity ordering is inverted: the healthy transient's worst residual (28) exceeds the He/H= 1 poisoning event (2.1×10^{-2}) by three decades. No severity threshold separates a recovering cold start from a poisoned run.

The rule, and the staged stop policy the measurement forced

Ranking per attempt (both branches, same rule): rank 2 = bounds and $\text{res} \leq 10^{-6}$ and solver-converged (accepted immediately); rank 1 = bounds and $\text{res} \leq 10^{-6}$ without solver convergence; the first bounded iterate failing the residual is kept aside as the state of last resort (the old rank-1 order), and among attempts that leave the simplex the closest is kept for projection. If no rank exists: the molecular branch projects the closest iterate onto the element budget and *recomputes the residual, which the nonlinear projection changes*; the atomic branch hands back the uncoupled ionization balance and measures its residual *under the full coupled system*. A projected or handback state within tolerance is a root (class 3). Everything else is a **class-4 non-root acceptance**: reported cell by cell (print-capped at 2000 lines, the counters uncapped), totaled in the end-of-run ioniz-eq acceptance WARNING line, and tracked per cell. A cell may rest on a non-root for at most $\text{ieq_nonroot_streak_stop} = 1000$ consecutive sweeps ($\sim 19\times$ the largest healthy streak measured); one more and the run stops with the full diagnostics of §11.5-A — cell, radius, temperature, densities, solver exit code, element violation, normalized residual, candidate fractions.

An unconditional stop on the first non-root — the letter of §11.5-A — was measured to be untenable: it would kill mol_metals and mol_ir_bands at step 0 while the run it was aimed at survives to step 0 with a *smaller* residual. What

does separate the two is persistence: a steady or pseudo-steady state re-evaluates the same cell every sweep, so a wind resting on a non-root fails the same cell indefinitely, while every measured healthy streak ends within 53 sweeps. A Δt -retry stage was considered and rejected on the same measurement: the failures live in the cold-start composition, not in the step size, and the largest events occur at step 0, where no Δt exists.

What the He/H= 1 arm now does, and what it deliberately still does not

Under the new rule the He/H= 1 molecular arm (the mol_diffusion gate with He/H= 1, 12000 steps) marches with **8 loudly-reported non-root acceptances instead of silent ones**: step 0, cell 387 (info = 1, 7.9×10^{-2} outside the simplex, projected, recheck 2.1×10^{-2} — the initiating acceptance of §11 of the memo), then isolated events at steps 55, 570, 6440 and four between steps 10,710 and 11,468 with residuals up to 1.4×10^6 in the collapsed region — each visible in the log the moment it happens, and summarized in the end-of-run WARNING with the maximum residual and longest streak. The run is no longer able to pass silently.

It does not abort, and the measurement says why no acceptance rule could have made it: every above-tolerance event is an isolated single-sweep excursion (longest streak 1), and the marched pathological state — the Mach ~ 224 base, 11.4 K first cells with ionized hydrogen — re-evaluates as *bounds-satisfying roots of the frozen network*: clamped states with element violations up to 0.64 recheck at $\text{res} \leq 5 \times 10^{-8}$, because at 11.4 K the network's rates are so small that almost any partition balances. The reaction residual is a necessary filter and it now catches the poisoning where it enters; it cannot condemn a state whose kinetics are frozen. That is the measured boundary of this change, and it is why the constrained solve of §11.5-B, the composition boundary of 11.5-C and the Mach validity check of 11.5-D remain necessary.

Verification

The order of proof was: instrument, measure, implement, re-verify.

- make check: **8/8 byte-identical**, twice — once for the instrumented build with the old acceptance (the measurement pass), once for the new rule; **16/16 files**, verified independently by cmp. The byte-identity of the second pass is a measured consequence, not luck: at 10^{-6} no golden decision flips (no solver-converged attempt above tolerance anywhere in the matrix), and in every case where the old rule kept a non-root the new rule keeps the same state — measured directly, zero events in which an attempt at the same cell and step had $\text{res} \leq 10^{-6}$ while the accepted state was above it (the “passed-over root” check, also zero in the He/H= 1 arm).
- No golden was refreshed in this stage: every golden mtime predates the implementation. Byte-identity is the structural statement that the new rule accepts exactly the states the matrix cases were already choosing — what changed is that the choice is now judged and reported, in a new end-of-run acceptance ledger printed by every run.
- run_fcheck.sh: **CLEAN** on the rebuilt -fcheck=bounds, do, mem binary (HD 209458 b, $\log_{10} \dot{M} = 9.60$, no runtime trap), the production build restored and its compiler verified (GCC 13.1.0); plus the same build run on the molecular arms (mol_metals gate and the He/H= 1 arm, 1500 steps each): no runtime trap.

- Cost: the instrumented 12000-step mol_ir_bands relaxation runs in 9 m 47 s under concurrent load against 9 m 51 s for the §112 build — one extra residual evaluation per solver attempt and per acceptance, against the hundreds a hybrd1 solve performs, is not measurable at run scale. Measured under concurrent load in this stage and not re-measured in the §114 series.

114. A constrained chemical solve, the $\text{He}^+ + \text{H}$ channel the network never had, and a dissociation guard that had been inverted (2026-08-31)

The judgement first. Stage 3 of the supersonic-molecular-base series (docs/supersonic_molecular_base.md §11.5-B) was to build a chemical equilibrium solve whose iterates cannot leave the physical composition domain, so that the poisoning §113 could catch but not condemn — cell 387 of the $\text{He}/\text{H} = 1$ arm, accepted at a normalized reaction residual of 2.072×10^{-2} — would stop entering at all. The solver was built and it works. **It is not what closed cell 387.** Two pieces of physics did: the non-radiative $\text{He}^+ + \text{H}$ electron capture, a channel the network never carried, and an asymptotic guard in the H_2 dissociation fit that had been inverted since the fit was transcribed, handing every cell below about 529 K a fully ATOMIC state labelled “the dense molecular limit”.

Three *numerical* explanations were advanced for that one cell across this series — an unavoidable floating-point cancellation of the two $\text{H} \leftrightarrow \text{He}$ charge-exchange rates, a finite-difference step mismatched to logarithmic unknowns, and wrong-basin convergence in warm cells — and **all three were measured and found not to be the cause**. The second was nonetheless a real defect and was fixed on its own evidence. The record of that sequence is kept in docs/charge_exchange_cancellation_limit.md, deliberately: two of the three were argued from plausible mechanisms before the failing state had been dumped and examined, and the state was diagnostic the moment it was.

The $\text{He}/\text{H} = 1$ arm now runs 12,000 steps with **zero** non-root acceptances, against nine in §113. $\log_{10} \dot{M} = 8.89$ is unchanged by every change in this section.

The constrained solve (stage 3a)

constrained_chemical_equilibrium.f90. The fraction systems solve for stage fractions and close the neutral stage as a remainder, so an iterate can leave the simplex and has to be projected back afterwards — and a projection is not a solve. This formulation removes the possibility instead of repairing it:

- the unknowns are $u_k = \ln n_k$, so every iterate has positive densities. Logarithmic composition variables for exactly this reason are the CEA method of Gordon & McBride (1994), NASA RP-1311;
- element conservation is a RESIDUAL ROW, one per element. HeH^+ carries one H and one He nucleus, and the oxygen carriers carry H as well as O, so each appears in two conservation rows at once and both hold at the root. That simultaneous satisfaction is the §11.5-B requirement, and it is what rescaling stage fractions cannot do;
- the electron density is never an unknown: it is the charge-neutrality sum, so neutrality holds identically at every iterate.

The cell is bistable in its starting point, so a natural-parameter continuation (Allgower & Georg, *Introduction to Numerical Continuation Methods*, SIAM Classics 2003) tracks the root in the radiation field from $\lambda = 10^{-6}$ up to the cell's own field, each rung starting from the previous root and bisecting geometrically on failure. $\lambda = 0$ is a singular point of the path rather than an end of it: the trace ionization stages are linear in λ , so their logarithms run to $-\infty$ there.

Trace species below 10^{-10} of their element are held out of the Newton system and restored afterwards from their own balance — the treatment CEA uses for the same reason — because a species held at $\sim 10^{-25}$ of its element has a null

Jacobian direction and MINPACK grows its trust region against it. Rung acceptance is over the FULL row set, held rows included at their placed values; a Newton solve over the reduced system says nothing about the rows taken out of it.

The solver accepts nothing. It returns a candidate, and the caller judges it with the same test every other candidate faces (ionization_fractions_physical and normalized_reaction_residual against ieq_res_tol, §113). Promotion happens only where the existing hybrd1 ladder fails to obtain a root (best_rank = 0), so healthy cells keep their path exactly, and a state it produces is accepted as a new acceptance class 5.

The finite-difference step (stage 3b)

fdjac1 chooses $h = \sqrt{\varepsilon}|x_j|$, falling back to $h = \sqrt{\varepsilon}$ only at exactly zero. The constrained unknowns are logarithmic DEPARTURES, which sit at or near zero at every rung start and every converged state, so that rule swings between $h \sim 1.5 \times 10^{-8}$ and $h \rightarrow 0$: at $u = 10^{-8}$ the step is 1.5×10^{-16} and the column is noise. Measured against a central-difference reference on the saved cell-387 state, MINPACK's own rule got the HeH⁺ Jacobian column wrong by 86% and the H₃⁺ column by 70%.

The fix holds the unknowns near 1 (u_shift), so a rung starts at $v = 1$ and $h = \varepsilon|v| = \varepsilon$ there instead of collapsing. That is what repaired the measured columns, and it was obtained without touching fdjac1 or hybrd, so no other caller of either changes.

It is not, however, the absolute floor the review asked for, and this section does not claim it is. MINPACK sees $v = u_shift + \ln(n/n_{ref})$, so its rule is $h = \varepsilon|1 + u|$: the step still collapses where v passes through zero, which is now $u = -1$, i.e. $n = 0.368n_{ref}$ — an ordinary value to traverse during a solve. The singularity has been translated from the rung start, where every iterate began, to a point the path may cross, which is why the measured columns improve. A genuine floor of the form $h_j = \eta \max(1, |v_j|)$ requires a difference routine belonging to the constrained solver, and that is recorded as later work. The residual sensitivity reported below and this unfinished floor are stated side by side; no causal link between them has been measured. The step is then set by $epsfcn = fd_eta^2$ with $fd_eta = 10^{-4}$, chosen on a measured sweep: column agreement improves monotonically 8.6×10^{-1} , 6.8×10^{-2} , 1.1×10^{-2} , 4.6×10^{-3} for $\eta = 10^{-7} \dots 10^{-4}$, so the value sits on a plateau rather than at a point. (These derivative measurements are from the stage-3b diagnostic on the saved state, not re-measured here.)

For context on what this fix is: none of the three reference codes in the workspace builds its Jacobian by finite differences at all. Photochem uses forward-mode automatic differentiation by default, VULCAN generates the Jacobian symbolically, and Equilibrate's CEA matrix is itself the analytic linearization. Photochem's finite-difference opt-out uses $R(j) = epsj * abs(usol(i, j))$ with no absolute floor — the same pathology described above — so the precedent is that the answer to a bad difference step is to stop differencing, not to tune the step. u_shift reaches the same destination while keeping MINPACK, and it is a targeted repair rather than the standard: an analytic or automatically differentiated Jacobian is the durable form and is recorded as such for later work.

Section 3.4 of the review is also closed: u had no lower bound, so $\exp(u)$ could underflow to exactly zero against the module's own positivity claim. The bound is now two-sided.

The $\text{He}^+ + \text{H}$ channel, and two labels that named the wrong physics (stage 3c)

The labels first, because they are why the gap was invisible. Row B2 of the charge-exchange table is the RADIATIVE charge transfer of Stancil, Lepp & Dalgarno (1998) ApJ 509, 1, Table 1 row (19), whose source is Zygelman, Dalgarno, Kimura & Lane (1989) Phys. Rev. A 40, 2340; row B1 is the NON-RADIATIVE collisional rate of Kimura et al. (1993). A photon-emitting channel has no collisional reverse, so the ratio of the two is under no obligation to satisfy detailed balance, and the factor $1.05 \times 10^6 / T$ by which it departs from it ($921 \times$ at 1140 K, $210 \times$ at 5000 K, $105 \times$ at 10^4 K, evaluated here from the two coded expressions) is a consequence of comparing two different channels, not evidence of a defect in either fit. The caveat in `mol_rates.f90` calling this a detailed-balance violation was wrong and is corrected, as is §4.3 of `docs/molecular_chemistry_audit_he_rich.md`. An independent 2018 network (Ziegler, A&A 620, A81, Table A.1 rows 7 and 8) pairs exactly the same two sources.

What was missing. Zygelman et al. (1989) computed BOTH channels — its title is “Radiative and nonradiative charge transfer in $\text{He}^+ + \text{H}$ collisions at low energy” — and the code carried only the radiative one. The non-radiative channel is adopted from Kingdon & Ferland (1996) ApJS 106, 205, Table 1, whose He^+ row (7.47(−6) 2.06 9.93 −3.89, range 6×10^3 – 10^5 K) references that same Zygelman calculation, so the two channels come from one source and adding them is a total removal rate, not double counting. Their shapes confirm they are different physics: the radiative rate goes as $T^{0.25}$ (nearly flat), the non-radiative as $t_4^{2.06}$ (steeply rising). Measured, the added channel is $0.360 \times$ the radiative one at 1140 K, $1.158 \times$ at 3020 K and $2.992 \times$ at 10^4 K, so total $\text{He}^+ + \text{H}$ removal rises by $1.360 \times$, $2.158 \times$ and $3.992 \times$.

Validity and extrapolation are recorded at the code site. Kingdon & Ferland state that their fits “are only valid within the given temperature range; extrapolation significantly outside this range is not recommended”. Below the range the expression is evaluated as fitted — it decays as $t_4^{2.06}$ toward low temperature, so downward extrapolation under-weights the channel and cannot make it spuriously large — and above it the temperature is capped at the published ceiling, because the fit grows without bound there. The asymmetry follows a published precedent: Ziegler (2018) §2.1 checks every coefficient for “unphysical behavior or unboundedness ... in the asymptotic limits” and, for the Kingdon & Ferland $\text{C}^+ + \text{H}$ rate, sets it to zero below 5390 K and floors it at 10^9 K.

An external cross-check was attempted and there is nothing to check against, which is itself a measurement: VULCAN carries helium only as a zero-rate spectator and Photochem’s reaction network has no ions at all, so neither code contains $\text{He}^+ + \text{H}$ or $\text{He} + \text{H}^+$ in any form. Neither cites Visscher et al. (2006) or Koskinen et al. (2022) either; both obtain their equilibrium by free-energy minimization instead. No independent implementation of the adopted channel or of the dissociation fit exists in the workspace.

B2 is kept. Courtney et al. (2021) ApJ 919, 70 recomputed the same process (their Table 1 reaction 9) and report that the smooth backgrounds “agree very well with the previous results of Zygelman et al. (1989)”. They give no closed-form fit for it, so there is nothing to substitute. Two of their findings sharpen the validity note: the LTE / NLTE difference comes from narrow resonances and is negligible above 1000 K, and stimulated processes matter only above about 50,000 K, so neither correction is needed from 1140 K upward. What remains a caveat is the extrapolation beyond the published tabulation itself, which stops at 1000 K.

The B1 high-temperature branch. Glover & Jappsen (2007) ApJ 666, 1 fit R27 in two branches, $1.26 \times$

$10^{-9} T^{-0.75} \exp(-127,500/T)$ below 10^4 K and $4.0 \times 10^{-37} T^{4.74}$ above it; only the low branch was carried. The code's form, written with $t_r = T/300$, expands to $1.2614 \times 10^{-9} T^{-0.75} \exp(-127,500/T)$, so the route through Huang et al. (2023) Table 4 and the original agree on that branch and there is nothing to adjudicate. The branches join to 0.4% at the switch (3.661×10^{-18} against 3.648×10^{-18}), and above it the low branch runs high — $1.895 \times$ at 1.1×10^4 K, $13.1 \times$ at 2×10^4 K, $7.07 \times$ at 4×10^4 K — so carrying the published high branch LOWERS B1 there. 66 of the 504 cells of the `wasp_full` golden exceed 10^4 K, reaching 1.082×10^4 K.

Finally, `he_h_cx_rates` now passes through the same $[0, 10^{-6}]$ admissible band `cx_set_cell` imposes on every other charge-exchange row; group B reaches the systems only through that routine and was the one pair that skipped it.

The dissociation guard that had been inverted

`q_h2_equilibrium` (`lower_column.f90`) transcribes Koskinen et al. (2022) ApJ 929, 52 Eq. 11, quoting Visscher et al. (2006), read from the published paper:

$$q_{\text{H}_2} = \frac{1.9845 + 10^u - \sqrt{10^u (3.9690 + 10^u)}}{2.3670}, \quad u = -\frac{23672}{T} - \log_{10} p + 6.2645.$$

u runs to $-\infty$ as the gas gets COLD, and the expression tends to $1.9845/2.3670 = 0.83840$ there — the fit's approximation of the fully molecular solar value $0.5/(0.5 + \text{He}/\text{H}) = 0.863$ — and to 0 as it gets HOT. Both asymptotic guards were the wrong way round: $u > 30$ returned 1.0 (“fully H_2 ”) where the fit gives 0, and $u < -30$ returned 0.0 (“fully atomic”) where it gives 0.8384. The routine's own header stated the correct intent (“very cold $\rightarrow$ 1, very hot $\rightarrow$ 0”), so the code contradicted its comment as well as its equation, and the expression is flat at 0.838403 for every $u \leq -18$, which makes the cold guard a discontinuity of 0.84 in mixing ratio at its own threshold.

Only the cold branch is reachable ($u < -30$ is $T \lesssim 529$ K at 3.7×10^{-8} bar; $u > 30$ needs $p < 10^{-24}$ bar). There it handed every cold cell a fully atomic composition as its “dense molecular limit” — and that value is the $\lambda \rightarrow 0$ seed of the constrained continuation and the molecular-basin retry seed of `ioniz_eq`, i.e. precisely the two places that decide which basin a cold cell is solved from.

Scope, measured. The other three call sites are the base boundary condition, the initial condition and the Tier-1 column, all of which need a base below ~ 560 K to trip it. Of the 942 LHS 1140 b run directories — the only ones in the tree at $T_{\text{eq}} = 226$ K — 554 hand the lower atmosphere over as a profile and 5 supply `q_H2_base`, and **none of the remainder has molecular chemistry enabled**, so no archived run reached the guard through those paths. No matrix golden has a cell below 560 K (the coldest is 863.6 K), so the matrix cannot reach it either.

What closed cell 387, and what did not

The first four items below rest on the stage-3b Phase-A/B diagnostic run on the saved cell-387 state and were not re-measured for this section; they are marked because the conclusion they support — that none of the three numerical explanations was the cause — is the load-bearing claim here. The rate magnitudes, the channel ratios and the guard analysis were re-derived independently.

- **Not the charge-exchange cancellation.** At the cell's own temperature the two coded rates are 1.6991×10^{-60} and 1.7452×10^{-15} — 36 orders apart, so there is nothing to cancel. The H^+ row is carried by hydrogen recom-

bination, ten orders above either term. Deleting the pair outright moves the scaled Jacobian's condition number from 1.17131×10^{11} to 1.17130×10^{11} and the full residual from 2.2255×10^{-4} to 2.2261×10^{-4} .

- **Not residual precision.** Summing every signed term of the H^+ and He^+ rows in `real*16` and comparing with `binary64` gives relative summation errors of 1.6×10^{-16} and 1.3×10^{-16} .
- **Not the finite-difference step**, although that was a real defect: with it corrected the cell still failed, and it still failed with the step varied by a decade.
- **Not a warm wrong-basin convergence.** The 59 warm cells whose H_2 was exactly zero were never promoted to the constrained solver at all; the same states appear under the §113 binary, which has no such solver; and they are 100% IONIZED — $n_{HI} = 0$, every hydrogen nucleus in H^+ — so $H_2 = 0$ is the exact root of its own balance row, and the thermochemical fit, which describes a neutral gas in equilibrium, does not apply to them.
- **The missing channel and the inverted guard closed it.** With both in place the arm's step-0 event is gone and the run carries zero non-root acceptances.

What is NOT resolved, and is handed on

Cells in the $He/H = 1$ arm still reach the temperature floor of the semi-implicit energy update (0.01 in code units; 11.4 K at this arm's $T_0 = 1140$ K). The trajectory shows two distinct populations: a cold-start transient that RECOVERS (127 cells on the floor at step 10, 3 by step 2000) and a late re-collapse that does not (1 at step 4000, 8 at 6000, 31 at 8000, 38 at 12000), tracking the growth of the supersonic layer over the same interval (293, 325, 349 and 385 of 502 cells supersonic; base Mach 88 to 108). The order of events is measured and is the opposite of the natural guess: in every cell tracked across its promotion the temperature is already collapsing while the gas is still fully molecular (82.1 K, 38.1 K and 39.9 K at the last sample before promotion, with $n_{H_2} \sim 1.8 \times 10^{13}$), and the constrained solve fires afterwards. **The promotion is a symptom of the collapse, not its cause**, and what drives the collapse is not identified here. It belongs to the boundary-condition work of §11.5-C/D, not to the chemistry.

A cell resting on that floor has not converged to a physical temperature: the update overshoot and the clamp absorbed it, and at 11.4 K the network is frozen, so its composition is not constrained by its reaction balance either. The clamp was silent. It is now counted and reported at the end of every run — activations, distinct cells, first and last step — for the same reason §113 made non-root acceptances loud.

A caveat that applies to every arm number in this section. The $He/H = 1$ arm carries `q_H2_base` 0.75, and the largest H_2 mixing ratio its composition admits is $0.5/(0.5 + He/H) = 0.3333$. The input is therefore $2.25\times$ over its own ceiling and the code has been silently capping it, so this arm was not a clean probe of the window. Nothing in the preceding subsections depends on it — the guard bug, the adopted channel, the cell-387 diagnosis and the matrix goldens are all independent — but the floor-cell trajectory, the η sweep and the collapse ordering reported here were all measured on that arm and are qualified accordingly. Re-specifying the arm with an admissible `q_H2_base` and re-measuring is the first item of the window block.

A residual sensitivity remains and is stated rather than hidden: with the guard fixed, the floor population of this arm is 38, 36 and 27 cells at $\eta = 10^{-4}$, 10^{-5} and 10^{-6} , against 62, 27 and 31 before the fix. The spread narrows from a factor

2.30 to 1.41 but does not close, so which already-collapsing cells end on the floor still depends on a derivative step inside its own accuracy plateau. It does not touch the mass-loss rate: $\log_{10} \dot{M}$ is 8.89 in all three, and in every stage variant measured.

No acceptance test against the dissociation equilibrium was added, and the measurement is the reason. Such a test would have to establish that a cell is shielded, neutral AND chemically equilibrated before the thermochemical fit is a valid yardstick. At step 10 of this arm the fit calls all 59 fully ionized cells wrong and would force H₂ onto gas with no neutral hydrogen at all, and 47 of them remain atomic to the end of the run. A criterion that cannot cheaply establish those three conditions is a new defect, not a guard.

Verification

- **The guard fix changes nothing in the matrix**, as predicted from the fact that no golden has a cell below 560 K: the eight cases were re-run with it and compared file by file against the same eight produced without it — **16/16 byte-identical**.
- **Goldens refreshed once, at the end of the series**. Provenance, because every later comparison rests on it: they were produced by EXHALE . x built from a single compiler, GCC (Ubuntu 13.1.0-8ubuntu1~18.04) 13.1.0, verified by the version strings in the binary itself, running all eight cases single-threaded. A stray module compiled by a second gfortran was found in build/ during this work; it was written by a make that aborted before linking, so it never entered a binary, and the tree was cleaned and rebuilt from one compiler before anything was snapshotted.
- **The refreshed goldens reproduce from an independently clean build**: re-running the matrix with a freshly cleaned single-compiler build, and with the temperature-floor counter added, gives **8/8 byte-identical, PASS**, in 67 m 20 s. That one run establishes both that the counter does not perturb any result and that the snapshot itself was not taken from a contaminated build.
- **He/H= 1 arm**, 12000 steps: zero non-root acceptances (nine in §113), every promotion accepted as a root, $\log_{10} \dot{M} = 8.89$. **He/H= 10 arm**: 5 non-root acceptances, longest streak 1 sweep, $\log_{10} \dot{M} = 8.84$. Element budgets close to 4×10^{-16} in both.
- **Cost of the promotion path**: 485 cells promoted and 3191 field solves in 0.084 s over a 12000-step arm; on the matrix, mol_metals promotes 48 cells (1887 field solves, 0.202 s) and mol_ir_bands 26. Five of the eight cases promote nothing at all. Matrix wall time is unchanged within the spread of the machine (67–69 minutes across the runs of this series).
- run_fcheck.sh: **CLEAN** (HD 209458 b, $\log_{10} \dot{M} = 9.60$, no runtime trap, production build restored). The same -01 -fcheck=bounds,do,mem build was additionally run on the two molecular configurations for 1500 steps each — the mol_metals gate and the He/H= 1 arm — with **no runtime check fired in either**.
- **The temperature-floor counter behaves as intended**: silent on mol_metals, which never reaches the floor, and loud on the He/H= 1 arm, which closes 1500 steps with energy floor WARNING: 176356 activation(s) in 257 distinct cell(s), steps 0 to 1499. That population is the cold-start transient above seen from the other side: it was there before, and until now nothing said so.

115. Making the chemistry solver consistent with itself: detailed balance for H_2 , a finite-difference step with a floor, and a constant block that was losing digits (2026-09-01)

The judgement first. Section 114 closed the third stage of the supersonic-molecular-base series with the chemistry solver working but two of its inputs still inconsistent with each other, and with a measuring instrument — the $\text{He}/\text{H} = 1$ arm — whose floor-cell population depended on an arbitrary derivative step. This section cleans the instrument before the boundary condition is measured with it. Four changes, each measured on its own:

- **H_2 dissociation is now derived from recombination by detailed balance** rather than fitted independently. The two fits were the same channel in opposite directions with validity ranges that do not overlap (77–300 K and 2500–8000 K), and we evaluate in the gap.
- **The constrained solver’s finite difference has a real absolute floor.** The previous `u_shift` offset had only moved the collapse point, not removed it.
- **Every real*8 constant carries a d exponent.** Twenty of thirty were default-real literals, and six were losing digits — including the polytropic index and π .
- **The Lyman-Werner band fraction is evaluated in closed form** from the published fit rather than reconstructed by quadrature, and its warning now says what is actually true.

$\log_{10} \dot{M}$ is unchanged by all four, everywhere it was measured.

H_2 dissociation from detailed balance

R15 ($\text{H} + \text{H} + \text{M} \rightarrow \text{H}_2 + \text{M}$) and R12 ($\text{H}_2 + \text{M} \rightarrow \text{H} + \text{H} + \text{M}$) are the same channel, so detailed balance applies to them exactly — unlike the $\text{H} \leftrightarrow \text{He}$ charge-exchange pair of §114, whose two directions are different channels. But they were independent fits: Ham, Trainor & Kaufman (1970) J. Chem. Phys. 53, 4395 for recombination, measured over **77–300 K**, and Baulch et al. (1992) J. Phys. Chem. Ref. Data 21, 411 p. 550 for dissociation, recommended over **2500–8000 K**. The molecular layer sits at roughly 1000–2000 K, in the gap between them, so nothing forced the pair to be consistent and it was not.

R12 is now $k_{\text{diss}}(T) = k_{\text{rec}}(T)/K_{\text{eq}}(T)$.

The anchor is the recombination side, which makes a blocking question moot. Anchoring on dissociation would require knowing whether Baulch’s coefficient is a vibrationally equilibrated rate, since that decides between branches differing by orders of magnitude. Anchoring on recombination removes the question: the H_2 partition function inside K_{eq} contains the vibrational excitation by construction. The premise it does require — that the gas is vibrationally equilibrated — holds by five to nine orders: critical densities for vibrational equilibration are 10^4 – 10^8 cm^{-3} at 1000 K, against a measured molecular-layer density of $5.4 \times 10^{13} \text{ cm}^{-3}$ in the matrix cases and 7.2×10^{14} in the arms.

The primary source settles it anyway. Breshears & Bird (1973), the shock-tube measurement behind Baulch’s recommendation, describe their coefficient as “the rate corresponding to conditions of vibrational equilibrium in the

thermal sense but with negligible extent of dissociation”, applicable “during the ‘steady state’ which follows the induction period”. More to the point, **they impose detailed balance themselves**: their rate law is written with an explicit $k_d/k_r = K_{eq}$. What this section does to the pair is what the original measurement was interpreted with.

The recombination coefficient is Cohen & Westberg (1983) J. Phys. Chem. Ref. Data 12, 531, $2.8 \times 10^{-31} T^{-0.6} \text{ cm}^6 \text{ s}^{-1}$ over 50–5000 K, rather than Ham’s $8.0 \times 10^{-33} (300/T)^{0.6}$. The criterion is validity range: theirs covers the layer, Ham’s requires extrapolating three to seven times above its measured range. The two agree on the exponent, and Cohen & Westberg say why — “the low temperature (77–300 K) data for $M = \text{H}_2$ are in reasonable agreement and suggest a temperature dependence of $T^{-0.6}$ ” — so the choice keeps Ham’s physics and drops only the extrapolation. Their amplitudes differ by 14%; Baulch p. 495 independently recommends $2.7 \times 10^{-31} T^{-0.6}$ over 100–5000 K, 3.6% away, and both reference networks in the workspace carry that value.

K_{eq} is evaluated from statistical mechanics — an explicit rovibrational sum over the Huber & Herzberg levels with ortho:para 3:1 nuclear-spin weights and $D_0 = 36118.11 \text{ cm}^{-1}$ — tabulated over 100–20000 K and interpolated, since the direct sum per cell would cost order 10^9 exponentials per run. Measured interpolation error is at most 4.7×10^{-4} .

Three independent confirmations of K_{eq} , sharing no route:

against	1000 K	2000 K	2500 K	3000 K	5000 K
the code’s own q_h2_equilibrium, dilute		0.993	1.029	1.082	
NIST-JANAF Shomate	1.010 (600 K)				1.101
Cohen & Westberg’s own two directions	0.966	1.063	1.118		

and D_0 gives $\Delta H^\circ(298.15 \text{ K}) = 435.8 \text{ kJ/mol}$ against Baulch’s 436 and Cohen & Westberg’s 436.00 ± 0.01 .

The strongest evidence is what the derived rate does against Baulch inside Baulch’s own range. It reproduces it there and departs only outside:

$T \text{ [K]}$	1000	1500	2000	2500	3000	5000	8000
derived / Baulch	0.11	0.36	0.61	0.82	0.97	1.18	1.05

A construction that merely fitted something would not land on the evaluation across its whole validity range and diverge only below it.

A pitfall recorded, because it produced a wrong answer twice before it was caught. Comparing a dissociation rate against another network’s is meaningless unless the third body matches. An $M = \text{H}$ rate and an $M = \text{H}_2$ rate differ enormously at low temperature, and a comparison made across that mismatch produced an apparent factor of 200 in one direction, while the matched comparison puts our value between the two existing determinations. The honest collider spread, inside the range where the measurements live, is $2\text{--}9\times$ — and for RECOMBINATION atomic H is the *more* efficient collider, $1.98\times \text{H}_2$ at 1000 K. Both ratios are forced to agree by detailed balance, and they do.

Validity, stated at the code site. What is derived is the rate for third body H_2 , because both underlying fits are H_2 -as-third-body. EXHALE passes the total density as M, so every collider is assumed to behave like H_2 — best in the molecular base where H_2 dominates, weakest above the H_2 front where atomic H does.

A finite-difference step with an actual floor

fdjac1 chooses $h = \sqrt{\varepsilon}|x_j|$, falling back to $\sqrt{\varepsilon}$ only when x_j is exactly zero. The constrained solver's unknowns are logarithmic DEPARTURES, so §114's `u_shift=1` merely moved the point where that rule collapses — from the rung start to $u = -1$, i.e. $n = 0.368 n_{\text{ref}}$, a value a solve traverses routinely. Section 114 said so; this section fixes it.

A `unit_step_floor` selector is now threaded through fdjac1 and hybrd. hybrd1 passes false and keeps MINPACK's rule bit-for-bit; the constrained solver passes true and gets $h_j = \text{fd_etamax}(1, |v_j|)$. Both rules live in one internal function so the dense and banded branches cannot drift apart. No global flag was used: it would have to be saved and restored around every call and would not be safe under the OpenMP-parallel ionization sweep.

The verification has a trap in it worth recording. The obvious byte-identity gate is vacuous here: `wasp_full` has no molecular chemistry, and `mol_base_handoff` never promotes a cell, so neither ever enters the constrained solver and neither can exercise the new argument. The test that means something is the He/H=1 arm, which promotes 503 cells and spends 3345 field solves in the routine; run with the pre-change binary and with a build passing false from the constrained solver, its output is byte-identical. A verification design has to be checked against the path it claims to cover.

u_shift is kept, and its stated reason was wrong. Removing it costs 22 cells at the temperature floor against 0. But it does not carry the difference step — the floor does that now, and sweeping the step over two decades moves neither count. What it carries is MINPACK's TERMINATION: hybrd stops on $\delta \leq \text{xtol} \cdot \text{xnorm}$ with $\text{xnorm} = \|\text{diag} \cdot x\|$, a relative test that cannot fire for departure variables which start and end near zero, so without the offset the solve runs past its own root to a slow-progress exit.

What this change did not do. It was aimed at an η -dependence in the arm's floor-cell population — 38, 36 and 27 cells at $\eta = 10^{-4}, 10^{-5}, 10^{-6}$. That dependence was already gone before the floor was implemented: the detailed-balance work above had driven the population to zero. The floor is justified because the step rule was wrong as written, which is an argument about correctness and independent of the symptom.

A separate defect fixed in passing: hybrd declared its `params` dummy not at all, making it an implicit single-precision scalar, while the arrays actually passed carry 40 to 60 elements; fdjac1 declared `params(11)` and hybrd1 `params(25)`. All three are now `params(*)`. Benign under address passing, but wrong as written and invisible to the compiler.

Constants that were losing digits

Twenty of the thirty `real*8`, parameter declarations in `src/modules/init/parameters.f90` carried no `d` exponent, so their literals were parsed in single precision and the extra digits discarded despite the declared type. Most were harmless, quoted to seven digits or fewer. Six were not: π , the polytropic index $g = 5/3$, `erg2eV`, `c_light`, `parsec` and `AU`. π and g enter the equation of state and the geometric factors directly, so a relative error of 2.4×10^{-8} in g and 2.8×10^{-8} in π propagated silently.

All thirty now carry a `d` exponent — including those already within single precision, because one kind of thing gets one rule — and g is written as the exact rational `5.0d0/3.0d0`, which also states the intent. Values are otherwise unchanged; this is a notation correction. A sweep of the other modules found no further lossy literal.

The Lyman-Werner band fraction in closed form

The fraction of the 912–1110 Å band that H₂ lines remove was built by integrating $\sigma_{\text{pump}} f_{\text{shield}} n_{\text{H}_2}$ along the star-ward ray. That integral has a published closed form — Draine & Bertoldi (1996) eq. (39), the analytic integral of the same f_{shield} (their eq. 37) the code already uses — so it is now evaluated directly from the cell’s H₂ column and Doppler parameter. The quadrature array and its column integration are gone, and with them the grid dependence of a quantity that has an exact expression.

Three things were investigated and deliberately NOT changed, because each would have been wrong:

- **The normalization is correct.** $\sigma_{\text{pump}} = \sigma_{\text{lw}} / p_{\text{diss}} = 2.5570 \times 10^{-17} \text{ cm}^2$ against 2.5579×10^{-17} implied by their Table 1 and Table 2 row (0,1) — a ratio of 1.00036. An independent check on the unshielded rate agrees to 2.3%. Dividing by p_{diss} is required by the difference between their eq. (35) and eq. (40), and the two are a matched pair from one table row: changing either alone would make the product 13% wrong.
- **$1 - A$ is exact, not a first-order expansion.** A is a summed dimensionless equivalent width, not an optical depth, so for non-overlapping lines across a flat band the removed fraction is A . Replacing it with e^{-A} would have been a regression, treating discrete saturated lines as a uniform continuum.
- **No line-overlap term was added.** The fit already contains one: their eq. (29) saturates the summed width at $W_{\text{max}} = 0.2$, and the factor 5 in the archived $\exp(-5 \sum W)$ is $1/W_{\text{max}}$. Adding another would double-count.

What the warning was actually reporting. The old message said the lines had removed the whole band and that the fit “does not bound the photon budget”. Neither is true. The eq. (39) asymptote is $1.05(1 + 0.0117 b_5)$, i.e. 1.084–1.105 at our Doppler parameters — the authors’ own fit slack. Evaluated up to the largest column at which they compared the fit against an exact calculation, $3 \times 10^{21} \text{ cm}^{-2}$, the band fraction stays below unity (0.956); the excess appears only beyond that, and the mol_lyman_werner case reaches 5.3×10^{21} . The warning now says that, printing the column and that bound. Note that DB96 state no explicit validity interval for this function: “ $10^{14} < N < 3 \times 10^{21}$ ” is not a published range, and the code previously described it as one. A is clamped at 1 as a safety net rather than as the fix.

A residual that this change does not remove. DB96 cap the overlap-corrected width at $W_{\text{max}} = 0.2$, which in band-fraction units is 1.018, while the eq. (39) asymptote reaches 1.084–1.105 over our Doppler range — the fit overshoots the ceiling the same paper sets, by 6.5 to 10.5%. That is an internal inconsistency of the published fit, not of our use of it, and it survives this change because it is the same fit. An earlier draft quoted a different figure here, that we sit “about 12% high” against DB96’s own tabulated widths; that comparison could not be reproduced and is withdrawn.

A measurement discipline this section had to learn

The constants change is the cleanest perturbation available — 2.4×10^{-8} on one constant, no physics touched — and it was measured twice:

case	state	output response
wasp_full	converged, $du = 1.0 \times 10^{-3}$	1.9×10^{-7} , same step count, same $\dot{M}$
mol_metals	12000-step snapshot, $du = 2.8$	5.1×10^{-3}

The converged case responds at the order of the input. The snapshot amplifies it by about 10^5 , because a non-converged transient grows small perturbations. **Six of the eight regression cases are such snapshots.** That makes them excellent byte-identity detectors — any change is amplified into visibility — and useless as magnitude measurements. Several change sizes quoted in §114 came from snapshots and are transient amplification rather than physical change; where they are quoted there they now say so.

The sharpest demonstration this series produced. On the He/H=1 arm, with the Lyman-Werner path inactive and everything else identical, the same 2.4×10^{-8} perturbation sent the run to an entirely different 12000-step state: $\log_{10} \dot{M}$ 8.89 against 11.04, median temperature 981 K against 2875 K, du 13.5 against 121.

Neither of those states is a solution, and the wording matters. Both du values stand four to five orders above the 10^{-3} convergence threshold, so what was measured is not a choice between two steady solutions but a transient that has not converged going two different places. The code prints “steady-state Mdot” for both and the premise of that label holds in neither. Whether this configuration possesses two genuine steady solutions is NOT settled here; settling it means driving each state to Newton convergence, a stage-4 item. Section 93 records the same split at He/H=1 — 11.06 for one build, 8.91 for another — but that table carries no du , so the convergence state of those runs is unrecorded and is not assumed here.

What this does not explain, checked before it was assumed. The η -dependence that motivated this block was not this effect. All six of those runs sit in the 8.89 state with the same median temperature and the same du to four digits; only the floor count moved. B1 likewise removed the floor cells within that state rather than by selecting between states. Measured state dependence: projected/handback acceptances (0 in both), promotion-to-root rate (479/481 against 424/425) and converged-root totals (0.015% apart) are insensitive to which state a run lands in; energy-floor activations (218,722 in 331 cells against 62,604 in 263) and roots-without-convergence (1488 against 611) are sensitive. Acceptance statements therefore carry no condition; thermal statements carry the state as one.

The consequence for how the arm may be used is blunt. It is a legitimate probe of *acceptance behavior*, which is what §§114–115 used it for. It is **not** an instrument for any quantitative statement, because an arbitrarily small perturbation relocates it. Its $\dot{M}$ records where a non-converged run happened to be at step 12000; it measures nothing. Taken with the fact that this arm’s q_{H2_base} is 2.25 times its own composition ceiling and has been silently capped throughout, no number from it should be quoted as a physical result until it is re-specified and driven to convergence.

Verification

- **The matrix moves, and that is the point of the change.** Against the §114 goldens all eight cases differ, as they must: B1 alters a reaction rate every molecular case uses, and the constants correction touches the equation of state in all of them. Goldens were refreshed once, at the end of the block.
- **The refreshed goldens reproduce from a clean build: 8/8 byte-identical, PASS**, 17 of 17 files.
- **A build-contamination incident, and how it was caught.** Between the pre-refresh matrix and the snapshot, a separate worker ran `make` without `PATH=/usr/bin:$PATH`; the conda gfortran 15.2.0 was selected, the link failed on a LAPACK mismatch, and it left EXHALE.x deleted and 15.2.0 objects in `build/`. The matrix outputs were unaffected — every one is timestamped 07:30–08:28 against a contamination at 10:20 — so they remained

valid to snapshot. Recovery was make `distclean`, a rebuild with the correct compiler, and verification from the binary's own version strings that only 13.1.0 appears. The same worker had edited two source files claiming comment-only changes; since a `git diff` could not separate those from the block's own edits, the post-refresh check was used as the test, and its 8/8 byte-identical result is what establishes the claim was true. This is the second time in this series that concurrent work threatened a refresh, the first being a duplicated section in the $\LaTeX$ source; both were caught by verification steps rather than by noticing the interference.

- **run_fcheck.sh: CLEAN** (HD 209458 b, $\log_{10} \dot{M} = 9.60$, production build restored), plus the same `-O1 -fcheck=bounds,do,mem` build on the `mol_metals` gate and the $\text{He/H} = 1$ arm for 1500 steps each — no runtime check fired in either.
- **The three- η sweep, for the record.** On the final build the arm closes with zero cells at the temperature floor at every η , against 38, 36 and 27 before this block. The population is now η -independent because it is empty, and that was achieved by the detailed-balance work rather than by the finite-difference floor.
- **Each change measured on its own.** B1: the H_2 front moves one grid cell, $\max n(\text{H}_3^+) \times 1.02$, $\log_{10} \dot{M}$ unchanged; the arm's $n(\text{H}_3^+) \times 2.43$ and its energy-floor activations fall 37%. B2: byte-identical on a run that actually exercises the new argument. Constants: 1.9×10^{-7} on the converged `wasp_full`. Lyman-Werner: byte-identical on `mol_lyman_werner`, the only case that reaches it.

What this series taught, which is not about any one change

Four defects in this series were diagnosed as numerical and were not. Cell 387 of the $\text{He/H} = 1$ arm was attributed in turn to a floating-point cancellation, to a finite-difference step, and to wrong-basin convergence in warm cells; it was closed by a missing reaction channel and an inverted asymptotic guard. Thirty-seven per cent of the thermal-collapse pressure in that arm came off from a rate correction that leaves $\dot{M}$ untouched. The η -dependence that motivated the finite-difference floor had already been removed by detailed balance before the floor existed.

Every numerical fix in the series is independently justified and each one corrected something genuinely wrong. But **not one of them fixed the symptom that motivated it**. Three of the numbers that motivated work in this series did not survive measurement — a claimed $1.3\text{--}1.7\times$ inconsistency, a $5801\times$ collider spread, and an attribution of the η -dependence to a change of state — and each is recorded as withdrawn where it was quoted. The risk is not doing the work; the work was right. The risk is letting the symptom's number into the record as though it were established. A residual that resists the solver is first evidence that the equations are not the right equations, and only then evidence about how they are being solved.

116. H₂ self-shielding: the fit we were using is calibrated for cold gas, and our molecular layer is not cold (2026-09-01)

Superseded in part by Sec. 132. (2026-09-02). The rate no longer multiplies a constant cross section by a shielding factor: the level-resolved cross section itself is tabulated, and `sigma_lw`, `p_diss_lw` and `sigma_lw_pump` have been removed from the module.

[SUPERSEDED 2026-09-02 by section 122..] The replacement of DB96 by Richings, decided here on the authors' 100 K comparison, was measured against a level-resolved CLOUDY calculation at the temperature of our own layer and did not survive it: the rate now reads a table of that calculation and neither fit sets it. What this section establishes and section 122. keeps — the 100 K premise itself, the thermal-only Doppler width, and the band share A staying on DB96 — all stand.

The judgement first. `h2_self_shielding_factor` implemented Draine & Bertoldi (1996) eq. (37), which is calibrated for gas in which only the lowest rotational states of H₂ are populated. Our molecular layer runs at 865–2724 K. Richings, Schaye & Oppenheimer (2014) MNRAS 442, 2780 compared that function against a level-resolved CLOUDY calculation, found it wrong for exactly this reason, and published a temperature-dependent replacement. Adopting it changes the H₂ photodissociation rate by up to a factor of 4 in the deep molecular layer. This is a correction of wrong physics, not an addition of new physics, so it is default-on and the goldens move.

The correction is **not one-signed**, which is the first thing to say because the headline number is misleading alone. Where the column is high the old function over-shielded and we were over-dissociating; near $N_{\text{H}_2} \sim 10^{16} \text{ cm}^{-2}$ the new function shields less, and we were under-dissociating. The realized range across the cells of the affected case is 0.72 to 4.00.

Why DB96 is wrong here, in the authors' words

Richings et al. compared DB96 and the Wolcott-Green et al. (2011) modification against CLOUDY's level-resolved "big H2" model and report: "We find that neither function produces satisfactory agreement with CLOUDY. For example, both overestimate $S_{\text{self}}^{\text{H}_2}$ compared to CLOUDY by a factor of ~ 3 at H₂ column densities $N_{\text{H}_2} \gtrsim 10^{17} \text{ cm}^{-2}$ in gas with a temperature $T = 100 \text{ K}$ ". The stated reason applies to us directly: DB96 is accurate only for cold gas in which only the lowest rotational states are populated. At 100 K that is nearly true and the error is already a factor of 3; at our 865–2724 K the rotational ladder is populated far beyond the states the fit assumes.

What was adopted

Their equations (3.12)–(3.15), read from the published paper:

$$S_{\text{self}} = \frac{1 - w(T)}{(1 + x/b_5)^{\alpha(T)}} e^{-5 \times 10^{-7}(1+x)} + \frac{w(T)}{\sqrt{1+x}} e^{-8.5 \times 10^{-4} \sqrt{1+x}}, \quad x = \frac{N_{\text{H}_2}}{N_{\text{crit}}(T)}, \quad b_5 = \frac{b}{10^5 \text{ cm s}^{-1}}$$

with $w(T) = 0.013 [1 + (T/2700)^{1.3}]^{1/1.3} \exp[-(T/3900)^{14.6}]$, $\alpha(T) = 1.4$ below 3000 K, and $N_{\text{crit}}(T)/10^{14} = 1.3 [1 + (T/600)^{0.8}]$ below 3000 K; the higher-temperature branches are carried but unreachable here.

Verified against an independent evaluation at four points, thermal Doppler width, DB96 against Richings: 9.632×10^{-7} / 1.785×10^{-7} at 945 K and $N_{\text{H}_2} = 4.36 \times 10^{21}$ (ratio 5.40); 2.242×10^{-7} at 1140 K (4.30); 2.669×10^{-7} at 1307 K (3.61); and 5.351×10^{-5} / 1.917×10^{-5} at 1140 K, $N_{\text{H}_2} = 10^{20}$ (2.79). The compiled routine reproduces all four.

Two decisions, both settled from the paper rather than by judgement

The Doppler parameter stays thermal. Richings includes turbulence at $b_{\text{turb}} = 7.1 \text{ km s}^{-1}$ for interstellar gas, and it would have been easy to copy. Appendix B says not to: “The agreement between our best-fitting self-shielding function and CLOUDY is poorer when turbulence is included, as it was fitted to the purely thermal Doppler broadening case.” Thermal-only is therefore inside the fit’s calibration, which is a better reason than the one otherwise available (that a 1-D planetary wind carries no turbulence model and 7.1 km s^{-1} is a molecular-cloud value).

The Elwert radiation-field exponent was not adopted, and it is not a rival correction. CLOUDY’s default replaces DB96’s fixed exponent 2 by $-0.05 \log G_0 + 2.25$, and our G_0 is far above interstellar, so it looked relevant. Measured, it is inoperative here: that exponent belongs to DB96’s FIRST term, which supplies 96% of the function at $N_{\text{H}_2} = 10^{14}$ but 1.2×10^{-7} of it at our 4.4×10^{21} . At our columns the Elwert form and DB96 agree to five decimal places, and its largest effect anywhere is 2.4% near $N_{\text{H}_2} = 10^{16}$. The two corrections act on different terms in different column regimes: the radiation-field one at low columns, the temperature one where we operate. Adopting Richings leaves no Elwert-shaped gap.

Two shielding functions now coexist, deliberately

The band fraction A that H_2 removes from the shared 912–1110 Å beam — the quantity H_2O and OH need in order to receive the remainder — is DB96’s own equivalent-width integral (their eq. 39), and its normalization reproduces their Table 1 and Table 2 values to 0.03%. Richings publishes no corresponding equivalent-width form. **A therefore stays on DB96 and only the dissociation rate moves to Richings.** Each function is used where it was calibrated: Richings for the rate he fitted to level-resolved CLOUDY, DB96 for the band share that is that fit’s own integral. The module states this so it cannot be mistaken for an oversight, and the band share is reached only when Oxygen chemistry is enabled, which no regression case enables.

Validity, stated plainly

The paper says its fit “agrees with CLOUDY to within 30 per cent at 100 K for $N_{\text{H}_2} < 10^{21} \text{ cm}^{-2}$, and to within 60 per cent at 5000 K for $N_{\text{H}_2} < 10^{20} \text{ cm}^{-2}$ ”. **Our conditions — 865–2724 K, columns to 4.4×10^{21} — are outside both stated ranges.** The justification for adopting it anyway is not that it is verified where we work, because it is not. It is that DB96 is verified to be *wrong* where we work, by a factor of 3 at 100 K and for a stated physical reason that grows worse with temperature. Replacing a function known to be wrong with one fitted to the right calculation, and recording that we are beyond its tested range, is the honest position; claiming a verified rate would not be.

And this choice has not been decided by measurement, which is worth saying so that the next person does not re-argue it. Neither fit is verified at our conditions: one is known wrong, the other is outside its stated range. That is a choice made on the better evidence available, not a determination. It can be determined, by the route Richings took:

their fit was made against a level-resolved CLOUDY calculation, and that calculation is available to us. Running it over our own (T, N_{H_2}, b) grid — 900–2700 K, 10^{16} – 5×10^{21} cm $^{-2}$, thermal width, no dust, at our radiation-field strength — yields the self-shielding factor directly, and both fits can then be measured against it rather than compared with each other. Matching G_0 makes the same run settle the Elwert axis. It needs no new code in this tree, only a calculation run outside it, and it would also close the two-fit arrangement above by supplying the calibration A currently depends on DB96 for.

What it changes

Only one regression case can reach this code. `F_LW_star` is set by the `Stellar LW flux` key alone and defaults to zero, and of the 32 case directories exactly one — `mol_lyman_werner` — supplies it. The other seven matrix cases were measured byte-identical across the change, which is the structural statement that the correction is confined to the path it belongs to.

quantity	DB96 (before)	Richings (after)
H ₂ front, largest r with $n_{\text{H}_2} > 10^{10}$	1.167902 R_p	1.165804 R_p
max N_{H_2}	4.3612e21	4.4112e21
max $n(\text{H}_3^+)$	6.69637e4	6.69649e4
minimum temperature	945.4 K	884.8 K
final	count=12000, du=2.2398	count=12000, du=2.2057
$\log_{10} \dot{M}$	10.57	10.58

The He/H = 1 arm is **byte-identical in all seven output files** — it carries no Lyman-Werner flux, so this correction does not touch the arm’s thermal state, and none of the arm results of §§114–115 are affected by it.

The realized size of the correction, cell by cell, is what the headline number does not convey. Across the 504 cells the ratio DB96/Richings runs from **0.72 to 4.00**: the old function over-shielded by up to a factor of 4 in the deep layer, where we were over-dissociating H₂, and under-shielded near $N_{\text{H}_2} \sim 10^{16}$, where we were under-dissociating by up to about 1.4. Restricted to the 248 cells that hold H₂ the range is 1.83 to 4.00. The 5.4 of the spot comparison requires 945 K at 4.36×10^{21} , and the converged deep cells sit nearer 1200 K.

The changes to observable structure are small — the front moves by one grid cell and $\log_{10} \dot{M}$ by 0.01 — which is worth stating plainly: this correction matters because the rate was wrong, not because it moved the answer far. Its importance is that the molecular layer’s photodissociation now rests on a fit calibrated for gas of the right temperature, which is a precondition for the base-boundary work that follows rather than a result in itself.

Verification

- **Matrix (the P18 run; its log is named `blockC_check1.log`, which predates the decision to take P18 ahead of the C block — it is the P18 matrix).** Against the §115 goldens, seven of eight cases are byte-identical and only `mol_lyman_werner` moves, matching the reachability analysis exactly. All eight ran on one binary: the last source edit was 13:29:49, the rebuild 13:34:48, and the first case output 13:48:29, so no mid-matrix rebuild occurred.

- **Goldens refreshed once**, with the pre-P18 set preserved at backup/regression/golden_blockB_20260901/ beforehand. The refreshed set reproduces: make check returns **PASS, all cases byte-identical, 17/17 files, EXIT 0**.
- **run_fcheck.sh: CLEAN**, production build restored and verified to carry only GCC 13.1.0. The same -O1 -fcheck=bounds, do, mem build was run on the mol_metals gate and the He/H = 1 arm for 1500 steps each: **no runtime check fired in either**.
- **Unit check** against an independent evaluation, four points: the compiled routine reproduces DB96/Richings = 5.3967, 4.2954, 3.6089 and 2.7921 against 5.40, 4.30, 3.61 and 2.79 computed separately.
- **The f_shield column written by the run reproduces the independent Richings evaluation to 8×10^{-6} relative**, and the pre-change column reproduces DB96 to 4×10^{-6} — what the code uses is what was intended, not merely what was intended to be called.
- **Stale outputs, flagged not regenerated**: examples/18_oxygen_chemistry/ and examples/19_molecular_ir_bands/ carry a Lyman-Werner flux, so their stored results predate this correction. Two documents still describe DB96 as the rate's shielding function — docs/EXHALE_user_manual.tex and docs/molecular_hydrogen_treatment.tex. The second is the memo this section answers and needs a substantive rewrite rather than a term substitution, so it was deliberately left rather than patched.

117. The base H₂ fraction becomes the composition of the inflowing gas, and an impossible one is refused rather than capped (2026-09-01)

The judgement first. q_H2_base was an equation-of-state scalar: it removed the H nuclei bound into H₂ from the base particle count and did nothing else. The species state at the same ghost was solved independently, and the two disagreed. Measured on the molecular regression cases, the equation of state believed 92.51% of the base hydrogen nuclei were bound into H₂ while the chemistry delivered 99.97%, a 6.4% disagreement in the base particle count. That disagreement was not hidden in an internal variable: the ghost pressure is $(n_{\text{tot_bc}} + dp_{\text{bc}}) T_0$ at the pinned density, so a base that had been asked to sit at $T_0 = 1140$ K was marching at **1213.35 K**. The isothermal lower boundary condition was not isothermal, and had not been for as long as the molecular base has existed.

Two changes close it, and they act on the same boundary:

- **C1.** When a lower-atmosphere handoff states the base H₂ fraction, that value is imposed on the inflowing ghost species, and the base particle count is taken from the resulting species state. One number now describes the base, so the equation of state and the chemistry cannot describe different gas.
- **C2.** A requested q_H2_base above $0.5/(0.5 + \text{He}/\text{H})$ is refused at startup instead of being silently reduced to that ceiling.

Why the handoff wins over the local solve, argued without the fixture. The inflowing gas is not a piece of atmosphere the base cell's radiation field determines; its molecular partition was set lower down, where photodis-

sociation and mixing act and where the base cell cannot see. The measurement that shows this is in the tree already: the same VULCAN network, cut at two handoff levels, gives $q_{\text{H}_2} = 0.4237$ at 10^{-6} bar and 0.8020 at 10^{-4} bar (vulcan_work/pc_compare_p1/exhale_hd209/). The shallower level is far more dissociated. A shielded base cell solving its own local balance knows none of that, and drives H_2 to the fully molecular limit — which is exactly the 99.97% measured above. Upstream information therefore has to win, and that is what C1 implements.

What C1 changed, and the numbers it moved

`q_H2_base` is now the single definition of how molecular the base is, held in `base_h2_nuclei_fraction()` (`composition.f90`) as the fraction x_2 of base hydrogen nuclei bound into H_2 . The equation of state removes $x_2/2$ particles per H nucleus through `h2_bound_fraction`, and the species state imposes the same x_2 on both lower ghosts. The imposition reuses the row the molecular systems already had for a partition this cell does not own: the H_2 balance row is replaced by $x(4) - x_{\text{H}_2, \text{fix}}$ and every other row is solved against it, so the ionization stages stay consistent with the imposed partition rather than being imposed themselves. Both lower ghosts are pinned, because they are the same reservoir and `Rec_BC` builds face states from each. The particle count then comes back from the same species state, beside the electron count `dp_bc` that was already taken that way, through `calc_ntot` so that the counting policy keeps its single definition.

Measured on `mol_base_handoff`, base ghost, before against after:

quantity	before	after	intended
x_2 , H nuclei bound into H_2	0.999678	0.925114	0.925114
particles per (H+He) nucleus	0.536886	0.571429	$\text{ntot_bc} = 4/7$
base ghost temperature [K]	1213.3461	1140.0000	$T_0 = 1140$

The three lines are one statement: the composition mismatch was the cause and the 73.3 K temperature error was its symptom, and closing the first closes the second exactly. The end-of-run `ntot_bc_per_H` is $5.714285703774987 \times 10^{-1}$ against the analytic $4/7 = 5.714285714285714 \times 10^{-1}$ — they now agree to nine digits, the residual being the trace molecular ions the analytic expression omits, where they disagreed by 6.4% before.

Which branch imposes and which does not. Only a stated handoff imposes the composition. Without one the partition comes from the chemical-equilibrium fit, which is the solver’s own local estimate rather than upstream information; imposing that would hand the solver back its own answer. That branch reaches consistency the other way, through the particle count alone. The particle count is taken from the species state for every run that solves the molecular network, so “one particle count” is a single rule across both branches. A passive molecular base — EOS-only, species left atomic — keeps the analytic count, because taking it from atomic species there would silently undo the molecular base.

Reach: five cases move, three do not, and why

case	result	why
mol_base_handoff	moves	handoff-imposed base composition
mol_metals	moves	”
mol_lyman_werner	moves	”
mol_diffusion	moves	”
mol_ir_bands	moves	”
wasp_full	byte-identical	atomic; never enters the molecular path
wasp_he23off	byte-identical	”
lower_profile	byte-identical	molecular base off (particles/nucleus 1.000764)

The three invariant cases are what bounds the change: it reaches exactly the runs whose base is molecular and handoff-specified, and nothing else. The five that move are the size of the approved interface revision.

C2: the refusal, which changes nothing it does not refuse

For a mixture ratio $q_{\text{H}_2} = n_{\text{H}_2}/(n_{\text{H}_2} + n_{\text{H}} + n_{\text{He}})$ the attainable maximum at fixed He/H is $q_{\text{H}_2, \text{max}} = 0.5/(0.5 + \text{He}/\text{H})$, reached when every H nucleus is bound into H_2 . Above it the request is not a large value but an impossible one. `h2_bound_fraction` used to clamp x_2 to 1 silently, so an impossible input became a fully molecular base and the run continued on a state nobody had asked for. The clamp is gone and the refusal is at startup, on the resolved value whichever source produced it, with the requested value, the ceiling, He/H and the source printed, and with the requested q_{H_2} , the ceiling and the implied x_2 written to `EXHALE_resolved.out`.

- **8 of 8 regression cases byte-identical.** C2 refuses and does nothing else, which was its acceptance criterion.
- **The refusal fires** on the He/H= 1 diagnostic arm: requested 0.750000 against a ceiling of 0.333333, $2.25\times$ over, silently capped until now.
- **The boundary case is accepted.** LHS1140b/examples/scalar_base_cno sits at 99.88% of its own ceiling (0.19265465 against 0.19289121), because a handoff written from a nearly fully molecular column sits just under the ceiling by construction. The comparison therefore carries a round-off tolerance only (10^{-12}); anything looser would refuse a legitimate handoff.
- Every `base.inp` tracked in the tree was checked against the ceiling beforehand: **27 of 27 pass**, the largest ratio being the LHS1140b one above.

The fit that hides behind the cap: a solar-composition function used off-composition

Removing the clamp exposed something more general than one bad input. `q_h2_equilibrium` has a FIXED cold asymptote, $1.9845/2.3670 = 0.83840$, which is that fit’s approximation of the fully molecular mixing ratio AT SOLAR COMPOSITION. The attainable ceiling is not fixed: $0.5/(0.5 + \text{He}/\text{H})$ falls as helium is added. The two cross at

$$\text{He}/\text{H} = 0.5/0.83840 - 0.5 = 0.0964,$$

and above that the fit returns a value the element ratio cannot supply. Measured ratios of fit to ceiling: 0.971 at solar $\text{He/H} = 0.0793$, 1.000 at 0.0964, 2.515 at $\text{He/H} = 1$, 17.6 at $\text{He/H} = 10$. The $\text{He/H} = 1$ case is not a calculation but a direct run: with `q_H2_base` deleted the fit returns 0.838388 against a ceiling of 0.333333 and startup refuses it.

Three things follow.

- **Solar composition is inside the fit by only 3%.** That is the whole margin at the composition the fit was built for.
- **The entire $\text{He/H} = 1$ –1000 diagnostic ladder is outside it.** A molecular base anywhere on that ladder has to state `q_H2_base` explicitly; there is no fit to fall back on. This is a constraint on the window re-measurement, not only on one arm.
- **The silent clamp had been concealing exactly this** for as long as the molecular base has existed, turning an out-of-range fit into a fully molecular base without a word. This is the part of C2's value that goes beyond refusing one arm.

The validity range is now stated at the code site, above `q_h2_equilibrium` in `lower_column.f90`, because that is where the next reader will meet it.

The arm's replacement value follows from this, and is not another representative number. The $\text{He/H} = 1$ arm needs an admissible `q_H2_base`, and the choice is 0.300861, fixed by requiring that the fraction of H nuclei bound into H_2 be **the same** $x_2 = 0.9251143$ **the matrix cases carry** (the arm reproduces it to seven digits, at 0.9026 of its ceiling). The reason is that a He/H ladder is supposed to vary He/H . If the base molecularity moves along the ladder as well, the two variables are confounded and a measured trend cannot be attributed to either. Holding x_2 fixed makes the ladder an experiment in He/H alone. For contrast, the `q_H2_base = 0.30` used in the D2 experiment of `docs/supersonic_molecular_base.md` was chosen only for being “below the maximum 1/3” — nearly the same number, with no reason attached to it.

A separate defect: one flag was fixing three carriers

Found while implementing C1 and fixed as its own item. `ieq_cell%carriers_fixed` read as though it fixed the H_2 partition, and in `System_HeH_mol` it did. In `System_HeH_mol_metals` and in the constrained solver's layout and seed it fixed **H_2 , OH and H_2O together**. Reusing it for the base ghost would have imposed the oxygen carriers there as well, at whatever `x_oh_fix` and `x_h2o_fix` happened to hold — zero — and so would have zeroed the oxygen carriers of the base ghost in any run with oxygen chemistry on. That is reachable: `examples/18_oxygen_chemistry` and `examples/19_molecular_ir_bands` both enable it.

The flag is now two, `x_h2_fixed` and `x_ox_fixed`, named for what is imposed rather than for the one caller that first needed them. Both existing uses map onto it exactly — carrier transport sets both — and the base ghost sets only the hydrogen one. No regression case enables oxygen chemistry, so this fix moves nothing in the matrix; it removes a trap rather than a symptom.

Interaction with the §113 acceptance test

A pinned row does not satisfy a reaction balance, so the residual acceptance test of §113 could have condemned every base ghost as a non-root. It does not, and by construction rather than by luck: `normalized_reaction_residual` evaluates the residual by calling the same system routine the cell solve used, and that routine replaces the pinned row with $x - x_{\text{fix}}$, which is exactly zero at the solution. The 50-step probe reports no non-root acceptance and no warning of any kind.

What may not be claimed from the regression cases

The five molecular cases carry `q_H2_base = 0.75`, and their own `base.inp` header states what that value is: “No VULCAN run exists for this planet, so 0.75 is a representative photochemical value rather than a computed one”. It was chosen deliberately to differ from the equilibrium fit so that the regression detects a loss of precedence. As a golden and as a test of the mechanism it is valid, and the mechanism is what these cases guard.

It is not a basis for any physical claim about the base composition. Before C1 the value set a particle count; after C1 it sets the base species state, so anything that depends on how molecular the base is — the position of the H_2 front, $n(\text{H}_3^+)$, the layer structure — must be stated from a configuration whose handoff is `COMPUTED`. Every tracked production run is such a configuration: 22 VULCAN-written `base.inp` files under `vulcan_work/` and 5 under `LHS1140b/`, all written by the adapters from a photochemical column. The representative value appears only in the regression fixtures. This is recorded as a precondition on P23, the three-layer structure gate, which compares exactly those quantities.

Statements this change made false were corrected; dated records were not. C1 turned several descriptions of `q_H2_base` into falsehoods — most directly `input_read.f90`’s “It is an EOS anchor, NOT a composition pin: nothing holds H_2 at that value” — and those were fixed where they describe CURRENT behaviour: `parameters.f90`, `input_read.f90`, `docs/input_schema.md`, `docs/EXHALE_user_manual.tex` and `docs/composition_restart_and_base_handoff.tex`. The same wording survives in this changelog’s earlier sections and in the dated plan and review memos (`docs/oxygen_chemistry_plan_code_review.md`, `docs/a2_oxygen_option_design`), and was deliberately left there: those are records of what was true when they were written, and rewriting them would destroy the history rather than correct it.

The pathology of §11.3 does not appear

Under the imposed partition the ghost ionization stages are physical: $\text{H I } 6.94 \times 10^{12}$, $\text{H II } 674$, $\text{He I } 7.35 \times 10^{12}$, $\text{H}_2 4.286 \times 10^{13}$, $\text{H}_3^+ 5.26 \times 10^4 \text{ cm}^{-3}$. Hydrogen is not ionized; the fully ionized 11–132 K base recorded in `docs/supersonic_molecular_base.md` §11.3 does not recur here. This is one probe on one case and is not a claim about the $\text{He/H} = 1$ arm, which is refused by C2 as configured and is re-specified in the window re-measurement block.

C3: the base boundary now says when it is over-specified

A subsonic inflow carries one outgoing characteristic, so one piece of interior information belongs in the ghost and the other two are the reservoir’s to state. A SUPERSONIC inflow carries none: all three characteristics enter the domain,

every ghost variable is the reservoir's, and a boundary condition that copies an interior velocity into it is over-specified — it feeds the domain a state the domain itself produced. The interior downstream of such a boundary is not a solution of the stated problem, whatever the chemistry is doing. §2 of docs/supersonic_molecular_base.md established this; what was missing was that a run never said when it had crossed.

check_base_inflow_is_subsonic now measures the base ghost Mach number once per marching step, using the code's own adiabatic sound speed — the same expression eval_dt uses for the CFL condition, so the two agree on what sonic means — and warns on the first supersonic inflow, with a run-end summary of the count, the first step and the maximum. Inflow is $v > 0$ here: the coordinate increases outward, so gas entering at the base moves outward. An outflowing base is a different condition and is measured but not warned on.

Reported, not enforced. §11.5-D of docs/supersonic_molecular_base.md is explicit that this must not be called the chemistry fix, and it is not: the crossing is evidence about the configuration, not a numerical fault to repair in place. It also does not make Base velocity: massflux redundant, nor the reverse — that option removes the copied-velocity feedback, but a mass-flux base can still go supersonic if the reservoir state and the flux constant disagree, which is why the check reads whatever velocity the boundary produced rather than assuming which branch produced it.

Both branches were exercised. The molecular gate stays deeply subsonic and says so (inflow stayed subsonic, max Mach 6.622E-03); the re-specified He/H=1 arm crosses at step 6 and the warning fires. The arm figure is a statement that the branch runs, not a physical result: that run is an unconverged 1500-step relaxation, and whether the phenomenon survives re-specification and convergence is the window re-measurement block's question.

The check reads W and changes nothing, so the goldens must not move, and that is its acceptance criterion rather than a hope.

C5 is specified and approved, and deliberately not implemented

The block's fifth item was the handling of the frozen region — what to do where the reaction residual carries no information because the network has stopped. The design was worked out against both precedents read in their own source (Equilibrate's removal of undecidable unknowns, CHIMES's cheap test before the solver) and is recorded as an approved specification in TO_BE_DONE.md (P27). It is not implemented, and the reason is a measurement one rather than a scheduling one: its effect can only be measured on a converged arm, which does not exist until the arm is re-specified and driven to Newton convergence. Implemented today it would be default-off code that does nothing and that nothing verifies. The acceptance criteria are fixed now so that they are not negotiated later: **zero activations in the five molecular gates**, and an **ON/OFF pair the matrix cannot distinguish**.

Two points of the specification are worth repeating here because they are the ones an implementer can get wrong silently. What a frozen cell holds is the FRACTIONS, not the absolute number densities — hydrodynamics keeps changing ρ while the partition stays put. And f_sp is not in the hydrodynamic flux path, so holding the previous fractions is holding them in an Eulerian cell, which equals freezing a fluid parcel's composition only where the flow crosses a cell slowly compared with the step. The base layer is such a place; that is the validity range, and it belongs at the code site.

The design does not fix the late thermal collapse. It answers what to do where the residual is uninformative, not what drove a cell into that state.

Verification

- **C2 alone: 8 of 8 byte-identical**, measured on its own build before C1 was compiled, which is what makes “the refusal changes nothing it does not refuse” a measurement rather than an expectation.
- **C1: five cases move and three do not**, matching the reachability analysis exactly — the three that do not are the atomic pair and the one molecular-base-off case.
- **Goldens refreshed once**, for the five cases that moved only, with the pre-block set preserved at backup/regression/golden_block beforehand. The refreshed set reproduces.
- **A cross-proof that fell out of the sequence, worth recording.** The goldens were snapshotted from outputs written by the 18:19 binary, and the confirming matrix ran on the 19:47 binary, rebuilt after the documentation corrections to `parameters.f90`, `input_read.f90` and `lower_column.f90`. Those corrections were comment-only, and the confirming run is what proves it: a comment-only claim is otherwise only as good as the reader’s eye, and here byte-identity across two builds established it without anyone having to trust the claim.
- **The arm is refused as configured**, which is the intended outcome and not a failure: `q_H2_base 0.75` at `HeH_base 1.0` against a ceiling of 0.3333. Its re-specified value is 0.300861.

118. Secondary ionization in a molecular gas: the H_2 target the prescription never had, the heating fraction that went to zero in a neutral gas, and the two heat returns that were blocked on unsourced constants (2026-09-02)

The judgement first. EXHALE’s photoelectron partition is the Shull & van Steenberg (1985, ApJ 298, 268) prescription, and **it has no H_2 term in either direction**: molecular hydrogen was a source of photoelectrons in PH_heat_HHe but never a target, so in a layer where 92% of the hydrogen nuclei are bound into H_2 the energy that should ionize H_2 was handed to H I, and the heat that comes back when a photoelectron dissociates H_2 was not returned at all. The routine’s own docstring said so. That is missing physics, not a missing option.

Repairing it turned up three more defects in the same prescription, and this section records all four: the missing H_2 target, a heating fraction evaluated eight to twelve decades outside the range its authors state (and going to *zero* where the true value is 0.11–0.26), ionization channels applied without the energy axis their sources say they need, and an “ionized fraction” that was a count of H and He ions rather than the electron density both sources define. Each was built and measured on its own build, because all four move the same integral.

The source is Dalgarno, Yan & Liu (1999, ApJS 125, 237), *Electron Energy Deposition in a Gas Mixture of Atomic and Molecular Hydrogen and Helium*. Every coefficient below was read from the published paper, and the two tables the change stands on (their Tables 5 and 7) and their equations (9) and (10) were checked against the *rendered pages* rather than a text extraction, because the extraction of this paper turns + and x into the same character: their eq. (17) reads $a(x) = 0.5 + (x/10^4)^{0.15}$ in the extracted text and $a(x) = 0.5 \times (x/10^4)^{0.15}$ on the page.

What was missing, and how large it is

Three separate defects, of which this section repairs the first.

1. **No H_2 ionization channel.** The hydrogenic share of the secondary ionization energy went entirely to H I.
2. **No molecular heat return.** A photoelectron in H_2 loses energy to rotational excitation, to dissociation through the triplet states, to vibrational excitation and to the Lyman and Werner bands. In a gas dense enough to quench the internal excitation most of that comes back as heat. None of it was carried.
3. **The heating amplitude is evaluated far outside its own stated range.** Shull & van Steenberg state their fits for $0.0001 < x < 1.0$ — “the fit to the heating curve, which is reproduced to $\sim 2\%$ over the range $0.0001 < x < 1.0$ ” — and their heating form goes to *zero* as $x \rightarrow 0$. Our molecular layer sits at $x = 10^{-16}$ to 5×10^{-5} , up to twelve decades below the range they state, and the true heating fraction of a neutral gas is not zero but 0.11–0.26 (Dalgarno et al. Table 7; Shull & van Steenberg’s own Monte Carlo gives 0.1358 ± 0.0088 at their lowest $x = 10^{-4}$, against 0.1114 for their fit there). Repairing that amplitude moves every run, molecular or not, and is deliberately *not* in this section.

The size is best given as a property of the prescription evaluated on a composition, not as the movement of an answer

— the molecular gates are fixed-step relaxation snapshots and their movement is admissible only as a bit-identity test. Taking the composition of `mol_base_handoff` and a 100 eV photoelectron:

	at the base	mid-layer	top of the layer
$n(\text{H}_2)/n(\text{HI})$	6.2	31	0.011
before	0.0015	0.0105	0.0910
after this change	0.0720	0.0812	0.0916
with defects 2 and 3 also repaired	0.259	0.268	0.164

so this change carries about a third of the total and the two deferred repairs carry the rest. The last column is a cell with almost no H_2 left, where the whole remaining difference is defect (3).

One collision weight in place of two mean energies

Dalgarno et al. give the three-species mixture as two separate statements, their eqs. (9) and (10),

$$W_m(\text{H}^+) = W_q(\text{H}^+) \left[1 + 1.89 \frac{n(\text{H}_2)}{n(\text{H})} \right], \quad W_m(\text{H}_2^+) = W_q(\text{H}_2^+) \left[1 + 0.53 \frac{n(\text{H})}{n(\text{H}_2)} \right].$$

The product of the two coefficients is $1.89 \times 0.53 = 1.0017$. The two statements are therefore, to 0.17%, a single one: the hydrogenic ionization energy divides between the two targets in the ratio $n(\text{H}) : 1.89n(\text{H}_2)$, with H_2 counted 1.89 times per molecule. **That identification is not made in the paper;** it is read off here from the two published coefficients, and it is recorded as ours both here and in `electron_energy_degradation.f90`.

It is worth having rather than being a curiosity. Written as two mean energies, each expression divides by a density that vanishes at one end of the composition range; written as the weight

$$u(r) = \frac{1.89n(\text{H}_2)}{n(\text{H}) + 1.89n(\text{H}_2)},$$

the same physics is bounded in $[0, 1]$, exact at both ends, and needs no guard. It is also what makes the whole change collapse to nothing when $n(\text{H}_2) = 0$.

Frelikh & Murray-Clay (2026, ApJ 996, 96) weight by $n(\text{H}_2)/[n(\text{H}) + n(\text{H}_2)]$, citing Glassgold, Galli & Padovani (2012) eq. (11); read at the source, that equation is a statement about a different quantity and is not a rival prescription for THIS split. Glassgold et al.’s eq. (11) is the dissociation heating per ion pair,

$$Q_{\text{diss}} = \frac{x(\text{H}_2)}{x(\text{H}) + x(\text{H}_2)} \frac{D_0}{1 + Cx_e^\alpha}, \quad D_0 = 2.14 \text{ eV},$$

one term of the heat budget of their eq. (3) and taken from Dalgarno et al.’s Table 5; they apply the same plain number fraction to their vibrational heating in eqs. (13) and (14). Frelikh & Murray-Clay’s eqs. (27), (28) and (35) apply it to exactly those heating terms, so the attribution is accurate, and neither paper restates the ionization-energy split of Dalgarno et al.’s eqs. (9) and (10).

Where the two weights do meet is the dissociation heat, which `photoelectron_energy_partition` also weights by u (`f_diss` and `f_fluor`, the “dissociation heat” and “heating amplitude” subsections of this section). There the difference is real — 0.65 against 0.50 at $n(\text{H}_2)/n(\text{H}) = 1$, and 0.159 against 0.091 at 0.1 — and we follow Dalgarno et

al., whose Figure 20 states the dissociation yield as a number of dissociations *per* H_2^+ ion: it tracks the H_2^+ production that eq. (10) weights, rather than the number fraction of molecules. Their own eq. (11) is a third thing again — the vibrational yield $Y = Y_{\text{HeH}_2} r / (r + a)$, which is what `f_vib` uses.

The ionization channels

`svs85_secondary_branching` already renormalized the H/He split onto the cell's own neutral ratio, leaving the total ionization energy at the Shull & van Steenberg value. The generalization adds a third target to the same construction, with H_2 entering the weight at 1.89 per molecule:

$$\begin{aligned} n_{Hw} &= n_{\text{HI}} + 1.89 n_{\text{H}_2}, & w_{\text{tot}} &= n_{Hw} \eta_{\text{ref}} f_{\text{HI}} + n_{\text{HeI}} f_{\text{HeI}}, \\ c_{\text{HI}} &= f_{\text{ion}} \eta_{\text{ref}} f_{\text{HI}} / w_{\text{tot}}, & c_{\text{H}_2} &= 1.89 c_{\text{HI}}, & c_{\text{HeI}} &= f_{\text{ion}} f_{\text{HeI}} / w_{\text{tot}}, \end{aligned}$$

with $\eta_{\text{ref}} = 0.1$ the neutral He/H ratio the fits were measured at. The construction conserves energy identically,

$$n_{\text{HI}} c_{\text{HI}} + n_{\text{H}_2} c_{\text{H}_2} + n_{\text{HeI}} c_{\text{HeI}} = f_{\text{ion}},$$

for every composition. That is the property that makes it safe to move energy between the channels: the H_2 channel is fed by taking from HI, not by inventing.

Checked against its source. In a neutral, fully molecular gas the construction implies a mean energy per H_2^+ ion of $E_{\text{th}}(\text{H}_2) / f_{\text{HI}}(0) = 15.4 / 0.3908 = 39.4$ eV, against the **41.9 eV** of Dalgarno et al. Table 4 (H_2 and He mixture, 1 keV). The 6% is the same size as the disagreement between the two sources' own mean energy per H^+ (Dalgarno et al. section 6 quote 37.2 eV for Shull & van Steenberg against their own 39.5 eV) and is well inside Dalgarno et al.'s stated 15%. An arithmetic aside, because it explains that 37.2 eV. Shull & van Steenberg's fitted amplitude gives $13.598 / 0.3908 = 34.8$ eV at $x = 0$, not 37.2. Their Monte Carlo at their lowest tabulated point, $x = 10^{-4}$, gives $\phi(\text{HI}) = 0.3656$, i.e. $13.598 / 0.3656 = 37.2$ eV exactly. Dalgarno et al. quoted the *data*, not the fit; the fit's extrapolation to $x = 0$ overshoots its own Monte Carlo by 7%.

The rate enters the chemistry through `P_H2`, the channel that already drives both the H_2 destruction row and the H_2^+ production row of `System_HeH_mol` — the same route the primary photoionization takes — so no parallel path was added.

How much energy the new channel actually takes. This is exact and needs no run: the split is a function of the cell's neutral densities alone. Evaluated on the composition of `mol_base_handoff`, as fractions of the Shull & van Steenberg ionization budget f_{ion} :

$n(\text{H}_2)/n(\text{HI})$	HI	HeI	H₂	sum/ f_{ion}
736	0.0006	0.1063	0.8930	1.0000000
347	0.0014	0.1063	0.8923	1.0000000
85	0.0056	0.1063	0.8881	1.0000000
6.8	0.0648	0.1061	0.8291	1.0000000
6.2 (base)	0.0705	0.1059	0.8236	1.0000000
0.011 (top)	0.8793	0.1024	0.0183	1.0000000

Through the molecular layer, 82–89% of the secondary ionization energy now goes to H₂ rather than to HI. Before this change all of it went to HI — the docstring of the routine it replaces said so in as many words. The last column is the energy identity holding to every printed digit at every composition.

The dissociation heat, and where Frelikh’s version overshoots

Dalgarno et al. section 5.3 gives the kinetic energy released when H₂ is dissociated through a triplet state as a mean of **5.4 eV** (“cannot be much different from a mean of 5.4 eV corresponding to a vertical transition”). Frelikh & Murray-Clay apply that 5.4 eV to the whole Dissociation yield of Table 5. But that yield is the total, and its footnote b says so: it excludes only the dissociative excitation channel, not the singlet predissociation. The singlet route releases about 0.4 eV, not 5.4, and its share is fixed by the same paper — section 5.3 gives radiative dissociation probabilities of **0.27** for $B^1\Sigma_u^+$ and **0.036** for $C^1\Pi_u$.

Separating the two reproduces the answer Dalgarno et al. state directly (the dissociation heat per H₂⁺ ion, 11 eV at 30 eV and 1.7 eV at 1 keV, their sections 5.3 and 6) far better than not separating them:

E_0	Frelikh $5.4E/W_{\text{diss}}$	with the singlet split	Dalgarno’s own value
30 eV	4.67 eV	4.31 eV	4.23 eV
1 keV	58.3 eV	45.4 eV	40.6 eV

10% and 44% against 2% and 12%. The split is what the code does.

The dissociation heat carries **no critical-density factor, and should not**: the 5.4 eV and the 0.4 eV are translational from the moment they are released. The vibrational and Lyman–Werner returns are internal excitation and do need one; they are not in this change.

The seam between the two sources is an energy axis, not a disagreement

The H₂ target channel above was staged behind the heating amplitude, so that the first step could be checked by bit-identity on the atomic cases. In that step the heating fraction was carried as a *ratio*,

$$f_{\text{heat}} = f_{\text{SvS85}}(x) \frac{\eta_{\text{mix}}(x, E_0, r)}{\eta_{\text{H,He}}(x, E_0)} + u(r) f_{\text{diss}}(x, E_0),$$

with η_{mix} from Dalgarno et al. eq. (12): the ratio keeps Shull & van Steenberg as the amplitude and is exactly 1 in a gas without H₂. Repairing the amplitude, defect (3), is what this subsection decides and the next one implements.

The first design considered was to use Dalgarno’s η below $x = 0.1$, where his fits are stated, and Shull & van Steenberg’s above it, and to join them. The offset at the join was measured before committing to that plan, and it did not come out where it was expected to:

E_0	$\eta_{\text{Dalgarno}}(0.1)$	$f_{\text{SvS85}}(0.1)$	offset
30 eV	0.9625	0.6400	+33.5%
50 eV	0.9078	”	+29.5%
100 eV	0.8384	”	+23.7%
200 eV	0.7749	”	+17.4%
500 eV	0.7075	”	+9.5%
1000 eV	0.6691	”	+4.4%

Read as two determinations of one quantity that is a disagreement far larger than either source’s stated error.

It is not one. The footnote to Shull & van Steenberg’s Table 1 says their fractional energy depositions are “of a 3 keV primary electron”, and their Figure 4, from which Table 2 is fitted, is captioned “Limiting values ($E_0 > 100$ eV)”. Their fit is therefore a single high-energy asymptote with no E_0 axis at all, while Dalgarno et al.’s Table 7 resolves E_0 and finds η at $x = 0.1$ changing by 25% between 100 eV and 1 keV — 0.8384 to 0.6691. (The 44% figure that stood here is the 30 eV to 1 keV range; 30 eV is outside the $E_0 > 100$ eV range Shull & van Steenberg claim, so it cannot be used to argue against them. 25% is enough, and it is the honest number.) **The column above is not two calculations disagreeing; it is one E -resolved calculation compared against the other’s 3 keV asymptote.** At the one energy where the two describe the same thing — Dalgarno’s highest row, 1 keV — the offset is 4.4%, and Shull & van Steenberg’s own Monte Carlo at $x = 0.1$ is 0.6524 ± 0.0421 , which contains it.

The right test is therefore against the *data* rather than against the other fit. Judged on Shull & van Steenberg’s Table 1, over $x > 0.1$:

prescription	max relative deviation	max deviation / 1σ
Shull & van Steenberg’s own fit	3.9% ($x = 0.2$)	1.05 ($x = 0.3$)
Dalgarno eq. (14) at 1 keV, extrapolated	9.2% ($x = 0.95$)	8.45 ($x = 0.95$)
Dalgarno + the neutral-fraction closure	5.6% ($x = 0.15$)	1.01 ($x = 0.15$)

and below $x = 0.1$ the Dalgarno branch stays within 0.90σ everywhere while Shull & van Steenberg’s fit reaches 2.77σ at $x = 10^{-4}$ and collapses below it. **The right design is therefore one expression over the whole range: Dalgarno’s η everywhere, extrapolated above $x = 0.1$, with a closure that supplies the $x = 1$ endpoint his fitting form does not have.** It tracks Shull & van Steenberg’s own Monte Carlo about as well as their fit does over the range where the fit was made, and much better outside it.

A note on the third row. It was first recorded as 6.8% and 1.89σ , for a closure described in words but preserved neither in code nor in a script. That closure could not be reconstructed, and the row now carries the measurement of the closure actually implemented, which is the better of the two on both measures. The first two rows are reproduced exactly by re-evaluating the two published fits against Table 1.

The heating amplitude, and the closure that supplies the endpoint

The judgement: Shull & van Steenberg’s heating fit is the wrong function in a weakly ionized gas, and it is not a matter of degree. Their Form 1, $C[1 - (1 - x^a)^b]$, was chosen “because of their appropriate limits at $x = 0$ and $x = 1$ ” —

and its $x = 0$ limit is *zero*. A neutral gas does not convert none of a photoelectron's energy to heat: it converts 0.117 of a 1 keV primary and 0.259 of a 30 eV one (Dalgarno et al. Table 7), and Shull & van Steenberg's own Monte Carlo says 0.1358 ± 0.0088 at their lowest tabulated point. In our molecular layer, at $x = 10^{-16}$ to 5×10^{-5} , their fit was returning 0.1% to 1.5%.

The heating fraction is now Dalgarno et al.'s, at every x :

$$f_{\text{heat}}(x, E_0) = 1 - [1 - \eta_{\text{mix}}(x, E_0, r)](1 - x) + f_{\text{diss}} + f_{\text{vib}} + f_{\text{fluor}},$$

with η_{mix} their eq. (12). Shull & van Steenberg's heating fit is no longer called.

The $(1-x)$ factor is a closure and is ours. Dalgarno et al. state eq. (14) “for $x \leq 0.1$ ”, and its form leaves $(1-\eta_0)/(1+c)$ of the primary energy unthermalized at $x = 1$ — 10% of it at 1 keV — which is impossible in a gas with no neutral left to ionize or excite. The factor says that the channels competing with heat need a neutral to act on, so their share must fall in proportion to the neutral fraction of the H and He nuclei, which is exactly $1-x$ when x is the electron density over those nuclei. It has no free parameter, it is 1 to within 10^{-4} over the whole molecular layer, and it forces $f_{\text{heat}}(1) = 1$ exactly.

What this does *not* justify. Evaluating eq. (14) above $x = 0.1$ is extrapolation beyond the range its authors state, and forcing the endpoint is not the same as the fit being right in between. The extrapolation is accepted on a measurement instead, against Shull & van Steenberg's Table 1 — their Monte Carlo, 18 points with 1σ widths, not their fit:

x	Table 1	1σ	SvS85 fit	dev	$\eta_D(1 \text{ keV})$	dev	closed	dev
0.95	0.9895	0.0108	0.9936	+0.4%	0.8983	-9.2%	0.9949	+0.5%
0.90	0.9847	0.0123	0.9882	+0.4%	0.8949	-9.1%	0.9895	+0.5%
0.80	0.9726	0.0138	0.9738	+0.1%	0.8873	-8.8%	0.9775	+0.5%
0.70	0.9592	0.0161	0.9548	-0.5%	0.8781	-8.5%	0.9634	+0.4%
0.50	0.9196	0.0211	0.9014	-2.0%	0.8521	-7.3%	0.9260	+0.7%
0.30	0.8459	0.0291	0.8154	-3.6%	0.8044	-4.9%	0.8631	+2.0%
0.20	0.7784	0.0355	0.7481	-3.9%	0.7594	-2.4%	0.8075	+3.7%
0.15	0.7247	0.0401	0.7019	-3.1%	0.7238	-0.1%	0.7652	+5.6%
0.10	0.6524	0.0421	0.6400	-1.9%	0.6691	+2.6%	0.7022	+7.6%
0.05	0.5296	0.0426	0.5435	+2.6%	0.5678	+7.2%	0.5894	+11.3%
0.02	0.4015	0.0369	0.4348	+8.3%	0.4341	+8.1%	0.4454	+10.9%
0.01	0.3256	0.0297	0.3658	+12.3%	0.3460	+6.3%	0.3526	+8.3%
0.005	0.2696	0.0249	0.3070	+13.9%	0.2756	+2.2%	0.2792	+3.6%
0.002	0.2148	0.0166	0.2429	+13.1%	0.2099	-2.3%	0.2115	-1.5%
0.001	0.1852	0.0140	0.2031	+9.7%	0.1775	-4.2%	0.1783	-3.7%
5×10^{-4}	0.1642	0.0120	0.1697	+3.3%	0.1558	-5.1%	0.1562	-4.9%
2×10^{-4}	0.1441	0.0098	0.1336	-7.3%	0.1383	-4.0%	0.1384	-3.9%
10^{-4}	0.1358	0.0088	0.1114	-18.0%	0.1304	-4.0%	0.1305	-3.9%

(“closed” is the prescription now in the code, at Dalgarno's 1 keV row — his highest, since Table 1 is a 3 keV

calculation.) Over $x > 0.1$ the closure is 5.6% and 1.01σ at worst against 3.9% and 1.05σ for Shull & van Steenberg’s own fit; over $x \leq 0.1$ it is 11.3% and 1.40σ against their 18.0% and 2.77σ , and their fit keeps falling below that while ours does not. **There is no seam:** one expression covers the whole domain, so the question of where to join the two sources does not arise.

Measured on the two converged cases that fire the coupling (wasp_full_newton, wasp_he23off_newton): escaping mass flux +3.6%, $\log_{10} \dot{M}$ 13.20 to 13.22, photoheating +22% at $r = 1.075$ and the temperature +9% at $r = 1.086$, and under 0.5% above $r = 1.6$ where x is already 0.99.

E_sec_ion: 40 eV was another code’s number

E_sec_ion is the photoelectron energy above which the partition is applied at all. It was 40 eV, and the code site said where that came from: “matching the wind_ae X-ray cutoff”. That is a numerical convention of a different code, not a physical threshold, and it sat between the two sources rather than on either.

It is now **30 eV**, the lowest primary energy at which any coefficient of the partition is *published* — Dalgarno et al. tabulate 30, 50, 100, 200, 500 and 1000 eV. What is still discarded is written at the code site: a 13.6–30 eV photoelectron can ionize hydrogen, and in the hot-Uranus band those photons carry 18% of the incident XUV energy, but no source consulted resolves the partition there.

Measured, on its own build: escaping mass flux -0.17% and -0.13% , $\log_{10} \dot{M}$ unchanged at 13.22, photoheating -1.6% at $r = 1.3$. The sign is right — those photoelectrons now spend part of their energy ionizing instead of all of it heating — and the size is what TO_BE_DONE.md (P32) predicted.

The ionization channels get the energy axis they were missing

Shull & van Steenberg’s $\phi(\text{H I})$ and $\phi(\text{He I})$ have no E_0 dependence: their Table 2 footnote says “These formulae are only accurate for $E_0 > 100$ eV” and Table 1 is a 3 keV calculation. Dalgarno et al.’s Table 4 supplies what is missing, so the amplitude is now theirs in E and Shull & van Steenberg’s in x :

$$f_{\text{H I}}(x, E_0) = \phi_{\text{H I}}(x) W_{\text{H}^+}(x, 1 \text{ keV}) / W_{\text{H}^+}(x, E_0),$$

$$f_{\text{He I}}(x, E_0) = \phi_{\text{He I}}(x) W_{\text{He}^+}(x, 1 \text{ keV}) / W_{\text{He}^+}(x, E_0).$$

The ratio is 1 at the top of the table, so the high-energy limit is exactly Shull & van Steenberg’s. Below it a photoelectron makes fewer ion pairs per unit energy, and for helium the effect is large: W_{He^+} rises from 487 eV at 1 keV to 6740 eV at 30 eV, so a 30 eV photoelectron puts 14 times less of its energy into He I ionization than the unresolved fit asserts — which is physical, 30 eV being barely above helium’s 24.6 eV threshold. For hydrogen the factor is 1.51. The collision weight 1.89 is deliberately *not* given an energy dependence: Dalgarno et al. state their eqs. (9) and (10) coefficients once, not per energy.

Plumbing. $E_0 = h\nu - E_{\text{th}}$ depends only on the spectral bin and the absorber’s own threshold, both fixed for a run, so the bracket index and interpolation weight are built once at startup (photoelectron_energy_grid, called from set_energy_vectors) and each cell then costs one gather and one linear interpolation per bin. Without that, the heating-amplitude change alone cost 49% in run time (3 m 40 s to 5 m 27 s on wasp_full_newton); with it the same case runs in 3 m 51 s, 5% above the original while doing strictly more work.

Measured: escaping mass flux +0.64% and +0.53%, $\log_{10}\dot{M}$ unchanged, temperature +3.8% at $r = 1.13$, photoheating +3.3% to +5.2% at $r = 1.25$, and $n(\text{H}^+)/n(\text{H})$ down 13% at $r = 1.10$ — the weakly ionized shell fed by the softest photoelectrons.

xion was not the electron fraction

Dalgarno et al. section 7 defines the ionization fraction of the H-H₂-He mixture as “the number density ratio of the electrons to the hydrogen and helium nuclei”. EXHALE built it at three sites — `ionization_equilibrium.f90`, `util_ion_eq.f90`, `post_process_adv.f90` — as $(n_{\text{H}^+} + n_{\text{He}^+} + n_{\text{He}^{2+}})/(n_{\text{H}} + n_{\text{He}})$, a count of H and He *ions*: it omitted every molecular-ion electron and every metal electron, and counted He III once although it releases two. In the metals-on molecular gate the two differ by up to nine decades.

The numerator is now the total free-electron density, `calc_ne`’s own value; the denominator stays the nuclei. **The metal electrons are an extension beyond both sources’ composition** and are marked as such at the code site: neither Monte Carlo had a metal in it, and the reason for including them anyway is that what degrades a photoelectron is Coulomb scattering off free electrons, which an electron released by a metal does as well as one released by hydrogen. It also lets the ratio exceed one, which is why every entry point clamps into [0, 1]. One asymmetry is recorded rather than removed: on the post-process path the molecular-ion electrons are still absent, because that path does not carry the molecular densities at all.

Measured: escaping mass flux +0.94% and +0.81%, $\log_{10}\dot{M}$ unchanged, photoheating +8.6% at the base itself ($r = 1.000$) and the base temperature +2.1% at $r = 1.02$ — where the metal electrons dominate the free-electron budget of the shielded gas.

The two heat returns that were blocked on constants, and their sources

What Dalgarno et al.’s heating efficiency is *not*. Their section 5.3 defines it as “the heating efficiency, defined as the fraction of the primary energy that is converted into heat from elastic scattering and rotational excitations”. Vibrational excitation is outside it, explicitly: “The fraction of the energy E lost in vibrational excitation is illustrated in Figure 11. If the vibrationally excited H₂ radiates the excitation energy is lost. However, if it is collisionally quenched, the excitation energy is converted into heat.”

(a) **Direct vibrational excitation**, $v = 1$ and $v = 2$. Table 5’s own rows for those levels give the mean energy per excitation in the H₂-He mixture; the level energies are the Roueff et al. (2019) term values the code already carries for the infrared lines, 0.51592 eV for ($v=1, J=0$) and **1.00266 eV** for ($v=2, J=0$) — not the harmonic twice-0.516, which is 2.8% high. The yield is transferred to a mixture containing atomic hydrogen by eqs. (11), (16) and (17), which are one expression, $Y = Y_{\text{HeH}_2} r/(r+a)$ with $r = n(\text{H}_2)/n(\text{H})$.

A typographical correction, made and stated. The paper prints $a(x) = 0.5 \times (x/10^4)^{0.15}$ for the middle range. With the exponent as printed, eq. (17) gives $a = 0.0316$ at $x = 10^{-4}$ where eq. (16) requires 0.5, and $a = 0.0112$ at $x = 10^{-7}$ where eq. (11) requires 0.177 — two discontinuities, **both by the same factor 15.85**. With 10^{-4} in place of 10^4 the expression meets eq. (16) exactly at $x = 10^{-4}$ and eq. (11) to three digits at $x = 10^{-7}$ (0.1774 against their 0.177). Since $15.85 = (10^8)^{0.15}$ is exactly the ratio between the two readings, the sign of the exponent is the whole discrepancy, and

the minus sign is taken to have been lost in typesetting.

(b) The fluorescence return. Of the B and C excitations, 0.27 and 0.036 dissociate; the rest fluoresce back into *bound*, vibrationally excited levels of the ground state. Black & van Dishoeck (1987), pp. 412–413: “The initial absorptions are followed by fluorescence to the vibrational continuum of the ground state with a typical average probability of 0.10. ... The remaining 90% of the fluorescent transitions populate various bound excited vibration-rotation levels of the ground state.” The energy left behind per pump is **2.0 eV**, from Burton, Hollenbach & Tielens (1990), Appendix A, p. 635: “The process is approximated as per Paper I, except that the effective heating per pump, E_* , has been taken as 2.0 eV.” Their eq. (1) prefactor 3.2×10^{-12} erg is exactly that.

Two limits are recorded with it. It is an *adopted* value in Burton et al., revised from their Paper I without a derivation; and **Black & van Dishoeck do not state it**, contrary to how it is usually attributed — what they supply is the branching above and a line-by-line intensity table. Applying it to *electron-impact* excitation of B and C, as well as to photon pumping, is a further assumption: the upper states and the cascade are the same, the distribution over upper vibrational levels is not.

(c) The collisional-quench fraction. Hollenbach & McKee (1979), p. 586: “The term $(1 + n_{\text{cr}}/n)^{-1}$ expresses the fraction of the excitation energy going into heat via collisional de-excitation rather than into radiation.” $n_{\text{cr}} = A/\gamma$ (their p. 576) with $A_{10} = 8.3 \times 10^{-7} \text{ s}^{-1}$ (p. 583, *J*-averaged, from Turner, Kirby-Docken & Dalgarno 1977) and the rate coefficients of their eq. (6.29). The code carries it as $f = q/(q + A_{10})$ with $q = \gamma_{\text{H}} n_{\text{H1}} + \gamma_{\text{H2}} n_{\text{H2}}$, in `h2_vibrational_relaxation.f90` — one file, so the two drivers cannot drift apart.

Note the direction: this factor is not a suppression. It is the statement that at high density the excitation energy *does* become heat. In our layer it is 1.0000 to four digits everywhere, because $n_{\text{cr}}(v=1)$ is $5.0 \times 10^4 \text{ cm}^{-3}$ against atomic H and $1.1 \times 10^6 \text{ cm}^{-3}$ against H_2 at 950–1300 K, while the layer carries $n(\text{H}_2) = 4.4 \times 10^9$ to $4.6 \times 10^{13} \text{ cm}^{-3}$. It is written out anyway, because a shallower base would not be.

This also fixes an uncited number elsewhere in the tree. Both `molecular_infrared_cooling.f90` and `molecular_hydrogen_treatment.tex` used “ $n_{\text{crit}}(\text{H}_2 v=1) \sim 10^9\text{--}10^{11} \text{ cm}^{-3}$ ” as a validity argument for LTE level populations, with no citation. Evaluated from Hollenbach & McKee it is $2\text{--}6 \times 10^6 \text{ cm}^{-3}$ (H_2 collider) or $5\text{--}8 \times 10^4 \text{ cm}^{-3}$ (H collider) over 900–1350 K — three to five decades lower, which *widens* the LTE margin. Both sites now carry the sourced value. Neither source gives a **helium** collision rate for $\text{H}_2 v = 1$, so a helium-dominated gas is outside both; helium is left out of the collider sum rather than given a borrowed rate, which errs towards radiating the energy away rather than towards claiming it as heat.

How large the two returns are, exactly and needing no run, on the `mol_base_handoff` layer:

	base	mid-layer	top
$n(\text{H}_2)/n(\text{H1})$	6.2	31	0.011
f_{heat} at 30 eV, without (a) and (b)	0.235	0.244	0.264
f_{heat} at 30 eV, with them	0.394	0.407	0.265
f_{heat} at 1 keV, without (a) and (b)	0.098	0.103	0.125
f_{heat} at 1 keV, with them	0.196	0.204	0.126

The two returns roughly double the molecular heat fraction, and they vanish where the H_2 does, as they must.

(d) **The photon-driven half of the same physics.** `lyman_werner.f90` already knew that only 13.5% of Lyman–Werner pumps dissociate, and `ionization_equilibrium.f90` deposited the 0.4 eV those dissociations give the fragment pair. The other 86.5% were accounted as energy that does not come back. At the density of a molecular base that is wrong, and the missing term is $(k_{\text{LW}}/p_{\text{diss}})(1 - p_{\text{diss}})n(\text{H}_2)E_*f_q$, whose prefactor is $(0.865/0.135) \times (2.0/0.4) = 32$ times the fragment term that was there. It uses the same two constants from the same file as the electron-driven channel.

And it is not a correction, it is the channel. Measured on `mol_lyman_werner`, the only default case that carries a stellar Lyman–Werner field, from its own `Heating_breakdown.txt`:

	value
Lyman–Werner heating at the base cell	$1.216 \times 10^{-8} \text{ erg cm}^{-3} \text{ s}^{-1}$
total photoheating at the base cell	$2.492 \times 10^{-8} \text{ erg cm}^{-3} \text{ s}^{-1}$
Lyman–Werner share at the base	48.8%
Lyman–Werner share over $r = 1.0\text{--}1.2$	66.2%
peak Lyman–Werner heating	3.568×10^{-8} at $r = 1.078$

With only the 0.4 eV fragment term the same cell would carry 3.685×10^{-10} in that channel and a total of 1.313×10^{-8} : **the base photoheating of a molecular layer under a Lyman–Werner field was low by a factor of 1.9.** The column-integrated FUV ledger of `write_output.f90` moves with it — the 86.5% of pumped photon energy it used to book as `bond_col`, energy that never returns, is now the largest single heating term of the layer — and the run’s mean photoheating over $r < 1.2$ rises 20.5% once the wind relaxes to it. (`mol_lyman_werner` is a 12000-step snapshot, so that last number is a direction and an order, not a converged answer.)

What is still not here

- **Dissociative ionization of H_2** , for which Dalgarno et al. give a mean energy of $22W_m(\text{H}_2^+)$: about 4.5% of the H_2 ionizations. It was absent from the primary photoionization as well. **Both sides are now carried; see Sec. 130.**, which also corrects the branching the primary side would have used.
- **The $\text{Ly}\alpha$ excitation channel**, still treated as escaping line radiation rather than coupled to the $\text{Ly}\alpha$ field of the excited-hydrogen model.
- **Electrons as a collider** for the H_2 vibrational quench: Burton et al.’s Table 7 gives an $e\text{--H}_2$ rate about two orders above the $\text{H}_2\text{--H}_2$ one, but the molecular layer runs at an electron fraction of 10^{-16} to 5×10^{-5} .
- **A helium-dominated gas.** Neither degradation source samples one, and neither vibrational-quench source gives a helium rate. This is the validity limit that matters most for the $\text{He}/\text{H} = 1\text{--}10$ arms, and it is in the header of both new modules.

Structure and verification

Two names were corrected before anything else was touched, and the rename alone was verified byte-identical on all eight regression cases: `photoelectron_partition_cell` became `photoelectron_energy_partition(_cell`

names a place in the code, not a physical quantity), and `dal_eta_node / dal_W_node` became `dalgarno_heating_efficiency / dalgarno_mean_energy_per_ion_pair`.

`svs85_secondary_branching` computed a partition blind to both the molecular content and the photoelectron energy, and its name named a source rather than a physical quantity. It is replaced by `src/modules/radiation/electron_energy_degradation` holding both sources' fits and three entry points: `photoelectron_energy_grid` (once at startup), `photoelectron_energy_partition` (once per cell) and `photoelectron_shares` (once per absorber). The leaf functions keep their source names (`svs85_fion_HI, ...`) in the convention `rr_badnell` and `h2_self_shielding_richings` already set. `h2_vibrational_relaxation` holds the two constants and the quench fraction the electron-driven and photon-driven channels share.

The split into a per-cell part and a per-energy part is what keeps the cost down: the powers x^a are evaluated once per cell at the six tabulated energies, and each absorber then costs one gather and one interpolation per spectral bin, on *its own* excess energy $h\nu - E_{\text{th}}$ rather than a single shared one, with the bracket and weight built once at startup.

Verification.

- The module was compiled standalone against an independent Python implementation of the same equations and agrees **to every printed digit** over 24 combinations of ionized fraction (0, 10^{-8} , 10^{-5} , 10^{-2}) and $n(\text{H}_2)/n(\text{H})$ (0, 0.1, 1, 10^3) at all six tabulated energies. Re-run after each of the four changes.
- Energy conservation, $n_{\text{H}1}c_{\text{H}1} + n_{\text{H}2}c_{\text{H}2} + n_{\text{He}1}c_{\text{He}1} = f_{\text{ion}}$, holds **exactly** at every one of those points and at every tabulated energy.
- At 1 keV and $n(\text{H}_2) = 0$ the partition reduces to Shull & van Steenberg's own per-atom values, $c_{\text{H}1} = 0.3908$ and $c_{\text{He}1} = 0.554$ at $n(\text{He}1)/n(\text{H}1) = 0.1$ — the check that the energy ratios are normalized where they should be.
- The vibrational and fluorescence channels leave the two converged atomic cases **byte-identical**, which is the evidence that the H_2 terms are zero when they should be rather than merely small.
- Rebuilt with `-fcheck=bounds,do,mem` and run bounded on three cases — HD 209458 b (1500 steps), `mol_sec_ion` and `mol_lyman_werner` (800 each, so that both new molecular paths are entered) — with **no trap**. That matters here because the interpolation table `sec_kE/sec_wE` is indexed by an absorber number that runs past the five fixed edges into the metal ions.
- make check on the refreshed goldens is **byte-identical on all nine cases**, re-run once more after the optimized rebuild that followed the runtime-checked one.

And the molecular gates could not test any of it, which is a finding in its own right. `Secondary_ionization` defaults to STAGED: the coupling is switched on only after the wind first converges without it. Every molecular regression case is a fixed-maxsteps relaxation snapshot that never converges — `mol_base_handoff` ends at $du = 2.48$ — so `sec_ion_active` never flips, `sec_on` is false for the whole run, and the entire partition path, old and new, is by-passed. A sweep of every case's run. Log found the line "secondary ionization activated" in exactly four: `wasp_full`, `wasp_full_newton`, `wasp_he23off`, `wasp_he23off_newton`. None of the molecular gates, and not `lower_profile`. So a molecular gate coming back byte-identical under a change to this physics is **not** evidence that the change is small. It is evidence that the channel is switched off in that case, and that **the regression matrix had never exercised secondary ionization in a molecular gas at all**.

mol_sec_ion closes that hole and is now a default case. It is mol_base_handoff with Secondary_ionization: Immediate and nothing else changed, at the same 12000 steps, so it isolates the switch. It is the only case in the matrix in which a photoelectron is partitioned in molecular gas, and it is what guards the H₂ target, the dissociation heat and the two vibrational returns. It was recorded as (P28) in TO_BE_DONE.md, which this closes.

Two consequences for how the rest of this section reads. **Quantitative movement is quoted only from wasp_full_newton and wasp_he23off_newton**, the two converged cases in which the coupling actually fires; the molecular gates are relaxation snapshots and are used only for bit-identity and for the sign and order of a change. And every “measured” line above comes from its own build: the four changes were made and measured one at a time, because all four move the same integral.

A case that does exercise the path. mol_base_handoff was re-run with Secondary_ionization: Immediate added and nothing else changed, at the same 12000 steps, so the comparison is the switch and not the code version. (This A/B was taken on the code with the H₂ target in and the three later repairs not yet made; mol_sec_ion on the finished code ends at $du = 2.46$, the same order again.) It completes with no NaN: du rises through a transient peak near step 2900 and relaxes back to 2.56 at step 12000, against 2.48 for the staged run, i.e. to the same order. **Both are relaxation snapshots, not converged solutions, and nothing quantitative should be read off them** — what they establish is that the path is live and stable, and the direction:

	secondary off (staged)	secondary on
max n_e/n_H in the molecular layer	5.0×10^{-5}	9.7×10^{-5}
peak $n(H_3^+)$ [cm ⁻³]	6.69×10^4 at $r = 1.032$	2.84×10^5 at $r = 1.016$
H ₂ front, $f(H_2) = 0.5$	$r = 1.158$	$r = 1.143$
layer temperature range [K]	873–1309	905–1304

The electron fraction roughly doubles and H₃⁺ rises rather than falls. **That is the opposite of the direction Frelikh & Murray-Clay report**, where the extra electrons destroy H₃⁺ faster than it forms. The two are consistent if the difference is the weighting: our H₂ channel takes 82–89% of the ionization energy in this layer and feeds H₂⁺, which converts to H₃⁺ through R8, while their Glassgold weighting gives H₂ about half that share. Whether that is the whole explanation is not settled here, and the comparison it belongs to is (P23), which requires a converged solution.

The whole block, end to end

Four changes, each built and measured on its own, on the two converged cases that fire the coupling. Escaping mass flux $\rho v r^2$ at the escape radius, wasp_full_newton:

step	mass flux	$\log_{10} \dot{M}$	largest local change
starting point (rename only, byte-identical)	6.3388×10^{15}	13.20	—
heating amplitude + closure	6.5682×10^{15} (+3.62%)	13.22	heat +22% at $r = 1.075$
E_sec_ion 40 $\rightarrow$ 30 eV	6.5567×10^{15} (−0.17%)	13.22	heat −1.6% at $r = 1.31$
ionization channels resolved in E_0	6.5986×10^{15} (+0.64%)	13.22	T +3.8% at $r = 1.13$
xion = electron fraction	6.6604×10^{15} (+0.94%)	13.22	heat +8.6% at $r = 1.000$
vibrational and fluorescence returns	6.6604×10^{15} (0.00%)	13.22	byte-identical (no H ₂)

Cumulative: +5.07% in the escaping mass flux, +0.02 dex in $\log_{10} \dot{M}$, with wasp_he23off_newton giving +4.91% and the same 13.20 to 13.22. The mass flux moves by five percent while the reported $\dot{M}$ moves by two hundredths of a dex, which is the usual reminder that a two-decimal $\dot{M}$ is not the quantity to judge a change by.

The signature is where the changes land, not their size in the total: every one of them is concentrated between $r = 1.0$ and $r = 1.3$, the weakly ionized shell just above the base, and every one is under half a percent above $r = 1.6$ where x is already 0.99. That is the region where Shull & van Steenberg’s fit was being evaluated eight to twelve decades outside its stated range, and it is the region the molecular layer sits in.

The regression matrix, and what its pass/fail pattern proves. The goldens were refreshed once, at the end of the block, with the previous set preserved in backup/regression/golden_blockL_20260902/. Which cases moved is itself the evidence for (P28):

case	verdict	why
wasp_full	moved	fires the coupling; $\dot{M}$ 13.23 to 13.26, 14091 to 11590 steps
wasp_he23off	moved	fires the coupling; $\dot{M}$ 13.23 to 13.26, 14068 to 11269 steps
mol_lyman_werner	moved	carries a stellar Lyman–Werner field
mol_sec_ion	new	the only case that partitions in molecular gas
mol_base_handoff	byte-identical	staged coupling never fires, no LW field
mol_metals	byte-identical	same
mol_diffusion	byte-identical	same
mol_ir_bands	byte-identical	same
lower_profile	byte-identical	same

Five of the nine cases came back bit for bit unchanged under a block that rewrote the heating fraction, the ionization channels, the ionized fraction and two heat returns. **That is not a statement about the size of the change; it is the demonstration that the staged default was keeping this physics switched off in every one of them.** wasp_full’s $\dot{M}$ is quoted here only because it is the golden’s own number: it is a du-stop rather than a Newton finish, and the du-stop’s path dependence is several percent, which is why every quantitative line above uses the _newton pair instead.

119. A segmentation fault that only appeared on more than one thread: a residual histogram indexed by $\log_{10}$ of a NaN (2026-09-02)

The defect. A molecular run with Solver: Newton died with SIGSEGV as soon as the steady solver was entered, at the first residual evaluation after the (JFNK) below `r_esc` line, whenever `OMP_NUM_THREADS` was greater than one. The identical run on one thread did not crash. `OMP_STACKSIZE=1G` changed nothing, so it was not the size of a thread stack. Reproduced on backup/regression/armD_D2_newton (8 threads), on the same case with the large stack, and on armD_D2_LW_newton (6 threads).

The cause is one bin selector: `src/modules/radiation/ionization_equilibrium.f90`, the `residual_decade` expression at line 2195 as it stood. Rebuilt with `-g -fcheck=bounds,do,mem`, the fault is a bounds violation rather than a stack overflow, and it names its own line:

At line 1267 of file `src/modules/radiation/ionization_equilibrium.f90`

Fortran runtime error: Index '-2147483633' of dimension 1 of array

'hist_uncv' below lower bound of 0

`residual_decade(res)` returned `min(15, 16 + int(floor(log10(res))))`. NaN fails every comparison, so a NaN residual reached that expression; `int()` of NaN is the integer indefinite value -2^{31} , and the whole expression then evaluates to exactly `-2147483633` on this compiler (checked with a standalone program). The value is used directly as the index of `hist_conv` / `hist_uncv` at lines 1264/1267 and 1494/1497, both inside the OpenMP-parallel cell sweep, so the increment writes about 8.6 GB below the array.

Why the residual is NaN, and only under the steady solver. `jv_product` (`src/modules/time_step/steady_newton.f90:679-700`) forms Jv by evaluating the full residual at $Y + \epsilon v$ with no admissibility test on that state at all, and the line-search trial test (`steady_newton.f90:944-945`) covers only density and pressure positivity. A probe state can therefore drive a balance row of the molecular network to NaN, and `normalized_reaction_residual` returns it. In the marching phase of this arm it never happens: the `-fcheck` build runs 2002 marching steps and trips on the first JFNK residual evaluation.

Why the thread count decides whether it crashes. `hist_conv` and `hist_uncv` are `reduction(+: . . .)` arrays of the parallel sweep, so each thread increments its own private copy. On a worker thread that copy sits on an 8 MB pthread stack, and the address 8.6 GB below it is unmapped, so the write faults; on the master thread the same wild write does not fault. Measured with one and the same -O3 binary: `OMP_NUM_THREADS=1` completes the JFNK attempt (`info=2`) and resumes marching, `OMP_NUM_THREADS=8` dies in `__ionization_equilibrium_MOD_ioniz_eq._omp_fn.1`.

The fix. `residual_decade` now selects the top bin from the bracket instead of clamping a computed index, so $\log_{10}$ is never evaluated on a residual that is not finite:

```
if (res .le. 1.0d-16) then
  ib = 0
else if (res .lt. 1.0d-1) then
  ib = 16 + int(floor(log10(res)))
else
  ib = 15
```

```
endif
```

For every finite residual this returns exactly what the old expression returned: below 10^{-1} the `min(15, . . .)` was never binding, and at or above 10^{-1} it always returned 15. NaN and an overflowed residual now land in the top bin, which is where a residual that large belongs. The bins are diagnostic counters; no physics reads them.

Verification, three things.

1. *The reproduction, 8 threads.* `armD_D2_newton` now enters the steady solver, runs its JFNK iterations to the no-descent abort, exits `info=2` (this arm does not converge; the criterion here is the absence of the crash) and finishes the run.
2. *Serial versus parallel byte-identity*, re-checked after the change on the two cases that exercise the atomic and the molecular equilibrium path: `wasp_full_newton` and `mol_sec_ion`, each run at `OMP_NUM_THREADS=1` and `=8`. `Hydro_ioniz.txt` and `Ion_species.txt` are identical in both pairs, and `mol_sec_ion` also reproduces its golden.
3. *The regression matrix.* `make check`, all nine cases byte-identical. A thread-safety repair cannot move a single-threaded result, so any movement here would have meant the repair was wrong.

Two things this section does not repair, recorded where they were found.

The steady solver's probe states reach run-level accumulators. The species fractions do not leak – `jv_product` and `build_banded_jac_full` restore `f_sp` from a base copy before every probe, and the line search adopts `f_sp_j` only for an accepted trial (`steady_newton.f90:959`). But `ieq_nonroot_streak` (`ionization_equilibrium.f90:189`, written at 2223 and 2227) is module state that no probe path saves or restores, and its limit error stops the run at 1000 consecutive sweeps. Its meaning during a JFNK phase is not “consecutive marching sweeps on a non-root”: probe sweeps both increment it and reset it. The acceptance statistics (`ieq_acc_n`, `ieq_acc_resmax`, `ieq_hist_conv` / `ieq_hist_uncv`, the constrained-solve totals and the print budget, lines 203–210 and 1655–1664) are accumulated the same way, so the end-of-run report is dominated by states the run never adopted. Measured on `armD_D2_newton`: not one non-root acceptance in the 2002 marching steps before the hand-off, and 18339 over the JFNK phase and the eight marching steps that followed it, largest residual 1.74×10^{97} , with 18373 cells promoted to the constrained continuation solve of which 34 were accepted.

The pathological arm is not serial-versus-parallel byte-identical in its marching phase either. `armD_D2_newton` at 1 and at 8 threads diverges in the last bit at step 1267, well before the steady solver is reached. It is reproducible at each thread count and it is present in the code as it stood before this section (built with the same default flags), so it is not introduced here; the `-O3 -g` build of the same source does not show it, which makes it a marginal decision flipping on a rounding difference rather than a data race. It is not diagnosed here. The regression cases stay byte-identical, as verification 2 and 3 above show.

120. He recombination photons had no competing absorber, and an initial condition that clipped H_2 at 1 handed them a cell with no hydrogen (2026-09-02)

Judgment first. Two expressions were physically wrong, independently of each other, and they combined into a runaway.

1. `he_rec_coupling` implements Draine’s (2011) on-the-spot treatment: a He II recombination photon above 13.6 eV is absorbed inside the cell that emitted it and ionizes hydrogen. Only the ≥ 24.6 eV ground-capture continuum asked *who* absorbs the photon. Every other channel — four of them in the He 2^3S branch, the case-B cascade in the atomic one — gave every photon to H I unconditionally and then divided by $\max(n_{HI}, 10^{-99})$. In a molecular gas that is wrong on the physics before it is wrong on the arithmetic: H_2 photoionizes from 15.4 eV with a cross section 1.2–3.7 times that of H I over the 16–25 eV band these photons occupy, so in a molecular layer H_2 takes essentially all of them, and the share it takes belongs in the H_2 photoionization rate.
2. `set_IC` clipped the initial H_2 nucleus fraction at $x_2 = 1$ rather than at the element-ratio ceiling $q_{H_2} \leq 0.5/(0.5 + \text{He}/H)$ — the same silent cap §117 removed from the base particle count, left behind in the initial condition. Above $\text{He}/H = 0.0964$, where the solar-composition fit exceeds that ceiling (item P26), the clip produced $f_{\text{sp}}(H\text{ I}) = 0$ *exactly*, which is the one input that makes (1) diverge.

The rate is now split among the absorbers present, at each channel’s own photon energy, and the H_2 share is returned as an H_2 photoionization rate; the initial condition takes its H_2 fraction from the same source the inflowing ghost does. A third defect the same audit turned up — an EOS-only molecular base whose ghost silently sits at $n_{\text{tot_bc}} \times T_0$ — is refused at startup from this section on; it is its own subsection below because the refusal invalidates existing runs and the decision to write it was the user’s. The measured consequence is that the “molecular base runs supersonic near $\text{He}/H \sim 1\text{--}10$ ” window of `docs/supersonic_molecular_base.md` is not a physical phenomenon: the same arm now relaxes to a subsonic, warm, fully molecular base.

What the defect did

In the $\text{He}/H = 1$ hot-Uranus arm the first physical cell was fully ionized ($f(H\text{ II}) = 1$) after the single `ioniz_eq` pass of step 0. With $n_{HI} = 0$ the photon production rate $P_{\text{add}} \sim 1.5 \times 10^{-5} \text{ cm}^{-3} \text{ s}^{-1}$ became a rate coefficient of $1.5 \times 10^{94} \text{ s}^{-1}$, and $x(H\text{ II}) = 1$ is then the exact root of the hydrogen balance row — not a failed solve. The energy solve of the same step took that cell from 1169 K to the 11.4 K floor on the recombination and free-free cooling of the false electron reservoir, the particle count and hence the pressure jumped by a factor 54, and the legacy velocity boundary copied the resulting interior velocity: supersonic base inflow by step 6.

Three properties made this survive every guard built in §§113–117:

- *the residual test cannot see it.* The acceptance test of §113 normalizes each balance row by its own turnover, and the hydrogen row’s turnover (`s(1)` in `System_HeH_mol.f90`) is built from the same P_{HI} . Numerator and denominator both carried the 10^{94} : normalized residual 2.7×10^{-17} , raw residual -1.3×10^{16} ;

- *the rate and the heat of one channel disagreed structurally.* The heating of the same photons is $n_{\text{HeII}}n_e\alpha$ times an energy, a volumetric rate, and stayed at a perfectly ordinary $2.6 \times 10^{-10} \text{ erg cm}^{-3} \text{ s}^{-1}$. A channel whose rate diverges while its heat does not is not a physical channel;
- *no regression case can reach it.* The five molecular cases run at $\text{He}/\text{H} = 0.0793$, where the fit gives $x_{2,\text{ic}} = 0.984 < 1$, the clip never engages, and n_{HI} never reaches zero.

The correction

For each channel c the photon of energy E_c is divided among the absorbers whose threshold lies below E_c ,

$$w_s(E_c) = \frac{n_s \sigma_s(E_c)}{\sum_{s'} n_{s'} \sigma_{s'}(E_c)},$$

with s over H I (13.6 eV), H₂ (15.4 eV) and, for the ≥ 24.6 eV ground-capture continuum only, He I. Five channels, each with its own energy:

channel	E_c [eV]	absorbers	$\sigma_{\text{H}_2}/\sigma_{\text{HI}}$
ground capture continuum	24.6	H I, H ₂ , He I	3.66
$2^1P \rightarrow 1^1S$ 584 Å resonance	21.2	H I, H ₂	3.54
$2^3S \rightarrow 1^1S$ line	19.8	H I, H ₂	3.46
2^1S two-photon continuum	16.110	H I, H ₂	1.20
case-B cascade (atomic branch, lumped)	19.741	H I, H ₂	3.45

The H I share becomes dP_HI , the H₂ share a new output dP_H2 that the callers add to P_H2 (which already drives the H₂ destruction row and the H_2^+ formation row of the molecular system), and the He I share is excluded from the effective He II recombination coefficient, which generalizes from $\alpha_B + y\alpha_1$ to $\alpha_B + (w_{\text{HI}} + w_{\text{H}_2})\alpha_1$. The photoelectron of each share is deposited at its own threshold, 13.6 eV for H I and 15.4 eV for H₂. Where the cell holds no absorber for a channel the photon leaves and that channel returns zero, so the He II recombination reverts to its uncoupled value instead of dividing by the floor.

Two properties of the implementation are worth recording because they are not free choices:

- *the shares are written in the ratio form* $w_{\text{HI}} = 1/(1 + R_{\text{He}}n_{\text{HeI}}/n_{\text{HI}} + R_{\text{H}_2}n_{\text{H}_2}/n_{\text{HI}})$, not as $n_{\text{HI}}\sigma_{\text{HI}}/\sum$. With $n_{\text{H}_2} = 0$ the added term is exactly 0.0 and w_{HI} is exactly 1.0, so multiplying an atomic run's channel by it is the identity and the three atomic regression cases stay byte-identical. Rearranging to the algebraically equivalent $P_c\sigma_{\text{HI}}/\sum$ would not have been;
- *the rates stay finite as* $n_{\text{HI}} \rightarrow 0$. Dividing the H I share by n_{HI} cancels the n_{HI} in w_{HI} , leaving the trace-particle limit — the rate a lone H I atom would see in a sea of H₂ — rather than a divergence. The same cancellation gives dP_H2 without dividing by n_{H_2} . This is the property the earlier form lacked, and it is a consequence of writing the physics correctly, not a guard bolted on top.

Approximations, stated at the code site. The on-the-spot assumption itself is unchanged and its optical depth is not tested, so a cell thin to these photons keeps them in the local budget and both the ionization and the heating are

overestimated. The trace metals are not in the absorber competition (item P34; they were added in section 129., where the share they take is measured). Each channel is collapsed onto one representative photon energy; for the 2^1S two-photon continuum, which is not a line, that energy is the mean of the in-band photons, $13.598 + E_{e_2q_HeI} = 16.110$ eV. Its H_2 -ionizing photon *count* should be the integral of the Drake, Victor & Dalgarno (1969) shape over 15.4–20.62 eV rather than the 13.598–20.62 eV integral $f_{2q_HeI} = 0.5564$ the code carries; that shape is not in the code, only its two integrals, so it cannot be re-integrated and f_{2q_HeI} is used for H_2 as well. This overestimates one third of one of five channels, on the channel with the smallest $H_2/H\ I$ cross-section ratio.

The initial condition

set_IC's molecular seed now has one rule: **the initial x_2 comes from the same source the ghost does.** When a lower-atmosphere handoff states the partition (base.inp key `q_H2_base`, or a profile's value at the matching level), `base_h2_nuclei_fraction()` — the §117 ghost value — is used for the whole column. Otherwise the chemical-equilibrium fit gives the local $q_{H_2}(p, T_0)$, clipped at `h2_mixing_ratio_ceiling()` with a one-line startup notice rather than silently at $x_2 = 1$. Clipping at the ceiling still gives $x_2 = 1$, which is what a fully molecular mixture means, and with the coupling corrected that is harmless.

This moves the five molecular regression cases: all five carry a handoff, so their IC H_2 fraction changes from the fit's 0.984 to the handoff's own value, and their goldens were refreshed for it. `lower_profile` does not solve the molecular network at all, so its seed block never runs and it stays byte-identical.

A third defect the same audit found, and the decision it needed

Molecular base: True with Molecular chemistry: False is an EOS-only molecular base: the H nuclei bound into H_2 leave the base particle count (`comp_ntot_bc`) while the species arrays stay atomic. The `ntot_bc` recomputation at the end of `ioniz_eq` that reconciles the two is gated on the network being solved, so with the network off it never fires and the base ghost pressure $(ntot_bc + dp_bc)T_0$ sits on gas that counts one particle per nucleus. **The isothermal lower boundary is then not isothermal: it sits at $ntot_bc \times T_0$.** Measured on backup/regression/armD_D1, `ntot_bc` = 0.768722 and the ghost is at 876.34 K where Equilibrium temperature: 1140.0 was asked for. Section 11.5-C of docs/supersonic_molecular_base.md had already written down the required response — “the code must instead refuse a mismatch between that EOS state and the species state used at the same ghost”.

It was not written immediately, because a startup stop changes what an existing input file means and the grep said 31 run directories use the combination. **The user chose refusal on 2026-09-02**, and `input_read` now stops with the resolved q_{H_2} , x_2 , `ntot_bc`, the implied ghost temperature against the requested T_0 , and the two ways out: turn the network on so the H_2 the particle count assumes is actually carried, or drop Molecular base: True and `q_H2_base` if the base is meant to be atomic.

What the refusal invalidates. backup/regression/armD_D1 (a diagnostic arm, input deliberately left as it was so the refusal reproduces there) and 30 HD 209458 b VULCAN handoff runs under `vulcan_work/`, all at $T_0 = 1450$ K:

group	ntot_bc range	base ghost [K]
hd209_wind_response/ (10 runs)	0.556–0.994	806.2–1441.3
pc_compare_p1/exhale_hd209/ (20 runs)	0.555–0.994	804.8–1441.3

The lowest values come from the deep 10^{-4} bar handoffs ($q_{\text{H}_2\text{base}} \sim 0.80$, $x_2 = 0.977$) and the highest from the two shallow chemical-equilibrium-fit arms ($q_{\text{H}_2} = 0.006$). None has been re-run — that is a separate instruction. Each run directory carries an `INVALID_BASE_TEMPERATURE.md` with its own numbers, the two parent directories carry the summary tables above, and the seven documents that quote these runs carry a stale marker at the quoting passage (`base_composition_handoff_plan.md` section 11, `newton_scaling_and_base_wall.md`, `viscosity_conduction.md` G3, `oxygen_chemistry_new_plan.md`, `vulcan_photochem_comparison.md` in two places, `TO_BE_DONE.md` item (A), and the section 117 passage above, whose two quoted q_{H_2} values are `VULCAN` handoff *inputs* and are unaffected).

A diagnostic for the failure mode

`n_cells_he_rec_photoionization_dominant` counts cell-steps in which the coupling's `dP_HI` exceeds the direct stellar photoionization integral `P_HI` by more than 1000. The coupling is a *correction* to that integral, so three decades of dominance is a statement that the local photon budget has run away. It is reported at the end of a run only when it is nonzero, like the positivity counters, and it is zero on every case measured here.

Verification

1. **Byte-identical** on the three cases with no molecular hydrogen, which are the ones the bitwise property above is designed to protect: `wasp_full` (He 2^3S on, the TR branch of the routine), `wasp_he23off` (the atomic branch) and `lower_profile`. `Hydro_ioniz.txt` and `Ion_species.txt` identical to the goldens in all three.
2. **The arm, 12000 steps** (backup/regression/armD_D2, the hot Uranus at $\text{He}/\text{H} = 1$ with $q_{\text{H}_2\text{base}} 0.300861$, `OMP_NUM_THREADS=1`): base inflow subsonic throughout at max Mach 8.64×10^{-2} , zero energy-floor activations, zero activations of the new counter, $\log_{10} \dot{M} = 10.91$. The first physical cell at $r = 1.000193 R_p$ holds $n(\text{H}_2) = 2.49 \times 10^{13}$, $n(\text{HI}) = 2.28 \times 10^{10}$, $n(\text{HII}) = 1.4 \times 10^3 \text{ cm}^{-3}$ at 1165.7 K, against $n(\text{HI}) = 0$, $n(\text{HII}) = 5.0 \times 10^{13}$, 132.1 K before. Two controlled runs, each cutting one defect alone, reach the same solution: `He_rec_coupling: False` gives max Mach 8.76×10^{-2} and $\log_{10} \dot{M} = 10.89$, and the IC clip alone gives 8.64×10^{-2} and 10.91.
3. **The H_2 share is real.** In the molecular base of `mol_diffusion`, where $n_{\text{H}_2}/n_{\text{HI}} = 1.7 \times 10^3$, the returned rates are $\text{dP}_{\text{HI}} = 3.65 \times 10^{-23}$ and $\text{dP}_{\text{H}_2} = 1.10 \times 10^{-22} \text{ s}^{-1}$ per particle, so H_2 absorbs $n_{\text{H}_2} \text{dP}_{\text{H}_2} / (n_{\text{HI}} \text{dP}_{\text{HI}}) = 5.2 \times 10^3$ times as many of these photons as HI does: 99.98% of them. All of that used to go to HI.
4. `make check`, 9 of 9. `wasp_full`, `wasp_he23off` and `lower_profile` pass against the existing goldens; the five molecular cases move, for the two reasons above, and their goldens were refreshed at the end of this block. **How far they move**, measured against the old goldens before the refresh: $\log_{10} \dot{M}$ falls by 0.00 to 0.02 dex in every case ($10.57 \rightarrow 10.56$, $10.58 \rightarrow 10.58$, $10.58 \rightarrow 10.56$, $10.57 \rightarrow 10.56$, $10.58 \rightarrow 10.57$, and $10.54 \rightarrow 10.52$).

for `mol_sec_ion`), the sign being the same everywhere. In the molecular layer, $r < 1.3R_p$, the largest relative changes are 4–7% in T , 9–14% in density, and up to 80% in $n(\text{H}_2)$ at the H_2 front itself, which moves *outward* by $0.0003R_p$ ($1.0178 \rightarrow 1.0181$). The outer wind of these 12000-step snapshots moves more, but they are relaxation snapshots at $du \sim 3$ rather than converged solutions, and their outer cells are still in the cold-start transient.

5. `run_fcheck.sh` (bounds/do/mem checking build, one bounded HD 209458 b run): clean, no runtime check fired. It needed two repairs to run at all, neither related to this section’s physics and both in the script: it is the only harness that compiles from scratch, so it was picking up whichever gfortran the PATH resolved to (a conda toolchain here, whose LAPACK links against a different libgfortran and fails with undefined `_gfortran_*@GFORTRAN_1.0`), and its `-g` build hit a linker that cannot read gfortran 13’s compressed `.debug_info`. It now prefers `/usr/bin/gfortran` when there is one and passes `-gz=none`, and it prints the tail of the build log when the build fails instead of only “BUILD FAILED”.

121. What the steady solver’s probe states were allowed to write, and what they were allowed to be (2026-09-02)

Two defects, both found while diagnosing the segmentation fault of section 119. and neither repaired there. They are numerical hygiene of the steady (JFNK / PTC) solver, not physics: no rate, cross section or balance is touched, and the marching path is unchanged.

(a) The equilibrium sweeps of probe states wrote the run’s own counters. `ioniz_eq` keeps run-wide totals — the acceptance classes of section 113. and their largest normalized reaction residual, the residual-decade histograms, the root-validation counters, the constrained-solve totals, the print budget — and one array with authority over the run, `ieq_nonroot_streak`, whose limit error stops it. The steady solver calls `ioniz_eq` on three different kinds of state: its current iterate; the finite-difference and Krylov probes $Y + \epsilon v$ that build the Newton model; and line-search trials, which are candidates until one is adopted. All three wrote the same counters as the marching loop.

The species fractions did *not* leak — `jv_product` and `build_banded_jac_full` restore `f_sp` from a base copy before every probe, and the line search adopts `f_sp_j` only for an accepted trial. The counters did. Measured on `backup/regression/armD_D2_newton` at 8 threads: not one non-root acceptance in the 2002 marching steps before the hand-off, and 18,339 over the JFNK phase and the eight marching steps after it, largest residual 1.74×10^{97} — an end-of-run report dominated by states the run never adopted. `ieq_nonroot_streak` was worse than noisy: probe sweeps both raised it and cleared it, so “consecutive sweeps resting on a non-root”, the quantity the stop is defined on, counted evaluations that were thrown away in the next statement.

(b) Nothing tested whether a probe state could be described at all. The line-search trial test was density and pressure positivity (`steady_newton.f90`, `minval(Wtry(1,1:N)) > 0` .and. `minval(Wtry(3,1:N)) > 0`). `jv_product` evaluated the full residual at $Y + \epsilon v$ unconditionally, with v a Krylov direction of no physical meaning and `gm = 40` such probes per outer iteration; `build_banded_jac_full` probed $Y + \sqrt{\epsilon}D$ the same way. A probe state can drive a balance row of the molecular network to NaN or to overflow, `normalized_reaction_residual` returns it, and that residual then entered the Jacobian, the Arnoldi recursion and the merit comparison. The SIGSEGV of section 119. was

one symptom; the residual itself was the disease.

The design

Three ledgers, tagged at the call site. Each equilibrium sweep is tagged `ieq_state_marching`, `ieq_state_steady_iterate` or `ieq_state_steady_candidate` (`set_ioniz_eq_sweep_state_kind`, called from outside any parallel region), and its totals go to the ledger of that tag. The counters that used to be a dozen separate module variables are now the fields of one derived type, `ioniz_eq_ledger`, held in three instances; the end-of-run report prints one block per instance and stays silent for an instance nothing was recorded in, so an ordinary marching run prints exactly what it printed before. The tag also decides two things that are not statistics. A candidate sweep does not touch `ieq_nonroot_streak` — it neither increments nor resets it, and the stop is not evaluated — so the streak once again counts consecutive sweeps of a state the run holds. And a candidate sweep's non-root acceptances are not reported cell by cell and do not spend the print budget: a probe acceptance is a property of a state the solver invented to differentiate at, and tens of thousands of them bury the events that matter. They are not lost; they are in the third ledger, with the class populations and the largest residual of each.

The line-search trial that *is* adopted was evaluated as a candidate, so its sweep is counted in the candidate ledger; the next outer iteration re-evaluates the same state as the iterate and counts it there. The one state this leaves in the candidate ledger only is the final accepted trial of a converged solve, which is never re-evaluated.

Admissibility, in two layers.

The signal. `ioniz_eq` takes an optional `sweep_ledger` argument and returns this sweep's own totals in it, including three new fields: the number of cells whose accepted state carries a normalized reaction residual that is **not a finite number**, the number of cells at which no starting point stayed inside the element simplex, and the largest element-budget violation of any iterate that left the simplex (whether or not a later starting point of the same cell then found a root). Two related repairs went in with it, both consequences of NaN failing every comparison: a non-finite `acc_res` used to pass the class-3 recheck (`acc_res > ieq_res_tol` is false for a NaN) and be recorded as a *root*, and it used to enter `acc_resmax`, where a single occurrence would have destroyed the largest-residual statistic of the whole run. A non-finite residual is now class 4 by construction and is kept out of the maximum. The atomic branch also measures the element violation of the iterate it rejects, which it previously reported as zero.

`eval_residual` combines that signal with a scan of the assembled residual vector and returns one logical, `admissible`. Two independent statements are required because they catch different things: the sweep count names the cell whose chemistry broke, which is where the diagnosis is; the row scan catches anything non-finite that reached the residual by another route. The argument is optional, so the marching path never asks and is not affected.

The line search. A trial that is not admissible is rejected — λ is halved — exactly as a negative density is, in both `solve_steady_jfnk` and `solve_steady_ptc`. This is not only explicitness: a non-finite merit failed the Armijo test by accident of IEEE ordering rather than by decision, and a trial whose chemistry broke in a few cells can still produce a finite, smaller merit and be adopted.

The temperature band was considered and not added. The only temperature limit in the code is the $0.01 T_0$ clamp inside the Newton iteration of `energy_semi_implicit`, which belongs to the marching energy update and has no upper counterpart; there is no band in the steady path. The brief for this work was to use a band if one existed and

not to invent one, and none exists. Positivity of the pressure already excludes a non-positive trial temperature, and a temperature that breaks the network shows up in the admissibility signal.

The element-budget violation is measured and reported, and does not reject. The accepted state of every acceptance class is inside the simplex by construction — the molecular branch clamps, the atomic branch hands back the uncoupled balance — so the violation describes the iterate that was rejected, not the state returned. Whether a large one should disqualify a trial cannot be decided without measuring where healthy solves sit, and the healthy cold start of section 113. legitimately clamps. It is therefore carried in the ledgers and printed, and no threshold is asserted on it.

The probes: sample closer, and never invent the derivative. A Jacobian column and a Krylov product are directional derivatives *of the residual at Y* . The point $Y + hv$ at which they are sampled is not physics: h is ours to choose and v has no physical meaning. So when the sample point leaves the set of states the code can describe, the honest answer is to sample closer to Y — the step is halved and the probe repeated, at most `n_probe_step_halvings` = 6 times, over which the finite-difference step falls by 64 and stays far above the round-off floor of the residual.

If even the smallest of those steps is inadmissible, the boundary runs through Y itself along v and there is no derivative to sample. Then none is invented:

- `pgmres` stops the cycle at the directions it did build and returns the step from that subspace. A shorter GMRES cycle is a valid approximate solve, and the line search judges the step it produces.
- If not one direction could be sampled, the step is zero and there is no Newton model at all for that outer iteration. The zero step is then kept *out of the line search*, and the iteration is recorded as a no-descent iteration. This is not cosmetic: the non-monotone test compares a trial against the *worst* merit of the last five iterates, so $Y_{\text{try}} = Y$ passes it whenever the current iterate is not that worst one — measured on `armD_D2_newton`, consecutive outer iterations with `gm` = 0 were being recorded as accepted steps at $\lambda = 1$, resetting the no-descent counter and letting the solve spin to `maxit` instead of aborting. With the zero step rejected the solve reaches its stagnation abort, which is the honest outcome.
- `build_banded_jac_full` leaves the columns of an unresolvable color at zero. Those columns are a *preconditioner*; zeroing them degrades the treatment of those unknowns to plain relaxation and costs iterations, not correctness. If the base point Y itself is inadmissible no column is built at all.

Both events are counted and printed, per solve. Nothing non-finite can reach the Arnoldi recursion or the band matrix: `jv_product` returns $Jv = 0$ with `ok` = `.false.` and the caller is required to stop using v rather than use the value.

What the code now says, measured

The verification arms are two builds of the **same source tree** — the tree as it stood after section 120. went in — one of them with this change and one without, so the comparison isolates it. Single thread unless stated.

The regression matrix and the two Newton cases are byte-identical. All nine matrix cases plus `wasp_full_newton` and `wasp_he23off_newton`: the data lines of `Hydro_ioniz.txt` and `Ion_species.txt` agree in all 22 file pairs, and the nine matrix cases reproduce `backup/regression/golden/` exactly in both arms. `wasp_full_newton` is one of the two cases that runs the JFNK to convergence (`info` = 0, $\|R\| = 5.615 \times 10^{-4}$ in 18 outer iterations), so its identity is

the statement that matters: **no line-search trial that the new rule rejects was ever adopted by the old code in that solve.** Its new ledger reads 4241 marching sweeps, 19 steady-solver iterate sweeps, 388 probe sweeps.

Serial versus parallel. `wasp_full_newton` with the final build at `OMP_NUM_THREADS = 1` and `= 8`: both output files identical.

The He/H = 1 arm, restarted from a healthy state and forced through the JFNK. The arm of section 120. (hotUranus_diffusion, He/H = 1, molecular chemistry and He/H diffusion on) was restarted from its step-168,550 state — $du = 1.735 \times 10^{-2}$, monotonically converging — with Load IC? True, `du_th [PLM,WEN03]: 1.0e9 1.0e-3` and Solver: Newton 100.0, which puts the solver into the JFNK immediately. This is a healthy state, so the criterion is the opposite of a failure report: **the solve has to converge, and the new hygiene must not be what stops it.** (The restart was necessary: that marching run does not reach `du_th` by itself — its du bottoms out at 9.96×10^{-3} at step 251,287 and then rises again.)

At one thread the two arms are **byte-identical**, output files and JFNK trace alike. The run marches 2002 steps to $du = 3.1094 \times 10^{-2}$, enters the JFNK three times, and every one of them returns `info = 0`; the last leaves $\|R\| = 3.264 \times 10^{-4}$ and the run closes as *converged: JFNK steady solution*, at

$$\log_{10} \dot{M} = 10.32 \text{ g s}^{-1},$$

which is the first quotable quantitative He/H = 1 value since P17. Not one hygiene event fired in the whole run: no GMRES cycle truncated, no Jacobian color zeroed, no trial rejected as inadmissible. The three ledgers are clean throughout — marching 2007 sweeps with 1,011,495 converged roots, 15 roots without solver convergence and 18 constrained-continuation roots and **no non-root acceptance at all**; 22 iterate sweeps, all roots; 396 probe sweeps, 199,584 converged roots and again no non-root, no non-finite residual and no off-simplex cell in any of the three.

At 8 threads both arms also converged, `info = 0`, but at $\log_{10} \dot{M} = 10.35$ (old) and 10.33 (new) against the serial 10.32. The traces separated at the very first JFNK line — the starting $\|F_s\|_2$ differed in its third digit, before any hygiene decision can act — so that spread was not this change; the serial identity above is what settles it.

It was a defect of its own, and it is now repaired (item (P37) of `TO_BE_DONE.md`, `docs/supersonic_molecular_base.md` section 12.6). Traced to the constrained element-conserving continuation: its seed floored the species listed in `n_unknown / species_of_unknown`, which are rung-partition state built later in the same solve, so on entry to a cell they still described whichever cell that thread had handled last — and on a thread's first cell they were never initialized at all. A cell's chemical equilibrium has to be a function of that cell's state, so the serial answer was order-contaminated in the same way rather than correct. The seed now uses the cell's own species set and the rung context is cleared on entry; `heh1_jfnk`, `mol_sec_ion` and `mol_metals` are byte-identical at 1 and 8 threads under `static`, `static,1` and `dynamic,8`, and the He/H = 1 rate is $\log_{10} \dot{M} = 10.32$ at every thread count. The six molecular matrix cases that promote a cell to the continuation move with it (the three that do not — `wasp_full`, `wasp_he23off` and `lower_profile` — stay bit for bit); $\log_{10} \dot{M}$ is unchanged to two decimals in all six and the largest relative profile change is 2.0×10^{-2} . Their goldens are refreshed at the end of the block, not here.

The contamination itself, measured on the old pathological restart. The `backup/regression/armD_D2_newton` state — the pathology that preceded section 120., a state the code no longer produces — is what defect (a) was measured on, at 8 threads. Old code, one report for the whole run: 20,704 non-root acceptances under the relaxation

amnesty, largest residual 1.01×10^2 , longest cell streak 17 sweeps, and 17,156 molecular clamps. New code, the same run split by state kind:

Table 2: Acceptance ledgers of one armD_D2_newton run (8 threads), separated by the kind of state the equilibrium sweep was evaluating.

ledger	sweeps	non-root (max res)	off-simplex cells	non-finite residual
marching states	46,047	0	0	0
steady-solver iterates	22	0	0	0
steady-solver probes	1,254	21,789 (2.14×10^2)	12,695	9,093

The entire non-root population, and every non-finite reaction residual, belonged to states the run never adopted; the marching phase it actually walked is clean, which the merged report had made impossible to see. The JFNK on that state fails in both arms (`info` = 2 after the 12 no-descent iterations, then marching resumes and the run ends at the same $\log_{10} \dot{M} = 17.68$), and the new code says why: probe states without an admissible residual: 0 Jacobian color(s) zeroed, 14 GMRES cycle(s) truncated. (That arm is not thread-reproducible in its marching phase, so its counter totals differ from run to run; the JFNK block and the final mass-loss rate do not.)

Why the element-budget violation is not a rejection criterion, measured. The healthy matrix cases carry off-simplex iterates of order unity in their cold start: `mol_metals` 27 cells with a largest violation of **9.55**, and `mol_ir_bands` 3 cells at **0.423**, both reproducing their goldens. There is no threshold that separates those from a probe excursion — the same finding, in the same direction, that section 113. made for the residual severity of a cold start — so the violation is measured and printed and nothing is decided on it.

The boundary of this section

- **Marching is untouched.** Every new argument is optional or is set only from the steady solver, and the two default paths of a marching run are the ones it had. The marching ledger prints the same lines, word for word, that the old counters printed.
- **The state the solver hands back is now consistent in the frozen background too.** `f_sp` is restored with the best iterate and `nmol_eq` / `nox_eq` are rebuilt from `f_sp` at the top of every sweep, so those were always consistent. `bg_cell` — the copy of each cell's coefficients that the carrier transport (`diffusive_photochemistry`) reads — was not: it is written by every molecular sweep, probe sweeps included, so on return it described whatever state was evaluated last, which is a *rejected* line-search trial whenever the final iteration found no descent, and the best-iterate restore could move the returned state further still. Both solvers now keep a copy of it at each state they adopt (`bg_cell_adopted`, `bg_cell_best` in `ionization_equilibrium`) and install the right one before returning, so the invariant is that every cell state derived from the returned state was evaluated at the returned state.

Measured, and the scope is narrower than it looks. On the converging $\text{He}/\text{H} = 1$ solve the repair is a no-op: all three JFNK phases end on an accepted trial with no best-iterate restore, so the background the solver was

already leaving matched the adopted state in all 504 cells. On the failing path it is not: restarted on the old `armD_D2_newton` state, which ends `info = 2` after the twelve no-descent iterations, the background the old code left differed from the adopted state in **all 504 cells, by up to a factor 4 in temperature** (largest relative T_K difference 3.01; sum over cells 6.90×10^5 against 7.32×10^5). *No consumer saw it*, however: EXHALE_main calls `ioniz_eq` at the returned state immediately after the solver returns and before the carrier transport or any resumed marching step, which overwrites `bg_cell` in every cell — measured, the post-solver sweep and the newly installed background agree to 3.4×10^{-5} relative. The repair is therefore a correctness invariant of the solver’s own boundary, not a change of any result; an earlier draft of this section asserted that a molecular run resumed its transport from the rejected trial, and that was wrong.

- **What “admissible” covers.** A non-finite normalized reaction residual in the equilibrium sweep, and a non-finite row of the assembled residual vector. It does not cover a state that is finite everywhere and physically absurd; positivity of density and pressure is still the only other filter, and no temperature band was added because the code has none to use.
- **The six halvings are a bound, not a measurement.** Nothing here measures how often a probe becomes admissible on the second or third halving; the counters record only how often all six failed. On the healthy He/H = 1 arm that was never; on the old pathological restart it was 14 GMRES cycles and no Jacobian color.

122. H₂ self-shielding: the fits were measured against a level-resolved calculation, and the rate is now a table of it (2026-09-02)

Superseded in part by Sec. 132. (2026-09-02). The rate no longer multiplies a constant cross section by a shielding factor: the level-resolved cross section itself is tabulated, and `sigma_lw`, `p_diss_lw` and `sigma_lw_pump` have been removed from the module.

Superseded by Sec. 135. (2026-09-03). The tabulated rate was CLOUDY’s H₂g sub-reservoir and line overlap was absent; both are fixed there.

The judgement first. Neither closed-form fit reproduces a level-resolved CLOUDY calculation at the temperature of our molecular layer, and the failure is not an offset that a rescaling would fix: the calculated $f_{\text{shield}}(N_{\text{H}_2})$ has a trough near 10^{16} – 10^{17} and a shoulder near 10^{19} cm⁻² that a two-term algebraic form of this kind cannot carry. Draine & Bertoldi (1996) runs from $1.85\times$ high at 900 K to $2.3\times$ low at 2700 K over $10^{18} \leq N_{\text{H}_2} \leq 5 \times 10^{20}$ cm⁻²; Richings, Schaye & Oppenheimer (2014), which section 116. adopted, is low by 3.3 – $4.3\times$ at every temperature of the layer. The physically correct quantity is the level-resolved calculation, so it is tabulated and the rate reads the table. **This reverses the earlier choice, and it does so on measurement rather than on a re-reading of the papers.**

122.1 The premise of the earlier section is not what failed

Richings et al. adopted their fit because DB96 overestimates the shielding factor by about 3 at $T = 100$ K for $N_{\text{H}_2} \gtrsim 10^{17}$ cm⁻². That is reproduced here from our own deck — DB96/CLOUDY = 2.27–3.38 over $10^{17} \leq N_{\text{H}_2} \leq 5 \times 10^{20}$ cm⁻²

at 100 K in interstellar-like conditions — so the premise stands and the pipeline is calibrated against the very comparison that motivated the change. What could not be known then is that **the sign of the DB96 error reverses between 100 K and 900 K**: a fit built to remove the 100 K excess keeps removing it after it has gone, and our layer runs at 865–2724 K.

122.2 What was calculated

CLOUDY c25.00, DATABASE H2, plane-parallel slab irradiated on one face: pure H + He at He/H = 0.0793, no dust, no metals, cosmic-ray background as the only ionization source, constant temperature, thermal Doppler width. The incident field is flat F_λ ($f_\nu \sim \nu^{-2}$), DB96’s own eq. (24) spectrum, normalized to $F = 343 \text{ erg cm}^{-2} \text{ s}^{-1}$ over 912–1110 Å, i.e. our mol_lyman_werner field and $G_0 = 1.25 \times 10^6$ Habing.

Twenty-seven runs, $T = 700, 900, 1100, 1300, 1600, 1800, 2200, 2700, 3200 \text{ K}$ at $n_{\text{H}} = 10^{12}, 10^{13}, 10^{14} \text{ cm}^{-3}$, each carried to $N_{\text{H}_2} = 5.01 \times 10^{21} \text{ cm}^{-2}$. f_{shield} is the ground-state Solomon dissociation rate divided by the band-integrated field, normalized to its own value at the illuminated face of the same run.

Two independent checks of the setup, both in docs/h2_self_shielding_cloudy.md: CLOUDY’s own G(TH85) at the face is 1.25×10^6 , matching the Habing conversion done by hand, and its band photon flux agrees with $F/\langle h\nu \rangle$ as lyman_werner.f90 forms it to 0.3%.

122.3 The table

src/modules/lower_atmosphere/h2_self_shielding_table.f90, generated by src/utls/h2_shielding_table_from_cloudy from the decks in src/utls/h2_shielding_cloudy_decks/: $\log_{10} f_{\text{shield}}$ on 39 columns $\times$ 9 temperatures $\times$ 3 densities = **1053 doubles**, trilinear in $(\log_{10} T, \log_{10} n_{\text{H}}, \log_{10} N_{\text{H}_2})$, every axis clamped rather than extrapolated.

The column axis starts at 10^{12} cm^{-2} , not at 10^{16} . The plan was 10^{16} with $f = 1$ below it; the calculation says $f_{\text{shield}}(10^{16})$ is 0.067 at 700 K and 0.24 at 3200 K, so a table starting there would have needed an invented rule underneath it. At 10^{12} the measured factor is 1.000 in every run (0.9989 in the worst), which is the optically thin limit, so the bottom clamp asserts only what the runs show. The cost is 432 extra doubles.

122.4 Where the table stops being a value

CLOUDY shields one line at a time — RT_continuum_shield_fcn applies the Federman, Glassgold & Kwan (1979) function per transition — so overlap between distinct H₂ lines is absent from it, and overlap is what DB96 §5.2 say produces “the rapid falloff ... for $N_2 > 10^{20} \text{ cm}^{-2}$ ”, suppressing the pumping rates by a factor ~ 10 at 3×10^{21} .

The size of that gap is measured inside the runs rather than assumed. Build the summed equivalent width of the pumping lines from CLOUDY’s own curve with DB96’s eq. (38) construction, $A(N) = \sigma_{\text{pump}} \int_0^N f_{\text{shield}} dN'$, using the sigma_lw_pump the module already carries. A is a fraction of the band and cannot exceed 1. It does, and the column where it reaches 1 is h2_shielding_overlap_column(T, n_H): $1.4\text{--}1.8 \times 10^{21} \text{ cm}^{-2}$ at 700 K falling monotonically to 2.5×10^{20} at 3200 K — the hotter the gas, the more lines share the band and the earlier the un-overlapped sum breaks down. **Above that column the table is an upper bound on the shielding, hence a lower bound on the rate.**

No correction is applied there: inventing the missing transfer would be worse than recording its absence. The run reports the crossing once, through the warning that used to quote the two fits' published ranges.

Our base sits at $N_{\text{H}_2} = 4.4 \times 10^{21}$, i.e. above it, so the deepest cells of a molecular run carry that bound. Extending the grid with a code that treats line overlap is an open item.

122.5 What the band share keeps, and why

A stays on DB96 eq. (39), and the same calculation is now the reason rather than the absence of an alternative. Measured off the 1300 K run, the band-removal cross section is $\sigma_{\text{pump}} = 2.610 \times 10^{-17} \text{ cm}^2$ against the 2.557×10^{-17} the module carries — **2 per cent**. The two departures behind it cancel: CLOUDY's dissociation cross section is 34% above σ_{LW} and its $\langle p_{\text{diss}} \rangle = 0.177$ is 31% above the coded 0.135. So the normalization the equivalent width rests on is confirmed at our temperature, and the earlier decision to leave A on DB96 is now measured instead of argued.

Two constants are recorded as measured but *not* changed, because changing either alters the unattenuated rate rather than its shielding and belongs with its own verification: $\sigma_{\text{LW}} = 3.452 \times 10^{-18} \text{ cm}^2$ is 23–60% low against CLOUDY over 900–2700 K (mostly a line-data difference, not temperature — it is already 37% low at 100 K), and p_{diss} is not constant with depth (0.177 at the face, 0.217 at $N_{\text{H}_2} = 5 \times 10^{21}$).

122.6 Two axes closed by measurement

Density. The table carries n_{H} because it can, but the axis is nearly flat: f_{shield} moves 3–7% over $n_{\text{H}} = 10^{12}$ – 10^{14} below $N_{\text{H}_2} = 10^{19}$ and by at most 40% at 10^{21} . The absence of a density argument in either fit is not what is wrong with them.

Radiation field, i.e. the Elwert axis. Dropping F_{LW} by four decades (G_0 from 1.25×10^6 to 125) moves f_{shield} by at most 9% anywhere and by under 5% below $N_{\text{H}_2} = 10^{20}$. The earlier section rejected the Elwert exponent by arithmetic; this is the same conclusion by measurement, and it is why the table has no radiation-field axis.

H Lyman lines inside the band, checked while there: running with Database H-like Lyman pumping off moves f_{shield} by 2–11%, always upward. `lyman_werner.f90` §4 flags their neglect as the least controlled of the three absorbers it drops; it is a few per cent.

122.7 What changed in the code

- `h2_self_shielding_table.f90` is new and generated; `h2_shielding_table_from_cloudy.py` and the 27 decks are in `src/utils/` so the table can be rebuilt.
- `lyman_werner_dissociation_rate` takes n_{H} and reads the table. `h2_self_shielding_richings` and `h2_self_shielding_draine_bertoldi` are **kept and are no longer called by the rate**; the header says so. The second is still live for the band share.
- `fuv_lw_photon_field` takes the total hydrogen nucleus density and returns the largest ratio of the star-ward column to `h2_shielding_overlap_column`; the `f_shield` column of `output/Lyman_Werner.txt` is the table's value, i.e. still the factor the rate actually carries.

- The extrapolation warning in `ionization_equilibrium` and `write_output` no longer quotes the Richings range, which is no longer used; it reports the crossing of N_{overlap} and the DB96 range that the band share still rests on. `NH2_richings_max` is gone, `NH2_db96_max` stays.

Verification

Two changes share this refresh, and they are separable. The section 121. constrained-continuation seed fix landed in the tree first and had already moved the six molecular cases off their goldens; this section's table moves one of them further. Which case moved for which reason was measured, not inferred: the current tree source was built a second time with this section's patch reverse-applied, and the six molecular cases were re-run against it.

- `wasp_full`, `wasp_he23off`, `lower_profile` — no Lyman–Werner field; PASS throughout, moved by neither change.
- `mol_base_handoff`, `mol_metals`, `mol_diffusion`, `mol_ir_bands`, `mol_sec_ion` — no Lyman–Werner field; moved by the seed fix, and **byte-identical with and without the table**.
- `mol_lyman_werner` — the one case that sets Stellar LW flux; moved by both.

That is the structural statement this change should make: `F_LW_star` defaults to zero and exactly one matrix case supplies it, so exactly one matrix case can reach the new code at all.

The size of the table's own move, from the pair above (same source, same 12000 steps, shielding the only difference; per item P17 these are relaxation snapshots, so these are not converged change sizes): the H_2 front, the largest r with $n_{\text{H}_2} > 10^{10}$, goes from 1.165804 to 1.149903 R_p ; the minimum temperature over the molecular cells from 837.4 to 806.1 K; $n(\text{H}_2)$ over those cells changes by a factor 0.0040 to 1.0714; `final` from `du = 3.1452` to 2.5958 at the same step count. The sign is the one the measurement predicts — over the columns the layer occupies the table shields *less* than Richings, so the rate is higher, the front is pushed inward, and the outermost molecular cells lose H_2 by up to a factor 250.

- **Matrix before the refresh:** the three no-LW cases PASS byte-identical, the six molecular cases FAIL — exactly the split the two changes predict.
- **Goldens refreshed once**, the previous set preserved at `backup/regression/golden_blockD_20260902/` beforehand. `make check` returns **PASS, all cases byte-identical, 9/9, EXIT 0**.
- **`run_fcheck.sh`: CLEAN**, production build restored, and the matrix re-run on that restored binary returns **PASS 9/9** again.
- **The runtime-checked build was additionally run on `mol_lyman_werner`**, which the `fcheck` case (HD 209458 b, atomic) does not reach: the same `-O1 -fcheck=bounds,do,mem` binary, 1500 steps, **no runtime check fired**. That is the only run in which the new table is traversed under bounds checking.
- **Unit check of the table against what generated it:** a probe program linked against `h2_self_shielding_table.o` reproduces the Python-side value at all 1053 grid points to 1.2×10^{-9} relative, and N_{overlap} at all 27 to the same,

i.e. the Fortran interpolation returns the CLOUDY numbers themselves at the nodes. Off the nodes, at the mid-points of the 1300 K column axis, the table reproduces the underlying run to 0.6–1.9%.

- **The warning fires and says the right thing:** `mol_lyman_werner` reports the star-ward column reaching $4.99 \times 10^{21} \text{ cm}^{-2}$, 3.30 times the column at which the table becomes an upper bound.
- **Not verified:** nothing was re-run outside the matrix. The stored results of `examples/18_oxygen_chemistry/` and `examples/19_molecular_ir_bands/` carry a Lyman–Werner flux and now predate this change as well as the earlier one.

123. The heating dump left the oxygen carriers’ hydrogen out of the ionized fraction, and the H/He nucleus totals are now one routine (2026-09-02)

Section 118. fixed what the ionized fraction handed to the photoelectron partition is — the *total* free electron density over the hydrogen and helium **nuclei**, the quantity Dalgarno, Yan & Liu (1999) section 7 define — and made the sites that build it agree. The oxygen chemistry (sections 108. and 109.) then introduced two further carriers of hydrogen nuclei, OH and H₂O, and one of those sites was not taught about them.

123.1 What was different

`ionization_equilibrium` builds the nucleus totals as

$$n_{\text{H}} = n(\text{HI}) + n(\text{HII}) + 2 [n(\text{H}_2) + n(\text{H}_2^+)] + 3 n(\text{H}_3^+) + n(\text{HeH}^+) + n(\text{OH}) + 2 n(\text{H}_2\text{O}),$$

$$n_{\text{He}} = n(\text{HeI}) + n(\text{HeII}) + n(\text{HeIII}) + n(\text{HeH}^+),$$

while `write_heat_breakdown_eq`, the diagnostic that writes `output/Heating_breakdown.txt`, stopped at the molecular carriers: OH and H₂O were absent from its n_{H} . The dump is not short of the information — it fills its `nox` array from `f_sp_in` twenty lines earlier, because it needs those densities for the FUV photolysis heating channel — it simply did not count their nuclei. So its n_{H} was too small in exactly the cells the oxygen chemistry populates, which is the molecular base.

Two quantities inside the dump read that n_{H} , and neither of them is part of the solved state:

- $x_{\text{ion}} = n_e / (n_{\text{H}} + n_{\text{He}})$, which the photoelectron partition of every photoheating channel reads, so every `heat_*` column of the dump carries it;
- the hydrogen nucleus density handed to `fuv_lw_photon_field`, which is the density argument of the H₂ self-shielding table of section 122., so the `heat_H2_LW` column carries it as well.

123.2 One definition, not one more copy

The remedy is the standing rule that a quantity is defined in exactly one place. `hydrogen_helium_nuclei_density` is new in `utils (src/modules/functions/utilities.f90)`, beside `calc_ne` and `calc_ntot`, and every carrier is

counted with its nucleus multiplicity: H_2 and H_2^+ two H, H_3^+ three, HeH^+ one H and one He, OH one H, H_2O two, CO none. Its molecular and oxygen arguments are optional, so a caller that does not track those carriers omits them rather than writing a shorter sum of its own.

Four sites now call it and no site builds the totals itself:

site	carriers passed
ionization_equilibrium	molecular, oxygen
write_heat_breakdown_eq (util_ion_eq.f90)	molecular, oxygen
post_process_adv, before the advection sweep	none
post_process_adv, after it	none

The two post-process sites pass neither, which is the composition that pass already had and its module header already states: the advection correction carries only the H/He columns, so the molecular and oxygen carriers are not available to it. What they gain is the same statement order as the other two.

Verification

Byte-identity, and where it deliberately stops. The four sites now evaluate the same statements in the same order, so the change is arithmetically inert everywhere except the two places it is meant to move.

- **make check: PASS, 9/9, all cases byte-identical.** No matrix case runs the oxygen chemistry, so no matrix case can reach the corrected term at all; what the matrix proves is that the refactor of the other three sites did not move anything.
- **The oxygen case was run explicitly**, since the matrix does not cover it: examples/18_oxygen_chemistry/ (HD 209458 b, molecular and oxygen chemistry with a Lyman–Werner flux), 2000 steps, single-threaded, the same source built with and without the patch. Hydro_ioniz.txt and Ion_species.txt are **byte-identical** and the marching trace agrees step for step, i.e. the solved state does not see this.
- **What did move, in that run**, all of it inside the diagnostic: n_{H} at the base ($r = 1.0002$) was low by 2.55×10^{-4} and is corrected upward; x_{ion} there was high by 2.35×10^{-4} and is corrected downward; the heat_HI, heat_HeI, heat_HeII and heat_H2 columns move by up to 9.4×10^{-6} relative, downward; heat_H2_LW by up to 1.27×10^{-5} ; heat_metals by up to 1.9×10^{-7} ; and heat_total by up to 1.1×10^{-9} . The sign is the one the correction predicts: adding hydrogen nuclei lowers the ionized fraction, and a less ionized gas gives the photoelectron less of its energy as heat. The size is set by how much hydrogen the oxygen carriers hold at a solar oxygen abundance, which is 2.5×10^{-4} of the nuclei even where H_2O peaks — this is a diagnostic reading corrected by a fraction of a per mil, not a physics change waiting to be found.
- **Cooling_breakdown.txt and Lyman_Werner.txt are byte-identical** in the same run; neither builds the nucleus totals.
- **The _adv files move at round-off**, by at most 1.8×10^{-14} relative (Ion_species_adv.txt), because the post-process previously summed $n(\text{HeI singlet}) + n(\text{HeII}) + n(\text{HeIII})$ and added the triplet last, and now adds the two

He I populations first, as the other three sites do. The two orders are the same sum; nothing else about that pass changed.

124. The ladder, converged: He/H from 0.0793 to 30 on one hot Uranus (2026-09-02)

Judgment first. Every rung of the He/H ladder has a Newton-converged solution, the JFNK returns `info = 0` on all of them, and the ladder is monotonic and undramatic: raising He/H from 0.0793 to 30 thins the molecular layer, pulls the hydrogen ionization front inward, heats the wind and leaves the mass-loss rate almost unchanged. There is no window and no crossing anywhere on the ladder, which is what section 120. predicted when it removed the two defects that had made the He/H ~ 1 arm look supersonic. The full account, with the convergence recipe and the run ledger of each rung, is docs/supersonic_molecular_base.md section 13; this section is its record here.

The planet is the hot Uranus of the Tier-2 gate ($0.0457 M_J$, $0.49 R_J$, $T_{\text{eq}} = 1140$ K, 0.048 au, power-law spectrum), metals off, no opacity.inp, Secondary_ionization at its STAGED default, He_diffusion: True except where noted. Each rung changes He/H number ratio and the base.inp HeH_base / q_H2_base and nothing else, q_H2_base being set to hold the H₂ nucleus fraction at $x_2 = 0.9251143$ on every rung so that the ladder varies He/H alone. Every q_H2_base is below its element-ratio ceiling $0.5/(0.5 + \text{He/H})$, by 8 to 13%.

124.1 The rungs

Seven `info = 0` states (the six rungs plus the second He/H = 0.0793 rung that carries the arm's He_diffusion), OMP_NUM_THREADS=8:

He/H	0.0793	0.0793 (diff.)	0.3	1	3	10	30
$\log_{10} \dot{M}$ [g/s]	10.42	10.49	10.35	10.32	10.36	10.42	10.39
final JFNK residual	9.89e-4	9.16e-4	7.97e-4	6.57e-4	6.02e-4	9.21e-4	4.95e-4
H ₂ front $f_{\text{H}_2} = 0.5 [R_p]$	1.1012	1.1200	1.0681	1.0469	1.0467	1.0357	1.0250
H ₃ ⁺ peak $[R_p]$	1.0226	1.0200	1.0167	1.0014	1.0017	1.0015	1.0027
ioniz. front $x_{\text{HII}} = 0.5 [R_p]$	2.573	2.755	2.040	1.593	1.396*	1.333*	1.241*
T maximum [K]	2751	2698	3642	4967	5864	6328	6660
boundary outflow speed [km/s]	8.97	8.97	8.28	7.34	6.85	6.74	6.61

* measured before the coupling repair of section 125.; the corrected values are in section 125..

Read across, the structure is monotonic in everything: more helium means a thinner molecular layer, a hotter wind, an ionization front closer to the planet and a slower outflow, which is the sign helium's extra opacity and its 24.6 eV photoelectrons predict.

The rate is the exception: it is flat. $\log_{10} \dot{M}$ spans 10.32 to 10.49 over a factor 378 in He/H, and it is not even monotonic — it has a shallow minimum at He/H = 1. The 0.17 dex spread includes the 0.07 dex the He_diffusion switch alone moves the 0.0793 rung, so the He/H dependence of the rate over this range is at most a factor 1.5 and probably less.

What converges the rungs is not the marching. Every rung reaches $du = 2 \times 10^{-2}$ on its own and the staged secondary ionization fires there, so the wind has to cross the switch a second time; the JFNK entered at that second crossing fails (info = 2) on every rung that got there inside 250,000 steps, and three rungs never got there. What converges all six is a restart of the marched state with Load IC? True, du_th [PLM,WENO3]: 1.0e9 1.0e-3 and Solver: Newton 100.0, which arms the hand-off immediately and gives the solver the state instead of making it march to the state. Five of six converge on the first such restart, in under two minutes each, against hours of marching.

124.2 Marching does not leave the converged solution

The He/H = 1 state was restarted with the JFNK hand-off put out of reach (Solver: Newton 1.0e-30) and marched **50,000 steps with no steady solve at all**. du has one excursion — from 2.58×10^{-4} up by a factor 20 at step 3968 and back — and then holds between 2.58 and 2.63×10^{-4} for the last 30,000 steps. **The steady residual ends below where the JFNK left it**, 2.07×10^{-4} against 6.57×10^{-4} , $\log_{10} \dot{M}$ is 10.32 at step 50,000, and handing the marched state back to the JFNK closes it immediately at info = 0. The cells that move during the excursion are at $r = 2.12\text{--}2.19 R_p$, not at the base; below $1.1 R_p$ nothing moves by more than 0.5%.

This answers the two-solution question of item (P17) for this arm: the marching branch and the JFNK branch are not two steady solutions. It does not show that no second solution exists elsewhere.

124.3 The P23 gate does not pass

Compared against Koskinen et al. (2022) Model A — whose mass, gravity at the escape base, orbital distance, base temperature and XUV flux all match ours to within a few percent — and Frelikh & Murray-Clay (2026), both read from the publisher PDFs in references/:

- **agreeing:** the rate (10.42 against their 10.28, a factor 1.4), the outflow speed (8.97 against 9.8 km/s), and the wind being mostly neutral — $n(\text{HI})$ exceeds $n(\text{HII})$ out to $2.57 R_p$, which is the property both papers single out;
- **disagreeing:** our H_2 is truncated (four decades between 1.10 and $1.27 R_p$) where theirs “remains significant at all altitudes”; our H_3^+ is base-confined, nine decades over half a planetary radius, where theirs is present throughout; and we are colder, 2751 K peak against their 4400 K at $3 R_p$.

The explanation P23 proposed for the disagreement is refuted by this measurement. P23 reasoned that the difference from an H-and-He-only model was most likely the metal electrons our metals-on gates carry. This ladder is metals-off and converged, and H_3^+ is still base-confined, so composition is not the explanation. Nor is temperature, which is the mechanism Frelikh give for their own result: our H_2 front sits at 765 K against the 48,350 K barrier of the thermal-dissociation reaction both codes take from Baulch et al. (1992), so **our H_2 is not being thermally dissociated; it is being removed by something else, at a temperature where their argument says it should survive.** Which channel that is — H_2 photoionization at 15.4 eV, the $\text{H}_2^+ \rightarrow \text{H}_3^+ \rightarrow$ dissociative-recombination route, or charge exchange with H^+ — is the next measurement, and it is not one this campaign made. Lyman–Werner photodissociation is not a candidate: it is off in these runs and Koskinen’s Table 1 does not carry it either.

124.4 Ledger

None of the six converging runs admitted any base inflow — the base boundary: line, printed whenever a run has one, is absent from all six logs — and none activated the energy floor. Root acceptance is clean for $\text{He}/\text{H} \leq 10$: zero non-root acceptances, zero non-finite reaction residuals, zero projected or handback roots. **He/H = 30 is not clean:** 34 non-root states accepted under the relaxation amnesty, 153+6+52 projected/handback roots, and 165 cells with no in-simplex starting point. Every printed non-root cell sits at $r = 2.12\text{--}2.22 R_p$ at 5700–5820 K — the ionized outer wind, which is where the defect of section 125. lived. The largest element-budget violation grows along the ladder: none at 0.0793, $1.2\text{e-}3$ at 0.3, $2.1\text{e-}1$ at 1, $1.5\text{e-}1$ at 3, 2.8 at 10, $1.1\text{e}1$ at 30.

Item (P17) may be lifted for the six converged states, with the asterisked front and x_{HII} entries taken from section 125. and the He/H = 30 row carrying its ledger.

125. The He recombination photons are now only as absorbed as the cell is thick (2026-09-02)

Judgment first. The expression was physically wrong, and the divergence item (P40) reported is that wrongness showing itself rather than a numerical accident. The on-the-spot assumption of section 120. — every He recombination photon above 13.6 eV is absorbed inside the cell that emitted it — is a statement that the cell is optically thick to those photons. It was never tested, and in the thin outer wind of a helium-rich run it is false by 160 orders of magnitude: a cell holding no hydrogen and 2.1×10^{-151} H_2 molecules per cm^3 has a cell optical depth of 10^{-160} at 16–21 eV and keeps none of its photons, while the code handed all of them to the H_2 residue and returned a photoionization rate of $2.6 \times 10^{100} \text{ s}^{-1}$. The repair is not a floor on the absorber density. It is the closure the assumption was standing in for: the cell keeps the fraction $1 - e^{-\tau_c}$ of each channel's photons and the rest leave.

Opened as items (P33) (the untested optical depth) and (P40) (the divergence it produces); both are closed here. Item (P34), the trace metals missing from the absorber competition, is untouched here and is closed in section 129..

125.1 The closure

For each channel c , emitting at its representative energy E_c into the absorbers whose threshold lies below E_c ,

$$\tau_c = \left(\sum_s n_s \sigma_s(E_c) \right) \Delta r_j, \quad f_{\text{abs}} = 1 - e^{-\tau_c}, \quad w_s = \frac{n_s \sigma_s(E_c)}{\sum_{s'} n_{s'} \sigma_{s'}(E_c)},$$

and the rate per particle of absorber s , its heating, and the effective He II recombination coefficient become

$$\begin{aligned} dP_s &= \frac{P_c f_{\text{abs}} w_s}{n_s} = P_c \sigma_s(E_c) \Delta r_j \frac{1 - e^{-\tau_c}}{\tau_c}, \\ H_c &= P_c f_{\text{abs}} \sum_s w_s (E_c - E_{\text{th},s}), \\ \alpha_{\text{eff}} &= \alpha_B + \alpha_1 (1 - f_{\text{abs}} w_{\text{HeI}}) \quad [\text{TR branch: } \alpha_1 (1 - f_{\text{abs}} w_{\text{HeI}}) + 0.25 \alpha_B]. \end{aligned}$$

The second form of dP_s is the one that shows there is no singularity anywhere. The bracket is 1 at $\tau \rightarrow 0$ and $1/\tau$ at $\tau \rightarrow \infty$, so the rate runs from the optically thin limit $P_c \sigma_s \Delta r_j$ — what a single absorber in a transparent cell sees, which is finite whether that absorber exists or not — to the thick limit $P_c \sigma_s / \sum n \sigma$, which is exactly the expression section 120. wrote. **The thick limit is the old expression, so the correction is a physical closure of it, not a replacement of it.** The He recombination coefficient changes for the same reason the rates do: a ground-capture photon that leaves the cell does not re-ionize helium either, so only the He I share of the photons that *stay* is excluded from the net recombination.

Implementation. `he_rec_coupling` in `src/modules/radiation/util_ion_eq.f90` keeps section 120.'s ratio form for the shares w and multiplies each channel by its own f_{abs} ; the five channels have five different optical depths, because they have five different energies and (for the ≥ 24.6 eV ground-capture continuum alone) different absorber sets. f_{abs} comes from a new module function `absorbed_photon_fraction(tau)`, which uses the series $\tau \left[1 - \frac{\tau}{2} \left(1 - \frac{\tau}{3}\right)\right]$ below $\tau = 10^{-4}$, where $1 - e^{-\tau}$ loses all but a few digits to cancellation, and returns zero for a non-positive τ (the ionization solve can hand it a small negative trace density). The recombination coefficient is written $(1 - f_{\text{abs}}) + f_{\text{abs}} (w_{\text{HI}} + w_{\text{H}_2})$ rather than $1 - f_{\text{abs}} w_{\text{HeI}}$ so that $f_{\text{abs}} = 1$ reproduces section 120. bit for bit. Δr_j is the cell width of the code's own grid, defined on the ghost cells as well, and `sigma` returns 10^{-18} cm^2 , the unit the stellar optical depth of `PH_heat_HHe` also carries.

Where the two forms actually differ. f_{abs} is exactly 1.0 in double precision for $\tau > 37$, so wherever the cell is that thick the returned numbers are unchanged to the last bit. Measured on the existing goldens:

case	τ at the base cell (gnd / casc)	$f_{\text{abs}} < 1$ from	f_{abs} at the outer boundary
<code>wasp_full</code> , <code>wasp_he23off</code> (WASP-121 b, atomic)	16.5 / 20.2	the base cell	2.0e-3 / 2.4e-3
<code>mol_sec_ion</code> (hot Uranus, molecular)	174 / 240	$r = 1.102 / 1.116 R_p$	7.0e-4 / 7.7e-4
<code>mol_base_handoff</code>	174 / 240	$r = 1.106 / 1.119 R_p$	8.1e-4 / 8.9e-4
<code>lower_profile</code> (HD 209458 b, profile handoff)	327 / 403	$r = 1.037 / 1.041 R_p$	7.6e-7 / 1.1e-6

The molecular base is thick and is not touched; the atomic base is not thick. The hot Uranus keeps its photons out to 1.10 – $1.12 R_p$ and HD 209458 b to $1.04 R_p$, but WASP-121 b's base cell has $\tau = 16.5$ in the ground-capture channel, so no cell of the two atomic cases was ever optically thick to these photons and the coupling there was overestimated everywhere — by a factor 2 at $1.15 R_p$, 10 at $1.25 R_p$ and 500 at $1.5 R_p$.

125.2 The atomic cases: converged, and how far they moved

`wasp_full_newton` and `wasp_he23off_newton` (WASP-121 b, He 2^3S on and off, metals on, Newton finish), each run before and after with the same binary except for this change, `OMP_NUM_THREADS=1`:

	wasp_full_newton	wasp_he23off_newton
JFNK	info = 0 before and after	info = 0 before and after
$\log_{10} \dot{M}$ (run report)	13.22 $\rightarrow$ 13.21	13.22 $\rightarrow$ 13.22
$\log_{10} \dot{M}$ (re-measured)	13.2414 $\rightarrow$ 13.2338	13.2422 $\rightarrow$ 13.2342
largest T change	0.96% at $r = 1.087$	0.94% at $r = 1.087$
largest $n(\text{HI})$ change	26% at $r = 1.231$	27% at $r = 1.230$
largest $n(\text{He I})$ change	18% at $r = 1.218$	18% at $r = 1.217$
largest $n(\text{He } 2^3\text{S})$ change	9.1% at $r = 1.163$	—
largest heating change	11% at $r = 1.180$	12% at $r = 1.178$

The rate moves by 0.008 dex and the structure moves at the ionization front, 1.15–1.25 R_p , which is where f_{abs} falls from 0.5 to 0.1. Below 1.05 R_p nothing moves by more than a few times 10^{-3} .

125.3 The ladder rungs the defect contaminated

The three states item (P40) measured, each restarted from its converged rung and re-converged with the corrected coupling (info = 0 in all three), against the He_rec_coupling: False controls the item reported. The controls are not the target: a corrected channel still ionizes something, so the fixed run should land *between* the contaminated run and the control, near the control.

	He/H = 1	He/H = 3	He/H = 30
n_cells_he_rec_photoionization_dominant before / after	0 / 0	1.50e5 / 0	3.03e5 / 0
ionization front, contaminated	1.586	1.394	1.238
ionization front, coupling off (control)	—	1.641	1.500
ionization front, corrected	1.758	1.618	1.444
x_{HII} at the boundary, contaminated	0.9499	1.0000	1.0000
x_{HII} at the boundary, control	—	0.9112	0.9197
x_{HII} at the boundary, corrected	0.9008	0.9138	0.9236
$\log_{10} \dot{M}$ before / after	10.32 / 10.29	10.35 / 10.35	10.39 / 10.37
H ₂ front [R_p] before / after	1.047 / 1.047	1.046 / 1.046	1.025 / 1.025

The saturation is gone. $x_{\text{HII}} = 1.000$ exactly, the signature the item identified, becomes 0.914 and 0.924, within 0.003 and 0.004 of the coupling-off controls, and the front lands 0.023 and 0.056 R_p inside them — the size of the channel that is left once the escaped photons are removed. The diagnostic counter, which counted 150,000 and 303,000 cell-steps of runaway, is silent on both.

The He/H = 1 rung moves too, and it never fired the counter. Its front goes from 1.586 to 1.758 R_p and its boundary x_{HII} from 0.950 to 0.901, a larger move than the He/H = 3 rung's. That is the point of the correction and not a surprise: the counter only sees a *runaway*, and the ordinary overestimate of a thin cell — a factor of a few hundred in the outer wind, where f_{abs} is 10^{-3} — never trips a threshold written at 1000 times the stellar rate. Item (P40) was one visible symptom of item (P33), not the whole of it.

The molecular layer is untouched on all three rungs: the H₂ front does not move at the fourth decimal, and $\log_{10} \dot{M}$ moves by at most 0.03 dex.

125.4 Goldens

make check is 9 of 9 against goldens refreshed at the end of this block. All nine move, and the two reasons are the two rows of the table above: the atomic cases have no optically thick cell at all, so they move from the base outward, and the molecular cases are thick below $1.10R_p$ (1.04 for lower_profile), so their molecular layer does not move and their wind does. **How far, measured against the goldens before the refresh:**

case	$\log_{10} \dot{M}$ golden $\rightarrow$ new	molecular layer, $T / n(\text{HI})$	wind above it, $T / n(\text{HI})$
wasp_full	13.2807 $\rightarrow$ 13.2744 (−0.006)	no optically thick cell	2.1% / 33%
wasp_he23off	13.2829 $\rightarrow$ 13.2745 (−0.008)	no optically thick cell	0.46% / 27%
mol_base_handoff	10.6110 $\rightarrow$ 10.6001 (−0.011)	5.2e-5 / 6.2e-5	12% / 19%
mol_metals	10.6226 $\rightarrow$ 10.6127 (−0.010)	6.4e-4 / 1.2e-2	13% / 21%
mol_lyman_werner	10.6153 $\rightarrow$ 10.5971 (−0.018)	3.5e-5 / 2.0e-5	13% / 23%
mol_diffusion	10.6110 $\rightarrow$ 10.6001 (−0.011)	5.2e-5 / 6.1e-5	12% / 19%
mol_ir_bands	10.6226 $\rightarrow$ 10.6126 (−0.010)	6.1e-5 / 6.6e-4	12% / 19%
mol_sec_ion	10.5728 $\rightarrow$ 10.5607 (−0.012)	3.4e-4 / 1.3e-4	9.5% / 16%
lower_profile	9.7334 $\rightarrow$ 9.7204 (−0.013)	1.2e-5 / 3.4e-5	65% / 74%

The “molecular layer” column is the largest relative change below the radius at which f_{abs} leaves 1.0 for that case ($1.10R_p$ for the six hot-Uranus cases, $1.04R_p$ for lower_profile), and the “wind” column is the largest above it. $\log_{10} \dot{M}$ is re-measured from the profiles, not the two-decimal run report.

The molecular cases are 12000-step relaxation snapshots rather than converged solutions, so their outer cells are still in the cold-start transient and move more than a converged wind would. What is worth reading in them is the split: in mol_base_handoff the whole molecular layer below $1.10R_p$ moves by at most 7.5×10^{-4} relative in every quantity, against 11–35% above it, and in mol_sec_ion by at most 1.9×10^{-3} against 9–51%. The one exception is the velocity of the base cells, which goes from -0.72 to -9.7 cm/s at $1.048R_p$ in mol_sec_ion — the base breathing this code has always had, five orders of magnitude below the 11 km/s wind and not a change in the solution.

Why the molecular layer moves at all. The coupling term itself is bit for bit unchanged there, because f_{abs} is exactly 1.0. What reaches it is the star-ward column: the outer cells’ neutral densities change, so the attenuated stellar field at the base changes. Measured on a one-step run of mol_base_handoff, the base cells differ by 10^{-7} relative after the single ioniz_eq pass of step 0 — and the ghost and the first physical cell, whose state the boundary condition pins, are identical.

The previous goldens are kept at backup/regression/golden_blockP40_20260902/.

run_fcheck.sh (bounds/do/mem checking build, one bounded HD 209458 b run): clean, no runtime check fired, $\log_{10} \dot{M} = 9.60$. It was run after the 9-of-9 check; it rebuilds from the same sources with the same flags and restores the production build, so the check was not repeated after it.

126. The Newton direction at the hand-off state points uphill, and the solver had no direction that uses the gradient (2026-09-02)

Judgment first. The suspected cause was wrong and the real one is simple. The WENO3 smoothness weights are not what makes the JFNK hand-off fail on the $\text{He}/\text{H}=0.3$ arm: the inner Newton model already freezes them, and freezing them in the line search too changes the merit at the rejected step by seven parts in ten thousand — the step still ascends. What ascends is the Newton/PTC direction itself, at every damping including the $\Delta\tau \rightarrow 0$ limit, because the scaled Jacobian is indefinite there ($s^T A s < 0$); and `solve_steady_jfnk` had only that one direction, so backtracking shortened an uphill step twenty times, the pseudo-time floor made the next iteration identical, and the solve counted to twelve and gave up. The merit gradient is not zero there — $\|A^T s\| = 12.6$ — and the damped Gauss–Newton (Levenberg–Marquardt) step built from the transpose of the banded Jacobian, which is a descent direction for every damping by construction, cuts the merit in half in one step. With that escape the hand-off closes at `info = 0` where it used to abort. What it does NOT do is reproduce the state the restart trick reaches, and section 126.6 says so with numbers: the two `info = 0` states differ by 24% in the L_2 norm of (ρ, v, p) and by 0.22 dex in the rate, and the code’s convergence measure rates the worse one better. That is a defect of the measure, and it is left open, not papered over.

Opened as item (P41) of `TO_BE_DONE.md`, which is closed here except for the part promoted to a new item.

126.1 The state, and how it is reached in ten seconds

The hot Uranus of the Tier-2 gate at $\text{He}/\text{H}=0.3$ with `He_diffusion: True` and `Secondary_ionization: Immediate`, restarted from its own archived output with `Load IC? True`. The failing hand-off is reproduced with no marching at all by the direct steady route,

```
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x
```

which lands on $\|R\| = 1.259 \times 10^{-2}$, merit $\|F/D\|_2 = 5.140363464 \times 10^{-1}$, worst cell $j = 304$ at $r = 1.3412$ in the mass row, and returns `info = 2` — the in-run failure to every printed digit. Every measurement below is made at that point, on the first outer iteration.

126.2 The direction report

The merit is $f_2 = \|F/D\|_2$ over the whole domain, the quantity the line search minimizes. `weno_mode = 0` recomputes the smoothness weights at the trial state (what the line search does); `weno_mode = 2` reuses the weights stored at Y (what the Jacobian probes and the Krylov products do). Entries are $(f_2^{\text{trial}} - f_2)/f_2$, so **positive is uphill**.

direction	step	weights recomputed	weights frozen at Y
Newton/PTC (GMRES, $\Delta\tau_0 = 1$)	$\lambda = 1$	$+2.2249 \times 10^{-2}$	$+2.2242 \times 10^{-2}$
	$\lambda = 0.5$	$+9.1531 \times 10^{-3}$	$+9.1528 \times 10^{-3}$
	$\lambda = 0.1$	$+1.5086 \times 10^{-3}$	$+1.5091 \times 10^{-3}$
	$\lambda = 10^{-2}$	$+1.4355 \times 10^{-4}$	$+1.4361 \times 10^{-4}$
	$\lambda = 10^{-4}$	$+1.4275 \times 10^{-6}$	$+1.4281 \times 10^{-6}$
	$\lambda = 10^{-6}$	$+1.4274 \times 10^{-8}$	$+1.4280 \times 10^{-8}$
scaled steepest descent, Cauchy point	$\lambda = 1$	-3.1886×10^{-5}	-2.6393×10^{-5}
	$\lambda = 0.5$	-2.0839×10^{-5}	-1.9783×10^{-5}
	$\lambda = 10^{-2}$	-5.1275×10^{-7}	-5.2475×10^{-7}
	$\lambda = 10^{-6}$	-5.1471×10^{-11}	-5.2735×10^{-11}
	$\lambda = 10$	$+1.4369 \times 10^{-3}$	
	$\lambda = 100$	$+1.7001 \times 10^{-1}$	

Auxiliary numbers at the same point: $\|A^T s\| = 1.2563 \times 10^1$ (the merit gradient in the scaled space, from the banded Jacobian), Cauchy step length $t = \|g\|^2 / \|Ag\|^2 = 8.8215 \times 10^{-8}$, $|\delta Y_{\text{Newton}}|/|Y| = 1.5322 \times 10^{-3}$, $|\delta Y_{\text{Cauchy}}|/|Y| = 4.1409 \times 10^{-8}$.

Three things follow, and the first one kills the hypothesis this section was opened to test.

1. **The WENO weights are not involved.** The two columns agree to four significant figures at every λ . The suspicion was that the nonlinear weights make F non-smooth at the fixed point and that the finite-difference directional derivatives cross the kink; they do not, and in any case the inner model has frozen weights already (weno_mode = 1 stores them at Y , the Jacobian colors and the Krylov products reuse them at weno_mode = 2; section 121. and newton_scaling_and_base_wall.md section 10 record why the line search deliberately does not). Freezing the line search too, the one part that was still free, moves the rejected step by 7×10^{-4} relative and leaves it uphill.
2. **The Newton direction is uphill in the limit, not merely overshooting.** The excess merit falls linearly with λ over six decades and never changes sign, so the directional derivative is strictly positive. Backtracking cannot fix a direction; it can only make an uphill step short.
3. **A descent direction exists.** The scaled steepest-descent direction $-A^T s$ descends, its measured directional derivative (-2.65×10^{-5} per unit λ , read at $\lambda = 10^{-6}$) matches the value $-t\|g\|^2/\|s\|$ the banded Jacobian predicts to 2%, and the Cauchy point is close to the minimum along that ray ($\lambda = 10$ is already back uphill). The gradient is therefore accurate and nonzero, and the solve simply never built anything from it.

126.3 Why the Newton/PTC direction can point uphill at all

The solve minimizes $m(x) = \frac{1}{2}\|s\|^2$ with $s = F/D$ and $Y = Dx$; its gradient is $A^T s$ with $A = D^{-1}JD$. The step it takes solves $(I/\Delta\tau + A)\delta Z = -s$, which is an implicit Euler step of the pseudo-transient continuation and is not a minimizer of m . In the $\Delta\tau \rightarrow 0$ limit that step is $\delta Z = -s$, whose directional derivative is $-s^T A s$; the heavily damped step at $\Delta\tau_0 = 1.36 \times 10^{-4}$ (the CFL value) raises the merit, so $s^T A s < 0$ at this state and even the shortest pseudo-time step

climbs. Tightening the GMRES tolerance from 10^{-1} to 10^{-4} (gm 1 $\rightarrow$ 4) gives a better-resolved solution of the same wrong problem and still climbs, to $+1.771 \times 10^{-2}$. Nothing in the family $(I/\Delta\tau + A)\delta Z = -s$, over any $\Delta\tau$, is guaranteed to descend when A is indefinite.

The damped Gauss–Newton (Levenberg–Marquardt) step is

$$(A^T A + \mu \text{diag}(A^T A)) \delta Z = -A^T s.$$

The matrix is symmetric positive definite for every $\mu > 0$, so $\nabla m^T \delta Z = -g^T(\cdot)^{-1}g < 0$ **for any damping** — descent is a property of the family, not of a lucky parameter. $\mu \rightarrow \infty$ recovers the Cauchy direction and $\mu \rightarrow 0$ the Gauss–Newton step, so μ is the trust parameter, and choosing it is a one-dimensional search along a curve rather than along a ray. Measured on this state (relative merit change, weights recomputed):

μ	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-6}
$(f_2 - f_2^0)/f_2^0$	-4.69e-5	-3.64e-4	-1.88e-3	-1.15e-2	-7.65e-2	-3.21e-1	-4.93e-1	+4.25e-2
$ \delta Y / Y $	2.7e-8	2.4e-7	1.3e-6	4.5e-6	1.5e-5	3.0e-5	7.5e-4	4.7e-3

One step at $\mu = 10^{-4}$ takes the merit from 0.514 to 0.260. The curve falls monotonically to that point and rises again below it, which is what a trust parameter passing through its optimum looks like.

126.4 What changed

All of it in `src/modules/time_step/steady_newton.f90` except the last item.

(a) The transpose the solver never had. `band_matvec_transpose` computes $J^T x$ from the band-stored Jacobian, and `normal_equations_band` forms $J^T J$ (half-bandwidth `2kl_jac = 16`, LAPACK general-band layout, leading dimension `ldab_normal = 49`). Both act on the ALREADY SCALED band `ab` that the outer iteration builds, so what they return is $A^T s$ and $A^T A$ in the scaled space with no further bookkeeping. `levenberg_marquardt_step` adds μdiag and solves by one banded LU. The cost is one factorization of a 16-wide band per μ , against 17 full residual evaluations for the Jacobian itself, so the escape is cheap next to the iteration that calls it.

(b) The escape. `levenberg_marquardt_descent` sweeps μ down by decades from 10^2 , measures every candidate with the TRUE residual (weights recomputed, exactly as the line search does), stops when the merit turns back up, and returns the best step. It is called from `solve_steady_jfnk` when the pseudo-time step is at its floor ($\Delta\tau \leq \Delta\tau_0$) and either the backtracking search rejected every step, or it accepted one that did not lower the true merit. The second case is the non-monotone window: the test compares a trial against the WORST of the last five iterates, so a sideways step passes it, and on this state 487 of 500 iterations were such acceptances before the escape covered them. The $\Delta\tau \leq \Delta\tau_0$ guard is what makes the escape a repair rather than a change of algorithm: while the pseudo-time step is still above its floor, reducing it does change the next iteration, and the PTC ramp is left alone.

(c) The bit-identical repetitions are gone. When the line search fails at $\Delta\tau = \Delta\tau_0$, `dtau = max(dtau*0.25d0, dtau0)` returns $\Delta\tau_0$, so the preconditioner, the Krylov direction and the search repeat exactly — eleven times, before `n_no_descent_max = 12` fired. Now the escape is taken at the first such iteration, and if it too finds nothing the solve stops there and prints $\|\nabla m\|$, which is the number that says whether the point is stationary or the model is wrong. That

path was exercised during development, when a band-layout error made every Levenberg–Marquardt solve fail: the solve aborted at iteration 1 reporting $\|\nabla m\| = 1.256 \times 10^1$ instead of repeating eleven identical iterations.

(d) The hand-off is no longer given up for the rest of the run. `src/EXHALE_main.f90`: a failed hand-off records the step, holds the next attempt for `N_stall` steps of further marching, and disables the Newton finish only after `n_newton_attempt_max = 3` attempts. The failure is a property of the state the solver was handed, not of the run. No input key is added.

126.5 What was measured

Everything single-threaded where byte-identity is claimed, `OMP_NUM_THREADS=8` otherwise.

Which source each item was run on. The archived hand-off states of items 1, 2, 5 and 6 are fixed points of the source that produced them, which is this tree minus the two files `section 125. changed (util_ion_eq.f90, ionization_equilibrium.f90)`; loaded into the current code they are no longer near-steady and the failure this section repairs cannot be posed. Those four items were therefore run on that snapshot plus this change, with the same before/after comparison on both sides. Items 3 and 4, which are the byte-identity claims, were run on the current tree, and make check on the tree passes 9 of 9 against the goldens refreshed for section 125..

- 1. The control that reproduced the failure now converges.** The $\text{He/H}=0.3$ arm restarted single-stage (Reconstruction scheme: WENO3, no PLM step), marching to the hand-off at step 105: before, `info = 2` at $\|R\| = 1.258 \times 10^{-2}$ after eleven bit-identical iterations at $\lambda = 9.54 \times 10^{-7}$; after, `info = 0` at $\|R\| = 5.614 \times 10^{-4}$ in 38 outer iterations, 36 of which took the escape, in 16 s.
- 2. The two rungs whose hand-off never engaged converge, and the escape is not what does it.** The $\text{He/H}=3$ and $\text{He/H}=30$ marched states (250,000 steps, ended at $du = 3.9 \times 10^{-2}$ and 4.4×10^{-2} with the hand-off never fired), restarted single-stage: `info = 0` at $\|R\| = 6.042 \times 10^{-4}$ and 3.157×10^{-4} , $\log_{10} \dot{M}$ 10.35 and 10.40 against the ladder's 10.36 and 10.39. Before and after are byte-identical and the escape never fires — those rungs were never blocked by an ascending direction, only by a du that stayed above the switch.
- 3. The two atomic Newton cases are byte-identical.** `wasp_full_newton` and `wasp_he23off_newton`, `OMP_NUM_THREADS=1`: `Hydro_ioniz.txt` and `Ion_species.txt` identical to the same tree without this change, `info = 0` at $\|R\| = 5.536 \times 10^{-4}$ and 6.341×10^{-4} , $\log_{10} \dot{M}$ 13.21 and 13.22, escape count 0.
- 4. The regression matrix of nine is byte-identical** to the goldens, and the escape fires in none of them. **Note what that does and does not guard:** `wasp_full` and `wasp_he23off` carry no `Solver: key` — they stop on $du < du_{th}$ and never reach the JFNK at all — and the seven molecular cases are fixed-step relaxation snapshots that do not reach it either. The Newton guards for this change are the two `*_newton` cases of item 3 and the restarts of items 2 and 5.
- 5. The $\text{He/H}=1$ reference is unmoved:** $\log_{10} \dot{M} = 10.32$, output byte-identical, escape count 0.
- 6. The retry of (d) works, and it is what shows the escape has not stalled.** The same control run with `Resid tol: 1.0e-8`, a target it cannot meet, makes three hand-offs 2000 marching steps apart and reaches $\|R\| = 2.324 \times 10^{-5}$,

1.271×10^{-5} and 7.697×10^{-6} before the finish is given up, against 1.258×10^{-2} where the old code aborted. So the escape does not merely move the iterate off the repeat: it goes on reducing the residual by three further decades. What that does NOT buy is the next subsection.

126.6 What is NOT repaired: two info = 0 states the convergence measure cannot tell apart

The other half of item (P41) was that the restarts of the ladder converge only because `in_plm_stage = (du_th_plm .gt. du_th)` is evaluated without reference to `do_load_IC`, so every two-stage restart takes one PLM step before switching to WENO3, and that step is what unblocks the solve. Making the restart honest — letting a loaded state whose flux spread is already below `du_th_plm` continue in WENO3 — was implemented, measured, and **held back**, because the measurement says the accident is doing something the escape does not.

On the same arm, from the same wind:

	one PLM step (restart as it is today)	no PLM step, escape only	the same, driven to $\ R\ = 7.7 \times 10^{-6}$
final $\ R\ $	8.818×10^{-4}	5.614×10^{-4}	7.697×10^{-6}
$\log_{10} \dot{M}$	10.34	10.56	10.46
flux spread of $\rho v r^2$, $r > 1.2 R_p$	1.134×10^{-2}	1.132×10^{-1}	1.203×10^{-1}
flux spread, $r > 2 R_p$	3.540×10^{-3}	2.521×10^{-3}	2.470×10^{-3}
T maximum [K]	3650	3398	3492

The state handed to the solver has a flux spread of 1.129 over $r > 1.2 R_p$ and 2.0×10^{-2} over $r > 2 R_p$: the marching flattened the escaping wind and left the region between 1.2 and $2 R_p$ an order of magnitude from steady. The PLM-kicked solve flattens that region by a factor 100; the escape flattens it by a factor 10 and stops, because it has reached the floor of the merit basin it started in and $\|R\|$, which is measured over $[j_{\min} : N]$, i.e. $r > r_{\text{esc}}$ only, is by then 5.6×10^{-4} and the run is declared converged. The two solutions differ by 24% in the L_2 norm of (ρ, v, p) (20% for the third column), and the ladder's own value for this rung, 10.35, is the PLM-kicked one.

The third column is the decisive one. Driving $\|R\|$ down by a further factor 73, through three hand-offs, moves the inner flux spread not at all — it ends at 1.20×10^{-1} , no better than the 1.13×10^{-1} the loose tolerance reached. $\|R\|$ and the state of the wind between 1.2 and $2 R_p$ are simply not measuring the same thing, and no amount of the former fixes the latter.

So the convergence measure is the next defect, not the restart rule. A descent method cannot leave a merit basin, and the PLM step is a perturbation that leaves it; removing the perturbation before $\|R\|$ can see the region it repairs would silently replace the ladder's rates with the rates of a state that is not a steady wind between 1.2 and $2 R_p$. The restart rule is therefore left as it is and the pair is recorded as a new open item.

127. The convergence measure now covers the whole column: what the wind does below the escape radius is part of $\|R\|$ (2026-09-02)

Judgment first. The measure was wrong, and it was wrong in the one way that matters for a steady wind: it did not look at most of the wind. $\|R\|$ was a relative residual over $[j_{\min} : N]$ alone, i.e. over $r > r_{\text{esc}}$, so the column between the base and the escape radius — for the hot Uranus of the ladder, everything below $2R_p$, which is 391 of 500 cells and where the molecular layer, the H_2 front and the transonic acceleration all live — was controlled by nothing. A steady state is steady in every physical cell or it is not a steady state; measuring it in a fifth of them is not a tolerance choice, it is the wrong quantity. $\|R\|$ is now the larger of two relative residuals, one over the wind and one over the layer below the escape radius, each taken on the scale that region's own physics sets. The two solves of section 126.6 that the old measure could not tell apart are now separated by a factor 5.7 in $\|R\|$, in the right direction: re-scored on the same converged states, the one with a flux spread of 7.6×10^{-2} above $1.2R_p$ goes from 4.93×10^{-4} to 4.00×10^{-3} , while the one with a spread of 3.0×10^{-3} stays at 6.97×10^{-4} . The nine-case matrix is byte-identical, `run_fcheck.sh` is clean, and both WASP-121b Newton cases still converge — one of them from a state whose layer the old measure had left at 12% of ρg while reporting $\|R\| = 6.9 \times 10^{-4}$. This closes item (P42).

127.1 Where the window came from: it was inherited, not chosen

$j_{\min}$ is upstream ATES's: `define_grid.f90` sets it to the first cell with $r \geq r_{\text{esc}}$, and `ATES/ATES-Code-main/ATES_main.f90` lines 286–298 use `mom(j_min:N)` for the marching `du` and `dtu` monitors, because r_{esc} is where ATES expects $\rho v r^2$ to have become constant. The steady residual inherited that range on the day it was written: `git show 10c248a:src/modules/time_step/steady_residual.f90` (2026-06-11, the commit that introduced it) already carries the comment “max relative residual rate over the wind $[j_{\min}:N]$ ”, and `resid_relnorm` in `steady_newton.f90` says in its own header that it uses “the same definition the marching monitor uses”. So the window is not a repair of anything; it is the marching monitor's range, adopted without the question being asked.

It was asked once, on 2026-08-15, and answered on a scale that has since been replaced. Section 61. measured the layer's residual on $|u|$ for all three rows, found converged states reading 0.2 to 10 there, and concluded that widening the window on that norm would report `info = 2` for every configuration that converges. That conclusion was correct for that norm and is void for this one: section 62.3, the same series, replaced the momentum row's $|\rho v|$ — which collapses in a layer moving at 10 cm/s — by the gravitational force density, and re-measured the same converged states at 10^{-5} to 1.4×10^{-4} of ρg . The obstacle was removed and the criterion was never revisited.

What was NOT the reason: a base null mode. The 2026-08-11 and 2026-08-13 experiments recorded in `solve_steady_jfnk` went the other way — they narrowed the line-search *merit* to $[j_{\min} : N]$ and broke two solves that converge with the whole-domain merit. The merit has covered every physical cell $1..N$ since it was written; it was never windowed in space, and nothing in it was a work-around for the base. The counter printed as `window-only` accepts was about the Grippo non-monotone window in *time* — acceptances that beat the worst merit of the last five iterates without beating the current one — and had nothing to do with a spatial range. It is now printed as

non-monotone accepts, so the name cannot be read as the other thing.

127.2 The change

(a) **One definition of the row scale.** `residual_row_scale(k, j, u)` in `src/modules/time_step/steady_residual.f90` is now the only place in the code that says how a residual row is made dimensionless:

- mass and energy rows, everywhere: $|u(k, j)|$. Both are positive definite and collapse nowhere, so $|R|/|u|$ is that cell's fractional drift rate, in $1/t_s$.
- momentum row at $r \geq r_{\text{esc}}$: $|u(2, j)| = |\rho v|$, the momentum density the row transports.
- momentum row at $r < r_{\text{esc}}$: $|u(1, j)| |G\phi_i(j) - G\phi_i(j-1)|/dr_j(j)$, the gravitational force density discretized exactly as the momentum source term is (Source.f90 line 33). The layer is quasi-hydrostatic, so this is the balance the row expresses there and the ratio is the fractional violation of hydrostatic balance.

This is the expression section 62.3 introduced for the below-escape print. It is not a new scale; it moved from a print-only routine into the measure, and `resid_relnorm_below_escape` now reads it from the same place instead of carrying its own copy.

(b) **One norm, evaluated on two regions.** `relnorm_over_cells(Res, u, ja, jb, volume_weighted, rc)` computes the relative residual of each row over a cell range, volume-weighted ($V = r^2 dr$, the default) or as a ratio of maxima. `residual_norms`, `residual_norms_vol` (both in `steady_residual.f90`) and `resid_relnorm` (the steady solver's convergence measure, `steady_newton.f90`) each call it twice — once on $[j_{\min} : N]$, once on $[1 : j_{\min} - 1]$ — and return the componentwise **maximum** of the two.

Why the maximum and not a sum. The two regions' rows are made dimensionless by different physics, so their numbers are not addends of one quantity. Each is already a dimensionless statement whose target is “small”, and the maximum says that every part of the column is steady on the scale its own physics sets. A volume- or cell-count-weighted sum would not: on the ladder grid the region above $2R_p$ carries most of $\sum r^2 dr$, which is exactly how a wind an order of magnitude from steady between 1.2 and $2R_p$ came to be scored better than one that was not.

One number to read with care. The criterion is stated on the volume-weighted form, which is the default (`resid_vol`) and the one the JFNK tests. The legacy L-inf form — a ratio of maxima — is selected by its own key, `Resid norm:` `Linf` (`Resid norm: vol` is the default); it is *not* a fourth word of `Resid tol:`, whose parser reads only the number and silently drops anything after it. Over the layer the L-inf form inherits the $1/dr$ bias its own header already warns about: the smallest cells sit at the base, so on the hot Uranus of the ladder a converged state reads 1.3×10^{-2} on it against 7.0×10^{-4} volume-weighted. **How much that costs is planet-dependent**, because it depends on whether the L-inf maximum falls in the layer or in the wind: on WASP-121b the layer carries it and the two forms differ by about a factor 20 (4.66×10^{-1} L-inf against 2.30×10^{-2} volume-weighted on the `wasp_full_newton` state), while on HD 189733 b the wind carries it and the L-inf number is the *smaller* of the two (7.50×10^{-3} against 2.63×10^{-2} on the `jfnk_hd189` state, both re-read in PLM, the scheme that state was converged in). Neither ratio is a property of the state; both are properties of the L-inf norm on that grid, and they are why the volume-weighted form is the default.

(c) **Everything else is comments and one diagnostic.** The EXHALE_NEWTON_TEST=1 self-test compared $\max |F| / \max |u|$ over $[j_{\min} : N]$ against the EXHALE_RESIDUAL=1 numbers by open-coding the old formula; it now packs F into the residual array and calls `residual_norms`, so the two sides of the self-test cannot drift apart again. The stale comments in `solve_steady_jfnk` that explained the merit/rnorm region mismatch are rewritten: the two now cover the same cells and remain different functionals (a 2-norm of F/D with D the cell's own signal scale, against a maximum of volume-weighted ratios on the physical scales), which is still why the iterate with the smallest $\|R\|$ need not be the last one.

What is deliberately NOT changed. The marching du is still the flux spread of $\rho v r^2$ over $[j_{\min} : N]$. It is ATES's and CETIMB's flux-based stop, its threshold is calibrated against that range, and every golden case stops on it; moving it is a separate change with its own rebaseline. The consequence is recorded here: a run that stops on du alone still says nothing about the column below r_{esc} , and only the Newton finish does.

127.3 The two states of section 126.6, re-scored

Both states re-read with EXHALE_RESIDUAL=1 (volume-weighted, the default), same binary for the row, so the only difference is the measure:

state (He/H = 0.3 arm)	flux spread, $r > 1.2 R_p$	old $\ R\ $	new $\ R\ $
two-stage restart (one PLM step)	3.02×10^{-3}	6.97×10^{-4}	6.97×10^{-4}
single-stage restart (no PLM step)	7.59×10^{-2}	4.93×10^{-4}	4.00×10^{-3}

The old measure rated the second state the better of the two while its inner wind was 25 times less flat. The new measure rates it 5.7 times worse, and the row that carries it is the layer's energy row. On the state that is flat inside, the wind row is the larger half and the number does not move at all.

127.4 What the two paths do under the new measure

The hand-off state of section 126.6, restarted the two ways, on the current tree before and after this change (OMP_NUM_THREADS=8). The *before* column is not identical to section 126.6's, because that measurement was made before section 125. and the q_H2_base ghost-composition change and the archived state is no longer the same fixed point; the defect reproduces at 0.16 dex instead of 0.22.

	two-stage before	two-stage after	single-stage before	single-stage after
info of the finish	0	0	0	1, three attempts
final $\ R\ $	4.540×10^{-4}	5.642×10^{-4}	4.927×10^{-4}	1.601×10^{-3} (best)
$\log_{10} \dot{M}$	10.31	10.27	10.47	10.43 (marched stop)
flux spread, $r > 1.2 R_p$	3.02×10^{-3}	2.456×10^{-3}	7.59×10^{-2}	5.115×10^{-3}
flux spread, $r > 2 R_p$	2.76×10^{-3}	2.455×10^{-3}	2.53×10^{-3}	2.510×10^{-3}
T maximum [K]	3641	3707	3439	3475

The measure now does its job, and it exposes a second thing that it is not. The two-stage path converges and its solution is flat over the whole column — the spread above $1.2R_p$ equals the spread above $2R_p$ to three digits, which is what a steady wind looks like. The single-stage path no longer converges: all three Newton attempts return `info = 1` (2.076×10^{-3} , 1.677×10^{-3} , 1.601×10^{-3} against a target of 10^{-3}), the run falls back to the `du` stop, and it reports a marched state at 10.43. That is the honest outcome — the state is not steady and the code now says so instead of reporting `info = 0` on it — but it is not convergence, and the two paths still do not meet: they differ by 33% in the L_2 norm of (ρ, v, p) , against 22% before.

What blocks the single-stage solve is measured, and it is not the base. The term the merit is dominated by — the largest $|F/D|$ over the whole column, which `solve_steady_jfnk` prints on every outer iteration — is the **energy row** in 493 of the 500 iterations of the third attempt, and it sits at $j = 198$ or 199 , $r = 1.090$, in 485 of them; the base cell $j = 1$ holds it 7 times. Driving the same state directly with `EXHALE_PTC=1 EXHALE_PTC_JFNK=1` and 3000 iterations instead of 500 does not change the answer: $\|R\|$ falls 4.00×10^{-3} (start) $\rightarrow 2.27 \times 10^{-3}$ (201) $\rightarrow 1.89 \times 10^{-3}$ (601) $\rightarrow 1.74 \times 10^{-3}$ (1001) $\rightarrow 1.62 \times 10^{-3}$ (1601), and ends at `info = 1`, 1.437×10^{-3} , after 3000 iterations of which 2994 were non-monotone acceptances — each damped Gauss–Newton step buying about one part in ten thousand of the merit, and the worst cell walks slowly down from $r = 1.134$ to $r = 1.066$ — i.e. onto the H_2 front of this arm, which sits at $1.0681 R_p$. It is a slow mode of the energy equation at the molecular front, not a null mode of the base contact or of the ghost temperature, and it is asymptotic rather than budget-limited: no practical iteration count reaches 10^{-3} from there. Repairing it is solver work in the line of item (P41), and the measure is not what stands in its way.

The restart rule of (P41)(c) — letting a loaded state whose flux spread is already below `du_th_plm` continue in WENO3 instead of silently taking one PLM step — is still not applied. The reason for holding it back has changed: it is no longer that nothing would notice, because the measure now does; it is that the path it would take is the one that does not converge.

127.5 The atomic Newton cases: the rate does not move, the layer moves by orders

`wasp_full_newton` and `wasp_he23off_newton`, `OMP_NUM_THREADS=1`, before and after. The three layer numbers are the volume-weighted mass, momentum/ ρg and energy rows of [1 : 473] at the end of the solve.

	wasp_full_newton		wasp_he23off_newton	
	before	after	before	after
<code>info</code>	0	0	0	0
final $\ R\ $	9.847×10^{-4}	8.006×10^{-4}	6.861×10^{-4}	4.763×10^{-4}
layer mass row	1.51×10^{-3}	1.65×10^{-6}	9.70×10^{-2}	4.37×10^{-5}
layer momentum / ρg	8.32×10^{-5}	1.51×10^{-7}	6.33×10^{-3}	6.78×10^{-6}
layer energy row	4.38×10^{-3}	8.01×10^{-4}	1.23×10^{-1}	4.76×10^{-4}
$\log_{10} \dot{M}$	13.21	13.21	13.22	13.22
L_2 change of (ρ, v, p)	—	1.0×10^{-4}	—	1.05×10^{-2}
largest local change	—	1.2% in v at $r = 1.0002$	—	46% in v at $r = 1.0002$

wasp_he23off_newton is the case that shows what the old criterion was accepting: it returned $\text{info} = 0$ at $\|R\| = 6.9 \times 10^{-4}$ on a state whose below-escape layer was **12% out of energy balance and 0.6% out of hydrostatic balance**, because the measure was not looking. With the layer in the measure the same case converges to 4.8×10^{-4} with the layer three orders lower, the mass-loss rate is unchanged to the printed digit, and what moves is the base: half the base velocity, 3% of the base density, 3.4% of the base temperature. Both cases converge with the layer inside the measure; neither oscillates, and the layer's rows fall monotonically from the 1.1–1.3 they start at.

127.6 The ladder, re-measured on the new criterion

Three rungs of the section 124. ladder, restarted from their own marched states the way that section prescribes, before and after, OMP_NUM_THREADS=8. All six solves return $\text{info} = 0$.

He/H	0.0793	1	30
$\log_{10} \dot{M}$, section 124. (earlier source)	10.42	10.32	10.39
$\log_{10} \dot{M}$, current tree, old measure	10.33	10.29	10.37
$\log_{10} \dot{M}$, current tree, new measure	10.28	10.29	10.37
final $\ R\ $, old measure	8.557×10^{-4}	2.156×10^{-4}	3.890×10^{-4}
final $\ R\ $, new measure	8.925×10^{-4}	7.579×10^{-4}	5.362×10^{-4}
flux spread $r > 1.2R_p$, old	9.267×10^{-3}	2.579×10^{-3}	1.192×10^{-2}
flux spread $r > 1.2R_p$, new	2.386×10^{-3}	2.588×10^{-3}	2.858×10^{-3}
flux spread $r > 2R_p$, new	2.386×10^{-3}	2.588×10^{-3}	2.812×10^{-3}
L_2 change of (ρ, v, p)	27%	14%	4.4%

The signature of the repair is the two flux-spread rows. Under the new criterion the spread below $2R_p$ equals the spread above it on every rung: the column is flat everywhere to the same $2.4\text{--}2.9 \times 10^{-3}$, where before the inner region was 4 to 5 times worse than the outer one on two of the three rungs. The rate moves by at most 0.05 dex and the structure below r_{esc} moves by 4 to 27% in L_2 , which is the part of the solution the old criterion did not determine. The section 124. rates were obtained under the old measure *and* on an earlier source; the current-tree numbers in the third rate row are the ones to quote.

127.7 What was checked

- make check, nine cases, OMP_NUM_THREADS=1: **9/9 byte-identical**. wasp_full and wasp_he23off carry no Solver: key and stop on du, and the seven molecular/profile cases are fixed-step snapshots, so none of them enters the JFNK — which is the point: the change touches the steady solver's acceptance and nothing on the marching path.
- backup/regression/run_fcheck.sh: -fcheck=bounds, do, mem rebuild and the bounded HD 209458 b case, **CLEAN**.
- The two *_newton cases, the three ladder rungs and the two restarts above, all measured.

- NOT checked: the cases that set `Resid tol: (jfnk_*, ptc_*, tpm_*, resid_on in backup/regression/)` are not in the nine-case matrix and were not re-run. `residual_norms/residual_norms_vol` now measure the whole column, so the marching `Resid tol` gate and the `[diag] ||R|| (ref)` line are stricter for those cases in exactly the way described here.

Files: `src/modules/time_step/steady_residual.f90`, `src/modules/time_step/steady_newton.f90`, `src/EXHALE_main.f90`.

Documentation: `T0_BE_DONE.md` item (P42), `docs/supersonic_molecular_base.md` section 13.2.

128. The molecular carriers are transported without the oxygen chemistry, and H₂ is the only one that needs to be (2026-09-02)

Judgment first. EXHALE solved H₂ in local chemical equilibrium, and section 13.6 of `docs/supersonic_molecular_base.md` measured that closure to be violated on its own solution: across the H₂ front of the converged He/H=0.0793 hot Uranus the omitted advection term is 44 to 117 times the whole net chemical rate. The transport operator that closes that gap existed, but ran only with the oxygen chemistry on, which the case does not have. It now runs on the molecular network alone, with H₂ as its one carrier. It is *default off* for a molecular-only run: the transported problem has a front that is still propagating when the steady solver is handed it, so whether a steady transported solution exists on that planet is not yet answered, and the default does not change until it is. Both comparison papers carry the term. Koskinen et al. (2022) Appendix B equation (B1) is a continuity equation for each species, with an advection and a diffusion velocity, and they attribute their H₂-at-all-altitudes to it: “outflow brings fresh H₂ up from deeper layers of the atmosphere and primarily balances the total loss rate”. Frelikh & Murray-Clay (2026) plot “advection of H₂” as a term of their Figure 7. Neither balances H₂ chemically; we did, because that was all the closure could do.

128.1 H₂ is the only carrier that needs transporting, and that is measured

The transported set is a property of the run, so “which species” has to be answered with numbers. Evaluated on the converged He/H=0.0793 state, against the flow time $r/|v|$:

$r [R_p]$	$\tau(\text{H}_2^+)$	$\tau(\text{H}_3^+)$	$\tau(\text{HeH}^+)$	r/v	$\tau(\text{H}_3^+)/(r/v)$	$3n(\text{H}_3^+)/n_{\text{H}}$
1.02	2.1×10^{-5}	9.1×10^1	2.7×10^{-5}	7.5×10^8	1.2×10^{-7}	1.9×10^{-8}
1.10	9.1×10^{-4}	7.9×10^{-1}	9.1×10^{-4}	1.2×10^8	6.9×10^{-9}	2.5×10^{-9}
1.15	6.1×10^{-3}	2.8×10^{-1}	4.4×10^{-3}	1.1×10^7	2.7×10^{-8}	1.7×10^{-10}
1.25	1.3×10^{-1}	4.8×10^{-1}	1.2×10^{-1}	4.0×10^5	1.2×10^{-6}	5.1×10^{-12}
2.00	1.1×10^0	1.3×10^0	4.8×10^0	2.5×10^4	5.1×10^{-5}	3.7×10^{-13}

The three molecular ions are local by four to seven decades everywhere in the domain and hold at most 2×10^{-8} of the H nuclei. H₂’s own ratio, from the section 13.6 table, is 0.02 to 1.3 over the same radii. So the carrier is H₂ and the molecular ions stay local — the same statement the module header already made for HD 209458 b, now measured

on the planet the gate is about. `n_carrier` is therefore a run-time count (1, or 4 with the oxygen cycle) set once by `carrier_set_init`, with `n_carrier_max` dimensioning the arrays; the oxygen path is unchanged and byte-identical.

128.2 The key was named for the wrong thing

Oxygen transport switched an operator whose first carrier is H_2 and which runs with or without the oxygen cycle. It is renamed `Molecular_carrier_transport: True|False`, and the old key is *refused at startup with a message*, not silently aliased: a stored input file that says Oxygen transport describes a switch that no longer exists, and reading it as if it did would hide that. No `input.inp` in the repository carried the key and no regression case sets it, so nothing stored breaks.

The default is not a constant. Left unstated it is `thereis_oxchem`: the oxygen cycle exists to compute a base partition a local steady state cannot produce, so it turns the transport on, exactly as before; a molecular run without it keeps the closure it has always had.

128.3 The base face of an inflowing carrier

Design decision D6 gave the carrier partition a zero-flux base, on the argument that the split of an element among its carriers is what the oxygen option exists to compute and imposing it would make that option's own gate circular. That argument holds exactly when nothing upstream states the partition. When a handoff does — `base.inp`'s `q_H2_base`, or a profile at the matching level — the partition is incoming data, and section 117 already pins it on the ghosts of the ionization sweep; leaving the base cell free while the ghost below it is pinned puts the defect section 117 removed one cell higher up.

So where the handoff speaks and the base face is an *inflow*, the advective term at that face carries the ghost's composition. Only the advective term: face 0 still carries no diffusive flux, because the handoff supplies the composition of the gas that flows in, not a mixing rate across a boundary whose gradient is set by the ghost spacing and by a K_{zz} that describes an unresolved region. The two branches cannot both apply — decision D7 refuses `q_H2_base` while the oxygen chemistry is on.

Measured on `mol_base_handoff` (HD 209458 b, 12000-step snapshot, $v = +97.7 \text{ cm s}^{-1}$ at the base, `q_H2_base = 0.75`):

	x_2 ghost	x_2 base cell 1	jump	$f(\text{H}_2) = 0.5$ at	$\log_{10} \dot{M}$
transport off	0.925114	0.999674	0.0746	1.15467	10.56
on, D6 zero flux	0.925114	0.999052	0.0739	1.15862	10.55
on, handoff Dirichlet	0.925114	0.939813	0.0147	1.15813	10.57

It closes 80% of the jump and it does not pin the cell, and it cannot. This is a condition on the *face*, not on the cell: the remaining 0.0147 is the cell's own chemistry rebuilding H_2 faster than the inflow sweeps it out. Pinning cell 1 itself would take a physical cell out of the solution. Both numbers are reported by every run, so the size of the remaining jump is visible rather than assumed.

Whether the base face is an inflow is decided by the mass flux the wind carries, not by the base cell's velocity, and that distinction was forced by a measurement. On the converged hot Uranus the base cell's v is -250 cm s^{-1} and

its cell-centred $\rho v r^2$ is 196 times the wind's flux inward — while the face flux the scheme actually transports there, reconstructed by integrating its own steady mass residual down from the smooth wind, is +0.36 to +0.98 times the wind flux, *outward*, in every converged configuration measured (docs/p44_base_sawtooth.md). The cell-centred velocity at that base is a collocated odd-even mode, not infall: ρ , p and T are monotone through the same cells, the layer carries 288 cells per scale height, and no dissipative remedy tried — Shapiro filter, Low-Mach damping, Base velocity: massflux, a continuous ghost temperature — changes $v(1)$ by more than 3 cm s^{-1} . A condition gated on $\text{sign}(n_{\text{tot}}v(1))$ reads the mode; one gated on the wind's flux reads the state. The ghost velocity is not usable either: the $+0.94 \text{ cm s}^{-1}$ it carries in the converged state is exactly the softplus floor of the smooth valve.

So `carrier_base_state` computes $\sum \rho v r^2$ over $[j_{\min} : N]$ — the same constant and the same average Base velocity: massflux uses for the ghost velocity, seeded in `init.f90` and updated each step in `EXHALE_main.f90` — and its sign decides both the composition condition and the advective direction of cell 1. In a steady spherical wind that flux is one number at every face, so using it at the base face states the steady state rather than approximating it. The interior keeps the cell-velocity upwind, for the reason its own header gives (the element operator's frozen helium hole); only cell 1 is decided on the face, and only because the measurement says the cell velocity there is not a flux.

20000 marching steps, hot Uranus	x_2 ghost	x_2 cell 1	jump	$f = 0.5$	10^{-4}
transport off	0.925114	0.998668	0.0736	1.1008	1.2698
on, condition on $\text{sign}(n_{\text{tot}}v(1))$	0.925114	0.998518	0.0734	1.1016	1.3072
on, condition on the wind flux	0.925114	0.957446	0.0323	1.1017	1.3073

The condition now fires on this planet, where the cell-centred test left it permanently off, and it closes 56% of the composition jump. It does not move the front ($1.1016 \rightarrow 1.1017$), which is the right outcome: this is a boundary condition on the partition, not on the wind. On `mol_base_handoff`, where the cell-centred velocity happens to agree in sign, the arm is nearly unchanged and slightly better — the jump goes from 0.0147 to 0.0090 — because the cell-centred test also flipped transiently during the run and the flux test does not.

On the oxygen option it is not a no-op, and that is the point. With the transport on and no handoff, cell 1's advective term used to be taken from cell 2 whenever $n_{\text{tot}}v(1) < 0$, and on `examples/18_oxygen_chemistry` that quantity is 1.5×10^6 times the wind flux inward — the same mode, worse. With the direction taken from the face, the base cell's H_2 partition moves from 0.2593 to 0.3411 of the H nuclei (+32%), base H_2O by +120%, and $\log_{10} \dot{M}$ by +0.02 dex over a 2000-step snapshot. The base partition is what A2 exists to compute, so the correction lands on its own subject. Both states are 2000-step snapshots from a cold start, so this is a direction and an order of magnitude, not a result.

The base cell's own velocity, and what it is, are `TO_BE_DONE.md` item (V).

128.4 A defect the transported problem found: two definitions of the same budget

`carrier_source` closed atomic H by sharing the element with the frozen ion stages,

$$n_{\text{HI}} = n_{\text{H,free}} - n_{\text{HII}} - 2n_{\text{H}_2^+} - 3n_{\text{H}_3^+} - (\text{what the carriers hold}),$$

while `limit_to_element_budget` let the carriers take the whole of $n_{\text{H,free}}$. The two disagreed by exactly the H sitting in the frozen ions, and that gap is what the transported hot Uranus walked into: 80 of 500 cells between 1.032 and

1.156 R_p reached $2n(\text{H}_2)/n_{\text{H}} = 1$, the write-back's remainder for HI, HII, H_2^+ and H_3^+ went to zero, n_e collapsed to 3.7 cm^{-3} , and the next ionization sweep stopped with “persistent non-root chemical equilibrium” at $r = 1.112$.

The two sites now call one function, `hydrogen_available_to_carriers`. The comment that permitted the looser bound has been corrected where it stood: “the next sweep re-solves the stages” is true of the sweep and false of the step in between, because this step's write-back does not re-solve HI, HII, H_2^+ and H_3^+ — it rescales them in proportion to what they already held.

Measured effect on the oxygen option: none, and the reason is measured too. On `examples/18_oxygen_chemistry` (2000 steps) all twelve output files are byte-identical, because the carriers there reach only 0.358 of the hydrogen available to them and no cell has $n(\text{H}_3^+) = 0$. The fix bites only where the transported H_2 approaches the whole hydrogen budget, which on that planet it does not. On the hot Uranus it removes the electron collapse ($n_e 3.7 \rightarrow 1.8 \times 10^4 \text{ cm}^{-3}$) and cuts the saturated cells from 80 to 54.

128.5 A scale for the carrier row, beside the others

The steady solver never sees the carrier equation: it holds the partition fixed and solves the wind, and an outer Picard iteration relaxes the carriers at that wind. Their common fixed point is a steady state of both, but only the hydrodynamic half of it is in $\|R\|$. `carrier_steady_residual` reports the other half — the same residual the transport step assembles, with the time term switched off — made dimensionless by `carrier_row_scale`, which lives beside `residual_row_scale` in `steady_residual.f90` so that every row scale in the code has one definition site.

The scale is $n_{\text{H}}(|v| + c_s)/\Delta r$, the flux divergence a cell would carry if its whole hydrogen inventory moved at its own fastest signal speed. The unknown $n(\text{H}_2)$ cannot be its own scale for the same reason $|\rho v|$ cannot be the momentum row's in the quasi-hydrostatic layer: it falls four decades across the front, and $|R|/n(\text{H}_2)$ would then be large wherever the carrier is scarce whether or not the state is right.

128.6 What the steady route does with it, and why the default is off

On the converged $\text{He}/\text{H} = 0.0793$ state with the transport on, the outer loop does not converge, and the measurements say why.

- **The wind solve is not the blocker.** The first JFNK converges, `info` = 0, $\|R\| = 8.7 \times 10^{-4}$ — the same grade as with the transport off.
- **The relaxation then moves the composition by order unity, at the front.** Outer drift 0.536, worst at cell 247, $r = 1.168$; carrier steady residual 1.06×10^{-2} at $r = 1.129$. That is not a numerical artefact: it is the size of the correction the closure was missing.
- **Bounding the pass does not bound the move.** Capping how far one pass may advance the carriers at 1.0 and at 0.1 of a cell flow time gave drifts of 0.536 and 0.533. At the front, one tenth of a flow time is about sixty cell crossings; the front is advected, and advection is not slow on any scale such a bound can be written in. The bound was therefore *not* kept — a bound that controls nothing reads as a control to the next person.
- **The front propagates outward at the gas speed, pass after pass.**

outer pass	$f = 0.5$	10^{-2}	10^{-4}	$n(\text{H}_3^+) > 1$ to	cells with $n(\text{H}_3^+) = 0$	x_2 at $1.5 R_p$
1	1.1664	1.2257	1.2921	1.2211	52	2.44×10^{-5}
2	1.2482	1.3122	1.4051	1.3053	33	1.89×10^{-5}

$0.082 R_p$ per pass of 0.1 flow times is about 2050 cm s^{-1} against a gas speed of 3000 cm s^{-1} there. The far wind ($1.5, 2.0 R_p$) does not move, and the saturated cell count *falls* as the front leaves them. This is a front filling a region, not a runaway.

- **A real feedback reinforces it, and the frozen background is the conservative side of it.** Within one relaxation pass He^+ , n_e , T and the photoionization rates are frozen — `relax_photochemical_composition` never calls `ioniz_eq` and `carrier_write_back` never writes a helium index — so the pass advances the front using the *old*, larger He^+ . Between passes the sweep re-solves it, and where H_2 has arrived it collapses:

$r [R_p]$	$n(\text{He II})$ entry	after pass 1	after pass 2
1.100	1.23×10^3	9.56×10^4	1.05×10^4
1.150	1.77×10^6	2.25×10^6	1.85×10^4
1.200	1.07×10^7	8.35×10^6	1.99×10^5
1.250	1.32×10^7	1.11×10^7	2.85×10^6

A factor 50 to 100 at $1.15\text{--}1.20 R_p$, where section 13.6 measured $\text{He}^+ + \text{H}_2$ to be 95% of the whole net H_2 loss. So the sink that would stop the front is removed by the front itself. Whether that makes two steady states or one slow approach is *not* settled here: the front had not stopped after two passes.

- **The blocking mode is the front's energy row, and it moves with the front.** Over all JFNK iterations after the first relaxation the worst row is the energy row in 674 to 765 of about 1030, at $r = 1.113\text{--}1.137$ when the front is at $1.15\text{--}1.28$ and at $r = 1.162\text{--}1.195$ when the front is at $1.17\text{--}1.25$. With the transport off that mode is not the worst cell at all — the base is, row $k = 2$, $r = 1.000$ — and the solve converges. This is the slow mode of section 127., enlarged by the transport and no longer at a fixed radius.
- **Marching does not fail.** 20000 marching steps with the transport on, from the same state, run clean (no ERROR STOP) and move the front outward monotonically: $f(\text{H}_2) = 10^{-4}$ from 1.2698 to $1.3072 R_p$, $n(\text{H}_3^+) > 1$ from 1.1810 to $1.1976 R_p$, $\log_{10} \dot{M} = 10.40$ either way. What fails is asking a steady-state routine to follow a propagating front.

Hence the default. Molecular carrier transport stays off for a molecular-only run until a marching run long enough to stop the front says where it stops, and a steady solve started from that state returns `info = 0`. Nothing above is a converged solution and nothing above is quoted as a rate.

128.7 What was checked

- `make check`, nine cases, `OMP_NUM_THREADS=1`: **9/9 byte-identical**. The transport is off in all of them, which is what the default says.

- examples/18_oxygen_chemistry, 2000 steps, the transport *on*: byte-identical against the pre-change binary through the headroom change, and then deliberately *moved* by the base-face condition, by +32% in the base H₂ partition and +0.02 dex in the rate. The two were measured separately so that each is attributable.
- backup/regression/run_fcheck.sh: -fcheck=bounds, do, mem rebuild and the bounded HD 209458 b case.
- The six molecular regression cases were additionally run pairwise against the pre-change binary outside the harness, at four separate stages of this change (carrier count, gates and key, base condition, headroom): identical every time.
- NOT checked: mol_diffusion with the transport *on*, i.e. the helium element operator and the carrier operator in the same split step. They are ordered element_diffusion_step then photochemical_transport_step, and the second reads ntot, T_K and the rate coefficients from the bg_cell the sweep *before* the diffusion left, so it lags by one operation on the marching path (not on the Picard path, where ioniz_eq runs between them). Nothing measures the size of that lag yet.

Files: src/modules/lower_atmosphere/diffusive_photochemistry.f90, src/modules/radiation/ionization_equilibrium.f90, src/modules/time_step/steady_residual.f90, src/modules/files_IO/input_read.f90, src/modules/files_IO/write_set.f90, src/modules/files_IO/write_output.f90, src/modules/init/parameters.f90, src/EXHALE_main.f90. Documentation: docs/supersonic_molecular_base.md sections 13.6 and 13.7, docs/a2_oxygen_option_design.md, TO_BE_DONE.md items (P43) and (V).

129. The trace metals now compete for the He recombination photons, and nitrogen at depth notices (2026-09-02)

Judgment first. The expression was physically incomplete, and the size of what it left out was never measured. A He recombination photon of energy E_c is absorbed by every species whose ionization threshold lies below E_c , in proportion to $n_s \sigma_s(E_c)$. Sections 120. and 125. wrote that competition with H I, H₂ and (above 24.6 eV) He I in it, and gave the metals' share to hydrogen. The metals are 10^{-4} to 10^{-3} of the gas by number, but their cross sections in the 16–25 eV band are *larger* than hydrogen's — 11.9 (O I), 12.2 (N I) and 4.9 (C II) against 1.24 for H I at 24.6 eV — and in the ionized wind the H I that used to take those photons is itself a trace. Measured before the change: the metal share of the absorption coefficient runs from 3×10^{-3} at the base to 2.8×10^{-2} at $2.8 R_p$. That is a share of the photons, and it was going to hydrogen. Item (P34), opened with section 120., is closed here.

129.1 The size, measured first

For each channel c the metal share of the absorption coefficient is

$$\frac{\sum_m n_m \sigma_m(E_c)}{n_{\text{HI}} \sigma_{\text{HI}} + n_{\text{H}_2} \sigma_{\text{H}_2} + n_{\text{HeI}} \sigma_{\text{HeI}}},$$

evaluated on the pre-change solutions with the same `sigma_*` functions the code uses. The five channels are the ground-capture continuum (24.6 eV), the 584 Å resonance (21.2), the 2³S line (19.8), the 2¹S two-photon continuum (16.110) and the case-B cascade average (19.741).

case and radius	24.6	21.2	19.8	16.11	19.74	who
wasp_full_newton, 1.00 R_p	4.4×10^{-3}	3.6×10^{-3}	2.8×10^{-3}	1.1×10^{-3}	2.8×10^{-3}	O I, 84%
wasp_full_newton, 1.61 R_p	2.2×10^{-2}	2.3×10^{-3}	1.7×10^{-3}	6.9×10^{-4}	1.7×10^{-3}	C II, 89%
wasp_he23off_newton, 1.61 R_p	2.2×10^{-2}	2.3×10^{-3}	1.7×10^{-3}	6.9×10^{-4}	1.7×10^{-3}	C II, 89%
mol_metals, 1.00 R_p	3.2×10^{-3}	2.9×10^{-3}	2.4×10^{-3}	3.2×10^{-3}	2.4×10^{-3}	O I; C I at 16.11
mol_metals, 4.81 R_p	6.0×10^{-3}	4.4×10^{-3}	3.5×10^{-3}	1.4×10^{-3}	3.4×10^{-3}	O I, 54%
lower_profile, 1.00 R_p	4.7×10^{-3}	4.8×10^{-3}	4.0×10^{-3}	2.0×10^{-3}	3.9×10^{-3}	O I
lower_profile, 2.82 R_p	2.8×10^{-2}	5.8×10^{-4}	4.7×10^{-4}	2.1×10^{-4}	4.6×10^{-4}	C II, 98%

It is above 10^{-3} everywhere, in every channel, in every metals-on case. The threshold the measurement was to be judged against was 10^{-3} , so the metals belong in the competition and were implemented.

Two features of the table are worth reading. The ground-capture channel is the one that grows outward, because it is the only channel whose H I competitor thins out while a metal ion that can still absorb at 24.6 eV — C II, threshold 24.38 eV — does not. And the two-photon channel at 16.110 eV is the one the molecular cases pick out, because C I (threshold 11.26 eV, $\sigma = 13.4$ there against H I's 4.0) is neutral in a molecular layer.

129.2 The change

`he_rec_coupling` (`src/modules/radiation/util_ion_eq.f90`) now runs the competition over H I, H₂, He I *and* every photo-ionizable metal ion of `species_table` whose `mion_ethr` lies below the channel energy:

$$\begin{aligned}
 \tau_c &= \left(n_{\text{HI}} \sigma_{\text{HI}} + n_{\text{H}_2} \sigma_{\text{H}_2} [+n_{\text{HeI}} \sigma_{\text{HeI}}] + \sum_m n_m \sigma_m \right) \Delta r_j, \\
 w_s &= n_s \sigma_s(E_c) / \sum_{s'} n_{s'} \sigma_{s'}(E_c), \\
 \delta P_m &= P_c f_{\text{abs}} w_m / n_m = P_c \sigma_m(E_c) \Delta r_j [1 - e^{-\tau_c}] / \tau_c, \\
 H_c &= P_c f_{\text{abs}} \sum_s w_s (E_c - E_{\text{th},s}), \quad E_{\text{th},m} = \text{mion_ethr}(m), \\
 \alpha_{\text{eff}} &= \alpha_B + \alpha_1 \left[(1 - f_{\text{abs}}) + f_{\text{abs}} (w_{\text{HI}} + w_{\text{H}_2} + \sum_m w_m) \right].
 \end{aligned}$$

The metals enter in four places and each one is the same statement: they take a share of the photons. They add opacity to the cell optical depth, so the cell keeps more of them; they take a share out of the H I and H₂ shares; the share they take ionizes them, and is returned in a new output `dP_m` that the callers add to the metal photoionization rate `P_m`; and the share they take is not available to re-ionize helium, so it enters the effective He II recombination coefficient with the other absorbed shares.

The shares stay in the ratio form of section 120., $w_{\text{HI}} = 1 / (1 + R_{\text{He}} n_{\text{HeI}} / n_{\text{HI}} + R_{\text{H}_2} n_{\text{H}_2} / n_{\text{HI}} + \sum_m R_m n_m / n_{\text{HI}})$ with $R_m = \sigma_m(E_c) / \sigma_{\text{HI}}(E_c)$ evaluated once at entry, and the metal rate is $P_c f_{\text{abs}} w_{\text{HI}} R_m / n_{\text{HI}}$ so that the metal density of the share cancels rather than appearing in a denominator — the same property that makes `dP_H2` finite as $n_{\text{H}_2} \rightarrow 0$. A run with no metals adds *exactly* 0.0 to every one of those sums, which is why it is bitwise unchanged; and `metal_photoion_sigma` returns zero below an ion's own threshold, so an ion that cannot absorb a channel drops out of it with no test.

The ions that participate are the 17 photo-ionizable stages, i.e. every stage but the inert top one of each element. At 24.6 eV that is 15 of them (O II at 35.12 eV and N II at 29.60 eV are above the edge); at 16.110 eV it is 12.

129.3 One place that says what column of the cross-section table is which

Adding the metals meant evaluating their cross sections at the five channel energies, which are not on the `e_v` grid the `sigma_tab` table lives on. The 17 fits were listed once, by hand, in `set_energy_vectors`. A second hand-kept list in `he_rec_coupling` would be a copy that can drift, so the ordering was moved into one function instead: `metal_photoion_sigma(k,E)` in `Cross_sections` dispatches on the photo-table column index `k`, and `set_energy_vectors` fills `sigma_tab` through it in a loop. The ordering now exists in one place, and the coupling reads the same function the table does.

This is a refactor with no intended physics change, and it is byte-identical on a metals-*on* case: `wasp_full` run with this change alone reproduces its golden `Hydro_ioniz.txt` and `Ion_species.txt` exactly.

129.4 What moved

make check, nine cases, `OMP_NUM_THREADS=1`: **the four metals-off cases are byte-identical** (`mol_base_handoff`, `mol_lyman_werner`, `mol_diffusion`, `mol_sec_ion`) and the five metals-on cases move.

case	$\Delta \log_{10} \dot{M}$	$\max \delta T/T$	at r	$\max \delta n_{\text{HI}}/n$	at r	largest ion move
<code>wasp_full</code>	-0.00001	2.0×10^{-4}	1.144	1.4×10^{-3}	1.258	N II 5.4×10^{-2} at 1.110
<code>wasp_he23off</code>	-0.00002	2.8×10^{-4}	1.133	1.8×10^{-3}	1.257	N II 8.3×10^{-2} at 1.095
<code>mol_metals</code>	+0.00001	2.1×10^{-3}	1.193	1.8×10^{-3}	1.193	H ₃ ⁺ 4.8×10^{-2} at 1.193
<code>mol_ir_bands</code>	-0.00016	5.4×10^{-2}	1.193	4.7×10^{-2}	1.193	H ₃ ⁺ 6.9×10^{-1} at 1.190
<code>lower_profile</code>	-0.00000	1.5×10^{-3}	2.797	4.3×10^{-3}	2.797	N II 1.7×10^{-2} at 1.097

$\Delta \log_{10} \dot{M}$ is the outer-cell mass flux $r^2 \rho v$, re-measured from the profiles. The molecular layer is again not touched, for the reason section 125. gives — it is optically thick to these photons, f_{abs} is exactly 1.0 below $1.10 R_p$, and only the star-ward column reaches it: below $1.10 R_p$ the temperature moves by 2.1×10^{-5} in `mol_metals` and 8.0×10^{-4} in `mol_ir_bands`, against 2.1×10^{-3} and 5.4×10^{-2} above it. `mol_ir_bands` moves the most of the nine because its infrared cooling channels sit exactly where the change lands, and because all six molecular cases are 12000-step relaxation snapshots at $du \sim 4$ rather than converged solutions.

The converged cases, before and after with the same binary except for this change (`wasp_full_newton` and `wasp_he23off_newton`, WASP-121 b, metals on, JFNK info = 0 before and after):

	wasp_full_newton	wasp_he23off_newton
$\log_{10} \dot{M}$ shift	$+1 \times 10^{-5}$ dex	-6×10^{-5} dex
largest T change	0.02% at 1.317	0.03% at 1.113
largest heating change	0.19% at 1.245	0.14% at 1.228
largest $n(\text{H I})$ change	0.22% at 1.315	0.20% at 1.274
largest $n(\text{He } 2^3\text{S})$ change	0.11% at 1.333	—
largest $n(\text{C II})$ change	0.16% at 1.217	0.14% at 1.212
largest $n(\text{N II})$ change	5.2% at 1.085	8.4% at 1.074

The wind does not move and the nitrogen ionization balance does. That is the correction doing what the cross sections said it would: N I has $\sigma = 12.2$ at 24.6 eV, ten times H I's 1.24, so at $1.085 R_p$ — deep enough that the stellar field above 24.6 eV is attenuated but the He recombination photons are produced locally — the He recombination channel is a 5% addition to the N I photoionization rate. There $n(\text{N II})$ is 3% of $n(\text{N I})$ and rises from 1.447×10^5 to $1.522 \times 10^5 \text{ cm}^{-3}$, with $n(\text{N I})$ falling by 0.17% to pay for it. No other element moves by more than 0.25%: carbon and oxygen are already ionized by the stellar field at that depth, and iron and magnesium have small cross sections at these energies.

129.5 What was checked, and what was not

1. **Byte-identical without metals**, which is the property the ratio form is written for, checked on both branches of the routine and on the molecular path: the four metals-off molecular regression cases in make check, plus wasp_full and wasp_he23off re-run without their metals.inp (the TR and the atomic branch of he_rec_coupling, respectively) against a binary built from the pre-change sources. Identical in Hydro_ioniz.txt and Ion_species.txt in all six.
2. **Byte-identical with metals for the cross-section-table refactor alone**, wasp_full.
3. make check: 4 of 9 byte-identical, 5 metals-on cases move as tabulated. The goldens are refreshed at the end of the block this change belongs to.
4. The two converged WASP-121 b cases, before and after, info = 0 both times.
5. n_cells_he_rec_photoionization_dominant is zero on every case measured here, as it was in sections 120. and 125..
6. **A bounds/do/mem-checking build**, the flags run_fcheck.sh uses (-O1 -fopenmp -g -gz=none -fcheck=bounds,do,mem), on three bounded 1500-step *metals-on* runs, because run_fcheck.sh's own case is HD 209458 b and the arrays this change adds are only touched when metals are on: wasp_full (TR branch), mol_metals (molecular branch) and wasp_full with pp_metals 2, which is the only configuration that exercises the metal rate correction in the *_adv post-process. Clean, no runtime check fired, and each produced its *_adv.txt files. It was run in a scratch copy of the tree rather than through run_fcheck.sh, which begins with a make distclean in the repository.

7. **NOT checked.** How much the *_adv.txt profiles themselves move: the regression compares only Hydro_ioniz.txt and Ion_species.txt, so the post-process correction is exercised but its size is not measured.

Approximations, unchanged or newly stated at the code site. The photoelectron of a metal share is deposited entirely as heat, at $E_c - \text{mion_ethr}(m)$, without passing through the Shull & van Steenberg secondary-ionization partition the stellar photoelectrons go through — the same treatment the H I and H₂ shares of these photons already had. The ions that actually carry the metal share leave 11.0 eV (O I), 10.1 (N I), 8.4 (Fe II) and 0.2 (C II) at the ground-capture channel and 4.9 (C I) at the two-photon one; the largest of the set, 20.3 eV for K I, belongs to an ion whose cross section there is 0.048 against H I's 1.24 and which therefore carries almost none of it. The σ_m are the same single-shell Verner et al. (1996) fits sigma_tab carries, with the Fe I caveat already recorded at sigma_FeI. The metal cross sections are evaluated at the channel energy with their (small) kT dependence neglected, as the H I, H₂ and He I ones already were. Files: src/modules/radiation/util_ion_eq.f90, src/modules/functions/cross_sec.f90, src/modules/init/set_energy_v src/modules/radiation/ionization_equilibrium.f90, src/modules/post_process/post_process_adv.f90. Documentation: TO_BE_DONE.md item (P34).

130. Dissociative photoionization of H₂ is the second product channel of a cross section the code already had, and the branching comes from the measurement rather than from a summary of it (2026-09-02)

Judgment first. The photon channel $\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}^+ + e^-$ (threshold 18.076 eV) was missing, and it was missing on both sides: from the primary photoionization and from the secondary-electron partition of Sec. 118.. That is not a small omission dressed up as a large one, and it is not a new opacity either. sigma_H2 is the *total* photoionization cross section, so the channel was already inside the H₂ destruction rate the code applied — it was simply all being credited to H₂⁺. In the shielded molecular base that is the wrong ion: the dissociative channel makes a proton and an H atom, does not feed the H₃⁺ chain (R8), and is the dominant photon-driven proton source there, 39 times the H I photoionization at 1.004 R_p on the converged He/H = 0.0793 hot Uranus. Item (P31), opened with Sec. 118., is closed here.

130.1 The two channels

$$(a) \quad \text{H}_2 + h\nu \rightarrow \text{H}_2^+ + e^-, \quad E_{\text{th}} = 15.4 \text{ eV}, \quad (43)$$

$$(b) \quad \text{H}_2 + h\nu \rightarrow \text{H} + \text{H}^+ + e^-, \quad E_{\text{th}} = 18.076 \text{ eV}. \quad (44)$$

Both consume one H₂. Since (b) is a share of the same sigma_H2, the H₂ destruction rate, the opacity and the absorbed energy are unchanged; what moves is the product and the threshold each channel is charged. The code carries the whole rate as P_H2 and the dissociative part as P_H2_di, a SUBSET of it, and the coupled molecular system reads row (4), H₂: the whole P_H2, as before; row (5), H₂⁺: P_H2 - P_H2_di; row (1), H⁺: +P_H2_di n_{H_2} . The H atom of the pair is supplied by the H-nucleus closure $n_{\text{HI}} = (1 - x_1 - x_4 - x_5 - x_6 - x_7)n_{\text{H}}$, exactly as the two Lyman–Werner fragments are, so no row was added.

The photoelectron of (b) carries $h\nu - 18.076$ eV, not $h\nu - 15.4$: the 2.68 eV between the thresholds goes into breaking the bond and is not available as heat, the rule every other absorber in PH_heat_HHe follows. It is therefore its own absorber in the secondary-ionization energy grid (iabs_H2_di, n_abs_fixed 5 $\rightarrow$ 6), because its photoelectron lands elsewhere in the Dalgarno energy table and thermalizes a different share.

130.2 The secondary side is one sentence of the source

Dalgarno, Yan & Liu (1999), the paragraph after their Eq. (10): “The production of H^+ ions from dissociative ionization of H_2 is given by a mean energy per ion of $22W_m(H_2^+)$. The harmonic mean of $22W_m(H_2^+)$ and $W_m(H^+)$ is the mean energy for the production of H^+ ions in the gas mixture.” A mean energy 22 times larger is a yield 22 times smaller, and the harmonic mean makes the two H^+ sources ADD rather than share, so the fast electrons make one dissociative proton for every 22 H_2^+ ions they make. The 4.5% is not deducted from the Shull & van Steenberg amplitude the module borrows for the ionization budget, because Dalgarno et al.’s W is the primary energy divided by the number of ions and already contains every loss of the degradation; deducting it would double-count.

130.3 The branching, and a correction to the summary we would otherwise have used

The obvious source is the paper the cross section itself comes from. Yan, Sadeghpour & Dalgarno (1998), ApJ 496, 1044, Sec. 4: “The cross section for dissociative ionization, $H_2 + h\nu \rightarrow H + H^+ + e^-$, is complicated by the contribution of resonance states ... Its ratio to the total photoionization cross section increases from zero at the threshold of 18.08 eV to 0.284 at 76 eV and then decreases slowly to 0.258 at 124 eV.”

Those two numbers are not the ratio to the total. They are the H^+/H_2^+ column of Chung, Lee, Masuoka & Samson (1993), J. Chem. Phys. 99, 885, Table II, whose caption reads “The partial cross sections $\sigma(H^+)$, $\sigma(H_2^+)$, and the ratio H^+/H_2^+ for dissociative photoionization of H_2 ”: a ratio to the H_2^+ *partial* cross section. The ratio to the total is $r/(1+r)$, which is 0.221 at 76 eV and 0.205 at 124 eV. An implementation built on the quoted sentence — which is what this work first built, and ran — overstates the channel by a factor 1.26–1.28 above 70 eV; and because the real curve rises to a 0.02 plateau within 0.03 eV of threshold instead of ramping linearly from it, the same implementation understates it by up to a factor 20 below 20 eV and misses the Q_1 Rydberg autoionization resonance entirely.

The code therefore reads Chung et al.’s Table II directly, as $f_{di}(E) = \sigma(H^+)/[\sigma(H^+) + \sigma(H_2^+)]$ at their 69 energies from 18.076 to 124 eV, linearly interpolated. Two checks were run on the transcription: their two partial columns reproduce their own ratio column to better than 0.5% at every row, and the partial columns sum to their $\sigma(\text{abs})$ column exactly except at the ten rows carrying their footnote b (33–41 eV), where the photoionization yield is below unity and the deficit is their Table I neutral cross section ($1.29 - 1.20 = 0.09$ against their 0.092 at 37 eV).

The flux-weighted branching of an unattenuated E^{-1} power law over 15.5–1240 eV is 0.0239 against the ramp’s 0.0224: the two agree in the mean and disagree everywhere in detail, which is exactly the case in which a mean-preserving error survives a coarse test.

Uncertainty, as the source states it. “An estimate of the total uncertainty in the data is about $\pm 4\%$ – 5% ”; the ionization yields used over 18–28 eV are unity “within an accuracy of about $\pm 5\%$ ”. Against other determinations Chung et al.’s H^+ data lie 0%–5% below Backx, Wight & van der Wiel over 18–33 eV, growing to 26% at 70 eV; they agree within

Table 3: The branching of Chung et al. (1993) Table II against the three-point ramp the Yan et al. (1998) sentence implies.

E [eV]	Chung Table II	three-point ramp	ramp / Chung
18.1	0.0022	0.0001	0.045
19	0.0161	0.0045	0.28
20	0.0196	0.0094	0.48
22	0.0216	0.0192	0.89
25	0.0237	0.0339	1.43
30	0.0483	0.0584	1.21
34	0.0995	0.0781	0.79
35	0.1084	0.0830	0.77
37.5	0.0909	0.0952	1.05
40	0.1089	0.1075	0.99
45	0.1342	0.1320	0.98
50	0.1537	0.1565	1.02
60	0.1982	0.2055	1.04
74	0.2219	0.2742	1.24
76	0.2212	0.2840	1.28
110	0.2096	0.2656	1.27
124	0.2048	0.2580	1.26
> 124	0.2048	0.2580	1.26

5% with Kossmann et al. over 25–75 eV and lie 11%–20% below them over 75–110 eV.

Above 124 eV, the top of the measured range, f_{di} is held at its 124 eV value, 0.2048. The source quotes no higher-energy data. The curve is nearly flat over its last 50 eV (0.2219 at 74 eV to 0.2048 at 124 eV), so this is a mild extrapolation — but it is an extrapolation, and it is the value the whole X-ray grid uses.

130.4 Two things the branching carries that its name does not

Both are stated by the source, both are recorded at the code site, and both are carried as open items — (P31b) and (P31c) of TO_BE_DONE.md, which give the size of each and what removing it would take.

1. **Double ionization.** Chung et al.’s Eqs. (3)–(4) note that “the branching ratio $H^+/(H^+ + H_2^+)$ is understood to include both single and double ionization where appropriate”, and their discussion of Fig. 4 that “by 80 eV about 20% of $\sigma(H^+)$ comes from double ionization. This percentage remains fairly constant towards higher energies.” Their $\sigma(H^+)$ counts PROTONS, and a double ionization yields two, so using f_{di} as a share of the H_2 ionization rate gives the PROTON source rate correctly; what it mis-assigns is the co-product, since those events leave no H atom and no H_2^+ . Above 80 eV that is about 4% of the H_2 ionizations, in a band where sigma_H2 is already two decades below its threshold value.
2. **Neutral dissociation over 33–41 eV.** sigma_H2 follows Samson & Haddad’s ABSORPTION cross section there — it is 1.76 against Chung et al.’s $\sigma(\text{abs})$ 1.74 and their ionization sum 1.699 at 34 eV — so the H_2 destruction rate the code applies in that band includes $H_2 + h\nu \rightarrow H + H$, up to 7% of it at 37.5 eV, and the $1 - f_{\text{di}}$ branch hands that share to H_2^+ . This is a property of the cross-section fit, which predates this change, not of the branching.

Not split, deliberately: the He recombination photons. dP_H2_hrc (he_rec_coupling) sits at 19.8–24.6 eV, where f_{di} is 1.9%–2.3%. Splitting off that much of a channel that is itself a correction would be well inside the ± 4 –5% the branching carries, so the whole of it still makes H_2^+ , and the code says so where it is applied.

130.5 What moved

The three atomic cases are byte-identical. wasp_full, wasp_he23off and lower_profile carry no H_2 , so nh2 is never passed, s_h2_di is allocated and never read, and every H_2 term is identically zero. Their Hydro_ioniz.txt and Ion_species.txt reproduce the pre-change tree bit-for-bit. They do *not* reproduce the goldens, and neither did the tree before this change: Secs. 128. and 129. moved five of the nine cases and the goldens are refreshed once at the end of the series, so the comparison that means anything is against the tree’s own output immediately before the change, which is what the table below is.

The six molecular cases move. Each is a fixed-step relaxation snapshot, not a converged solution, so a per-cell extreme is noisy; the entry below is the largest relative change over the cells where that species is within three decades of its own peak.

The signature is the one the physics predicts: **H_2 barely moves** (the total destruction rate is unchanged by construction), **H_2^+ falls, H_3^+ falls with it** because R8 is fed by H_2^+ , and **H^+ rises**. The H_2 front moves inward in the fifth decimal, i.e. not at all. The largest temperature change of the six is 4.2% (mol_ir_bands) and the smallest 5.7×10^{-4}

Table 4: P31 against the tree immediately before it.

case	H ₂ front [R_p]	H ⁺	H ₂ ⁺	H ₃ ⁺	H ₂
mol_base_handoff	1.15471 → 1.15462	+13.8%	-16.7%	-10.9%	-1.25%
mol_metals	1.15628 → 1.15622	-13.3%	-21.5%	-48.0%	-0.65%
mol_lyman_werner	1.12886 → 1.12886	+12.9%	-15.4%	-9.0%	+0.19%
mol_diffusion	1.15471 → 1.15462	+13.8%	-16.7%	-10.9%	-1.25%
mol_ir_bands	1.15626 → 1.15618	+37.2%	+42.6%	+150.7%	-1.30%
mol_sec_ion	1.14036 → 1.14007	+38.7%	-6.1%	-12.3%	+3.17%

(mol_lyman_werner). mol_ir_bands and mol_metals carry the opposite sign at their extreme cell; both are the same hot-Uranus gate at 12000 fixed steps with a base that is still breathing, and neither case is converged.

Where the channel actually matters, measured with the diagnostic build. On mol_sec_ion, $P_{\text{H}_{2,\text{di}}}/P_{\text{H}_2}$ is 4.67% in the shielded base and rises to 6.5% at the top of the molecular layer as the surviving spectrum hardens (8.5% on the He/H = 0.0793 run) — the 4.5% order the secondary side of Dalgarno et al. predicts, arrived at independently through the photon branching. The proton source ratio $P_{\text{H}_{2,\text{di}}}n_{\text{H}_2}/(P_{\text{HI}}n_{\text{HI}})$ is 20.5 at $1.004R_p$, 3.1 at 1.04, and falls through unity at $1.064R_p$; on the converged He/H = 0.0793 run the same ratio is 38.9 at $1.004R_p$ and crosses unity at 1.036. Inside the molecular layer this channel, not the H I photoionization, is what the protons come from.

Table 5: The He/H = 0.0793 hot Uranus of Sec. 124., re-converged from its own restart; JFNK info=0 in every arm.

	$\log_{10} \dot{M}$	H ₂ front [R_p]	max $n(\text{H}_3^+)$	base $n(\text{H}^+)$
before	10.5984	1.05388	3.703×10^5 @ 1.0234	6.471×10^3
P31, Chung Table II	10.5991	1.05388	3.684×10^5 @ 1.0193	7.720×10^3
P31, the Yan-quoted ramp (rejected)	10.5891	1.05780	3.597×10^5 @ 1.0190	7.735×10^3

The escaping flux does not care (+0.16%), the H₂ front does not move, the H₃⁺ peak loses 0.5% and shifts inward by $0.004R_p$, and **the base proton density rises by 19%**. The third row is worth keeping because it measures what the mis-read branching would have cost: a 2.1% error in $\dot{M}$ and a spurious $0.004R_p$ outward shift of the H₂ front, both from a curve that is $1.28\times$ too high in the band that heats the wind.

130.6 What was checked, and what was not

1. **Byte-identity of the atomic path**, against the tree's own pre-change output: wasp_full, wasp_he23off, lower_profile, both files. This is the guard that n_abs_fixed going $5 \rightarrow 6$ did not shift the metal absorbers in the secondary-ionization energy grid, which it would have done silently.
2. **The six molecular cases**, table above.
3. **Runtime checks**. An `-01 -g -fcheck=bounds,do,mem` build run on mol_sec_ion for 1200 steps: clean, no trap. Run in a scratch copy of the tree rather than through run_fcheck.sh, which begins with a `make distclean`

in the repository and would have collided with the other work in flight; and that script's own case is HD 209458 b, which is atomic and would not have entered this code at all.

4. **Thread determinism.** `mol_sec_ion` at `OMP_NUM_THREADS=8` reproduces the single-threaded run bit-for-bit, which is what the three new `!$OMP PRIVATE` entries are for.
5. **The transcription of Chung et al.'s Table II**, by their own two redundant columns (Sec. 130.3).
6. **NOT checked: the `_adv` post-process.** `post_process_adv` calls `PH_heat_HHe` without the new optional argument and does not re-solve the molecular network, so the channel reaches the `*_adv` profiles only through the equilibrium state they are built on. Unchanged in kind from every other molecular channel; not measured.
7. **NOT checked: any planet run outside the matrix.** The paper and poster figures predate this change, along with Secs. 128. and 129..

Files: `src/modules/functions/cross_sec.f90`, `src/modules/init/parameters.f90`, `src/modules/init/set_energy_vectors`, `src/modules/radiation/electron_energy_degradation.f90`, `src/modules/radiation/util_ion_eq.f90`, `src/modules/radiation/ionization_equilibrium.f90`, `src/modules/nonlinear_system_solver/ion_cell_state.f90`, `src/modules/nonlinear_system_solver/System_HeH_mol.f90`, `src/modules/nonlinear_system_solver/System_HeH_mol_m`, `src/modules/nonlinear_system_solver/constrained_chemical_equilibrium.f90`, `src/modules/lower_atmosphere/diffus`
Documentation: `T0_BE_DONE.md` items (P31), (P31a), (P31b), (P31c); `docs/molecular_hydrogen_treatment.tex`.

One correction made on the way, unrelated to the physics. The `params` layout (1-18 . . .) comment block at the head of `System_HeH_mol.f90` documented a numbered argument array the module has not read since `ion_cell_state` replaced it: `ion_system_HeH_mol's params(40)` is the MINPACK callback signature and no element of it is touched. The block now says where the cell state comes from. Comment only; the binary is unchanged.

131. What the below-escape rows are measured against, a “PLM+WENO3” that could not reconstruct, and a residual gate that switched off every other stop (2026-09-02)

Judgment first, in three parts.

(1) **The layer's energy and mass rows are measured against the wrong thing, and neither of the two proposed replacements survives measurement.** The claim this item was opened on — that $|u(3, j)|$ collapses in the quasi-hydrostatic layer the way $|\rho v|$ does — is **refuted**: $u(3, j)$ is the energy *density*, not an energy flux, and it collapses nowhere (in the layer it is 0.16 of $|\text{heat}| + |\text{cool}|$ on WASP-121b and 69 times it on HD 189733 b). What is nevertheless wrong is real: $|R|/|u|$ is a fractional drift *rate* in units of the code's global time t_s , and near the base the cell's own timescale is three to four orders shorter than t_s , so a layer that is steady to one part in 10^7 of its own terms reads 10^{-3} on that rate. Measured on the converged `jfnk_hd189` state, the layer holds $\|R\|$ at 9.915×10^{-4} while the wind is at 1.836×10^{-6} — a factor 540 — and 806 of the solve's outer iterations go into the layer alone. **Candidate (a)**, $|\text{heat}| + |\text{cool}|$, is **disqualified**: on the hot Uranus the radiative terms are not the dominant balance (layer $|\text{heat}| + |\text{cool}| = 9.4 \times 10^{-5}$

against $|dF_3| = 8.7 \times 10^{-5}$), and converged states read 2.3×10^{-2} to 5.4×10^{-1} on it. **Candidate (b)**, $|p \operatorname{div} v| + |\rho v g|$, is **disqualified**: both terms carry v and collapse in the layer exactly as $|\rho v|$ does; converged states read 1.8×10^{-1} to 1.9 on it. **The one scale that survives** is the signal-speed transport bound $|u(k, j)|(|v_j| + c_{s,j})/dr_j$, the same construction `cell_state_scales` uses for the momentum unknown and `carrier_row_scale` for the carriers; on it every converged state above reads 7×10^{-9} to 4×10^{-6} in both regions and both rows. **It is not applied, because applying it is a re-calibration of a user-facing tolerance and, measured, it costs section 127..** With the layer's mass and energy rows on that bound, `jfnk_hd189_tight` reaches `info = 0` easily (its binding row becomes the layer momentum row at 1.8×10^{-7}) — and the two hot-Uranus states of section 127. swap places again: the wind rows become the maximum, and the state whose inner flux spread is 7.6×10^{-2} scores 4.8×10^{-4} against 7.0×10^{-4} for the one whose spread is 3.0×10^{-3} . The discrimination itself is scale-independent (the bad state is 5.7 times worse than the good one in the layer energy row on *every* candidate); what the re-scaling changes is whether the layer is ever the larger half. **So the two goals cannot both be met by one max-combined residual with one threshold, and the resolution is a design decision, proposed in section 131.4 and not taken here.**

(2) **“PLM+WENO3” with an empty PLM stage crashed on the first step. Fixed.** The parser left `rec_method` at the input word `'PLM+WENO3'`, which is the name of a two-stage recipe and not of a reconstruction; when the stage-1 threshold did not exceed the stage-2 one, `EXHALE_main`'s `in_plm_stage` was false, the string was never normalized, and the first Reconstruct fell through to case `default` and aborted with unknown reconstruction scheme: `PLM+WENO3`. `rec_method` is now always `'PLM'` or `'WENO3'`; the two-stage intent travels in `recon_two_stage` alone; and an empty PLM stage resolves to single-stage WENO3 with a message saying so. Three regression cases were affected — `wasp_hybrid_finish`, `crit_warm` and `wasp_localdt_cont`, all with `du_th [PLM,WENO3]: -1.0 ...` — and all three now run.

(3) **A Resid tol gate switched off every other stop, including the stall net. Fixed.** With `resid_th > 0` the marching loop set `is_stalled = .false.` unconditionally, so a run that could not reach its residual gate marched to `count_max = 1e6`. Measured: `jfnk_cold` reaches $du = 1.5 \times 10^{-4}$ and `ptc_warm` $du = 5.5 \times 10^{-5}$ — flux-converged by any ordinary standard — and neither crossed its gate in 40,000 steps. A residual gate is one more stop condition, not a licence to remove the others; the plateau detector now stays live under `Resid tol`, and the stop it produces is reported as what it is, a stall with the residual target unmet.

131.1 What the layer rows actually measure

$R(3, j) = dF(3, j) - (\text{heat}_j - \text{cool}_j)$ — the source $S(3, :)$ is identically zero, the gravity work being folded into the energy flux (`RK_rhs.f90`) — so the energy row has exactly two terms. Volume-weighted sums over the layer, in code units, on four states (`EXHALE_RESIDUAL=1` on the converged output, WENO3 except HD 189733 b which is re-read in the PLM it was converged in):

layer sums	WASP-121b	HD 189733 b	hot U, two-stage	hot U, single-stage
$ dF_3 $	1.404×10^{-1}	1.964×10^{-4}	8.720×10^{-5}	1.349×10^{-4}
$ \text{heat} $	1.054	1.083×10^{-3}	8.141×10^{-5}	9.384×10^{-5}
$ \text{cool} $	9.138×10^{-1}	1.522×10^{-3}	1.302×10^{-5}	7.488×10^{-5}
$ \text{heat} - \text{cool} $	1.399×10^{-1}	5.310×10^{-4}	8.416×10^{-5}	1.577×10^{-4}
$ R_3 $	1.161×10^{-3}	5.664×10^{-4}	1.015×10^{-5}	9.130×10^{-5}

Which term dominates is planet-dependent. On WASP-121b the radiative pair is an order above the flux divergence; on the hot Uranus it is not ($|\text{heat}| + |\text{cool}| = 9.4 \times 10^{-5}$ against $|dF_3| = 8.7 \times 10^{-5}$). That is why $|\text{heat}| + |\text{cool}|$ cannot serve as the scale: it is not a bound on the row's terms where the layer is cold and shielded. The ratio $(|\text{heat}| + |\text{cool}|)/(E(|v| + c_s)/dr)$ over the layer has median 5.1×10^{-3} (WASP), 7.1×10^{-5} (HD 189733 b), 1.1×10^{-4} and 3.4×10^{-6} (hot Uranus) and never exceeds 0.15, so the radiative pair never dominates the signal-speed transport bound either.

131.2 The candidates, side by side

Volume-weighted, per region, on the states the solver itself holds (printed by the JFNK at the end of the solve; the hot-Uranus single-stage row is the EXHALE_RESIDUAL=1 re-read of its section 127. output).

state / region	mass $/ u $	mass $/\rho \text{ dyn}$	energy $/ u $	energy $/(h+c)$	energy $/(h+c+E \text{ dyn})$
jfnk_hd189 layer	7.183×10^{-4}	9.07×10^{-8}	9.915×10^{-4}	9.736×10^{-3}	1.344×10^{-7}
jfnk_hd189 wind	3.907×10^{-6}	1.62×10^{-8}	1.836×10^{-6}	4.760×10^{-6}	6.772×10^{-9}
wasp_full_newton layer	6.766×10^{-7}	1.516×10^{-10}	9.886×10^{-4}	2.541×10^{-5}	2.534×10^{-7}
wasp_full_newton wind	1.501×10^{-8}	8.681×10^{-12}	7.666×10^{-4}	1.651×10^{-5}	4.321×10^{-7}
hot U two-stage layer	1.779×10^{-4}	7.087×10^{-8}	1.761×10^{-4}	2.315×10^{-2}	6.629×10^{-8}
hot U two-stage wind	1.362×10^{-4}	8.894×10^{-7}	5.642×10^{-4}	7.937×10^{-4}	3.772×10^{-6}
hot U single-stage layer	1.064×10^{-3}	3.871×10^{-7}	4.001×10^{-3}	5.411×10^{-1}	1.464×10^{-6}
hot U single-stage wind	4.757×10^{-4}	3.116×10^{-6}	1.264×10^{-4}	1.522×10^{-4}	8.582×10^{-7}

Read down the columns: $/(h+c)$ ranges over four decades and is worst on the states that are converged, so it is not a criterion. $/|u|$ is what the code uses and is 10^{-3} to 10^{-4} on every converged layer, i.e. it saturates the default tolerance by construction. $/(h+c+E \text{ dyn})$ is 7×10^{-9} to 4×10^{-6} everywhere, on both planets and in both regions, which is what a scale that bounds its own row looks like.

131.3 Why this cannot simply be adopted

jfnk_hd189_tight (Resid tol: $1.0\text{e-}4$) is blocked by the layer, and by both rows, not only the energy one: at convergence of the 10^{-3} run the layer reads mass 7.183×10^{-4} , momentum/ ρg 1.845×10^{-7} and energy 9.915×10^{-4} , so re-scaling the energy row alone leaves the mass row at 7.2×10^{-4} and the 10^{-4} target still out of reach. Re-scaling both makes the layer's binding row the momentum one at 1.8×10^{-7} , and the target is met with room to spare.

But the same re-scaling removes what section 127. was written for. The discrimination between the two hot-Uranus states is present on every candidate — the single-stage state is 5.7 times worse than the two-stage one in the layer energy row and 2.4 times worse in the layer mass row, on $|u|$ and on the signal-speed bound alike — yet $\|R\|$ is a maximum, so what decides the verdict is which half is larger:

policy	hot U two-stage	hot U single-stage	ordering
section 127. as it stands	6.97×10^{-4}	4.00×10^{-3}	correct
energy row re-scaled in the layer only	6.97×10^{-4}	1.06×10^{-3}	correct
mass and energy re-scaled in the layer	6.97×10^{-4}	4.76×10^{-4}	inverted
every row re-scaled, both regions	4.68×10^{-6}	3.12×10^{-6}	inverted

131.4 The proposal, not implemented

Separate the two statements the one number is being asked to make.

1. **Steadiness of the equations** $\rightarrow \|R\|$ on the signal-speed bound in both regions and all three rows, with the layer’s momentum row keeping the gravitational force density of section 62.3 (it is that row’s dominant-term bound). Every row is then divided by a bound on its own largest term, the number is dimensionless everywhere, and the two regions become commensurable — which would also retire the “not addends of a common quantity” caveat that section 127. had to make for the maximum. `Resid tol` would need a new default: converged states measure 7×10^{-9} to 4×10^{-6} , so 10^{-5} is the natural equivalent of today’s 10^{-3} , and every existing `Resid tol` in the repository changes meaning.
2. **Constancy of the mass flux** $\rightarrow$ a separate gate on the radial spread of $\rho v r^2$ over the *whole* column rather than over $[j_{\min} : N]$, which is the quantity that actually separates the two hot-Uranus states (3.0×10^{-3} against 7.6×10^{-2} , a factor 25, against 1.9 to 5.7 in any residual norm) and which item (P42) named in the first place.

That is a change to the meaning of a user input, so it is proposed and not taken.

131.5 What was checked

- **The nine regression cases are byte-identical to what the tree produced before this change.** The comparison is against a snapshot of the tree’s own current outputs (`treeC`), not against the `goldens`, because the `goldens` predate sections 128–132 and every case failed them until the refresh below. All nine match `treeC` bit for bit in both `Hydro_ioniz.txt` and `Ion_species.txt`, which is what the two fixes predict: none of the nine carries a `Resid tol`: key, and the three cases the reconstruction crash hit are not in the matrix.
- **The three cases that could not start now run**, each printing the new “the PLM stage is empty $\rightarrow$ single-stage WENO3” line. **Corrected 2026-09-03 (Sec. 140.6): the rates quoted here were 30-step probes, written as though they were the cases’ results.** Run to their own stopping criteria, `wasp_hybrid_finish` and `crit_warm` — the same case under two names — give $\log_{10} \dot{M} = 13.23$ ($du = 9.9997 \times 10^{-4} < du_{\text{th}}$ at step 22 935) and `wasp_localdt_cont` gives 13.36 at a 40 000-step cap ($du = 4.97 \times 10^{-5}$), so not a converged rate. The probe values were 13.40, 13.40 and 13.24.
- **The A_{10} provenance comment is comment-only:** the -03 object of `h2_vibrational_relaxation.f90` has the same md5 before and after.

- NOT checked: whether the README_HOWTO.md molecular prescription (Solver: Newton 5.0e-2 + Resid tol: 1.0e-5) is reachable. That question depends on the decision of section 131.4; under the measure as it stands the answer is set by the layer’s energy row, which saturates 10^{-3} on every converged state measured here, so a 10^{-5} target is not reachable through it — and the prescription’s own text (“ $\|R\|$ is set by the two or three cells just above the base”) is describing exactly that saturation. The 150,000-step run was started and stopped at step 27,746 ($du = 0.28$) when the tree regression needed the cores.
- NOT checked: the Resid tol cases themselves (jfnk_*, ptc_*, tpm_*, resid_on) are not in the nine-case matrix and were not re-run after the stall-net change; by construction they can now stop earlier than count_max, which is the intent.

131.6 The block-end golden refresh

The goldens carried the state of the tree before sections 128–132 and every one of the nine cases failed against them. They are refreshed here, at the end of the block; the previous set is preserved as backup/regression/golden_blockE_20260903. **What the refresh carries, and what moved.** The movement is measured as the largest relative change of each field between the old and the new golden, over the physical cells. Attribution is by *scope* — which cases a section’s physics can reach — and not by an A/B of each section separately, except for the one row that was measured that way.

section	what it changes	cases it can reach	measured
128	molecular carriers without the oxygen chemistry	the seven molecular / profile cases	scope
129	metals compete for the He recombination photons	metals-on: wasp_full, wasp_he23off, mol_metals, lower_profile	scope
130	dissociative photoionization of H ₂	the seven molecular / profile cases	scope
131	this section	none	A/B: all nine byte-identical
132	level-resolved Lyman–Werner rate, computed E_{bound}	mol_lyman_werner	scope

case	$\log_{10} \dot{M}$ old $\rightarrow$ new	$\max \Delta\rho/\rho $	$\max \Delta T/T $	at $r [R_p]$
wasp_full	13.26 $\rightarrow$ 13.26	5.12×10^{-4}	2.04×10^{-4}	1.246
wasp_he23off	13.26 $\rightarrow$ 13.26	6.49×10^{-4}	2.77×10^{-4}	1.244
mol_base_handoff	10.56 $\rightarrow$ 10.56	1.26×10^{-2}	9.36×10^{-3}	2.003
mol_metals	10.57 $\rightarrow$ 10.57	2.53×10^{-2}	2.58×10^{-2}	1.193
mol_lyman_werner	10.55 $\rightarrow$ 10.55	2.61×10^{-1}	1.69×10^{-1}	4.423
mol_diffusion	10.56 $\rightarrow$ 10.56	1.27×10^{-2}	9.37×10^{-3}	2.003
mol_ir_bands	10.57 $\rightarrow$ 10.57	1.08×10^{-2}	9.88×10^{-3}	1.193
mol_sec_ion	10.51 $\rightarrow$ 10.51	2.38×10^{-2}	1.75×10^{-2}	1.193
lower_profile	9.56 $\rightarrow$ 9.56	1.60×10^{-3}	1.54×10^{-3}	2.797

No rate moves at the printed digits, and the two converged WASP-121b cases move by $5\text{--}6 \times 10^{-4}$ in density. The molecular rows move by 1 to 26%, and per item (P17) those are 12,000-step relaxation snapshots whose magnitudes are transient amplification, not physical change: the largest velocity ratios in the same comparison run to 7×10^3 at near-zero-velocity cells. Read the molecular rows as “this case moved”, not as a size.

The parse corpus is re-snapshotted with it. `write_setup_report` dumps `rec_method`, and the parser now resolves it to PLM or WENO3 where it used to echo the input word PLM+WENO3, so `backup/regression/run_parse_corpus.sh` reported a diff for every two-stage case. The corpus was re-snapshotted (`run_parse_corpus.sh` golden, then check: **1092 cases, 0 diff, PARSE CORPUS CHECK OK**); the previous set is preserved as `backup/regression/parse_golden_blockE_20` with its `parse_skip_blockE_20260903.txt`. **What the refresh carries is mostly not this section:** of the 1076 changed dumps, 956 change only in variables this section does not touch (`R0`, `Mp`, `Mstar`, `a_orb`, `t_s`, `q0`, `b0`, `atilde`, `Mrapp`, `R_star`, `r_max`, and the new `carrier_transport / oxygen_transport` keys of section 128), and 120 additionally carry `rec_method`. An A/B against a binary with only this section’s parser change reverted isolates the contribution exactly: `rec_method` goes `PLM+WENO3  $\rightarrow$  PLM` on the 120 two-stage cases, and on the three empty-PLM-stage cases (`wasp_hybrid_finish`, `crit_warm`, `wasp_localdt_cont`) it goes to WENO3 with `use_plm/use_weno3` flipped with it. Nothing else in any dump is this section’s. Sixteen dumps are new cases that post-date the previous snapshot, and none was lost.

One thing the re-snapshot exposes and does not fix: the corpus skip list grows from 2 to 33. The 31 added cases (`backup/regression/armD_D1` and 30 `vulcan_work/` directories) all stop in the parser with `ERROR: a molecular base particle count with an atomic species state` — the section 117 validation, reached because their `input.inp` carries a molecular base with `Molecular chemistry` off. That is a property of those stored inputs, not of this change, and they are recorded as skipped rather than repaired here.

Verification of the refresh. `run_check.sh` golden, then `run_check.sh` check: **9/9 PASS, byte-identical**. Then `backup/regression/run_fcheck.sh` (`a -fcheck=bounds,do,mem` rebuild from scratch and a 1500-step bounded HD209458 b run): **CLEAN**, $\log_{10} \dot{M} = 9.60$. `run_fcheck.sh` restores the production build, and `run_check.sh` check on that restored binary is **9/9 PASS** again.

Files: `src/modules/files_I0/input_read.f90`, `src/EXHALE_main.f90`, `src/modules/lower_atmosphere/h2_vibrational_rel` (comment only).

132. The Lyman–Werner rate is now the level-resolved cross section itself, one pump has two branching ratios, and the 2.0 eV of the heat return is computed instead of adopted (2026-09-02)

Superseded in part by Sec. 135. (2026-09-03). The tabulated rate was CLOUDY’s H₂g sub-reservoir and line overlap was absent; both are fixed there. The absolute cross section, the two branching ratios and the computed heat return all stand; the table they are read from was rebuilt.

Judgment first. The two Lyman–Werner band constants did not disagree with the CLOUDY calculation of Sec. 122. for the reason that note recorded. The line data agree: rebuilt from CLOUDY’s own Abgrall, Roueff & Drira (2000) files, the band-averaged dissociation cross section for a thermal H₂ population at 100 K is $3.479 \times 10^{-18} \text{ cm}^2$ against the $3.4520 \times 10^{-18} \text{ cm}^2$ the module took from Draine & Bertoldi (1996) — 0.8 per cent — and the pump cross section agrees to 0.02 per cent. The factor 1.15–1.37 between them is LINE TRAPPING of the fluorescent decay photons, which CLOUDY’s converged pass carries and DB96’s unshielded tables do not. The defect was the mixture: an untrapped constant multiplied by a shielding factor whose own normalization is trapped. The rate now comes from the absolute cross section, tabulated. Item (P39) is closed here.

132.1 What the measurement was

CLOUDY does not write the untrapped branching out — its Solomon rate carries the escape probability inside $D/(D + \sum A_{ul}P_{\text{loss}})$. A four-line diagnostic was added to a scratch build of c25.00 to print, at full precision, the fresh pump rate of the H₂g population, the dissociation rate that pump gives with $D/(D + \sum A_{ul})$, the rate it gives with CLOUDY’s own branching, and the band field. The patch is in `src/utls/h2_shielding_cloudy_decks/` and it is inert: the 27 decks re-run with it reproduce the unpatched f_{shield} curve with a maximum relative difference of exactly zero, and the table built from them is byte-identical to the one Sec. 122. installed.

With it the attribution is a measurement. At the illuminated face of the 1300 K run the single-pump branching is 0.1466 *on both passes* — there is no escape probability in it, so it cannot move — while the effective one goes $0.1466 \rightarrow 0.1771$ between the first pass and the converged one. At fixed $T = 1300 \text{ K}$ the converged face cross section moves 13 per cent between $n_{\text{H}} = 10^{12}$ and 10^{14} cm^{-3} while the untrapped value is identical to five digits, and atomic data cannot depend on density.

132.2 One absorption, three counts

σ_{diss}	dissociations per unit INCIDENT band photon fluence [cm^2]; self-shielding and trapping both inside it — the RATE.
p_{single}	dissociations per PUMP with every fluorescent photon free to escape, $D/(D + \sum A_{ul})$ — the HEAT RETURN.
p_{eff}	dissociations per photon REMOVED FROM THE BEAM, i.e. per fresh pump, $D/(D + \sum A_{ul}P_{\text{loss}})$ — the PHOTON LEDGER.

p_{single} and not p_{eff} is what the fluorescence heat needs, because the trapping cancels out of that count exactly. Per fresh pump the chain ends in dissociation with probability p_{eff} and contains $(D+A)/(D+AP_{\text{loss}})$ pump events, so

$$\text{pumps per dissociation} = \frac{D+A}{D} = \frac{1}{p_{\text{single}}}, \quad (45)$$

independent of P_{loss} , and the decays per dissociation are $(1-p_{\text{single}})/p_{\text{single}}$. Physically, a trapped decay and the re-absorption that follows it move one molecule down and another up and deposit nothing. CLOUDY's own bookkeeping says the same: the return flux into $X(v'', J'')$ is $\text{Pop}_{\text{up}} A_{ul} P_{\text{loss}}$ and the excited-state population is $n_{\text{in}}/(D + \sum A_{ul} P_{\text{loss}})$, so the arrivals in vibrationally excited X per dissociation are A_+/D , with no escape probability in them. The beam photon count is the other one: only fresh pumps take photons out of the stellar beam, so `write_output's ph_lw` divides by p_{eff} . The two differ by 20–40 per cent through the layer.

132.3 What the table carries now

`h2_self_shielding_table.f90` holds three cubes on the same $39 \times 9 \times 3$ grid in $(\log N_{\text{H}_2}, \log T, \log n_{\text{H}})$ — $\log_{10} \sigma_{\text{diss}}$, $\log_{10} p_{\text{single}}$, $\log_{10} p_{\text{eff}}$ — generated by the same script from the same 27 decks. `h2_self_shielding_level_resolved` survives as the ratio $\sigma_{\text{diss}}(N)/\sigma_{\text{diss}}(N_{\text{min}})$, a diagnostic, which is what `output/Lyman_Werner.txt` reports. At $n_{\text{H}} = 10^{13} \text{ cm}^{-3}$:

$N(\text{H}_2) [\text{cm}^{-2}]$	$\sigma_{\text{diss}} (1300 \text{ K})$	p_{single}	p_{eff}
10^{12}	4.630×10^{-18}	0.147	0.177
10^{17}	3.806×10^{-20}	0.225	0.293
4.3×10^{18}	1.060×10^{-21}	0.175	0.238
4.8×10^{20}	9.844×10^{-23}	0.147	0.220
5.0×10^{21}	3.105×10^{-23}	0.151	0.215

132.4 The 2.0 eV of the heat return is now computed

Burton, Hollenbach & Tielens (1990) Appendix A read: “the effective heating per pump, E_* , has been taken as 2.0 eV”, an adopted value with no derivation, and their Eq. (A1) multiplies it by the collisional de-excitation rate of a UV-excited pseudolevel, not by the pump rate and not by the dissociation rate. Black & van Dishoeck (1987), pp. 412–413, supply only the branching — “fluorescence to the vibrational continuum of the ground state with a typical average probability of 0.10 ... The remaining 90% of the fluorescent transitions populate various bound excited vibration-rotation levels of the ground state” — not a mean energy.

The same quantity is computable. Summing the Abgrall, Roueff & Drira (2000) transition probabilities over every decay of every in-band pump, weighted by the $X(v'', J'')$ energy the decay lands on, gives the mean landing energy per BOUND fluorescence:

$T [\text{K}]$	700	1300	1800	2700	3200
$E_{\text{bound}} [\text{eV}]$	2.060	2.085	2.102	2.121	2.129
per pump [eV]	1.769	1.779	1.786	1.795	1.799
against 2.0 eV	+3.0%	+4.3%	+5.1%	+6.1%	+6.4%

`h2_energy_per_bound_fluorescence_eV(T)` replaces the constant, a quadratic reproducing the nine points to 0.03 per cent and clamped at 700–3200 K. The form stored is per BOUND fluorescence, not per pump, because it then belongs to the cascade alone and not to the branching of whatever excited the molecule: the Lyman–Werner path multiplies by $(1 - p_{\text{single}})$, the photoelectron path by its own state-resolved $(1 - p_{\text{diss},B})$, $(1 - p_{\text{diss},C})$ (0.27, 0.036). The sixth of the bound fluorescences that land back on $v'' = 0$ and leave only rotational energy is already inside these means and must not be discounted a second time.

132.5 Three constants removed

With the rate and both branchings on the table, `sigma_lw`, `p_diss_lw` and `sigma_lw_pump` entered no number the code produced. They are gone from `lyman_werner.f90`; the values (DB96 Table 1–2, $\chi = 1$, the $T_r = 100$ K row: $F = 1.208 \times 10^7 \text{ cm}^{-2} \text{ s}^{-1}$, $\zeta_{\text{pump}} = 3.09 \times 10^{-10} \text{ s}^{-1}$, $\langle p_{\text{diss}} \rangle = 0.135$, hence $\sigma_{\text{diss}} = 3.452 \times 10^{-18}$ and $\sigma_{\text{pump}} = 2.557 \times 10^{-17} \text{ cm}^2$) are stated in the `h2_band_equivalent_width` header as what DB96’s Eq. (39) is normalized to, since that fit is still what computes the band share $A(N)$ the H_2O and OH continua see. The band share itself is unchanged.

132.6 What moved

Nine matrix cases, single-threaded, against the same tree without this change: **seven byte-identical**; `mol_lyman_werner` and `mol_sec_ion` move, and nothing else does. In `mol_lyman_werner` the unattenuated rate rises by 1.274 at the base (1.15–1.37 was the predicted band), the H_2 half-abundance radius moves inward from 1.13116 to 1.12652 R_p , and `heat_H2_LW` integrates 1.16 times higher from the rate and a further 1.040 from E_{bound} ; `heat_total` moves by 3 per cent. `mol_sec_ion` is the only case in which the photoelectron fluorescence channel is live, and there E_{bound} moves `heat_total` by 0.05 per cent integrated.

132.7 Where it stops being a value

- **The trapping inside σ_{diss} and p_{eff} is slab geometry.** CLOUDY computes it from escape probabilities in a plane-parallel slab illuminated on one face and closed on the other. Our layer has the same thick-inward side, which is the side that matters, but it is spherical and open outward, and the size of the enhancement depends on the column *behind* the point — a property of the run, not of H_2 . p_{single} is free of this.
- **E_{bound} is an unshielded average.** With shielded weights the same sum gives 2.17–2.25 eV at $N(\text{H}_2) = 10^{18} - 10^{21} \text{ cm}^{-2}$, +4 to +9 per cent, because self-shielding removes the strongest lines first and the survivors land higher. The face value is used, so the term is a lower bound inside the shielded layer by about as much as the correction it applies.
- **Above `h2_shielding_overlap_column` everything here is an upper bound**, for the line-overlap reason Sec. 122. gives. That item is unchanged.
- Sources and arithmetic: `docs/p39_lw_cross_section_sources.md`.

133. Two gates: the residual on each row's own largest term, and the mass flux on the wind (2026-09-03)

Judgment first. One number was being asked to say two things, and it could say neither well. $\|R\|$ is a statement that nothing is changing; the constancy of $\rho v r^2$ is a statement that the wind carries one mass flux at every altitude. They are not the same statement — the two solves of section 126.6 differ by a factor 25 in the flux spread and by only 1.9 to 5.7 in every residual norm that was tried — and section 131. measured that no single max-combined residual with one threshold can make both. **They are now two gates, and a steady solve is accepted only when BOTH are met.** The residual's row scale becomes, at every cell and in every row, a bound on that row's own largest term: the cell's state scale times the rate at which a signal crosses the cell, $D_k(j)(|v_j| + c_{s,j})/dr_j$. That is the same D_k the line-search merit uses, so the merit and the acceptance test are now the same scaling of the same residual, differing only by each cell's signal-crossing time. `Resid tol` therefore changes meaning by a measured factor 10^{-2} and every value in the repository is renormalized. The result is what the pair was for: the two restarts of section 126.6 now land on the same wind — $\log_{10} \dot{M}$ 10.30 and 10.32 against 10.47 and 10.27 — and `jfnk_hd189`, which used to spend 806 iterations on a layer the measure was mis-scaling, converges on both gates.

This closes item (P42).

133.1 The residual row scale, and what the measurement chose

Section 131. left one candidate standing, $|u(k, j)|(|v| + c_s)/dr_j$, and one question: whether the momentum row's gravitational scale of section 62.3 should be replaced by it or combined with it. The question is settled by measuring which of the two collapses, on the four converged states of section 131., volume-weighted, in each region:

momentum-row candidate	layer, four states	wind, four states
$ R_2 /\rho d\Phi/dr $ (section 62.3)	1.5×10^{-6} – 2.5×10^{-4}	9.0×10^{-7} – 1.3×10^{-2}
$ R_2 / \rho v (v + c_s)/dr$	4.4×10^{-5} – 2.5×10^{-3}	5.2×10^{-10} – 6.8×10^{-5}
$ R_2 /\rho(v + c_s)^2/dr$	1.9×10^{-8}–1.5×10^{-6}	2.5×10^{-10}–1.0×10^{-5}

The literal reading of $|u(2, j)| = |\rho v|$ collapses in the layer exactly as it always did — it is 40 to 120 times smaller than the gravitational scale there — so it cannot be the bound. What does not collapse anywhere is $\rho(|v| + c_s)$, which is `cell_state_scales`' own momentum scale, times the signal-crossing rate. **Measured, it exceeds the gravitational scale at every cell of every state: by a factor 20.8 at the tightest cell and 21 to 3600 in the volume-weighted sums.** So gravity never wins, and the maximum is kept only so that the bound is a bound by construction rather than by measurement on four states.

The row scale is therefore, with no region switch at all,

$$s_k(j) = D_k(j) (|v_j| + c_{s,j})/dr_j, \quad D = (\rho, \rho(|v| + c_s), E)$$

$$s_2(j) = \max \left(s_2(j), \rho_j |G\phi_i(j) - G\phi_i(j-1)|/dr_j \right).$$

D_k now lives in one place, `state_scales_of_cell` in `src/modules/time_step/steady_residual.f90`, and `cell_state_scales` in the solver calls it. The convergence measure is the merit's own scaling of the same residual,

multiplied by the cell's signal-crossing time: the merit is a rate, the acceptance test the dimensionless fraction of the row's largest term.

The two-region maximum of section 127. stays, but its justification changes. It was there because the wind's and the layer's numbers were not commensurable; they are now. It is kept because the outer region carries almost all of $\sum r^2 dr$, so any single volume-weighted norm would let it average the inner column away.

133.2 The flux gate

New key, parsed in `input_read.f90`, defaulted in `parameters.f90`:

```
Flux spread tol: <tol> [<r_flux [R_p]>] default 5.0e-3, 1.2
```

`flux_spread_of_state(steady_residual.f90)` returns $(\max F - \min F)/|\bar{F}|$ with $F = \rho v r^2$ over $[j_{\text{flux}} : N]$, i.e. over $r \geq r_{\text{flux}}$; j_{flux} is set in `define_grid` beside j_{min} and carries the same empty-window guard. `steady_gates_met` in `steady_newton.f90` is the only acceptance test either solver uses, and it is $\|R\| < \text{resid_tol}$ AND `spread` < `flux_spread_th`. A tolerance ≤ 0 disables the flux gate. When a solve ends unaccepted the log names the gate that is left:

```
(JFNK) the residual gate is met; the FLUX gate is the one left: spread 1.874E-02 >=
5.000E-03
```

Why the window is not $[1 : N]$, measured. Mass conservation does make $\rho v r^2$ constant everywhere in a steady state, but the base boundary condition deliberately admits a small inflow through the smooth valve, and just above it the wind is still being launched at $|v| \sim 1\text{--}10$ cm/s, eight orders below the wind speed. On a hot Uranus state that is flat everywhere else, the spread over all cells is 9.9×10^2 and F changes sign, so $\max/\min$ is not even defined; over $r \geq 1.03$ it is 7.5×10^{-2} , over 1.05 4.6×10^{-2} , over 1.10 3.3×10^{-3} and over 1.20 2.5×10^{-3} . Including the launch region would measure the base boundary condition, not the wind.

Why the defaults are 5×10^{-3} and $1.2 R_p$. Over $r \geq 1.2$ every converged state in hand is flat: 2.4×10^{-3} (hot Uranus, two-stage restart), 2.4×10^{-3} (He/H=0.0793), 2.6×10^{-3} (He/H=1), 2.9×10^{-3} (He/H=30), 4.8×10^{-4} (WASP-121b, both Newton cases). The state section 126.6 showed the old measure could not reject reads 7.6×10^{-2} there. A threshold of 5×10^{-3} clears the worst converged state by 1.7 and rejects the bad one by 15. $1.2 R_p$ is the smallest fixed radius at which all of them are flat — at 1.10 the He/H=30 rung reads 2.1×10^{-2} — and it is the inner edge of the region sections 126.6 and 127. identified as the one the old measure could not see.

133.3 The renormalization of Resid tol

The scale changed, so the number does. Measured on the same states, in the solver, before and after:

state	old $\ R\ $ (sec. 131 scale)	new $\ R\ $	ratio
jfnk_hd189 (JFNK solution)	9.915×10^{-4}	9.340×10^{-7}	1060
wasp_full_newton	9.886×10^{-4}	9.337×10^{-6}	106
wasp_he23off_newton	—	9.024×10^{-6}	—
hot Uranus, two-stage restart	5.642×10^{-4}	9.092×10^{-6}	62
He/H = 1 reference	3.264×10^{-4}	6.729×10^{-6}	49

The ratio is not one number — it is 50 to 1000, because which row bound the old measure differed by planet. What *is* one number is where converged states now sit: 9.3×10^{-7} to 9.3×10^{-6} , where they used to sit at 4.9×10^{-4} to 9.9×10^{-4} . The old default 10^{-3} stood a factor 1 to 2 above that band; 10^{-5} stands in the same relation to the new one, so every tolerance is multiplied by 10^{-2} :

was	now	where
1.0e-2	1.0e-4	ptc_wasp, resid_on
1.0e-3	1.0e-5	jfnk_cold, jfnk_hd189, ptc_warm, tpm_hd189, tpm_wasp, valve_sens/eps5, examples/04_newton_from_state, and the built-in default in EXHALE_main.f90
5.0e-3	5.0e-5	LHS1140b/examples/scalar_base_kzz1e9
4.0e-4	4.0e-6	LHS1140b/examples/photochem_profile_base, scalar_base_cno
1.0e-4	1.0e-6	jfnk_hd189_tight, valve_sens/eps5
1.0e-5	1.0e-7	examples/15_molecular, and the molecular prescription of README_HOWTO.md and this manual

Fifteen lines in fourteen canonical files, each with a # Resid tol renormalized 2026-09-03, Update_EXHALE.md section 133 (was ...) line above it, plus the two built-in defaults in EXHALE_main.f90, the input_schema.md entry and eleven places in docs/EXHALE_user_manual.tex. **932 further input.inp files under LHS1140b/exhale/ were NOT changed:** they are stored records of past runs, paired with the outputs those tolerances produced, and rewriting them would make each directory inconsistent with its own result. Re-running one of them now asks for a tolerance 100 times looser than intended, which is recorded here rather than repaired silently.

133.4 What was measured

Every steady solve below is the JFNK with both gates; the flux spread quoted is the gate's own, at the accepted iterate.

1. jfnk_hd189 reaches both gates. Resid tol: 1.0e-5 (renormalized), direct PTC/JFNK route: info = 0 at $\|R\| = 9.340 \times 10^{-7}$ and flux spread 4.645×10^{-3} , in 520 outer iterations. Under section 131. the same state took 806 iterations to reach 9.915×10^{-4} and its flux spread was 2.1×10^{-2} — the solve now keeps working until the flux is flat, and the extra work is what flattens it.

2. jfnk_hd189_tight reaches the residual gate and not the flux gate. Resid tol: 1.0e-6: $\|R\|$ falls to 2.830×10^{-7} , a factor 3.5 inside the target, and stops improving at iteration ~ 1500 ; the flux spread ends at 1.874×10^{-2} and the solve returns info = 1 with the log naming the flux gate. **The two gates pull apart at that depth on this planet:** the state

that satisfies the residual to 2.8×10^{-7} is four times *less* flux-flat than the one that satisfies it to 9.3×10^{-7} . That is a property of HD 189733 b's solve, not of the gates, and it is the first thing the split makes visible.

3. The two restarts of section 126.6 land on the same wind.

	two-stage restart	single-stage restart
info	0	0
$\ R\ $	9.092×10^{-6}	5.610×10^{-6}
flux spread (gate)	1.535×10^{-3}	2.825×10^{-3}
$\log_{10} \dot{M}$	10.32	10.30
L_2 difference of (ρ, v, p)	4.8%	

Against section 127.'s 10.27 and 10.43 (and section 126.6's 10.34 and 10.56) with an L_2 difference of 36%. The single-stage path was expected to be *rejected* by the flux gate; what it does instead is converge, because the gate keeps the solver working on the region the residual alone was satisfied with. That is a better outcome than the one the split was designed for.

4. The atomic Newton cases. `wasp_full_newton`: info = 0, $\|R\| = 9.337 \times 10^{-6}$, flux spread 1.858×10^{-3} , $\log_{10} \dot{M} = 13.21$ (unchanged). `wasp_he23off_newton`: info = 0, $\|R\| = 9.024 \times 10^{-6}$, flux spread 7.457×10^{-4} , $\log_{10} \dot{M} = 13.21$ against 13.22 — one in the last printed digit.

5. The He/H = 1 reference. info = 0, $\|R\| = 6.729 \times 10^{-6}$, flux spread 3.543×10^{-3} , $\log_{10} \dot{M} = 10.31$ against 10.32.

6. make check, nine cases, single-threaded: 9/9 byte-identical. None of the nine enters the JFNK — `wasp_full` and `wasp_he23off` stop on du and the seven molecular / profile cases are fixed-step snapshots — which is the point: everything here is inside the steady solver's acceptance.

133.5 What was not checked

- Whether `examples/15_molecular` reaches its renormalized `Resid tol: 1.0e-7`. Converged states measure 9.3×10^{-7} to 9.3×10^{-6} , so 10^{-7} is below the band; the old 10^{-5} was equally far below the old band and was reported as reachable for that configuration (section 61.), so the renormalization is consistent, but it is not re-measured here. It is a 150,000-step run.
- The `Resid tol` cases other than `jfnk_hd189*` (`jfnk_cold`, `ptc_*`, `tpm_*`, `resid_on`, `valve_sens`) were renormalized but not re-run.
- The 932 stored `LHS1140b/exhale/` inputs, as above.

133.6 The state the gates accept is the state that is written; the file's ghost rows are what differed

Section 133.5 recorded that the flux spread the solve accepted (4.6×10^{-3} on HD 189733 b) did not match the spread computed from the written `Hydro_ioniz.txt` (1.0×10^{-2}), and left it open. **The cause is not a state change. It is that the output file carries ghost cells and the offline measurement counted them.**

Nothing between the solve's return and the write touches `u`. Traced, file:line:

step	what it does
EXHALE_main.f90:1593 solve_steady_jfnk (or :1596 solve_steady_ptc)	return
:1599 U_to_W(u,W); :1600 rho,v,p,E re-read	reads
:1601 get_species_densities, :1603 comp_T_from_p, :1604 excited_H_update, :1605 ioniz_eq, :1606 get_species_densities	comp
:1608/:1609 exit (no diffusion, no carriers)	—
:1169–:1171 is_mom_const, newton_finished	—
rest of the marching-loop body to :1260	only t
:1262–:1449 (contact_mode_dissipation_magnitude, excited_H_update, ...)	read u
:1445 write_output(rho,v,p,...)	writes

There is no hydro step after the Newton finish — the RK stages are at :592–:760, before it — so $\rho v r^2$ cannot move, which is what the physics says too.

What did differ. write_output.f90:101 writes do $j = 1-N_g$, $N+N_g$: the file has $N+4$ rows with $N_g = 2$ **ghost cells at each end**. The upper two are filled by Apply_BC from a zero-gradient / WENO3 extrapolation, so their $\rho v r^2$ is a boundary value, not part of the solution, and an offline $r \geq 1.2$ selection over the raw file picks them up. Dropping them reproduces the gate exactly:

state	all file rows, $r \geq 1.2$	physical cells	the gate's value
jfnk_hd189	1.0486×10^{-2}	4.6447×10^{-3}	4.6447×10^{-3}
wasp_full_newton	2.3148×10^{-3}	1.8581×10^{-3}	1.8581×10^{-3}
He/H = 1 reference	3.5434×10^{-3}	3.5434×10^{-3}	3.5434×10^{-3}

The factor is state-dependent — 2.3 on HD 189733 b, 1.25 on WASP-121b, 1.00 on the hot Uranus — which is why it looked like a state change on the one case it was first noticed on. examples/exhale_io.py has no ghost handling either, so every Python consumer of these files has the same trap.

Both halves are closed, in the two places they belong.

The file now says which rows are physical. write_output emits a third # header line,

```
# rows 504: 2 ghost cells at each end; physical cells are rows 3 to 502
```

It is a comment, so no numeric parse and no golden changes (run_check.sh compares with `grep -v '^ *#'`).

The run now asserts that the written state is the accepted one. assert_written_state_is_the_accepted_one (EXHALE_main.f90:1545) is called immediately after both final write_output calls — :475 on the direct steady route and :1450 on the marching route — and re-measures the flux gate on the state as written, printing it beside the value the solve accepted and warning loudly if they differ:

```
(EXHALE_main) flux gate: accepted 4.6447E-03 as written 4.6447E-03
```

The table above is that line on the three states; none of them raised the warning. The check is an assertion rather than a correction: the trace says the two are the same object, and the line is what makes that a measured statement on every run instead of a claim in a document. It costs one pass over the wind window.

What is asserted and what is not. The flux gate is a function of u alone, so the assertion is exact. $\|R\|$ is not: it depends on u *and* on the equilibrium composition, and `ioniz_eq` is called once more at :1605 after the solve returns. That call re-solves the same equilibrium at the same (T, ρ) , so the residual of the written state is the accepted one to the ionization solver's own tolerance rather than bit for bit. It is not re-evaluated here, because doing so costs a full extra residual evaluation and would mutate `f_sp` after the state has been written.

Checked. `make check`, nine cases, single-threaded: 9/9 byte-identical — the header line is a comment and the assertion is output only. The three reference solves above were re-run end to end on this build: `jfnk_hd189 info = 0`, `wasp_full_newton` $\log_{10} \dot{M} = 13.21$, $\text{He/H} = 1$ reference $\log_{10} \dot{M} = 10.31$, all unchanged from section 133.4. **Not changed:** `examples/exhale_io.py` still returns the ghost rows to its callers. Fixing it would change what every notebook and the transit tool see, which is its own change with its own verification.

Files: `src/modules/time_step/steady_residual.f90`, `src/modules/time_step/steady_newton.f90`, `src/modules/init/parameters.f90`, `src/modules/init/define_grid.f90`, `src/modules/files_IO/input_read.f90`, `src/modules/files_IO/write_setup_report.f90`, `src/modules/files_IO/write_output.f90`, `src/EXHALE_main.f90`. Documentation: `docs/input_schema.md`, `README_HOWTO.md`, `docs/EXHALE_user_manual.tex`, `TO_BE_DONE.md` item (P42).

134. The transported H_2 front was being carried by the scheme: second-order limited advection, and a Lyman–Werner flux the run computes for itself (2026-09-03)

Judgment first. The runaway H_2 front of section 128. had two causes and the larger one was ours. First-order donor-cell advection gave the carrier operator a numerical diffusivity 8 to 22 times the molecular one at the front, and it was pushing the front outward; second-order limited advection removes it and buys, on a 500-cell grid, what doubling the grid bought. The missing Lyman–Werner sink is real and is the second cause — it is the *dominant* sink where the front now sits — and it is no longer optional to know it: a run with a numerical spectrum integrates its own band flux. With both fixed the front is grid-converged to 0.2% and nearly stationary. It is *not* a steady state yet, and the carrier transport stays default off until it is.

134.1 What was measured before anything was changed

Two candidates were separated by controlled runs.

The grid. Two refinements were tried and only one of them is a test. Refining the base region (`Base grid [dr, cells]: 1.0e-4 100`) halves the smallest cell, which halves the global CFL step — measured, $\min \Delta r / (|v| + c_s)$ goes from 2.382 s to 1.180 s, a ratio 0.495 — so equal step counts are no longer equal physical times. Doubling the *total* cell count (`Grid cells: 1000`) leaves the base spacing alone: 2.428 s against 2.382 s, ratio 1.019, so step counts *are* comparable. That is the test:

100000 steps, cold start, LW off	$f = 0.5$	10^{-2}	10^{-4}	$n(\text{H}_3^+) > 1$
first order, 500 cells	1.1658	1.2979	1.3967	1.2979
first order, 1000 cells	1.1594	1.2211	1.2689	1.2252
second order, 500 cells	1.1641	1.2269	1.2468	1.2267

The first-order 500-cell front sits $0.15 R_p$ outside the resolved one, and the second-order scheme on the coarse grid lands where the fine grid does. The mechanism is not inferred: donor-cell carries $D_{\text{num}} = |v|\Delta r/2$, and against the molecular diffusivity of H_2 that is

	$r = 1.15$	1.30	1.50
$D_{\text{num}}/D_{\text{H}_2}$, 500 cells	8.5	20	22
$D_{\text{num}}/D_{\text{H}_2}$, 1000 cells	0.74	5.3	7.5

with $K_{zz} = 0$ in these runs, so the molecular value is the whole of the physical transport coefficient.

Lyman–Werner. It was off in every section-128. run, and Frelikh & Murray-Clay carry it always on. Switched on it roughly halves the advance, but on the first-order grid it does not stop it. Its size depends on where the front is, and that is why the first measurement of it was misleading: in the *runaway* states LW is 25–34% of the He^+ channels, because the front has reached low density where He^+ is abundant; in the states where the front holds at $1.22 R_p$ it is the dominant sink by 4 to 12 times.

134.2 Second-order limited advection, by deferred correction

The residual now carries, for $v \geq 0$,

$$\left. \frac{\partial f}{\partial r} \right|_j = \frac{f_j - f_{j-1}}{h} + \frac{\sigma_j - \sigma_{j-1}}{2},$$

and its mirror image for $v < 0$, with σ a limited slope. On a linear profile the two slopes cancel; on a quadratic they cancel the donor-cell truncation term exactly. **With $\sigma = 0$ this is exactly the first-order operator the module had,** on the same non-uniform spacing, so the correction is attributable on its own.

σ is **van Leer**, $2ab/(a+b)$, zero when the one-sided differences disagree in sign. Not minmod, and the reason is about this solver rather than about accuracy in general: minmod has a kink wherever $|a| = |b|$, and a kink in the source of a Newton residual is the defect section 126.6 spent a campaign on, while van Leer is differentiable except at the sign change — and it is the less diffusive of the two.

The correction is deferred: σ is frozen over one Newton solve and refreshed at every transport step. That keeps the Jacobian block-tridiagonal — the correction reaches $j-2$ and $j+2$, which `block_thomas` cannot carry — and keeps the residual smooth inside a solve. At the fixed point of the marching loop, and of the fixed-wind relaxation, consecutive steps agree, so the state the code converges to is the second-order one. `block_thomas` is untouched. Cell 1 keeps the face-flux upwind of section 128. and carries no correction.

Order of accuracy, measured. A Gaussian mixing ratio on the code’s own spacing convention, the discrete derivative against the analytic one in L_1 over the pulse, using the module’s own `carrier_slope`:

N	L_1 first order	L_1 corrected	p first order	p corrected
100	1.070×10^{-1}	1.834×10^{-2}	—	—
200	5.356×10^{-2}	4.786×10^{-3}	0.998	1.938
400	2.679×10^{-2}	1.229×10^{-3}	0.999	1.962
800	1.340×10^{-2}	3.122×10^{-4}	1.000	1.976
1600	6.699×10^{-3}	7.886×10^{-5}	1.000	1.985

The test lives in the campaign scratch directory as `unit/adv_order.f90`; it is written in centimetres because `carrier_slope` floors its spacing at 1 cm, the code's own units, and a test on a unit-length grid measures the floor instead of the scheme.

134.3 The Lyman–Werner band flux is now computed, not remembered

H₂ photodissociation is not an option of the physics: the band exists whenever the star does. What was optional was our knowing the number, and a run with a numerical spectrum knows it. `lyman_werner_band_flux_from_sed` integrates the SED over 912–1110 Å and `input_read` uses it when the run carries a spectrum and states no Stellar LW flux. **A stated key always wins**: it is the more specific statement, and a band-integrated measurement can be better than a trapezoid over whatever grid the file happens to have.

No dilution is applied, and that is not an omission. EXHALE's own SED file is at the planet — `read_sed`'s header says so — while the $(R_*/a)^2$ step of the documented prescription belongs to the stellar-surface files the value used to be produced from by hand. Both routes were checked against each other: Gueymard's solar spectrum integrated over 91.2–111.0 nm at the stellar surface and diluted to 0.048 AU behind a $1.155 R_\odot$ star gives $328.96 \text{ erg cm}^{-2} \text{ s}^{-1}$ against the 343.0 the molecular cases carry, i.e. **the prescription reproduces its own number to 4%**. And the new routine against an independent integral of the LHS 1140 b spectrum: 5.980×10^{-1} from the code against 5.9796×10^{-1} , four significant figures.

With neither a key nor a spectrum the flux stays zero and the setup report says what that means, because a silently absent channel is what the runaway was partly made of.

134.4 The two together, and what is left

Cold start on the hot Uranus, the H₂ front at $f(\text{H}_2) = 10^{-4}$, with the advance per 20000 steps in brackets:

step	second order, LW off	2nd order, LW on, 500	2nd order, LW on, 1000
20000	1.2186 (0.0092)	1.2134 (−0.0141)	1.2157 (−0.0065)
60000	1.2304 (0.0052)	1.2168 (0.0034)	1.2148 (0.0025)
100000	1.2458 (0.0091)	1.2223 (0.0024)	1.2199 (0.0023)
150000	—	1.2300 (0.0034)	1.2275 (0.0033)

The front is grid-converged. At 150000 steps the two grids put it at 1.2300 and $1.2275 R_p$, 0.2% apart, and they drift at the same physical rate, 0.0034 against $0.0033 R_p$ per 20000 steps. A residual motion that does not change when the

grid is halved is not a discretization artefact; it is the state still relaxing, which du agrees with at $3.1\text{--}3.3 \times 10^{-2}$ and falling. LW's own share: at 100000 steps it holds the front $0.024R_p$ further in and cuts the advance rate by 3.8.

The sinks where the front now sits, on the final states [$\text{cm}^{-3} \text{ s}^{-1}$]:

run	r	T	$n(\text{H}_2)$	R17	R20	LW	LW/(R17+R20)
LW on, 500	1.15	1029	1.93×10^{11}	2.62×10^2	2.80×10^1	1.75×10^3	6.0
LW on, 500	1.22	986	5.18×10^8	5.49×10^2	7.46×10^1	5.26×10^3	8.4
LW on, 1000	1.22	1046	3.45×10^8	5.40×10^2	5.27×10^1	7.00×10^3	11.8
LW on, 500	1.30	925	5.38×10^5	7.74×10^0	1.54×10^0	3.72×10^1	4.0

What is not closed, and why the option stays off. The front is nearly stationary at about $1.23R_p$ but it is not stationary: 0.85 to 1.26 cells per 20000 steps on the coarse grid, which is *at* the stopping criterion rather than below it. The wind is not steady either, and the steady solver cannot finish it: restarted from the 150000-step state with `Secondary_ionization: Immediate` (the STAGED default reloads a different state — du jumps from 0.03 to 9), the JFNK stalls at $\|R\| = 2.78 \times 10^{-2}$ and exits `info = 2` with no descent along either the Newton or the damped Gauss–Newton direction, and its worst cell is *the energy row at $r = 1.186$, inside the H_2 front*. That is the slow mode of section 127., and neither change removes it. It is now `TO_BE_DONE.md` item (Y).

134.5 What was checked

- `make check`, nine cases: byte-identical. The carrier transport is off in all of them and no regression case carries a `Spectrum` file, so neither change can reach them — which is why `backup/regression/mol_carrier` now exists.
- `examples/18_oxygen_chemistry`, 2000 steps, the transport *on*: moved, as it must be. The base cell's H_2 partition goes from 0.5634 to 0.3634 of the H nuclei, base H_2O falls by about half, and $\log_{10} \dot{M}$ moves +0.03 dex. Eleven of twelve output files change; `IC_dump.txt` does not.
- The order-of-accuracy test above, and the two independent checks of the band integral.
- NOT checked: the second-order operator on a *converged* oxygen state. The `examples/18` comparison is a 2000-step snapshot from a cold start, so it is a direction and an order of magnitude.

One line that belongs to every profile file, carried in one place. The row-layout header of section 133.6 — `# rows N: Ng ghost cells at each end; physical cells are rows a to b` — was written by `Hydro_ioniz(_adv).txt` alone, which made it a property of one file rather than of the convention, while `Ion_species`, the cooling and heating breakdowns, `Excited_H`, `Lyman_Werner`, `OI_levels`, `Oxygen_chemistry` and `FUV_bands` span the same ghost rows and said nothing. It is now `write_row_layout_header` in `utils`, called by all nine writers and defined once. `residual_profile.txt` and `diffusion_pass_profiles.txt` do *not* get it, because they write `1:N` and the sentence would be false there. The regression strips `#` lines, so this is byte-identical by construction and measured to be.

Files: `src/modules/lower_atmosphere/diffusive_photochemistry.f90`, `src/modules/radiation/sed_read.f90`, `src/modules/files_IO/input_read.f90`, `src/modules/files_IO/write_setup_report.f90`, `src/modules/init/parameters.`

src/modules/functions/utilities.f90, src/modules/files_IO/write_output.f90, src/modules/radiation/util_ion_eq.f90, src/modules/radiation/excited_hydrogen.f90. Regression: backup/regression/mol_carrier. Documentation: docs/molecular_hydrogen_treatment.tex, docs/input_schema.md, TO_BE_DONE.md items (P43) and (Y).

135. The H₂ Lyman–Werner table was a rate for a sub-reservoir of H₂ and it left out line overlap; the two errors had opposite signs and hid each other (2026-09-03)

The judgement first. Two independent errors sat in the H₂ photodissociation rate. The first made it too small, the second too large, they crossed near $N_{\text{H}_2} = 2 \times 10^{20} \text{ cm}^{-2}$, and below that column they very nearly cancelled — which is why neither showed up when the table was built (Sec. 122.) or when it was turned into an absolute cross section (Sec. 132.). Both are now gone: the table is rebuilt from a line-by-line calculation in which every Lyman and Werner line absorbs every other line’s beam, and it is a rate for **all** H₂ rather than for the part of it CLOUDY calls H2g.

135.1 Error one: the tabulated rate belonged to a sub-reservoir

CLOUDY splits H₂ into two chemical species at ENERGY_H2_STAR, set to 4100 cm^{-1} (5899 K) where the H₂ model is constructed (h2.cpp:10, diatomics h2("h2", 4100., ...)): **H2g** is $v = 0, J \leq 8$ — the top of it, (0,8), is at 4051.9 cm^{-1} — and **H2s** is every level above. H2_Solomon_rate accumulates the dissociation of the two separately and divides each by its **own** density (mole_h2_etc.cpp:60-92, Solomon_dissoc_rate_g /= SDIV(H2_den_g)), and the Shield(H2) column that Sec. 122. tabulated is Solomon_dissoc_rate_g/G(TH85) (mole_h2_io.cpp:1580). EXHALE carries one H₂ species and applied that rate to all of it.

The two are not interchangeable at depth. The H2s lines start from levels 100 to 1000 times less populated, are therefore far less saturated, and keep dissociating long after the H2g lines have gone black: Solomon_dissoc_rate_s/Solomon_dissoc_rate_g reaches 165 (1300 K) and 615 (900 K) at $N_{\text{H}_2} = 10^{19}$. Weighting the two by their densities — read from the runs, not assumed — raises the tabulated factor by

$N_{\text{H}_2} [\text{cm}^{-2}]$	900 K	1300 K	1800 K	2700 K
10^{17}	1.24	1.94	2.98	4.16
10^{18}	2.15	3.94	6.16	9.27
10^{19}	2.43	4.15	6.89	11.10
10^{20}	1.60	2.42	3.51	5.34
10^{21}	1.23	1.49	1.78	2.34
4.4×10^{21}	1.12	1.27	1.42	1.73

At the illuminated face the correction is +0.02 per cent (700 K) rising to +12.7 per cent (3200 K), so the 0.5 per cent agreement with Draine & Bertoldi’s Table 1–2 normalization that Sec. 132. rests on — measured at 100 K, where H2s is empty to five digits — is untouched. It is the depth dependence that was wrong.

How it was found. Two calculations that shield one line at a time, the Meudon PDR code with `f_ShieldAlgo = 0` and a line-by-line calculation written for the purpose, agreed with each other to 15–35 per cent at 100 K and at 1300 K and both sat 5–9× **above** the table over $10^{19} \leq N_{\text{H}_2} \leq 10^{21}$. The reservoir split accounts for essentially all of it: with the weighting undone the table reproduces the line-by-line no-overlap branch to 1–4 **per cent up to** $N_{\text{H}_2} = 10^{17}$ and 28–36 per cent at 10^{18} .

135.2 Error two: line overlap, which is the dominant term at our columns

CLOUDY applies the Federman, Glassgold & Kwan (1979) shielding function to each transition using that transition’s own optical depth; nothing couples one line to another. Draine & Bertoldi’s sec. 5.2 say overlap is what produces “the rapid falloff ... for $N_2 > 10^{20} \text{ cm}^{-2}$ ”, and at the columns of a molecular base it is not a correction. Two calculations differing **only** in whether the lines absorb each other’s beam give

$N_{\text{H}_2} [\text{cm}^{-2}]$	900 K	1300 K	1800 K	2700 K
10^{19}	0.73	0.62	0.47	0.30
10^{20}	0.45	0.33	0.22	0.096
10^{21}	0.080	0.042	0.019	0.0040
4.4×10^{21}	0.0047	0.0016	0.00056	0.000048

i.e. a factor 24 at 10^{21} and 630 at the 4.4×10^{21} of the base, at 1300 K. Sec. 122. knew the table was an upper bound above `h2_shielding_overlap_column` and said so; what it could not know is that the boundary was about a decade optimistic — at the column that function returned, the table already stood 7 to 10 times above the overlapping-line answer.

135.3 What the table is now

The table is now `h2_self_shielding_table.f90` in `src/modules/lower_atmosphere/`, generated by the new `h2_shielding_table_line_by_line.py` in `src/utis/`. Same three quantities on the same grid (39 columns $\times$ 9 temperatures $\times$ 3 densities), same axes, same accessors — `h2_lw_dissociation_cross_section`, `h2_lw_dissociation_per_pump`, `h2_lw_dissociation_per_absorbed_photon`, `h2_self_shielding_level_resolved`.

The physics behind each number changed:

- **σ_{pump} and p_{single} are line-by-line, with overlap.** Plane-parallel slab, normally incident beam, flat F_λ normalized to $F = 343 \text{ erg cm}^{-2} \text{ s}^{-1}$ over 912–1110 Å, $\tau(\nu, N) = N \sum_i x_i \sigma_i(\nu)$ summed over every Lyman and Werner transition between 911.75 and 1200 Å on a 4×10^5 -point frequency grid, Voigt profiles with $b = (2kT/m_{\text{H}_2})^{1/2}$, LTE populations. The line data are Abgrall, Roueff & Driera (2000) — the same measurements CLOUDY’s `dissprob_*/transprob_*` files and the Meudon PDR code carry, so no line-list difference enters.
- **The fluorescent trapping is still CLOUDY’s.** $p_{\text{eff}}/p_{\text{single}}$ is read node by node from the same 27 runs and multiplied in. Reimplementing CLOUDY’s escape probabilities would be a second, unvalidated version of the same thing. **Its geometry is plane-parallel, illuminated on one face and closed on the other;** our layer has the same

thick-inward side, which is the side that matters, but it is spherical and open outward. That is the one ingredient of the table carrying a geometry our layer does not have, and it is a new item of `TO_BE_DONE.md`.

- **h2_shielding_overlap_column is gone**, and with it the `h2_shield_log_col_overlap` plane. There is no longer a column above which the value stops meaning anything, because overlap is inside it. `h2_shield_max_column()` replaces it: the top of the tabulated column axis, $5.0 \times 10^{21} \text{ cm}^{-2}$, above which the edge value is returned. The two run-time warnings that quoted the old boundary now report that extrapolation instead.

135.4 Verification

- **The line-by-line calculation against a published overlap-including curve.** At 100 K it reproduces the Draine & Bertoldi (1996) self-shielding function to 2–9 **per cent over** $10^{16} \leq N_{\text{H}_2} \leq 10^{21}$, and its no-overlap branch departs from DB96 exactly where DB96 say overlap sets in ($3.3 \times$ at 10^{21} , $12 \times$ at 4.4×10^{21}).
- **Against a second code that carries overlap.** The Meudon PDR code (Le Petit et al. 2006) run with `f_ShieldAlgo = 1`, its exact UV line transfer — 30 H_2 levels in full transfer, 20 global iterations, pure H/He at $n_{\text{H}} = 10^{13} \text{ cm}^{-3}$ — agrees to 5–22 **per cent over** $10^{19} \leq N_{\text{H}_2} \leq 10^{21}$ at 900, 1300 and 1800 K.
- **Absolute quantities.** σ_{pump} at the bottom of the column axis is 2.59×10^{-17} (700 K) to $2.88 \times 10^{-17} \text{ cm}^2$ (3200 K) against DB96’s own $3.4520 \times 10^{-18} / 0.135 = 2.557 \times 10^{-17}$ (1–13 per cent); p_{single} there is 0.1425 to 0.1592, and at 1300 K it is 0.1475 against 0.1465 from the CLOUDY runs (0.7 per cent); σ_{diss} there agrees with the H2g-corrected CLOUDY table to 1.3–3.6 **per cent**, which is the statement that the two differ in the shielding and not in the normalization.
- **The trapping factor that came from CLOUDY** is 1.13–1.38 at the *bottom* of the column axis ($N_{\text{H}_2} = 10^{12} \text{ cm}^{-2}$, the optically thin end) across the whole grid, reproducing the 1.15–1.37 Sec. 132. records at the illuminated face, and it rises to 1.25–3.07 at the top of the axis, $5 \times 10^{21} \text{ cm}^{-2}$.
- **Fortran against Python** at all 1053 grid points: max relative difference 3.1×10^{-9} on σ_{diss} , 8.4×10^{-10} on p_{single} , 9.6×10^{-10} on p_{eff} .
- **Regression, both stages, in a scratch copy of the tree first.** The baseline reproduces the golden 9/9 byte-identical. With the H2g weighting alone, and again with the overlapping-line table, **exactly one case moves, mol_lyman_werner, and the other eight are byte-identical** — which is the structural statement the change should make, since `F_LW_star` defaults to zero and one matrix case supplies it. Sizes (12000-step relaxation snapshots, so not converged change sizes):

quantity	golden	H2g weighting	+ overlapping lines
H_2 front, largest r with $n_{\text{H}_2} > 10^{10}$	$1.146163 R_{\text{p}}$	1.142513	1.144327
f_{shield} at the deepest cell	—	8.165×10^{-6}	1.716×10^{-7}
$n(\text{H}_2)$ over the golden molecular cells	1	0.150–1.0005	0.147–1.013
final du	2.6075	2.7629	2.7689

The front barely moves while the deepest cell's shielding falls $48\times$, because the front sits at moderate columns where the two errors nearly cancel and the deep cells are already almost fully molecular.

Which gate these were measured on. The four rows above are the 9 microbar hot-Uranus gate, the one the goldens were written against. Re-running the matrix in the tree afterwards gives a larger move — H_2 front $1.064744 R_p$, N_{H_2} reaching 2.877×10^{20} rather than 4.296×10^{21} — and none of that difference is the table: section 136 re-pinned the `base.inp` of the six hot-Uranus cases to `p_base 1.0e-06`, `q_H2_base 0.84`, i.e. a 1 microbar base, and the molecular layer above a 1 microbar gate is an order of magnitude thinner to begin with. The two sets of numbers are therefore not to be compared with each other, and neither needs re-running: this section's are the change in the table at fixed gate, and the tree's are the same table on the gate section 136 installed.

- **Goldens were NOT refreshed.** `make check` reports FAIL on `mol_lyman_werner` by design; the refresh is a separate decision.
- **CLOUDY was patched a second time** to make the reservoir weighting possible: four more columns of `save h2` rates (`pump_s`, `diss_single_s`, `den_g`, `den_s`), the H_2 s counterparts of the four Sec. 132. added. The 27 decks were re-run with it; nothing else about them changed.

Files: `src/modules/lower_atmosphere/h2_self_shielding_table.f90`, `src/modules/lower_atmosphere/lyman_werner.f90`, `src/modules/radiation/util_ion_eq.f90`, `src/modules/radiation/ionization_equilibrium.f90`, `src/modules/files_I0/write_output.f90`, `src/utls/h2_shielding_table_from_cloudy.py`, `src/utls/h2_shielding_table_line_by_line.py`. Documentation: `docs/p38_line_overlap_shielding.md`, `docs/h2_self_shielding_cloudy.md`, `docs/molecular_hydrogen_treatment.tex`, `TO_BE_DONE.md` (P38 closed, one new item).

136. The base level was stated twice and the two statements disagreed everywhere; `p_base` now fixes it, and the hot-Uranus gate moves to 1 μ bar (2026-09-03)

The judgement first. The lower boundary is one level, and EXHALE let two inputs state it independently: `Log10 lower boundary number density` in `input.inp`, and `p_base` in a `base.inp` handoff. Nothing compared them. Every one of the 52 `base.inp` files in this tree that carries `p_base` disagreed with its own density key — the hot-Uranus gate by 10%, the He/H ladder by 1.4–55%, the `vulcan_work` HD 209458 b runs by factors of 7 to 32. The key had never actually located the base level anywhere. `p_base` now fixes the level, the pair is refused when it disagrees by more than 1%, and the hot-Uranus gate moves to the 1 μ bar level its own base radius already corresponded to.

136.1 What was wrong

`docs/p44_base_sawtooth.md` (Secs. 8 and 9 there) arrived at this from the other end. The hot-Uranus gate carries a base velocity artifact at cell 1 ($v(1) \approx -250 \text{ cm s}^{-1}$, a first-order-in- dr boundary discretization error) whose effect on the base-face mass flux scales with the base density: -21 , -203 and -589 times the wind flux at base pressures of 1, 9

and 28 μbar . On the default 9 μbar base the converged molecular layer is not even reproducible across restart cycles — three info = 0 states of the same configuration put the H_2 front anywhere in $1.053\text{--}1.089 R_p$. At 1 μbar the same configuration reproduces the front to five digits, converges from a cold start in 27 minutes where the 9 μbar base did not converge in 1 h 41 m, and leaves $\log_{10} \dot{M}$ where it was.

And 1 μbar is where that gate's boundary belongs. Koskinen et al. (2022, ApJ 929, 52, Sec. 3.2) state their lower boundary as

“The lower boundary of the model is at $r_0 = 1.34 R_p$ ($p_0 = 10^{-6}$ bar, $T_0 = 1,140$ K) where the boundary conditions are consistent with our models of the lower and middle atmosphere”

and $1.34 R_{1\text{bar}} = 3.425 \times 10^9$ cm with their $R_{1\text{bar}} = 2.5559 \times 10^9$ cm is the gate's own Planet radius: $0.49 R_J = 3.4256 \times 10^9$ cm, to 0.2%. The gate had the right radius and the wrong pressure at it, because $n_0 = 10^{14}$ was chosen by hand.

The earlier reading, that $T_0 = 1140$ K was a 1 μbar value applied nine times deeper along a falling profile, is **withdrawn**: the same paper's Sec. 2.2 says their lower and middle atmosphere is isothermal at the effective temperature, so 1140 K is their number at any depth below their boundary. The inconsistency was never the temperature.

136.2 What changed

input_read.f90. A `base.inp` carrying `p_base` is a handoff written AT that pressure — its composition, its q_{H_2} and its temperature all refer to that level — so the level is the handoff's to state, and

$$n_0 = \frac{p_{\text{base}}}{k_B T_0 n_{\text{tot,bc}}}, \quad (46)$$

the relation Base BC: pressure already used. Three refusals go with it:

- Log10 lower boundary number density is now **optional**. Given alongside `p_base`, the two must agree to 1%; otherwise startup stops and prints both pressures, the ratio, the n_0 each implies, $n_{\text{tot,bc}}$, and the two ways to fix it (delete one or the other).
- `p_base` and Base BC: pressure together must name the same level to round-off, or startup stops.
- Neither given: startup stops and lists the three keys that can state it.

A lower-atmosphere **profile** sets `p_base_bar` from its own matching pressure, so a profile run owns the level itself and is outside the rule (`lower_profile` stays byte-identical).

write_setup_report.f90 gained one line naming the source and both numbers:

```
- Base level: p = 1.0000E-06 bar (from base.inp p_base) -> n0 = 1.1690E+13 cm^-3
- Base level: n0 = 3.0903E+12 cm^-3 (from the density key) -> p = 1.0069E-06 bar
```

The gate and the arms. The six molecular regression cases and the sixteen diagnostic arms now carry `p_base` $1.000\text{e-}06$ with the density key deleted from `input.inp`, and the gate's `q_H2_base` becomes 0.84 — the same paper's value at the same level (Sec. 3.2: “The volume mixing ratio of H_2 at the lower boundary is about $q_0 = 0.84$ while H is

a minor species with $q_0 = 0.026''$), which sits at 97.3% of the ceiling $0.5/(0.5 + \text{He}/\text{H}) = 0.863110651$ this He/H can supply. The He/H ladder keeps its own rule — hold the fraction of H nuclei bound into H_2 fixed and vary only He/H — so its values were recomputed from $x_2 = 2q(1 + \text{He}/\text{H})/(1 + q) = 0.985447826$ (it was 0.925114286):

He/H	q_H2_base	ceiling	fraction of ceiling
0.0793 (gate)	0.840000000	0.863110651	97.3%
0.3	0.610353658	0.625000000	97.7%
1.0	0.326896922	0.333333333	98.1%
3.0	0.140486207	0.142857143	98.3%
10.0	0.046893592	0.047619048	98.5%
30.0	0.016151029	0.016393443	98.5%

136.3 What it moves

make check: `wasp_full`, `wasp_he23off` and `lower_profile` **PASS, data identical** — they carry no `p_base`, so the code path is untouched, and this was also confirmed directly by building the same source snapshot with and without the change and running all three cases with both binaries. The six molecular cases **FAIL, i.e. move**, which is the point; the row count is unchanged (504), so the goldens are re-snapshotted rather than restructured, and there is no NaN in any of them.

case	ρ_{bc} old $\rightarrow$ new	r_{front}	peak $n(\text{H}_3^+)$	$N(\text{H}_2)$ new/old	$\log_{10} \dot{M}$
mol_base_handoff	1.220e14 $\rightarrow$ 1.427e13	1.1546 $\rightarrow$ 1.0741	6.68e4 $\rightarrow$ 4.40e4	0.094	10.56 $\rightarrow$ 10.42
mol_metals	1.234e14 $\rightarrow$ 1.441e13	1.1562 $\rightarrow$ 1.0748	3.89e1 $\rightarrow$ 4.66e1	0.095	10.57 $\rightarrow$ 10.41
mol_lyman_werner	1.220e14 $\rightarrow$ 1.427e13	1.1243 $\rightarrow$ 1.0350	6.67e4 $\rightarrow$ 4.42e4	0.067	10.55 $\rightarrow$ 10.51
mol_diffusion	1.220e14 $\rightarrow$ 1.427e13	1.1546 $\rightarrow$ 1.0741	6.68e4 $\rightarrow$ 4.40e4	0.094	10.56 $\rightarrow$ 10.42
mol_ir_bands	1.234e14 $\rightarrow$ 1.441e13	1.1562 $\rightarrow$ 1.0748	3.89e1 $\rightarrow$ 4.66e1	0.095	10.57 $\rightarrow$ 10.41
mol_sec_ion	1.220e14 $\rightarrow$ 1.427e13	1.1401 $\rightarrow$ 1.0622	2.89e5 $\rightarrow$ 1.68e5	0.085	10.51 $\rightarrow$ 10.39

The base is 8.6 times lighter, the H_2 column falls to 7–10% of what it was, the front moves in by $0.08\text{--}0.09 R_p$, and $\dot{M}$ falls by 0.04–0.16 dex. All of it follows from putting the base nine times higher; no physics changed.

The 12000-step snapshot convention survives the move: at the same 12000 steps the retargeted gate reaches a *smaller du* (2.56 against 3.75 on `mol_base_handoff`, 2.23 against 4.68 on `mol_sec_ion`) and runs faster (5 m 35 s against 6 m 59 s single-threaded), so the lighter base costs it no physical time.

136.4 What this does not fix, and one thing it exposes

The cell-1 velocity artifact itself is unchanged in magnitude — it is a first-order-in- dr discretization error and only base refinement removes it. What the lighter base removes is its *consequence*: the base-face mass-flux error falls with ρ_{bc} , from 203 to 21 times the wind flux.

The 30 `base.inp` files under `vulcan_work/` were **not** retargeted — they are records of past VULCAN comparison runs, not part of the matrix — but each of them states a `p_base` that is wrong by a factor of 7 to 32 against the level its density key actually used, and each will now be refused if it is re-run. That is a finding about those runs.

137. The output readers returned the file's ghost rows as solution cells; the transit depths move by up to 3.8 per cent (2026-09-03)

The judgement first. Every Python consumer of an EXHALE profile was reading the file's boundary rows as if the solver had converged them, and that is wrong whatever it costs: the two rows at each end are a fixed base state below and a zero-gradient / WENO3 extrapolation above, both written by `Apply_BC`. Sec. 133.6 closed the Fortran half of this by making the file say which rows are physical; this closes the Python half. `examples/exhale_io.py` now returns the physical cells and takes `ghost=True` for the file as written, and `EXHALE_transit.py` and `EXHALE_plots.py`, which bypassed the reader with their own `np.loadtxt` calls, go through it. Item (P48) of `TO_BE_DONE.md` closes with this.

137.1 The rule, and why the fallback is not a legacy path

`exhale_io.physical_cell_rows` has two ways of knowing, in order.

1. **The header, which is the contract.** Since Sec. 133.6 the writer emits `# rows 504: 2 ghost cells at each end; physical cells are rows 3 to 502 (1-based, inclusive)`. It is a comment, so no numeric parse changes.
2. **The layout,** for a file written before that header existed — the regression goldens and every saved output*/directory. $N_g = 2$ is a Fortran parameter and every writer loops $1-N_g \dots N+N_g$, so two rows at each end are ghosts unconditionally, for every grid type. What the file can still be asked is whether it has that layout at all, and the base radius answers it: `define_grid` puts a cell center exactly at $r = 1$ and that center is a *ghost* — $r(1-N_g) = 1$ for the Uniform and Stretched grids, $r(2-N_g) = 1$ for Mixed — so $r = 1$ falls on row 1 or row 2 and never later. A file whose radius column increases monotonically and equals 1 to roundoff at row 1 or row 2 has the ghost layout, and two rows come off each end. Checked against all 9214 profile files in this working copy: every one satisfies it. A file that satisfies neither rule is returned whole with a warning, which is the pre-2026-09 behavior.

The two routes were checked against each other on `backup/regression/mol_diffusion/output/Hydro_ioniz.txt`, which has the header: with the header line removed the layout rule returns the same (2,502) and the same 500 rows.

The fallback is not a legacy path. `write_output` emits the `# rows` header for `Hydro_ioniz(_adv).txt` *only*; `Ion_species`, `Cooling_breakdown`, `Heating_breakdown`, `Excited_H`, `Lyman_Werner` and `OI_levels` carry no such line even in a run made today, so they take rule 2 on every read. Giving the other writers the same header is a one-line change per file in `write_output.f90 / util_ion_eq.f90 / excited_hydrogen.f90` and is left as an open item — the Fortran was not touched here.

137.2 What it changes, measured

The mechanism is at the top of the domain. The transit integrates chords out to $R_{\text{ib}} = \min(r_{\text{last}}, R_{\star}/R_{\text{p}})$, so dropping the two upper ghost rows shrinks the outermost annulus, and the area lost is $(r_{\text{top,phys}}/r_{\text{top,ghost}})^2 - 1$: -1.04 per cent on WASP-121b, -4.6 per cent on HD 189733 b and -3.6 per cent on the hot Uranus. That is the size of the effect, not a prediction of it: a line opaque all the way to the boundary loses the whole annulus, one that is not loses part of it, and the lower-level column densities change with the same two rows.

state	rows	first r	last r
WASP-121b (wasp_full_newton)	504 $\rightarrow$ 500	0.999804 $\rightarrow$ 1.000196	1.60976 $\rightarrow$ 1.60138
HD 189733 b (HD189733b/output)	504 $\rightarrow$ 500	0.999904 $\rightarrow$ 1.000096	4.42773 $\rightarrow$ 4.32553
hot Uranus (He/H = 0.0793)	504 $\rightarrow$ 500	0.999807 $\rightarrow$ 1.000193	4.81412 $\rightarrow$ 4.72534

MPLBACKEND=Agg python3 EXHALE_transit.py on each of the three states, before and after, from a copy directory with output/ symlinked and the repository's own script named through a symlink. Excess absorption in per cent, from the theoretical curve of tpm_<line>.txt: line center by interpolation at the laboratory wavelength, band as the mean over $\lambda_0 \pm 2 \text{ \AA}$ clipped to the model window.

WASP-121b (metals on, He 2 ³ S, Newton-finished)						
line	line center [%]			4 Å band [%]		
	before	after	rel.	before	after	rel.
He I 10830	3.7248	3.6639	−1.64%	1.6046	1.5741	−1.90%
Ly α	3.7531	3.6897	−1.69%	1.6464	1.6377	−0.53%
H α	3.4571	3.4144	−1.23%	1.0071	1.0014	−0.56%
H β	2.5042	2.4912	−0.52%	0.4524	0.4478	−1.03%
Mg II 2796	3.7531	3.6897	−1.69%	0.3216	0.3164	−1.64%
Ca II K	3.7246	3.6641	−1.62%	0.2480	0.2425	−2.22%
Na I D ₂	1.0135	1.0127	−0.07%	0.0568	0.0567	−0.17%
O I 1302	3.1800	3.1577	−0.70%	0.0765	0.0760	−0.71%
HD 189733 b (HD189733b/output)						
He I 10830	2.8387	2.8333	−0.19%	0.3989	0.3987	−0.06%
Ly α	37.9394	36.9732	−2.55%	1.6858	1.6746	−0.66%
H α	0.6053	0.6053	−0.01%	0.1008	0.1013	+0.48%
H β	0.3661	0.3660	−0.01%	0.0368	0.0370	+0.54%
Mg II 2796	0.5060	0.4985	−1.48%	0.0374	0.0371	−0.72%
Ca II K	0.3104	0.3112	+0.25%	0.0114	0.0113	−0.16%
Na I D ₂	0.3736	0.3740	+0.10%	0.0283	0.0282	−0.31%
hot Uranus (molecular base; metals off)						
He I 10830	9.2358	9.0559	−1.95%	1.1888	1.1592	−2.49%
Ly α	27.8690	26.8069	−3.81%	3.8440	3.7534	−2.36%
H α	1.0801	1.0782	−0.18%	0.0823	0.0822	−0.14%
H β	0.6203	0.6200	−0.05%	0.0388	0.0388	−0.01%

backup/regression/jfnk_hd189 is the state Sec. 133.6 measured, but it holds no *_adv.txt pair and the transit tool reads only those, so the converged HD189733b/output stands in for it. The H β band on the hot Uranus and, less severely, its H α band are clipped by the auto-sized model window (70 and 95 per cent of the 4 Å width); they are means over what the window holds.

Largest single move: the hot Uranus Ly α line center, −3.81 per cent. Two further numbers the run prints move by more than the depths do: the peak $n = 2$ hydrogen density of the hot Uranus falls from 1.794×10^5 to $2.693 \times 10^4 \text{ cm}^{-3}$ (a factor 6.7 — the maximum sat in the lower ghost row, in the fixed base state, behind the planetary disk, which is why the H α depth barely notices), and the He 10830 red-peak depth of the same state goes from 8.989 to 8.812 per cent. EXHALE_plots.py on the same states shows the change as an axis range: the radial axis now starts at the first physical cell, and on the hot Uranus the ion-density panel loses the two decades of empty base ghost that used to set the bottom of its logarithmic scale. The curves themselves lie on top of each other.

mdot_log10 was fixed at the same time and for the same reason. The Fortran evaluates the mass-loss rate at the physical cell $j = N - 20$ (EXHALE_main.f90:1514); the reader evaluated it at index $\text{len}(r) - 20$ of an array that still held the

ghost rows, i.e. three cells further out. It is now $\text{len}(r) - 1 - j_{\text{from_top}}$ of the ghost-free array, the same cell the run reports. On the WASP-121b state both give $\log_{10} \dot{M} = 13.2224$ — the mass flux is flat there, which is what the convergence gate asserts — so this is a correctness fix with no measured effect on that state.

137.3 Every consumer, and the two that would have double-counted the fix

Eighteen files besides the reader itself name `exhale_io`; seventeen import it (LHS1140b/make_memo_figures.py only mentions it in a comment). Going through the reader, they pick the change up automatically — but two of them cut the ghost rows a second time and one mixed the two conventions:

- `src/utis/collisional_validity.py` held its own `N_GHOST = 2` and applied `slice(N_GHOST, -N_GHOST)` to what `load_run` returned. Both the constant and the slice are gone; the Knudsen diagnosis now runs on the 500 cells it meant to. Its $\log_{10} \dot{M}$ for the WASP-121b state reads 13.222, the run's own value.
- `LHS1140b/exhale/metastable_step_diagnosis/knudsen_scale.py` sliced with `CV.N_GHOST` for the same reason; removed.
- `LHS1140b/heh_series/audit_checks.py` read `Hydro_ioniz`, `Ion_species` and the two breakdown files with a bare `np.loadtxt` and compared them column by column with the ion densities `exhale_io.load_ions` returned. After the change those are 504 rows against 500 and its charge-neutrality check raises `IndexError`; it now reads all four through `exhale_io.loadtxt_cells`, and runs.
- `poster/make_poster_figs.py` read one `Hydro_ioniz_adv.txt` directly while importing the reader for everything else; it goes through `loadtxt_cells`.

Not changed, deliberately: `LHS1140b/make_memo_figures.py` reads profiles with about twenty bare `np.loadtxt` calls and never imports the reader, so nothing there breaks and nothing there is corrected either — its figures still include the ghost rows. Whether to convert it is tied to re-making the LHS 1140 b memo figures, which is its own decision.

137.4 What was checked, and what was not

Checked: the two selection routes against each other and against all 9214 profile files in the working copy; the three transit runs above, end to end, before and after; `EXHALE_plots.py` on two of them; `src/utis/collisional_validity.py` and `LHS1140b/heh_series/audit_checks.py` run to completion after the fix.

Not checked, and not run: `examples/make_figures.py` and the figures under `docs/figures/`, every notebook, and the paper and poster figure scripts. All of them read through the reader (`make_poster_figs.py` now included), so their output *will* move by the amounts tabulated above the next time they are made; nothing under `docs/figures/` was regenerated here. No Fortran source was touched, so `make check` and the goldens are untouched by this change.

Files: `examples/exhale_io.py`, `EXHALE_transit.py`, `EXHALE_plots.py`, `src/utis/collisional_validity.py`, `LHS1140b/exhale/metastable_step_diagnosis/knudsen_scale.py`, `LHS1140b/heh_series/audit_checks.py`, `poster/make_poster_figs.py`. Documentation: `README_HOWTO.md`, `TO_BE_DONE.md` ((P48) closed, one new item).

138. The Newton direction at the H_2 front pointed uphill because the residual is discontinuous there: the positivity guard was a switch, and it is now a limiter (2026-09-03)

The judgement first, and it retires item (Y) by contradicting both of the causes that item proposed. The energy row at the front is smooth in the temperature: its finite-difference directional derivative converges to six digits over four decades of step size, a two-sided scan of it is antisymmetric down to $h = 10^{-9}$, and the residual's effective $d \ln \text{cool} / d \ln T$ there is 0.26, not an exponential. What has no descent direction is the state as a whole, because $F(Y)$ is genuinely *discontinuous* at the iterate. The WENO3 reconstruction at the face of cell 273 ($r = 1.2324 R_p$) returned a left density of 2.50×10^{-15} against cell averages of 10^{-4} — eleven orders below, and still positive by a hair — so the iterate sat exactly on the switching surface of `positivity_limited_faces`. Raising the density of the neighbouring cell by one part in 10^9 tipped that face through zero, the guard replaced the whole face pair by the first-order cell averages, the face density jumped by a factor 1.7×10^9 , and the merit jumped from 202.5 to 945.1 — by the *same* factor 4.67 at every step size from 10^{-2} to 10^{-9} , which is what a jump looks like and what a slope does not. Every direction the solver could build put its largest scaled component at that face, so every one of them ascended: the Newton/PTC step at all six decades of damping and all ten damped Gauss–Newton steps, by factors 4.7 to 27.8. The escape of Sec. 126. was not defeated by an indefinite Jacobian; it was asked to descend a function that is not continuous. The guard is now the continuous limiter of Zhang & Shu (2010), and a trial state whose chemistry the equilibrium sweep could not certify is refused the way Sec. 121. refuses one it cannot describe at all. On the reproduction case the abort is gone: it is printed once in a whole run instead of ending every hand-off, and the solve that used to stop at $\|R\| = 2.775 \times 10^{-2}$ having excavated a 65000 K hole in the layer now meets the residual gate at 2.95×10^{-6} and is held only by the flux gate.

This closes item (Y) of `TO_BE_DONE.md`. The full measurement is `docs/p49_front_energy_mode.md`.

138.1 What the measurement found, and what it ruled out

Everything below is on the hot Uranus of Sec. 134. — 500 cells, second-order carrier advection, Stellar LW flux: 343.0, 150 000 marching steps, restarted with `Secondary_ionization: Immediate` and `Solver: Newton 4.0e-2` — reproduced to every printed digit of item (Y): `info = 2` at $\|R\| = 2.775 \times 10^{-2}$, worst cell $j = 255$, $\|\nabla \text{merit}\| = 4.583 \times 10^{11}$, 46 non-monotone accepts.

The blocking row, term by term. At the abort the worst *scaled* energy residual is cell 259, $r = 1.1952$, at $|R|/D = 67.6$. Its row is $R_3 = dF_E - (\text{heat} - \text{cool})$ with cool at 97.7% of $|R_3|$; from `Cooling_breakdown.txt` at that state, 93% of that cooling is H_3^+ infrared emission at 65 173 K, from an $f(H_3^+) = 4.5 \times 10^{-3}$ the chemical solve did not certify (acceptance class 4, reaction residual 2.4×10^{-3}). H_3^+ does not survive at 65000 K. Cells 256–259 sit exactly at the radiative/advective transition, $E/|\text{heat} - \text{cool}|$ over $dr/|v| = 0.75, 0.65, 0.97, 0.63$; from cell 260 outward the class-4 acceptance sets $n(H_3^+) = x(\text{HII}) = 0$ exactly, the cooling collapses by five orders and the ratio is 715 to 2319.

Candidate (a) of item (Y), the temperature entering through n_{tot} : real, and not the blocker. Scanning the energy unknown of cell 259 alone in steps of 10^{-3} relative, the accepted composition does jump between two branches at

isolated points — $x(\text{HII}) = 0$ exactly, $f(\text{H}_3^+)$ a factor 38 lower, the reaction residual from 3.2×10^{-3} to 1.4×10^{-5} — and returns in between, which is seed-dependent root selection rather than a threshold. The energy row does not notice: cool reads 1.0859×10^{-2} on the jumped points against 1.0857×10^{-2} on the trend.

Candidate (b), the H_3^+ and H_2 rates: measured, and it is the residual that cannot see them. On the same scan cool has $d\ln\text{cool}/d\ln T = 0.263$ while the composition's own $f(\text{H}_3^+)$ has 1.728. Both cannot belong to a rate that is exactly proportional to $n(\text{H}_3^+)$, and they do not: within one `ioniz_eq` the molecular densities the cooling uses are built at the top of the routine from the composition handed *in* (`ionization_equilibrium.f90:570--573`), `eval_cool` is called with them, and the per-cell sweep that changes the composition runs afterwards. So the residual's temperature derivative carries the emissivity's 0.26 and none of the density's 1.728 — about one seventh of the self-consistent response. That lag is not a defect and is not corrected (Sec. 138.5).

Candidate (c), the frozen-radiation Jacobian: not the blocker either. The matrix-free Jv of the full residual along the energy unknown of cell 259 converges to six digits from $h = 10^{-4}$ to 10^{-10} , and a two-sided merit scan on cells 255, 259 and 260 is antisymmetric to nine digits with the WENO weights recomputed and frozen alike. Descent directions along the front's energy rows exist.

What does block it. A forward-difference gradient of the merit itself over cells 244–274 has one component that is the whole norm — cell 272, the mass row, 7.43×10^8 against 10^0 – 10^3 everywhere else. Stepping along $+g$ changes the merit by +3.6668 at every step size from 10^{-2} to 10^{-9} , constant to seven digits. Dumping both sides at $h = 10^{-9}$: every cell's temperature, class, heating, cooling and composition is identical to every printed digit, and what changes is the reconstruction — faces dropped to first order $2 \rightarrow 3$, the left density at face 273 $2.50 \times 10^{-15} \rightarrow 4.16758 \times 10^{-6}$, the mass-row residual of cell 273 $|R|/D$ 9.50 $\rightarrow$ 918.3.

138.2 The positivity repair is continuous

`positivity_limited_faces` (`src/modules/states/Reconstruction.f90`) no longer swaps a face pair between two schemes. Each face state is scaled toward *its own* cell average by the largest factor that keeps ρ and p above a floor,

$$W_{\text{face}} \leftarrow W_{\text{avg}} + \theta (W_{\text{rec}} - W_{\text{avg}}), \quad \theta = \min_{\rho, p} \frac{q_{\text{avg}} - \varepsilon}{q_{\text{avg}} - q_{\text{rec}}},$$

which is the positivity-preserving limiter of Zhang & Shu (2010, J. Comput. Phys., 229, 8918; doi:10.1016/j.jcp.2010.08.016). The publisher PDF is not in references/, so only the bibliography is recorded. Two functions carry the arithmetic, `positivity_scaling` and `positive_variable_scaling`.

The floor. The switch this replaces tested $q > 0$, so its own floor was zero, and zero is what the change was asked to keep. It cannot be: with $\varepsilon = 0$ the scaling puts the face value at exactly zero and hands `Num_Fluxes` a $\sqrt{\gamma p/0}$. The floor used is the smallest one that is not a chosen number — one unit in the last place of the cell average, $\varepsilon = \text{epsilon}(1.0d0) q_{\text{avg}}$. No dimensional constant and no tuning enter, and the residual jump that survives at the crossing is 2.2×10^{-16} of the cell average against the factor 1.7×10^9 the switch made.

Byte-identity. A face with $\theta = 1$ is left untouched rather than rewritten as $W_{\text{avg}} + 1 \cdot (W - W_{\text{avg}})$, which is not bitwise W . Every run in which the guard never fires is therefore unchanged to the bit, and Sec. 138.4 measures that on eight of the ten matrix cases. The counter now counts face *states*, not face pairs, because the limiter scales the offending side only.

138.3 A trial whose chemistry is not certified is refused

`eval_residual` (`src/modules/time_step/steady_newton.f90`), inside the admissibility of Sec. 121.. A state whose equilibrium sweep accepted a cell as class 4 (the relaxation amnesty) or class 5 (the constrained continuation) is a residual computed on a composition the chemistry does not describe, and its value says nothing about how far the hydro is from steady. The count and the worst reaction residual are reported by the solve, separately from the `ioniz_eq` ledgers.

Two properties of the test, both forced by measurement rather than chosen.

1. **It is a comparison, not a threshold.** Refusing any trial with an uncertified cell was written first and measured: on this state the *iterate* carries 11 such cells, so all 17 Jacobian colours and every Krylov direction were refused, the solve had no Newton model at all ($\|\nabla \text{merit}\| = 0$) and aborted at iteration 1 with $\|R\|$ unchanged. A rule that rejects the neighbourhood of the point it stands on cannot be used to leave that point. The test is $n_{\text{uncert}}(\text{trial}) \leq n_{\text{uncert}}(\text{iterate})$, the reference refreshed from the iterate at the top of every outer iteration; at a fully certified iterate it is the same rule as the absolute one.
2. **It applies to states that could be adopted, not to derivative samples.** With the comparison in place the solve ran 18 iterations and stopped again with 13 of 17 colours zeroed: the refusal had moved onto the finite-difference probes. A Jacobian colour and a *Jv* sample are points where the residual is read, never states anyone keeps, and a model built from an imperfect derivative is still a model. `eval_residual` takes `may_be_adopted` (default `.true.`); `build_banded_jac_full` and `jv_product` pass `.false.`

138.4 The regression, and which section moved which case

The matrix is now ten cases: `mol_carrier` (added by Sec. 134.) gets its first golden here. Attribution was measured by building the tree as it stood before this change and running all ten plus the two `*_newton` cases on both binaries, single-threaded, comparing the numeric content.

case	pre vs. new	guard firings, pre $\rightarrow$ new	trials refused
<code>mol_base_handoff</code>	byte-identical	0 $\rightarrow$ 0	0
<code>mol_metals</code>	byte-identical	0 $\rightarrow$ 0	0
<code>mol_lyman_werner</code>	byte-identical	0 $\rightarrow$ 0	0
<code>mol_diffusion</code>	byte-identical	0 $\rightarrow$ 0	0
<code>mol_ir_bands</code>	byte-identical	0 $\rightarrow$ 0	0
<code>mol_sec_ion</code>	byte-identical	0 $\rightarrow$ 0	0
<code>mol_carrier</code>	byte-identical	0 $\rightarrow$ 0	0
<code>lower_profile</code>	byte-identical	0 $\rightarrow$ 0	0
<code>wasp_full</code>	moves	1195 $\rightarrow$ 1300	0
<code>wasp_he23off</code>	moves	1208 $\rightarrow$ 1302	0
<code>wasp_full_newton</code>	moves	1195 $\rightarrow$ 1300	0
<code>wasp_he23off_newton</code>	moves	1208 $\rightarrow$ 1302	0

The two counts are not directly comparable: the old one counted face pairs, the new one face states. refused = 0 everywhere, so the whole move is the limiter; the chemistry refusal changes nothing outside the molecular arm. Only the two WASP-121b cases fire the guard at all, which is why only they move.

case	$\log_{10} \dot{M}$ pre $\rightarrow$ new	info, $\ R\ $, spread	max rel. change, $r > 1.2 R_p$
wasp_full	13.26 $\rightarrow$ 13.26	stops on du	ρ 4.6e-5, v 4.8e-5, T 5.9e-6
wasp_he23off	13.26 $\rightarrow$ 13.26	stops on du	ρ 2.2e-5, v 2.4e-5, T 2.9e-6
wasp_full_newton	13.21 $\rightarrow$ 13.21	0, 9.337e-6, 1.858e-3 $\rightarrow$ 0, 7.398e-6, 1.357e-3	ρ 1.7e-2, v 1.7e-2, T 2.1e-3
wasp_he23off_newton	13.21 $\rightarrow$ 13.21	0, 9.024e-6, 7.457e-4 $\rightarrow$ 0, 5.353e-6, 1.226e-3	ρ 6.5e-3, v 4.0e-3, T 5.8e-4
mol_carrier	10.41 $\rightarrow$ 10.41	—	byte-identical
lower_profile	9.56 $\rightarrow$ 9.56	—	byte-identical

The rate is unchanged to the printed digit in every case, both Newton solves still meet *both* gates and reach a smaller residual than before. The largest single relative change anywhere in wasp_full is 2.4×10^{-1} in the velocity of the base cell at $r = 1.00235$, where $v = -1.9 \text{ cm s}^{-1}$: that is the collocated odd–even mode of docs/p44_base_sawtooth.md, not a change in the wind.

Block-F golden refresh. The previous goldens are preserved at backup/regression/golden_blockF_20260903. The refresh carries six changes and each one’s cases were isolated by the worker that made it.

sec.	change	cases it moves
133.	the two gates, and the signal-speed row scale	none: no matrix case enters the JFNK
134.	second-order carrier advection, computed LW flux	none; <i>adds</i> mol_carrier
135.	level-resolved H ₂ self-shielding table	mol_lyman_werner
136.	base level stated once, gate at 1 μ bar	the six mol_* cases
137.	the output readers	none (Python only)
138.	the continuous positivity limiter	wasp_full, wasp_he23off

lower_profile is moved by none of them and passed against the *old* goldens throughout. Verification, in the order the harness requires: check before the refresh reports 9 FAIL and lower_profile PASS; golden re-snapshots the ten; check again is **10/10 byte-identical**; run_fcheck.sh (-fcheck=bounds,do,mem rebuild plus the bounded HD 209458 b case) is **clean**; and check on the restored production binary is 10/10 again.

138.5 What this does to the reproduction case, and what is still open

On the tree, with the two gates of Sec. 133. and both changes above, the same hot Uranus (its base.inp p_base removed, so the base level is the density key the archived initial condition was built at — Sec. 136. refuses the two statements together):

	first solve	after the carrier pass
$\ R\ $	$1.273 \times 10^{-4} \rightarrow 2.369 \times 10^{-6}$	$2.664 \times 10^{-5} \rightarrow 2.950 \times 10^{-6}$
flux spread	5.391×10^{-4}	7.594×10^{-3}
info	0, both gates	1, residual gate met, flux gate left
trials refused	0	37 (worst 4 cells, worst residual $1.33\text{e-}6$)

“no descent direction exists” is not printed at all, where before it ended the hand-off. What holds the second solve is the flux gate at 7.6×10^{-3} against a threshold of 5×10^{-3} , a factor 1.5 — and that is the propagating front of docs/supersonic_molecular_base.md Sec. 13.7, not a solver failure: the carrier pass moves the composition by 0.553 and leaves a carrier steady residual of 3.92×10^{-2} at $r = 1.203$, and over the marching that follows the flux spread grows monotonically from 1.24×10^{-1} at step 14000 to 4.29×10^{-1} at step 21 500, where the run stops on the persistent-non-root guard of Sec. 113..

The three-layer table is not produced and nothing is quoted for it. The first solve’s state meets both gates, but the carrier pass overwrites it before anything is written, and there is no key that caps the Picard outer passes — the bound is `merge(20, 1, ...)` at EXHALE_main.f90:1628. Capturing it needs either an output write at the accepted return or that bound made settable, and both are changes of their own. Until then the rule of item (Y) still applies to the *carrier-relaxed* molecular wind: it is quoted on the marching criterion, and the fact is stated where it is quoted. What has changed is that the wind solve itself is no longer the obstacle.

The lagged cooling is judged, not corrected. At the fixed point the composition the cooling uses and the composition the sweep returns are the same object, so the lag is a property of the iteration and not of the solution, and re-evaluating `eval_cool` after the sweep would change no converged answer. What it costs is the Newton model — the factor 7 above — and adding the analytic $d \ln \text{cool} / d \ln T$ of the dominant channel to the banded Jacobian’s energy diagonal is the repair for that. It is not done here; it is held until the two changes above are shown to be insufficient.

138.6 What was not checked

- The three-layer table, as above.
- The $\text{He}/\text{H} = 1$ reference and the ladder rungs were not re-run; they carry molecular chemistry with the carrier transport off, so the guard does not fire in them, but that is an inference from the matrix rather than a measurement on those cases.
- Whether the reproduction case reaches both gates after the carrier alternation converges. It does not converge (0.553, then a growing flux spread), and that is Sec. 13.7’s open question, untouched here.
- The `Resid tol` cases outside the ten (`jfnk_*`, `ptc_*`, `tpm_*`, `resid_on`, `valve_sens`) were not re-run.

Files: `src/modules/states/Reconstruction.f90`, `src/modules/time_step/steady_newton.f90`, `src/EXHALE_main.f90`.
Documentation: `docs/p49_front_energy_mode.md`, `TO_BE_DONE.md` ((Y) closed).

139. The carrier/wind alternation was pointing the wrong way, its convergence measure understated the imbalance by 64, and the two are now solved together (2026-09-03)

The judgement first, in three parts.

(a) **The alternation of Sec. 128. does not converge because it cannot.** Its drift gate of 10^{-3} stands *below* the smallest move the transport operator can make: one step at a cell's own crossing time moves the largest H_2 mixing ratio by 1.1×10^{-2} , measured. A gate that cannot fire reads as a control to the next person, so it is gone. What replaces it is the test the wind is already held to — is the row steady on the scale of its own largest terms.

(b) **That test could not be applied, because the carrier row's scale was not its own largest term.** `carrier_row_scale`, $n_H(|v| + c_s)/dr$, is a *bound* on the row; measured against the sum of the terms the row actually balances it is $10^{2.7}$ to $10^{6.4}$ too large. A wind the steady solver accepted at `info = 0` reported a carrier residual of 1.6×10^{-4} while carrying **0.60 of its own terms** at the H_2 front and 1.07×10^{-2} volume-weighted — a factor 64 on the volume-weighted measure and 1850 on the worst cell. Sec. 133. already made “a bound on that row's own largest term” the principle for every hydrodynamic row; the carrier row now follows it, using the quantity `carrier_residual` has always computed for its own Newton.

(c) **The splitting was not merely slow — it had the sign wrong.** The alternation moves the H_2 front OUTWARD about one cell in each pass, at 83 per cent of the local gas speed. Solving the carriers and the wind *together*, from the same state, moves it INWARD: $f(H_2) = 0.5$ from 1.1421 to 1.0618 R_p , $f(H_2) = 10^{-2}$ from 1.2192 to 1.1488. The reason is a feedback a fixed wind cannot see, and it closes exactly: H_2 costs particles, fewer particles at the wind's own pressure means a hotter cell, and a hotter cell destroys H_2 . At the front the predicted temperature from the particle count alone is 1915 K against 1915 K measured, and 1958 against 1961.

Molecular carrier transport and Coupled carrier solve both stay DEFAULT OFF. Nothing below is a converged solution and nothing below is quoted as a rate.

139.1 What was measured, before anything was changed

Reproduction is cheap and that is worth recording: the alternation's failure does not need the 64000-step hand-off of Sec. 138.. The FIRST steady solve of that run, at marching step 2, already returns `info = 0`, and the carrier pass that follows it throws $\|R\|$ from 5.0×10^{-4} to 8.2×10^{-2} . Dumping that accepted wind and restarting from it reproduces the failure in three seconds.

On that state, one fixed-wind carrier pass, undamped:

$r [R_p]$	x_2 entry	x_2 exit	$f_c(\text{H}_2)$ entry	$f_c(\text{H}_2)$ exit
1.050	0.9425	1.0000	7.75e-1	8.22e-1
1.150	0.4447	1.0000	2.60e-1	5.84e-1
1.200	0.0963	1.0000	4.67e-2	4.85e-1
1.227	6.3e-4	1.0000	2.90e-4	4.63e-1
1.400	5.9e-5	4.4e-5	2.75e-5	2.05e-5
1.835	2.7e-5	1.1e-5	1.24e-5	5.07e-6

$x_2 = 1.0000$ is not a rounding: **123 of 500 cells between 1.047 and 1.227 R_p sit exactly on the hydrogen element ceiling** that `limit_to_element_budget` imposes (entry: none). The state the pass returns is pinned by the limiter over a quarter of the grid, which is a solution of nothing. The far wind moves by 7×10^{-6} in f_c , four decades below.

The drift measure was not the problem, and that was checked. It normalizes by the largest H_2 mixing ratio anywhere on the grid — one number, not the cell’s own value — so it is already absolute. On the LATE hand-off, where Sec. 138.5 reported the worst drift “in the wind” at $1.795 R_p$, the H_2 mixing ratio there goes from 0.252 to 0.507: a quarter of the gas to a half. There is no trace-species artefact; the wind is molecular to the top of the domain, which is the runaway itself.

139.2 The alternation, repaired as far as it can be

Two changes, both to what the outer loop does with a pass.

The under-relaxation blend is gone. It is measurably worse than either state it interpolates: carrier steady residual 1.6×10^{-4} at entry, 2.2×10^{-3} at the relaxed exit, **8.6×10^{-3}** for the $\omega = 0.5$ blend, which then threw $\|R\|$ to 6.7×10^{-2} against 1.9×10^{-2} for the relaxed state handed over whole. A blend of two states solves neither equation. Reducing ω does not help and is measured to hurt ($\omega = 1.8 \times 10^{-3}$ gave 2.0×10^{-1}).

A composition trust region replaces it. The pass advances the carriers and stops as soon as it has moved the composition by `carrier_trust`, and that state is handed over whole. Sec. 128. tried a bound in FLOW TIMES and measured that it controls nothing; a bound on the COMPOSITION does, because the operator’s smallest step is already 1.1×10^{-2} of it. Where the boundary is, measured on the same hand-off:

bound on one pass	passes returning info = $\emptyset$	$\ R\ $ handed on
the full relaxation, blended at $\omega = 0.5$ (old)	0	6.7e-2, abort
the full relaxation, whole	0	1.9e-2, abort
0.4	1	6.7e-4 then 8.2e-3, abort
0.2	8+	2.8e-4
0.05	19	2.4e-4
10^{-2} (adopted)	20 of 20	2.4e-4 to 3.1e-4

So `carrier_trust =  $10^{-2}$` stands a factor 20 inside the measured boundary. **And the alternation still does not reach a fixed point:** with it, the drift creeps 1.13×10^{-2} to 1.34×10^{-2} over 20 passes while the front walks from 1.203 to

$1.250R_p$, one cell in each pass. Each pass is one transport step at the cell's own advective crossing time — dt_{phys} equals $dr/|v|$ exactly there, 799 s at $1.227R_p$ — and $0.047R_p$ in 20 such steps is $1.03 \times 10^4 \text{ cm s}^{-1}$ against a gas speed of 1.24×10^4 . The alternation is time integration wearing a fixed point's clothes.

139.3 The measure, and the floor that a steady evaluation had killed

`carrier_steady_residual` now divides by `row_terms`, the sum of the row's own term magnitudes, and reports the volume-weighted ratio of sums over the layer and the wind separately, combined by the larger — the same construction `relnorm_over_cells` uses for the hydrodynamic rows. The retired scale is still computed and printed beside it while the two are being compared.

A defect found on the way. `carrier_residual`'s absolute floor, 10^{-20} of the element, was converted to the row's units by dividing by the step. The steady evaluation substitutes $dt = 10^{30}$ to kill the time term, and the floor went with it: the far wind was being measured with no floor at all. It is now converted by $\max(1/dt, (|v| + c_s)/dr)$ — the same statement written without a step, so it survives $dt \rightarrow \infty$ and agrees with the step form whenever the step is the cell's own crossing time, which is what the relaxation uses.

The gate is `carrier_resid_th = 10^{-3}` and it is NOT `Resid tol`. Sec. 133. renormalized `Resid tol` against a row scale that BOUNDS each hydrodynamic row's largest term rather than being it, so the two numbers are not on the same footing. What this one says is the plain statement: the carrier row balances to a thousandth of the terms it is made of.

139.4 The coupled solve: $n(\text{H}_2)$ as a fourth Newton unknown

Coupled carrier solve: True|False, default false, meaningless without Molecular carrier transport: True.

- **Unknowns** are four per cell, $(\rho, pv, E, n(\text{H}_2)/n_0)$, the last in the code's own density units and not in cm^{-3} : the Krylov step size is $\sqrt{\epsilon}(1 + \|Y\|)/\|v\|$, a norm of the whole vector, so a component thirteen decades from the others would set that step by itself.
- **The chemistry coupling already existed.** With the carrier transport on, `ioniz_eq` holds H_2 fixed at the partition it is handed and solves everything else around it — that is what made the splitting possible. The coupled solve changes only where that number comes from: the Newton unknown instead of the last transport step. So the particle count, the temperature that follows from it, and every heating and cooling term are functions of the carrier unknown, and the Jacobian sees it. **No new chemistry was written.**
- **The limited slope is frozen inside the model**, on `weno_mode` — the switch Sec. 121. already uses for the WENO3 weights, and for the same reason: it reaches $j-2$ and $j+2$, which the banded stencil does not carry.
- **The element budget is a CONSTRAINT, not a row.** A trial whose $n(\text{H}_2)$ leaves the hydrogen its cell has is refused where admissible is asked for, the way the positivity of ρ and p is refused in the line search. It is not clamped: a clamp puts a kink in the residual, which is the defect Sec. 138. removed, and it is exactly what made the fixed-wind relaxation return a state pinned on the ceiling. The test is a COMPARISON against how many cells

the ITERATE leaves outside, for that section's reason: a rule that rejects the neighbourhood of the point it stands on cannot be used to leave that point.

- **Band geometry** is set at run time: $|\text{row} - \text{col}| \leq 4 \times 2 + 3 = 11$, so $k_l = k_u = 11$ and 23 colours against 8 and 17. The carrier row itself reaches only ± 1 — its second-order correction is a frozen source — so the WENO3 reach still sets the bandwidth and the fourth variable only widens the stride.
- **Acceptance is a THIRD GATE, not a row of $\|R\|$** , and that is Sec. 133.'s own judgement applied again: one number cannot say two things. `Resid_tol` was renormalized there against a row scale that BOUNDS each hydrodynamic row's largest term, and converged states sit at 10^{-6} on it; the carrier row is measured against the terms THEMSELVES, where a converged row sits near 10^{-3} . Folding it into $\|R\|$ was tried first and holds the carrier row to a tolerance meant for a different scale — on the 1 microbar gate the solve then works against a target it cannot reach. So a steady solve is accepted on three gates: $\|R\| < \text{Resid_tol}$, the flux spread, and the carrier row below `carrier_resid_th` (10^{-3}), and the log names whichever is left. `carrier_resid_th` lives in `parameters.f90` and the outer loop reads the same number.
- `solve_steady_ptc` **refuses** the coupled unknown and says so: it builds its Jacobian from `frozen_residual`, which does no equilibrium sweep and so has no carrier row.

139.5 Three defects of the coupled formulation, each found by measurement

(i) **One scale vector cannot scale both the carrier row and its unknown.** The solver scales rows and columns with the same D , which is right when the residual has the units of the unknown over a time — true of every row here. It is still not enough, because the carrier row is STIFF: its natural rate is the chemical one, about 10^3 per code time at the front against the hydrodynamic rows' 10^{-6} .

carrier scale used for both	merit at the hand-off	outcome
$n(\text{H}_2) + \text{row_terms} / ((v + c_s) / dr)$ (state-like)	89.8	abort at iteration 2
$\text{row_terms} \times (R_0 / v_0)$ (the row's own terms)	2.51	abort at iteration 2
row terms for the ROW, state-like for the COLUMN	2.51	runs

With the state-like scale on the row the line search sees nothing but the carrier and lands on a state whose chemistry cannot be certified; with the row scale on the column the Newton proposes carrier steps **84 times the cell's own** $n(\text{H}_2)$ and the line search cuts to $\lambda = 4.9 \times 10^{-4}$. The fix is a two-sided scaling, `cell_row_scales` beside `cell_state_scales`.

(ii) **The constraint was moving with the trial it constrained.** The element headroom depends on the very unknown it bounds ($n_{\text{H_free}}$ counts H_2 's own two nuclei). Refreshed at every state that is not a Jacobian probe, the last line-search trial leaves its own headroom behind and the Gauss–Newton escape then finds the ITERATE outside it: every backtracked step is refused down to $\lambda = 5 \times 10^{-4}$ and the solve aborts holding a descent direction. It is now frozen at the outer iterate.

(iii) **The Gauss–Newton escape had no line search, and with the carrier row it needs one.** The escape used to evaluate the full step and nothing else, which suffices while the only way it can fail is to ascend. With the carrier

unknown a second failure mode appears: the full step LOWERS the merit ($2.28 \rightarrow 1.50$) and puts 73 cells’ molecular equilibrium outside the physical simplex, so the trial is refused and the solve aborts with a descent direction in hand. It now backtracks up to 12 times. **The three-unknown route keeps the single evaluation it has always had.**

139.6 (P51): the best iterate stops moving, and the search keeps accepting

The non-monotone (Grippe) line search compares a trial against the WORST merit of the last five iterates, so it accepts sideways steps. That is what lets the solve leave a bad neighbourhood, and it is also what lets it circle: on the coupled hand-off the best iterate is reached at outer iteration 10 and **490 further iterations produce no improvement at all** while the search accepts 61 sideways steps.

So when the best iterate has not improved for `n_stall_best` iterations the solve GOES BACK TO IT and switches to a monotone Armijo acceptance; if that buys nothing in another `n_stall_best`, it stops and says it STAGNATED, which is a different statement from “no descent direction exists”.

Where 20 comes from. Over every solve of this campaign that ended `info = 0` on a configuration that survived measurement, the largest gap between two improvements of the best iterate is **6** outer iterations (both WASP-121 b Newton cases); on the coupled hot Uranus it is 3. Two accepted solves show gaps of 68 and 85 and both are configurations that were measured and rejected. So 20 stands a factor 3.3 above anything a surviving route has needed, and 40 — where the solve gives up — stands a factor 6.7 above it and below every stall observed (tails of 278 to 729). Measured effect on the coupled hand-off: the solve now stops at **49 iterations** instead of running to its 500-iteration limit, with the same best iterate and an explicit reason.

139.7 What the coupled solve does, and where it stops

From the accepted wind ($\|R\| = 1.454 \times 10^{-4}$ on three rows), with the carrier row in the unknowns and in $\|R\|$, and both gates:

	at entry	at exit
$\ R\ $ with the carrier row folded in (the form first tried)	1.221e-2	4.073e-3
carrier row, worst cell / own terms	6.04e-1	1.36e-1
carrier row, volume-weighted	7.72e-3	4.07e-3
info	—	2, STAGNATED at 49 iterations

$\|R\|$ at entry is 1.221×10^{-2} rather than 1.454×10^{-4} because the carrier row is now in it: the wind that solve accepted was steady in three rows and not in the fourth.

It does not converge, and where it stops is not where the alternation stopped. The binding row is the carrier row, and resolving its volume-weighted sum by radius says the imbalance is **spread through the molecular layer**, not concentrated at the front the alternation could not move: the wind above $2R_p$ reads 1.17×10^{-4} , the layer 6.5×10^{-3} , and the single largest contributor is the BASE cell at $r = 1.000$ with 19 per cent of the layer’s numerator. Excluding the launch region the way the flux gate does makes the number worse, not better (1.97×10^{-2} over $r \geq 1.03$), because the

denominator is dominated by the same cells: the layer is 0.6 to 4 per cent out of carrier balance throughout, against a 0.1 per cent gate.

The front, and the three layers, against the state the solve was handed. **The returned state is four times the carrier tolerance, so this is a direction and a size, not a result:**

		hand-off	coupled solve		
$f(\text{H}_2) = 0.5$ at		1.1421	1.0618		
$f(\text{H}_2) = 10^{-2}$ at		1.2192	1.1488		
$f(\text{H}_2) = 10^{-4}$ at		1.2300	1.2327		
$\log_{10} \dot{M}$		10.413	10.412		

$r [R_p]$	$T [\text{K}]$	hand-off $\rightarrow$ coupled	$f(\text{H}_2)$	$n(\text{H}_3^+)$	n_e
1.05	711 $\rightarrow$ 867		0.943 $\rightarrow$ 0.886	9.7e4 $\rightarrow$ 3.7e4	1.18e6 $\rightarrow$ 2.60e6
1.15	871 $\rightarrow$ 766		0.445 $\rightarrow$ 9.8e-3	8.5e2 $\rightarrow$ 6.6e0	7.17e7 $\rightarrow$ 8.29e7
1.25	1567 $\rightarrow$ 1562		8.1e-6 $\rightarrow$ 5.9e-5	7.9e-4 $\rightarrow$ 5.7e-3	6.74e7 $\rightarrow$ 6.74e7
1.50	2538 $\rightarrow$ 2538		5.4e-5 $\rightarrow$ 2.5e-5	1.3e-3 $\rightarrow$ 6.2e-4	6.47e7 $\rightarrow$ 6.48e7
2.00	2691 $\rightarrow$ 2693		1.8e-5 $\rightarrow$ 7.6e-6	4.7e-5 $\rightarrow$ 1.9e-5	4.43e7 $\rightarrow$ 4.43e7
3.01	2206 $\rightarrow$ 2211		4.1e-6 $\rightarrow$ 2.4e-6	1.8e-6 $\rightarrow$ 1.1e-6	1.87e7 $\rightarrow$ 1.87e7

The wind above 1.25 R_p barely moves and the rate does not move at all (10.413 $\rightarrow$ 10.412). Everything happens between 1.05 and 1.20 R_p , and it happens in the direction the alternation could not go.

139.8 Cost

Both from the same hand-off, 20 outer iterations, OMP_NUM_THREADS=8:

	3 unknowns	4 unknowns
colours	17	23
full residual evaluations	64 (converged in 6)	778
wall	1.00 s	14.51 s

Per outer iteration, 11 residual evaluations against 39. Six of the extra come from the wider colouring; the rest are the damped Gauss–Newton escape, which now backtracks and fires in almost every iteration. **Sec. 128.’s cost objection is not what stands in the way:** the alternation spends about 1 s in each of 20 passes and does not converge; the coupled solve spends 14.5 s and reaches a residual four times the tolerance with a qualitatively different front.

139.9 What was checked

- make check, **10 of 10 byte-identical** — wasp_full, wasp_he23off, mol_base_handoff, mol_metals, mol_lyman_werner, mol_diffusion, mol_ir_bands, mol_sec_ion, mol_carrier, lower_profile — run

twice, once with the alternation change alone and once with the coupled route on top of it. **And that had to be worked for.** The two-sided row scaling is the SAME arithmetic when there is no carrier row — Drow holds exactly D 's values — but not the same instruction sequence, and a reduction summed in a different order differs in its last bits: without a branch, `wasp_full_newton` and `wasp_he23off_newton` moved by 1.2×10^{-9} and 4.0×10^{-8} relative, with $\log_{10} \dot{M}$ unchanged to six decimals and, on the first, an identical iteration log. **The run-time band geometry is not the cause and that was tested:** a build with `kl_jac` and the rest back at compile time is bit-identical to the run-time one. Every site that divides by the row scale therefore branches on `nvar_jac`, with the ORIGINAL expression character for character in the three-unknown branch.

- The carrier arrays are allocated only when the coupled route is on, and Drow last and on its own, so every array the three-unknown solve has always had keeps the address it has always had.
- **NOT checked:** the coupled route on any planet but the He/H = 0.0793 hot Uranus, on any state but that hand-off, or with helium diffusion also on.
- **NOT checked:** whether the band half-width 11 is exact rather than sufficient. It is derived, and no column-by-column comparison against a dense Jacobian was made.
- **The 1 microbar gate (`mol_carrier`) was run, and it does not produce a quotable steady state either — for a reason that is about the case, not about the solver.** That case is a 12000-step relaxation SNAPSHOT, not a converged wind: restarted from its own output with the coupled solve on, the marching takes 21000 further steps to reach the JFNK hand-off and the solve then stagnates at $\|R\| = 2.30 \times 10^{-3}$ with the carrier row binding at 2.30×10^{-3} and, decisively, a **flux spread of 0.244 against a gate of 5×10^{-3}** . The wind is not carrying one mass flux at every altitude, so no gate can be met and no three-layer table from it would describe a steady state. A quotable coupled solution needs a converged wind on that planet first, which is the same thing item (P43) has been waiting for.
- **NOT checked:** the Eld hand-off in the tree. It is a 9 microbar state whose `base.inp` and `Log10_lower` boundary disagree, and Sec. 136. made `p_base` authoritative, so the tree would set up a different base than the campaign's pre-136 working copy did; the comparison would not be like-for-like.

Files: `src/modules/lower_atmosphere/diffusive_photochemistry.f90`, `src/modules/time_step/steady_newton.f90`, `src/EXHALE_main.f90`, `src/modules/init/parameters.f90`, `src/modules/files_I0/input_read.f90`. Documentation: `docs/p50_carrier_wind_alternation.md`, `docs/input_schema.md`, `docs/molecular_hydrogen_treatment.tex`, `TO_BE_DONE.md` ((P43) closed for the alternation; (P51) recorded).

140. Six ways a run could not say why it stopped, and an acceptance test that existed in two versions (2026-09-03)

The judgement first, in three parts. (1) The marching loop's `Resid tol stop` wrote its verdict into the `du` flag, so a run that stopped on the residual printed converged: `momentum constant (du < du_th)` at $du = 3.156 \times 10^{-3}$

with $du_{th} = 10^{-3}$: the reported reason and the reason that fired were different statements and only one of them was true. (2) There were two definitions of an accepted steady state — `steady_gates_met` (the residual *and* the flatness of the mass flux, Sec. 133.) inside the JFNK, and the residual alone in the marching loop — so for the same `Resid tol` key the marching accepted a WASP-121 b state whose flux spread was 9.879×10^{-2} , twenty times the default threshold, while the JFNK was refusing 1.874×10^{-2} . (3) A key stated twice was resolved by file order with nothing saying so, and one file in the repository was stated twice. All three are fixed; the acceptance test now exists once, in `steady_residual.f90`, and every route that may stop a run on “the equations are satisfied” calls it.

This closes item (P52) of `TO_BE_DONE.md`.

140.1 The stop that was reported was not the stop that fired

`EXHALE_main.f90` carried three stop flags — `is_mom_const`, `is_zero_dt`, `is_stalled` — and printed a fixed sentence for each. The `Resid tol` branch had no flag of its own and borrowed the first, `is_mom_const = is_resid_ok`, so the run’s own log said $du < du_{th}$ about a stop that had nothing to do with du . Measured on `resid_on` (WASP-121 b, `Resid tol: 1.0e-4`, `du_th [PLM,WEN03]: 0.5 1.0e-3`) before the change, the loop stopped at step 14 000 printing

```
-> converged: momentum constant (du < du_th)
final: count=14000 du= 3.1560E-03 dtu= 2.4942E-05
```

with du three times its own threshold, because what had happened was $\|R\| = 8.066 \times 10^{-5} < 10^{-4}$.

Every stop condition now has its own flag and its own line, and the flags are named for what they measure rather than for the order they were added in:

flag (<code>parameters.f90</code>)	the condition it holds
<code>mass_flux_converged</code>	$du < du_{th}$: the radial spread of $\rho v r^2$ over the wind
<code>step_change_converged</code>	$dtu < dtu_{th}$: the largest relative change of u in one step
<code>du_plateaued</code>	du settled on a plateau (the stall net)
<code>steady_gates_converged</code>	<i>both</i> steady gates met on the marched state
<code>hit_max_steps</code>	the <code>EXHALE_MAXSTEPS</code> cap
<code>newton_finished</code>	the JFNK finish returned <code>info = 0</code>

`newton_finished` used to set `is_mom_const` to leave the loop; it is now a stop condition of the loop in its own right, so nothing has to borrow a flag to exit. The report tests the flags in the order the loop leaves on them, with the NaN scan first because a NaN invalidates any convergence a flag may also be carrying.

140.2 One acceptance test, in one place

`steady_gates_met` — $\|R\| < \text{resid_tol}$ *and* the mass-flux spread over $r \geq r_{\text{flux}}$ below `flux_spread_th`, Sec. 133. — was a private function of `steady_newton.f90`, so only the steady solvers could call it and the marching loop’s

Resid tol stop tested the residual alone. **The two routes were reading the same user key and applying different acceptance criteria.** On resid_on the state the marching accepted at step 14000 had a gate-window flux spread of 9.879×10^{-2} — the diagnostic line was already printing it — against a default Flux spread tol of 5.0×10^{-3} and against the 1.874×10^{-2} the JFNK is on record refusing on jfnk_hd189_tight.

The function moves to steady_residual_mod (steady_residual.f90), beside the residual and the flux functional it combines, because neither solver owns the definition. steady_newton.f90 imports it; the marching monitor calls it at the same N_resid cadence it already used, in place of its bare flux_spread_of_state call. Where a Resid tol run stops without meeting the pair, the log names the gate that is left, in the same terms the JFNK uses:

```
-> stopped: du plateau detected (stalled 500 steps) with the steady gates NOT met
    the residual gate is met: ||R|| 7.153E-05 < 1.000E-04
    the FLUX gate is the one left: spread 4.218E-02 >= 5.000E-03
```

140.3 The step cap is a stop, and now says so

EXHALE_MAXSTEPS=N leaves the loop from inside the body, so none of the three flags was set and the count_max branch tested only the hard 10^6 cap: a run that ended on the cap printed no stop line at all, going straight from Time integration done. to final: count=.... Every fixed-step case in the regression matrix ends this way, and so does wasp_localdt_cont. The exit now sets hit_max_steps and the report prints -> stopped: reached EXHALE_MAXSTEPS = 12000 steps WITHOUT convergence, with the unmet gates listed after it when the run asked for Resid tol. The Do only PP exit and the NaN exit likewise now name themselves.

140.4 A key states one quantity

The parser resolved a repeated key to its LAST occurrence. That is a rule about file order, not about intent: the run proceeds on a value nobody chose and nothing in its log says a choice was made. backup/regression/valve_sens/eps5/input.inp carried Resid tol twice, and its own stored run.log shows the parser echoing both and the solve taking the second (tol = 1.00E-03, then tol = 1.00E-04).

refuse_duplicate_keys (input_read.f90) now stops the run before any line is consumed, printing both line numbers and both values. Each non-blank, non-# line is attributed to the LONGEST key it matches, because the key set is not prefix-free (Lower atmosphere is a prefix of Lower atmosphere profile, Wind-AE seed of Wind-AE seed out) and the longest match is the key the parser itself consumes. The same routine covers base.inp. metals.inp gets its own check, keyed on the QUANTITY rather than on the token, because there the labels alias: C and CI both set X_C, so a token-level test would miss the pair that matters.

The repository was swept for the same defect. All 1889 settings files under the repository were attributed key by key with the parser's own matching rule (1126 input.inp, 650 base.inp, 113 metals.inp, including the 942 stored LHS1140b/ inputs and the 32 under vulcan_work/). **Exactly one file carried a duplicate**, the valve_sens/eps5 above, and it is repaired here. Which of its two values was meant is settled by its own run log, not by preference: the solve ran on the second, 10^{-4} on the pre-Sec. 133. scale, which the renormalization there makes 10^{-6} .

140.5 The direct steady route printed none of the run's counters

EXHALE_PTC=1 solves from the initial condition and stops. The run-counter ledger — the ionization-solver acceptance report, the energy-floor, positivity-limiter and dt -bisection counts, the base-inflow Mach, the He-recombination coupling warning, the low-Mach damping magnitude — sat inline in the marching route below its stop line, so this route reported none of it. It is now one contained routine, `write_run_counter_report`, and both routes call it: the marching loop where the block used to be, and the direct route before its stop. Measured on a WASP-121 b PTC/JFNK solve from the `ptc_warm` initial condition, the same build with and without the call: the outputs are byte-identical and the run gains 846 equilibrium sweeps' worth of acceptance statistics it was discarding.

A second diagnostic-only defect travels with it. `write_heat_breakdown_eq(util_ion_eq.f90)` called `PH_heat_HHe` without `f_vibq/e_vibq`, the two arrays carrying the collisional quench fraction of an H_2 vibrational excitation and the internal energy a B/C fluorescence leaves behind. They are optional arguments, so their absence switched the two H_2 vibrational heat channels off silently and only in the dump — the solver's own call in `ionization_equilibrium.f90` has always passed them. Measured on `mol_base_handoff`, the dump's photoheating total rises by up to a factor 1.97 (at $r = 1.0002 R_p$), with `heat_H2` nearly doubling there, and by 0.49% volume-integrated. `Hydro_ioniz.txt` and `Ion_species.txt` are untouched.

140.6 Regression-directory hygiene, and the numbers of Sec. 131.

`backup/regression/valve_sens/eps4` holds two output files and no `input.inp`, so it is not a case any harness can run; nothing in `run_check.sh` or in `docs/` refers to it. It is left as found with a `NOTE.txt` recording that, checked 2026-09-03. Three pairs of case directories are the same case under two names, with byte-identical `input.inp` and `metals.inp`: `crit_warm = wasp_hybrid_finish`, `ptc_wasp = resid_on`, `solver_newton_cold = wasp_full_newton`. Each of the six gets a `NOTE.txt` recording the identity; none is deleted or merged, because which name to keep is a judgement about what the stored outputs are records of.

The rates quoted in Sec. 131. are corrected. They were 30-step probes reported as though they were the cases' results. Run to their own stopping criteria:

case	stop	$\log_{10} \dot{M}$
<code>wasp_hybrid_finish = crit_warm</code>	$du = 9.9997 \times 10^{-4} < du_{th}$, step 22 935	13.23
<code>wasp_localdt_cont</code>	the 40 000-step cap, $du = 4.97 \times 10^{-5}$	13.36

The second row is not a converged rate, and until this section its run printed no line saying so.

140.7 What was checked

- **The ten regression cases are byte-identical, and to two references.** The tree was being edited concurrently, so the comparison is an A/B between two builds made from *one* frozen snapshot of it: the snapshot as it stood, and the same snapshot with the six files of this section changed and nothing else. All ten agree with the control bit for bit in `Hydro_ioniz.txt` and `Ion_species.txt`, and all ten agree with `backup/regression/golden/` as well.

What changed is the stop line: `wasp_full` and `wasp_he23off` now print converged: `mass flux rho*v*r^2` constant over the wind (`du= 9.999E-04 < du_th 1.000E-03`), and the eight fixed-step cases print stopped: `reached EXHALE_MAXSTEPS = 12000 steps WITHOUT convergence` where they used to print nothing at all.

- **resid_on no longer stops at step 14 000, and what it does instead is worth having.** With both gates it marches on, and the staged secondary-ionization flip — which fires on the first stop condition the loop reaches — now fires at step 37 500 on a state with $\|R\| = 3.808 \times 10^{-6}$ and a flux spread of 4.793×10^{-3} , i.e. on a state that meets *both* gates, instead of at step 9500 on one that meets the residual alone. The run then reaches the 40 000-step cap and reports it. Over the 80 monitor evaluations the residual gate is met at 52 and the pair at exactly one; the flux spread at those 52 ranges over 4.793×10^{-3} to 1.290×10^{-1} , so **the residual gate on this planet says nothing about whether the mass flux is flat**. $\log_{10} \dot{M}$ is 13.26 either way: the premature stop was not costing accuracy here, it was costing the statement.
- **The other two Resid tol cases, both under a 40 000-step cap.** `ptc_warm` stops on the cap with $\|R\| = 1.653 \times 10^{-5} \geq 10^{-5}$ and spread $1.070 \times 10^{-2} \geq 5 \times 10^{-3}$ ($\log_{10} \dot{M} = 13.23$); `jfnk_hd189` with 3.865×10^{-5} and 2.304×10^{-1} (9.93). Neither reaches both gates by marching alone, and both now say so in one line each.
- **The duplicate-key refusal, on all three file kinds.** The unrepaired `eps5` stops naming lines 33 and 36; a second `T_base` in a `base.inp` stops naming lines 32 and 38; a CI line beside an existing C line stops on `X_C`, which is the alias case a token-level check would miss. The repaired `eps5` parses and reports `tol = 1.00E-06`.
- **The heat-breakdown dump, and what it does NOT close.** Its own closure stays exact ($\max |\sum \text{channels}/\text{heat_photo-1}| = 1.8 \times 10^{-15}$). What the fix does *not* do is make the dump agree with the `Hydro_ioniz.txt` heat column on the molecular gates: that ratio is 0.23 on `mol_base_handoff` before *and* after, because those cases are 12 000-step relaxation snapshots and the dump re-solves the equilibrium at the final (T, ρ) while the heat column is the marching state's own. On the converged `wasp_full` the same measurement gives 1.0000 at every cell, which is what the dump's header already claims ("up to convergence"), so a 1.000 closure on a molecular case has to be measured on a converged one.
- **Runtime checks.** A `-fcheck=bounds,do,mem` rebuild of the changed tree and a 1500-step bounded HD 209458 b run: **clean**, $\log_{10} \dot{M} = 9.60$, and the run now prints stopped: `reached EXHALE_MAXSTEPS = 1500 steps WITHOUT convergence`.
- **NOT checked:** whether any `Resid tol` case reaches both gates given more than 40 000 marching steps (all three were capped); the remaining `Resid tol` cases (`jfnk_cold`, `jfnk_hd189_tight`, `tpm_*`); `ifort / ifx` builds; and the 932 stored `LHS1140b/exhale/` inputs, which were scanned for duplicate keys and have none but were not re-run.

Files: `src/modules/time_step/steady_residual.f90`, `src/modules/time_step/steady_newton.f90`, `src/EXHALE_main.f90`, `src/modules/init/parameters.f90`, `src/modules/files_IO/input_read.f90`, `src/modules/files_IO/`

metals_input_read.f90,src/modules/radiation/util_ion_eq.f90,backup/regression/valve_sens/eps5/input.inp.
Documentation: docs/input_schema.md, TO_BE_DONE.md ((P52) closed, (P53) recorded).

141. The adiabatic index of the molecular layer, and the chemical heat the network releases (2026-09-03)

The judgement first, in two parts. (1) The code carried a two-part equation of state whose halves disagreed: the thermal half has been composition exact for a long time, $p = (n_{\text{tot}} + n_e)kT$ with the species count of the cell, but the caloric half, $e = p/(\gamma - 1)$, used the monatomic $\gamma = 5/3$ everywhere. That asserts that every particle stores $\frac{3}{2}kT$; H_2 at 400–1300 K stores about $\frac{5}{2}kT$, so where the gas is 80 per cent H_2 the specific heat was wrong by a factor 1.67. (2) Looking for the thermal H_2 dissociation-energy sink that `lyman_werner.f90` records as an open item turned up something larger: none of the COLLISIONAL reactions of the H_2/He network deposits its chemical energy, and on the converged $\text{He}/\text{H} = 0.0793$ rung that sum is 81 per cent of the total heating rate at $1.02 r_{\text{base}}$. Both are fixed. The caloric EOS is structural, so an atomic run is byte-identical; the reaction heat defaults ON, because in a molecular gas the ionization energy comes back to the GAS through dissociative recombination instead of leaving as a photon.

This closes item (Z) of `TO_BE_DONE.md` and opens item (AB). The census of every place γ entered the code, the sizing experiment and the design are in `docs/p53_h2_adiabatic_index.md`.

141.1 The rename

The polytropic index was the single-letter global `g` (`parameters.f90`), in a module almost every other module uses whole, and five routines already declared a local or dummy `g` that shadowed it: the merit gradient of `levenberg_marquardt_step` (whose `mu` shadowed the global hydrogen-atom mass as well), a Jacobian entry in `charge_exchange`, the local gravitational acceleration and mean molecular weight in `lower_column`, a rate-coefficient function result in `h2_vibrational_relaxation` and a table abscissa in `h2_self_shielding_table`. None was a bug, but this is the shadowing class that cost hours twice before (`t0/T0`, `G2s/g2s`).

The global is now `gamma_ad` and the five locals are named for what they hold: `grad_merit`, `mu_damp`, `dR_dn`, `grav_accel`, `mu_mean`, `gamma_deex`, `axis`. The whole regression matrix is byte-identical across the rename.

141.2 The caloric equation of state

`src/modules/states/caloric_eos.f90` holds the map between internal energy density, pressure and temperature for a mixture in which every atom, ion and free electron carries $\frac{3}{2}k$ and H_2 carries its own rovibrational heat capacity:

$$e = \frac{3}{2}(n_{\text{tot}} + n_e)kT + n(\text{H}_2)ku_{\text{rv}}(T), \quad \gamma_{\text{eff}} = 1 + \frac{n_{\text{tot}} + n_e}{\frac{3}{2}(n_{\text{tot}} + n_e) + n(\text{H}_2)c_{\text{rv}}(T)}. \quad (47)$$

u_{rv} and c_{rv} are evaluated from the COMPLETE bound rovibrational ladder of $\text{H}_2 X^1\Sigma_g^+$ — 302 levels up to 51 966 K, the 4.478 eV dissociation limit — which the code already carried in `molecular_infrared_data` (Roueff et al. 2019,

A&A 630, A58, table 2, via VizieR J/A+A/630/A58). Using the observed ladder is not a refinement for its own sake: it is the only H_2 level data in the tree, so there is no second copy of the spectroscopic constants to drift, and a rigid rotor plus a harmonic oscillator gives $c_v(\text{H}_2)/k$ smaller by 1.4 per cent at 200–900 K, 2.3 per cent at 1300 K and 7.3 per cent at 3000 K. Because the sum runs over one partition function with the nuclear-spin weights inside it, this is the ortho/para EQUILIBRIUM mixture; a frozen 3 : 1 mixture agrees to four decimals at and above 300 K.

H_2^+ , H_3^+ and HeH^+ are counted with $\frac{3}{2}k$ each. Measured over every cell where H_2 exceeds 10^{-3} of the particle count ($r = 1.000$ to $1.101 r_{\text{base}}$), the largest ratios to H_2 are 1.7×10^{-7} , 1.2×10^{-7} and 3.2×10^{-8} , so giving all three the full H_2 heat capacity would move C_V by less than 10^{-6} .

The table, and why the inverse is solved rather than lagged. The 302-level sum costs one exponential per level per evaluation, so u_{rv} is tabulated once on a 4096-point logarithmic grid from 1 K to 5×10^4 K and interpolated by a CUBIC HERMITE in $\ln T$ built on exact nodal values of both u_{rv} and $du_{\text{rv}}/d\ln T$, so the interpolant is C^1 and its own derivative is exact at the nodes. Against the direct sum over 200–5000 K the largest relative error is 10^{-12} in u_{rv} and 9×10^{-10} in c_{rv} . With the vibrational ladder in c_v the map from internal energy to temperature is a genuinely nonlinear scalar equation; it is inverted to full precision at every conversion by a bisection-safeguarded Newton on the bracket $0 < T \leq u/1.5$, NOT lagged at a previous temperature, because a residual that depended on the previous state would blur the fixed point of the steady solver and the finite-difference Jacobian alike. The round trip $T \rightarrow u \rightarrow T$ is exact to 1.1×10^{-15} over 24 000 points and six H_2 fractions.

Where the local index is used, and where it is not. Every $E \leftrightarrow p$ conversion: UW_conversions, Source, the HLLC and Lax–Friedrichs total energies and the physical flux in Num_Fluxes, the low-Mach dissipation pressure, the heating/cooling temperature update in energy_semi_implicit, the heat capacity of the conduction tridiagonal and the two energy rebuilds in viscous_conduction, the JFNK state scales in steady_residual, and the post-process energy equation T_equation. The sound speeds of eval_dt, the base Mach diagnostic of Apply_BC and the carrier-row signal rate of diffusive_photochemistry use each cell’s own γ_{eff} .

The reconstruction needs no index: it works in primitive variables. Each face state’s energy and sound speed are built with the heat capacity of the cell it was reconstructed from, which is well defined because both stored composition quantities are ratios of number densities to ρ — functions of the mass fractions alone — and so are unchanged by the density difference between a cell average and its face value. The one place a single index is still needed is the Roe average $a_{\text{avg}} = \sqrt{(\gamma - 1)(H_{\text{avg}} - v_{\text{avg}}^2/2)}$ and the adaptive p_* estimate of speed_estimate_ROE, both derived for one ideal gas on both sides; they are given the arithmetic mean of the two face values. That is a WAVE-SPEED ESTIMATE, not a definition of the steady state: the converged solution is the root of a residual assembled from the conservative fluxes, and those carry each side’s own EOS. The HLLC branch needs no such average, its Davis–Einfeldt speeds taking one sound speed per side already.

One module was deleted on the way. speed_estimate_HLLC (Toro’s adaptive p_* estimate) was used by Num_Fluxes and never called. It has been removed with its use and its SRC line; nothing links to it, so the change is byte-identical.

Byte identity is structural, never a limit of the formula. With $n(\text{H}_2) = 0$ the mixture expression evaluates $1 + 1/1.5$, which is not the same double as $5/3$. Two gates: a run with molecular chemistry off leaves every map on the legacy expression verbatim, and inside a molecular run a cell with exactly zero H_2 takes the same legacy expression.

141.3 What it does to the answer

The measurement is the converged He/H = 0.0793 hot-Uranus rung ($p_{\text{base}} = 10^{-6}$ bar, $q_{\text{H}_2, \text{base}} = 0.84$, molecular chemistry on, metals off), restarted from its own converged output with Load IC? True, Solver: Newton 100.0, du_th [PLM, WENO3]: 1.0e9 1.0e-3, Secondary_ionization: Immediate. **All three runs reached info = 0**; nothing below is a relaxation snapshot.

	A: $\gamma = 5/3$	B: caloric EOS	C: B + reaction heat
JFNK info	0	0	0
$\ R\ $	1.494e-6	6.572e-6	2.031e-6
flux spread ($r > 1.2$)	2.966e-3	4.638e-3	2.106e-3
$\log_{10} \dot{M}$ [g s ⁻¹]	10.30	10.30	10.34
$\Delta \log_{10} \dot{M}$ vs A	—	+0.003	+0.046
layer T minimum [K]	410.2	417.5	525.2
its radius [r_{base}]	1.0622	1.0639	1.0701
H ₂ front, $f_2 = 0.5$	1.0418	1.0420	1.0429
H ₂ front, $f_2 = 0.1$	1.0590	1.0598	1.0633
T at the $f_2 = 0.5$ front [K]	532.0	545.6	632.4
H ₃ ⁺ peak radius	1.0048	1.0048	1.0029
H ₃ ⁺ peak density [cm ⁻³]	1.771e5	1.755e5	1.469e5

r/r_{base}	1.05	1.10	1.15	1.20	1.30	1.50	2.00	3.00
T (A) [K]	457	667	1250	1722	2320	2766	2668	2120
ΔT (B–A) [K]	+16.5	–13.9	–17.7	–15.3	–10.7	–5.3	–2.1	–1.9
ΔT (C–A) [K]	+130.7	–43.9	–193.2	–211.7	–172.4	–74.3	+19.0	+43.2
γ_{eff} (B)	1.601	1.6661	1.6666	1.6666	1.6667	1.6667	1.6667	1.6667

Reading. The caloric EOS alone is small and confined to the layer: $\dot{M}$ moves by +0.003 dex (0.7 per cent) and T by at most 18 K, warmer at the base and cooler by 1.4 per cent through 1.10–1.30, with the H₂ front unmoved to 0.02 per cent. **The term that responds is the advection term**, exactly as item (Z) predicted: $p \text{ div } v$ rises by 5 per cent at 1.06 to 26 per cent at 1.01, while the advection term, which carries $u = e/\rho$, changes SIGN over 1.03–1.06 (at 1.04, from $+4.7 \times 10^{-9}$ to -2.5×10^{-8}). Above the front, where γ_{eff} is 5/3 to four decimals, nothing moves.

Against the sizing experiment. A whole-domain $\gamma = 7/5$ put $\log_{10} \dot{M}$ at $10.30 \rightarrow 10.23$ (a factor 0.85) and T down by up to 97 K, and was called an upper bound. The local treatment is inside that bound by a factor 20 in $\dot{M}$ and 5 in T , and *it does not even share its sign for $\dot{M}$* . The bound held; the extrapolation from it did not, because the wind’s energetics are set at and above the sonic transition, where γ_{eff} is exactly 5/3, so a global hack moves a region the correct treatment leaves alone.

141.4 The chemical heat the molecular network releases

What the code already deposited. `ionization_equilibrium` adds the photoelectric heating of H I, He I, He II, He(2³S), H₂ and the metals; the Balmer photoelectric heating of H(*n* = 2); the He-recombination coupling; the He(2³S)+H Penning exothermicity (6.2 eV); the He(2³S)+H₂ Penning exothermicity (4.4 eV); the Lyman–Werner fragment kinetic energy (0.4 eV per pair) and the fluorescence heat; and the FUV photolysis heat of H₂O and OH. `eval_cool` carries the radiative-recombination and collisional-ionization sinks. **No collisional reaction of the molecular network deposited anything.**

Why that is not a small omission, and why the default is ON. Follow one closed cycle a photon drives:

```
H2 + hv -> H2+ + e    photon pays I(H2) = 15.43 eV; the photoelectron keeps
                        hv - I(H2), which PH_heat_HHe already deposits
H2+ + H2 -> H3+ + H    + 1.70 eV
H3+ + e -> H2 + H      + 9.25 eV, DISSOCIATIVE: into the fragments, not a
                        photon
```

The cycle returns to H₂ having turned one H₂ into H+H and left $I(\text{H}_2) - D_0(\text{H}_2) = 15.43 - 4.48 = 10.95 \text{ eV}$ in the gas. There is no photon anywhere in it to carry that away, and that is exactly what differs from the atomic case, where the ionization energy leaves as the Lyman photon of a radiative recombination and is therefore correctly absent from the heating. Sized on the converged constant- γ state, the sum of the collisional reaction heats is 2.37 times the whole photoelectric heating rate over 1.00–1.05 r_{base} , crossing 1 at 1.053 and falling to 1 per cent by 1.10; solved self-consistently, the run's own `Heating_breakdown.txt` puts the new `heat_mol_chem` column at **81 per cent of the total heating rate** at 1.02 r_{base} , 55 per cent at 1.05, 5 per cent at 1.10 and 1 per cent above. So `Molecular reaction heat:` defaults to True and the key exists to turn the term OFF for A/B work; an atomic run is gated out (`thereis_mol`) and is byte-identical either way.

The 4.478 eV term alone would not have been right. Sized on the same state, $D_0(\text{H}_2) \times (\text{three-body formation} - \text{collisional dissociation})$ is 1.06 times the whole photoelectric heating in the molecular base, and every other molecular channel is 4 to 45 per cent of it by rate (R8 0.45, R7 0.44, R6 0.19, R9 0.17, R20 0.15, R13 0.13, R19 0.12, R18 0.046, R17 0.039), so the term the open item asked for is not separable from the rest of the network.

How it is built. `src/modules/lower_atmosphere/molecular_reaction_heat.f90` computes the heat from ONE table of species enthalpies — the energy to build each species out of ground-state H, ground-state He and free electrons at rest — so that the heat of $A + B \rightarrow C + D$ is $h(A) + h(B) - h(C) - h(D)$ and a cycle that returns to its starting species releases exactly zero. A reaction-by-reaction list has no such property. Excluded, so that nothing is counted twice: the photon-driven reactions (H₂ photoionization and its dissociative branch, Lyman–Werner photodissociation, the H and He photoionizations), the radiative recombinations of H⁺ and He⁺, the collisional ionizations of H and He, the Penning channels, and the H/He charge exchange. The three-body association carries Hollenbach & McKee's (1979, ApJS 41, 555, p. 586) collisional branching $(1 + n_{\text{cr}}/n)^{-1}$, evaluated by the module that owns n_{cr} ; in this layer that factor is 1 to a part in 10³.

constant	value	checked against
$D_0(\text{H}_2)$	36118.11 cm^{-1}	not repeated: read through mol_rates, the code's single definition (Huber & Herzberg 1979)
$I(\text{H})$	$13.598434599702(12) \text{ eV}$	NIST Atomic Spectra Database
$I(\text{He})$	$24.587389011(25) \text{ eV}$	NIST Atomic Spectra Database
$I(\text{H}_2)$	$15.42593(5) \text{ eV}$	NIST Chemistry WebBook
$D_0(\text{H}_3^+ \rightarrow \text{H}_2 + \text{H}^+)$	$35076 \pm 2 \text{ cm}^{-1} = 4.3489 \text{ eV}$	read in the PUBLISHED text of Mizus, Polyansky, McKemmish & Tennyson (2019), Mol. Phys. 117, 1663 (references/Mizus_2019MP_117_1663.pdf)
$D_0(\text{HeH}^+ \rightarrow \text{He} + \text{H}^+)$	14882.16 cm^{-1}	DERIVED from the published text of Coxon & Hajigeorgiou (1999), J. Mol. Spectrosc. 193, 306 (references/Coxon_1999JMS_193_306.pdf): their Table 9 $D_e(^4\text{HeH}^+) = 16448.84$ minus their Table 8 $G_0(^4\text{HeH}^+) = 1566.6764$

$D_0(\text{H}_3^+)$ is a 0 K dissociation energy and deliberately not the proton affinity of H_2 , which NIST gives as $422.3 \text{ kJ mol}^{-1} = 4.377 \text{ eV}$ at 298 K — the wrong quantity for a 0 K ledger, and 0.028 eV away from the right one.

The HeH^+ constant had to be corrected, and that is why the citation matters. Coxon & Hajigeorgiou determine a D_e , never a D_0 : $16456.24 \pm 0.1 \text{ cm}^{-1}$ for the Born–Oppenheimer potential and 16448.84 for the $^4\text{HeH}^+$ isotopomer. The value $14874.215 \text{ cm}^{-1}$ that the secondary literature attributes to them appears nowhere in the paper. What the paper supports is $D_0 = D_e - G_0$ with $G_0 = 1566.6764 \text{ cm}^{-1}$ from their Table 8, i.e. 14882.16 cm^{-1} , and their own sentence fixes the channel: “Since the dissociation limit of the ground state of HeH^+ is $\text{He} + \text{H}^+$ ”. The difference from the quoted number is 7.95 cm^{-1} ($9.9 \times 10^{-4} \text{ eV}$, 0.053 per cent); it moves $h(\text{HeH}^+)$ and the four HeH^+ reaction heats by that much, which is bitwise visible and was folded in before the golden refresh. The converged measurement of Sec. 141.3 was re-run with the corrected constant and did not move: $\text{info} = \emptyset$, the same $\|R\|$, flux spread, $\log_{10} \dot{M}$, layer temperature minimum and H_2 front, and the same ΔT against the control to 0.1 K at every radius in the table.

What it does, on the converged rung (column C above). The photoheating rate at $1.02 r_{\text{base}}$ goes from 5.6×10^{-8} to $1.9 \times 10^{-7} \text{ erg cm}^{-3} \text{ s}^{-1}$, a factor 3.3; the layer temperature minimum rises $410 \rightarrow 525 \text{ K}$; 1.15 to $1.50 r_{\text{base}}$ falls by 74 to 212 K; $\log_{10} \dot{M}$ rises by 0.046 dex; the H_2 front moves out from 1.0418 to $1.0429 r_{\text{base}}$ and the H_3^+ peak density falls by 17 per cent. The run converges to $\text{info} = \emptyset$ with a flux spread BETTER than the control's (2.1×10^{-3} against 3.0×10^{-3}). Neither change is a candidate for the Koskinen temperature gap in the direction it needs: the caloric EOS moves the layer minimum by +7 K, and the reaction heat moves it by +115 K while taking $1.15\text{--}1.50 r_{\text{base}}$ DOWN by 74–212 K, so it is a redistribution rather than a closing of the gap.

141.5 A lag the constant index was hiding

The steady residual fills the ghosts (Apply_BC) BEFORE its ionization sweep runs, so the ghost conservative state it writes carries the composition of the previous sweep while the fluxes assembled afterwards carry this one. Under a

constant index the primitive-to-conservative round trip returns the pressure the boundary condition asked for and the two cannot disagree; under the caloric EOS they can, by as much as the ghost composition moved in one sweep. The gap vanishes at the fixed point and is largest during relaxation.

It is not repaired here. The obvious fix — a second `Apply_BC(u)` after the last composition refresh — is not free, because `Apply_BC` is not idempotent even today: its interior round trip evaluates $0.5 * W(1) * W(2) ** 2$ where the input was $0.5 * u(2) * u(2) / u(1)$, and those are not the same double, so a second call perturbs every interior cell at the last bit and moves the whole atomic regression matrix — the byte-identity gate this change is accepted on. The lag is stated at the head of `caloric_eos.f90` and recorded as `TO_BE_DONE.md` item (AB). `n_part_cell1` already had the same lag and the code already said so; what changed is that it now reaches the ghost pressure itself. The marching loop does not have it: there `Apply_BC` follows the refresh.

141.6 The golden refresh

Both columns come from the SAME binary: the right-hand one is the default, the left-hand one is that binary with `Molecular reaction heat: False` appended to each input, i.e. stage 2 alone. Each entry is $\Delta \log_{10} \dot{M} / \max |\Delta T| / T$ over the physical cells / the largest ΔT below $1.2 r_{\text{base}}$, against the goldens in place before this change (preserved as `backup/regression/golden_blockG_20260903`).

case	stage 2 only	default (stage 2 + reaction heat)
<code>wasp_full</code>	byte-identical	byte-identical
<code>wasp_he23off</code>	byte-identical	byte-identical
<code>lower_profile</code>	byte-identical	byte-identical
<code>mol_base_handoff</code>	+0.0143 / 3.5e-1 / -65 K	+0.0168 / 3.7e-1 / -65 K
<code>mol_metals</code>	+0.0159 / 1.9e-1 / -73 K	+0.0183 / 2.0e-1 / -73 K
<code>mol_lyman_werner</code>	-0.0349 / 4.4e-1 / -40 K	-0.0323 / 4.5e-1 / -39 K
<code>mol_diffusion</code>	+0.0143 / 3.5e-1 / -65 K	+0.0168 / 3.7e-1 / -65 K
<code>mol_ir_bands</code>	+0.0159 / 1.9e-1 / -73 K	+0.0183 / 2.0e-1 / -73 K
<code>mol_sec_ion</code>	-0.0003 / 4.4e-1 / -57 K	+0.0025 / 4.4e-1 / -57 K
<code>mol_carrier</code>	+0.0082 / 4.6e-1 / -68 K	+0.0111 / 4.7e-1 / -68 K

The largest relative temperature change always falls at the OUTERMOST cells, where the two builds sit at different points of the same transient; the layer column is the one that means something.

The two columns are close only because of what these cases are. Every `mol_*` case is a pinned-step relaxation snapshot, so it stops long before a change in the heating rate can restructure the layer. The reaction heat is 81 per cent of the base heating rate on a CONVERGED state and moves that state's layer temperature minimum by +115 K (Sec. 141.4); here it adds about +0.003 dex and no measurable ΔT . These numbers are the golden refresh, not the physics.

141.7 What was checked, and what is still open

Regression: `wasp_full`, `wasp_he23off` and `lower_profile` are byte-identical against the previous goldens both with the reaction heat on and with it off (`lower_profile` never sets `Molecular chemistry: True`, so the module-level gate never opens); the seven `mol_*` cases move and their goldens were refreshed, with the previous block preserved as `backup/regression/golden_blockG_20260903`. Neither the Roe nor the Lax–Friedrichs branch is in the matrix — every case runs HLLC — so `wasp_full` was run for 300 steps under each against the pre-change build: byte-identical on both output files for both, which is also the test of the deleted module. The Roe branch was then run for 400 steps on `mol_base_handoff`, where the two face values genuinely differ: clean, no NaN. A `-fcheck=bounds,do,mem` build ran 600 steps each of `wasp_full`, `mol_base_handoff` and `mol_base_handoff` with the reaction heat off, with no trap. Still open: nothing on the constants — both dissociation energies were read in the publisher PDFs — except that $D_0(\text{HeH}^+)$ is DERIVED from its paper rather than printed in it, so a later determination would mean redoing the derivation; the associative branch of the $\text{He}(2^3\text{S})$ ionizations, $\text{He}(2^3\text{S})+\text{H} \rightarrow \text{HeH}^++\text{e}$ (8.1 eV), left out because the metastable density in the layer is $2 \times 10^{-7} \text{ cm}^{-3}$, which makes the channel 10^{-13} of the sum; the $\text{O}(^1\text{D})+\text{H}_2 \rightarrow \text{OH}+\text{H}$ exothermicity, already recorded as an omission elsewhere; item (AB) above.

142. P51: the coupled solve's step ascends because the pseudo-transient term puts it there, and the flux gate cannot see how far the non-flatness reaches (2026-09-03)

The judgement first, in four parts. (1) Of the three things item (P51) left open, two are excluded by measurement. The banded model's missing non-local column coupling is REAL — perturbing $n(\text{H}_2)$ in one front cell changes the residual of every cell BELOW it, 215 of them, far outside the band half-width 11 — and it is not the obstruction: that tail carries 0.53 per cent of the response, its largest entry outside the band is 3.5×10^{-4} of the diagonal one, and the matrix-free Jv has it already. The Krylov cycle reaches its tolerance in 1 to 2 directions at $\text{rtol} = 10^{-1}$ and 2 to 4 at 10^{-3} , never truncating; tightening it changes neither the direction nor where the solve stops. The frozen limited slope is not the obstruction either: refreshing it at every residual evaluation leaves the same sign change at the same outer iteration, and along the Newton ray the inner model's residual and the true one agree to 5–6 significant figures. **(2) The stagnating iterates are not stationary points:** the merit falls along $-\delta Y$, 0.9287 to 0.8717 at $t = -0.1$, and along the ray it is smooth over eight decades of t with no kink and no noise, so this is not the non-differentiable residual of §138. either. **(3) The pseudo-transient term is what makes the step ascend.** An exactly solved Newton system gives slope $= -f_2$ whatever the Jacobian is; this solver solves $(I/\Delta\tau + J)\delta Y = -F$ under the two-sided scaling of §139.5(i), and

$$\left. \frac{df_2}{dt} \right|_0 = -f_2 - \frac{1}{\Delta\tau} \frac{\sum_k g_k (D_k/D_k^{\text{row}}) x_k}{f_2}, \quad g = F/D^{\text{row}}, \quad (48)$$

whose second term carries no fixed sign once D and D^{row} differ. **(4) A rule that raises $\Delta\tau$ until the slope is negative was built, measured and NOT adopted:** it resolves every predicted ascent and converges neither solve, because the arc over which the raised direction descends is far shorter than the step. **The controlling quantity is the step LENGTH,**

not the pseudo-time. What did go into the code is one reporting line — the mass-flux spread over $r \geq 1.03$ and $r \geq 1.10$ printed beside the gate’s own window — which changes no number and immediately found two things.

The full measurement is docs/p51_coupled_stagnation.md and docs/p23_transport_on_state.md. Molecular carrier transport and Coupled carrier solve stay default off and nothing here is quoted as a rate.

142.1 The three candidates, closed

Everything is measured on the 1 microbar gate — mol_carrier plus Coupled carrier solve: True, Secondary_ionization: Immediate, Solver: Newton $4.0\text{e-}2$ — from one state marched 36,001 steps past the golden snapshot, because §139.9 records that the E1d hand-off cannot be reproduced in the tree. The directional derivative of the merit along the Newton step, in units of $-f_2$ (an exact Newton direction gives -1):

outer iteration	1	2	3	4	5	6	8	10	12
the code’s own cycle, $\text{rtol} = 10^{-1}$	−0.502	−0.120	−0.019	−0.003	+0.018	+0.043	+0.487	+0.631	+0.858
tightened, $\text{rtol} = 10^{-3}$, $m = 120$	−0.503	−0.118	−0.016	+0.001	+0.021	+0.042	+0.494	+0.604	+0.817
the limited slope refreshed	−0.506	−0.119	−0.017	+0.001	+0.025	+0.009	+0.065	+0.280	+0.908

Two things at once: the direction is never a good Newton direction — it delivers half the descent an exact one would even at iteration 1 — and it changes sign at iteration 4 or 5 in every arm regardless of how the Krylov step is solved or whether the limiter is frozen. All three end at the same place, carrier row $1.02\text{--}1.03 \times 10^{-2}$ against a 10^{-3} gate.

142.2 The pseudo-time step, measured

The slope the linear model itself predicts, from the same Jv the Krylov space is built from, beside the $\Delta\tau$ the SER ramp was holding:

outer iteration	1	2	3	4	5	8	10	12
$\Delta\tau_0 = 1$ (the hand-off’s own), $\Delta\tau$	1.00	1.17	1.21	1.17	1.12	1.54	0.498	0.856
model slope	−0.928	−0.210	−0.031	−0.004	+0.025	+0.552	+0.657	+0.797
measured slope	−0.928	−0.211	−0.031	−0.004	+0.025	+0.547	+0.657	+0.797
$\Delta\tau_0 = 10^6$, model slope	−35.9	−29.1	−2.88	−8.96	−13.7			
$\Delta\tau_0 = 10^{12}$, model slope	−36.1	−30.3	−3.00	−9.00	−15.3			

The model slope and the measured slope agree to three or four digits at every iteration — the model is right about the function and wrong about the direction — and the $\Delta\tau_0$ rows say why: with the pseudo-transient term made negligible the SAME Jacobian, the SAME preconditioner and the SAME frozen model give a descent direction at every one of those iterations.

The escalation rule, measured and not adopted. Raise $\Delta\tau$ by 100 and re-solve whenever the model says the step ascends, until the slope is negative or $\Delta\tau$ reaches the ramp’s ceiling $10^{14}\Delta\tau_0$. Its directional derivative is free: pgmres can return $A_z x$ from its own Arnoldi relation $A_z M^{-1} V_k = V_{k+1} \bar{H}_k$, so no residual is evaluated.

	§139 hand-off		the P23 marching stop	
	baseline	with the rule	baseline	with the rule
best iterate $\ R\ $	1.147×10^{-4}	9.570×10^{-5}	1.582×10^{-5}	3.502×10^{-5}
carrier row, worst cell	0.983	0.123	2.30×10^{-2}	1.96×10^{-1}
flux spread	2.550×10^{-1}	2.471×10^{-1}	1.071×10^{-1}	1.674×10^{-1}
trials refused (element budget)	274	53	0	0
residual evaluations	2470	2472	21,344	1,050
raises	—	19 it., 22, 0 unresolved	—	10 it., 10, 0 unresolved
info	2	2	1 (budget)	2 (12 no-descent)

It helps one state and loses the other, and the reason is measured: after a raise the line search still cuts to $\lambda = 9.5 \times 10^{-7}$. Capping the raised step at the ascending one's length is necessary and not sufficient. So the repair belongs at the step length — a trust region sized by the arc the ray scan measures — and that is TO_BE_DONE.md item (P51 continuation).

142.3 What went into the code: the spread at three radii, not one

The gate is the mass-flux spread over $r \geq r_{\text{flux}}$, $1.2R_p$ by default (§133.2), and it cannot show how far the non-flatness reaches. flux_spread_above_radius takes the same spread over any inner radius, FOR REPORTING ONLY; flux_spread_of_state and steady_gates_met are untouched, so the acceptance test is the same test and no golden can move on this.

It earned itself immediately. On the 1 microbar hot Uranus the gate's whole 1.70×10^{-1} comes from its own window's inner edge — every cell above $1.4R_p$ is flat to 1 per cent. Run past its du plateau stop, that run falls to a gate-window minimum of **1.015×10^{-2}** at 180,500 further steps and **turns around**, reaching 4.417×10^{-2} by the time it converges on its own criterion at 212,706 further steps (du = 9.9998×10^{-4} against du_th = 10^{-3} , with $\|R\|$ at 3.5×10^{-6} , below its own gate); meanwhile $r \geq 1.10$ moves only $3.1 \rightarrow 2.5 \rightarrow 2.9 \times 10^{-1}$ and $r \geq 1.03$ $7.8 \rightarrow 6.9 \rightarrow 7.3 \times 10^{-1}$. So on this planet the marching's own convergence criterion and the flux gate are anti-correlated over the last third of the trajectory: **the state the run calls converged is not the flattest one it passed through, and neither meets the flux gate.** This is a finding about the du plateau detector and the du stop, not a proposal to move r_{flux} .

142.4 A reporting defect the same measurement exposed

The coupled Newton from either of those states meets the residual gate for the first time — $\|R\| = 1.547 \times 10^{-6}$ and 1.383×10^{-6} against 10^{-5} — and returns info = 2 on the other two. The old report then printed the CARRIER gate is the one left, which is wrong: the flux gate was also unmet, at 2.447×10^{-2} against 5×10^{-3} . The chain tested the residual and the flux gate TOGETHER first and only then fell through to the carrier row, so a state that met the residual gate and missed the other two named one of the two it missed; nothing before had met the residual gate with the carrier row in the unknowns. It is now one line per unmet gate, each asked separately.

142.5 What was checked

- make check, **10 of 10 byte-identical** against the goldens as refreshed for §141.. Only log lines change, and no golden carries them.
- Four Solver: Newton cases — wasp_full_newton, wasp_he23off_newton, solver_newton_cold, newton_rsw01 — byte-identical to the same cases run with the unpatched binary. The escalation rule, measured on the same fourteen before it was set aside, never once predicted an ascent in them.
- The Arnoldi slope against a direct Jv along the same step: **54 of 56** comparisons agree in sign, 1.5×10^{-7} relative at iteration 1. The two that disagree are where the merit turns over inside the probe step, which is the same curvature again.
- NOT checked: any planet but the He/H = 0.0793 hot Uranus and the WASP-121b of the matrix; whether the band half-width 11 is exact rather than sufficient; whether $r_{\text{flux}} = 1.2$ is the right window edge on this planet.

143. Every conservation row is measured against the largest term it contains (2026-09-03)

Judgement first. The residual's row scale was a bound on the wrong quantity, and the error is exactly quantifiable on each row. Dividing the mass residual by $\rho(|v| + c_s)/\Delta r$ gives identically the fractional change of the mass flux across the cell TIMES the local Mach number — verified to one per cent at ten radii — so a quasi-hydrostatic layer at $\text{Ma} = 5 \times 10^{-5}$ was invisible by a factor 2×10^4 . The momentum row's two terms are a near-cancellation that is 4.4×10^5 times the wind's mass flux in the same layer, so a row reading 10^{-8} on that scale was still displacing the wind measurably per crossing time. The energy row of the layer was out by one part in a thousand of the terms it balances, and cell 1 by 99 per cent, while both read 10^{-8} to 10^{-4} . **Each row is now divided by the largest term it itself contains** — the flux it differences, the source it is balanced against, and for the energy row the two radiative terms it is the difference of — all of which RK_rhs and the ionization sweep already produce. The 1 microbar hot-Uranus gate rung, accepted before at $\|R\| = 2.9 \times 10^{-6}$ with a mass flux 30 per cent out at $1.03 R_p$, now converges to a flux flat to 1.0×10^{-4} from $1.01 R_p$ outward and to residuals of 5×10^{-9} to 5×10^{-6} on every row above the first cell, in 14 JFNK iterations against 5, with $\log_{10} \dot{M}$ unchanged at 10.34. The Newton system and the line-search merit keep the state scales, deliberately.

Full measurement: docs/p54_base_layer_mass_flux.md (the mass row and the census), docs/p55_base_mode.md (the momentum row and the marching departure), docs/p54g23_row_scale_scan.md (all three rows compared). This closes item (AA) for the mass flux and opens (AD) and (AE) in TO_BE_DONE.md.

143.1 What the old scale measured

Section 133. made every row's scale a bound on that row's own largest term, the cell's state scale times the rate at which a signal crosses the cell, $D_k(j)(|v_j| + c_{s,j})/\Delta r_j$. It is a bound, but on the wrong quantity: nothing in any of these rows moves at c_s .

Mass. The row has no source (Source.f90, $S_1 = 0$), so it is

$$R_1(j) = \frac{F_{j+1/2} r_{j+1/2}^2 - F_{j-1/2} r_{j-1/2}^2}{\Delta V_j}, \quad (49)$$

and nothing else. Writing $\Delta V_j \simeq r_j^2 \Delta r_j$ and $F \simeq \rho v r^2$,

$$\frac{R_1}{s_1} \simeq \frac{\Delta(Fr^2)}{F} \frac{|v|}{|v| + c_s} \simeq \frac{\Delta F}{F} \text{Ma}. \quad (50)$$

Measured on the converged 1 microbar hot Uranus, the ratio of $|R_1|/s_1$ to $(\Delta F/F)\text{Ma}$ is 1.109, 1.004, 1.015, 1.007, 1.008, 1.010, 1.006, 0.979, 0.908, 0.656 at $r = 1.001, 1.004, 1.010, 1.023, 1.032, 1.048, 1.093, 1.177, 1.329, 2.119$ — one per cent everywhere the flow is subsonic. The flux change per cell that a tolerance of 10^{-5} still passed:

r	Ma	$\Delta F/F_0$ carried	$\Delta F/F_0$ passed at 10^{-5}
1.001	3.9×10^{-5}	1.5×10^{-2}	2.6×10^{-1}
1.010	5.2×10^{-5}	3.8×10^{-3}	1.9×10^{-1}
1.031	1.5×10^{-4}	9.9×10^{-4}	6.7×10^{-2}
1.093	4.5×10^{-3}	1.9×10^{-3}	2.2×10^{-3}
1.329	1.1×10^{-1}	2.1×10^{-5}	8.7×10^{-5}
2.119	5.4×10^{-1}	4.5×10^{-6}	1.9×10^{-5}

Momentum. In a quasi-hydrostatic layer the row is the near-cancellation of a flux divergence and gravity, both enormous beside the wind's mass flux: $\max(|dF_2|, |S_2|)/F_0$ is 4.41×10^5 at $1.005 R_p$, 1.12×10^5 at 1.03, 4.46×10^3 at 1.10 and 1.92×10^2 at 1.20 on the hot Uranus ($7.69 \times 10^3, 2.24 \times 10^3, 1.08 \times 10^2, 1.71 \times 10^1$ on WASP-121b). A residual ϵ of that row therefore displaces the mass flux by $\epsilon(|dF_2|/F_0)$ of the wind per crossing time, and docs/p55_base_mode.md §4 predicts the measured departure rate of an accepted state from exactly that product to **9 per cent**.

Energy. Same construction, same result: the accepted state's row reads 1.72×10^{-3} at $1.03 R_p$ and 4.68×10^{-3} at 1.05 against its own terms while reading 3×10^{-9} to 4×10^{-8} against $E(|v| + c_s)/\Delta r$.

143.2 It is not a diagnostic artifact

Before changing the measure, the obvious alternative was checked: that the cell-centred $\rho v r^2$ the diagnostics print is not what the finite-volume scheme conserves, and that the scheme's own face flux is flat. It is not.

$r \geq$	spread, cell centre	spread, Riemann face
1.002	3.247×10^{-1}	3.282×10^{-1}
1.030	3.105×10^{-1}	3.137×10^{-1}
1.100	2.920×10^{-2}	3.041×10^{-2}
1.200	2.460×10^{-3}	2.513×10^{-3}

The two agree to within one to four per cent at every window above $1.002 R_p$. The one place they differ is cell 1, where the centred product is $-21 F_0$ — negative, 21 times the wind flux — while both of its faces carry $+1.75$ and $+1.92 F_0$;

that is the odd-even mode of docs/p44_base_sawtooth.md, and it is why the face flux is single signed over the whole column where the cell-centred one changes sign. There is no mass source that could produce the rest: $S_1 = 0$, and the Shapiro filter, the low-Mach damping, the element diffusion, the carrier transport, the viscous and conduction operators and the semi-implicit energy step were each checked and none writes $u(1, :)$.

143.3 The change

residual_row_scale now dispatches to three functions, each named for the physics it scales:

$$s_1(j) = \max\left(|F_{j-1/2}|r_{j-1/2}^2, |F_{j+1/2}|r_{j+1/2}^2\right) / \Delta V_j \quad \text{mass_flux_row_scale} \quad (51)$$

$$s_2(j) = \max(|dF_2(j)|, |S_2(j)|) \quad \text{momentum_row_scale} \quad (52)$$

$$s_3(j) = \max(|dF_3(j)|, |S_3(j)|, \text{heat}(j), \text{cool}(j)) \quad \text{energy_row_scale} \quad (53)$$

plus the operator-split viscous and conduction sources where those are active. ΔV_j is the volume RK_rhs divides the flux difference by, so R_1/s_1 is exactly the fractional change of the mass flux across the cell. There is no fallback and no second definition: the terms are those assemble_residual has just had RK_rhs and the sweep produce, and a caller asking about a state they do not describe gets the Riemann solve done on the spot (refresh_row_terms). That path cannot rebuild heat/cool, which do not come from u ; the states that reach it are the solver's trial and finite-difference probes — *measured*, 15 of order 10^5 scale requests on the hot Uranus and 45 on WASP-121b, none of them a state the run adopts — and the code says so where it happens. The face-limiter ledger is restored across such a rebuild, because a measurement of a state is not a step of the run.

The §62.3 gravitational bound is no longer a special case. It existed because the old momentum scale could not see gravity; s_2 now takes gravity from S_2 , the same expression the residual subtracts, so the two are one term and not two.

The Newton system is NOT rescaled with any of this. cell_state_scales keeps D_k . Rescaling the merit and the scaled Jacobian by these quantities as well was tried on the mass row — it varies by a factor 9 across the first two cells — and the hot-Uranus solve then ground from 2.2×10^{-2} to a plateau at 7.8×10^{-4} , spent most of 179 iterations in damped Gauss-Newton escapes, and aborted on no descent direction exists for the banded model at this state at 8238 residual evaluations against 115 for the baseline. So §133's statement that the merit and the acceptance test are one expression apart now holds for neither, and that is the measured choice.

Resid tol changes meaning on all three rows, in the direction that makes it stricter: on the states measured the rows read 58 to 4000 times larger than before. No repository value is renormalized — the states that pass the new measure pass at the existing 10^{-5} — but *every golden that runs a steady solve or a residual gate moves*, and that is a deliberate refresh, not a byte-identical change. A run that stops on du alone is byte-identical.

143.4 What it does to the cases

All 34 named cases were run under both binaries — the 10-case run_check.sh matrix, every backup/regression/case carrying Solver: Newton, and the three Resid tol cases — from copies of their inputs, each keeping its own maxsteps and otherwise capped at 60,000 steps. Full table: docs/p54g23_census.md.

Seven goldens move; twenty-seven are byte-identical. The seven are `arm_heh1_x2matched`, `newton_rsw01`, `newton_rsw05`, `ptc_warm`, `solver_newton_cold`, `wasp_full_newton` and `wasp_he23off_newton` — exactly the cases that reach their answer through a steady solve or the residual gate. A run that stops on `du`, on `maxsteps`, or on a JFNK that aborts before its first accepted step never reads a row scale, and is unchanged: that is the check on the change, and it is why the default regression matrix is 10/10 identical.

The rate does not move. $|\Delta \log_{10} \dot{M}| \leq 0.01$ on every case that converged under both. The single exception is `arm_heh1_x2matched` at -0.19 , and it is an improvement: that case does not converge at all under the old measure — 60,000 steps and no acceptance — and under this one it converges, in 106 iterations against 40 and at 89 residual evaluations against 1127, i.e. twelve times cheaper. Its old number was an unconverged state's rate.

No case that converged stops converging. Three cases fail under this measure — `armD_D2_LW_newton`, `armD_D2_newton`, `armD_D2_newton_bigstack` — and all three fail under the old one too, at the same iteration and for the same reason (no descent direction exists for the banded model at this state, at flux spreads of 21 to 34, i.e. states nowhere near steady). The failure list is a strict subset of the old one; `docs/p54g23_nonconvergent.md` records which row and which cell blocks each.

case	iterations	residual evaluations	ratio
<code>newton_rsw01</code>	9 → 76	205 → 1679	8.2
<code>newton_rsw05</code>	20 → 71	471 → 1594	3.4
<code>solver_newton_cold</code> = <code>wasp_full_newton</code>	14 → 44	338 → 1000	3.0
<code>wasp_he23off_newton</code>	11 → 31	281 → 722	2.6
<code>arm_heh1_x2matched</code>	— → 106	1127 → 89	0.08
the hot-Uranus gate rung	5 → 14	115 → 321	2.8

Two and a half to eight times the residual evaluations where the old measure already converged, and the whole cost of a case it could not. In wall time the worst of these is a WASP-121b cold start at three minutes against two.

What the accepted states look like. Every case that converges under this measure ends with its mass flux flat to 5×10^{-5} to 9×10^{-5} over $r \geq 1.20$ — against 1.2×10^{-4} to 1.4×10^{-3} before — and to 6.8×10^{-5} to 7.7×10^{-5} over $r \geq 1.03$, where before it was 2.0×10^{-4} to 5.5×10^{-2} . On `solver_newton_cold` that last figure is $5.469 \times 10^{-2} \rightarrow 6.751 \times 10^{-5}$, a factor **810**.

Row by row on the accepted roots, every root measured with the same instrument:

hot Uranus, r	before: $R_1/R_2/R_3$	now: $R_1/R_2/R_3$
1.00019 (cell 1)	$-2.15 \times 10^{-1} / 1.42 \times 10^{-1} / -9.94 \times 10^{-1}$	$-2.16 \times 10^{-1} / 1.43 \times 10^{-1} / -9.94 \times 10^{-1}$
1.01005	$9.70 \times 10^{-4} / -1.54 \times 10^{-4} / -1.89 \times 10^{-1}$	$-3.13 \times 10^{-10} / -1.53 \times 10^{-8} / -3.89 \times 10^{-6}$
1.03016	$-6.68 \times 10^{-4} / -1.30 \times 10^{-5} / -5.53 \times 10^{-2}$	$4.67 \times 10^{-9} / 8.68 \times 10^{-9} / 4.70 \times 10^{-6}$
1.05012	$-2.70 \times 10^{-4} / -2.90 \times 10^{-4} / -3.88 \times 10^{-2}$	$-1.36 \times 10^{-9} / 5.00 \times 10^{-8} / -4.20 \times 10^{-6}$
1.20011	$-5.26 \times 10^{-5} / 4.17 \times 10^{-7} / 2.11 \times 10^{-2}$	$-3.15 \times 10^{-12} / -8.93 \times 10^{-12} / -7.63 \times 10^{-6}$

Cell 1 does not move, on either planet, under any variant tried. Its energy row is out by 99 per cent on the hot Uranus and 78 to 88 per cent on WASP-121b whichever measure is used, and the state passes only because the norm is volume weighted and that cell’s volume is negligible. That is item (V), and these are the sharpest numbers it has.

143.5 The marching stop and the flux gate measure the same quantity through different windows (decision pending, not implemented)

This is a separate finding of the census and it is **not implemented**: the code keeps du as it is. It is recorded because the census makes it unavoidable.

	marching du	flux gate
window	$[j_{\min} : N], r \geq r_{\text{esc}}$	$[j_{\text{flux}} : N], r \geq r_{\text{flux}} = 1.2$
functional	$(\max F - \min F) / \min F $	$(\max F - \min F) / \bar{F} $
sign	absolute value	signed
threshold	du_th, 10^{-3}	flux_spread_th, 5×10^{-3}
stops	the marching loop, the JFNK hand-off	the acceptance of a steady solve

r_{esc} is $2.0R_p$ on the hot Uranus and 1.5 on WASP-121b, so du is a statement about the outer wind only and says nothing about $1.2 \leq r \leq r_{\text{esc}}$ — a region the flux gate does cover. Not hypothetical: wasp_full, a matrix golden, stops as “converged” on $\text{du} = 9.999 \times 10^{-4} < 10^{-3}$ at an $r \geq 1.2$ spread of 3.13×10^{-2} , six times the flux gate’s own threshold and 1.37 at $r \geq 1.01$; P51’s transport-on hot Uranus stopped the same way at 4.4×10^{-2} . The census makes the same point on the other side: du and the gate’s spread are uncorrelated on the states these cases stop at — every armD_*/armHeH_* snapshot ends with du between 1.1 and 28 and a gate spread between 0.77 and 35.

Recommendation, physics first (a decision, not a change). A steady state has one mass flux at every altitude, not one mass flux above $2R_p$. Two definitions of the same statement, differing in window and in functional, is what the naming rule forbids: one kind of quantity, one rule. The du window should become the gate’s window and the functional the gate’s functional, so that du_th and flux_spread_th become one number. **This is a user decision and it is not in the code.** It is a different change with a different blast radius: it moves where every marching run stops, hence every golden that stops on du — which is most of them — for a reason independent of the row scale.

143.6 What is not settled

- **The first two cells.** No measure change moves them; they pass because the norm is volume weighted. Item (V).
- **The marching attractor.** A state this measure accepts is a zero of the steady residual and is not a fixed point of the marching loop. Reloaded and marched 5000 steps, the hot Uranus’s departure is halved by the mass row’s scale alone (peak-to-peak of $F(1.03)/F_0$ $1.127 \rightarrow 0.509$) and then moves only to 0.495 when the momentum and energy rows are added, while WASP-121b’s three roots — residuals spanning four orders of magnitude — all depart identically (4.277, 4.291, 4.292). *The departure is not controlled by the part of the residual this section removes*; it is a restart kick and a cell-1 problem (docs/p55_base_mode.md §7(2) and §7(3)). New item (AD).

- **The energy half of (AA)**, the factor-2 gap in heating = cooling + $p \operatorname{div} v$ + advection below $1.08 r_{\text{base}}$. This section is a statement about how a row is measured, not about what it contains.

Files: `src/modules/time_step/steady_residual.f90` (new `mass_flux_row_scale`, `momentum_row_scale`, `energy_row_scale`, `store_row_terms`, `refresh_row_terms`; `residual_row_scale` dispatches), `src/modules/time_step/steady` (comment only). Documentation: `docs/p54_base_layer_mass_flux.md`, `docs/p55_base_mode.md`, `docs/p54g23_row_scale_scar` `docs/input_schema.md` (K35, K35b), `docs/newton_scaling_and_base_wall.md` (§2 and §3), `TO_BE_DONE.md` ((AA) closed for the mass flux, (V) extended, (AD) and (AE) opened).

144. P55: the accepted root, the marching fixed point, and what a restart must carry (2026-09-03)

Judgement first. A state the steady solver accepts is not a fixed point of the marching loop, and the reason is mostly in the STATE, not in the operator. The spatial operator is the same by construction — `assemble_residual` calls the same `Reconstruct` and `RK_rhs` the marching calls — and the departure is not an instability: it starts at step 1 at a finite size and grows *linearly* in step number, at a rate the accepted state's own residual predicts to nine percent. Two things produce it, both measured; one is now fixed, and the other is what §143.'s item (AD) is about.

Full account and figures: `docs/p55_base_mode.md`, `docs/p55_base_mode.png`, `docs/p55_operator_terms.png`, `docs/p55_reload.png`.

144.1 The departure is a residual being integrated, not a mode growing

On the two `EXHALE_CONT_SCALE=gate` solutions of §143. (the $1 \mu\text{bar}$ hot Uranus and WASP-121b), reloaded and marched with `Solver`: removed:

- the flux at $r = 1.03$ moves **linearly in step number** for the first twenty steps — 4.71×10^{-4} of F_0 per step on the hot Uranus, against a state the solve left flat to 1.04×10^{-4} over $r \geq 1.01$. **One step undoes 4.5 times the flatness the solve achieved.**
- the state's own momentum-row residual there predicts 4.31×10^{-4} per step. Nine percent agreement, no free parameter. In a quasi-hydrostatic layer the momentum row is the near-cancellation of two terms worth 1.1×10^5 times the wind's mass flux, so one part in 10^5 of that row displaces the mass flux by order the wind itself.
- judged by its own largest term instead of by $\rho(|v| + c_s)/\delta r$, the accepted hot-Uranus state's energy row is out by 1.2×10^{-3} at $r = 1.03$ and 6.0×10^{-3} at 1.05, its momentum row by 2.5×10^{-5} , and cell 1 by 1.4×10^{-1} and 9.9×10^{-1} . That is what §143.'s row scales are for.

Not a discrete instability: quartering the time step changes the excursion by 3 percent at equal model time; first-order reconstruction at the base leaves WASP-121b unchanged and makes the hot Uranus 12 times worse; the Shapiro filter removes 8 percent on WASP-121b and multiplies the hot Uranus by **42**; freezing the first 20 cells does nothing on WASP-121b. *Not a physical instability* either: the rise is linear, and the same operator started from a state the marching itself produced departs 3 to 41 times less.

144.2 A restart did not run the physics its state was converged under

WASP-121b's input sets no `Secondary_ionization:` key, so the SvS85/Dalgarno coupling is STAGED. The run that WROTE the state armed it at step 2242 and the JFNK finished with it on — and every written state is a coupling-on state, because the post-loop arming precedes `write_output`. The run that RESTARTED it re-staged it, and the flip never fired again in 3000 steps. **Every marching experiment on WASP-121b in this series was therefore run at a photoionization rate the state was not converged under.** The first ionization sweep moved the particle count by 8.1×10^{-3} instead of 2.6×10^{-4} .

The state cannot carry that by itself, so the file now says it. `write_coupling_state_header` (`utilities.f90`) adds one # line to `Hydro_ioniz.txt`, which is the file a restart is fed as `Hydro_ioniz_IC.txt`:

```
# coupling: sec_ion=T sec_ion_step=2242 valve= 1.00000E-04 recon=WENO3 fluxconst=-1.00000E+00
```

It is the census of every switch EXHALE_main changes *inside* the marching loop:

switch	what changes it	field
sec_ion_active and the step it was armed at	the du-convergence flip, the pre-hand-off flip, the unconditional post-loop arming	sec_ion, sec_ion_step
valve_eps	the JFNK hand-off adopts 10^{-4} when the input left the valve off	valve
rec_method	the PLM+WENO3 stage switch at $du < du_{th,PLM}$	recon
base_flux_const	the 100-step average behind Base velocity: massflux	fluxconst
du_stop_armed, du_newton_armed	the first descending du crossing	already keyed to do_load_IC
weno_mode	frozen inside the JFNK, restored on exit	not run state outside the solver
*_active (viscosity, conduction, transport, low-Mach, caloric, CX)	nothing: pure functions of input parameters	not run state

`load_IC` parses the line and the run starts the coupling where the file says it was; `valve` and `recon` are properties of the run, which `input.inp` states, so they are reported when they disagree and never overridden. **A file with no # coupling: line starts staged exactly as it did before**, because nothing else can be inferred about it — and that is not a formality: on `jfnk_hd189`, whose IC predates the line, arming it blindly makes the composition mismatch *worse* ($4.12 \times 10^{-3} \rightarrow 6.26 \times 10^{-3}$) and produces a $du = 1.0 \times 10^2$ transient by step 5500. `write_setup_report` prints which of the three cases the run is in.

Measured, 3000 steps from the WASP-121b solution, peak-to-peak of $\Delta(Fr^2)/F_0$:

	$r=1.005$	1.03	1.10	1.20	$v(1)$ end	du end
before	7.9689	4.2961	1.1976	0.6349	5.766e-03	1.178e-02
composition equilibrated	8.0041	4.3197	1.1970	0.6341	5.880e-03	1.130e-02
with the header	0.0382	0.0174	0.0051	0.0040	3.150e-03	5.187e-05

du ends at 5.187×10^{-5} against the 5.03×10^{-5} the accepted state carries: **with the restart running the physics the state was converged under, the JFNK root IS a fixed point of the marching on WASP-121b**, to one part in 250 of what the re-staged run does. The hot Uranus is unchanged to five digits, as it must be: its input sets Secondary_ionization: Immediate.

144.3 A restart did not start from its own equilibrium composition

The marching step ends with $RK3 \rightarrow T = p/(n_{\text{tot}} + n_e) \rightarrow \text{ioniz_eq} \rightarrow p = (n_{\text{tot}} + n_e)T$, whose last stage rewrites the pressure at the particle count the sweep just produced. On any step of a relaxed run that is a rate. On the FIRST step of a restart it is a **projection of fixed size**, because the composition it corrects came from the file. The same first step at four time steps, as $\Delta E_{\text{chem}}/E$:

	CFL 0.6	0.06	0.006	0.0006
WASP-121b $r=1.10$, before	-6.2295e-03	-6.2297e-03	-6.2297e-03	-6.2297e-03
WASP-121b $r=1.10$, after	-3.2041e-06	-3.2067e-07	-3.2070e-08	-3.2072e-09
hot Uranus $r=1.03$, before	5.4549e-07	5.5078e-07	5.5131e-07	5.5136e-07
hot Uranus $r=1.03$, after	-5.8808e-09	-5.8712e-10	-5.8546e-11	-5.6966e-12

Four decades of δt and four identical values is a projection; one factor of ten per decade is a rate. Before the fix the jump reaches 6.2×10^{-3} of the cell energy in one step, twelve times that step's entire hydro contribution.

equilibrate_loaded_composition (EXHALE_main.f90) holds ρ , v and p — the primitive state the file carries — and iterates the composition to the fixed point of $f_{\text{sp}} \leftarrow \text{ioniz_eq}(\rho, T)$ with $T = p/(n_{\text{tot}} + n_e)$, using the same sweep the marching loop and the steady residual call, with the same secondary-ionization arming. It runs on the marching route and on the EXHALE_RESIDUAL route, capped at 64 sweeps, stopping at 10^{-10} (the floor the cell solve's own xtol sets, not a target). EXHALE_RELOAD_EQ=0 reproduces a run of the previous code bit for bit.

Writing the files with more digits would not have helped. They are written with list-directed output, i.e. full double precision already (0.99980405732338939), and the written triple is self-consistent to 10^{-8} at most radii. The gap is four to five orders above anything the format could cause.

Every named Load-IC case starts more than four parts in a thousand from the equilibrium of its own state; armD_D2_newton starts 7.127×10^{-1} away and armD_D2_LW_newton 5.000×10^{-1} (contracting at only 0.69 per sweep, needing 54). The bytes move; the answer does not: armD_D2_LW_newton at its 2100-step cap gives du $1.5463 \times 10^1 \rightarrow 1.5493 \times 10^1$ and $\log_{10} \dot{M} = 8.88$ either way, and at a matched step 40000 armD_D2_newton is 1.2 percent apart and jfnk_hd189 0.34 percent.

144.4 The temperature column belonged to a composition one sweep behind it

The refresh after the steady solve computed T from the particle count BEFORE its final ioniz_eq and never recomputed it, so the T column of a file written from there was not the temperature of the species columns beside it. One line closes it (call comp_T_from_p(p, n_tot, ne, T)); p is untouched, so (p, T, f_{sp}) becomes a consistent triple. Isolated on a converging hot-Uranus JFNK run, **only the T column moves, by at most 1.770×10^{-5} (at $r = 1.0488$); r , ρ , v , p ,**

heat, cool and all of `Ion_species.txt` are bit-identical and $\log_{10}\dot{M}$ is 10.34 either way. It cannot touch the ten default goldens: that code is reached only from the JFNK hand-off, and *none of the ten ever reaches one*.

144.5 Does the new root fix the departure? No

The hot Uranus was re-solved with the tree as it now stands (§143.'s row scales plus this section's restart handling), giving `info = 0`, $\|R\| = 9.958 \times 10^{-6}$, gate flux spread 2.501×10^{-4} , $\log_{10}\dot{M} = 10.34$ — and the new root's own flux is still 3.433×10^{-3} out over $r \geq 1.03$, fourteen times the gate's number. Marched 3000 steps from it with the tree binary:

root	p2p at 1.005	1.03	1.10	1.20	<i>du</i> end
mass row only	1.5476	0.5336	0.0710	0.0125	6.387e-04
mass row + this section's restart handling	1.5473	0.5296	0.0705	0.0126	6.389e-04
all three rows, tree binary	1.5481	0.5333	0.0719	0.0132	3.363e-04

The departure is unchanged to 0.06 percent. Making the momentum and energy rows honest halves *du* at the end ($6.39 \times 10^{-4} \rightarrow 3.36 \times 10^{-4}$) and moves the excursion not at all. So on the hot Uranus the two remaining causes of item (AD) are neither the restart (fixed here) nor the row measure: what is left is the layer's own imbalance in the terms those rows contain, and cell 1.

144.6 What is not settled

- **The hot Uranus root is still not a fixed point of the marching**, by the same excursion as before either change. Item (AD) stands, narrowed: it is not the restart and not the row scale.
- **The first two cells.** Untouched here; item (V).
- **The reloaded WASP-121b residual** improves 14 to 17 times in the hydro rows (mass L^∞ $4.406 \times 10^{-5} \rightarrow 2.604 \times 10^{-6}$, momentum $1.023 \times 10^{-4} \rightarrow 7.295 \times 10^{-6}$) but its energy row still reads 1.0×10^{-5} against the 3.138×10^{-6} the solve accepted. Not chased.
- **heat and cool are now one sweep behind the corrected T** in a file written from the steady solve; they are diagnostic columns, and running the full composition fixed point in that refresh was not done.
- **The *du* window** is measured over $r \geq r_{\text{esc}}$ while the flux gate measures $r \geq r_{\text{flux}}$; the measurement of that mismatch is `docs/p55_du_window_census.md` in the P55 scratch tree.

Files: `src/EXHALE_main.f90` (new `equilibrate_loaded_composition`; the arming rule reads the restart header; `comp_T_from_p` after the post-solve sweep; `sec_ion_armed_step` bookkeeping), `src/modules/functions/utilities.f90` (new `write_coupling_state_header`), `src/modules/files_IO/load_IC.f90` (new `parse_coupling_header`), `src/modules/files_IO/write_output.f90` (emits the line), `src/modules/files_IO/write_setup_report.f90` (reports the restore), `src/modules/init/parameters.f90` (`sec_ion_armed_step`). Documentation: `docs/p55_base_mode.md` (+ three figures), `docs/input_schema.md`, `docs/EXHALE_user_manual.tex`, `TO_BE_DONE.md` ((AD) narrowed, the roundtrip broken IC opened). Regression: 10/10 data identical; `run_fcheck.sh` CLEAN.

145. Acceptance measures what the code conserves: face mass flux, cellwise residual, one marching-stop label, the update-map diagnostic, provenance (2026-09-03)

Judgement first. Both halves of the acceptance test were measuring something other than what they were described as measuring. The flux gate read the cell-centred $\rho v r^2$, which is not the quantity the finite-volume scheme conserves; on a state whose conserved face flux is a single double-precision number over three hundred cells it reported 2.5×10^{-4} . The residual norm was either a volume-weighted ratio of sums or a ratio of two maxima taken in different cells; neither is $\max_j |R_j|/s_j$, which is what the routine's own comment says it is, and under it two accepted states carried cells 218 times above the tolerance. Both are now what they say. Neither change moves a mass-loss rate. Measurements: docs/p55_base_mode.md §11, docs/p55_update_map.md, docs/p55_du_window_census.md.

145.1 The flux gate reads the Riemann face mass flux

flux_spread_of_state and flux_spread_above_radius now both call one implementation, flux_spread_over_faces, which takes the spread of $F_{j+1/2} r^2$ — SIGNED, so a window in which the flow reverses is a large spread and not a small one — over the interior faces bounding the window's cells. The cached face flux is the one §143.'s refresh_row_terms maintains, so there is no second staleness rule and no second Riemann solve.

R_1/s_1 is exactly the fractional change of the face flux across a cell, so summing it bounds that flux's total variation. On the hot-Uranus root of §144.:

window	face-flux spread	$\sum R_1/s_1$ (bound)	cell-centred (what the gate read)
$r \geq 1.03$	1.21e-08	9.33e-08	3.433e-03
$r \geq 1.10$	0 (one value over 300 cells)	4.38e-09	2.375e-03
$r \geq 1.20$	0	3.17e-10	2.501e-04

The wind carries one mass flux to the last bit and the gate was reporting 2.5×10^{-4} : the number was the difference between two functionals, not a failure of conservation, and it was wrong by six orders. It also removes the cell-1 artifact for free.

This settles a comparison §144. got wrong. The 1.043×10^{-4} of P54b and the 3.433×10^{-3} of §144. are the same state measured by the two different functionals, not a change between P53 and now: the pre-P53 root's cell-centred spread is 3.523×10^{-3} , within 3 percent of the current root's. On the conserved functional the newer root is better by four orders.

145.2 The threshold, re-derived for that definition

flux_spread_th was 5.0×10^{-3} , calibrated against the cell-centred product; on the face flux it is not a gate. Face spread at $r \geq r_{\text{flux}}$ over every state available:

population	measured
accepted by the JFNK	1.02e-10, 3.89e-08, 3.97e-06, 3.83e-06
stopped on du only	1.14e-04, 1.04e-03, 3.17e-02, 3.18e-02
marching, 600 steps	4.00 to 1.38e+01

The two populations are separated by the interval 4×10^{-6} to 1.1×10^{-4} ; the new default is its geometric middle to one digit, 2.0×10^{-5} , five times above the loosest accepted state and six times below the tightest du stop. **No canonical input sets Flux spread tol** (checked over every `input.inp` outside LHS1140b/exhale/), so the default is the whole change. A value $\geq 10^{-3}$ draws a warning, not a refusal. One case is refused that used to be accepted, and is meant to be: `arm_heh1_x2matched` at 1.14×10^{-4} .

145.3 The residual norm is the maximum over cells of a cell's own scaled residual

`relnorm_over_cells` had two forms and neither was the local relative infinity norm the comments describe: a volume-weighted ratio of sums, and a ratio of two maxima taken in different cells (since $s \sim 1/\delta r$ is largest in the smallest cells at the base). There is now one form, $rc(k) = \max_j |R_{kj}|/s_{kj}$, and `resid_vol`, the `Resid norm` key and `residual_norms_vol` are gone with the second form they selected.

What the old norm hid, on the two roots of §144., both accepted at $\|R\| \sim 10^{-5}$: the hot Uranus carried 7 cells above 10^{-5} , all in cells 1–3 (mass 3.10×10^{-5} , energy 2.17×10^{-3} , 218 times the accepted number); WASP-121b carried 70, fifty of them in the *wind* between 1.30 and 2.00 R_p . **It is not the layer that was hidden:** between $r = 1.03$ and 1.30 not one cell of either row exceeds 10^{-5} on the hot Uranus.

	hot U, old	new	WASP, old	new
JFNK info	0	0	0	0
iterations	7	9	8	12
cell-max, mass	3.100e-05	2.021e-07	3.232e-07	4.374e-08
cell-max, energy	2.174e-03	2.002e-06	6.397e-05	9.764e-06
cells above 10^{-5}	7	0	70	0
$\log_{10} \dot{M}$	10.34	10.34	13.21	13.21
3000-step p2p at $r=1.03$	0.5333	0.5334	0.0103	0.0104

145.4 Both numbers, every time a residual is reported

Acceptance is the cellwise maximum, because that is the statement “no cell of any row is out”; it is not the statement “the column’s budget is closed”, which is what the integrated norm says, and the external review of 2026-09-03 (§4) asks for both to be visible before either is used alone. `write_residual_breakdown` prints, at the EXHALE_RESIDUAL diagnostic, the JFNK hand-back and every marching stop: per row the cellwise maximum with its cell and radius and the integrated ratio with its numerator and denominator; the numerator and denominator contributions of $r < 1.03$, 1.03–1.10, 1.10–1.20 and $r \geq 1.20$; and for the worst cell the signed dimensional residual with every term of its row.

145.5 A marching stop says that it is one, and the du window is not unified

The review (§5, item 7) judges the marching du stop, the JFNK hand-off threshold and the acceptance gates to be three concepts with three roles. **The proposal to unify the du window with the flux gate’s window is withdrawn**, and the census that measured the cost of it is kept as the reason: unifying changes no answer ($\log_{10} \dot{M}$ within 0.01–0.02 dex wherever both builds stop) but costs three WASP-121b Newton cases a factor 13.2 in steps and stops three other cases converging at all within 60000 steps.

Instead, every du , dtu , plateau and step-cap stop now prints THIS IS A MARCHING STOP, NOT AN ACCEPTED STEADY STATE with the three gates *measured* on the state being stopped at. Measured on a hot-Uranus run that used to print “converged: mass flux constant” at $du = 1.454 \times 10^{-4}$: its cellwise residual is **1.282** in the energy row of cell 7.

145.6 The complete update-map diagnostic, and what it decides

Review B2 asks for a diagnostic that applies one production step and compares $G_{\delta t} = [\Phi_{\delta t}(q) - q]/\delta t$ with the steady residual at three decreasing time steps, and says “this test decides item 3”. EXHALE_UPDATE_MAP does it, with $\Phi_{\delta t}$ the marching loop’s own body and not a copy. $|G_{\delta t} + R|$, max over cells:

$\delta t / \delta t_{\text{CFL}}$	hot U mass	momentum	energy	WASP mass	momentum
1	6.52e-01	5.33e-01	9.60e-01	9.63e-02	1.33e-01
0.1	8.93e-02	5.27e-02	1.40e-01	1.09e-02	1.69e-02
0.01	9.23e-03	5.17e-03	2.34e-02	1.10e-03	1.73e-03

$G_{\delta t}$ approaches the steady residual at first order in δt on both planets. **The production map and the residual share a fixed point, and the 42 percent discrepancy §144. measured at a CFL step is the splitting truncation error of an explicit step, not an operator mismatch.** The stage breakdown says where the rest is: Apply_BC’s conservative-primitive round trip moves the interior by a relative 4.2×10^{-16} , the semi-implicit energy step is a rate to five digits, and the composition pressure projection is a rate that converges on the hot Uranus and is 1.124 against a residual of 1.133 on WASP-121b.

With the Shapiro filter on the two provably cannot share a fixed point. With Shapiro filter: 0.5 1 the filter’s contribution to δu is the same at every time step — it is applied per step, not per unit time — so as a rate it is 1.187e+01, 1.189e+02, 1.188e+03 at the three δt , and $|G_{\delta t} + R|$ follows it exactly instead of converging.

A third trap cost the regression outright, and it is worth its own subsection: the snapshots were first placed between the *statements* of one RK3 stage. See below.

145.7 Provenance: what produced a profile file

Every profile file carries three further # lines: the revision and dirty flag stamped in by the Makefile into a generated build/build_stamp.f90 (*the run never calls git*) together with the *run* time from date_and_time — a build time compiled into the executable would make two builds of the same source different files, and the md5 of the binary is how a bitwise verdict is tied to a revision; the Makefile writes the stamp at parse time and only when its content changes, so make has a fixed point and run_check.sh can and now does run make -q after its make and refuse the

matrix if anything is still out of date; checksums of the input files as read, by `file_rolling_checksum`, two modular rolling sums that are exact in integer(8) on every compiler and deliberately not called a digest; and the discretization, lower boundary model, carrier model, restart schema version, residual definition version and grid size. Review §5, item 12.11.

145.8 The regression failed for a reason that was not the code

The tree binary built at 19:01 failed all ten regression cases at a relative 3×10^{-14} to 9×10^{-11} , with *du* and *dtu* at the stop agreeing to every printed digit. The first hypothesis was a stale object linked by an incremental build. It was not: `make distclean && make` reproduced the failure exactly. A bisect over clean builds — the pre-145 tree, the norm and gate group, the provenance group, and all of §145 without the update map — reproduced the goldens bitwise in every case, leaving one patch.

The arithmetic. The update-map snapshots were placed between the statements of one RK3 stage:

```
u1(k,:) = u(k,:) - dt_loc*(dF(k,:) - S(k,:))
if (upmap_n .gt. 0) u_umBCin = u1      ! <- here
call Apply_BC(u1)
```

a-bc is a fused multiply-add candidate and `-ffp-contract=fast` is gfortran's default, so inserting a statement changes the scheduling and with it whether the FMA is taken. An FMA rounds once where a multiply and a subtract round twice: one ulp per step, 10^{-11} over eleven thousand. **Every inserted statement was inside `if (upmap_n .gt. 0)` and did nothing with the diagnostic off.** The contraction setting is not being changed — it is the default the goldens were made under, and it is the faster and the more accurate of the two. What changes is where instrumentation may go: the boundary contribution is now taken once, out of line, in `update_map_begin_step`, the remaining snapshots sit between operator-split stages and never between the statements of one, and a comment at the first of them in `EXHALE_main.f90` says why. **Instrumentation that has to sit inside a stage must be measured out of line; an `if` that does nothing at run time is not guaranteed to leave the floating-point result alone, and the bitwise regression is what says so.**

The binary had stopped being a function of its source. §145's first build stamp compiled the BUILD TIME into a constant, so two `make distclean && make` of identical source produced different executables, differing in exactly one object (`utilities.o`) in exactly that string. Three workers comparing md5s therefore read “different program” from what was the same program. The stamp now carries the git revision and dirty flag only, and the run time is reported from `date_and_time` in the provenance header instead. Two clean builds of one source now give one binary.

And nothing checked that the binary matched the sources. The stamp had been generated by a rule with a phony prerequisite, so `make` never reached a fixed point and `make -q` reported “out of date” immediately after a successful `make` — the one cheap check that would catch a stale binary was useless. The Makefile now writes the stamp at parse time and only when its content changes, and `run_check.sh` runs `make -q` after its own `make` and refuses the matrix with exit code 2 if anything is still out of date. A bitwise verdict can no longer be taken from a binary the sources have moved past.

What an md5 of the binary identifies. Two measurements, because the guard leans on them. A comment does not change the code: same file name, one added comment line, the `-S` output is identical apart from the `.file` directive

and the object bytes match. But a *shifted line number* does change the object, because every I/O statement carries its own source line as an immediate for runtime error reporting — adding one comment line above a write turns `movl $0x4` into `movl $0x5`. The eight-line comment added to EXHALE_main.f90 at the end of this work moved the binary from e83b4ab4 to 62cc6c64 with no numerical change at all; both are bitwise identical to the goldens. **The md5 is a fingerprint of one source text, which is what the make -q guard and the # provenance: header need. It is not a semantic fingerprint, and two binaries with different md5s are not thereby different programs.**

Not established. Whether `fortdep.py` and the Makefile could ever have missed a rebuild before §145. Reading them found no mechanism: no duplicate source basenames, no module provided by two files, the dependency file remade whenever any source or `fortdep.py` changes, and a flag change forcing a full rebuild through the flag stamp. A recurrence test — a module-signature edit, an incremental build, then a clean build of the same source — gives identical binaries. No instance has been demonstrated, and because no earlier build recorded what it was built from (which is what the provenance header now fixes), the earlier builds of 2026-09-03 cannot be reconstructed either way.

145.9 What is not settled

- **The hot Uranus root is still not a fixed point of the marching**, by the same excursion as before any of this. What §145 removes is the possibility that the measure was hiding it. Item (AD), now pointing at the base boundary (review Phase C).
- **Cells 1–3.** The cellwise norm forces the solve to bring them below 10^{-5} and succeeds on both cases; nothing here says the state it reaches there is physical. Item (V).
- **The accepted state’s residual is not reproducible.** Re-running the ionization sweep on the same conserved state moves cells 1 and 2 from 10^{-5} to order unity, in the *mass* row, which reads no radiative term: what moves is ρ , which `ioniz_eq` rewrites through `calc_rho`. Every number above is therefore taken from the solver’s own residual vector. This is the next thing to diagnose.
- **The energy row of the update map on WASP-121b**, where the composition pressure projection is a rate as large as the residual itself.
- Goldens are NOT refreshed here and `run_fcheck.sh` is not run: items (AB) and (AF) follow immediately and do both once.

Files: `src/modules/time_step/steady_residual.f90`, `src/modules/time_step/steady_newton.f90`, `src/EXHALE_main.f90`, `src/modules/init/parameters.f90`, `src/modules/files_IO/input_read.f90`, `src/modules/functions/utilities.f90`, `src/modules/files_IO/write_output.f90`, `src/modules/files_IO/write_setup_report.f90`, `Makefile`.
Documentation: `docs/p55_base_mode.md` §11, `docs/p55_update_map.md`, `docs/p55_du_window_census.md`, `docs/input_schema.md`, `docs/EXHALE_user_manual.tex`, `TO_BE_DONE.md`.

146. The boundary condition writes the boundary, and the residual is a state function (2026-09-03)

Judgement first, in three parts. (1) Apply_BC was rewriting the interior it was called to leave alone. Its round trip — primitive out, boundary condition, conservative back over the WHOLE array — recomputed every interior cell's energy as $0.5 \cdot W(1) \cdot W(2) \cdot 2$ where the input held $0.5 \cdot u(2) \cdot u(2) / u(1)$. In an atomic gas that costs the last bit; in a molecular one the two conversions use different heat capacities and the projection is real: **measured, one step of mol_metals moves the interior pressure by 1.26×10^{-3} while the density it does not touch moves by 8.4×10^{-16} .** It now writes the ghost indices and returns the interior bit for bit. **(2) The steady residual filled the ghosts before its ionization sweep,** so the conservative ghost state the fluxes read carried the previous evaluation's composition (§141.5, item (AB)). It now fills them after, which is the order the marching loop has always had — state, then that state's composition, then the ghosts, then the fluxes — and the two are the same discrete operator, which §144. requires of them. This was only possible once (1) made a second call free. **(3) Neither of those is what makes the residual a state function, and the measurement that says so is new:** evaluate F at a state, at another state, then at the first again through one shared composition workspace and **1406 of 1500 entries differ, by 3.0×10^{-3} of the row scale — 300 times the acceptance tolerance.** Seed the sweep from one composition instead and the three evaluations are **bitwise identical**. The equilibrium sweep's starting guess is the whole of it, every call site in the solver already resets it, and what was missing is a test that says so.

On the 1 microbar hot-Uranus root the ordering fix moves the 3000-step marching departure at $r = 1.03$ from **0.4526 to 0.2517**, and at $r = 1.005$ from 1.6384 to 0.4443 — the first change in this campaign that moves item (AD) at all. $\log_{10} \dot{M}$ does not move on any case measured.

A note on which flatness number is quoted where. The departure figures above and in §146.5 were measured before §145. made the flux gate read the signed Riemann face flux; they are peak-to-peak excursions of that same face flux, so they carry over unchanged. The steady-state spreads in §146.4 are quoted in BOTH measures, because the gate's meaning changed under this work and item (AA) is written in the older one.

146.1 Apply_BC writes the ghosts

```
call U_to_W_interior(u,W)      ! interior only; ghosts are outputs, not inputs
call Apply_BC_W(W)            ! the boundary condition, in primitive variables
do k = 1,Ng                    ! WAS: call W_to_U(W,u), the whole array
  call W_to_U_comp(W(:,1-k), u(:,1-k), 1-k)
  call W_to_U_comp(W(:,N+k), u(:,N+k), N+k)
enddo
```

`W_to_U_comp` is the arithmetic `W_to_U` performs cell by cell, and for a gas with no molecules in it `energy_density_from_pressure` returns $p/(\gamma_{\text{ad}} - 1)$ verbatim, so an atomic ghost is the double it always was.

Two things follow. `Apply_BC` is *idempotent*, which is what item (AB) recorded as the obstruction to fixing the ghost lag. And the interior of a state stops changing under a routine whose subject is the boundary — a property nothing should have had to reason about, and one the caloric EOS of §141. turned from a rounding curiosity into a 1.26×10^{-3}

projection.

146.2 The residual’s order

```
unpack Y -> u(1:N)
get_species_densities      ! interior composition of Y (n_part_cell1)
Apply_BC(u)                ! provisional ghosts: U_to_W needs a density
U_to_W; get_species_densities; comp_T_from_p; excited_H; ioniz_eq
get_species_densities      ! the composition the residual will use
Apply_BC(u)                ! NEW: the ghosts, with THAT composition
assemble_residual          ! Reconstruct + RK_rhs read those ghosts
```

What is still one sweep behind, and deliberately: the ghost cells’ own composition was solved at the pressure the first Apply_BC wrote, not at the one the second writes. Iterating that pair to a ghost fixed point would make the residual something the marching loop is not, and being the same operator is the stronger requirement (§144.). No extra ionization sweep is needed; the cost is one call.

146.3 Is the residual a state function? Only because every call site is careful

EXHALE_RESID_DETERMINISM=1 evaluates $F(Y_0)$, then $F(Y_1)$, then $F(Y_0)$ again, then $F(Y_2)$, then $F(Y_0)$ again, and compares the three $F(Y_0)$.

how the sweep is seeded	entries differing (of 1500)	max, in row-scale units
one shared workspace, as newton_residual’s interface allows	1406 and 1400	2.968e-03 , 2.973e-03
the same, with this section’s ordering fix	1412 and 1399	4.239e-03, 4.280e-03
seeded from one composition every time	0 and 0	0

Measured on the converged 1 microbar hot Uranus, single-threaded. The tolerance these residuals are judged against is 10^{-5} .

So the carrier is the equilibrium sweep’s starting composition, and nothing else. That is a strong statement and the third row is what makes it: with the seed held fixed the residual reproduces to the bit, so no other module is holding state between evaluations.

The solver is already correct, by convention. Every one of the five places that call eval_residual copies the adopted composition into a workspace first ($f_sp_j = f_sp$ in the line searches, $fwork = f_sp_base$ in the banded Jacobian, the Gauss–Newton escape and the Krylov directional derivative), so what a solve compares are residuals seeded alike. The invariant holds and nothing about it is written down or tested. src/tests/residual_determinism/ is the test that pins it; it is what stops the sixth call site from being written without the reset.

What it does not license. newton_residual(Y, f_sp, F) still takes the composition as intent(inout), so the invariant is a property of the callers and not of the routine. Making it structural — the residual seeds from an adopted composition it owns, and returns the new one separately — is a change to a public interface and is left as item (AF).

146.4 What it does to the cases

The ten-case matrix, single-threaded, each case against the same case built without this section:

case	max relative difference, $\rho / v / p / T$	stop step and du	$\log_{10} \dot{M}$
wasp_full	1.4e-13 / 4.8e-10 / 1.1e-13 / 1.6e-13	11289, 9.9989e-04 both	13.26 both
wasp_he23off	1.4e-13 / 4.1e-10 / 9.5e-14 / 1.6e-13	11276, 9.8873e-04 both	13.26 both
mol_carrier	4.1e-13 / 4.3e-11 / 5.3e-13 / 4.4e-13	12000, 5.9036e-01 both	10.42 both
lower_profile	4.4e-12 / 6.1e-09 / 4.7e-12 / 4.2e-12	12000, 2.8228e+01 both	9.56 both
mol_diffusion	9.8e-06 / 8.6e-06 / 8.3e-07 / 9.3e-06	12000, 1.1470e+00 both	10.43 both
mol_base_handoff	1.9e-05 / 1.1e-05 / 2.8e-06 / 1.8e-05	12000, 1.1472e+00 both	10.43 both
mol_sec_ion	4.7e-05 / 2.0e-04 / 8.5e-06 / 4.5e-05	12000, 4.0963e-01 both	10.38 both
mol_lyman_werner	7.1e-05 / 8.0e-05 / 1.1e-04 / 4.4e-05	12000, 5.5044e-01 / 5.5038e-01	10.47 both
mol_ir_bands	4.8e-04 / 2.8e-04 / 2.1e-04 / 3.3e-04	12000, 1.2756e+00 / 1.2753e+00	10.43 both
mol_metals	8.2e-04 / 3.3e-04 / 4.1e-05 / 5.4e-04	12000, 1.2754e+00 both	10.43 both

All ten move and none of them changes an answer. The two atomic cases stop on du at the same step with the same du to five digits and the same rate: their trajectory changes only at the last bit, and after 11,289 steps that has grown to 10^{-13} . mol_carrier and lower_profile are the same story. The six molecular cases with the caloric EOS live move by 10^{-6} to 8×10^{-4} , which is not rounding: it is the projection of §146.1 removed.

That the molecular numbers are not monotone in step count is worth stating, because it says which they are. On mol_metals from its cold start:

steps	ρ	v	p	T
1	8.4e-16	1.8e-12	1.3e-03	5.4e-10
10	8.9e-03	3.9e-01	2.7e-02	5.2e-03
100	7.8e-03	7.1e-01	1.0e-02	4.1e-03
1000	1.3e+00	2.1e+01	1.6e+00	4.2e-01
12000 (the case)	8.2e-04	3.3e-04	4.1e-05	5.4e-04

The first row is the size of the defect: one step, one pressure, 1.3×10^{-3} . The middle rows are a cold-start transient amplifying it, and the last is the relaxation contracting it again. A golden refresh is required and no quoted number moves.

The named Newton cases. Both re-solve and both converge, at a cost:

	hot Uranus gate rung	WASP-121b (wasp_full_newton)
info	0 → 0	0 → 0
iterations	17 → 25	57 → 109
$\ R\ $ accepted	9.835e-06 → 3.176e-06	7.587e-06 → 8.138e-06
gate flux spread ($r \geq 1.2$)	5.970e-13 → 2.466e-14	1.352e-11 → 1.153e-11
face-flux spread over $r \geq 1.03$	1.228e-10 → 2.494e-12	2.225e-11 → 2.535e-11
cell-centred $\rho v r^2$ over $r \geq 1.03$	3.433e-03 → 3.432e-03	6.751e-05 → 6.751e-05
$\log_{10} \dot{M}$	10.34 → 10.34	13.21 → 13.21
$v(1)$	-15.014 cm s ⁻¹ → -251.875 cm s⁻¹	1372.504 → 1372.494 cm s ⁻¹

Two different flatness measures appear in that table and both belong there. The gate is the signed Riemann face flux (§145.); on the hot Uranus this section improves it by a factor 49 at $r \geq 1.03$ and 24 at the gate window. The cell-centred product does not move, because it is not what the solve is driving — and it is the quantity item (AA) was written about, so it is quoted beside the gate rather than instead of it.

The hot-Uranus accepted residual falls by a factor 3.1 when the ghosts are consistent with the composition, which is the lag being removed. The iteration count rises by 47 per cent there and by 91 per cent on WASP-121b.

146.5 Does it move item (AD)? On the hot Uranus, yes

Each root reloaded and marched 3000 steps with the JFNK hand-off removed. Peak-to-peak excursion of the face mass flux, in units of the root’s own F_0 :

root marched with	1.005	1.03	1.10	1.20	du at 3000
pre-146 root, pre-146 binary	1.6384	0.4526	0.1010	0.0190	3.2256e-04
pre-146 root, this section’s binary	1.6384	0.4526	0.1010	0.0190	3.2256e-04
this section’s root , pre-146 binary	0.4443	0.2517	0.0403	0.0084	3.2250e-04
this section’s root , this section’s binary	0.4443	0.2517	0.0403	0.0084	3.2250e-04

Read the rows in pairs: **the departure is a property of the root and not of the marching binary**, to four digits. The root this section produces departs **44 per cent less at $r = 1.03$ and 73 per cent less at 1.005**. §144. measured §143.’s row scales at 0.06 per cent and P55’s restart handling at 0.7 per cent on the same quantity; this is the first change that moves it.

What the two roots differ by is cell 1, and almost nothing else. Every column agrees to 10^{-3} or better except the base cell’s velocity, -15.0 cm s⁻¹ before and -251.9 cm s⁻¹ after; the flux spread over $r \geq 1.03$ is the same to five digits and $\log_{10} \dot{M}$ is the same. So the solve is landing at a different point along the near-null direction item (V) owns, and that point is the one whose ghosts the marching loop would have written anyway — which is what the ordering fix is for. **Whether the 44 per cent is that consistency or the cell-1 displacement is not separated here**, and the honest statement is that the two arrived together.

On WASP-121b there is nothing left to move: $0.0102 \rightarrow 0.0099$ at $r = 1.03$, a 3 per cent change on an excursion that P55’s restart handling had already reduced to one per cent of the hot Uranus’s.

146.6 What is not settled

- **The interface, not the invariant.** `newton_residual` still takes the composition as `intent(inout)`; the state-function property is a discipline its five callers keep. Item (AF).
- **The banded Jacobian's directional derivative** disagrees with the finite-difference ray by 1.0261×10^{-1} on the hot-Uranus hand-off, unchanged by this section (EXHALE_JAC_TEST=1, identical in both builds). It is a property of the banded model, and §142. is where it belongs.
- **The ghost cells' own composition** is still solved at the pressure the provisional `Apply_BC` writes, not at the final one. Deliberate; §146.2.
- **Which half of the hot-Uranus improvement is the consistency and which is cell 1** (§146.5).
- **The cost.** 36 per cent more iterations on the hot Uranus and 118 per cent on WASP-121b, not investigated. The residual is a different function now, so a different iteration count is expected; whether the extra iterations are the solve doing more work or the line search being handed a harder problem was not measured.

Files: `src/modules/states/Apply_BC.f90` (`Apply_BC` writes the ghosts), `src/modules/time_step/steady_newton.f90` (`eval_residual` re-fills the ghosts after its sweep). Test: `src/tests/residual_determinism/`. Documentation: `docs/p_ab_ghost_order.md`, `TO_BE_DONE.md` ((AB) closed, (AF) opened, (AD) narrowed again).

147. The equilibrium sweep's seed is an argument (2026-09-03)

Judgement first. The steady residual was a function of its argument only because five call sites remembered to **make it one**. `eval_residual` took the composition as one `intent(inout)` array that was both the seed the equilibrium sweep starts from in each cell and the place its answer is written, so whatever the caller's workspace last held became the next evaluation's initial guess. Measured (§146.3): evaluate F at a state, at two others, then at the first again through one shared workspace and **1406 of 1500 entries differ, by 3.0×10^{-3} of the row scale against an acceptance tolerance of 10^{-5}** . Every call site in the solver defended against that by copying the adopted composition into a workspace first — `f_sp_j = f_sp`, `fwork = f_sp_base`, five times — and nothing said so, tested it, or stopped a sixth call site from being written without it. **The seed is now an argument.** `eval_residual(Y, f_sp_seed, f_sp, ...)` takes the seed `intent(in)` and writes the swept composition to a separate `intent(out)` dummy; the copy happens once, inside; and because the two may not be the same array a caller cannot hand the sweep its own previous answer even by accident. The five copies are deleted. **The test now passes structurally**, the residual moving no more when another state is evaluated in between than it moves when nothing is (§147.3), and the solve is bitwise unchanged on every case measured — the convention was being kept, and it is now kept by the interface.

It is the same rule the chemistry of a single cell already states one level down: a cell's root may not depend on the cell solved before it (`seed_species_of_cell`, `constrained_chemical_equilibrium.f90`). This raises it from the cell to the sweep, and does not define it a second time.

147.1 The interface

```
subroutine eval_residual(Y, f_sp_seed, f_sp, Fvec, heat, cool, n_part, ...)
  real*8, ..., intent(in)  :: f_sp_seed      ! where the sweep starts
  real*8, ..., intent(out) :: f_sp           ! where its answer goes
  ...
  f_sp = f_sp_seed                      ! the first thing it does
```

newton_residual takes the same pair. Fortran forbids aliasing an intent(in) dummy with an intent(out) one, so the failure mode the measurement found is not expressible.

What the call sites look like now. The three that evaluate the iterate the solve is standing on read

```
call eval_residual(Y, f_sp, f_sp_j, F, heat0, cool0)
f_sp = f_sp_j
```

and that copy is an *adoption*, not a seeding convention: the seed is named in the call, and the assignment afterwards says the solve has taken the result as its new composition. The line searches and the damped Gauss–Newton escape pass (Ytry, f_sp, f_sp_j, ...) — seed from what is adopted, write somewhere else — and adopt only when the step is accepted, which they already did. The banded Jacobian, the Levenberg–Marquardt descent and the Krylov directional derivative pass (Y, f_sp_base, fwork, ...): those three already declared their seed intent(in) and copied it by hand, and now they simply hand it over.

147.2 What it changes: nothing numerical

	hot Uranus gate rung	WASP-121b wasp_full_newton
info	0 → 0	0 → 0
iterations	25 → 25	109 → 109
R accepted	3.176e-06 → 3.176e-06	8.138e-06 → 8.138e-06
gate flux spread	2.466e-14 → 2.466e-14	1.153e-11 → 1.153e-11
log ₁₀ $\dot{M}$	10.34 → 10.34	13.21 → 13.21
Hydro_ioniz.txt, Ion_species.txt	byte identical	byte identical

and the ten-case matrix is byte identical as well. That is the expected outcome and it is the point: the solver was keeping the convention, so moving it into the interface can only remove a way to break it.

147.3 The test

src/tests/residual_determinism/ drives EXHALE_RESID_DETERMINISM=1, which evaluates $F(Y_0)$ twice with nothing in between (the control), then $F(Y_0)$, a trial, $F(Y_0)$, another trial, $F(Y_0)$, **reusing one output workspace throughout** so that it carries each evaluation’s composition into the next call, and compares the three $F(Y_0)$ bit for bit.

	entries differing (of 1500)	max, row-scale units
before §147., one shared workspace	1406 and 1400	2.968e-03, 2.973e-03
before, with §146.'s ghost ordering	1412 and 1399	4.239e-03, 4.280e-03
§147.	6 and 3, within the control	1.374e-12, 6.253e-14
the control itself (nothing in between)	6	1.624e-12

The test is judged against its control, and that took a correction. As first written it demanded bitwise agreement, and on the tree it was installed in it therefore failed while the invariant held. The control — $F(Y_0)$ twice with nothing in between — differs in **6 entries, all in cells 1 and 2, row-scale max** 1.62×10^{-12} : the first call through a run builds tables the later ones reuse, the H_2 rovibrational table among them, and that is a one-time cost of the first evaluation rather than history passing between states. Re-evaluating $F(Y_0)$ after each of two trial states then differs by 1.37×10^{-12} and 6.25×10^{-14} — no more than the control, and seven orders below the 10^{-5} the solver accepts at. So the test asks the question it exists to ask: does F move MORE because another state was evaluated in between than it moves when nothing is? A 10^{-9} floor keeps that comparison meaningful when the control comes out clean, and the control line is printed rather than absorbed so that a future change which makes the first evaluation differ for a different reason stays visible.

147.4 What is not settled

- **The marching loop's `ioniz_eq` call is untouched**, and correctly so: there the seed IS the run's stored composition, which is what the sweep should start from, and `f_sp_io` being both is the honest contract. What this section removes is a caller passing a workspace and getting the last state's answer as its guess.
- **The 6-entry control difference** is attributed to first-call table building by its signature — cells 1 and 2 only, 10^{-11} , identical after the first evaluation — but the specific table was not identified.
- **Nothing here makes the sweep's root seed-independent.** A Newton solve in one cell, started from a different guess, may land on different bits, and the invariant this section establishes is that the guess is a stated function of the caller's argument, not that the root is independent of it. How far the root moves with the seed is not measured.

Files: `src/modules/time_step/steady_newton.f90` (`eval_residual` and `newton_residual` take the seed as an argument; the five copies deleted), `src/EXHALE_main.f90` (two call sites). Test: `src/tests/residual_determinism/`. Documentation: `TO_BE_DONE.md` ((AF) closed).

148. The PLM to WENO3 hand-off is a continuation between two discrete problems, not a switch (2026-09-03)

Judgement first, and it is two-sided. On the two cases that could be measured, the shipped one-step hand-off is a **harmless transient**, and that is a measurement rather than an assumption: the `dtu` excursion it causes is quantitatively the fixed-state operator difference (1.83 predicted against 1.71 measured on the first step), it decays below its pre-hand-off value within about fifty steps, `du` never stops descending, and no positivity guard fires. **But the two discretizations**

do not share a solution near the boundaries, which is the part that matters for the physics: over the wind PLM and WENO3 agree to a part in 10^2 to 10^4 , while in the first two or three cells above the base the WENO3 mass-row residual is **6.7 times the largest PLM residual anywhere on the grid**, and in the outermost cells the momentum rows differ by 62 per cent. A `wasp_full` run held on PLM does not converge in 15,000 steps and settles on a base 400–700 K hotter and up to 2.4 times denser, with the opposite sign of base velocity and $\log_{10} \dot{M}$ 0.08 higher. So the two-stage recipe is a continuation between two different discrete problems, and treating it as one is right even where the transient it produces is benign. The continuation is implemented and **default off**; $\lambda = 1$ reproduces the shipped code to the bit.

148.1 The instrument: an operator difference on a fixed state

`EXHALE_OPDIFF` assembles the PLM residual and the WENO3 residual for the same interior cells, each with its own ghosts and pressure split, and stops. On `mol_sec_ion` at the hand-off step (11,861):

row	max PLM /s	max WENO3 /s	max difference /s	at	difference / PLM
mass	1.519	10.245	10.232	$r = 1.00019$	6.74
momentum	1.678	2.379	1.036	$r = 4.681$	0.618
energy	1.924	1.925	1.202	$r = 1.00077$	0.625

The difference lives at the two boundaries and nowhere else, so its origin is the base ghost and pressure split at one end and the outer extrapolated ghost at the other, not the reconstruction over the wind. Predicted against measured across the hand-off, on the `dtu` window ($r \geq r_{\text{esc}}$):

	predicted	measured
PLM step dtu	5.979e-4	5.948e-4 (step 11,861)
WENO3 step dtu	1.095e-3	1.015e-3 (step 11,862)
ratio	1.831	1.706

148.2 The continuation

`reconstruction_continuation_rhs` assembles the whole right-hand side under each flag and combines them,

$$R_\lambda = (1 - \lambda) R_{\text{PLM}} + \lambda R_{\text{WENO3}}, \quad (54)$$

blending the outer ghosts of `Apply_BC_W` with the same λ . Both the three-stage marching step and `assemble_residual` go through it, so the marching map and the residual continue together. At $\lambda = 0$ and $\lambda = 1$ it is a single evaluation and bitwise equal to the pure scheme; on reaching $\lambda = 1$ it disarms itself and hands the flag to WENO3, **so the final gates are evaluated with the pure WENO3 operator**.

Input key, refused rather than ignored where it cannot apply:

Reconstruction continuation: `<dlambda>` [`<dtu_tol>`] [`fixed|adaptive`]

`dlambda` ≤ 0 or an absent key is the shipped one-step hand-off; `dtu_tol` (default 1.2) halves the λ step instead of advancing when a step exceeds that multiple of the `dtu` at the start of the ramp; `fixed` turns that control off. The

two optional fields are told apart by content rather than position, the mode being a word and the tolerance a number. Resolved at the end of `input_read` once `Reconstruction` scheme: and both `du_th` values are known: a run that is not two-stage, or whose PLM stage is empty ($\text{du_th_plm} \leq \text{du_th}$), gets the key cleared and a WARNING. The four `EXHALE_RECON_LAMBDA_*` environment variables of the measurement build are gone; `EXHALE_OPDIFF` stays one, with the other diagnostic probes that measure a state and change no result.

148.3 What the ramp does to the transient

`mol_sec_ion`, whose `dtu` before the hand-off is $5.948\text{e-}4$:

run	ramp steps	$\lambda = 1$ at	back-offs	dtu peak	vs before
shipped, one step	—	11,861	—	$1.060\text{e-}3$	1.763
<code>dlambda = 1</code>	—	11,861	—	$1.060\text{e-}3$	1.763
<code>0.02</code> , fixed	49	11,910	0	$7.501\text{e-}4$	1.248
<code>0.02</code> , adaptive	58	11,919	2	$7.213\text{e-}4$	1.200
<code>0.002</code> , adaptive	499	12,360	0	$5.948\text{e-}4$	0.990

A 500-step ramp removes the excursion entirely, and the cost is below the scatter of a shared machine (14,000 steps: 10 m 37 s shipped, 9 m 36 s with the ramp).

148.4 Why it is default off, and what would justify turning it on

Adopting a default ramp is not justified by these two cases: it buys a 20–25 per cent smaller `dtu` excursion that decays on its own within fifty steps. It would be justified by the configuration item (S) reports — HD 189733 b with molecular and oxygen chemistry, transport and IR bands, where the same hand-off is recorded as taking `du` to $\sim 10^5$ within 2000 steps — and that case was out of reach here (it is not in the matrix and one trajectory to it costs tens of thousands of steps). What this work supplies is the instrument to settle it in one run: `EXHALE_OPDIFF=2` on that state measures $R_{\text{WENO3}} - R_{\text{PLM}}$, and if the ratio is the ~ 7.5 its reported `dtu` jump implies then the hand-off is the cause, while ~ 1.8 like here would mean a WENO3 instability the hand-off merely uncovers, which a continuation will not fix.

148.5 A provenance defect, fixed at its single definition

The tree wrote `recon=trim(rec_method)` in two places, the coupling header and the provenance header, so a run stopped inside a ramp would have reported `recon=PLM` while its equations were part WENO3. `reconstruction_operator_label()` in `parameters.f90` is now the one definition and both headers call it:

```
# coupling: sec_ion=T sec_ion_step=0 valve=-1.00000E+00 recon=PLM+WENO3(lambda=0.0400) fluxconst=-1.00000E+00
# provenance: recon=PLM+WENO3(lambda=0.0400) base_bc=isothermal carrier=local restart_schema=3 resid_def=145 N
=500
```

Snapshots before the hand-off read `recon=PLM` and after the ramp `recon=WENO3`, in the same run.

148.6 Verification

$\text{dlambda} = 1$ is the stronger equivalence test than an absent key, since it drives the blended right-hand side, the blended ghosts, the λ -weighted face-pressure term, the controller and the disarm and still lands on the shipped result: `wasp_full` and `mol_sec_ion` are byte-identical to the unpatched control with the `du/dtu` traces agreeing digit for digit, and so is an absent key. The parse dump gains the three values **only when the key is used**, following the `if (do_read_sed)` precedent in the same file, so the 1122 files of `parse_golden/` do not move. The ten-case matrix is byte-identical: the key is off, and six of the ten never leave the PLM stage in the first place.

Files. `src/modules/init/parameters.f90` (the continuation state and `reconstruction_operator_label`), `src/modules/files_IO/input_read.f90` (the key and its refusal), `src/modules/files_IO/write_setup_report.f90`, `src/modules/functions/utilities.f90` (both headers, one label), `src/modules/states/Apply_BC.f90` (the blended outer ghost), `src/modules/time_step/RK_rhs.f90`, `src/modules/time_step/steady_residual.f90` (`reconstruction_continuation_rhs`), `src/EXHALE_main.f90` (the controller). Documentation: `docs/f_plm_weno_continuation.md`, `docs/input_schema.md`.

149. One routine counts the elements, and it is asserted around every refresh that moves them (2026-09-03)

Judgement: the code had no single statement of how many nuclei of each element a composition holds. `utilities.f90::hydrogen_helium_nuclei_density` covers H and He; C, N, O and the other metals had no counterpart, and their closure existed only in `src/utls/element_budget.py`, which reads a finished run. A run could therefore break an element budget in one refresh and only be found out several thousand steps later, by a post-processor that says a run failed but not which refresh failed it. `element_census.f90` is that statement, and the assertions built on it name the refresh.

149.1 What the routine counts, and from where

`element_nuclei_and_charge(rho, f_sp, n_nuc, n_chg, rho_comp)` returns the nucleus density of H, He and the ten metal elements, the positive charge density (the free electron density under neutrality), and the mass density the composition implies through `calc_rho`'s own weights.

No multiplicity is written in the module. It reads `bsp_nH`, `bsp_nHe`, `bsp_nO`, `bsp_nC`, `bsp_charge`, `bsp_mass`, `bsp_is_excited_level`, `mion_elem`, `mion_stage` and `melem_A` from `species_table`, so H_2 and H_2^+ carry two H nuclei, H_3^+ three, HeH^+ one H and one He, OH one H and one O, H_2O two H and one O, CO one C and one O, each metal ion stage one nucleus of its element, and the He 2^3S column is skipped because it is a level of He I and not an independent species. Adding a carrier is adding a row to `species_table`; this module follows without being edited.

`hydrogen_helium_nuclei_density` is left where it is. Its statement order is the floating-point add order the goldens were snapshotted with, and it is called from three places that must not disagree. The new routine is the metal counterpart it never had, and the two are held together by test E1, which requires the charge the census returns to equal `calc_ne` on the same state.

149.2 Where it is asserted, and what the statement is at each site

Off unless EXHALE_ELEMENT_ASSERT is 1 (report and continue) or 2 (report and stop), the convention EXHALE_CARRIER_DEBUG and EXHALE_DIFFUSION_CHECK already use. When off, nothing is allocated and nothing is computed.

site	statement
around carrier_write_back	absolute nucleus density of every element, per cell
around photochemical_transport_step	the same: the write-back restores the entry element the whole step is element-neutral
around every ioniz_eq sweep, tagged marching / steady iterate / steady candidate	every ratio $n_{\text{El}}/n_{\text{H}}$
around relax_photochemical_composition	absolute nucleus density of every element
around the restart equilibration	every ratio $n_{\text{El}}/n_{\text{H}}$
the accepted state of each steady outer pass, and immediately before each final write_output	against the reservoirs the run resolved

Why the ratio and not the density across ioniz_eq. n_{io} is `intent(inout)` and the sweep rewrites it from its own `calc_rho`, so every density it returns may move by one common mass-closure factor. That factor is reported separately — and the distinction matters, because “every element off by the same amount” is a moved n_{H} denominator while “one or two elements off” is a repartition defect. The failure report prints the worst departure of each of the twelve elements so the two read differently at a glance.

The tolerance is round-off, not abundance. Default 1.0×10^{-9} (EXHALE_ELEMENT_ASSERT_TOL overrides). These are exact bookkeeping identities: the operators rescale stage populations onto element totals, so the only error is floating point. Measured on `mol_carrier` at 300 steps, every operator closes at 3.5e-16 to 6.2e-16; the accumulated round-off of a finished marching state is $\leq 10^{-12}$; the cell solver’s own $\text{x}tol = \sqrt{\epsilon} = 1.5 \times 10^{-8}$ is the largest departure a converged cell solve can leave. 10^{-9} sits between the two.

On failure it prints the element, the cell, its radius, the before/after element and hydrogen totals, `rho*n0` and `rho(composition)` before and after, and every species carrying that element in that cell with its multiplicity.

Cost, `mol_carrier` at 300 steps single-threaded: 11.96 s with the assertions off against 11.77 s with them on and fatal (898 checks). Within noise.

149.3 Item 12.8 does not reproduce on the path it was reported on

`TO_BE_DONE.md` item (T) records the first converged A2 wind failing its element budgets near $r = 1.003\text{--}1.006$, carbon by +90.6 per cent and oxygen by +49.0 per cent, with the carrier nuclei counted once inside the carriers and once inside the free-stage columns “somewhere in the JFNK/Picard state refresh”.

Run with the assertions on, the A2 oxygen configuration restarted into the steady solver — 42 steady-iterate sweeps, 1230 probe sweeps, and, with the outer loop forced to take its carrier pass from an unconverged solve, 20 passes of `relax_photochemical_composition` — **breaks no element budget**. Every sweep, every write-back, every transport step and the restart equilibration close inside 10^{-9} , and the finished output closes H, He, C, N and O at 1.77×10^{-10} .

This is not the state item (T) was measured on: that state is a converged A2 wind reached from a ~25000-step march plus a JFNK finish at `info = 0`, and it does not exist in the tree. So this is a *not reproduced on the same code path*, not a

fixed. The likeliest reading is that the carrier rework of §128. removed it: that section is dated after item (T) was opened, it is in the same two routines, and what it repaired was exactly two disagreeing definitions of the element budget the carriers are held against.

What does survive: the steady/Picard path closes at 1.77×10^{-10} where the marching path closes at 2.6×10^{-13} , and the departure is the same number for all five elements — the hydrogen denominator, not the elements. Three decades inside the gate; reported, not chased.

149.4 A restart dump reported a molecular run as atomic, and under the wrong coupling

Both found by the new restart round-trip fixture, both in the EXHALE_DUMP_IC diagnostic rather than in load_IC:

- write_output takes the H_2 / H_2^+ / H_3^+ / HeH^+ / OH / H_2O / CO columns from nmol_eq and nox_eq, which only the equilibrium sweep fills. The dump hook runs before the first sweep, so on mol_base_handoff all four molecular columns came out *exactly zero* against the $5.34 \times 10^{12} \text{ cm}^{-3}$ of H_2 the restart file carried. molecular_carrier_densities_from_state(rho, f_sp) fills them from the loaded composition first: a state a diagnostic writes has to be the state it was given.
- The rule that adopts the file's # coupling: state runs about 280 lines after the dump hook, so the dump wrote sec_ion=F sec_ion_step=-1 over a file that stated sec_ion=T sec_ion_step=40. The same rule is now applied inside the dump block.

Neither touches a production path: both are inside the EXHALE_DUMP_IC block, which ends in stop.

149.5 Tests

make element_census_tests builds element_census_tests.x, on the diffusion_tests.x pattern: synthetic columns the driver sets itself, no input.inp, stop 1 on failure. 20 tests, 20 passing.

- E1 — the census reproduces the species-table stoichiometry on a hand-computed state, and its charge equals calc_ne on the same state.
- E2 — element closure through one carrier write-back: 0.0 for a generic perturbation and 1.9×10^{-16} for the configuration item 12.8 reports, a cell whose transported CO collapses by 2.5 decades and hands its carbon back to the free stages.
- E3 — the H_2 photoevent ledger. Over 15–200 eV of the Chung et al. (1993) branching, every final state the code applies carries two H nuclei and zero net charge once its electron is counted. The two channels the branching does *not* resolve are measured rather than assumed small and printed beside the result: double ionization is inside f_di and is about 4.3 per cent of the H_2 ionizations above 80 eV with its co-product mis-assigned, and neutral dissociation over 33–41 eV (up to 7 per cent of sigma_H2 at 37.5 eV) is handed to the H_2^+ branch by $1 - f_{\text{di}}$, which creates one H_2^+ and one electron the event does not make. Both are item 12.1's stated approximation; the gate for them belongs with explicit cross sections resolved by channel.

- E4 — the chemical-equilibrium branch of the base H_2 fraction (item 12.6): with no handoff, $q_{H_2} = 0.838401$ from the fit at (p_{base}, T_0) and the ghost H_2 nucleus fraction is 0.984428 at $He/H = 0.0793$; with a handoff, the handoff value and an imposed base composition; and the fit exceeds the attainable ceiling $0.5/(0.5 + He/H)$ at $He/H = 0.2$, which `input_read` refuses.

The restart round trip is a shell fixture rather than a Fortran test, because it needs two runs of the binary. `backup/regression/roundtrip` is rebuilt in place around it (item 12.4). The case that carried this name could not run at all — `output/Hydro_ioniz_IC.txt` was zero bytes, there was no `Ion_species_IC.txt`, `Load IC? True` made it abort in `load_IC`, its logs were from the ATES era, and it was in no `DEFAULT_CASES` list — because it had been built from a converged product that no longer exists.

It now builds its own state, in two halves that need no coordination:

- **stage A is an ordinary named case.** 40 deterministic steps of the hot-Uranus molecular gate (`input.inp`, `base.inp`, `maxsteps`), so `run_check.sh check roundtrip` bitwise-compares it exactly like a matrix case. 40 steps is short enough to run in seconds and long enough that the state is not the initial condition: the molecular network has solved and the `# coupling: line` carries `sec_ion=T sec_ion_step=40` rather than its default.
- **stages B and C are `roundtrip/roundtrip_check.sh`,** run after that. It hands the stage-A outputs back as the `_IC` pair, reloads with `EXHALE_DUMP_IC=1`, and requires every species column — matched BY LABEL, so a schema change reports a missing column instead of a numeric mismatch — together with `rho`, `v`, `p` and the `# coupling: header` to round-trip to 10^{-12} . It restores `input.inp` and the stage-A outputs on exit, so `run_check.sh golden roundtrip` still snapshots the right state.

The case stays OUT of `DEFAULT_CASES` and is described in the `run_check.sh` header, because its second half is not a bitwise comparison and needs its own invocation. `check_roundtrip_state.py` replaces `check_roundtrip.py`, which compared columns by POSITION and had this case as its only consumer; the retired contents are kept in `roundtrip/.superseded_20260903/` rather than deleted, since `backup/` is not in the git remote.

Measured: the rebuilt case FAILS on a binary without §149.4 (all four molecular columns zero, and the coupling header written as `sec_ion=F sec_ion_step=-1`) and PASSES with it, so it guards exactly the two defects it found.

149.6 Verification

- Byte-identical `output/Hydro_ioniz.txt` and `output/Ion_species.txt` against a build of the unmodified tree on `mol_base_handoff`, `mol_carrier` and `wasp_full`, 300 steps each single-threaded. The new module is inert unless the environment variable is set; the two restart fixes are inside a block that ends in `stop`.
- `element_census_tests.x: 20/20`.
- The rebuilt `backup/regression/roundtrip: run_check.sh check roundtrip` byte-identical against its own golden, and `roundtrip_check.sh` passing every species column, `rho`, `v`, `p` and the `# coupling: header` at 10^{-12} , leaving the case directory unchanged. The same case fails on a binary without §149.4.
- The A2 steady/Picard runs of §149.3, with `EXHALE_ELEMENT_ASSERT=1`.

150. E2: the Lyman–Werner geometry measured, and the one dayside factor a run is allowed to have (2026-09-03)

Judgement first, in two parts. (i) The plane-parallel trapping inside the H₂ self-shielding table is **physically wrong for a spherical layer, in a known direction and by a known amount: it over-traps**. Measured against a matched spherical calculation on the same line data, it over-predicts p_{eff} by 0.2 per cent at $r/H = 41$ and by up to 20 per cent at $r/H = 16$, one-signed, 6.5 per cent weighted by the dissociation rate over a hot-Uranus molecular layer. It is *not* rebuilt, because two escape-probability methods in the *same* slab differ by 20–40 per cent and not with one sign, so the method is the leading term and the geometry is not. The headers now carry the number instead of the words “an open item of TO_BE_DONE.md”, and the trapping carries the provenance name `slab_surrogate`. (ii) A separate defect found while measuring it: **a run could carry three dayside conventions at once**. `set_energy_vectors` halved F_{XUV} on an exact string match, `lya_rt` and `excited_hydrogen` halved the stellar Ly α beam on a substring test, and `fuv_band_flux` halved none of the five FUV bands — and the H₂ dissociation rate did not even read that function, taking `F_LW_star` directly. Under `Rate/2 + Mdot/2`, the setting of the whole molecular matrix, the Lyman–Werner rate stood **a factor 2 above the run’s own convention**. That is fixed: one definition, `global_parameters` `dayside_dilution()`, read by every stellar beam the code has.

Full account, design and figures: `docs/e2_lw_geometry.md`, `docs/e2_lw_geometry_factors.png`, `docs/e2_lw_geometry_rays.png`.

150.1 What the table’s geometry is, read out of the code

`sigma_pump` and `p_single` come from `lbl_table.absolute_curves` through `src/utils/h2_shielding_table_line_by_line.py`, a plane-parallel slab with one normally incident beam,

$$\tau(\nu, N) = N \sum_i x_i \sigma_i(\nu), \quad (55)$$

with N the column between the illuminated face and the point. $p_{\text{eff}}/p_{\text{single}}$ is read node by node from the 27 CLOUDY decks, whose geometry is plane-parallel and illuminated on one face. The column the code hands the table, `calc_column_dens_one(nH2, NH2col)`, is *purely radial* — accumulated from the outermost cell inward, no stellar zenith angle anywhere in the path.

That last identification is exact, not an approximation. For a parallel beam the substellar ray column *is* the radial column, in a sphere as in a slab. The slab table is therefore right for the substellar ray, and the geometry question is only about (a) the fluorescent trapping and (b) which ray a 1-D shell should carry.

150.2 The matched calculation

Slab and sphere are evaluated from one kernel built on the Abgrall, Roueff & Drira (2000) line list, LTE populations and the Voigt profiles that generated the installed table — `lbl.py` is imported directly, so no molecular datum can differ. An escape probability averaged over directions is linear in the transmission, so with

$$\sum_l A_{ul} \langle P_{ul} \rangle_{\Omega} = \sum_k w_k A_{\text{esc}}[u, N_k] \quad (56)$$

the two geometries differ *only* in the set of ray columns $\{N_k, w_k\}$. P_{ul} is a single-flight escape probability under complete redistribution against the full overlapping-line opacity.

Slab limit: sigma_pump and p_single reproduce lbl_table.absolute_curves to **seven significant digits**, and the installed Fortran table's p_single to 0.1–0.5 per cent from 10^{16} to 2×10^{21} at 700–2700 K. Resolution: angle 8 to 96 Gauss–Legendre points, $< 10^{-4}$; path sampling 500 to 16000, $< 3 \times 10^{-5}$; frequency window 2000 to 20000 Doppler widths, $< 2 \times 10^{-5}$; frequency grid 4×10^5 to 8×10^5 points, $< 10^{-5}$.

150.3 The numbers

r/R_p	$N(\text{H}_2)$	$p_{\text{eff slab}}$	sphere	f_{trap}	f_{curv}	f_{shell}
1.0002	3.90e20	0.2092	0.2088	0.998	1.058	0.204
1.0544	8.29e18	0.2540	0.2321	0.914	0.993	0.295
1.0710	1.21e18	0.2453	0.2210	0.901	0.992	0.320
1.1288	3.38e15	0.3016	0.2648	0.878	0.994	0.261
1.3629	9.91e12	0.2163	0.1728	0.799	1.201	0.589

H₂-column weighted / dissociation-rate weighted over the layer: f_{trap} 0.968 / 0.935, f_{curv} 1.034 / 1.013, f_{shell} 0.229 / 0.261.

The mechanism: escape is dominated by the near-radial directions, where the two geometries have the *same* column — which is why the effect is small — and the difference is the near-tangential rays, which a sphere caps at $\sqrt{\pi r/2H} = 8\text{--}12$ times the radial column where a slab lets them diverge, plus the lit sliver past the terminator. The controlling parameter is r/H , not the column, which is why the effect vanishes at the base ($r/H = 41$, the escape hemisphere closes exactly at $\mu = 0$) and is largest at the top of the layer ($r/H = 16$) where H₂ is already negligible.

$f_{\text{shell}} = 0.20\text{--}0.33$ is *not* a slab error. It is the ratio of the shell average a 1-D spherically symmetric H₂ budget needs to the substellar value the code evaluates. Its optically thin limit is 1/2, because the night side is dark; it falls below that because the slant columns away from the substellar point shield.

150.4 One dayside factor

global_parameters gains dayside_dilution(). set_energy_vectors, lya_rt, excited_hydrogen and fuv_band_flux all read it, and lyman_werner_dissociation_rate is called with fuv_band_flux(ib_LW) instead of F_LW_star, so the band has one owner again. The keys keep their meaning — the flux **at the planet's orbit**, whether stated or integrated from the spectrum file — and the dilution is applied where the beam is used, so the setup report and the resolved dump still state the number the run was given. The setup report gains one line naming the factor, and output/Lyman_Werner.txt a header line giving the beam that actually drove k_{LW} .

Two stale statements found in the same files and corrected: write_output described the f_shield column as “the Richings, Schaye & Oppenheimer (2014) temperature-dependent fit”, which §122. replaced by the level-resolved table; and the generator's docstring used a forbidden naming form.

150.5 What moved

Nothing that does not carry an FUV band flux. `wasp_full` (He 2³S, metals, Stellar Ly α flux: 1.0e5, Rate/2 + $\dot{M}$ /2, 400 steps) and `mol_base_handoff` (the molecular gate with no FUV band flux, 12000 steps) are **byte-identical** in every output file, `Excited_H.txt` and the `_adv` profiles included. The XUV refactor, the two ξ call sites and the new Ly α B2 dilution therefore cost nothing: `fuv_band_flux` reaches the solution only through the oxygen photolysis, which those runs do not have, and $F_{\text{XUV}} = \xi_{\text{day}} \cdot \dots$ reproduces the old literals exactly. The only change in those runs is one line of the setup report.

mol_Lyman_werner is the one matrix case that moves, and it moves by more than the factor 2 on the rate suggests, because the H₂ front is where the rate and the chemistry balance:

quantity	before	after	ratio
H ₂ front, $x(\text{H}_2) = 0.1$	1.04763 R_p	1.05546 R_p	1.0075
H ₂ front, $x(\text{H}_2) = 10^{-3}$	1.07299 R_p	1.08122 R_p	1.0077
$N(\text{H}_2)$ through the layer	2.857e20	3.241e20	1.134
$x(\text{H}_2)$ at the base	0.4715	0.4808	1.020
T at the base	1124.5 K	1134.4 K	1.009
$\dot{M}$ at 2 R_p	8.252e10 g s ⁻¹	7.917e10 g s ⁻¹	0.959
$\log_{10} \dot{M}$ (run report)	10.47	10.44	-0.03 dex

$x(\text{H}_2)$ rises by 1.02 at the base, 1.41 at $r = 1.039$ and **4.28** at $r = 1.081$, right through the front, and falls back to 1.04–1.38 in the wind above it. Halving the beam moves the front out by 0.8 per cent of R_p and puts 13 per cent more H₂ under it, and the extra molecular cooling and mass in the layer takes 4 per cent off the mass-loss rate.

The SED-derived branch moves the same way and by much less, for a reason that is about the star and not about the code. A hot-Uranus gate rerun with `Spectrum type: Load from file..` and the LHS 1140 SED (no Stellar LW flux key) integrates $F_{\text{LW}} = 0.598 \text{ erg cm}^{-2} \text{ s}^{-1}$ out of the spectrum — 570 times weaker than the 343 of the regression case — so Lyman–Werner is a minor H₂ sink there: after 3000 steps the front moves by 0.04 per cent and $N(\text{H}_2)$ by 0.18 per cent. What the case establishes is that the stated key and the integrated value take the *same* path: `input_read` assigns the SED value to `F_LW_star` and the dilution is applied downstream in `fuv_band_flux`, so neither can drift from the other. **The matrix has no case that exercises this branch;** the one above is a scratch construction.

The gate rung’s converged state is not re-established here. `mol_lyman_werner` is a fixed-step relaxation snapshot (`maxsteps 12000`), and uncapped runs of it from both binaries are still marching toward the JFNK hand-off at $du < 10^{-2}$: the baseline is at $du = 0.090$ after 30171 steps and the patched run at $du = 0.199$ after 15830, both falling at about 8.5×10^{-6} per step. On that trend the hand-off is thousands of steps away for the baseline and tens of thousands for the patched run. **The front and $\dot{M}$ above are therefore snapshot values at a common step count, not converged ones,** and a quantitative $\dot{M}$ for this case still needs the Newton finish.

150.6 Not done, deliberately

The trapping cube is not rebuilt in spherical geometry. No r/H axis is added: over the H₂-bearing layer it would span 0.90–1.00. The shell average is available as a **default-off** closure and is not the default, because it is calibrated on one

planet and changes the meaning of a converged solution.

150.7 The golden refresh

Measured on the eleven-case matrix with the change in: **ten cases byte-identical, mol_lyman_werner alone moved**, which is the case list the grep of §150.5 predicts, since it is the only one carrying an FUV band flux. Re-measured against the golden it replaces: ρ 7.3 per cent and T 4.5 per cent at their worst cells, p 10.8 per cent, $\log_{10} \dot{M}$ 10.47 $\rightarrow$ 10.44. The two large relative changes in velocity are both at velocity zeros and are not what they look like: at $r = 1.0093$ the old value is 63.9 cm s^{-1} , which is 5.7×10^{-5} of the domain maximum and the 46th smallest of 500 cells, and the change of 851 cm s^{-1} is 7.6×10^{-4} of the wind speed; the other, at $r = 1.200$, sits on the flux-gate window edge. Away from those, the wind above $1.5 R_p$ moves by 7 per cent in ρ , 11 per cent in v and 4 per cent in T , all of it at $r \sim 2.8$. These are positions along a transient, not converged physics: the case is a fixed-step snapshot, so what the numbers establish is the sign and the reach of the change, not its value at a steady state. The previous eleven goldens are preserved as backup/regression/golden_blockI_20260904/.

151. H₂ photoabsorption is a sum of four mutually exclusive channels, and the cross section they share is the measurement rather than a fit that does not meet itself (E1, E1b) (2026-09-03)

Judgement first. The code carried one H₂ photoabsorption cross section and split it with one branching fraction, and three different reactions were inside that split. Written as four mutually exclusive channels with one stoichiometric table, the H₂ destruction rate is unchanged — it always was right, because absorption destroys the molecule whatever it leaves behind — and the FRAGMENTS change: above 51.4 eV a double ionization makes two protons, two electrons and no H atom where the code made one proton, one electron and one H atom, and over 33–41 eV up to 7.4 per cent of the absorptions make two neutral H atoms and no ion at all where the code made an H₂⁺ and an electron. Both new channels default ON, because both reactions are measured and the code was making fragments that the events do not make. The channel split leaves the opacity untouched by construction — the four cross sections sum to sigma_H2 to 4.4×10^{-16} , machine precision, at every energy and under every model.

E1b is the second half and it is the larger one. sigma_H2 itself did not meet itself: the three-piece Yan fit steps by a factor 1.719 at 18 eV and 0.619 at 85 eV, a 72 per cent discontinuity in an opacity that the JFNK residual differentiates. It is now a MEASUREMENT everywhere the wind absorbs — Backx, Wight & Van der Wiel (1976) Table 1 over 15.4–18 eV and Samson & Haddad (1994) Table 1 over 18–300 eV — and no piece of the analytic fit survives below 300 eV. The worst of it was at the ionization threshold, where the fit fell to 0.17 against a measured 14.2, a factor 85. Three independent experiments now bound the cross section: they agree to a few per cent below 30 eV and diverge by 10 to 20 per cent above it, which is the real uncertainty and is wider than any one paper quotes. **The molecular goldens move and the atomic ones do not.**

151.1 What was wrong

sigma_H2 (Yan, Sadeghpour & Dalgarno 1998, Eqs. 17–19) is one number per photon energy, and set_energy_vectors split it in two:

```
s_h2_di = sigma_H2(E) * frac_H2_dissociative_ionization(E)    ! H + H+ + e-
s_h2     - s_h2_di                                           ! H2+ + e-
```

with frac_H2_dissociative_ionization the Chung, Lee, Masuoka & Samson (1993) Table II ratio. Three reactions were folded into those two branches.

- **Double ionization**, $\text{H}_2 + h\nu \rightarrow \text{H}^+ + \text{H}^+ + 2e^-$, threshold 51.4 eV. Chung et al.’s eqs. (3)–(4) state it: “The branching ratio $\text{H}^+ / (\text{H}^+ + \text{H}_2)$ is understood to include both single and double ionization where appropriate”, and their Fig. 4 discussion: “by 80 eV about 20% of $a(\text{H}^+)$ comes from double ionization. This percentage remains fairly constant towards higher energies.” Their $\sigma(\text{H}^+)$ is a PROTON count and a double event yields two, so the single dissociative branch was standing in for two reactions. Catalog item 12.2, TO_BE_DONE.md (P31b).
- **Neutral dissociation**, $\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}$, over 33–41 eV. sigma_H2 there follows Samson & Haddad’s ABSORPTION cross section, whose photoionization yield drops to 0.926 at 37.5 eV, so the $1 - f_{\text{di}}$ branch handed the neutral share to H_2^+ . Catalog item 12.3, (P31c).
- **The extrapolation above 124 eV**, where Chung et al.’s table stops and f_{di} was held flat. Catalog item 12.1, (P31a). This one is not closed here; it is now NAMED instead of silent (§151.6).

151.2 The four channels

src/modules/functions/h2_photo_channels.f90. One routine returns all four cross sections at a photon energy, and one table gives what each event does to the species counts. Nothing downstream writes those numbers out by hand.

	channel	threshold	H ₂	H ₂ ⁺	H	H ⁺	e [−]
M	$\text{H}_2 + h\nu \rightarrow \text{H}_2^+ + e^-$	15.4 eV	−1	+1	0	0	+1
S	$\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}^+ + e^-$	18.076 eV	−1	0	+1	+1	+1
D	$\text{H}_2 + h\nu \rightarrow \text{H}^+ + \text{H}^+ + 2e^-$	51.4 eV	−1	0	0	+2	+2
N	$\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}$	33 eV (window)	−1	0	+2	0	0

Every row conserves H nuclei ($2d\text{H}_2 + 2d\text{H}_2^+ + d\text{H} + d\text{H}^+ = 0$) and charge ($d\text{H}_2^+ + d\text{H}^+ - de = 0$); that is the ledger test of review item A3, and it is checked for all four channels in src/tests/e1_h2/.

How the split is built. With $f_n(E)$ the neutral share and $r_{pm}(E)$ the Chung Table II proton-to- H_2^+ ratio, and $d(E)$ the

share of the released protons that comes from the double channel,

$$\sigma_N = f_n \sigma_{H_2}, \quad (57)$$

$$\sigma_{\text{ion}} = (1 - f_n) \sigma_{H_2}, \quad (58)$$

$$\sigma_M = \sigma_{\text{ion}} / [1 + r_{pm}(1 - d/2)], \quad (59)$$

$$P = r_{pm} \sigma_M \quad (\text{the proton count}), \quad (60)$$

$$\sigma_S = (1 - d)P, \quad (61)$$

$$\sigma_D = dP/2. \quad (62)$$

With $d = 0$ and $f_n = 0$ this collapses to $\sigma_S = f_{\text{di}} \sigma_{H_2}$ and $\sigma_M = (1 - f_{\text{di}}) \sigma_{H_2}$, the previous arithmetic exactly, which is why the default build does not move.

A note on Chung et al.'s own normalization, because it does not close. Their Eqs. (3)–(5) set $\sigma(H^+) + \sigma(H_2^+) = \sigma_i = Y\sigma(\text{abs})$, which equates $(N_s + 2N_d) + N_m$ to an event count $N_m + N_s + N_d$ and so over-counts by N_d . This module does not adopt that identification: it takes their RATIO column as the measurement and imposes the event count separately. That is the only way four mutually exclusive channels can sum to the absorption.

Cross-check against the size bound TO_BE_DONE.md (P31b) predicted. That entry worked out, for $N_d/N_i = 0.0225$ above 80 eV, that the old arithmetic was low by 2.2 per cent in protons, H_2^+ and free electrons and high by 22 per cent in H atoms. Measured on the new construction at 500 eV with the chung80 model: protons +2.1 per cent, H_2^+ +2.1 per cent, H atoms –18.3 per cent. Same signs, same size.

151.3 The channel cross sections

sigma_H2 and the four channels, in units of 10^{-18} cm^2 . The last column is the construction's own closure: the four channels sum to the total absorption cross section at machine precision, so no consumer of the opacity changes. Below 51.4 eV the double channel is empty and all three models agree, so that range is tabulated once:

E [eV]	σ_{H_2}	M: $H_2^+ + e$	S: $H + H^+ + e$	D: $H^+ + H^+ + 2e$	N: $H + H$	sum/ $\sigma - 1$
16.0	1.311E+01	1.311E+01	0.000E+00	0.000E+00	0.000E+00	0.0E+00
18.0	9.850E+00	9.850E+00	0.000E+00	0.000E+00	0.000E+00	0.0E+00
18.5	9.239E+00	9.134E+00	1.047E-01	0.000E+00	0.000E+00	0.0E+00
20.0	7.650E+00	7.500E+00	1.500E-01	0.000E+00	0.000E+00	2.2E-16
25.0	4.310E+00	4.208E+00	1.019E-01	0.000E+00	0.000E+00	0.0E+00
30.0	2.710E+00	2.579E+00	1.309E-01	0.000E+00	0.000E+00	0.0E+00
34.0	1.740E+00	1.533E+00	1.693E-01	0.000E+00	3.800E-02	0.0E+00
37.5	1.228E+00	1.034E+00	1.034E-01	0.000E+00	9.086E-02	0.0E+00
41.0	9.110E-01	7.988E-01	1.052E-01	0.000E+00	7.008E-03	0.0E+00
50.0	4.770E-01	4.037E-01	7.334E-02	0.000E+00	0.000E+00	0.0E+00

Above the double-ionization threshold the three models part. N is zero throughout this range and the closure stays at or below 2.2×10^{-16} .

E [eV]	σ_{H_2}	both channels off			default: chung80			yan_rho		
		M	S	D	M	S	D	M	S	D
60.0	2.620E-01	2.101E-01	5.192E-02	0.000E+00	2.113E-01	4.909E-02	1.571E-03	2.112E-01	4.930E-02	1.453E-03
80.0	9.800E-02	7.640E-02	2.160E-02	0.000E+00	7.812E-02	1.767E-02	2.209E-03	7.779E-02	1.842E-02	1.784E-03
100.0	4.790E-02	3.770E-02	1.020E-02	0.000E+00	3.852E-02	8.337E-03	1.042E-03	3.885E-02	7.585E-03	1.463E-03
124.0	2.442E-02	1.942E-02	5.000E-03	0.000E+00	1.982E-02	4.083E-03	5.104E-04	2.011E-02	3.442E-03	8.679E-04
200.0	5.450E-03	4.334E-03	1.116E-03	0.000E+00	4.425E-03	9.115E-04	1.139E-04	4.463E-03	8.250E-04	1.621E-04
500.0	2.958E-04	2.352E-04	6.057E-05	0.000E+00	2.402E-04	4.947E-05	6.184E-06	2.404E-04	4.889E-05	6.509E-06

151.4 Against the Phase A photoevent ledger

Item A3 of the review asks for “an H2 photoevent ledger, requiring conservation of H nuclei and charge for every final-state channel”. Phase A implements it as test E3 on the TWO-way branching, and its own comment defers the gate here: “These are REPORTED, not gated: they are the known approximation of item 12.1, and the gate belongs with the explicit-channel cross sections of phase E1, not here.” So the two tests are a pair, not a duplicate — E3 sizes the two unresolved channels on the old branching, and src/tests/e1_h2/ gates them once they are resolved. The invariants are E3’s; this section does not restate them.

E3 prints two numbers. Both are reproduced by the channel construction, and one of them needs its weighting said out loud.

	Phase A E3	E1 channels	note
double ionization at 100 eV	0.04259	0.02176	E3’s is PROTON-weighted, $0.20f_{\text{di}}$; E1’s is the EVENT share
the same, as an event share	0.02129	0.02176	$0.10f_{\text{di}}$ against $\sigma_D/\sigma_{\text{H}_2}$; +2.2 per cent
double ionization, proton share	0.20 (stated)	0.20000	E1 reproduces Chung’s 20 per cent exactly
neutral dissociation at 37.5 eV	“up to 7%”	0.07398	$\sigma_N/\sigma_{\text{H}_2}$

The factor 2 between the first two rows is the defect itself, not a disagreement. Chung et al.’s 20 per cent is a share of $\sigma(\text{H}^+)$, which counts PROTONS, and a double event yields two of them; the share of the EVENTS is therefore half of it. E3 prints the proton-weighted figure because that is the form the source states; E1 has to carry the event form because its four channels are mutually exclusive events. The residual +2.2 per cent between 0.02129 and 0.02176 is the renormalization of the H_2^+ channel that appears once the event count is imposed separately from Chung et al.’s own closure (§151.2), and it is the same 2.2 per cent TO_BE_DONE.md (P31b) predicted for H_2^+ .

Under yan_rho instead, the double channel is 0.03055 of the events at 100 eV and 0.278 of the protons — against 0.02176 and 0.200 under chung80. That factor 1.4 is the model spread of §151.6.

151.5 What each channel is charged, and where its electrons go

The rows are only half of a channel; the other half is the energy. Each of the four is charged its own threshold and partitioned according to whether it makes a photoelectron.

channel	charged	photoelectron?	secondary ionization
M	15.4 eV	yes, one	yes, iabs_H2 partition
S	18.076 eV	yes, one	yes, iabs_H2_di partition
D	51.4 eV	yes, two sharing the excess	yes, iabs_H2_di partition (see below)
N	$D_0(\text{H}_2) = 4.478$ eV	no	none

- **N carries no fhv factor and no secondary term.** It is not an ionization. The bond energy is spent breaking the molecule and the whole remainder of the photon becomes kinetic energy of two H atoms, i.e. heat. This makes the 33–41 eV band heat MORE than it did before, because that share used to be charged the 15.4 eV ionization potential and then reduced by fhv.
- **D is partitioned like the other two ionizing channels, and it must be.** The channel exists only above 51.4 eV, which is where the secondary-ionization share is largest; charging its excess whole as local heat would overstate the heating and lose the ionizations its fast electrons make. $D_0(\text{H}_2)$ is taken from mol_rates, not redefined here.

The approximation in D, stated with its range. photoelectron_shares is indexed by absorber and returns the partition of ONE electron carrying the whole excess; D makes TWO that share it. The dissociative absorber's shares are used at that photon energy — the closest tabulated one, same molecular target, adjacent threshold. Because the partition varies slowly with electron energy above $E_{\text{section}} = 30$ eV, the error is second order in the difference between the mean electron energy and the full excess. **It is not zero**, and it biases toward too little heating and too much ionization, since a slower electron heats more. Removing it needs a partition evaluated at $(E - 51.4)/2$, which needs its own absorber row in electron_energy_degradation. That is not done here.

151.6 The double-ionization branching is a MODEL, and it is named as one

Nothing in the code's sources measures $d(E)$. Two published statements bracket it and they disagree, so both are implemented and neither is the default.

- chung80 — Chung et al. (1993), discussion of their Fig. 4: 20 per cent of $\sigma(\text{H}^+)$ above 80 eV, ramped from zero at the 51.4 eV threshold, held above 80 eV. The ramp is NOT in the source; “small” near threshold is all it says.
- yan_rho — Yan et al. (1998) section 4: “At 110 eV, the measured ratio of double to single ionization is 0.038. The asymptotic ratio is predicted to be 0.0225 (Sadeghpour & Dalgarno 1993).” The ratio is ramped from zero at 51.4 eV to 0.038 at 110 eV and relaxed to 0.0225 by 300 eV, then inverted to d at that energy's measured r_{pm} . Both the ramp and the relaxation are interpolations between two published numbers.

They differ by a factor 1.6 at 110 eV (chung80 gives a double-to-single ratio of 0.023 there, yan_rho 0.038). That is the uncertainty, and running both is how it gets reported. The review's requirement was to “either obtain a defensible model or expose the constant branching as an uncertainty option; do not silently present it as measured”; this is the second of those.

Above 124 eV the ratio r_{pm} is still held at its 124 eV value, 0.2048, because Chung et al. stop there. That extrapolation is unchanged by this work and remains catalog item 12.1 (P31a). What has changed is that it is now one named input to a documented construction instead of a property of an undifferentiated branching fraction.

151.7 What the channels are worth, measured with both switches

Both channels default ON, so the A/B is run by switching each OFF. mol_base_handoff at its pinned 12,000 steps, OMP_NUM_THREADS=1, on the final sigma_H2. **All five configurations print** $\log_{10} \dot{M} = 10.43$, so the table is the profile. Maximum relative difference against the default over the physical cells:

switched off	ρ	v	T	heat	H ₂	H ₂ ⁺	H ⁺	HI
double ionization off	2.2e-4	1.0e-3	1.1e-4	7.1e-4	3.0e-3	2.0e-2	2.7e-3	1.6e-3
neutral dissociation off	3.5e-5	2.5e-5	2.7e-5	4.0e-5	8.3e-4	2.2e-4	6.5e-6	2.7e-5
both off (the pre-E1 split)	2.2e-4	1.0e-3	8.7e-5	7.1e-4	3.3e-3	2.0e-2	2.7e-3	1.6e-3
yan_rho instead of chung80	8.3e-5	1.6e-4	4.8e-5	3.2e-4	2.2e-3	8.2e-3	7.8e-4	3.1e-4

The double channel carries the effect and it lands on H₂⁺. Turning it off moves $n(\text{H}_2^+)$ by 2.0×10^{-2} at the base, and turning the neutral channel off as well adds nothing to that (3.3×10^{-3} against 3.0×10^{-3} in $n(\text{H}_2)$ is the only visible change). The sign is the one the construction requires: resolving the double channel makes the H₂⁺ share LARGER, because the event count is imposed separately from Chung et al.’s own closure (§151.2), and it is the same 2.2 per cent TO_BE_DONE.md (P31b) predicted.

	$r = 1.00$	1.02	1.05	1.10	1.20
$n(\text{H}_2^+)$, both off vs default	-1.95e-2	-1.57e-2	-1.39e-2	-8.0e-4	+5.6e-5
$n(\text{H}^+)$	-2.44e-3	-2.57e-3	-2.32e-3	+7.2e-6	+5.2e-6
$n(\text{HI})$	-1.56e-3	-1.44e-3	-8.6e-4	-8.1e-5	-4.4e-5

The proton source moves by 0.26 per cent, against the “at most 0.5 per cent” bound TO_BE_DONE.md (P31b) derived for the shielded molecular base. The bound holds and the measurement is inside it.

The neutral channel is small here and that is a property of this atmosphere, not of the channel. It lives only over 33–41 eV, and in the shielded base that band carries little flux; its largest effect is 8.3×10^{-4} in $n(\text{H}_2)$. It would matter more where the 33–41 eV band is not shielded.

The model choice is worth 41 per cent of the double channel’s own effect. yan_rho against chung80 moves $n(\text{H}_2^+)$ by 8.2×10^{-3} where switching the channel off entirely moves it by 2.0×10^{-2} . That is the uncertainty of §151.6, measured on a run rather than on a cross section.

Everything is below 1.10 r_{base} : T moves by under 1.1×10^{-4} , v by under 1.0×10^{-3} , and $\log_{10} \dot{M}$ not at all to the printed digit.

151.8 Range of validity, stated at the code site

- f_n is measured over 33–41 eV only, twelve rows of Chung et al. Table 1, whose yields come from Backx, Wight & van der Wiel. It is set to zero outside that window. **That zero is the source’s assumption, not a measurement:** “No yield measurements were made above 28 eV because it was assumed that the ionization yield would be unity at higher energies.”
- A zero anchor is carried at 32.0 and 41.5 eV so f_n is continuous. Those two points are continuity anchors resting on the source’s own sentence that the Backx et al. yields “drop from 100% to 92%” between 32 and 41 eV; they

are not measurements. Without them f_n would step from 0 to 0.0078 at 33 eV, and a discontinuous cross section is differentiated by the JFNK residual.

- r_{pm} , the proton-to- H_2^+ ratio, is measured over 18.076–124 eV (Chung et al. Table II) and held flat above. That extrapolation is item 12.1 (P31a) and is unchanged by this work. It IS independently confirmed over 20–30 eV by Backx et al.'s own H_2^+ column, to 0.1–0.5 per cent (§151.10).
- d , the double-ionization share of the protons, is a MODEL. Default chung80; the two implementations disagree by a factor 1.6 at 110 eV (§151.6) and by 41 per cent of the channel's own effect on a run (§151.7).
- σ_{H2} is a measurement over 15.4–300 eV and a rescaled sum-rule tail above it (§151.10). Its real uncertainty is the spread between three experiments: a few per cent below 30 eV, 10 to 20 per cent above.
- The uncertainty Chung et al. quote on their own data: “An estimate of the total uncertainty in the data is about $\pm 4\%$ – 5% .”

151.9 A separate defect this work measured, and then fixed (E1b)

Building the continuity test exposed something that is not E1 and is larger than E1: **σ_{H2} itself is discontinuous at two of the three joins of the published piecewise fit.**

join	below	above	ratio
18 eV (Eq. 17 / Eq. 18)	5.693	9.787	1.719
85 eV (Eq. 18 / Eq. 19)	0.1281	0.0793	0.619

Both formulas were re-derived by hand from the coefficients in references/Yan_1998_ApJ_496_1044.pdf and both are transcribed correctly, so this is a property of the published fit as published, not a transcription error. Against Chung et al.'s Table II absorption column, Eq. 18 is right at its lower end (9.665 against their 9.75 at 18.1 eV, 7.659 against 7.65 at 20 eV) and drifts high toward its upper end, while Eq. 19 is right at 85 eV (0.0793 against their 0.08). So the 72 per cent step at 18 eV is Eq. 17 undershooting, and the 38 per cent step at 85 eV is Eq. 18 overshooting.

It changes the H_2 opacity directly and moves every molecular result, so it was first reported here as its own item. **§151.10 fixes it:** over the range Chung et al. measured, the measurement replaces the fit.

151.10 E1b: σ_{H2} is a measurement everywhere it is absorbed

The preceding subsection reported two joins in σ_{H2} as a defect. This subsection removes them, and it does so by reading the sources. No piece of the Yan analytic fit survives below 300 eV.

```

15.4 <= E < 18 eV : k_bx * Backx, Wight & Van der Wiel (1976) Table 1
18 <= E <= 300 eV : Samson & Haddad (1994) Table 1
E > 300 eV       : k_hi * Yan Eq. 19      (E^-7/2 sum-rule tail)

all tables interpolated in log sigma against log E
k_bx = 9.85      / Backx(18) = 1.016415
k_hi = 0.00154 / Eq19(300) = 0.885805

```

151.10.1 18–300 eV: Samson & Haddad (1994) Table 1

“Recommended Total Photoabsorption Cross Sections of H₂ from 18 eV to 300 eV”, 72 rows. Stated accuracy ± 2 to ± 3 per cent over 18–113 eV, the measured range, and ± 3 to ± 4 per cent from 113 to 300 eV. Yan Eq. 18 is not used at all.

The first version of E1b took this column from Chung et al. (1993) Table II. It is the same data — Chung’s text says “Values for $\alpha(\text{abs})$ were taken from the data of Samson and Haddad”, his reference 31 being this paper as “submitted”.

At the 50 energies the two tables share, 49 agree exactly and one differs in its last digit (62 eV, 0.237 against 0.236). The original is used because it reaches 300 eV where Chung’s stops at 124. It also settles the earlier draft’s 113 eV question: Chung’s rows at 115, 120 and 124 eV are Samson & Haddad’s own recommended values, not his extrapolation.

151.10.2 15.4–18 eV: Backx, Wight & Van der Wiel (1976) Table 1

J. Phys. B, 9, 315, “Oscillator strengths (10–70 eV) for absorption, ionization and dissociation in H₂, HD and D₂, obtained by an electron-ion coincidence method”, their Table 1, column $f^0(E)$. This is the experiment the other two normalize to: Chan et al. (1992) place their own spectrum “on an absolute scale by normalization at 18 eV” to it, and Chung et al. (1993) take the photoionization yields of their Table 1 — which set the neutral channel of §151.2 — from it.

Units and normalization, from the paper. Their Fig. 2 plots $f^0(E)$ in eV^{-1} with a $\times 10^{-1}$ axis multiplier and the table is in 10^{-2}eV^{-1} ; the absolute scale is the TRK sum rule, “normalized on an integral value of two (TRK sum rule) with a 1% correction for energies beyond 80 eV”. The conversion is the standard $\sigma[\text{Mb}] = 109.75 df/dE[1/\text{eV}]$, and it is **confirmed against the other measurement rather than assumed**: it puts Backx at 7.628 Mb at 20 eV and 2.700 at 30 eV against Samson & Haddad’s 7.650 and 2.710, i.e. 0.3 and 0.4 per cent. A factor 100 instead would put them 9 per cent low.

The join. Backx gives 9.691 Mb at 18 eV where Samson & Haddad give 9.850, so the branch is scaled by 1.0164. **That 1.6 per cent is a measured difference between two independent experiments**, inside both quoted accuracies, and not a fitted constant.

E [eV]	σ [10^{-18}cm^2]
15.4	14.20
15.5	14.06
16.0	13.05
16.5	12.05
17.0	11.27
17.5	10.69
18.0	9.85

How big the correction is. The branch this replaced was Yan Eq. 17 scaled by a continuity constant. **Eq. 17 falls to 0.10 (0.17 after scaling) at 15.4 eV, where the measurement is 14.2**: the analytic fit went to zero at the ionization threshold, which no photoionization cross section does. The error ran from **a factor 85 at threshold** to 1.7 at 18 eV.

151.10.3 The digitization, kept as a check and not used

Before Backx et al. arrived, this branch was digitized from Fig. 1 of Chan, Cooper & Brion (1992) — whose Tables 1–4 give only the discrete Lyman and Werner transitions, the continuum being in that figure alone. Method: page rendered at 600 dpi, energy axis calibrated on the 11, 17 and 19 eV ticks (residuals 0.3–1.2 px, 1 px = 0.0046 eV), trace height read per column outside the legend box and the H_2^+ threshold marker, **the read repeated 18 times** over three binarization thresholds, two search windows and three averaging widths.

E [eV]	digitized from Chan Fig. 1	Backx et al. Table 1	digitized / table
16.0	13.11	13.05	1.005
16.5	~10.9	12.05	0.90
17.0	10.38	11.27	0.921
17.5	~10.3	10.69	0.96
18.0	9.85 (anchored)	9.85	1.000

The digitization was good to 0.5 per cent at 16 eV and 8 per cent at 17 eV, against a quoted reproducibility of 1–4 per cent, so its stated error was optimistic in the middle of the band — a figure read cannot see a systematic tilt in its own axis calibration. It is recorded here because it was the evidence before the table existed, and it is not in the code.

A table beats a figure.

151.10.4 The spread between experiments is the real uncertainty

Three independent measurements at the energies they share (Mb):

E [eV]	Samson & Haddad 1994	Backx et al. 1976	Lee et al. 1976	Backx / S&H	Lee / S&H
18	9.850	9.691	9.072	0.984	0.921
20	7.650	7.628	7.298	0.997	0.954
22	6.030	6.047	5.771	1.003	0.957
25	4.310	4.302	4.018	0.998	0.932
28	3.200	3.172	3.055	0.991	0.955
30	2.710	2.700	2.533	0.996	0.935
33	1.930	2.085	2.000	1.080	1.036
35	1.570	1.701	1.582	1.084	1.008
38	1.170	1.416	1.222	1.210	1.045
40	0.990	1.087	1.000	1.098	1.010
44	0.722	0.779	0.792	1.079	1.097
50	0.477	0.527	0.515	1.104	1.080
56	0.331	0.351	0.365	1.061	1.104
60	0.262	0.263	0.323	1.005	1.234
66	0.191	0.230	0.273	1.207	1.431

Read the last two columns. **Below about 31 eV the three agree to a few per cent** — Backx within 1.6 per cent of Samson & Haddad, Lee 4 to 8 per cent low. **Above 32 eV they diverge:** Backx runs 6 to 21 per cent high, Lee 1 to 43

per cent high, and the disagreement grows with energy as the cross section falls. So σ_{H_2} is good to a few per cent below 30 eV and to 10–20 per cent above it, **whatever any single paper’s quoted accuracy says**. Samson & Haddad’s ± 2 to ± 3 per cent is the accuracy of one experiment, not of the quantity.

151.10.5 An independent check on the branching, from the same table

Backx et al. tabulate a second column, $f^{(i)}(\text{H}_2^+)$, the oscillator strength for producing H_2^+ specifically. Where the photoionization yield is unity, that column divided by the absorption column must be $1 - f_{\text{di}}$, and f_{di} in this code comes from an entirely different experiment (Chung et al. Table II).

E [eV]	Backx $f(\text{H}_2^+)/f_{\text{abs}}$	$1 - f_{\text{di}}$ (Chung)	ratio
20	0.9813	0.9804	1.0009
22	0.9800	0.9785	1.0016
24	0.9817	0.9770	1.0048
25	0.9770	0.9764	1.0007
26	0.9799	0.9754	1.0046
28	0.9723	0.9678	1.0047
30	0.9553	0.9517	1.0038

The two experiments agree on the dissociative branching to 0.1–0.5 per cent over 20–30 eV. That is an independent confirmation of f_{di} , which §151.2 uses to split three of the four channels. Above 32 eV the yield falls below unity and the identity no longer holds, and below 20 eV Backx’s ionization column carries their superexcitation feature, so the check is quoted only where it is valid.

What could NOT be checked this way. Chung’s Table 1 photoionization YIELDS, which set the neutral-dissociation channel, come from Backx’s ionization efficiency curve — their Fig. 4, which is not tabulated. So the neutral fraction f_n still rests on Chung’s transcription of a figure of Backx’s, and this work could not go behind it.

151.10.6 The high join is a real tension between a measurement and a sum rule

Eq. 19 agrees with Samson & Haddad to **0.33 per cent at 113 eV** (0.033270 against 0.033160), so over the measured range the sum-rule tail and the measurement are the same curve. They part above it: at 300 eV the table reads 0.00154 and Eq. 19 gives 0.00174. Yan et al. say why in their section 4 — they **raised** the 300 eV point from 1.54×10^{-21} to $1.75 \times 10^{-21} \text{ cm}^2$ “to achieve agreement with the sum rules”. So $k_{\text{hi}} = 0.886$ carries the measured normalization into the tail and gives up 12 per cent of their correction. **Continuity is chosen over the sum rule** because a step in an opacity is what this change exists to remove, and because $\sigma_{\text{H}_2}(300)$ is 1.5×10^{-3} against 14.2 at threshold. The 12 per cent is recorded, not resolved.

151.10.7 The cross section, before and after

E [eV]	new (measured)	old (Yan 3-piece)	new/old
16.0	1.3052E+01	4.2997E+00	3.035
17.0	1.1267E+01	6.3080E+00	1.786
17.9	1.0010E+01	5.7539E+00	1.740
18.1	9.7232E+00	9.6653E+00	1.006
20.0	7.6500E+00	7.6592E+00	0.999
25.0	4.3100E+00	4.3344E+00	0.994
30.0	2.7100E+00	2.5847E+00	1.048
40.0	9.9000E-01	1.0291E+00	0.962
60.0	2.6200E-01	2.3855E-01	1.098
84.9	7.9528E-02	1.2803E-01	0.621
85.1	7.8899E-02	7.8989E-02	0.999
100.0	4.7900E-02	4.7613E-02	1.006
124.0	2.4416E-02	2.5435E-02	0.960
300.0	1.5400E-03	1.7385E-03	0.886

Read down the last column: +204 **per cent at 16 eV** and +74 **to +79 per cent over 17–18 eV** (the measured continuum against a fit that collapsed toward threshold), within 5 per cent over most of 18–85 eV, –37.9 **per cent in the last few eV below 85**, and –11.4 per cent at 300 eV. The large numbers are the two joins being removed; the small ones are the measurement disagreeing with the fit inside its own range by about its stated ± 2 to ± 3 per cent.

151.10.8 Continuity, gated

src/tests/e1_h2/ scans sigma_H2 from 15.5 to 2000 eV in 10^{-3} eV steps and reports the largest fractional step. It is 1.7×10^{-4} , **at 17.501 eV**, i.e. a smooth cross section; either surviving join would have shown 4×10^{-1} or 6×10^{-1} , the scaled-shape version of this branch showed 8.6×10^{-3} and the digitized one 6.1×10^{-4} . The 15.4 eV ionization threshold itself is now a real step from 0 to **14.20**, as an ionization threshold must be — the fit stepped to 0.17 and then climbed, which was the wrong shape as well as the wrong size.

151.10.9 What this does not address

- The log–log interpolation between table rows is an assumption about the shape between measured points, not a measurement. Backx’s grid is 0.5 eV below 20 eV; Samson & Haddad’s is 1 eV to 40 eV and coarser above.
- Nothing here revisits the 33–41 eV window as an ABSORPTION cross section. It still is one; that is what makes the neutral channel of §151.2 necessary, and the two changes are consistent because the neutral share is taken as a fraction of whatever sigma_H2 is.
- Above 124 eV the proton/ H_2^+ ratio is still held flat (item 12.1, P31a). E1b changes the cross section there but does not touch the branching.

- Chung’s Table 1 photoionization yields, which set the neutral channel, come from a FIGURE of Backx’s (their Fig. 4) that is not tabulated, so f_n was not checked against the original.
- The 12 per cent measurement/sum-rule tension above 300 eV is recorded, not resolved.

151.11 Regression: the molecular cases move, the atomic ones do not

Nine cases at their pinned step counts, OMP_NUM_THREADS=1, against the goldens in the tree. Maximum relative difference over the physical cells.

case	ρ	v	T	heat	H ₂	H ₂ ⁺	H ⁺	HI
mol_base_handoff	4.4e-3	3.5e-3	2.7e-3	2.1e-2	4.8e-2	2.4e-1	6.1e-3	4.3e-2
mol_lyman_werner	7.3e-4	9.6e-4	8.3e-4	6.6e-3	7.0e-2	8.3e-2	7.3e-4	2.6e-2
mol_metals	4.7e-3	3.4e-3	2.9e-3	1.4e-2	6.8e-2	2.4e-1	6.5e-3	4.5e-2
mol_carrier	2.9e-3	5.1e-3	1.7e-3	2.0e-2	1.3e-1	1.5e-1	3.9e-3	4.2e-2
mol_diffusion	4.4e-3	3.5e-3	2.7e-3	2.1e-2	4.8e-2	2.4e-1	6.1e-3	4.3e-2
mol_ir_bands	4.7e-3	3.6e-3	2.9e-3	1.4e-2	6.8e-2	2.4e-1	6.5e-3	4.5e-2
mol_sec_ion	1.4e-3	see below	5.6e-4	1.3e-2	1.8e-2	2.3e-1	1.4e-3	1.5e-2
wasp_full	0	0	0	0	—	—	0	0
wasp_he23off	0	0	0	0	—	—	0	0

The two atomic cases are bitwise identical, which is the acceptance test for them: sigma_H2 is only evaluated where there is H₂. The molecular cases move, and this is not a default-off change — E1b changes the cross section itself, so their goldens are refreshed rather than defended.

Where the movement lands. $n(\text{H}_2^+)$ moves the most, up to 24 per cent, which is the same place the channel A/B of §151.7 puts it and the same place the cross section changed most (the 15.4–18 eV band, where H₂⁺ is made). $n(\text{H}_2)$ moves 2 to 13 per cent, $n(\text{HI})$ 1.5 to 4.5 per cent, the heating up to 2 per cent, and ρ , v and T by under 0.5 per cent.

The wind barely notices; the molecular layer does.

The one entry that needs a word. mol_sec_ion shows a maximum relative change in v of 3.35, at $r = 1.181$ — which is the velocity ZERO of that snapshot, the smallest $|v|$ of all 500 cells (14 cm s⁻¹, against 2.9×10^5 cm s⁻¹ at 2 R_p). A relative measure at a zero crossing is not informative. Over the wind window $r \geq 1.2$ the same case moves v by 9.9×10^{-2} , and its ρ and T move by 1.4×10^{-3} and 5.6×10^{-4} like every other case.

152. The lower boundary is a characteristic condition at the face (2026-09-03)

Judgement first. The ghost-cell closure was not inaccurate, it was inconsistent: it left a momentum imbalance at cell 1 that does not go away when the grid is refined. It wrote three primitive components into a ghost CELL and Rec_BC then reused the same three numbers as the state at the FACE, half a cell higher. That half cell is $\delta r/(2H)$ of hydrostatic stratification — 3.8×10^{-3} on the hot-Uranus base grid — and a state error of δr divided by δr to

make a flux difference is a fixed fraction of the local weight at every resolution. **Measured, on an isolated Euler-plus-gravity probe that links the production grid, reconstruction, Riemann flux, source and boundary modules verbatim: $R_{\text{mom}}(1)/(\rho g)$ reads 0.2514, 0.2507, 0.2503, 0.2502 over an $8\times$ refinement, against an interior truncation error of 4.9×10^{-6} .** The velocity it drives is first order in δr : $-247.1, -123.2, -61.3, -30.5 \text{ cm s}^{-1}$, which is docs/p44_base_sawtooth.md section 7's production ladder $(-248.3, -122.2, -60.0)$ reproduced to within 1–2 per cent **with no chemistry, no radiation, no molecules and no wind.** The base sawtooth was never a molecular-layer phenomenon; it was this boundary.

The replacement states two conditions at the face and takes one from the interior, which is the characteristic count for a subsonic inflow. The two are the lower atmosphere's pressure and specific entropy at the base level; the one is the outgoing acoustic invariant of cell 1,

$$p_b - \rho_i c_i v_b = p_i - \rho_i c_i v_i, \quad (63)$$

with c_i at cell 1's own γ_{eff} . The ghost cells become the volume averages of the hydrostatic isentrope through the face state continued below it. **On the same probe $R_{\text{mom}}(1)/(\rho g)$ falls to 1.75×10^{-4} and becomes first order in δr , and $v(1)$ falls from -247 cm s^{-1} to -0.28 cm s^{-1} and converges away.**

Four keys are retired and now REFUSED at startup: Valve eps, Hydrostatic base, Base ghost temperature, Base velocity. Each selected one component of a closure that no longer exists. Base BC: density|pressure is kept — it states the base LEVEL, not the closure — and its level is now carried to the face by the boundary itself, so refining the grid no longer moves the level the user stated.

Design note, every measurement, and the decisions: docs/phaseC_characteristic_base_bc.md. Patch: docs/phaseC_base_bc.diff

152.1 What the count is, and what the code had

At $r_{\text{edg}}(0)$ the speeds are $v - c$, v , $v + c$, and a wave enters when its speed is positive (radius increases outward, so base inflow is $v > 0$).

base state	entering	reservoir conditions	interior relations
subsonic inflow $0 < v < c$	2	2	1
flow reversal $-c < v < 0$	1	1	2
supersonic inflow $v > c$	3	3	0
supersonic outflow $v < -c$	0	0	3

The old default branch pinned ρ and p and valved the velocity, so its count was right; Base velocity: massflux pinned all three, which for subsonic inflow is one too many and is the combination Carlson (2011, NASA/TM-2011-217181, Table 1 footnote) names as inadmissible. What the code had none of was the reversal row: $\max(v_1, 0)$ clips a velocity, it does not change how many conditions are imposed.

152.2 The two defects were independent, and only one of them was the valve

Two probe measurements separate them, and they separate cleanly.

- **Hydrostatic maintenance is the pinned numbers.** Holding the same pair (ρ_{bc}, p_{bc}) with the interior-velocity copy REMOVED reproduces the legacy $R_{\text{mom}}(1)$, $R_{\text{mass}}(1)$ and $v(1)$ to six digits. The valve contributes nothing.
- **Acoustic reflection is the velocity.** The legacy boundary reflects $|R| = 0.953$ of a pulse leaving through it; freezing the ghost velocity gives 0.031.

docs/p44_base_sawtooth.md section 9 reached the same separation from the other side (raising T_0 closes the thermal gap and leaves the flux error untouched), and TO_BE_DONE.md item (W) records it.

152.3 What was built

One new module, src/modules/states/base_boundary.f90, with one entry point. base_boundary_states takes the first interior CELL AVERAGE and returns the face state, the ghost cell averages and the state at the face below the ghost, all at once; Apply_BC_W calls it and writes the ghosts, Rec_BC reads the face state. **That is the whole of the “one kind, one rule” repair:** the old code called BC_component_constrho from both paths, and the two paths handed it different data — the valve read the cell-1 average in Apply_BC_W and the velocity reconstructed at r_edg(1), a full cell higher, in Rec_BC.

The interior state at the face is cell 1 continued along **cell 1’s own hydrostatic isentrope**, not extrapolated linearly and not taken from the reconstruction stencil, which reaches into the ghost and would make the boundary read its own output. Continuing along the isentrope is what makes the condition well balanced: at rest the interior face pressure equals the reservoir’s and the compatibility relation returns $v_b = 0$ exactly, at any resolution.

The reversal handover is a cubic smoothstep of width base_face_mach_blend = 10^{-6} in face Mach number — the one designed-in constant, justified at its declaration by the physical base Mach number of the gate ($\rho_b v_b r^2 = F_{\text{wind}}$ gives $v_b \sim 1.6 \text{ cm s}^{-1}$ against $c \sim 2.6 \times 10^5$, i.e. 6×10^{-6}), so every intended operating point lies wholly inside the inflow branch. There is no valve_eps to arm: the residual is differentiable because the branch is a blend, not because a kink was rounded off.

Supersonic inflow is **reported, not repaired**. It would need three reservoir conditions and a (p, s) reservoir states two; the model has no lower-atmosphere velocity, so check_base_inflow_is_subsonic says the state is outside the stated problem rather than a velocity being invented for it.

152.4 Validation

Three tests on the probe, all new. (a) and (b) have baselines for the closure this replaces; (c) is new.

test	old closure	Phase C
(a) $R_{\text{mom}}(1)/(\rho g)$ at rest, $\delta r_c = 2 \times 10^{-4}$	2.5135e-1, zeroth order	1.749e-4, first order
(a) $v(1)$ at rest, $\delta r_c = 2 \times 10^{-4}$	-247.1 cm s ⁻¹	-0.278 cm s ⁻¹
(a) ladder $\delta r_c = 2 \times 10^{-4} \dots 2.5 \times 10^{-5}$	-247, -123, -61, -30 cm s ⁻¹	-0.28, -0.04, -0.007, -0.0001
(b) acoustic $ R $	0.953	0.956
(c) manufactured steady inflow, cell 1 vs interior	not run	mass $1.8 \times$, momentum $0.68 \times$

(c) is the test that earned its place. It drives an EXACT steady spherical isentropic solution ($\rho v r^2 = F$, $v^2/2 + h + \phi = B$, a solution of the Euler equations with gravity, so nothing is forced) through the boundary at base Mach 10^{-3} and reads the residual of that exact solution: cell 1 comes out at 1.8 times the interior truncation error in the mass row and 0.68 of it in the momentum row. The boundary is no longer a distinguished cell.

(b) did not improve, and section 7.3 of the design note shows it could not. Two characteristics enter a subsonic inflow face; if the reservoir states two STATIC quantities then the incoming acoustic amplitude is pinned and a wave arriving from inside is reflected. That is true of (p, s) , (ρ, s) and (ρ, p) alike — the pair does not matter, the staticness does. The alternative was implemented and measured (base_incoming_invariant_weight, state $p + \rho cv = p_{\text{res}}$ instead of p): $|R|$ falls to 0.0024, but on test (c) the cell-1 momentum residual rises to 1.32×10^{-1} , **460 times the interior**, because $p + \rho cv = p_{\text{res}}$ puts the face pressure below the reservoir's by a fraction M of the pressure where a real reservoir's Bernoulli drop is M^2 . It buys transparency by corrupting the steady state and is rejected; the constant stays at 0 with the measurement recorded at its declaration. A boundary that is transparent to sound without that cost needs a relaxation form, which is not a function of the instantaneous state and so has no fixed point the steady residual can express. That is an open item, in the design note as D6.

-fcheck=bounds, do, mem on the bounded HD 209458 b case: clean, $\log_{10} \dot{M} = 9.61$.

152.5 Production

The hot-Uranus gate rung, reloaded — the case (AB) measured, so the one comparison with a baseline. info = 0 both; **19 JFNK iterations** $\rightarrow$ **14**; $\|R\|$ 8.961e-06 $\rightarrow$ 7.848e-06; the flux gate 2.501e-04 unchanged to four figures, $r \geq 1.03$ 3.432e-03 $\rightarrow$ 3.402e-03, $r \geq 1.10$ 2.375e-03 $\rightarrow$ 2.372e-03; $\log_{10} \dot{M}$ 10.34 $\rightarrow$ 10.34; H_2 front 1.0449 R_p unchanged; H_3^+ column -0.9 per cent; $T(\text{cell } 1)$ 1116.68 $\rightarrow$ 1116.72 K. And $v(\text{cell } 1)$ -251.88 $\rightarrow$ +14.40 **cm s⁻¹**: the sawtooth is gone in the production code, and what replaces it is outward and of the size the wind's own mass flux through the base density gives.

The same rung from a cold start reaches the same root: info = 0 at $\|R\| = 1.405 \times 10^{-6}$, the same three gate numbers to four figures, the same $\log_{10} \dot{M}$, and a state agreeing with the reloaded root to 3.9×10^{-5} in ρ , 1.9×10^{-4} in v , 1.3×10^{-5} in T , $v(1) = +14.393$ against +14.395, and T at $r - R_p = 8 \times 10^{-3}$ equal to 818.01 K in both. **That repairs the reproducibility defect of docs/p44_base_sawtooth.md section 8.4**, which measured three info = 0 states of this configuration on this same default base grid putting the H_2 front anywhere in 1.053–1.089 R_p and that temperature anywhere in 583–949 K. The cold start costs 820 JFNK iterations across three attempts against the reload's 14; whether that is worse than the same cold start under (AB) is **not established**, because (AB) did not run one.

WASP-121b wasp_full_newton: info = 0 both, $\|R\|$ 9.026e-06 $\rightarrow$ 9.910e-06, flux gate 5.242e-05 $\rightarrow$ 5.238e-05, $\log_{10} \dot{M}$ 13.21 $\rightarrow$ 13.21, state moving by 5.9×10^{-3} in ρ and 3.2×10^{-3} in T , $v(\text{cell } 1)$ 1372.5 $\rightarrow$ 1705.9 cm s⁻¹, 96 $\rightarrow$ 108 iterations. Base Mach stayed subsonic at 0.331.

The reload-and-march departure is the negative result, and it is the one to read first. Peak-to-peak face mass flux over 3000 steps, in units of the root's own F_0 :

root, binary	1.005	1.03	1.10	1.20	du at 3000
(AB), rerun here as a control	0.4443	0.2517	0.0403	0.0084	3.2250e-04
Phase C	0.9284	0.3074	0.0473	0.0096	3.2904e-04

The control reproduces §146.5 exactly, so the harness is the same one. **Phase C does not repair item (AD) and makes it worse**, $2.09\times$ at $r = 1.005$ and $1.22\times$ at 1.03. That is consistent with §152.4 (b): the boundary still reflects what the base layer launches at it. One mechanism for the increase is that the retired one-way valve, by clipping v at zero, was also a nonlinear amplitude limiter on that oscillation; **that is a hypothesis, not a measurement**.

The complete update map says what the increase actually is, and it is not the boundary. On each build's own root (EXHALE_UPDATE_MAP=1, three steps at 1, 0.1, 0.01 of the CFL step): the accepted root's residual falls $9.916e-3 \rightarrow 6.067e-6$ (mass), $7.473e-3 \rightarrow 5.453e-6$ (momentum), $1.486e-2 \rightarrow 8.462e-6$ (energy) — **the base-cell residual floor, three orders of magnitude, measured on the production root** — while the defect $|G_{\delta t} + R|$ at $\delta t/100$ moves only $2.460e-5 \rightarrow 1.351e-5$, $2.643e-5 \rightarrow 9.840e-6$, and **1.892e-4 $\rightarrow$ 1.891e-4: the energy row does not move at all**, and it stops falling with δt in both builds, in the operator-split chemistry/energy columns. The root became $1600\times$ more accurate and the splitting error did not, so it now dominates: defect over residual at $\delta t/100$ goes from 2.5×10^{-3} , 3.5×10^{-3} , 1.3×10^{-2} to 2.2, 1.8, 22. The departure grew because the residual shrank, and what is left is Phase B's B2. Apply_BC moves the interior by a relative 0.000E+00 in both builds.

Regression matrix, ten cases against (AB), single-threaded, same caps: all ten move, and all ten flip the sign of $v(\text{cell } 1)$ — $-726, -756, -139, -139, -114, -139, -139, -601, -140, -582 \text{ cm s}^{-1}$ become $+2383, +2380, +49, +54, +43, +49, +54, +7, +30, +543$. Read the rest of that table with three qualifications: the eight molecular rows are step-12000 snapshots of a relaxation (du 0.4–1.3 in both builds), so their $\log_{10} \dot{M}$ $10.43 \rightarrow 10.52$ is a different point in a transient and not a rate; the two WASP rows are du-stops, whose several-per-cent path dependence covers $13.26 \rightarrow 13.24$ and whose Newton-finished sibling gives $13.21 \rightarrow 13.21$, though their 60 per cent longer march ($11289 \rightarrow 18026$) is a real cost; and lower_profile is not converged in either build (du 28.2 and 819) so its row compares two transients. Full table: docs/phaseC_characteristic_base_bc.md section 7.4.

152.6 What this does not do

This is a boundary repair, and the scope is exactly that. What it removes is the base cells' residual floor — the place every coupled finish has been stopping at $\|R\| \sim 1$ — together with the hydrostatic inconsistency that put it there. What it does NOT touch:

- **The H₂ front creep.** P51 measured, on the 200k-step marching state of the new physics, that FREEZING cells 1–5 at every step leaves the front's creep (1–3 cells per 20000 steps) unchanged to the fourth decimal (.../scratchpad/p51/docs/p23_transport_on_state.md section 9, read only, not remeasured here). The creep is therefore not driven by the base cells and no boundary condition addresses it; whether it is a numerical bias or physical is under separate diagnosis.
- It does not damp the base-layer ringing of item (AA): §152.4 (b), and the marching departure of §152.5 gets worse, not better.

- It does not change PLM's base layer, whose density drift is set by the second-order reconstruction and not by the boundary (6.5×10^{-3} against the old closure's 9.7×10^{-3} , and growing with refinement in both).
- It does not give the model a supersonic-inflow base.
- Goldens are not byte-identical, for any case. This is a model change.

153. Reserved: the coupled solve's step control (measured, not landed)

Two sections are reserved rather than written, so that the numbering matches the drafts they belong to. Both are step-control work on the COUPLED carrier solve, both are measured, and neither is in the tree:

- **153** — a scaled trust region (Powell dogleg on the Krylov and Cauchy legs, predicted reduction from the same Arnoldi relation the solve already builds, admissibility enforced inside the step). It works as a method: the ratio of actual to predicted reduction has median 0.93 with 42 of 47 iterations in $[0.5, 2]$, the finite-difference ray test refuses no model, and it never violates the element budget where the line search does 18 times.
- **154** — the merit and the predicted reduction measured in the ACCEPTANCE measure (the row scales of §143.) rather than in the state scales, at which the model becomes exact to three or four digits on the first iterations.

Neither closes the coupled solve, and the reason is why they are reserved and not applied: with both in place the two controls bottom out at the same place and the same number, $\|R\| = 1.00$ in BASE CELL $j = 1$, which is one cell's row off by 100 per cent of its own terms. That is the boundary imposing a cell instead of solving it, so the obstruction had left the solver before either section was written, and §152. is where it went. They are kept default-off pending a re-measurement on top of the characteristic boundary. Drafts and every measurement: the P51 notes under docs/.

155. The base cells reproduce, and the residual is not a function of the state alone (2026-09-03)

Measured 2026-09-03 on two builds:

- **Phase C** — `.../scratchpad/phaseC/src` copied read-only into `.../p55base/pc155` and rebuilt; the clean build reproduces Phase C's own binary bit for bit (583aaa98), so the boundary being measured is theirs. Roots: `phaseC/hotU_root` and `phaseC/wasp_newton`, restarted from their own output.
- **Pre-Phase-C** — the tree as of §145., copied into `.../base123`. Roots: the two accepted states of §144. (`s11/hotU_cellmax`, `s11/WASP_cellmax`).

Probes (scratch only, both env-gated, both off by default): EXHALE_SWEEP_CONSERVE reports what one sweep does to ρ , n_H and n_{He} ; EXHALE_RESID_REPRO= $\langle n \rangle$ repeats `eval_residual` on one accepted Y (negative n restores the composition before each repetition). Patch: `p155_selfconsistent.diff`.

155.1 The premise of item (4) does not reproduce

§11.6 of docs/p55_base_mode.md recorded that re-running the ionization sweep on an accepted state moves the mass residual of cells 1 and 2 from 9×10^{-6} to 2.7×10^{-1} and 8.6×10^{-1} , and attributed it to `ioniz_eq` rewriting ρ through `calc_rho`. **Neither half of that survives measurement.**

`calc_rho` does not move ρ . Over every sweep of both roots on both builds, the density the sweep writes back and the density it was handed differ by at most:

build	state	max abs $\delta\rho/\rho$	max abs $\delta n_{\text{H}}/n_{\text{H}}$	max abs $\delta n_{\text{He}}/n_{\text{He}}$	sweeps
Phase C	hot Uranus	8.13e-16	4.67e-16	4.32e-16	57
Phase C	WASP-121b	1.71e-15	2.22e-16	3.04e-16	86
pre-C	hot Uranus	9.68e-16	3.64e-16	4.42e-16	346

This is roundoff, and it is roundoff for a reason: the sweep's unknowns are fractions of the element totals n_{H} , n_{He} and $n_{\text{M,tot}}$ that it computes from the state it is handed, and every write-back is one of those totals times a fraction. Because the code's masses are the nucleus counts exactly ($\text{H}_2 = 2m_{\text{H}}$ with 2 H nuclei, $\text{H}_3^+ = 3$ with 3, $\text{HeH}^+ = 5 = 4 + 1$), conserving nuclei conserves mass identically. The `max(..., 0.0d0)` clamps on `nhi` and `nhei` in the molecular write-back CAN break this — they are the one place the element budget is not closed by construction — but on these states they do not bite.

The base cells' residuals are not $O(1)$. Repeating `eval_residual` on one accepted Y , cells 1–3, first evaluation against the twentieth:

build	state	cell	mass row, eval 1	eval 20	ratio
Phase C	hot Uranus	1	7.904e-05	7.765e-05	0.982
Phase C	hot Uranus	2	2.521e-05	2.470e-05	0.980
Phase C	WASP-121b	1	1.1725e-07	1.1725e-07	1.000
Phase C	WASP-121b	2	1.6359e-07	1.6359e-07	1.000
pre-C	hot Uranus	1	2.553e-07	2.491e-07	0.976
pre-C	hot Uranus	2	8.154e-08	7.762e-08	0.952
pre-C	WASP-121b	1	9.476e-09	5.163e-09	0.545
pre-C	WASP-121b	2	2.177e-08	2.010e-08	0.923

The largest excursion anywhere is a factor two, at WASP-121b cell 1 before Phase C. **Under Phase C that same cell is reproducible to four significant figures.** The base cells are not the problem, and to the extent they were one, Phase C's boundary closed it: it is the prescription that removed the base cell's dependence on the previous sweep's composition (no `n_part_cell1`, no valve).

155.2 What is real: the residual is a function of (state, composition)

`f_sp` is `intent(inout)` in `ioniz_eq` and the residual mutates it. It is not part of the Newton unknown Y . So `eval_residual(Y, f_sp)` is a function of two arguments, only one of which the state carries.

It is deterministic in both. With the composition restored before every repetition (EXHALE_RESID_REPRO=-4), all four states give bitwise identical residuals on every repetition. Nothing else — not the frozen chemical background, not the excited-hydrogen populations, not the stored WENO smoothness factors — contributes anything at all. That was worth establishing before looking further.

But the number reported at acceptance is evaluated at the WRONG composition. In the line search the accepted trial is evaluated as `eval_residual(Ytry, f_sp_j)` with `f_sp_j` copied from the PREVIOUS iterate, and `rnorm` is taken from that F . The state handed back and written to disk carries the composition the accepted trial LEFT. Evaluating the same Y at its own composition, and iterating the sweep at fixed Y until it stops moving:

build	state	reported $\ R\ $	own composition, 1 sweep	at the composition fixed point	sweeps	factor
Phase C	hot Uranus	2.211e-06	1.283e-05	1.257e-05	3	5.7
Phase C	WASP-121b	6.287e-06	3.865e-05	3.593e-05	9	5.7
pre-C	hot Uranus	4.141e-06	1.762e-05	1.722e-05	~10	4.2
pre-C	WASP-121b	6.264e-06	1.737e-04	3.539e-04	~14	56
pre-C	wasp_full_newton	8.873e-06	1.851e-04	3.412e-04	~10	38

The composition at fixed Y has its own fixed point and reaches it in 3 to 14 sweeps; one evaluation is still short of it by up to a factor two. **Phase C cuts the WASP-121b factor from 56 to 5.7 and leaves the hot Uranus at about 5.7.**

What this does and does not change about acceptance. It does not, today, change which states are accepted. Only 7 of the 63 regression cases set `Resid_tol` at all; without it `resid_th = -1`, the residual gate is off ($\|R\| < -1.000\text{E}+00$ in the acceptance line), and every one of the roots measured here was accepted on the FLUX GATE alone. The flux gate reads the conserved face mass flux of u and is composition-independent, so it is untouched by all of this.

What it changes is what the reported number means. $\|R\|$ is quoted in run logs, in docs/, and in item (AD)’s own measurements, and on these states it understates the residual of the state that was written by 4 to 56 times. Every one of the five states above is above 10^{-5} at its own composition, so a run that DID set `Resid_tol: 1.0e-5` would be accepting a state that does not meet it.

155.3 Prescription

(a) The gate and the report read the residual of the state that is written, at the composition that state carries. `p155_selfconsistent.diff` adds `resid_at_own_composition` (default `.false.`, `n_selfconsistent_max = 25`): at hand-back, re-evaluate at (Y, f_{sp}) and iterate the sweep at fixed Y until $\|R\|$ changes by less than 10^{-3} relative. Measured cost: 3 sweeps on the hot Uranus, 9 on WASP-121b, one-off. The coupled route already does a single re-evaluation for its carrier half (`nvar_jac` ≥ 4); this is the same idea taken to the fixed point and applied to all routes. Default off because turning it on makes every root accepted under `Resid_tol` report a residual 4 to 56 times larger, which is a physics decision about what “converged” means, not a refactor. The recommendation is to turn it on: a steady state is a joint fixed point of the hydrodynamic balance and the chemical equilibrium, and a measure that reads the hydro balance under last iterate’s chemistry is not measuring that.

(b) ρ should be an input to the sweep, not an output. The measurement says this changes no number today — the write-back is element-conserving on every state tested, to 10^{-15} . It is still the right shape: the sweep determines the

composition at a given (ρ, T) and the particle count that follows; mass is the hydrodynamics' variable. Making `n_io` `intent(in)` and normalizing `f_sp` by the input density would make that structural instead of accidental, and it would turn the `max(..., 0.0d0)` clamps in the molecular write-back — the one place the element budget is NOT closed by construction — into a detectable condition rather than a silent mass adjustment. It is a golden-refreshing change at the 10^{-15} level, so it belongs in a change series that is refreshing them anyway.

Not adopted here: anything that would make the composition part of Y . That is the coupled route's design and it exists (`nvar_jac` ≥ 4 , the carrier unknown); extending it to the full composition is a much larger question than this measurement supports.

155.4 Contact with Phase C

The patch touches `src/modules/time_step/steady_newton.f90` (the hand-back block, after `unpack_carrier/Apply_BC` and before `steady_gates_met`) and adds two parameters in `src/modules/init/parameters.f90`. **No boundary code is touched:** `Apply_BC.f90`, the new `base_boundary.f90`, `init.f90` and `utilities.f90` are exactly Phase C's. The one place the two meet is the hand-back `Apply_BC(u)` call, which Phase C rewrites and this patch only calls; if Phase C changes the call signature there, this patch follows it and nothing else.

The probes are separable and belong in `scratch`: `EXHALE_SWEEP_CONSERVE` in `ionization_equilibrium.f90` and `EXHALE_RESID_REPRO` in `steady_newton.f90`. The conservation probe is worth keeping in some form — as a cheap assertion that the sweep's write-back closed the element budget, reported once per run rather than printed per sweep.

155.5 What is not established

- One accepted state per planet per build, all with the residual gate off. A case that actually sets `Resid_tol` and converges under it has not been re-measured at its own composition.
- Whether the composition fixed point at fixed Y always exists and is unique. It was reached in 3 to 14 sweeps on all five states, monotonically on the two that start below it and with one overshoot on the three that do not (WASP-121b under Phase C: 3.86, 4.36, 3.79, $\dots 3.59 \times 10^{-5}$), but nothing here proves existence or uniqueness.
- Whether accepting on the self-consistent residual changes the mass-loss rate. It cannot change the state that is written — the patch only measures — but a run that keeps solving instead of stopping will end somewhere else.
- The `max(..., 0.0d0)` clamps: shown not to bite on these five states, not shown never to bite. The conservation probe is what would say.

155.6 What landed, and what the measure did to the one case that tests it

The own-composition residual is the DEFAULT (`resid_at_own_composition`), not an option to be opted into: the burden of a reason is on turning it off, and the only reason is reproducing a number from before this section. The fixed point is iterated explicitly through a separate output array, because §147. forbids the seed and the result of `eval_residual` being the same array — the patch as drafted assumed the older five-argument signature and did not build.

Two halves of the drafted patch were deliberately not taken, both being a second copy of a facility that had landed in the meantime: the EXHALE_SWEEP_CONSERVE probe in `ionization_equilibrium.f90`, which measures element conservation through the sweep that §149.’s `element_census` assertions already measure at every refresh that moves nuclei; and `residual_reproducibility_probe` (51 lines in `steady_newton.f90`), which asks the question `src/tests/residual_determinism/` answers as a test with a verdict. One facility, one rule.

Measured on `wasp_full_newton`, the only default case that runs a steady solve. It is the only one of the eleven that moves, which is the case list this change predicts:

	before	after
reported $\ R\ $ at hand-back	9.927e-06	5.677e-04
sweeps to the composition fixed point	—	7
flux spread (the gate that accepted it)	1.961e-11	1.961e-11
$\log_{10} \dot{M}$	13.21	13.21
state: $\rho / v / T$ max relative change	—	1.4e-15 / 0 / 4.2e-5

So the number the solver reports as the accepted root’s residual was **57 times too small**, and the state itself is untouched to roundoff — which is what this section claims: it does not move the solution, it stops misreporting it. The case still converges because it sets no `Resid tol` ($\|R\| < -1.000\text{E}+00$ in its log), so the residual gate was never tested and the flux gate accepted it; a run that does set one will now fail a tolerance it used to pass, and that is the intended consequence rather than a regression.

Not done. The other half of the prescription — making ρ an `intent(in)` of the sweep so that element conservation is structural rather than a property the arithmetic happens to have — is not implemented. It is worth doing and it changes no number (the sweep’s rewritten ρ differs from the one it was handed by 1.7×10^{-15} over 489 sweeps), so it belongs with the next change that refreshes goldens rather than on its own.

A Code-size summary: EXHALE vs. the original ATES (2026-06-07; updated 2026-08-30)

A line-by-line comparison of the Fortran source against the pristine upstream code (ATES/ATES-Code-main) quantifies the extent of the in-house extension. *Pure code lines* (SLOC) exclude blank lines and comments (the comment stripper is string-aware, so a `!` inside a quoted string is not treated as a comment); the line-by-line diff additionally normalizes whitespace and case, so reformatting and comment-only edits are *not* counted as changes. The measurement script is `src/utls/codesize.py`, which reprints every number of this appendix and the byte-identical list of Table 6. Updated 2026-08-30, i.e. after the binary element-diffusion layer and the lower-atmosphere profile handoff (sections 64–80) and the defect-fix and solver series of sections 81–93; the *core* rows exclude the ported Wind-AE solver (listed separately). Relative to the 2026-08-15 edition, `species_diffusion.f90` was deleted with the Phase-D rewrite and is superseded by `binary_element_diffusion.f90`.

	Fortran files	code lines (SLOC)
Original ATES (ATES-Code-main)	47	3 514
EXHALE core (excl. Wind-AE port)	75	16 272
net change (core)	+28	+12 758 ($\approx 4.6\times$)
Wind-AE IC-generator port (wind_ae/)	+36	+4 746
EXHALE total	111	21 018 ($\approx 6.0\times$)

Line-by-line diff, core (comments + blanks excluded; whitespace/case normalized): **13 457** code lines added in EXHALE (new modules + the new side of modified lines) and **699** removed from the original (deleted + the old side of modified lines), for **14 156** total changed lines. At the file level, of the 47 original files **34 were modified** and **13 are unchanged** (none were removed — the unchanged set is the 10 included MINPACK routines plus three small physics/setup pieces; the full list is Table 6), and **28 new core modules** were added:

New module	Purpose	SLOC
functions/binary_element_diffusion.f90	conservative binary H/He element diffusion (stage-resolved pair coefficients, donor-cell drift, metals slaved to H)	1164
tests/diffusion_tests.f90	test driver of the diffusion operator (analytic and conservation checks)	1025
time_step/steady_newton.f90	JFNK steady-state solve (PTC warm-up, line search, below-escape diagnostic)	636
files_IO/lower_atmosphere_profile.f90	lower-atmosphere profile reader: log- p interpolation, elemental reservoirs, $K_{zz}(p)$	367
radiation/charge_exchange.f90	charge-exchange network (Huang Table 4)	366
radiation/excited_hydrogen.f90	H($n=2$) model + Balmer feedback	288
time_step/viscous_conduction.f90	molecular viscosity + heat conduction (Crank–Nicolson)	264
nonlinear_system_solver/System_HeH_mol.f90	coupled H/He + molecular ($H_2/H_2^+/H_3^+/HeH^+$) system, turnover-scaled rows	210
radiation/opacity_models.f90	opacity dispatcher (A/C/P/T)	180
nonlinear_system_solver/System_HeH_metals.f90	coupled H/He+metals ionization (+analytic Jacobian)	150
nonlinear_system_solver/newton_solver.f90	analytic-Jacobian Newton (+hybrd1 fallback)	128
nonlinear_system_solver/ion_residual_core.f90	shared ionization-residual assembly of the System_* family	127
functions/composition.f90	single-point species/EOS policy	119
init/species_table.f90	metal + molecular species metadata table	112
files_IO/metals_input_read.f90	parse metals.inp	111
time_step/energy_semi_implicit.f90	semi-implicit energy solver	108
nonlinear_system_solver/System_HeH_mol_metals.f90	molecular + metals merged ionization system	105
lower_atmosphere/h3p_cooling.f90	H_3^+ infrared cooling (Miller et al. 2013)	104
lower_atmosphere/mol_rates.f90	molecular reaction rates (Koskinen et al. 2022, Table 1)	103
lower_atmosphere/lower_column.f90	analytic lower-atmosphere column (Koskinen et al. 2022)	90
radiation/lya_rt.f90	Ly α escape-probability RT (+ absorbing-bottom channel)	88
files_IO/opacity_input_read.f90	parse opacity.inp	85
flux/low_mach_dissipation.f90	gated fourth-difference contact-mode dissipation	79
nonlinear_system_solver/ion_cell_state.f90	named-field cell state of the ionization systems	77
radiation/hydrogen_n2_rates.f90	H($n=2$) rate coefficients (single source)	76
nonlinear_system_solver/System_HeH_TR_metals.f90	merged He 2^3S +metals system	68
time_step/steady_residual.f90	steady-residual assembly/norms	57
lower_atmosphere/lyman_werner.f90	H $_2$ Lyman–Werner photodissociation (Draine & Bertoldi 1996)	37
new-module total		6 324

So the 28 new core modules contribute 6 324 of the 13 457 added code lines; the remaining $\sim 7\,130$ added (and 699 removed) lines are edits spread across the 34 modified files — dominated by the cooling module (Cool_coeff.f90, 442 $\rightarrow$ 3 277 raw lines: CHIANTI metal-line fits, Fe tables, fine-structure statistical equilibrium, Penning and molec-

ular rate coefficients), followed by the ionization equilibrium and its diagnostics (ionization_equilibrium.f90, util_ion_eq.f90), the main driver, parameters/input parsing, IC construction, cross sections, and the post-processor, which wire the metals, molecules, opacity dispatcher, Roche potential, transonic/auto/Wind-AE IC, excited-H/Ly α physics, diffusion hooks, and the Brent/Newton solver upgrades into the existing solver. In total the EXHALE core is now $\approx 4.6\times$ the original code base (3 514 $\rightarrow$ 16 272 SLOC), touching most of the original files (34 of 47 modified, none deleted, plus 28 new); with the ported Wind-AE IC generator the full Fortran tree is 111 files / 21 018 SLOC ($\approx 6.0\times$). Beyond Fortran, the tree also carries the Python tooling (GUI, the EXHALE_transit.py transmission post-processor, plotting/analysis, the lower-atmosphere adapters and the elemental-flux closure driver) and $\approx 71\,000$ lines of documentation (94 md/tex files: the memos, the manual and this log) under docs/.

B Fortran source inherited unchanged from ATES

Of the 47 upstream Fortran files (ATES/ATES-Code-main), **13 are byte-for-byte identical** in EXHALE (re-measured 2026-08-30) — the numerics EXHALE reuses without touching: the included MINPACK nonlinear solver (the hybrid Powell hybrid method and its linear-algebra helpers, called by the ionization-equilibrium solver) and three small physics/setup pieces. Table 6 lists them with their role.

Five files left the byte-identical set since the 2026-07-02 edition of this table, each for a specific, documented reason rather than drift: time_step/RK_rhs.f90 gained the gated low-Mach dissipation hook (section 60); functions/UW_conversions.f90 and states/PLM_rec.f90 carry the state-array layout change (cells,3) $\rightarrow$ (3,cells) (2026-07 refactor); flux/Num_Fluxes.f90 gained a case default abort on an unknown flux name; and nonlinear_system_solver/System_implicit_adv_H.f90 was converted to the named-field cell state (ion_cell_state) with the rest of the System_* family. As before, the HLLC and ROE wave-speed estimators (flux/speed_estimate_HLLC.f90, speed_estimate_ROE.f90) are *effectively* untouched: their original logic is preserved verbatim and only a one-line defensive floor $p_{\min} = \max(p_{\min}, 1e-30 p_{\max})$ (plus its comment) was appended in the 2026-07-02 code review, so no original line was altered.

Table 6: The 13 Fortran files byte-identical between the original ATES (ATES-Code-main) and EXHALE (measured 2026-08-30), grouped by role.

File	Purpose
<i>MINPACK nonlinear-system solver</i> (Argonne; dir nonlinear_system_solver/)	
hybrd.f90	Powell hybrid method: find a zero of n nonlinear equations in n unknowns
hybrd1.f90	Simplified user-facing driver for hybrd
fdjac1.f90	Forward-difference approximation of the Jacobian
qrfac.f90	QR factorization via Householder transformations
qform.f90	Accumulate the orthogonal factor Q from the QR factorization
r1updt.f90	Rank-one update of the QR factorization
r1mpyq.f90	Apply the Givens-rotation sequence of the rank-one update
dogleg.f90	Dogleg (trust-region) step combining Gauss–Newton and steepest descent
enorm.f90	Euclidean norm of a vector (overflow/underflow safe)
dpmpar.f90	Machine floating-point precision / range constants
<i>Hydrodynamics, grid & radiation</i>	
states/Source.f90	Gravitational + geometric source terms
init/set_gravity_grid.f90	Pre-evaluate gravity at cell centers and interfaces
radiation/J_inc.f90	Energy-dependent incident stellar spectrum for photoionization